Species leads: plain-language explanations (v5, drafts for review)

AI-assisted drafts for review. Nothing here has been verified by a person.

How to read this. Explanations are built up but read down: the summary first, then each part from the whole model down, in plain words, each sentence followed by the claims it rests on (C01, C02, ...; every cited claim was computed on the model and checked). Numbers are changes in predicted risk for a typical patient. The technical record (original hypotheses, formulas, evidence tables with the test check, ranges and n) is at the end. The text describes what the model does, not the world; whether it is plausible is for the reviewer to judge.

What changed in v5e (figures). v5e: every diagram is a numbered figure with a caption (list of figures after the contents) and the text refers to it; the whole-network figure has its own landscape page, scaled to the page, with a short caption and its description in the text; an input used by several parts is drawn under each of them (same input, repeated for readability, dotted outline); and an appendix figure shows the trained network before any explanation (no parts, no splits).

What changed in v5c (tables and diagrams). v5c: the evidence tables are landscape, compact and shaded by row, with the header repeated on every page (longtable and pdflscape, used in this review PDF only; the thesis preamble has neither). In the diagrams each input is drawn once, in a shared row along the bottom (the explanation's split copies of an input are listed in the technical detail instead), every cluster has one output rectangle with its plain name, the inputs that feed the output directly meet in a drawn 'sum', and a one-line legend explains the edges.

What changed in v5b (diagrams). v5b uses one grouping everywhere, the annotation hierarchy: the summary's diagram is the same picture as the whole model's section, and a part that is itself a weighted sum (blood pressure with cholesterol) is a part of its own, described as one that adds up, so the text counts three parts throughout; the lint checks every 'N parts' against the hierarchy. A full-structure diagram follows the summary: every node and connection, nested by part, with the inputs that feed the output directly in their own cluster and the unused input columns listed. Each section from the top down adds its own region of that diagram, with stubs for connections from or to other parts; each connection is drawn in exactly one section. The gate's text now says that it lowers the risk for the patients it acts on, and that more vessels and higher cholesterol take that lowering away; its vessel statements are one per group.

What changed in v5 (read down). v5 reads the explanation down. The summary (one line, a diagram, one paragraph per block) is as in v4, except that the paragraph on what adds up is split into protective items, risk-raising items and rare results, and an input that also feeds a part is quoted with its direct contribution only (the rest is in that part's paragraph). Then the document walks the annotation hierarchy from the whole model down: for each part a plain hypothesis (what it does for the prediction), plain evidence (what was checked and what it showed; for a part made of others, what they are and how they combine), a diagram one level down and where to read on. The original hypotheses, closed forms and evidence tables are in 'Technical detail' at the end. Statements about structure (what a part is made of, a copy of another part, a formula that reproduces it) are now claims of their own, checked on the graph and the data.

v4. v4 follows the structure of the explanation. The summary opens with one line and a diagram at the same grouping, then gives one paragraph per top-level block of the annotation hierarchy: one for everything that adds up independently (the inputs that feed the model score directly, and parts that are a weighted sum), and one for each part that combines its inputs in another way: what it combines, how, and for whom. 'Parts in detail' follows the hierarchy below that, with a diagram one level down for each part that contains others. The prose no longer says 'clearly', 'slightly', 'consistently' or 'on the training data only': the numbers carry it, and the test check, the ranges and n stay in the evidence tables. Only genuinely small groups are marked, with their n. The lint rejects the confidence words and checks the block structure.

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383 The whole model: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. 406
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385 Fixed thallium defect with normal sugar: what feeds it, one level down. 407
386 Fixed thallium defect with normal sugar: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. 408

387 Fixed thallium defect with normal sugar: The predicted value for the typical case at each level, through this part (n = training patients; grey: fewer than 20). 408
388 Fixed thallium defect with normal sugar: This part's own output, unit h3 (the output of 'Fixed thallium defect with normal sugar'), on its own scale (its activation), with every other input at the typical case; it is active for 10 of 242. Its output for each combination of the levels it reads. 408
389 Fixed thallium defect with normal sugar: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom). 409
390 Chest pain offset: what feeds it, one level down. 409
391 Chest pain offset: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. 409
392 Chest pain offset: The predicted value for the typical case at each level, through this part (n = training patients; grey: fewer than 20). 410
393 Chest pain offset: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom). 410
394 Downsloping ST offset: what feeds it, one level down. 410
395 Downsloping ST offset: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. 410
396 Downsloping ST offset: The predicted value for the typical case at each level, through this part (n = training patients; grey: fewer than 20). 411
397 Downsloping ST offset: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom). 411
398 Maximum heart rate relay: what feeds it, one level down. 411
399 Maximum heart rate relay: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. 412
400 Maximum heart rate relay: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual patients (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks). 412
401 Maximum heart rate relay: This part's own output, unit h1 (the output of 'Maximum heart rate relay'), on its own scale (its activation), with every other input at the typical case; it is active for 190 of 242. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data. 413
402 Maximum heart rate relay: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom). 413
403 The network as trained, before any explanation, in full. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. Not used (no connection): atypical angina, normal ECG, LVH on ECG, no exercise pain, thallium code 4, reversible defect. 427
404 The model at the level of the summary: one box per part. 429
405 Every unit and connection of the explained model, nested by part. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. 431
406 The whole model: what feeds it, one level down. 432
407 The whole model: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. 432
408 The whole model: Each input's change in the predicted value for the typical case across the whole model, largest first (orange: raises it, teal: lowers it). 433
409 ST depression and thallium signal: what feeds it, one level down. 433
410 ST depression and thallium signal: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. 434
411 ST depression and thallium signal: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual patients (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks). 434
412 ST depression and thallium signal: How the effect differs between groups: one curve per group, for the typical case of each group. 435
413 ST depression and thallium signal: The predicted value for the typical case at each level, through this part (n = training patients; grey: fewer than 20). 435

414 ST depression and thallium signal: This part's own output, unit h1 (the output of 'ST depression and thallium signal'), on its own scale (its activation), with every other input at the typical case; it is active for 116 of 242. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data. . . . 435
415 ST depression and thallium signal: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom). . . . 436
416 Blood pressure step: what feeds it, one level down. . . . 436
417 Blood pressure step: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. . . . 436
418 Blood pressure step: The predicted value for the typical case at each level, through this part (n = training patients; grey: fewer than 20). . . . 437
419 Blood pressure step: This part's own output, unit h2 (the output of 'Blood pressure step'), on its own scale (its activation), with every other input at the typical case; it is active for 140 of 242. Left: its output over its two main inputs (orange line: where it switches on; dots: the data); right: slices at three values of the second input. . . . 437
420 Blood pressure step: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom). . . . 438
421 Extra age term: what feeds it, one level down. . . . 438
422 Extra age term: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. . . . 438
423 Extra age term: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual patients (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks). . . . 439
424 Extra age term: This part's own output, unit h3 (the output of 'Extra age term'), on its own scale (its activation), with every other input at the typical case; it is active for 85 of 242. Left: its output over its two main inputs (orange line: where it switches on; dots: the data); right: slices at three values of the second input. . . . 439
425 Extra age term: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom). . . . 440
426 The network as trained, before any explanation, in full. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. Not used (no connection): male, sugar above 120, ST-T abnormal ECG, LVH on ECG. . . . 452
427 The model at the level of the summary: one box per part. . . . 454
428 Every unit and connection of the explained model, nested by part. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. . . . 456
429 The whole model: what feeds it, one level down. . . . 457
430 The whole model: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. . . . 457
431 The whole model: Each input's change in the predicted value for the typical case across the whole model, largest first (orange: raises it, teal: lowers it). . . . 458
432 ST depression signal: what feeds it, one level down. . . . 458
433 ST depression signal: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. . . . 459
434 ST depression signal: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual patients (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks). . . . 459
435 ST depression signal: How the effect differs between groups: one curve per group, for the typical case of each group. . . . 460
436 ST depression signal: The predicted value for the typical case at each level, through this part (n = training patients; grey: fewer than 20). . . . 460
437 ST depression signal: This part's own output, unit h1 (the output of 'ST depression signal'), on its own scale (its activation), with every other input at the typical case; it is active for 109 of 242. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data. . . . 460
438 ST depression signal: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom). . . . 461

439 Blood pressure step: what feeds it, one level down. 461
440 Blood pressure step: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. 461
441 Blood pressure step: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual patients (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks). 462
442 Blood pressure step: The predicted value for the typical case at each level, through this part (n = training patients; grey: fewer than 20). 462
443 Blood pressure step: This part's own output, unit h2 (the output of 'Blood pressure step'), on its own scale (its activation), with every other input at the typical case; it is active for 145 of 242. Left: its output over its two main inputs (orange line: where it switches on; dots: the data); right: slices at three values of the second input. 463
444 Blood pressure step: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom). 463
445 Damping term: what feeds it, one level down. 463
446 Damping term: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. 464
447 Damping term: The predicted value for the typical case at each level, through this part (n = training patients; grey: fewer than 20). 464
448 Damping term: This part's own output, unit h3 (the output of 'Damping term'), on its own scale (its activation), with every other input at the typical case; it is active for 234 of 242. Bold: the typical case; band: 10th to 90th percentile across individuals; ticks: the data. 465
449 Damping term: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom). 465
450 Hypertrophy without exercise pain: what feeds it, one level down. 465
451 Hypertrophy without exercise pain: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. 466
452 Hypertrophy without exercise pain: The predicted value for the typical case at each level, through this part (n = training patients; grey: fewer than 20). 466
453 Hypertrophy without exercise pain: This part's own output, unit h4 (the output of 'Hypertrophy without exercise pain'), on its own scale (its activation), with every other input at the typical case; it is active for 74 of 242. Its output for each combination of the levels it reads. 466
454 Hypertrophy without exercise pain: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom). 467
455 High fasting sugar term: what feeds it, one level down. 467
456 High fasting sugar term: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. 467
457 High fasting sugar term: The predicted value for the typical case at each level, through this part (n = training patients; grey: fewer than 20). 468
458 High fasting sugar term: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom). 468
459 The network as trained, before any explanation, in full. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. Not used (no connection): male, atypical angina, ST-T abnormal ECG, flat ST, reversible defect. 484
460 The model at the level of the summary: one box per part. 486
461 Every unit and connection of the explained model, nested by part. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. 488
462 The whole model: what feeds it, one level down. 489
463 The whole model: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. 489
464 The whole model: Each input's change in the predicted value for the typical case across the whole model, largest first (orange: raises it, teal: lowers it). 490
465 ST depression signal: what feeds it, one level down. 490

466 ST depression signal: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. 491
467 ST depression signal: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual patients (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks). 491
468 ST depression signal: How the effect differs between groups: one curve per group, for the typical case of each group. 492
469 ST depression signal: The predicted value for the typical case at each level, through this part (n = training patients; grey: fewer than 20). 492
470 ST depression signal: This part's own output, unit h1 (the output of 'ST depression signal'), on its own scale (its activation), with every other input at the typical case; it is active for 105 of 242. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data. 492
471 ST depression signal: This part's own output, unit h2 (combines 2 non-linear units), on its own scale (its activation), with every other input at the typical case; it is active for 96 of 242. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data. 493
472 ST depression signal: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom). 494
473 Age term: what feeds it, one level down. 494
474 Age term: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. 494
475 Age term: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual patients (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks). 495
476 Age term: This part's own output, unit h3 (the output of 'Age term'), on its own scale (its activation), with every other input at the typical case; it is active for 148 of 242. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data. . . 495
477 Age term: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom). 496
478 The network as trained, before any explanation, in full. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. Not used (no connection): male, atypical angina, sugar above 120, ST-T abnormal ECG, LVH on ECG, no exercise pain, flat ST. 505

1 churn100-s2-c1

Genome `f188e353-beb3-4b23-9e19-b64aba8cb5ba`, explanation `6eb705c3-d493-4a02-ab0a-6e01b8c5d696`. Split: train primary (n=4000), test as a check (n=1000). Claims: 164, verified 148, not supported 16.
AI-assisted draft for review (unverified)

Reference (typical) customer: the typical customer: the median training customer for every number and the most common training category for every category (not a real customer). account length: 100; area code: area code 415; international plan: no international plan; number customer service calls: 1 calls; number vmail messages: 0 messages; state: West Virginia; total day calls: 100 calls; total day charge: 30.69 US dollars; total day minutes: 180 minutes; total eve calls: 100 calls; total eve charge: 17.12 US dollars; total eve minutes: 201 minutes; total intl calls: 4 calls; total intl charge: 2.78 US dollars; total intl minutes: 10 minutes; total night calls: 100 calls; total night charge: 9.02 US dollars; total night minutes: 200 minutes; voice mail plan: no voice-mail plan. Predicted chance that a customer leaves: 1%.

1.1 Summary

For a typical customer (in West Virginia, area code 415, no international or voice-mail plan, 1 call to customer service and median usage) the model predicts a chance of leaving of about 1%; it adds up seven parts: fixed-weight inputs, service calls, a voice-mail discount, heavy evening use, state units, a daytime-calls unit and a small state unit. [C056] Figure 1 shows the model at this level: one box per paragraph below.

How to read the diagrams. Edge colour is the sign of that one connection: blue positive, red negative; thicker is a larger weight. The outline and arrow on an input (and on a part's output box) is its overall effect on the final prediction, read from the verified claims: orange and up arrow raises it, teal and down arrow lowers it, grey and double arrow depends (it goes both ways, or differs between levels). A negative connection into a part that itself raises the prediction means that input lowers the prediction; through a hinge the overall effect can only be read from the claims, not from the edges. Up arrow means towards churn; down arrow, away from it. A double outline marks an input (or a part's output) that acts at a fixed rate everywhere: it adds the same amount to the score per unit, so it is explained by its rate alone (see 'Linear factors'); a dashed double outline, at a fixed rate on the observed data. The normal outline is an input whose effect switches on and off somewhere. For example, total daytime minutes has a negative connection into 'Service calls', a part that raises the predicted chance that a customer leaves, so total daytime minutes lowers it.

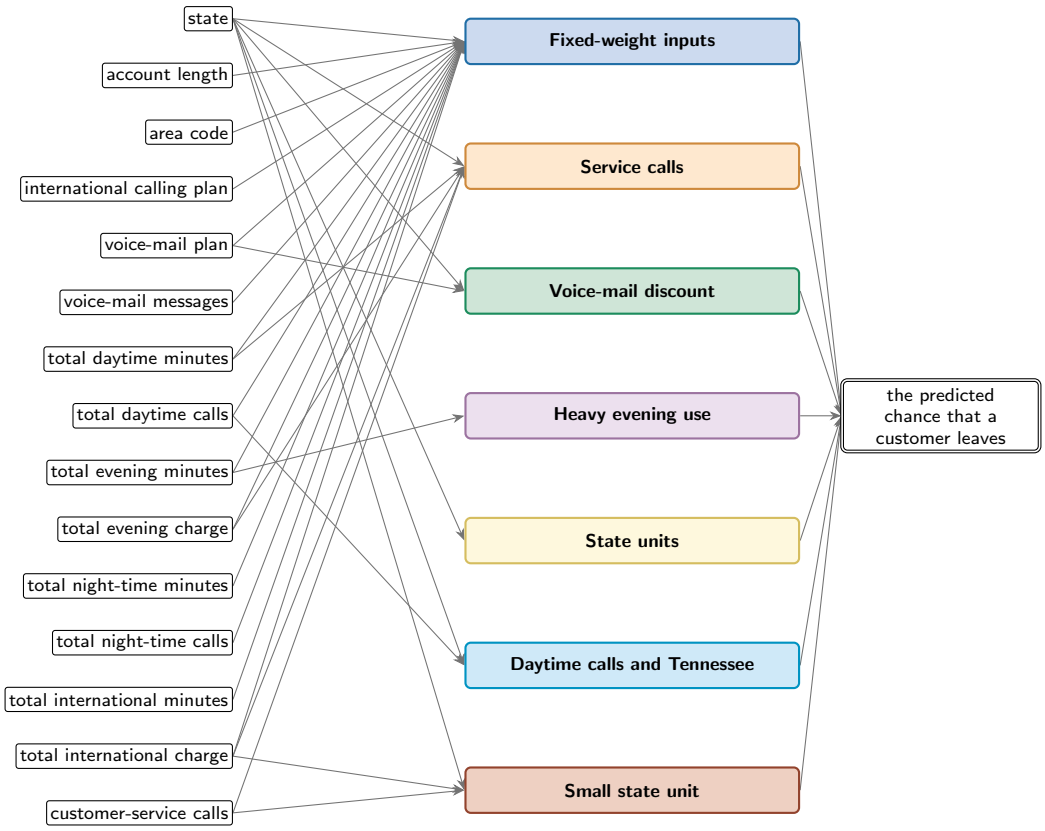


Figure 1: The model at the level of the summary: one box per part.

Fixed-weight inputs. This part adds up most inputs with fixed weights, and daytime minutes count most: they raise a typical customer's chance of leaving steadily, from under 1% at 112 to about 13% at 249 minutes. [C057] An international plan takes the chance from about 1% to about 13%, a voice-mail plan lowers it to under 1%, and more voice-mail messages raise it steadily, from about 1% at 0 to about 5% at 32 messages. [C058, C067, C059] A higher international charge, more evening minutes, a higher evening charge and more night-time minutes also raise it steadily, each by one to two points. [C062, C063, C064, C070] Through it, the state moves the chance from about 1% in West Virginia to about 4% in California and in Washington, and to under 1% in Hawaii, Vermont and Colorado (41 to 80 training customers per state). [C060, C061, C066, C068, C071]

Service calls. This part adds up two parts: customer-service calls combined with low daytime use, and a rise from 4 calls upwards. [C111] Through it, daytime minutes lower the chance only while they stay below about 170 minutes (between 147 and 216 minutes for different customers), in an uneven way: for a typical customer, from about 38% at 91 minutes to about 1% at 352 minutes. [C107] With an evening charge above 17.12 US dollars, the chance is lowest at about 154 minutes of daytime use and higher either side: from about 1% at 112 minutes to about 17% at 249 minutes, against about 2% to about 10% at or below that charge. [C103] Customer-service calls add to it from 4 calls upwards, in an uneven way: for a typical such customer, from about 3% at 0 calls to about 11% at 4 calls. [C108]

Voice-mail discount. This part reads the voice-mail plan and a few states: through it, a voice-mail plan lowers a typical customer's chance by less than a point, and customers in states other than Alabama, North Carolina and West Virginia get about 2% instead of about 1%. [C125, C126]

Heavy evening use. Through this part, evening minutes add nothing until about 239 minutes, then lower the chance steadily; it acts on 885 of 4000 training customers, 165 of whom left, and for the typical customer it acts on, the chance is about 5% without it and about 4% with it. [C135, C136]

State units. This part reads only the state and moves the chance by less than half a point: up in Arizona and Ohio, down in Nevada. [C138, C139, C140]

Daytime calls and Tennessee. This part reads the state and daytime calls: being in Tennessee raises a typical customer's chance from about 1% to about 2%, and daytime calls barely change it. [C144, C145]

Small state unit. This small part moves the chance by less than a tenth of a point, up in Arkansas and Utah. [C150, C153]

The whole network. Figure 2 (next page, landscape) draws every unit and connection of the explained model, nested by part: shaded clusters are the parts of the model and darker boxes the parts inside them; circles are hidden units (h1, h2, ...); each cluster's rectangle is its output, and a dashed 'sum' stands for an addition the next unit does (drawn, not a unit of the model); the hexagon is the model's output. An input used by several parts is drawn under each of them (same label, dotted outline, the first copy marked with the number of copies): it is the same input, repeated for readability. Not used by the model (input columns with no connection): state: 20 of 51 levels, total daytime charge, total evening calls, total night-time charge, total international calls.

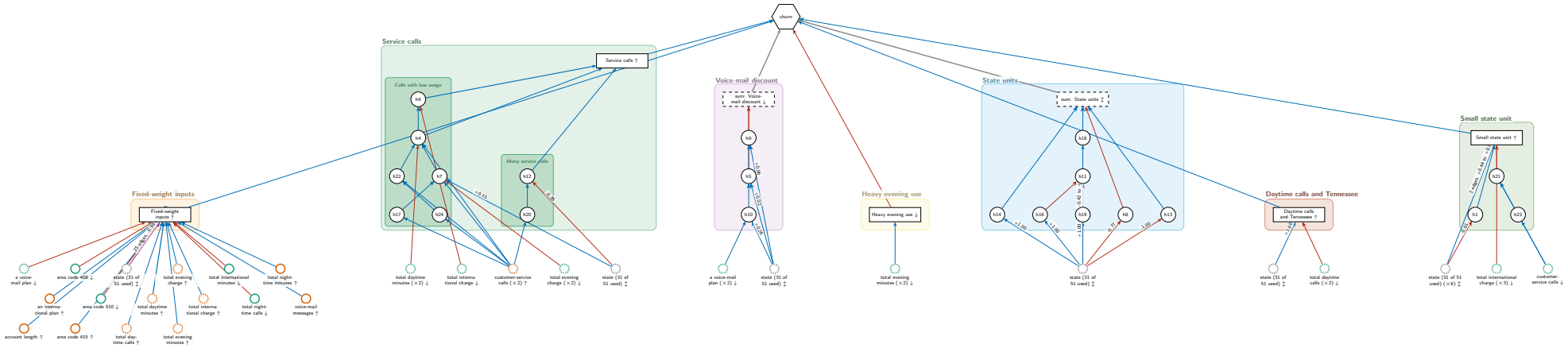


Figure 2: Every unit and connection of the explained model, nested by part. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Worth checking against what is known. Worth checking: the state moves the chance by several points (for example California and Washington about 6% and 5%, Hawaii under 1%), based on 41 to 122 training customers per state. [C003, C004, C016] Worth checking: the model treats a voice-mail plan as lowering the chance of leaving but more voice-mail messages as raising it. [C012, C005] Worth checking: the many-service-calls part is switched off for customers in West Virginia, the state of the typical customer. [C130] Worth checking: daytime minutes, the evening charge and the international charge raise the chance through the fixed-weight inputs but lower it through calls with low usage. [C057, C112, C064, C115, C062, C114]

1.2 The model, read from the top down

Each part of the explanation in turn, from the whole model down to the smallest parts: what it does for the prediction (hypothesis), what was checked (evidence), a diagram one level down, and where to read on. The technical record of each part is in "Technical detail" at the end.

1.2.1 The whole model

Figure 3 shows what feeds this part; Figure 4 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

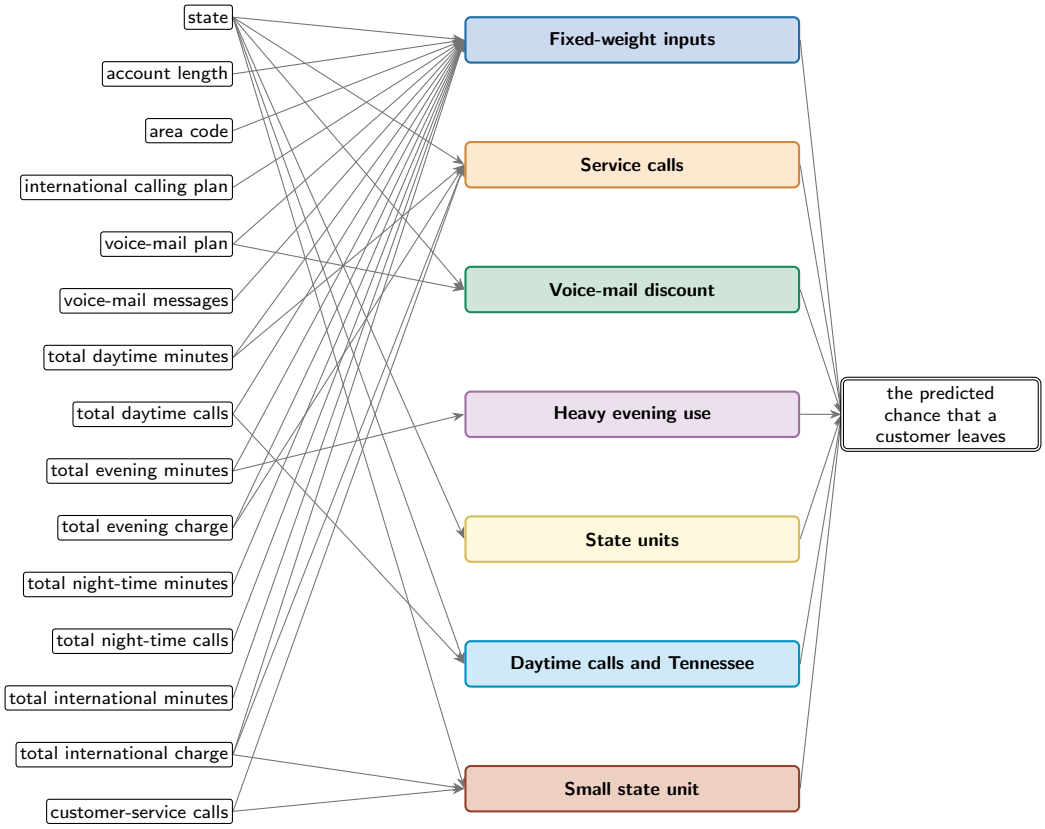


Figure 3: The whole model: what feeds it, one level down.

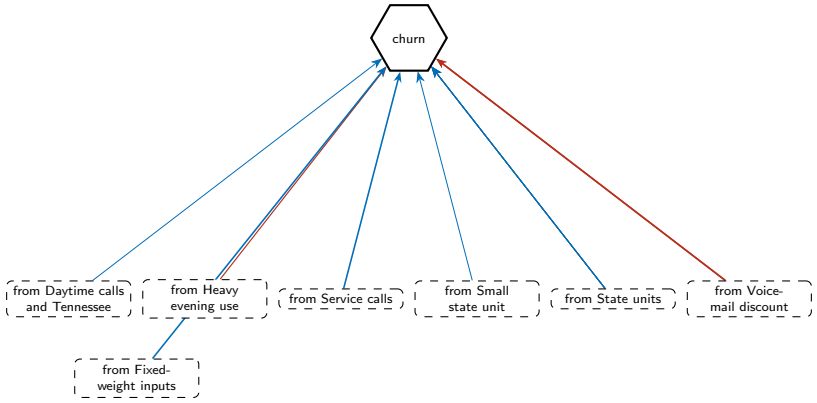


Figure 4: The whole model: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. The model adds up seven parts into one score: fixed-weight inputs, service calls, a voice-mail discount, heavy evening use, state units, a daytime-calls unit and a small state unit. [C056]

Evidence. It is made up of these seven parts (each explained below); the score is their sum, which was checked on all 4000 training customers. [C056] Changing one input at a time across the whole model, an international plan takes a typical customer's chance of leaving from about 1% to about 13%, and a voice-mail plan lowers it to under 1%. [C001, C012] More customer-service calls raise it in an uneven way, from about 1% at 0 calls to about 3% at 3 calls. [C008] The model ignores the daytime charge, evening calls, international calls and the night-time charge. [C052, C053, C054, C055]
Evidence plots: Figure 5, Figure 6.

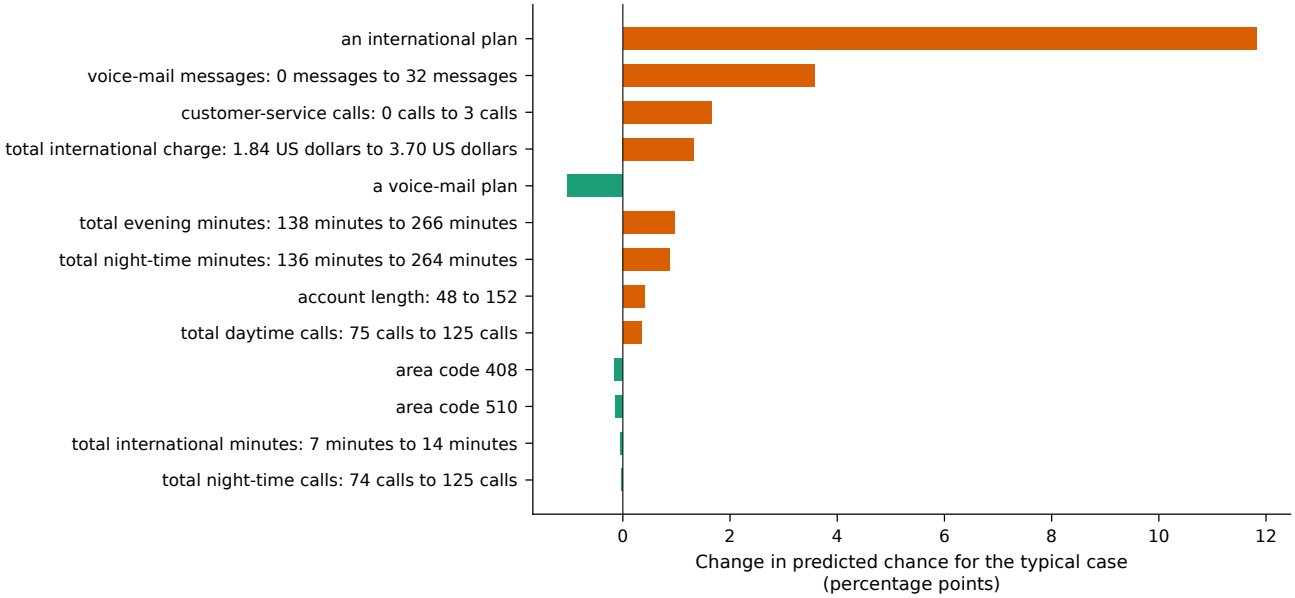


Figure 5: The whole model: Each input's change in the predicted value for the typical case across the whole model, largest first (orange: raises it, teal: lowers it).

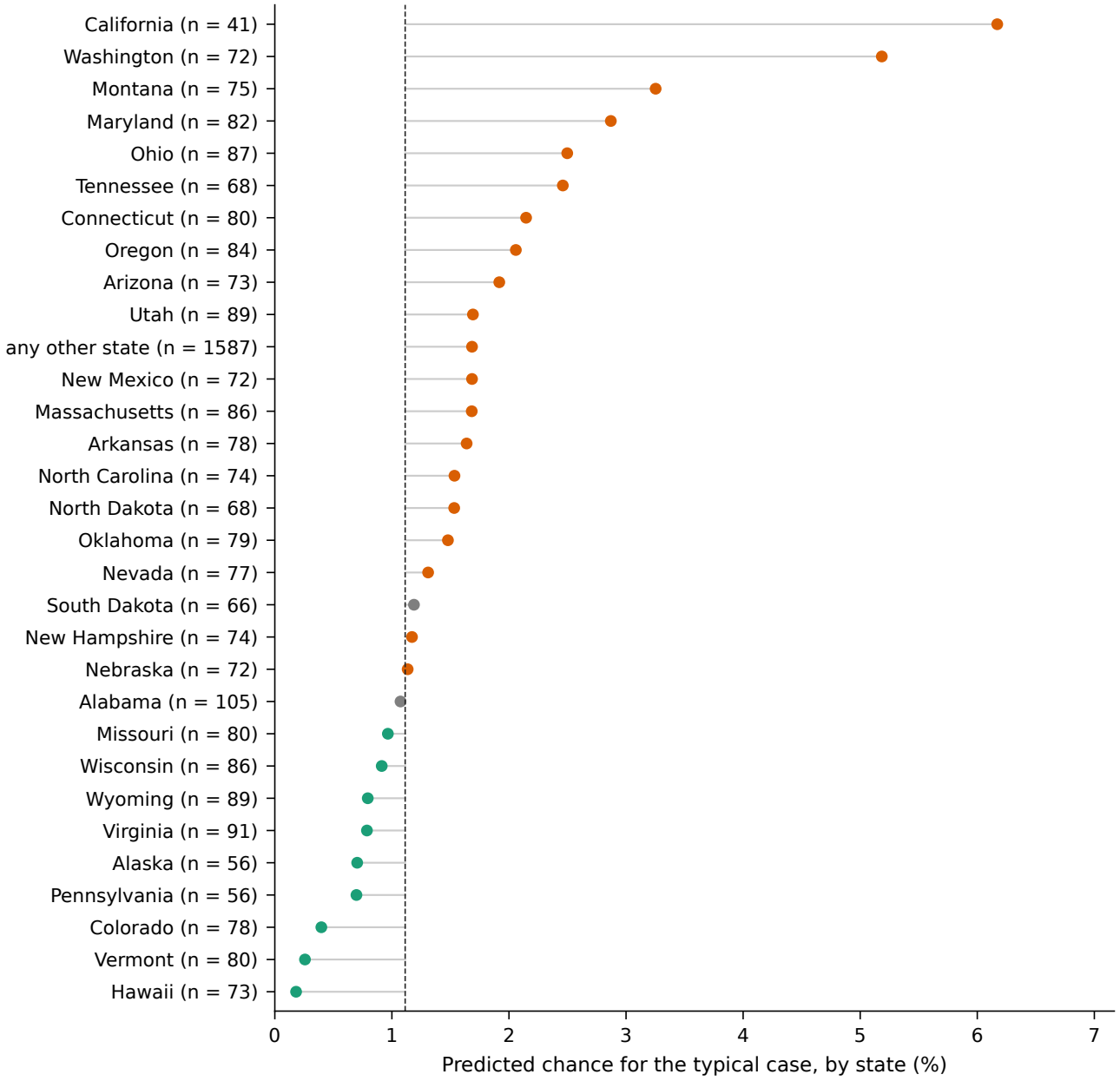


Figure 6: The whole model: Every level of state the model reads: the predicted value for the typical case at that level (dashed: the reference level); n = training customers; grey: few customers or not supported.

Dive deeper: Fixed-weight inputs (1.2.2); Service calls (1.2.3); Voice-mail discount (2.2.7); Heavy evening use (2.2.8); State units (2.2.3); Daytime calls and Tennessee (2.2.1); Small state unit (2.2.6). Technical detail: T.1.

1.2.2 Fixed-weight inputs

Part of: The whole model (1.2.1).
Figure 7 shows what feeds this part; Figure 8 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

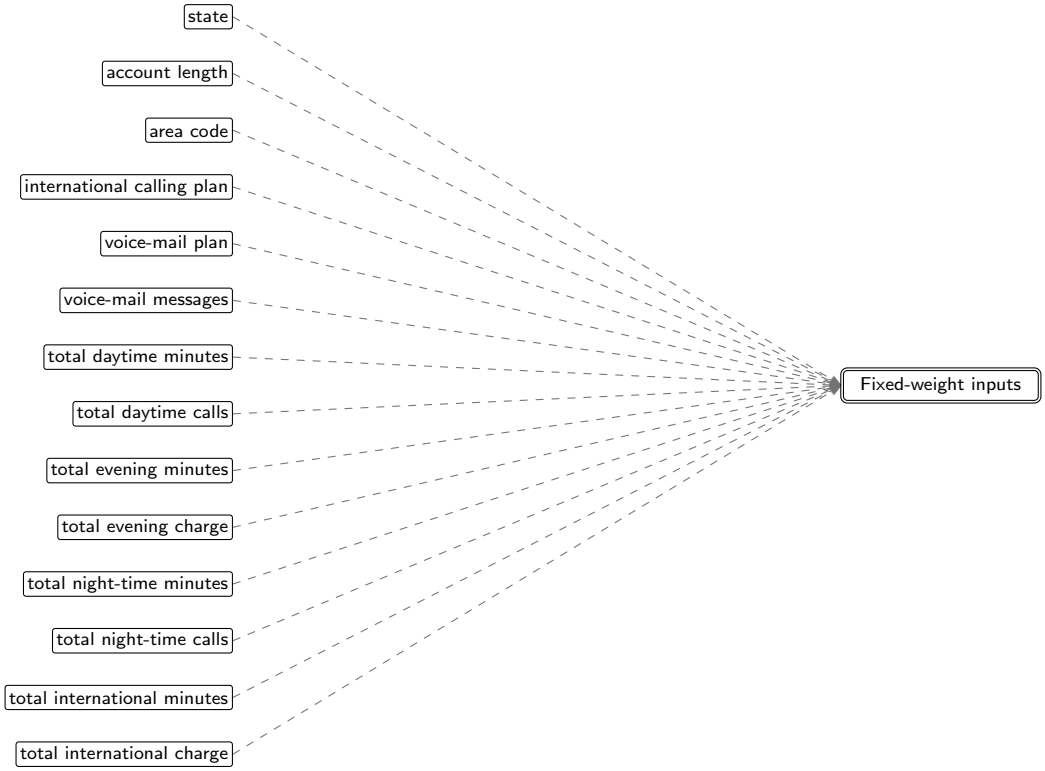


Figure 7: Fixed-weight inputs: what feeds it, one level down.

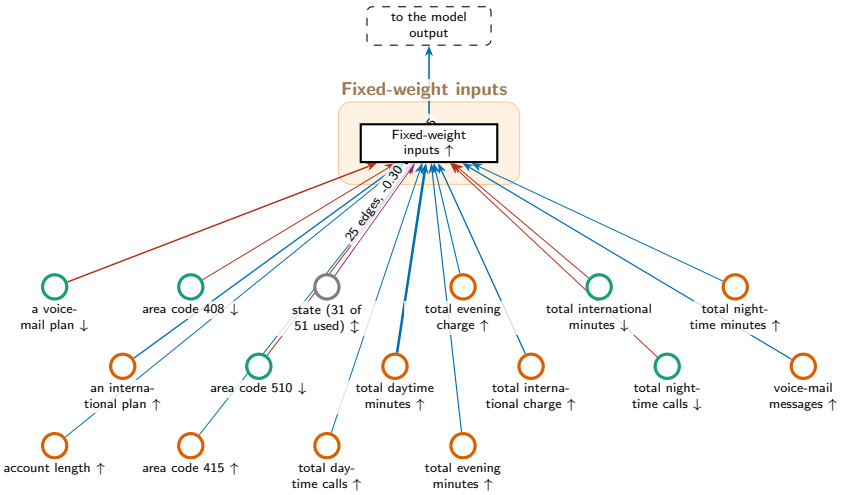


Figure 8: Fixed-weight inputs: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part adds up most inputs with fixed weights: daytime minutes, an international plan and voice-mail messages raise the chance of leaving, steadily for the minutes and messages, and a voice-mail plan lowers it. [C057, C058, C059, C067]

Evidence. Through it, daytime minutes take a typical customer steadily from under 1% at 112 to about 13% at 249 minutes, and an international plan from about 1% to about 13%; a voice-mail plan lowers the chance to under 1%. [C057, C058, C067] The state moves it from about 1% in West Virginia to about 4% in California and Washington, and to under 1% in Hawaii and Vermont. [C060, C061, C066, C068] Its formula reproduces the part exactly on all 4000 training customers. [C096]
Evidence plots: Figure 9, Figure 10, Figure 11, Figure 12.

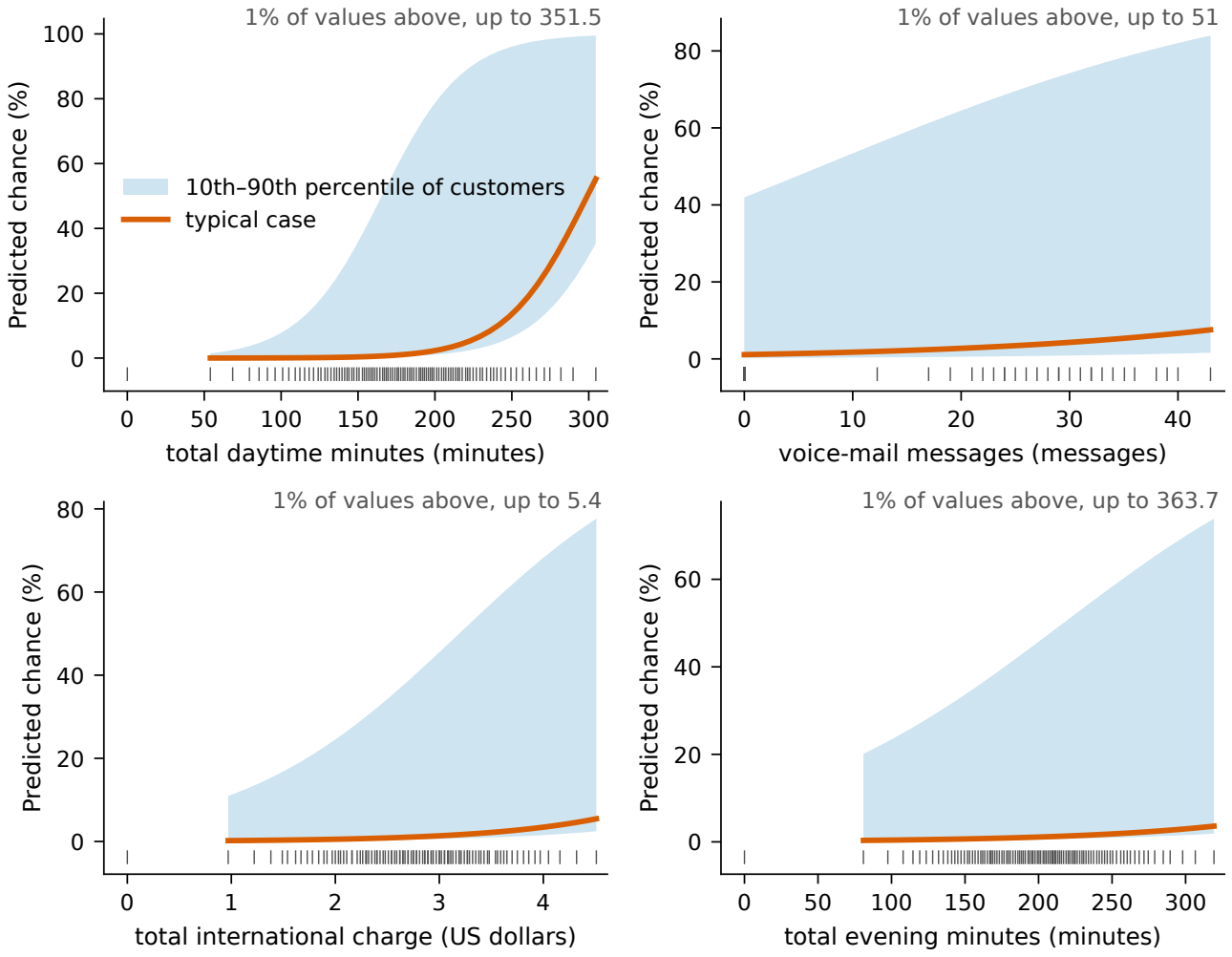


Figure 9: Fixed-weight inputs: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

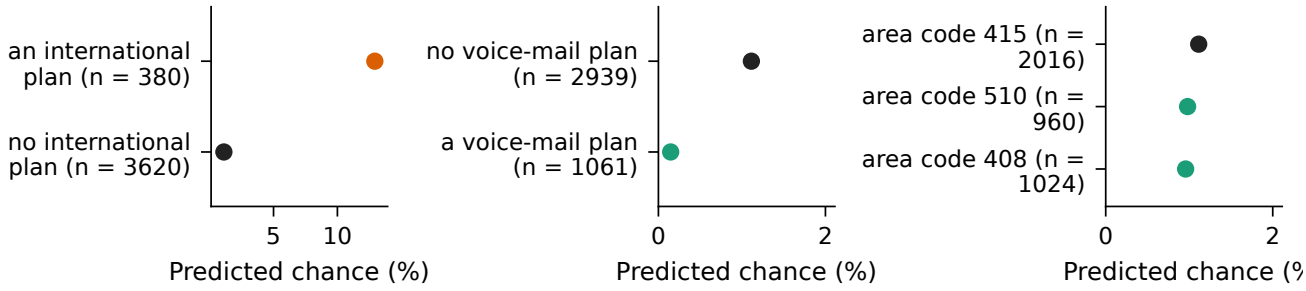


Figure 10: Fixed-weight inputs: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

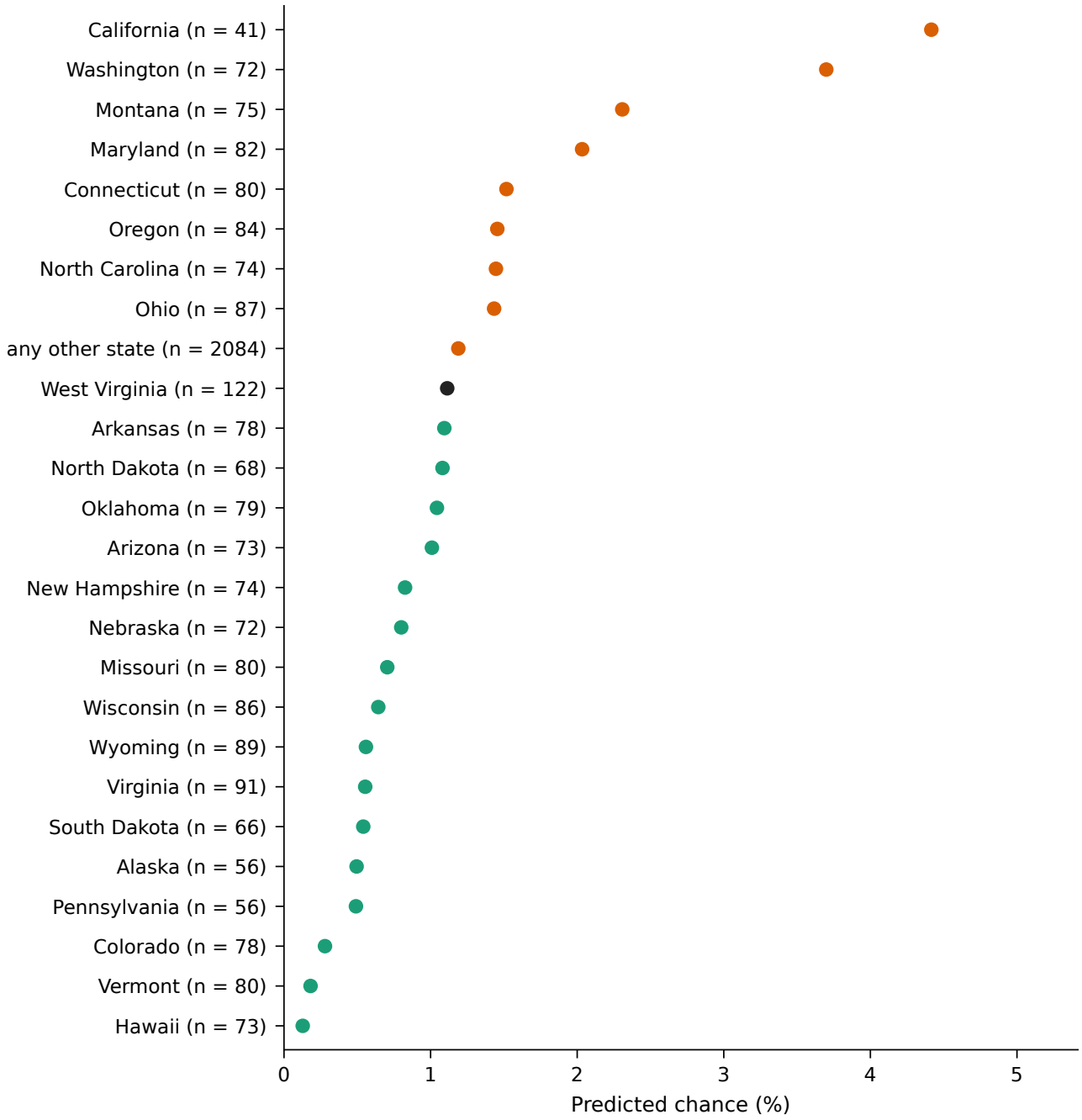


Figure 11: Fixed-weight inputs: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

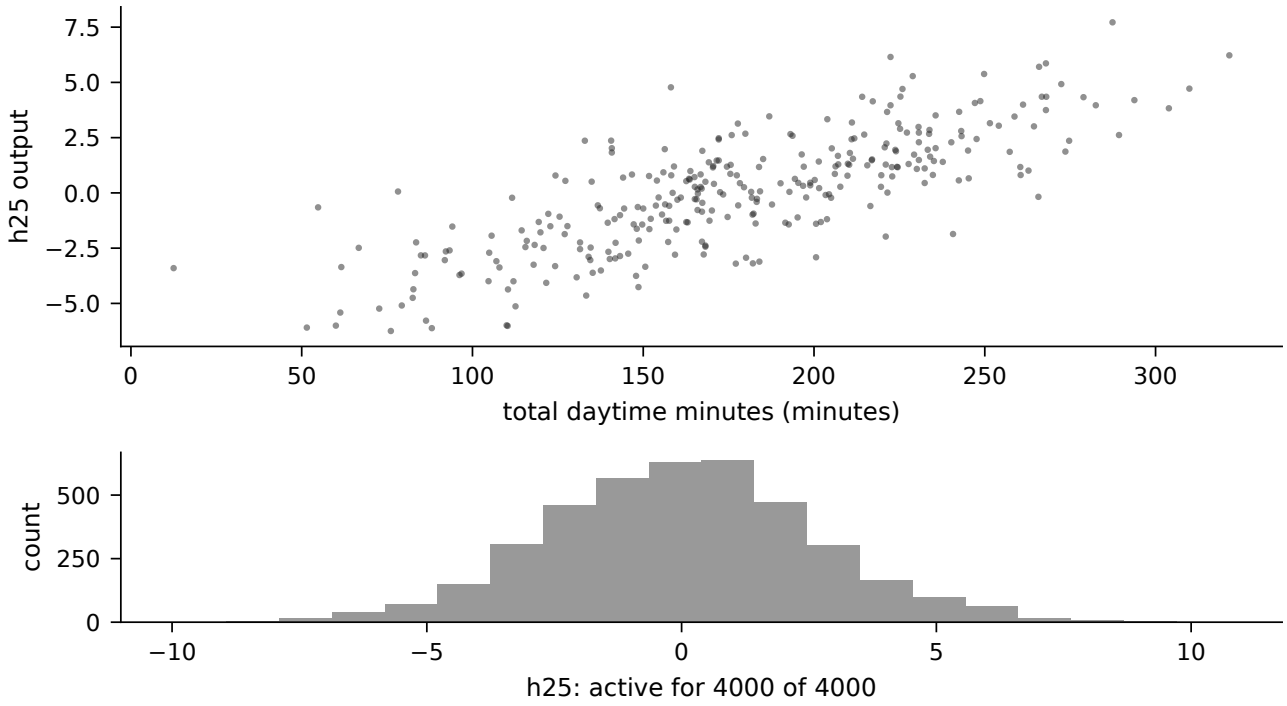


Figure 12: Fixed-weight inputs: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.2.

1.2.3 Service calls

Part of: The whole model (1.2.1).
Figure 13 shows what feeds this part; Figure 14 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

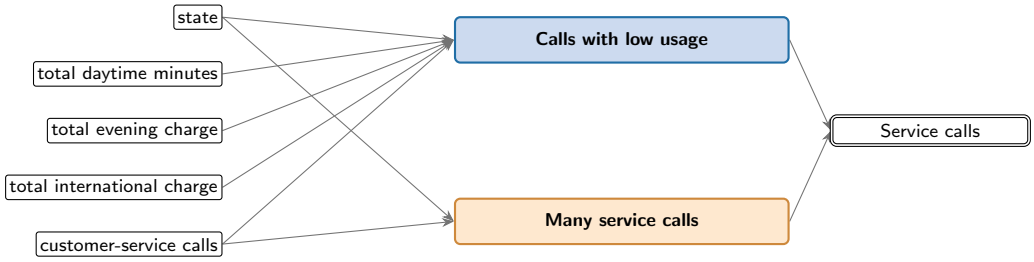


Figure 13: Service calls: what feeds it, one level down.

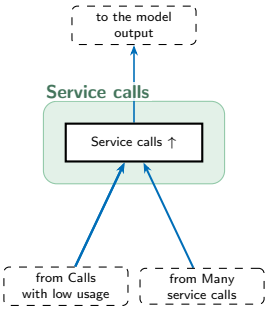


Figure 14: Service calls: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part raises the chance of leaving with calls to customer service, from 4 calls upwards and in an uneven way, and more so for customers with low daytime use; it adds up two parts. [C111, C108, C107]

Evidence. It is made of two parts that are added: calls with low usage, and many service calls. [C111] Through it, daytime minutes lower a typical customer's chance only while they stay below about 170 minutes (between 147 and 216 minutes for different customers), in an uneven way, from about 38% at 91 minutes to about 1% at 352 minutes. [C107] Customer-service calls add to it from 4 calls upwards, in an uneven way: for a typical such customer, from about 3% at 0 calls to about 11% at 4 calls. [C108] With an evening charge above 17.12 US dollars, the chance is lowest at about 154 minutes of daytime use and higher either side: from about 1% at 112 minutes to about 17% at 249 minutes, against about 2% to about 10% at or below that charge. [C103]
Evidence plots: Figure 15, Figure 16, Figure 17, Figure 18, Figure 19, Figure 20, Figure 21, Figure 22.
Figure 21 shows this part's own output, unit h26: over its two main inputs, with the line where it switches on.

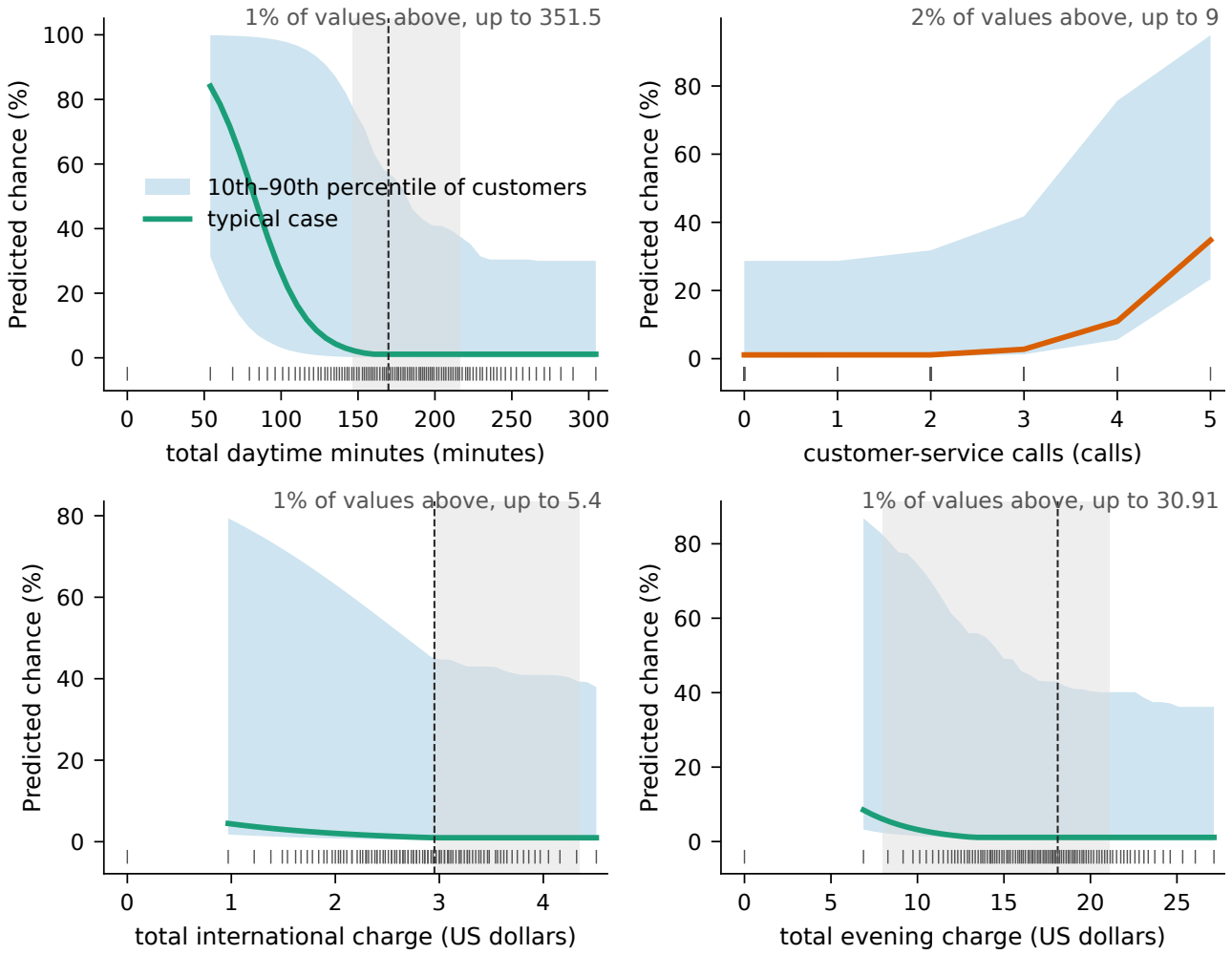


Figure 15: Service calls: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

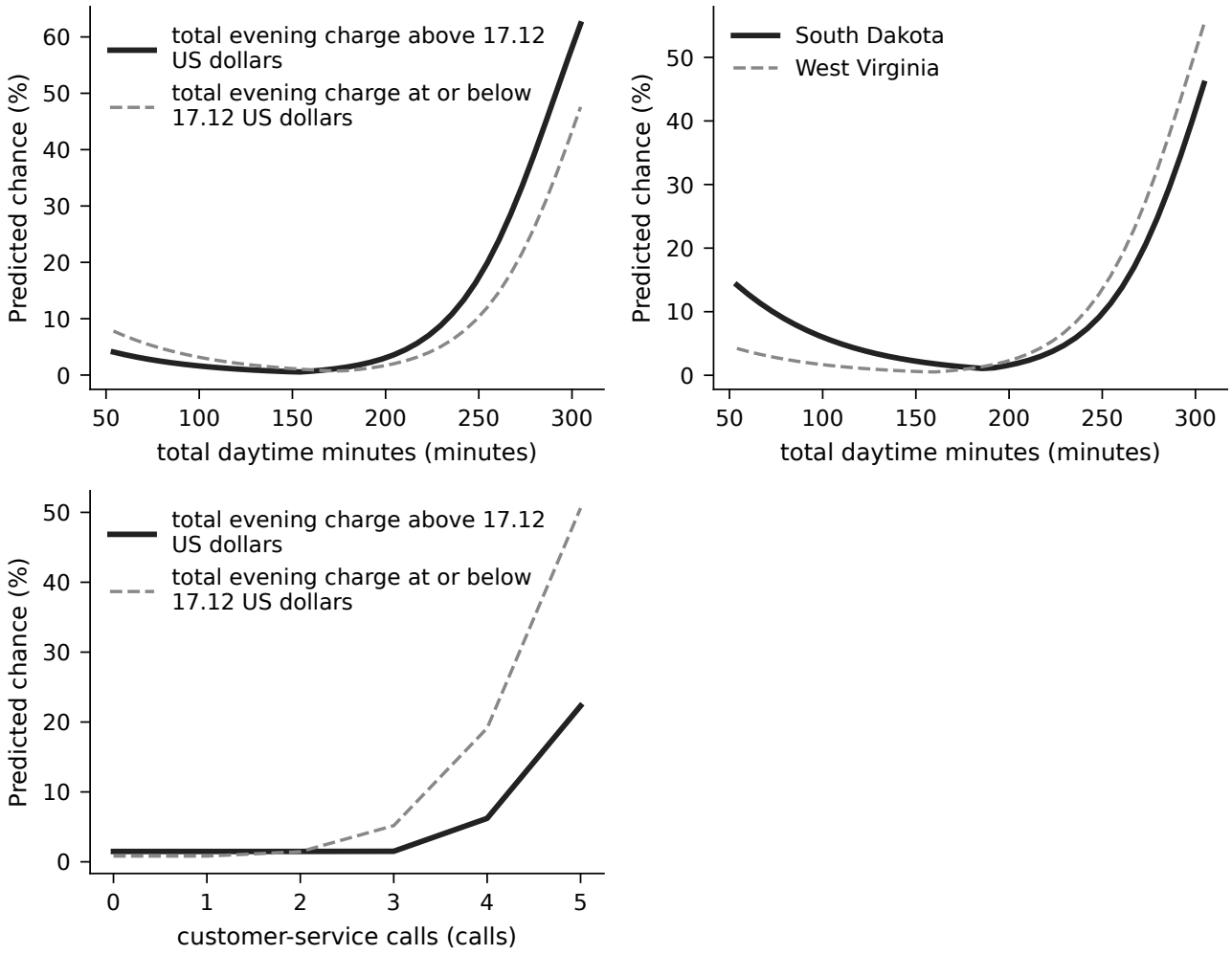


Figure 16: Service calls: How the effect differs between groups: one curve per group, for the typical case of each group.

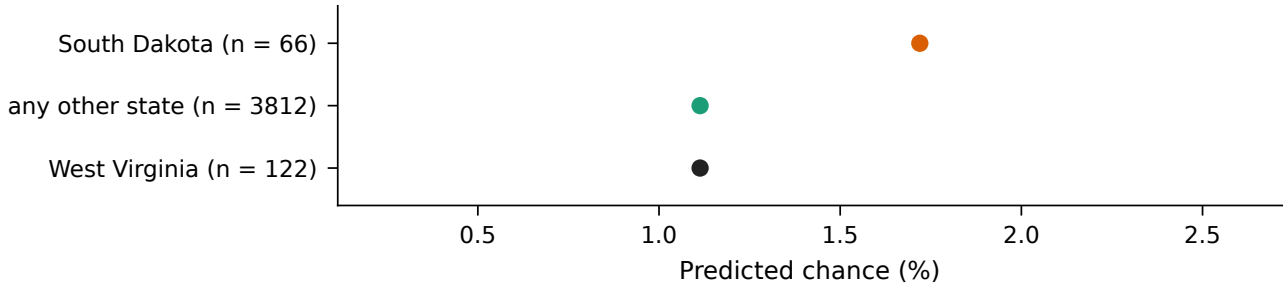


Figure 17: Service calls: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

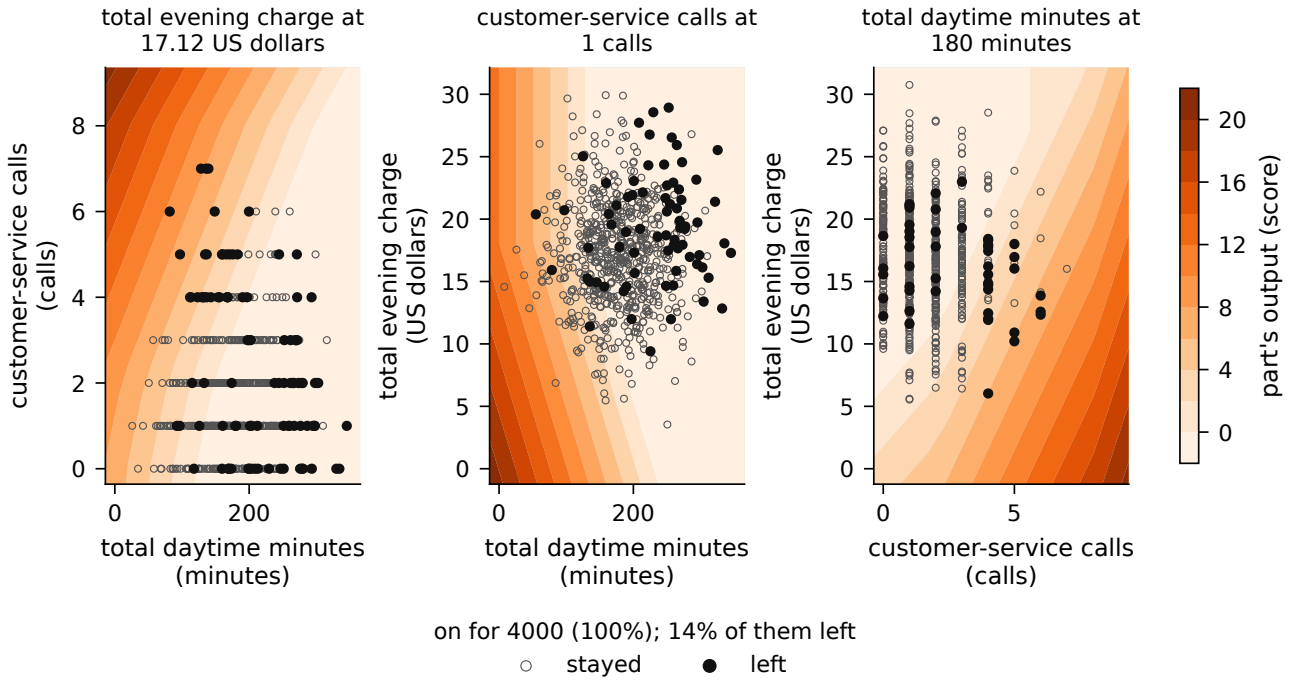


Figure 18: Service calls: The part's own output over each pair of its three main inputs (the third held at its median, named above each panel; the points are the training rows whose third input is between its 40th and 60th percentile), other inputs at the typical case: colour where it is on, flat grey where it is switched off, the black line where it switches on; training customers as hollow (no event) and filled (event) points, so the part can be seen catching the events (the predicted-chance overlay was harder to read). The switch-on line is traced on this map only; the boundary claim (C162) did not pass its check on the data, so no rule is stated.

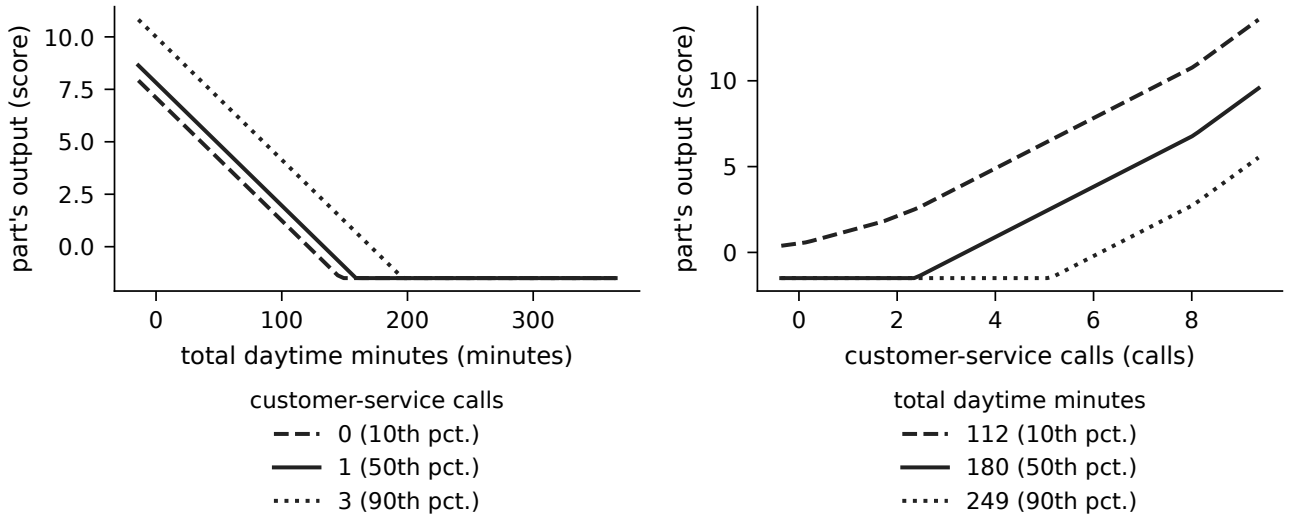


Figure 19: Service calls: Slices of the same map: the part's own output against each input at three values of the other, so the tilted boundary shows as a shifting hinge. The switch-on line is traced on this map only; the boundary claim (C162) did not pass its check on the data, so no rule is stated.

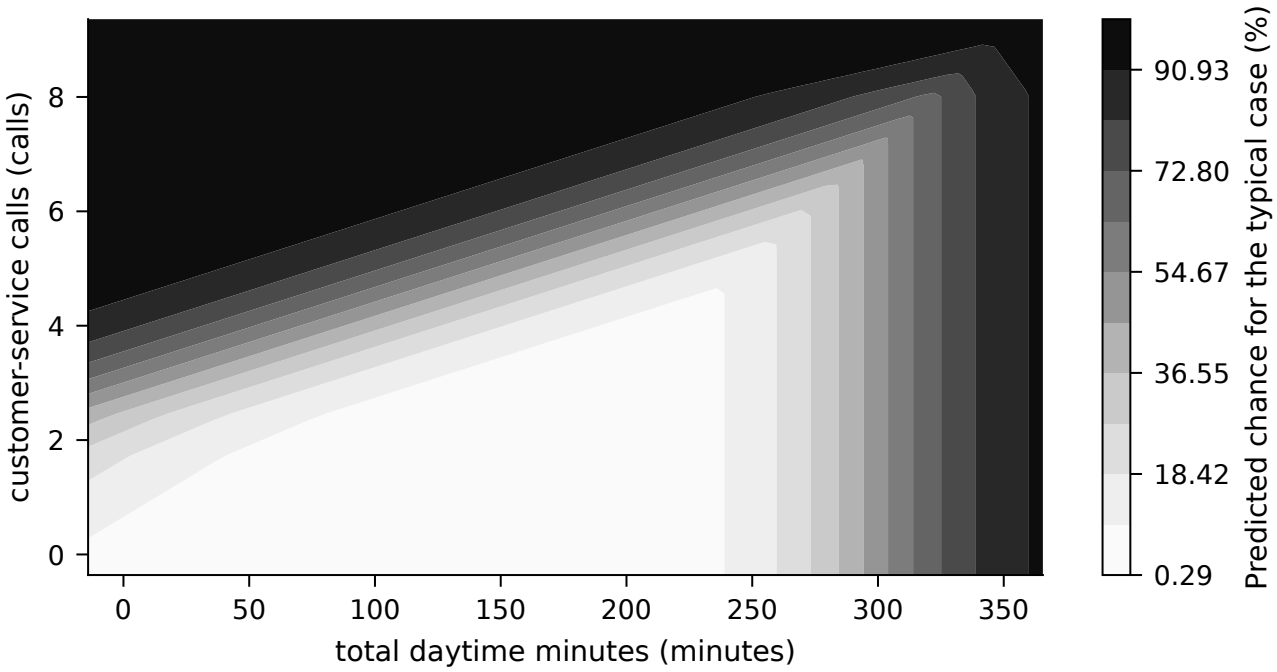


Figure 20: Service calls: The same map on the final scale: the predicted value for the typical case, with the switch-on line (orange). The switch-on line is traced on this map only; the boundary claim (C162) did not pass its check on the data, so no rule is stated.

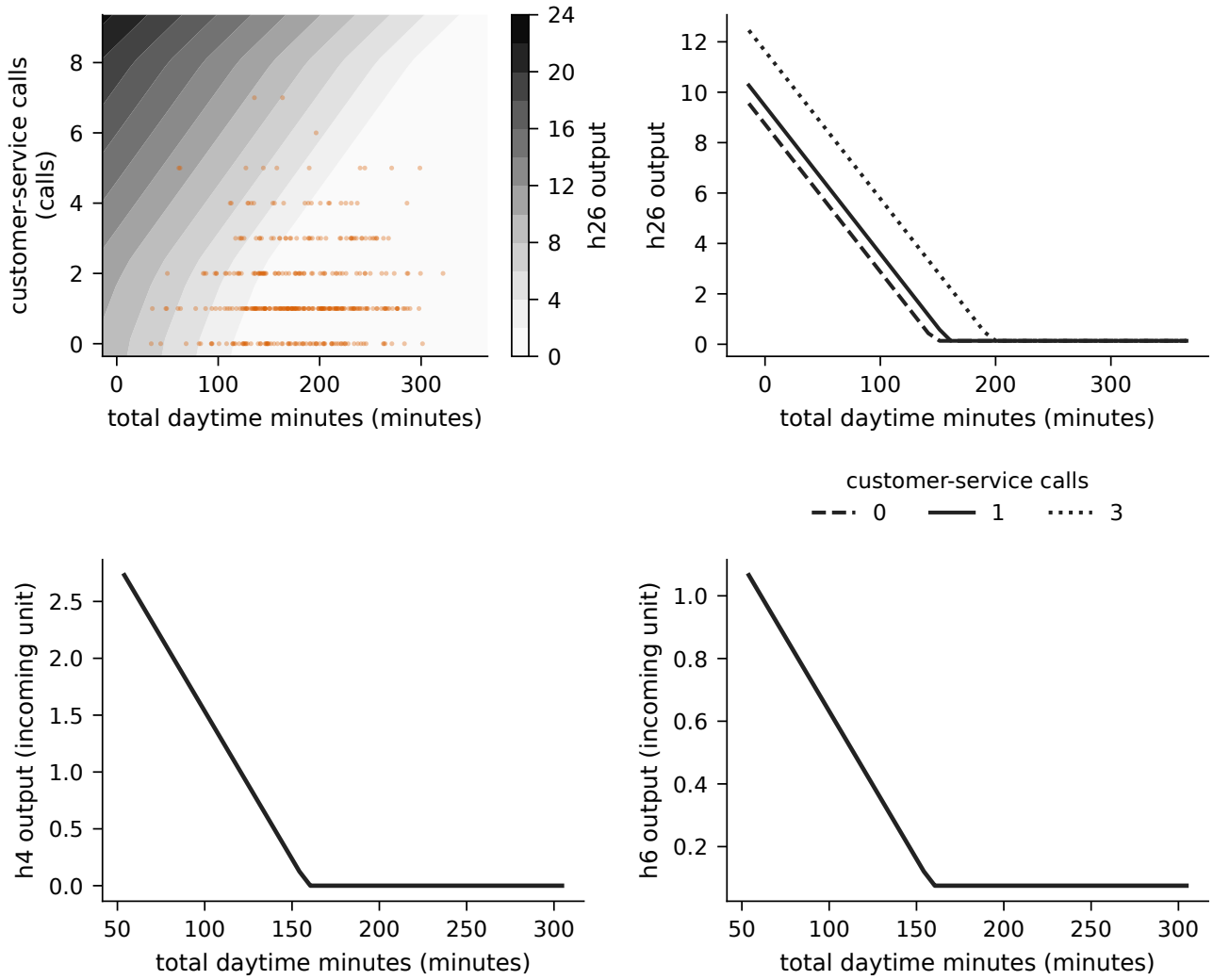


Figure 21: Service calls: This part's own output, unit h26 (combines 3 non-linear units), on its own scale (its activation), with every other input at the typical case; it is active for 3145 of 4000. Left: its output over its two main inputs (orange line: where it switches on; dots: the data); right: slices at three values of the second input; then each incoming non-linear unit against its own main input.

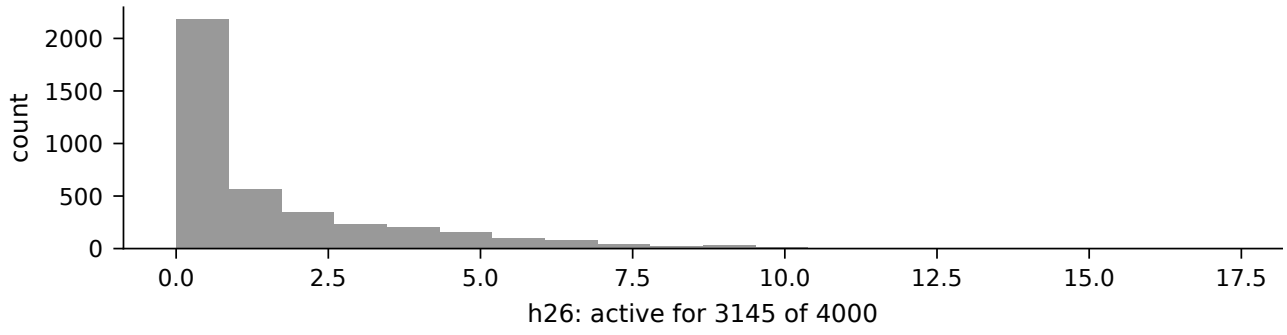


Figure 22: Service calls: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Dive deeper: Calls with low usage (2.2.5); Many service calls (2.2.9). Technical detail: T.3.

1.2.4 Calls with low usage

Part of: Service calls (1.2.3).
Figure 23 shows what feeds this part; Figure 24 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

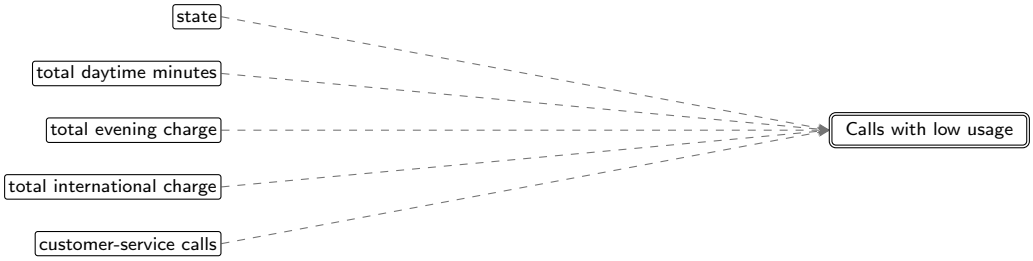


Figure 23: Calls with low usage: what feeds it, one level down.

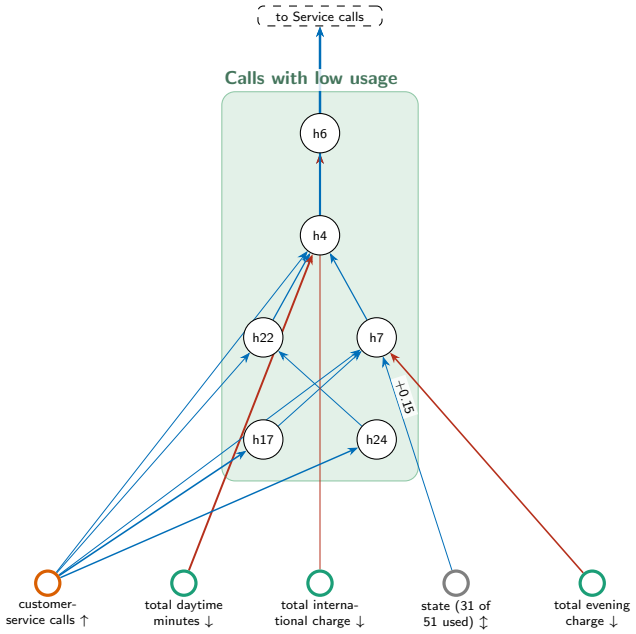


Figure 24: Calls with low usage: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part raises the chance of leaving for customers with low usage, more so when they call customer service: daytime minutes lower it only while they stay below about 170 minutes (between 147 and 216 minutes for different customers), and calls raise it, both in an uneven way. [C119, C113]

Evidence. It acts on 3121 of 4000 training customers, 402 of whom left; for the typical customer it acts on, it changes the chance by less than a point. [C123] Through it, a typical customer's chance falls in an uneven way from about 38% at 91 minutes of daytime use to about 1% at 352 minutes, with no further fall past about 170 minutes (between 147 and 216 minutes for different customers). [C119] A higher international charge lowers it only while it stays below about 2.95 US dollars (between 2.95 and 4.34 US dollars for different customers), in an uneven way, from about 3% to about 1%. [C121] Customer-service calls raise it in an uneven way, from about 1% at 0 calls to about 3% at 3 calls. [C113] Its formula reproduces the part exactly on all 4000 training customers. [C124]
Evidence plots: Figure 25, Figure 26, Figure 27, Figure 28, Figure 29, Figure 30, Figure 31, Figure 32, Figure 33.
Figure 31 shows this part's own output, unit h4: over its two main inputs, with the line where it switches on.
Figure 32 shows this part's own output, unit h6: over its two main inputs, with the line where it switches on.

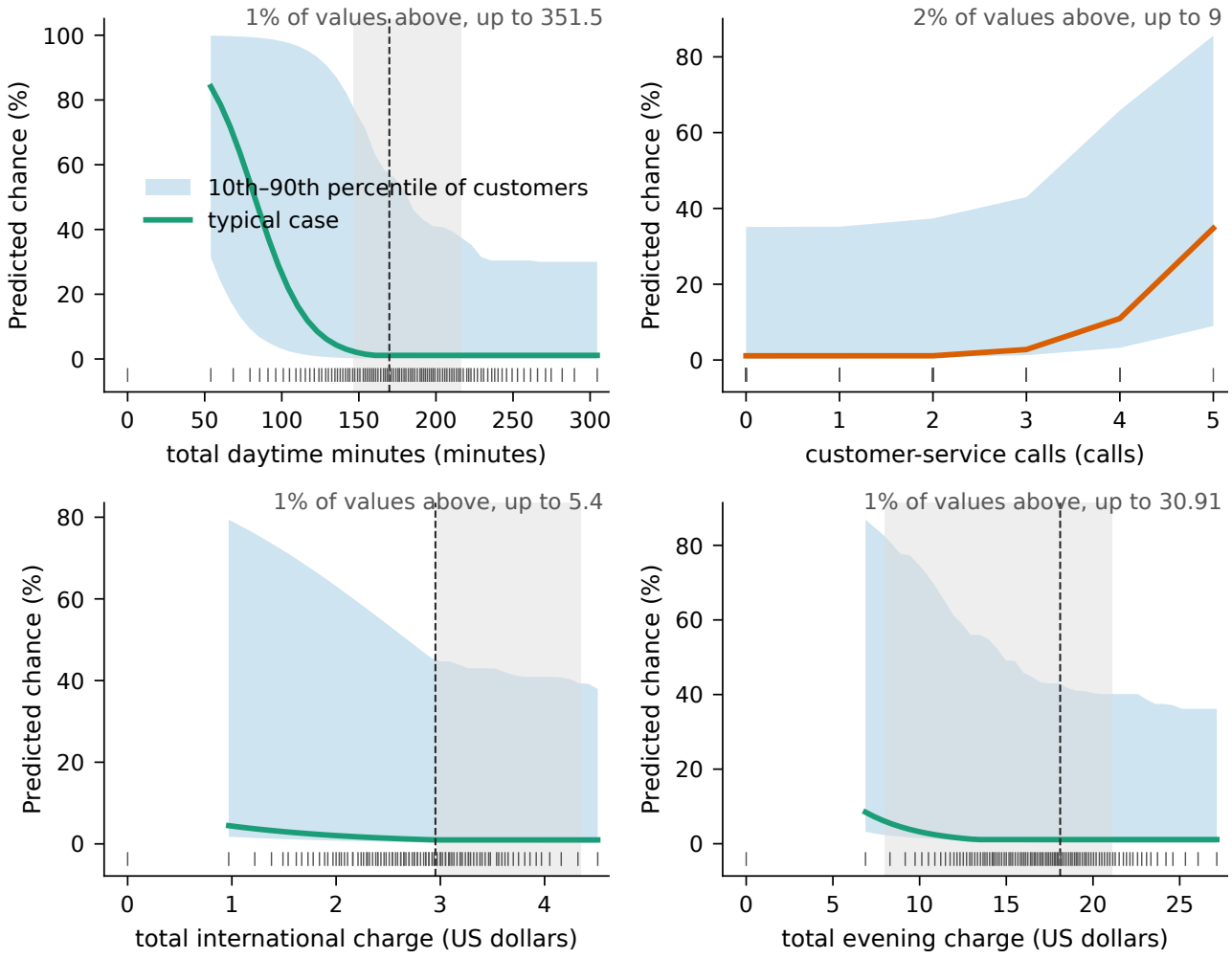


Figure 25: Calls with low usage: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

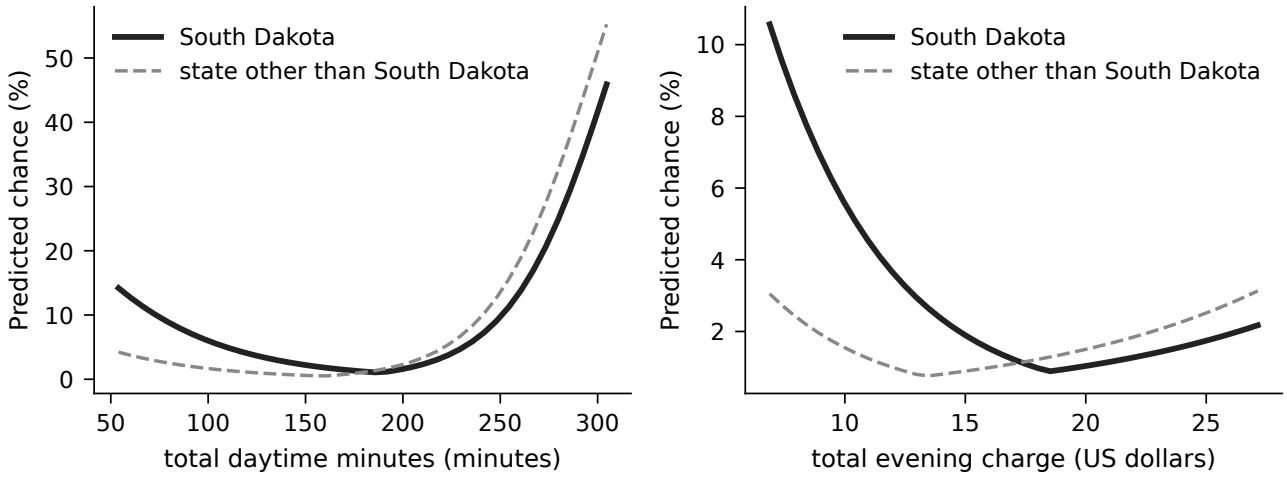


Figure 26: Calls with low usage: How the effect differs between groups: one curve per group, for the typical case of each group.

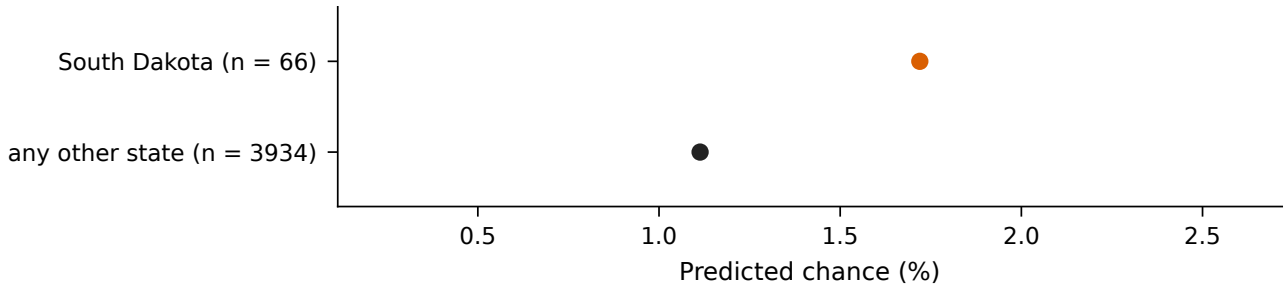


Figure 27: Calls with low usage: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

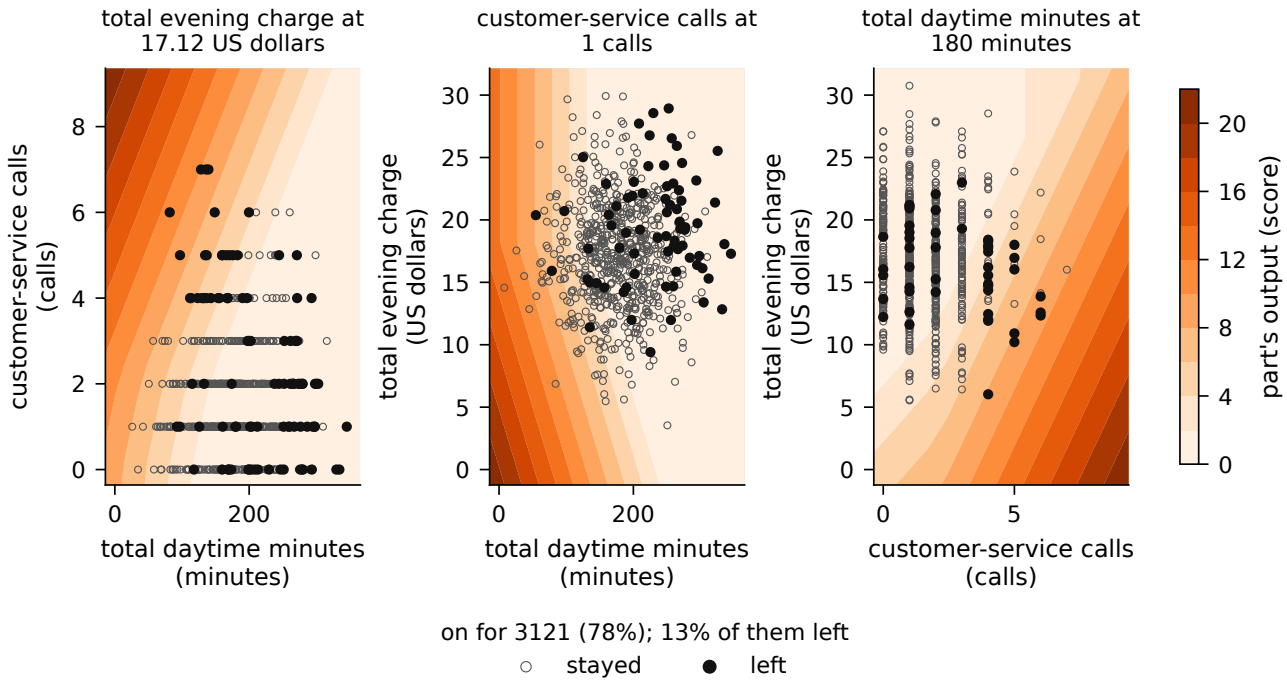


Figure 28: Calls with low usage: The part's own output over each pair of its three main inputs (the third held at its median, named above each panel; the points are the training rows whose third input is between its 40th and 60th percentile), other inputs at the typical case: colour where it is on, flat grey where it is switched off, the black line where it switches on; training customers as hollow (no event) and filled (event) points, so the part can be seen catching the events (the predicted-chance overlay was harder to read). The switch-on line is traced on this map only; the boundary claim (C163) did not pass its check on the data, so no rule is stated.

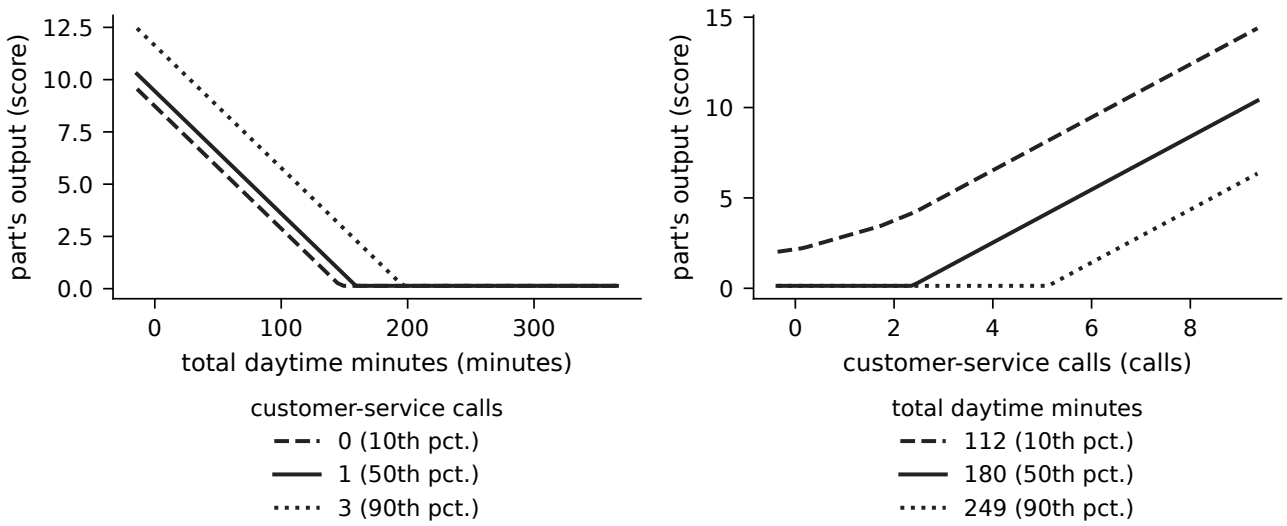


Figure 29: Calls with low usage: Slices of the same map: the part's own output against each input at three values of the other, so the tilted boundary shows as a shifting hinge. The switch-on line is traced on this map only; the boundary claim (C163) did not pass its check on the data, so no rule is stated.

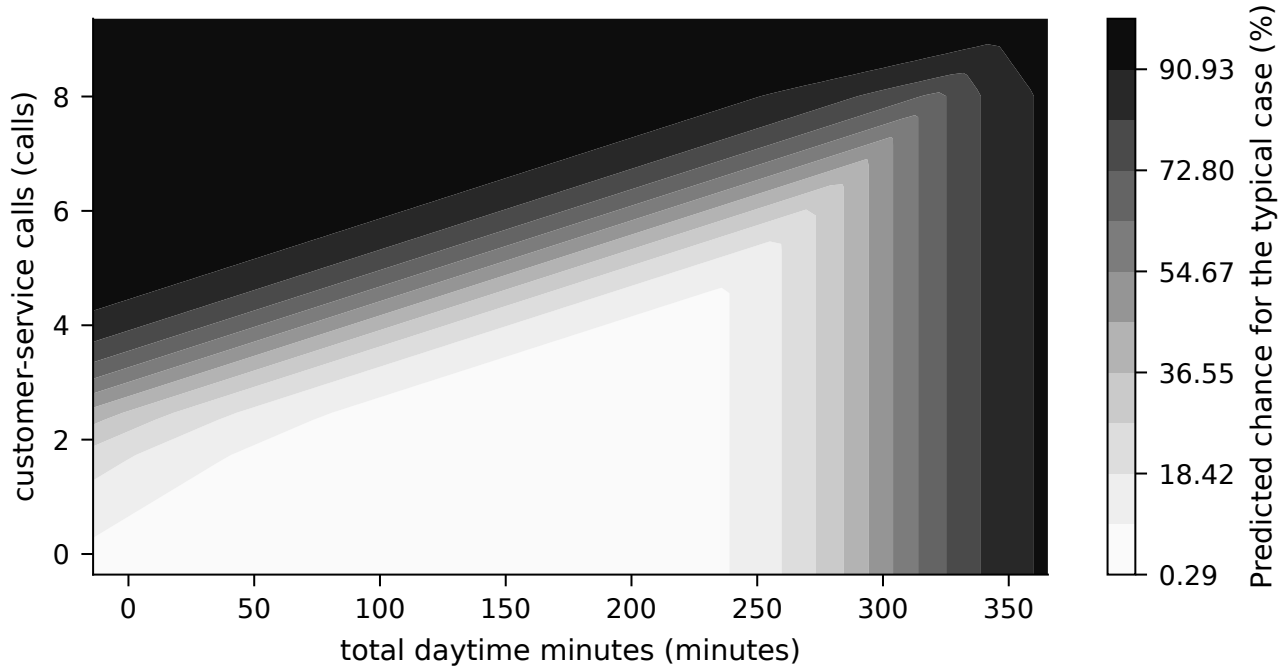


Figure 30: Calls with low usage: The same map on the final scale: the predicted value for the typical case, with the switch-on line (orange). The switch-on line is traced on this map only; the boundary claim (C163) did not pass its check on the data, so no rule is stated.

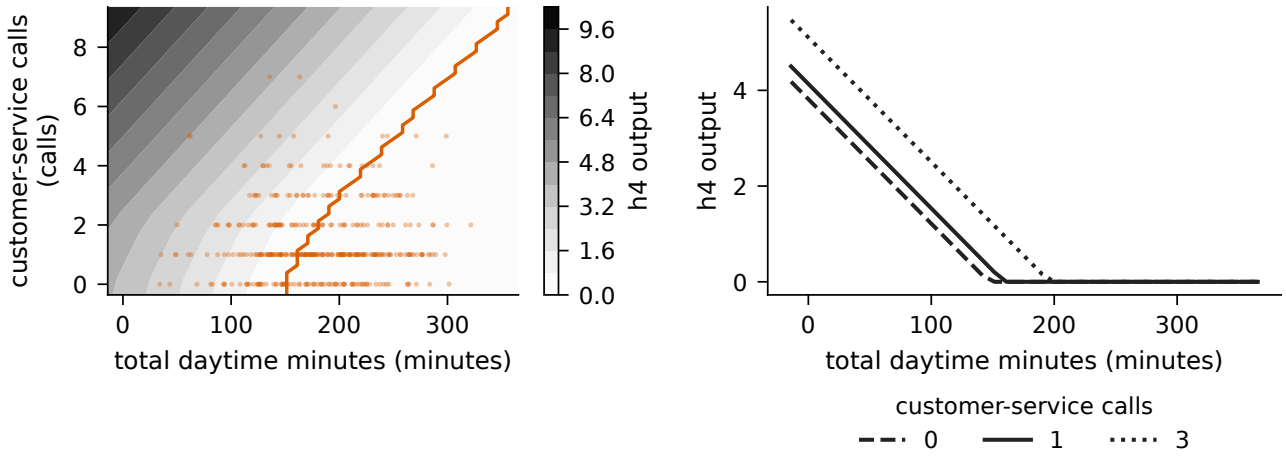


Figure 31: Calls with low usage: This part's own output, unit h4 (the output of 'Calls with low usage'), on its own scale (its activation), with every other input at the typical case; it is active for 1840 of 4000. Left: its output over its two main inputs (orange line: where it switches on; dots: the data); right: slices at three values of the second input.

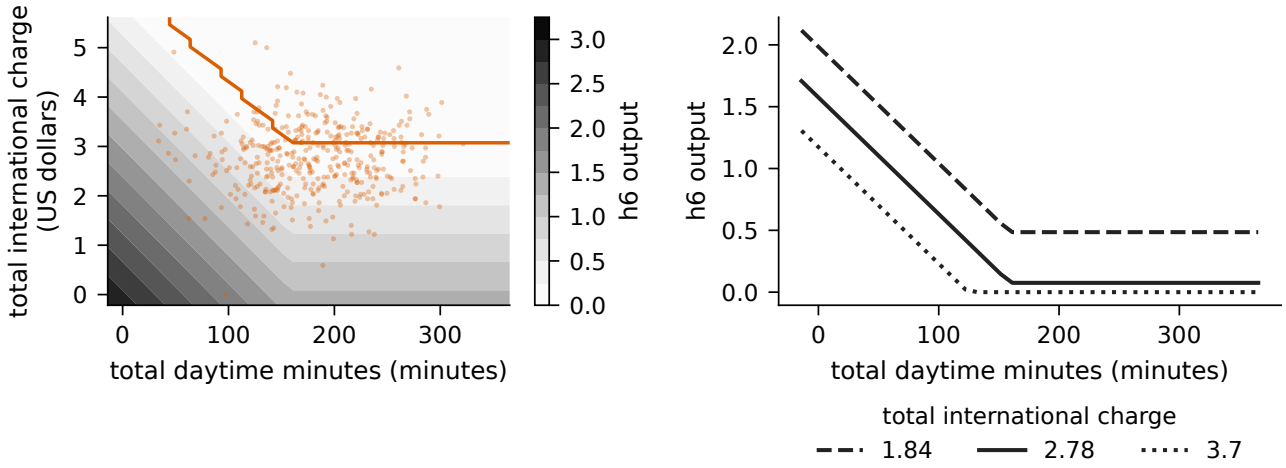


Figure 32: Calls with low usage: This part's own output, unit h6 (the output of 'Calls with low usage'), on its own scale (its activation), with every other input at the typical case; it is active for 2901 of 4000. Left: its output over its two main inputs (orange line: where it switches on; dots: the data); right: slices at three values of the second input.

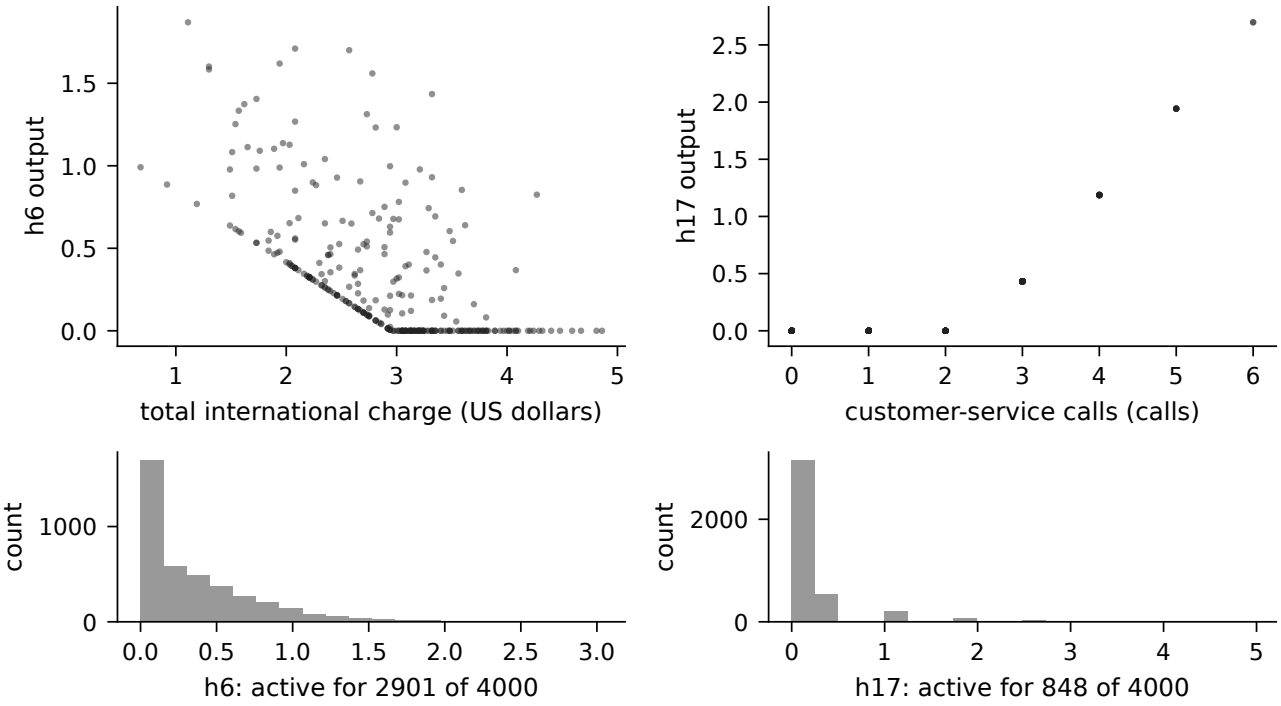


Figure 33: Calls with low usage: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.5.

1.2.5 Many service calls

Part of: Service calls (1.2.3).
Figure 34 shows what feeds this part; Figure 35 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

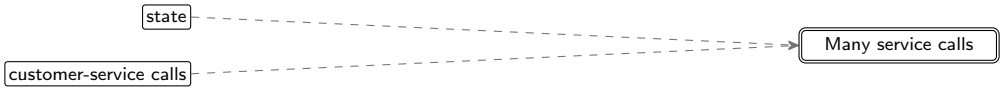


Figure 34: Many service calls: what feeds it, one level down.

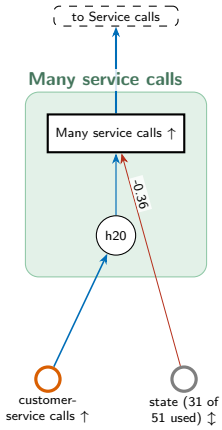


Figure 35: Many service calls: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part raises the chance of leaving with calls to customer service, adding nothing until about 3 calls, then steadily, except in West Virginia. [C131, C130]

Evidence. It acts on 306 of 4000 training customers, 154 of whom left; for the typical customer it acts on, the chance is about 16% without it and about 26% with it. [C132] It is switched off in West Virginia: for the typical customer it acts on, being in any other state takes the chance from about 16% to about 26%. [C130] Its formula reproduces the part exactly on all 4000 training customers. [C133]
Evidence plots: Figure 36, Figure 37, Figure 38.
Figure 37 shows this part's own output, unit h12.

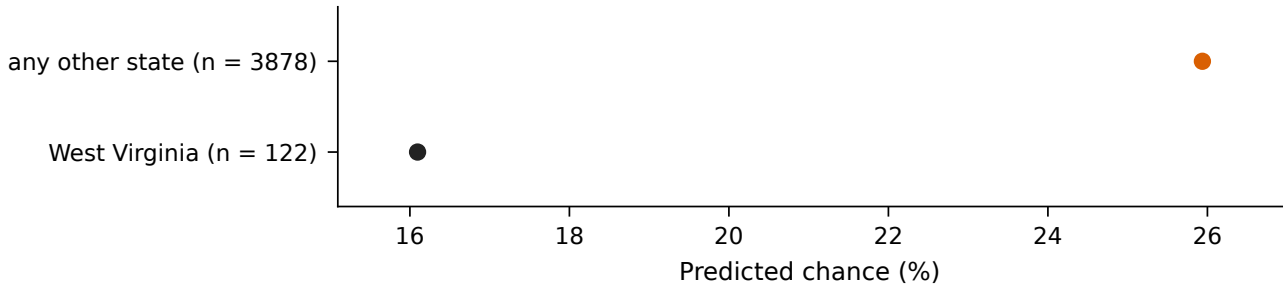


Figure 36: Many service calls: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

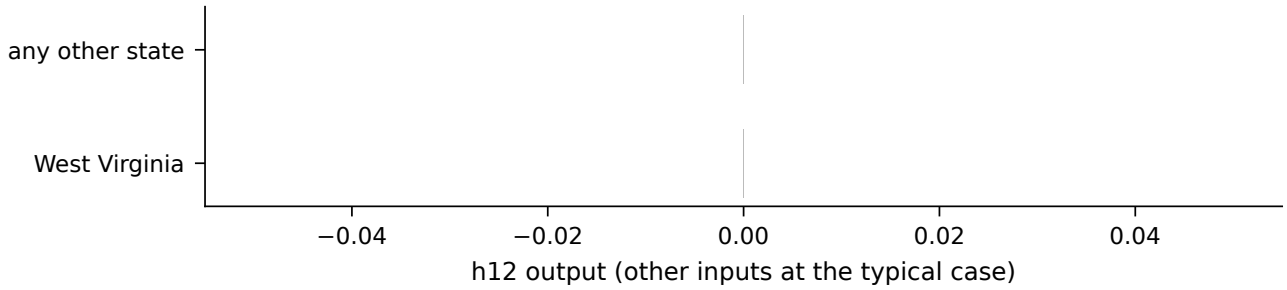


Figure 37: Many service calls: This part's own output, unit h12 (the output of 'Many service calls'), on its own scale (its activation), with every other input at the typical case; it is active for 306 of 4000. Its output for each combination of the levels it reads.

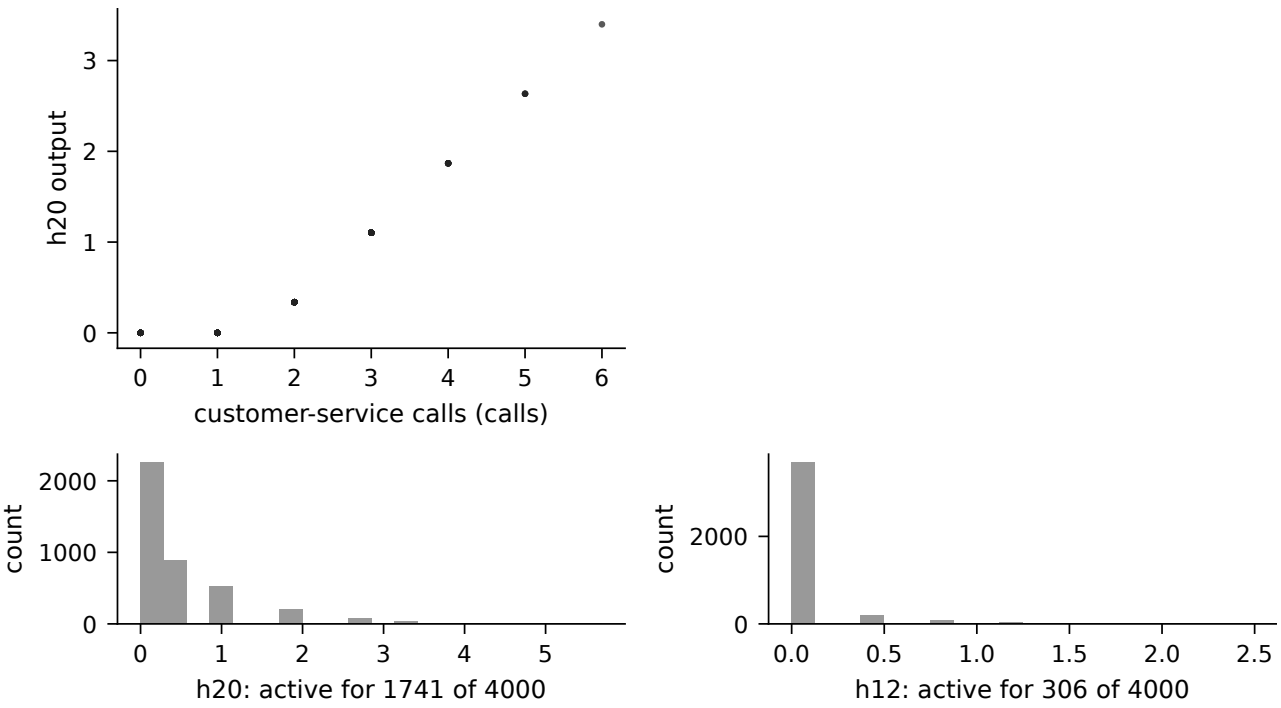


Figure 38: Many service calls: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.9.

1.2.6 Voice-mail discount

Part of: The whole model (1.2.1).
Figure 39 shows what feeds this part; Figure 40 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).



Figure 39: Voice-mail discount: what feeds it, one level down.

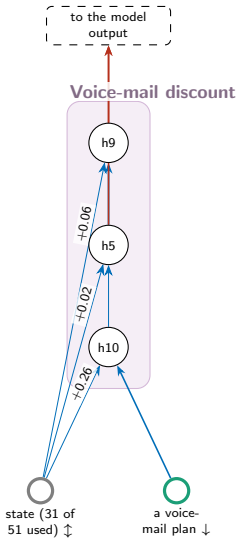


Figure 40: Voice-mail discount: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part gives customers with a voice-mail plan a small discount on the chance of leaving. [C125]

Evidence. Through it, a voice-mail plan lowers a typical customer's chance by less than a point, and customers in states other than Alabama, North Carolina and West Virginia get about 2% instead of about 1%. [C125, C126] Its formula reproduces the part exactly on all 4000 training customers. [C129]
Evidence plots: Figure 41, Figure 42.

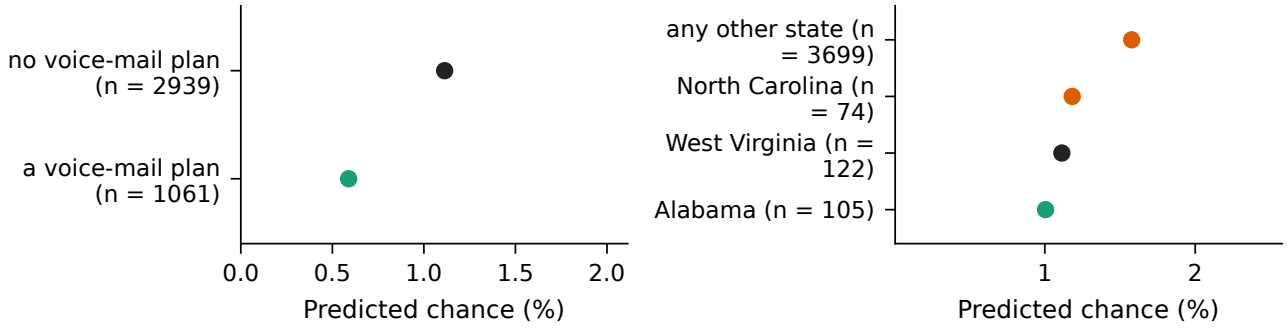


Figure 41: Voice-mail discount: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

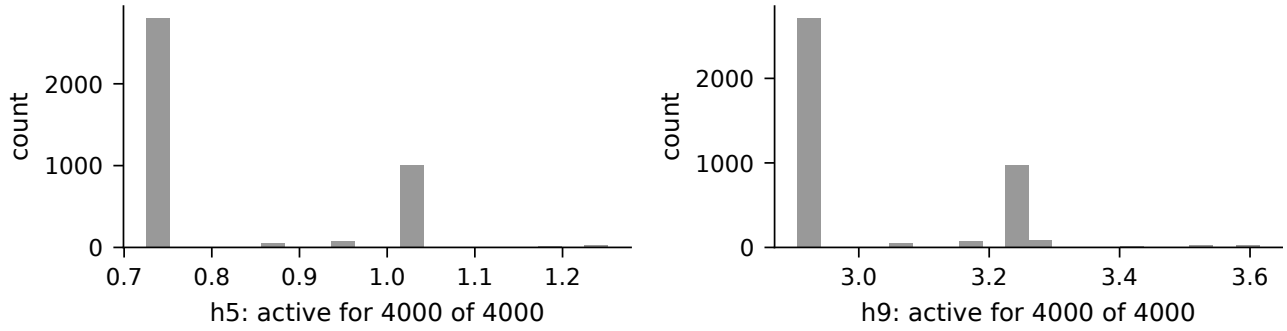


Figure 42: Voice-mail discount: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.7.

1.2.7 Heavy evening use

Part of: The whole model (1.2.1).
Figure 43 shows what feeds this part; Figure 44 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).



Figure 43: Heavy evening use: what feeds it, one level down.

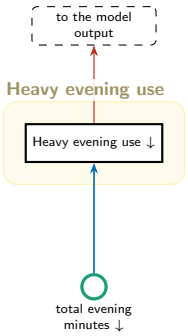


Figure 44: Heavy evening use: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part lowers the chance of leaving a little for customers with heavy evening use: evening minutes add nothing until about 239 minutes, then lower it steadily. [C134]

Evidence. It acts on 885 of 4000 training customers, 165 of whom left; for the typical customer it acts on, the chance is about 5% without it and about 4% with it. [C136] Through it, evening minutes change nothing until about 239 minutes, then lower a typical customer's chance steadily, from about 1% at 138 minutes to under 1% at 266 minutes. [C134] Its formula reproduces the part exactly on all 4000 training customers. [C137]
Evidence plots: Figure 45, Figure 46, Figure 47.
Figure 46 shows this part's own output, unit h3: it switches on at about 238.8 minutes of total evening minutes, and is zero below.

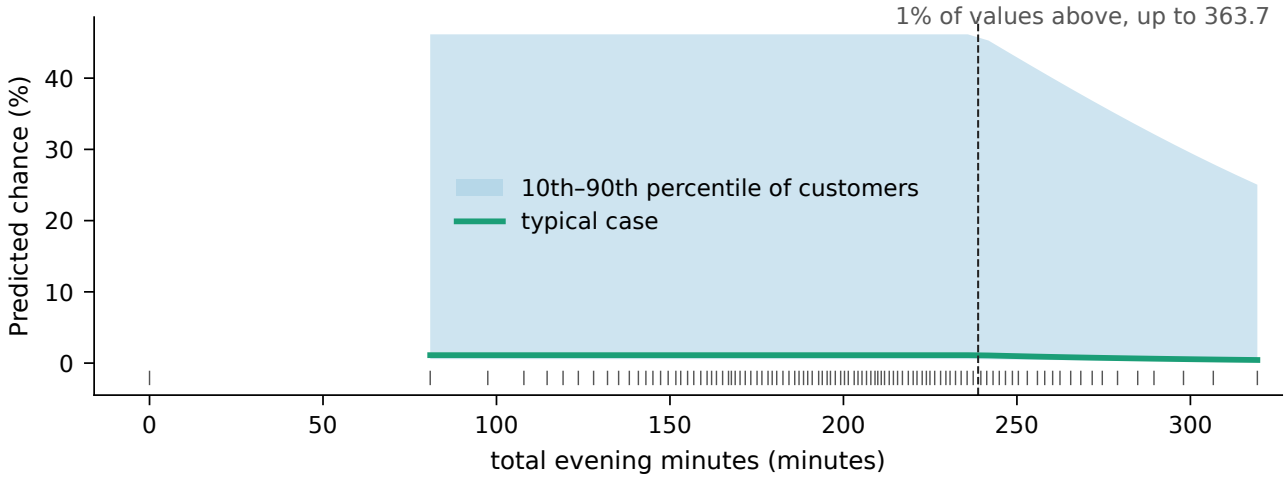


Figure 45: Heavy evening use: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

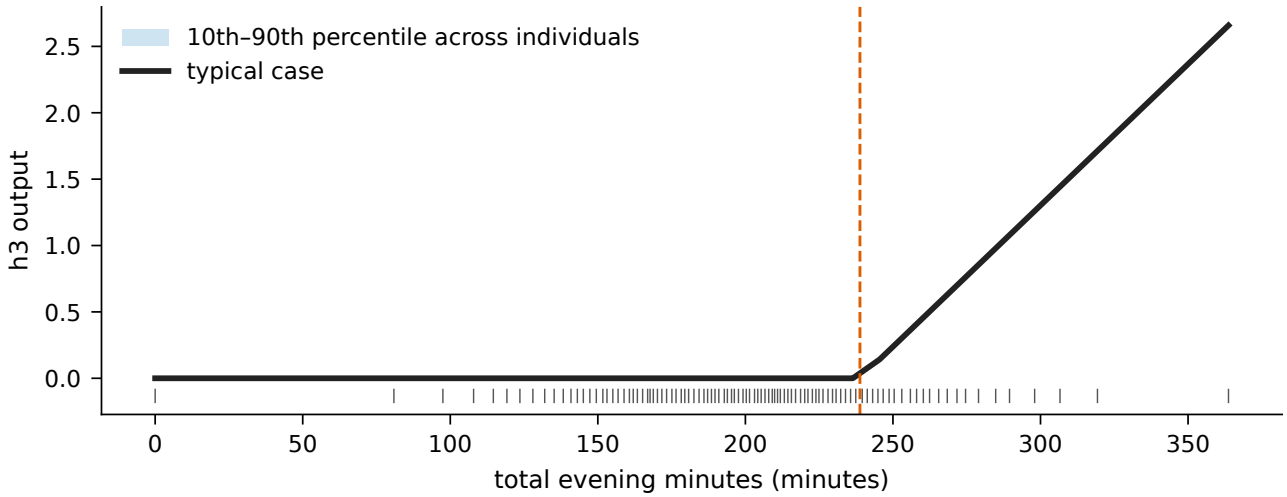


Figure 46: Heavy evening use: This part's own output, unit h3 (the output of 'Heavy evening use'), on its own scale (its activation), with every other input at the typical case; it is active for 885 of 4000. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

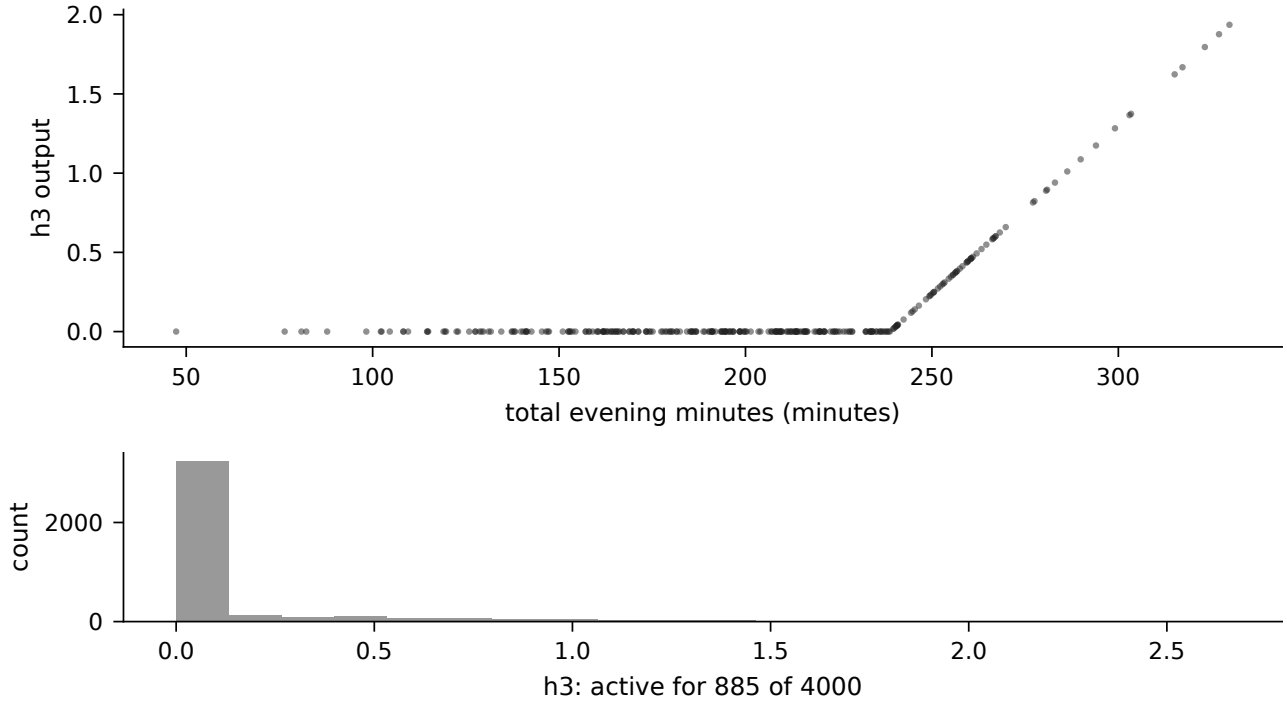


Figure 47: Heavy evening use: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.8.

1.2.8 State units

Part of: The whole model (1.2.1).
Figure 48 shows what feeds this part; Figure 49 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

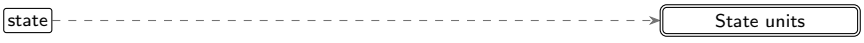


Figure 48: State units: what feeds it, one level down.

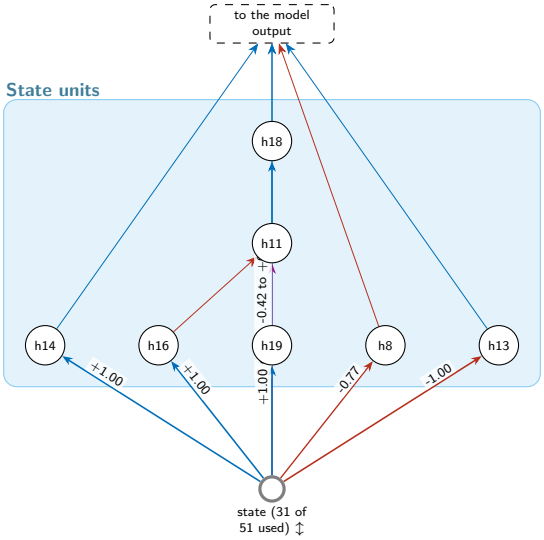


Figure 49: State units: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part adds small state offsets: up in Arizona and Ohio, down in Nevada. [C138, C139, C140]

Evidence. Through it, a typical customer's chance moves by less than half a point: up in Arizona and Ohio, down in Nevada, with Missouri and Massachusetts almost unchanged. [C138, C139, C140, C141, C142] Its formula reproduces the part exactly on all 4000 training customers. [C143]
Evidence plots: Figure 50, Figure 51.

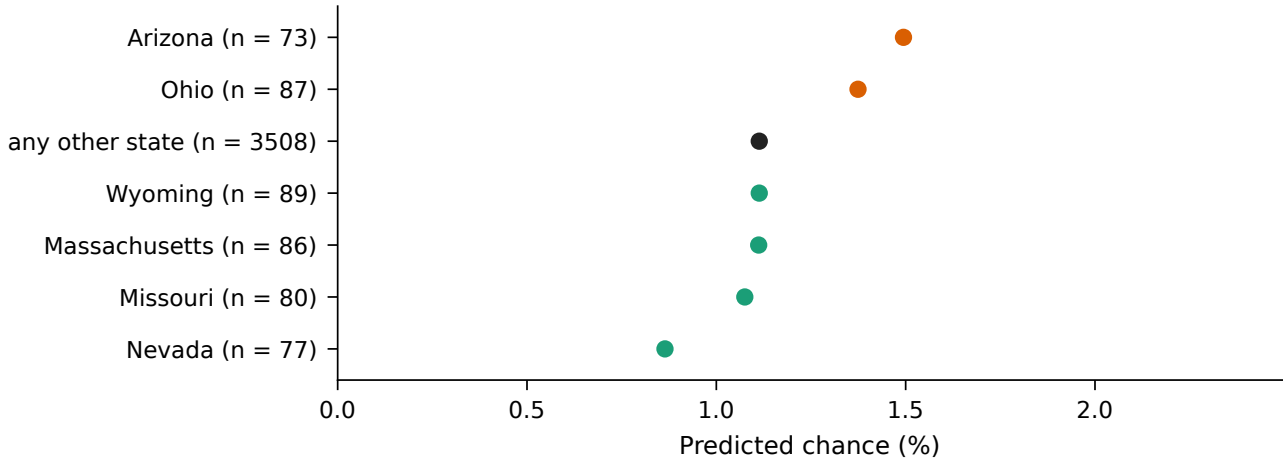


Figure 50: State units: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

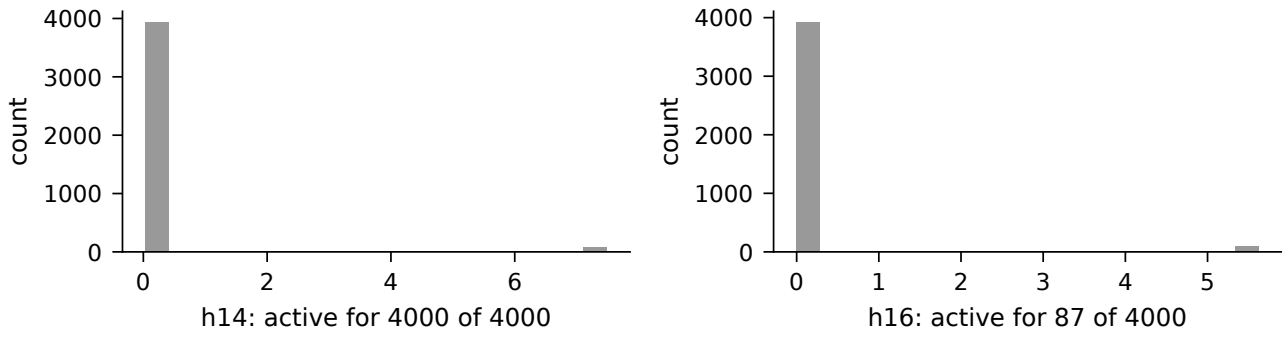


Figure 51: State units: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.3.

1.2.9 Daytime calls and Tennessee

Part of: The whole model (1.2.1).
Figure 52 shows what feeds this part; Figure 53 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).



Figure 52: Daytime calls and Tennessee: what feeds it, one level down.

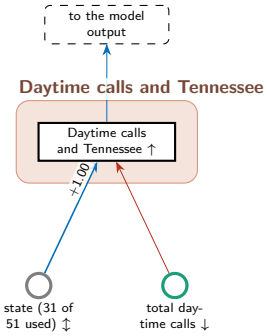


Figure 53: Daytime calls and Tennessee: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part gives customers in Tennessee a small rise in the chance of leaving; daytime calls barely change it. [C144, C145]

Evidence. Through it, being in Tennessee takes a typical customer's chance from about 1% to about 2%, and going from 75 to 125 daytime calls barely changes it. [C144, C145] Its formula reproduces the part exactly on all 4000 training customers. [C149]
Evidence plots: Figure 54, Figure 55, Figure 56, Figure 57.
Figure 56 shows this part's own output, unit h15: it switches on at about 113 calls of total daytime calls, and is zero below.

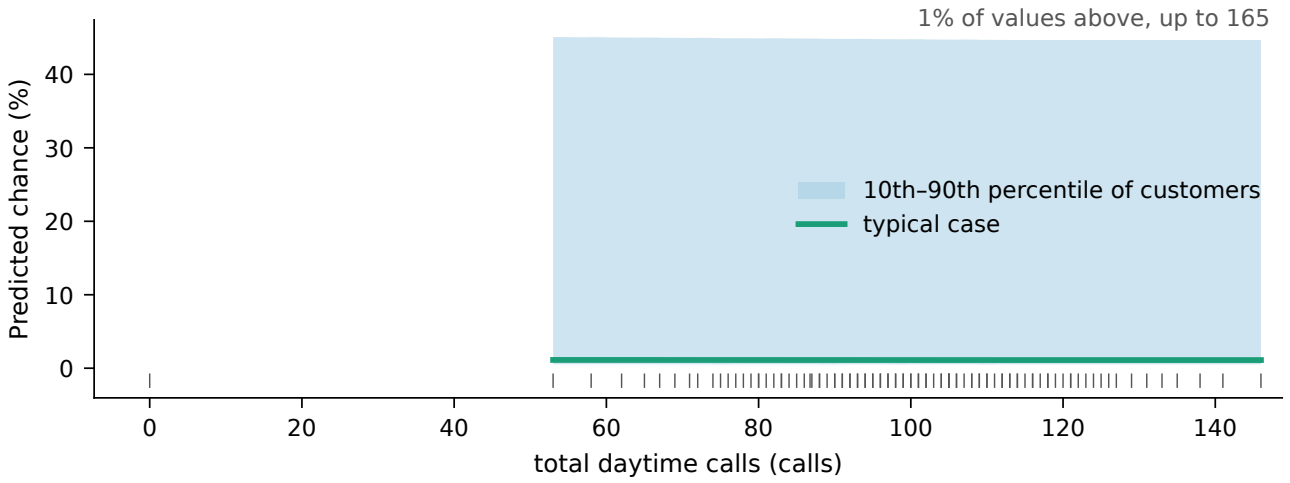


Figure 54: Daytime calls and Tennessee: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

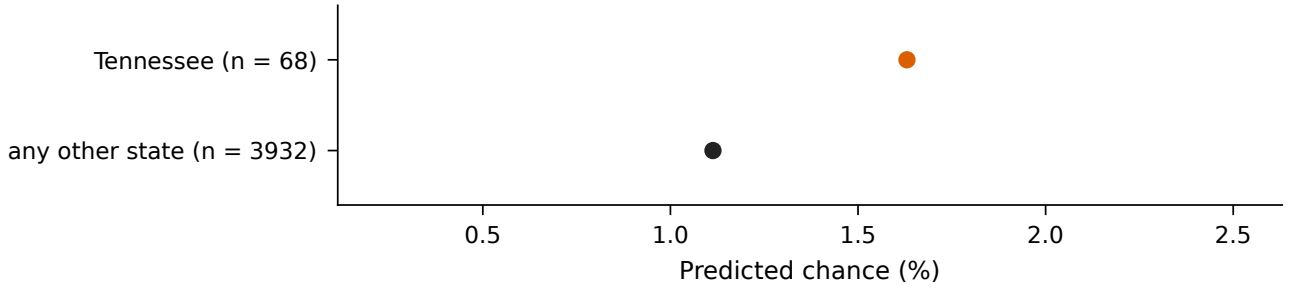


Figure 55: Daytime calls and Tennessee: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

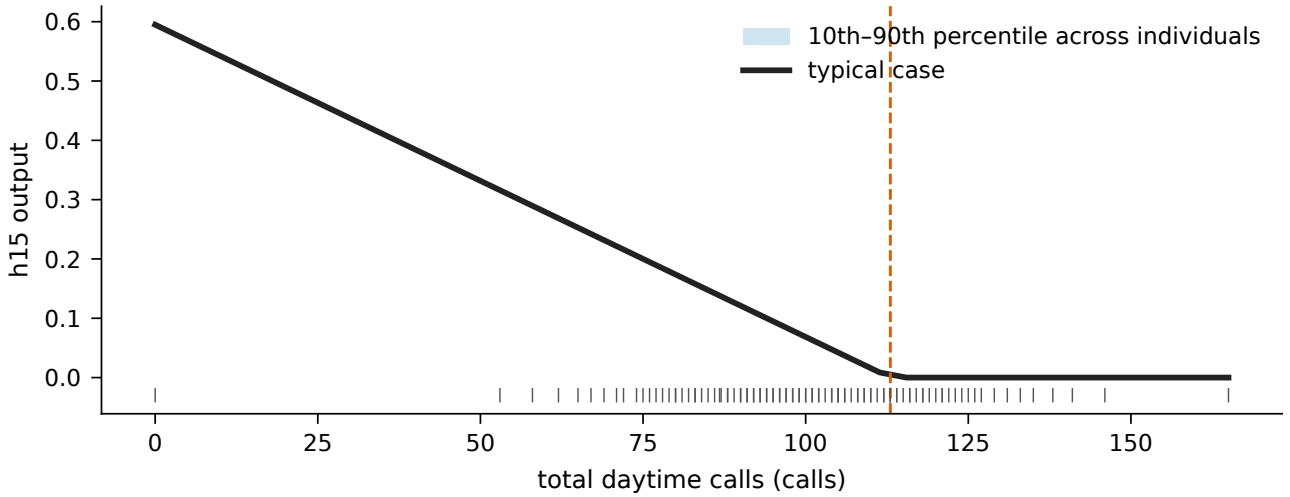


Figure 56: Daytime calls and Tennessee: This part's own output, unit h15 (the output of 'Daytime calls and Tennessee'), on its own scale (its activation), with every other input at the typical case; it is active for 2994 of 4000. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

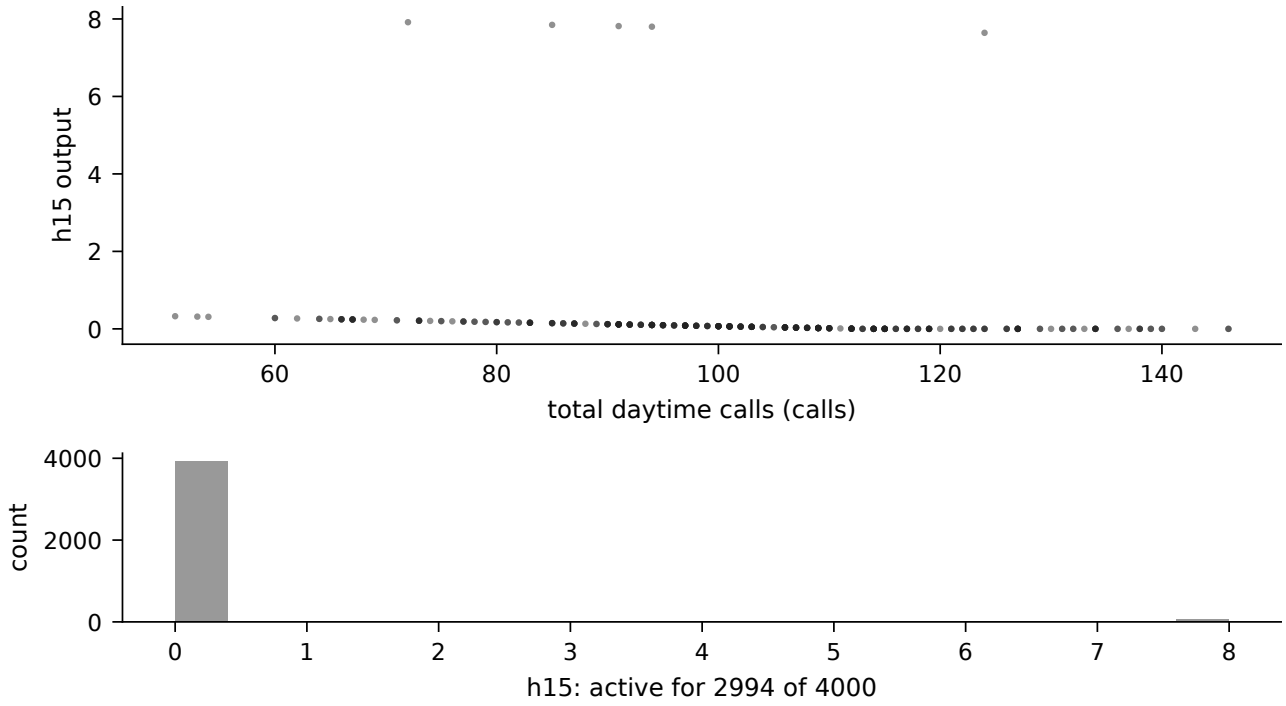


Figure 57: Daytime calls and Tennessee: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.1.

1.2.10 Small state unit

Part of: The whole model (1.2.1).
Figure 58 shows what feeds this part; Figure 59 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

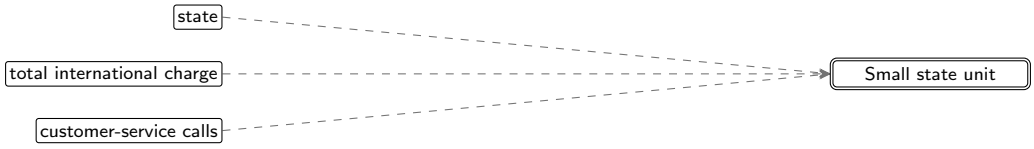


Figure 58: Small state unit: what feeds it, one level down.

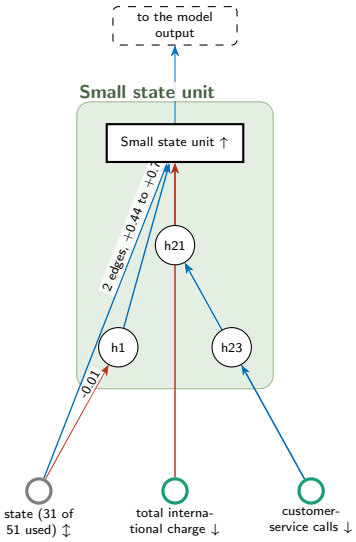


Figure 59: Small state unit: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This small part barely changes the chance of leaving: it adds less than a tenth of a point in Arkansas and Utah. [C150, C153]

Evidence. Through it, the chance moves by less than a tenth of a point in Arkansas and Utah, a higher international charge lowers it steadily by even less, and customer-service calls barely move it. [C150, C153, C151, C152] Its formula reproduces the part exactly on all 4000 training customers. [C161]
Evidence plots: Figure 60, Figure 61, Figure 62, Figure 63, Figure 64, Figure 65.

Worked example: reading this part. Figure 62 maps the part's own output over its main inputs. In the grey region it is switched off and adds nothing; past the black line it switches on, and the colour shows how much it adds to the score. The line is the rule in the claim (C164): it switches on when total international charge plus about 0.07 × customer-service calls below about 0.50 (for customer-service calls from about 0.00 to 1.73); total international charge plus about 0.14 × customer-service calls below about 0.62 (for customer-service calls from about 1.73 to 6.04). 153 of 4000 training customers lie on the 'on' side, and 12% of them left, against 14% of the others: the 'on' side holds mostly hollow points, so the part picks out low-risk cases. Figure 63 cuts the map into slices: the hinge moves as the other input changes, which is what a tilted line means. Figure 64 shows what this does to the final prediction for the typical case.

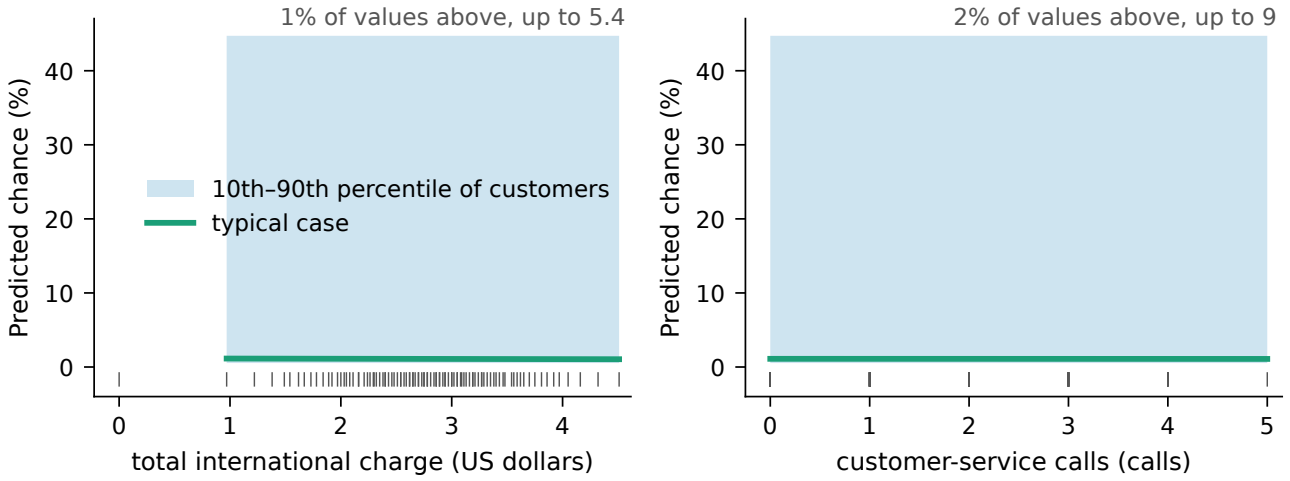


Figure 60: Small state unit: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

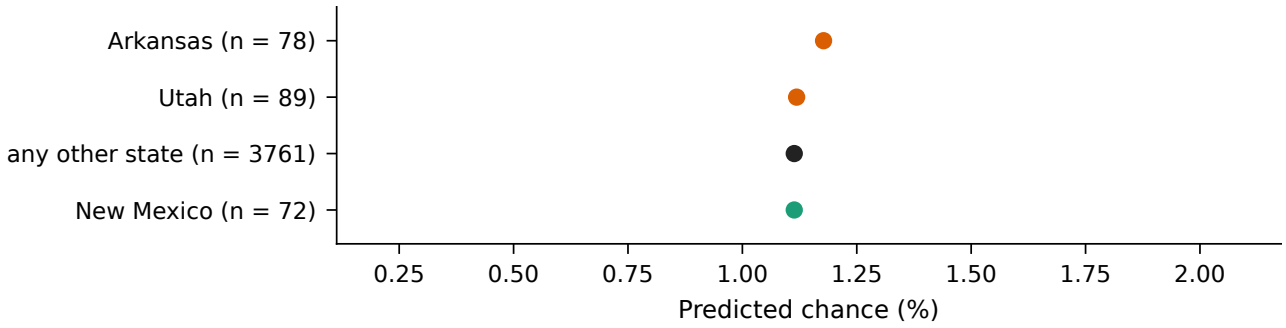


Figure 61: Small state unit: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

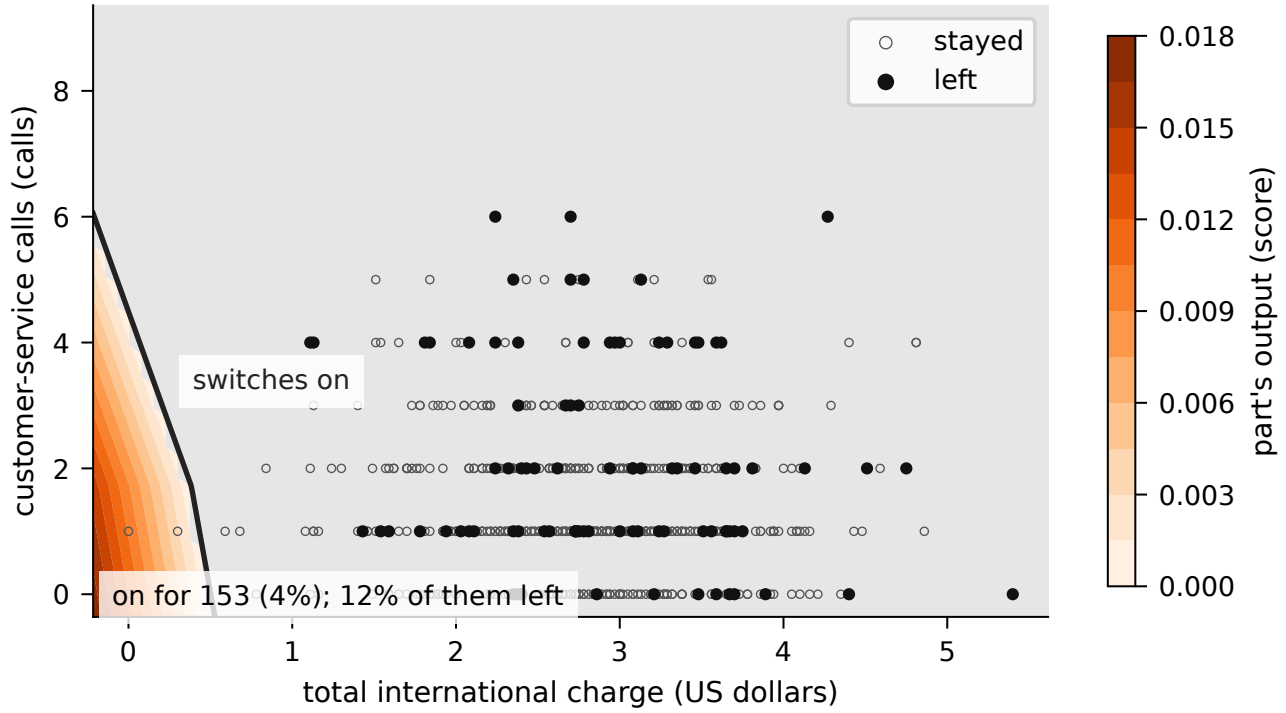


Figure 62: Small state unit: The part's own output over its two numeric inputs, other inputs at the typical case: colour where it is on, flat grey where it is switched off, the black line where it switches on; training customers as hollow (no event) and filled (event) points, so the part can be seen catching the events (the predicted-chance overlay was harder to read).

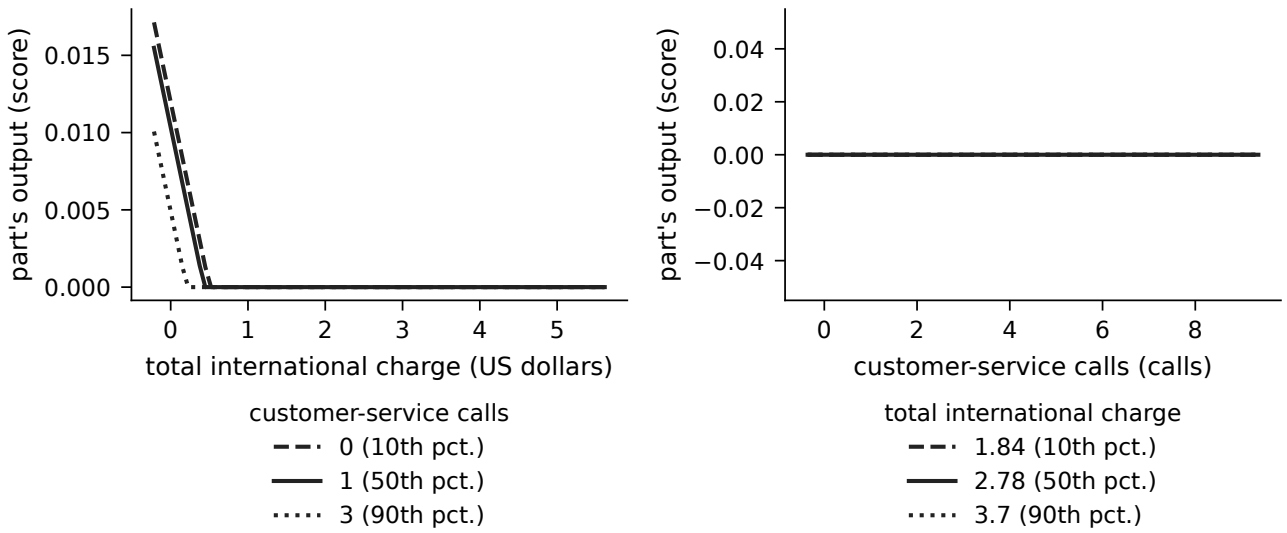


Figure 63: Small state unit: Slices of the same map: the part's own output against each input at three values of the other, so the tilted boundary shows as a shifting hinge.

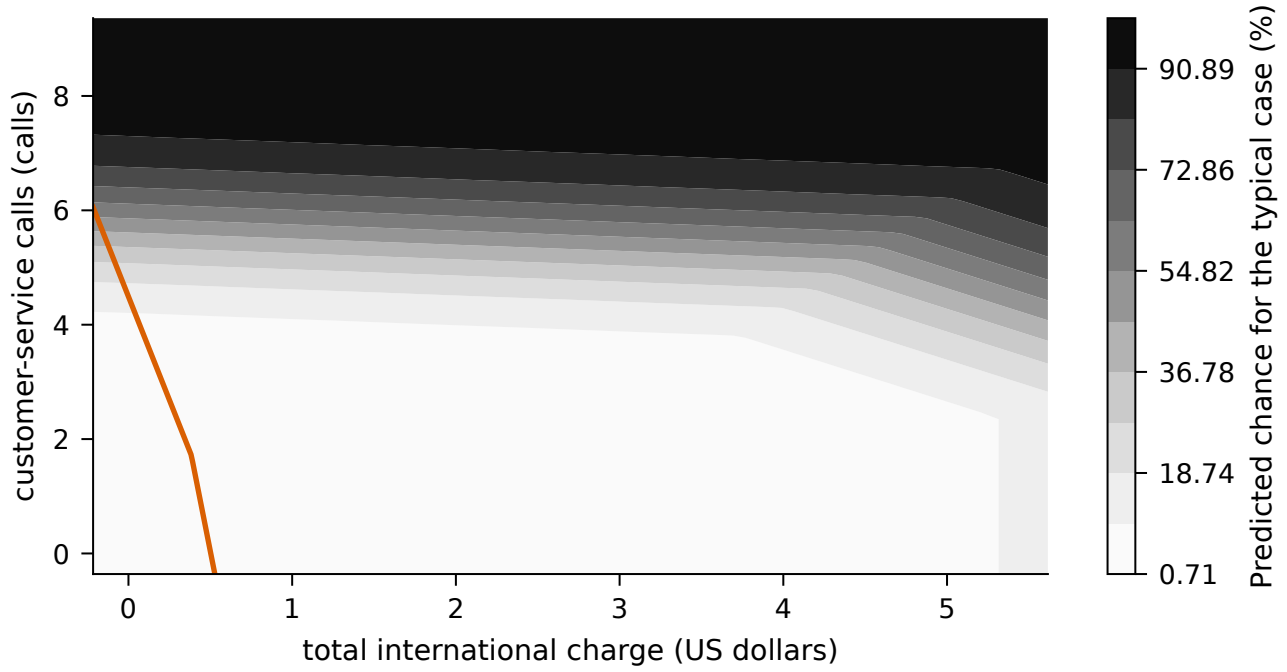


Figure 64: Small state unit: The same map on the final scale: the predicted value for the typical case, with the switch-on line (orange).

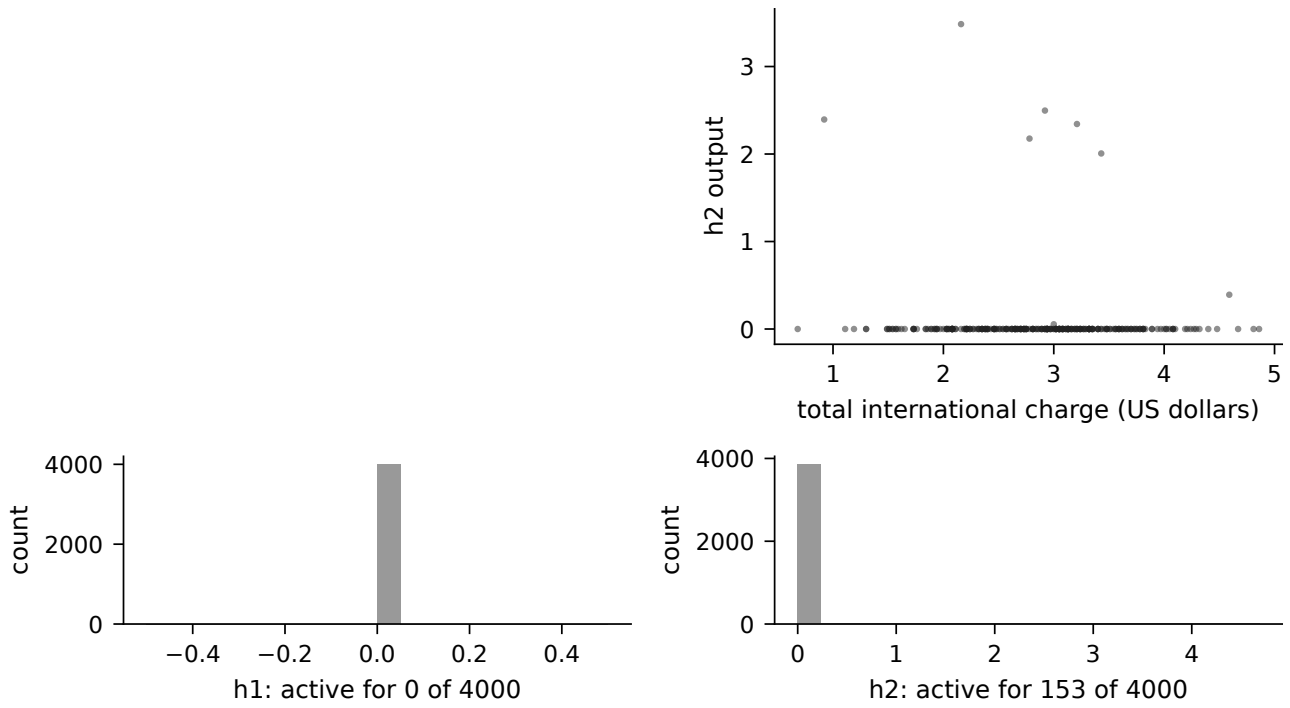


Figure 65: Small state unit: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.6.

1.3 Not supported by the model

- C002: direction claim about total daytime minutes in the whole model: not supported by the model (17% of patients have the opposite sign (allowed: 10%)).
- C017: direction claim about total evening charge in the whole model: not supported by the model (52% of patients have the opposite sign (allowed: 10%)).
- C041: direction claim about state in the whole model: not supported by the model (45% of patients have the opposite sign (allowed: 10%)).
- C044: direction claim about state in the whole model: not supported by the model (92% of patients have the opposite sign (allowed: 10%)).
- C146: interaction claim about total daytime calls and state in part ann_97: not supported by the model (the group difference is below 0.1 log-odds).
- C147: threshold claim about total daytime calls in part ann_97: not supported by the model (the fold changes the curve by only 0.014 log-odds compared with a straight line (needed 0.1); the effect past the fold is too small or not consistent across patients).
- C148: subgroup claim about state and total daytime calls in part ann_97: not supported by the model (its effect is negligible).
- C154: interaction claim about number of calls to customer service and state in part ann_92: not supported by the model (the part's own counterfactual difference has the opposite sign).
- C155: interaction claim about total international charge and state in part ann_92: not supported by the model (the group difference is below 0.1 log-odds).
- C156: interaction claim about number of calls to customer service and total international charge in part ann_92: not supported by the model (the group difference is below 0.1 log-odds; the part's own counterfactual difference has the opposite sign).
- C157: threshold claim about number of calls to customer service in part ann_92: not supported by the model (the fold changes the curve by only 0.015 log-odds compared with a straight line (needed 0.1); the effect past the fold is too small or not consistent across patients).
- C158: threshold claim about total international charge in part ann_92: not supported by the model (the effect past the fold is too small or not consistent across patients).
- C159: subgroup claim about state and total international charge in part ann_92: not supported by the model (its effect is negligible).

- C160: null claim about state and total international charge in part ann_92: not supported by the model (predictions move by up to 0.020).
- C162: boundary claim about total daytime minutes and number of calls to customer service in part ann_101: not supported by the model (no switch-on line could be traced on the first map to check against the data; no switch-on boundary in the observed range).
- C163: boundary claim about total daytime minutes and number of calls to customer service in part ann_91: not supported by the model (no switch-on line could be traced on the first map to check against the data; no switch-on boundary in the observed range).

Technical detail: churn100-s2-c1

The record behind each part: the original hypothesis, the closed form, and every claim with its estimate (log-odds and odds ratio), range across patients, n and test check.

T.1 The whole model

Annotation ann_102 (churn_logit). Reads: state, account length, area code, international calling plan, voice-mail plan, number of voice-mail messages, total daytime minutes, total daytime calls, total daytime charge, total evening minutes, total evening calls, total evening charge, total night-time minutes, total night-time calls, total night-time charge, total international minutes, total international calls, total international charge, number of calls to customer service.

Original hypothesis (unchanged). Root: logit P(churn) = -1.05 + linear_backbone + service_call_nonlinear (identity_svc = 1.60*807 + 1.81*1264 + 1.48*7208) - 0.95*1133 - 0.99*4051 (voicemail_chain) - 0.55*476 (eve_minutes_hinge) + state_offsets + 0.02*252 + 0.05*11427, purely additive in the logit; the sigmoid is the only coupling. 24 inputs are unconnected (total intl calls, night charge, evening calls, day charge and 20 state indicators) and have zero effect. Train AUC 0.901, accuracy 0.901. Performance sources (train): all children replaced by least-squares linear fits gives AUC 0.837 (an L2 logistic regression refit on all 71 inputs gets 0.843); restoring csc_usage_gate alone gives 0.898, csc_hinge alone 0.851, eve_minutes_hinge alone 0.834, the others 0.837 (exactly or nearly linear). So the gain over linear comes from the service-call x low-usage gate. Rare-category switches: state:11/46/5 offsets of -2.24/-1.89/-1.46 logit on 73-80 customers (empirical -1.35/-0.91/-0.51); state:4 +1.35 (41 rows, empirical +0.80); state:47 +1.16 (72 rows, +0.92); state:49 -0.42 offset, down to -2.87 for 12 customers with 4+ calls (turns off the 4+-calls hinge); state:35 +0.40 (empirical +0.06).

Closed form. Read through its parts (their formulas are in their own sections).

Split inputs. To annotate the parts, the explanation split these inputs into copies (a refactoring step that does not change what the model computes); the diagrams draw each input once: customer-service calls (3 copies); total daytime calls (2 copies); total daytime minutes (2 copies); total evening charge (2 copies); total evening minutes (2 copies); total international charge (3 copies); a voice-mail plan (2 copies).

State: every level the model reads. Whole-model change in the predicted chance for a typical customer when only the state changes from the reference; n = training customers in that level.

Claim	From	To	n	Chance	OK
C003	West Virginia	California	41	1% to 6%	yes
C004	West Virginia	Washington	72	1% to 5%	yes
C006	West Virginia	Montana	75	1% to 3%	yes
C007	West Virginia	Maryland	82	1% to 3%	yes
C009	West Virginia	Ohio	87	1% to 2%	yes
C010	West Virginia	Tennessee	68	1% to 2%	yes
C013	West Virginia	Connecticut	80	1% to 2%	yes
C015	West Virginia	Oregon	84	1% to 2%	yes
C020	West Virginia	Arizona	73	1% to 2%	yes
C022	West Virginia	Utah	89	1% to 2%	yes
C023	West Virginia	New Mexico	72	1% to 2%	yes
C024	West Virginia	state other than Alaska, Alabama, Hawaii, Massachusetts, ...	1587	1% to 2%	yes
C025	West Virginia	Massachusetts	86	1% to 2%	yes
C026	West Virginia	Arkansas	78	1% to 2%	yes
C027	West Virginia	North Carolina	74	1% to 2%	yes
C028	West Virginia	North Dakota	68	1% to 2%	yes
C032	West Virginia	Oklahoma	79	1% to 1%	yes
C037	West Virginia	Nevada	77	1% to 1%	yes
C041	West Virginia	South Dakota	66	1% to 1%	no
C042	West Virginia	New Hampshire	74	1% to 1%	yes
C045	West Virginia	Nebraska	72	1% to 1%	yes
C044	West Virginia	Alabama	105	1% to 1%	no
C039	West Virginia	Missouri	80	1% to 1%	yes
C036	West Virginia	Wisconsin	86	1% to 1%	yes
C035	West Virginia	Wyoming	89	1% to 1%	yes
C034	West Virginia	Virginia	91	1% to 1%	yes
C030	West Virginia	Alaska	56	1% to 1%	yes
C029	West Virginia	Pennsylvania	56	1% to 1%	yes
C021	West Virginia	Colorado	78	1% to <1%	yes
C019	West Virginia	Vermont	80	1% to <1%	yes
C016	West Virginia	Hawaii	73	1% to <1%	yes

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C001	direction: international calling plan: no international plan to an international plan (whole model) *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 13%	2.58	13.2	2.58 to 2.58	3620, 380	agrees; AUC gain 0.060 [0.035, 0.086]	yes
C002	direction: total daytime minutes: 112 minutes to 249 minutes (whole model) *shape: dip at 160 minutes; 100% of patients same; verified; risk saturates near 0%*	1% to 13%	1.55	4.72	-0.77 to 3.15	4000	agrees; AUC gain 0.092 [0.059, 0.128]	no
C003	direction: state: West Virginia to California (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 6%	1.84	6.30	1.76 to 1.76	122, 41	disagrees; AUC gain -0.016 [-0.030, -0.004]	yes
C004	direction: state: West Virginia to Washington (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 5%	1.66	5.24	1.58 to 1.58	122, 72	disagrees; AUC gain -0.016 [-0.030, -0.004]	yes
C005	direction: voice-mail messages: 0 messages to 32 messages (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 5%	1.48	4.37	1.48 to 1.48	4000	unclear; AUC gain 0.008 [-0.002, 0.019]	yes
C006	direction: state: West Virginia to Montana (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 3%	1.17	3.22	1.09 to 1.09	122, 75	disagrees; AUC gain -0.016 [-0.030, -0.004]	yes
C007	direction: state: West Virginia to Maryland (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 3%	1.04	2.83	0.96 to 0.96	122, 82	disagrees; AUC gain -0.016 [-0.030, -0.004]	yes
C008	direction: customer-service calls: 0 calls to 3 calls (whole model) *shape: accelerating; 84% of patients same; fallback; risk saturates near 0%*	1% to 3%	1.19	3.30	0.00 to 2.96	4000	agrees; AUC gain 0.097 [0.060, 0.139]	yes
C009	direction: state: West Virginia to Ohio (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 2%	0.90	2.45	0.82 to 0.82	122, 87	disagrees; AUC gain -0.016 [-0.030, -0.004]	yes
C010	direction: state: West Virginia to Tennessee (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 2%	0.88	2.41	0.80 to 0.81	122, 68	disagrees; AUC gain -0.016 [-0.030, -0.004]	yes
C011	direction: total international charge: 1.84 US dollars to 3.70 US dollars (whole model) *shape: accelerating; 87% of patients same; fallback; risk saturates near 0%*	1% to 2%	0.67	1.95	0.29 to 0.88	4000	unclear; AUC gain 0.005 [-0.001, 0.012]	yes
C012	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (whole model) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	1% to <1%	-2.68	0.07	-2.68 to -2.68	2939, 1061	agrees; AUC gain 0.033 [0.015, 0.053]	yes
C013	direction: state: West Virginia to Connecticut (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 2%	0.74	2.10	0.67 to 0.67	122, 80	disagrees; AUC gain -0.016 [-0.030, -0.004]	yes
C014	direction: total evening minutes: 138 minutes to 266 minutes (whole model) *shape: several pieces; 100% of patients same; fallback; risk saturates near 0%*	1% to 2%	0.98	2.66	0.98 to 0.98	4000	agrees; AUC gain 0.009 [0.002, 0.016]	yes
C015	direction: state: West Virginia to Oregon (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 2%	0.70	2.01	0.62 to 0.62	122, 84	disagrees; AUC gain -0.016 [-0.030, -0.004]	yes
C016	direction: state: West Virginia to Hawaii (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to <1%	-1.75	0.17	-1.82 to -1.82	122, 73	disagrees; AUC gain -0.016 [-0.030, -0.004]	yes
C017	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (whole model) *shape: dip at 13.46 US dollars; 75% of patients same; fallback; risk saturates near 0%*	1% to 2%	-0.05	0.95	-1.93 to 1.14	4000	agrees; AUC gain 0.012 [0.002, 0.024]	no
C018	direction: total night-time minutes: 136 minutes to 264 minutes (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	0.78	2.19	0.78 to 0.78	4000	agrees; AUC gain 0.006 [0.0005, 0.012]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C019	direction: state: West Virginia to Vermont (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to <1%	-1.40	0.25	-1.47 to -1.47	122, 80	disagrees; AUC gain -0.016 [-0.030, -0.004]	yes
C020	direction: state: West Virginia to Arizona (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 2%	0.63	1.87	0.55 to 0.55	122, 73	disagrees; AUC gain -0.016 [-0.030, -0.004]	yes
C021	direction: state: West Virginia to Colorado (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to <1%	-0.96	0.38	-1.04 to -1.04	122, 78	disagrees; AUC gain -0.016 [-0.030, -0.004]	yes
C022	direction: state: West Virginia to Utah (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 2%	0.50	1.65	0.42 to 0.46	122, 89	disagrees; AUC gain -0.016 [-0.030, -0.004]	yes
C023	direction: state: West Virginia to New Mexico (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 2%	0.50	1.64	0.42 to 0.42	122, 72	disagrees; AUC gain -0.016 [-0.030, -0.004]	yes
C024	direction: state: West Virginia to state other than Alaska, Alabama, Hawaii, Massachusetts, Arkansas, Maryland, Missouri, Montana, North Carolina, North Dakota, Nebraska, Arizona, New Hampshire, New Mexico, Nevada, Ohio, Oklahoma, Oregon, Pennsylvania, California, South Dakota, Tennessee, Utah, Virginia, Vermont, Washington, Wisconsin, West Virginia, Colorado, Wyoming, Connecticut (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 2%	0.50	1.64	0.42 to 0.42	122, 1587	disagrees; AUC gain -0.016 [-0.030, -0.004]	yes
C025	direction: state: West Virginia to Massachusetts (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 2%	0.49	1.64	0.42 to 0.42	122, 86	disagrees; AUC gain -0.016 [-0.030, -0.004]	yes
C026	direction: state: West Virginia to Arkansas (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 2%	0.47	1.59	0.37 to 0.43	122, 78	disagrees; AUC gain -0.016 [-0.030, -0.004]	yes
C027	direction: state: West Virginia to North Carolina (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 2%	0.40	1.49	0.33 to 0.33	122, 74	disagrees; AUC gain -0.016 [-0.030, -0.004]	yes
C028	direction: state: West Virginia to North Dakota (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 2%	0.40	1.49	0.32 to 0.32	122, 68	disagrees; AUC gain -0.016 [-0.030, -0.004]	yes
C029	direction: state: West Virginia to Pennsylvania (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 1%	-0.40	0.67	-0.47 to -0.47	122, 56	disagrees; AUC gain -0.016 [-0.030, -0.004]	yes
C030	direction: state: West Virginia to Alaska (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 1%	-0.39	0.68	-0.46 to -0.46	122, 56	disagrees; AUC gain -0.016 [-0.030, -0.004]	yes
C031	direction: account length: 48 to 152 (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	0.36	1.44	0.36 to 0.36	4000	disagrees; AUC gain -0.002 [-0.004, 0.0007]	yes
C032	direction: state: West Virginia to Oklahoma (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 1%	0.36	1.44	0.29 to 0.29	122, 79	disagrees; AUC gain -0.016 [-0.030, -0.004]	yes
C033	direction: total daytime calls: 75 calls to 125 calls (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	0.31	1.36	0.31 to 0.31	4000	disagrees; AUC gain -0.003 [-0.006, -8e-05]	yes
C034	direction: state: West Virginia to Virginia (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 1%	-0.28	0.76	-0.35 to -0.35	122, 91	disagrees; AUC gain -0.016 [-0.030, -0.004]	yes
C035	direction: state: West Virginia to Wyoming (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 1%	-0.27	0.77	-0.34 to -0.34	122, 89	disagrees; AUC gain -0.016 [-0.030, -0.004]	yes
C036	direction: state: West Virginia to Wisconsin (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 1%	-0.13	0.88	-0.20 to -0.20	122, 86	disagrees; AUC gain -0.016 [-0.030, -0.004]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C037	direction: state: West Virginia to Nevada (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 1%	0.24	1.27	0.16 to 0.16	122, 77	disagrees; AUC gain -0.016 [-0.030, -0.004]	yes
C038	direction: area code: area code 415 to area code 408 (whole model) *shape: categorical; highest to lowest: area code 415, area code 510, area code 408*	1% to 1%	-0.15	0.86	-0.15 to -0.15	2016, 1024	disagrees; AUC gain -0.002 [-0.003, -0.0001]	yes
C039	direction: state: West Virginia to Missouri (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 1%	-0.07	0.93	-0.14 to -0.14	122, 80	disagrees; AUC gain -0.016 [-0.030, -0.004]	yes
C040	direction: area code: area code 415 to area code 510 (whole model) *shape: categorical; highest to lowest: area code 415, area code 510, area code 408*	1% to 1%	-0.13	0.88	-0.13 to -0.13	2016, 960	disagrees; AUC gain -0.002 [-0.003, -0.0001]	yes
C041	direction: state: West Virginia to South Dakota (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 1%	0.48	1.62	-0.37 to 1.32	122, 66	disagrees; AUC gain -0.016 [-0.030, -0.004]	no
C042	direction: state: West Virginia to New Hampshire (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 1%	0.13	1.14	0.05 to 0.05	122, 74	disagrees; AUC gain -0.016 [-0.030, -0.004]	yes
C043	direction: total international minutes: 7 minutes to 14 minutes (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	-0.04	0.96	-0.04 to -0.04	4000	disagrees; AUC gain -0.0002 [-0.0007, 0.0003]	yes
C044	direction: state: West Virginia to Alabama (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 1%	0.04	1.04	-0.04 to -0.04	122, 105	disagrees; AUC gain -0.016 [-0.030, -0.004]	no
C045	direction: state: West Virginia to Nebraska (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 1%	0.09	1.10	0.02 to 0.02	122, 72	disagrees; AUC gain -0.016 [-0.030, -0.004]	yes
C046	direction: total night-time calls: 74 calls to 125 calls (whole model) *shape: flat; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	-0.02	0.98	-0.02 to -0.02	4000	disagrees; AUC gain -0.0001 [-0.0004, 0.0001]	yes
C047	magnitude: international calling plan: no international plan to an international plan, typical patient *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 13%	2.58	13.2		3620, 380	agrees; AUC gain 0.060 [0.035, 0.086]	yes
C048	magnitude: state: West Virginia to California, typical patient *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 6%	1.76	5.84		122, 41	disagrees; AUC gain -0.016 [-0.030, -0.004]	yes
C049	magnitude: voice-mail messages: 0 messages to 32 messages, typical patient *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 5%	1.48	4.37		4000	unclear; AUC gain 0.008 [-0.002, 0.019]	yes
C050	magnitude: customer-service calls: 0 calls to 3 calls, typical patient *shape: accelerating; 84% of patients same; fallback; risk saturates near 0%*	1% to 3%	0.93	2.52		4000	agrees; AUC gain 0.097 [0.060, 0.139]	yes
C051	magnitude: total international charge: 1.84 US dollars to 3.70 US dollars, typical patient *shape: accelerating; 87% of patients same; fallback; risk saturates near 0%*	1% to 2%	0.88	2.41		4000	unclear; AUC gain 0.005 [-0.001, 0.012]	yes
C052	null: model ignores total daytime charge (exact rule)		0.00			4000	train only	yes
C053	null: model ignores total evening calls (exact rule)		0.00			4000	train only	yes
C054	null: model ignores total international calls (exact rule)		0.00			4000	train only	yes
C055	null: model ignores total night-time charge (exact rule)		0.00			4000	train only	yes
C056	structure: made of linear_backbone, service_call_nonlinear, intl_state_unit, eve_minutes_hinge, voicemail_chain, state_offsets, day_calls_state42; added					4000	train only	yes

T.2 Fixed-weight inputs

Annotation ann_99 (linear_backbone). Reads: state, account length, area code, international calling plan, voice-mail plan, number of voice-mail messages, total daytime minutes, total daytime calls, total evening minutes, total evening charge, total night-time minutes, total night-time calls, total international minutes, total international charge.

Original hypothesis (unchanged). Direct input-to-output edges collected through identity_lin (identity). Exactly linear in 40 inputs: day minutes +2.04 per SD, voicemail plan -0.90, international plan +0.76, international charge +0.71, voicemail messages +0.62, evening minutes +0.51, evening charge +0.45, night minutes +0.31, state:11 -0.30, state:46 -0.27, state:5 -0.20, state:47 +0.15, account length +0.14, state:4 +0.14, day calls +0.13, and smaller state/area-code terms. Hypothesis: the additive backbone; day minutes and voicemail/international plan dominate. Evidence: closed form is exact (identity); sensitivities match the weights.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 3.55271e-15).

$$
\begin{aligned}
y_0 = {} & 0.154x_{state:47} + 0.135x_{state:4} + 0.623x_{numberVmailMessages} \\
& + 0.756x_{internationalPlan} + 0.0267x_{state:27_a} + 0.127x_{totalDayCalls_a} \\
& + 2.04x_{totalDayMinutes_a} + 0.0346x_{state:6} + 0.307x_{totalNightMinutes} \\
& + 0.0772x_{state:20} + 0.0915x_{state:26} + 0.45x_{totalEveCharge_a} \\
& + 0.511x_{totalEveMinutes_a} + 0.138x_{accountLength} + 0.705x_{totalIntlCharge_a} \\
& + 0.0454x_{areaCode:415} + 0.0276x_{state:35_a} + 0.0293x_{state:37} \\
& - 0.074x_{state:24_a} - 0.0272x_{areaCode:408} - 0.0115x_{state:49_a} \\
& - 0.0496x_{state:30} - 0.265x_{state:46} - 0.0184x_{state:36} - 0.0899x_{state:48} \\
& - 0.0124x_{state:28} - 0.201x_{state:5} - 0.112x_{state:50_a} - 0.0161x_{areaCode:510} \\
& - 0.0175x_{totalIntlMinutes} - 0.3x_{state:11} - 0.0223x_{state:3_a} \\
& - 0.0117x_{state:2_a} - 0.101x_{state:41_a} - 0.104x_{state:0} - 0.105x_{state:38} \\
& - 0.0533x_{state:29} - 0.899x_{voiceMailPlan_a} - 0.00711x_{totalNightCalls} \\
& - 0.115x_{state:45}
\end{aligned}
$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C057	direction: total daytime minutes: 112 minutes to 249 minutes (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	<1% to 13%	5.19	179	5.19 to 5.19	4000	agrees; AUC gain 0.188 [0.143, 0.234]	yes
C058	direction: international calling plan: no international plan to an international plan (through this part) *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 13%	2.58	13.2	2.58 to 2.58	3620, 380	agrees; AUC gain 0.060 [0.035, 0.086]	yes
C059	direction: voice-mail messages: 0 messages to 32 messages (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 5%	1.48	4.37	1.48 to 1.48	4000	unclear; AUC gain 0.008 [-0.002, 0.019]	yes
C060	direction: state: West Virginia to California (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 4%	1.41	4.10	1.41 to 1.41	122, 41	disagrees; AUC gain -0.012 [-0.025, -0.0006]	yes
C061	direction: state: West Virginia to Washington (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 4%	1.23	3.41	1.23 to 1.23	122, 72	disagrees; AUC gain -0.012 [-0.025, -0.0006]	yes
C062	direction: total international charge: 1.84 US dollars to 3.70 US dollars (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	<1% to 3%	1.76	5.79	1.76 to 1.76	4000	agrees; AUC gain 0.018 [0.006, 0.030]	yes
C063	direction: total evening minutes: 138 minutes to 266 minutes (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	1.30	3.66	1.30 to 1.30	4000	agrees; AUC gain 0.020 [0.009, 0.032]	yes
C064	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	1.14	3.12	1.14 to 1.14	4000	agrees; AUC gain 0.017 [0.007, 0.027]	yes
C065	direction: state: West Virginia to Montana (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 2%	0.74	2.10	0.74 to 0.74	122, 75	disagrees; AUC gain -0.012 [-0.025, -0.0006]	yes
C066	direction: state: West Virginia to Hawaii (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to <1%	-2.18	0.11	-2.18 to -2.18	122, 73	disagrees; AUC gain -0.012 [-0.025, -0.0006]	yes
C067	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	1% to <1%	-2.04	0.13	-2.04 to -2.04	2939, 1061	agrees; AUC gain 0.018 [0.004, 0.034]	yes
C068	direction: state: West Virginia to Vermont (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to <1%	-1.83	0.16	-1.83 to -1.83	122, 80	disagrees; AUC gain -0.012 [-0.025, -0.0006]	yes
C069	direction: state: West Virginia to Maryland (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 2%	0.61	1.84	0.61 to 0.61	122, 82	disagrees; AUC gain -0.012 [-0.025, -0.0006]	yes
C070	direction: total night-time minutes: 136 minutes to 264 minutes (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	0.78	2.19	0.78 to 0.78	4000	agrees; AUC gain 0.006 [0.0005, 0.012]	yes
C071	direction: state: West Virginia to Colorado (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to <1%	-1.39	0.25	-1.39 to -1.39	122, 78	disagrees; AUC gain -0.012 [-0.025, -0.0006]	yes
C072	direction: state: West Virginia to Pennsylvania (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to <1%	-0.83	0.44	-0.83 to -0.83	122, 56	disagrees; AUC gain -0.012 [-0.025, -0.0006]	yes
C073	direction: state: West Virginia to Alaska (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to <1%	-0.82	0.44	-0.82 to -0.82	122, 56	disagrees; AUC gain -0.012 [-0.025, -0.0006]	yes
C074	direction: state: West Virginia to South Dakota (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	-0.73	0.48	-0.73 to -0.73	122, 66	disagrees; AUC gain -0.012 [-0.025, -0.0006]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C075	direction: state: West Virginia to Virginia (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	-0.70	0.49	-0.70 to -0.70	122, 91	disagrees; AUC gain -0.012 [-0.025, -0.0006]	yes
C076	direction: state: West Virginia to Wyoming (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	-0.69	0.50	-0.69 to -0.69	122, 89	disagrees; AUC gain -0.012 [-0.025, -0.0006]	yes
C077	direction: state: West Virginia to Wisconsin (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	-0.55	0.57	-0.55 to -0.55	122, 86	disagrees; AUC gain -0.012 [-0.025, -0.0006]	yes
C078	direction: state: West Virginia to Missouri (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	-0.46	0.63	-0.46 to -0.46	122, 80	disagrees; AUC gain -0.012 [-0.025, -0.0006]	yes
C079	direction: state: West Virginia to Connecticut (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 2%	0.31	1.37	0.31 to 0.31	122, 80	disagrees; AUC gain -0.012 [-0.025, -0.0006]	yes
C080	direction: account length: 48 to 152 (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	0.36	1.44	0.36 to 0.36	4000	disagrees; AUC gain -0.002 [-0.004, 0.0007]	yes
C081	direction: total daytime calls: 75 calls to 125 calls (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	0.32	1.38	0.32 to 0.32	4000	disagrees; AUC gain -0.003 [-0.006, -8e-06]	yes
C082	direction: state: West Virginia to Oregon (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	0.27	1.31	0.27 to 0.27	122, 84	disagrees; AUC gain -0.012 [-0.025, -0.0006]	yes
C083	direction: state: West Virginia to North Carolina (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	0.26	1.30	0.26 to 0.26	122, 74	disagrees; AUC gain -0.012 [-0.025, -0.0006]	yes
C084	direction: state: West Virginia to Ohio (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	0.26	1.29	0.26 to 0.26	122, 87	disagrees; AUC gain -0.012 [-0.025, -0.0006]	yes
C085	direction: state: West Virginia to Nebraska (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	-0.33	0.72	-0.33 to -0.33	122, 72	disagrees; AUC gain -0.012 [-0.025, -0.0006]	yes
C086	direction: state: West Virginia to New Hampshire (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	-0.30	0.74	-0.30 to -0.30	122, 74	disagrees; AUC gain -0.012 [-0.025, -0.0006]	yes
C087	direction: area code: area code 415 to area code 408 (through this part) *shape: categorical; highest to lowest: area code 415, area code 510, area code 408*	1% to 1%	-0.15	0.86	-0.15 to -0.15	2016, 1024	disagrees; AUC gain -0.002 [-0.003, -0.0001]	yes
C088	direction: area code: area code 415 to area code 510 (through this part) *shape: categorical; highest to lowest: area code 415, area code 510, area code 408*	1% to 1%	-0.13	0.88	-0.13 to -0.13	2016, 960	disagrees; AUC gain -0.002 [-0.003, -0.0001]	yes
C089	direction: state: West Virginia to Arizona (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	-0.10	0.91	-0.10 to -0.10	122, 73	disagrees; AUC gain -0.012 [-0.025, -0.0006]	yes
C090	direction: state: West Virginia to state other than Alaska, Hawaii, Arkansas, Maryland, Missouri, Montana, North Carolina, North Dakota, Nebraska, Arizona, New Hampshire, Ohio, Oklahoma, Oregon, Pennsylvania, California, South Dakota, Virginia, Vermont, Washington, Wisconsin, West Virginia, Colorado, Wyoming, Connecticut (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	0.07	1.07	0.07 to 0.07	122, 2084	disagrees; AUC gain -0.012 [-0.025, -0.0006]	yes
C091	direction: state: West Virginia to Oklahoma (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	-0.07	0.94	-0.07 to -0.07	122, 79	disagrees; AUC gain -0.012 [-0.025, -0.0006]	yes
C092	direction: total international minutes: 7 minutes to 14 minutes (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	-0.04	0.96	-0.04 to -0.04	4000	disagrees; AUC gain -0.0002 [-0.0007, 0.0003]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C093	direction: state: West Virginia to North Dakota (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	-0.03	0.97	-0.03 to -0.03	122, 68	disagrees; AUC gain -0.012 [-0.025, -0.0006]	yes
C094	direction: total night-time calls: 74 calls to 125 calls (through this part) *shape: flat; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	-0.02	0.98	-0.02 to -0.02	4000	disagrees; AUC gain -0.0001 [-0.0004, 0.0001]	yes
C095	direction: state: West Virginia to Arkansas (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	-0.02	0.98	-0.02 to -0.02	122, 78	disagrees; AUC gain -0.012 [-0.025, -0.0006]	yes
C096	structure: its formula reproduces it on 4000 training patients (largest difference 3.55271e-15)		4e-15			4000	train only	yes

T.3 Service calls

Annotation ann_101 (service_call_nonlinear). Reads: state, total daytime minutes, total evening charge, total international charge, number of calls to customer service.

Original hypothesis (unchanged). Composition of csc_usage_gate and csc_hinge: their exits are summed by identity_svc = 1.60*807 + 1.81*1264 + 1.48*7208, which enters the logit with weight 1; they share only the customer-service-calls input (split copies). Hypothesis: all of the model's service-call effect is nonlinear (calls have no direct edge): risk jumps from 4 calls, and for frequent callers low day usage raises risk further (the gate). Evidence: mean-ablating both gives AUC 0.736, linear replacement of both 0.834 (all-linear model 0.837); the gate alone restores 0.898 of 0.901.

Closed form. Read through its parts (their formulas are in their own sections).

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C097	direction: total daytime minutes: 112 minutes to 249 minutes (through this part) *shape: hinge at 170 minutes; 71% of patients same; fallback; risk saturates near 0%; one point with C107 (median fold 170 minutes, typical patient 160 minutes, 10th-90th percentile 147 minutes to 216 minutes, wide)*	15% to 1%	-3.63	0.03	-5.95 to -2.04	4000	agrees; AUC gain 0.062 [0.035, 0.095]	yes
C098	direction: customer-service calls: 0 calls to 3 calls (through this part) *shape: accelerating; 84% of patients same; fallback; risk saturates near 0%*	1% to 3%	1.19	3.30	0.00 to 2.96	4000	agrees; AUC gain 0.097 [0.060, 0.139]	yes
C099	direction: total international charge: 1.84 US dollars to 3.70 US dollars (through this part) *shape: hinge at 2.95 US dollars; 84% of patients same; fallback; risk saturates near 0%; one point with C109 (median fold 2.95 US dollars, typical patient 2.92 US dollars, 10th-90th percentile 2.95 US dollars to 4.34 US dollars, wide)*	2% to 1%	-1.08	0.34	-1.46 to -0.88	4000	unclear; AUC gain 0.006 [-0.001, 0.014]	yes
C100	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (through this part) *shape: hinge at 18.11 US dollars; 65% of patients same; fallback; risk saturates near 0%; one point with C110 (median fold 18.11 US dollars, typical patient 13.46 US dollars, 10th-90th percentile 7.97 US dollars to 21.09 US dollars, wide)*	2% to 1%	-1.19	0.30	-3.06 to 0.00	4000	unclear; AUC gain 0.009 [-0.003, 0.020]	yes
C101	direction: state: West Virginia to South Dakota (through this part) *shape: categorical; highest to lowest: South Dakota, West Virginia, state other than South Dakota, West Virginia*	1% to 2%	0.86	2.36	0.00 to 1.70	122, 66	disagrees; AUC gain -0.002 [-0.004, -0.0001]	yes
C102	direction: state: West Virginia to state other than South Dakota, West Virginia (through this part) *shape: categorical; highest to lowest: South Dakota, West Virginia, state other than South Dakota, West Virginia*	1% to 1%	0.08	1.08	0.00 to 0.00	122, 3812	disagrees; AUC gain -0.002 [-0.004, -0.0001]	yes
C103	interaction: effect of total daytime minutes (112 minutes to 249 minutes) by total evening charge: total evening charge above 17.12 US dollars minus total evening charge at or below 17.12 US dollars; stronger in total evening charge above 17.12 US dollars *shape: dip at 154 minutes; 100% of patients same; verified; risk saturates near 0%*	total evening charge above 17.12 US dollars: 1% to 17%; total evening charge at or below 17.12 US dollars: 2% to 10%	1.53	4.60		1998, 2002	agrees; AUC gain 0.009 [-0.003, 0.020]	yes
C104	interaction: effect of total daytime minutes (112 minutes to 249 minutes) by state: South Dakota minus West Virginia; stronger in West Virginia *shape: dip at 160 minutes; 98% of patients same; verified; risk saturates near 0%*	South Dakota: 5% to 9%; West Virginia: 1% to 13%	-1.09	0.34		66, 122	agrees; AUC gain -0.002 [-0.004, -0.0001]	yes
C105	interaction: effect of customer-service calls (0 calls to 3 calls) by total evening charge: total evening charge above 17.12 US dollars minus total evening charge at or below 17.12 US dollars; stronger in total evening charge at or below 17.12 US dollars *shape: accelerating; 89% of patients same; fallback; risk saturates near 0%*	total evening charge above 17.12 US dollars: 1% to 2%; total evening charge at or below 17.12 US dollars: 1% to 5%	-0.81	0.45		1998, 2002	agrees; AUC gain 0.009 [-0.003, 0.020]	yes
C106	interaction: effect of total evening charge (11.76 US dollars to 22.57 US dollars) by state: South Dakota minus West Virginia; stronger in South Dakota *shape: dip at 18.53 US dollars; 68% of patients same; fallback; risk saturates near 0%*	South Dakota: 4% to 1%; West Virginia: 1% to 2%	-0.84	0.43		66, 122	unclear; AUC gain -0.002 [-0.004, -0.0001]	yes
C107	threshold: total daytime minutes: fold at about 170 minutes (stops) *shape: hinge at 170 minutes; 71% of patients same; fallback; risk saturates near 0%; one point with C107 (median fold 170 minutes, typical patient 160 minutes, 10th-90th percentile 147 minutes to 216 minutes, wide)*	38% to 1% (typical patient with a fold)	-4.88	0.008	-7.17 to -3.26	3999, 1839	agrees; AUC gain 0.062 [0.035, 0.095]	yes
C108	threshold: customer-service calls: fold at about 4 calls (starts) *shape: accelerating; 73% of patients same; fallback; risk saturates near 0%*	3% to 11% (typical patient with a fold)	1.84	6.28	0.60 to 3.83	2668, 577	agrees; AUC gain 0.097 [0.060, 0.139]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C109	threshold: total international charge: fold at about 2.95 US dollars (stops) *shape: hinge at 2.95 US dollars; 88% of patients same; fallback; risk saturates near 0%; one point with C109 (median fold 2.95 US dollars, typical patient 2.92 US dollars, 10th-90th percentile 2.95 US dollars to 4.34 US dollars, wide)*	3% to 1% (typical patient with a fold)	-1.42	0.24	-2.21 to -1.11	3851, 2752	unclear; AUC gain 0.006 [-0.001, 0.014]	yes
C110	threshold: total evening charge: fold at about 18.11 US dollars (stops) for customers in Colorado *shape: hinge at 18.11 US dollars; 75% of patients same; fallback; risk saturates near 0%; one point with C110 (median fold 18.11 US dollars, typical patient 17.01 US dollars, 10th-90th percentile 7.97 US dollars to 21.09 US dollars, wide)*	2% to <1% (typical patient with a fold (customers in Colorado))	-2.12	0.12	-3.60 to 0.00	71, 34	unclear; AUC gain 0.009 [-0.003, 0.020]	yes
C111	structure: made of csc_usage_gate, csc_hinge; added					4000	train only	yes
C162	boundary: boundary					4000	train only	no

T.4 Calls with low usage

Annotation ann_91 (csc_usage_gate). Reads: state, total daytime minutes, total evening charge, total international charge, number of calls to customer service.

Original hypothesis (unchanged). Service-call x low-day-usage gate. 807 = relu(-1.29 - 1.40 dayMin + 0.04 csc + 0.675 g(csc) + 0.622 relu(0.42 - 1.00 eveCharge + 0.36 csc + 0.34 relu(0.99 csc - 0.65) + 0.15 state:41)), with g(csc) = relu(1.01 + 0.28 csc + 0.33 relu(0.86 csc - 0.10)) always active (convex in csc, kink near 2 calls); 1264 = relu(0.08 + 0.36*807 - 0.33 intlCharge). The logit gets +1.60*807 + 1.81*1264. Hypothesis: 807 is a hinge that switches on at LOW day minutes, and its kink moves up with each customer-service call (median kink on train: 147 day minutes at 0 calls, 159 at 1, 174 at 2, 198 at 3, 220 at 4, 244 at 5, 276 at 6); lower evening charge also raises the kink. Below the kink the gate's slope is -3.15 logit per SD of day minutes, which more than cancels the direct +2.04, so the logit is V-shaped in day minutes (net -1.12 per SD below the kink). 1264 adds a second hinge on low international charge that is active on 1281 rows where 807 is off. Evidence (train, n=4000): CSC>=4 and day minutes < 160: 113 customers, empirical churn 0.885, model mean 0.655; CSC<4 and day minutes < 120: 476 customers, empirical 0.046, model 0.084. Mean churn by calls 0-6: 0.115/0.110/0.110/0.106/0.437/0.571/0.679, model 0.098/0.092/0.099/0.147/0.333/0.645/0.923; share of rows with 807 on rises 0.32 -> 0.96. Sensitivity: 807 mostly day minutes (0.64), service calls (0.26), evening charge (0.22). Ablation: mean-ablation AUC 0.764 (full 0.901); least-squares linear replacement 0.848 (R2 0.70/0.75); restored alone onto an all-linearised model 0.898 (all-linear 0.837). This is the model's main nonlinearity.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 1.77636e-15).

$$
\begin{aligned}
y_0 = \operatorname{relu}\big(&-1.29 + 0.622\operatorname{relu}\big(0.416 \\
&+ 0.154x_{state:41_b} + 0.342\operatorname{relu}\big(-0.649 \\
&+ 0.987x_{numberCustomerServiceCalls_abcfh}\big) \\
&+ 0.36x_{numberCustomerServiceCalls_abcfh} - 1.0x_{totalEveCharge_b}\big) \\
&+ 0.0405x_{numberCustomerServiceCalls_abcfh} + 0.675\operatorname{relu}\big(1.01 \\
&+ 0.325\operatorname{relu}\big(-0.0966 + 0.855x_{numberCustomerServiceCalls_abcfh}\big) \\
&+ 0.278x_{numberCustomerServiceCalls_abcfh}\big) - 1.4x_{totalDayMinutes_b}\big) \\
y_1 = \operatorname{relu}\big(&0.0759 + 0.363\operatorname{relu}\big(-1.29 \\
&+ 0.622\operatorname{relu}\big(0.416 + 0.154x_{state:41_b} \\
&+ 0.342\operatorname{relu}\big(-0.649 + 0.987x_{numberCustomerServiceCalls_abcfh}\big) \\
&+ 0.36x_{numberCustomerServiceCalls_abcfh} - 1.0x_{totalEveCharge_b}\big) \\
&+ 0.0405x_{numberCustomerServiceCalls_abcfh} + 0.675\operatorname{relu}\big(1.01 \\
&+ 0.325\operatorname{relu}\big(-0.0966 + 0.855x_{numberCustomerServiceCalls_abcfh}\big) \\
&+ 0.278x_{numberCustomerServiceCalls_abcfh}\big) - 1.4x_{totalDayMinutes_b}\big) \\
&- 0.326x_{totalIntlCharge_c}\big)
\end{aligned}
$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C112	direction: total daytime minutes: 112 minutes to 249 minutes (through this part) *shape: hinge at 170 minutes; 71% of patients same; fallback; risk saturates near 0%; one point with C119 (median fold 170 minutes, typical patient 160 minutes, 10th-90th percentile 147 minutes to 216 minutes, wide)*	15% to 1%	-3.63	0.03	-5.95 to -2.04	4000	agrees; AUC gain 0.062 [0.035, 0.095]	yes
C113	direction: customer-service calls: 0 calls to 3 calls (through this part) *shape: accelerating; 80% of patients same; fallback; risk saturates near 0%*	1% to 3%	1.19	3.30	0.00 to 2.96	4000	agrees; AUC gain 0.061 [0.035, 0.091]	yes
C114	direction: total international charge: 1.84 US dollars to 3.70 US dollars (through this part) *shape: hinge at 2.95 US dollars; 84% of patients same; fallback; risk saturates near 0%; one point with C121 (median fold 2.95 US dollars, typical patient 2.92 US dollars, 10th-90th percentile 2.95 US dollars to 4.34 US dollars, wide)*	2% to 1%	-1.08	0.34	-1.46 to -0.88	4000	unclear; AUC gain 0.006 [-0.001, 0.014]	yes
C115	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (through this part) *shape: hinge at 18.11 US dollars; 65% of patients same; fallback; risk saturates near 0%; one point with C122 (median fold 18.11 US dollars, typical patient 13.46 US dollars, 10th-90th percentile 7.97 US dollars to 21.09 US dollars, wide)*	2% to 1%	-1.19	0.30	-3.06 to 0.00	4000	unclear; AUC gain 0.009 [-0.003, 0.020]	yes
C116	direction: state: state other than South Dakota to South Dakota (through this part) *shape: categorical; highest to lowest: South Dakota, state other than South Dakota*	1% to 2%	0.78	2.19	0.00 to 1.70	3934, 66	disagrees; AUC gain -0.0007 [-0.002, 0.0001]	yes
C117	interaction: effect of total daytime minutes (112 minutes to 249 minutes) by state: South Dakota minus state other than South Dakota; stronger in state other than South Dakota *shape: dip at 160 minutes; 100% of patients same; verified; risk saturates near 0%*	South Dakota: 5% to 9%; state other than South Dakota: 1% to 13%	-1.26	0.28		66, 3934	agrees; AUC gain -0.0007 [-0.002, 0.0001]	yes
C118	interaction: effect of total evening charge (11.76 US dollars to 22.57 US dollars) by state: South Dakota minus state other than South Dakota; stronger in South Dakota *shape: dip at 18.53 US dollars; 68% of patients same; fallback; risk saturates near 0%*	South Dakota: 4% to 1%; state other than South Dakota: 1% to 2%	-0.90	0.41		66, 3934	unclear; AUC gain -0.0007 [-0.002, 0.0001]	yes
C119	threshold: total daytime minutes: fold at about 170 minutes (stops) *shape: hinge at 170 minutes; 71% of patients same; fallback; risk saturates near 0%; one point with C119 (median fold 170 minutes, typical patient 160 minutes, 10th-90th percentile 147 minutes to 216 minutes, wide)*	38% to 1% (typical patient with a fold)	-4.88	0.008	-7.17 to -3.26	3999, 1839	agrees; AUC gain 0.062 [0.035, 0.095]	yes
C120	threshold: customer-service calls: fold at about 4 calls (starts) for customers in North Carolina *shape: accelerating; 77% of patients same; fallback*	4% to 10% (typical patient with a fold (customers in North Carolina))	1.01	2.75	0.00 to 3.02	51, 8 (small n)	agrees; AUC gain 0.061 [0.035, 0.091]	yes
C121	threshold: total international charge: fold at about 2.95 US dollars (stops) *shape: hinge at 2.95 US dollars; 88% of patients same; fallback; risk saturates near 0%; one point with C121 (median fold 2.95 US dollars, typical patient 2.92 US dollars, 10th-90th percentile 2.95 US dollars to 4.34 US dollars, wide)*	3% to 1% (typical patient with a fold)	-1.42	0.24	-2.21 to -1.11	3851, 2752	unclear; AUC gain 0.006 [-0.001, 0.014]	yes
C122	threshold: total evening charge: fold at about 18.11 US dollars (stops) for customers in Colorado *shape: hinge at 18.11 US dollars; 75% of patients same; fallback; risk saturates near 0%; one point with C122 (median fold 18.11 US dollars, typical patient 17.01 US dollars, 10th-90th percentile 7.97 US dollars to 21.09 US dollars, wide)*	2% to <1% (typical patient with a fold (customers in Colorado))	-2.12	0.12	-3.60 to 0.00	71, 34	unclear; AUC gain 0.009 [-0.003, 0.020]	yes
C123	subgroup: who this part acts on: 3121 of 4000 patients	<1% to 1% (typical patient this part acts on (risk without the part, then with it))	2.00	7.39	0.18 to 5.02	3121, 879	agrees; AUC gain 0.141 [0.093, 0.190]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C124	structure: its formula reproduces it on 4000 training patients (largest difference 1.77636e-15)		2e-15			4000	train only	yes
C163	boundary: boundary					4000	train only	no

T.5 Many service calls

Annotation ann_93 (csc_hinge). Reads: state, number of calls to customer service.

Original hypothesis (unchanged). 7208 = relu(-0.67 + 0.54 relu(csc + 0.008) - 0.36 state:49), logit weight +1.48. Hypothesis: a hinge on customer-service calls that switches on from 4 calls upward (kink at 3.17 calls) and adds about +0.61 logit per extra call (contribution 0.60 at 4 calls, 1.22 at 5, 1.83 at 6, 2.45 at 7); state:49 switches it off (12 state:49 customers with 4+ calls, 5 of them churned). Active on 306 of 318 customers with 4+ calls. Evidence: line plot of 7208 on train rises from 0 to 2.5 (correlation with calls 0.66); mean ablation AUC 0.8996, linear replacement 0.8986; restored alone onto the all-linear model 0.851 (all-linear 0.837), but redundant once the gate is present (gate alone 0.898).

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 2.22045e-16).

$$y_0 = \mathrm{relu}\big(-0.673 + 0.544\,\mathrm{relu}\big(0.00784 + 1.0x_{numberCustomerServiceCalls_d}\big) - 0.359x_{state:49_c}\big)$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C130	direction: state: West Virginia to state other than West Virginia (through this part) *shape: categorical; highest to lowest: state other than West Virginia, West Virginia*	16% to 26% (typical patient this part acts on)	0.08	1.08	0.00 to 0.00	122, 3878	disagrees; AUC gain -0.001 [-0.003, 0.0002]	yes
C131	threshold: customer-service calls: fold at about 4 calls (starts) *shape: hinge at 3 calls; 100% of patients same; verified; risk saturates near 0%; one point with C131 (median fold 4 calls, typical patient 3 calls, 10th-90th percentile 3 calls to 3 calls)*	1% to 1% (typical patient with a fold)	0.58	1.79	0.60 to 0.60	4000, 306	unclear; AUC gain 0.002 [-0.001, 0.006]	yes
C132	subgroup: who this part acts on: 306 of 4000 patients	16% to 26% (typical patient this part acts on (risk without the part, then with it))	0.95	2.58	0.60 to 1.83	306, 3694	unclear; AUC gain 0.001 [-0.002, 0.005]	yes
C133	structure: its formula reproduces it on 4000 training patients (largest difference 2.22045e-16)		2e-16			4000	train only	yes

T.6 Voice-mail discount

Annotation ann_95 (voicemail_chain). Reads: state, voice-mail plan.

Original hypothesis (unchanged). 5577 = relu(0.99 + 0.99 voiceMailPlan + 0.26 state:1), 1133 = relu(0.68 + 0.14*5577 + 0.02 state:27), 4051 = relu(2.09 + 1.15*1133 + 0.06 state:49). All three are always active on train (minimum pre-activations 0.35, 0.73, 2.91), so the chain is exactly affine in its binary inputs (least-squares R2 = 1.0). Hypothesis: a linear voicemail-plan term (net logit -0.95*1133 - 0.99*4051 = about -0.28 per SD of voicemail plan on top of the direct -0.90) plus small state:1/27/49 offsets. Evidence: exact linear fit; linear replacement AUC 0.9009 (unchanged).

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 4.44089e-16).

$$
\begin{aligned}
y_0 = \operatorname{relu}\big(&0.68 + 0.137\operatorname{relu}\big(0.991 \\
&+ 0.255x_{state:1} + 0.991x_{voiceMailPlan_b}\big) + 0.0189x_{state:27_b}\big) \\
y_1 = \operatorname{relu}\big(&2.09 + 1.15\operatorname{relu}\big(0.68 \\
&+ 0.137\operatorname{relu}\big(0.991 + 0.255x_{state:1} + 0.991x_{voiceMailPlan_b}\big) \\
&+ 0.0189x_{state:27_b}\big) + 0.0611x_{state:49_b}\big)
\end{aligned}
$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C125	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	1% to 1%	-0.64	0.53	-0.64 to -0.64	2939, 1061	unclear; AUC gain 0.002 [-0.004, 0.006]	yes
C126	direction: state: West Virginia to state other than Alabama, North Carolina, West Virginia (through this part) *shape: categorical; highest to lowest: state other than Alabama, North Carolina, West Virginia, North Carolina, West Virginia, Alabama*	1% to 2%	0.35	1.42	0.35 to 0.35	122, 3699	disagrees; AUC gain -0.0005 [-0.002, 0.001]	yes
C127	direction: state: West Virginia to Alabama (through this part) *shape: categorical; highest to lowest: state other than Alabama, North Carolina, West Virginia, North Carolina, West Virginia, Alabama*	1% to 1%	-0.10	0.90	-0.10 to -0.10	122, 105	disagrees; AUC gain -0.0005 [-0.002, 0.001]	yes
C128	direction: state: West Virginia to North Carolina (through this part) *shape: categorical; highest to lowest: state other than Alabama, North Carolina, West Virginia, North Carolina, West Virginia, Alabama*	1% to 1%	0.06	1.06	0.06 to 0.06	122, 74	disagrees; AUC gain -0.0005 [-0.002, 0.001]	yes
C129	structure: its formula reproduces it on 4000 training patients (largest difference 4.44089e-16)		4e-16			4000	train only	yes

T.7 Heavy evening use

Annotation ann_94 (eve_minutes_hinge). Reads: total evening minutes.

Original hypothesis (unchanged). 476 = relu(1.07 eveMinutes - 0.81), logit weight -0.55. Hypothesis: above about 239 evening minutes (885 of 4000 training customers) this hinge flattens the direct evening-minutes slope (+0.51 per SD direct, -0.59 per SD through the hinge), so evening minutes stop adding risk above about 239 minutes (evening charge, +0.45 per SD, still adds). Evidence: line plot of 476 on train from 0 to 2.7; mean ablation AUC 0.9002, linear replacement 0.9005; restoring it alone onto the all-linear model lowers AUC (0.834 vs 0.837): a small fit-to-train detail.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 0.0).

$$y_0 = \mathrm{relu}\big(-0.805 + 1.07 x_{totalEveMinutes_b}\big)$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C134	direction: total evening minutes: 138 minutes to 266 minutes (through this part) *shape: hinge at 239 minutes; 100% of patients same; verified; risk saturates near 0%; one point with C135 (median fold 239 minutes, typical patient 236 minutes, 10th-90th percentile 239 minutes to 239 minutes)*	1% to 1%	-0.32	0.73	-0.32 to -0.32	4000	disagrees; AUC gain -0.004 [-0.009, 0.0002]	yes
C135	threshold: total evening minutes: fold at about 239 minutes (starts) *shape: hinge at 239 minutes; 100% of patients same; verified; risk saturates near 0%; one point with C135 (median fold 239 minutes, typical patient 236 minutes, 10th-90th percentile 239 minutes to 239 minutes)*	1% to 1% (typical patient with a fold)	-0.54	0.58	-0.54 to -0.54	4000, 885	disagrees; AUC gain -0.004 [-0.009, 0.0002]	yes
C136	subgroup: who this part acts on: 885 of 4000 patients	5% to 4% (typical patient this part acts on (risk without the part, then with it))	-0.35	0.70	-0.77 to -0.05	885, 3115	disagrees; AUC gain -0.004 [-0.009, 0.0002]	yes
C137	structure: its formula reproduces it on 4000 training patients (largest difference 0.0)		0.00			4000	train only	yes

T.8 State units

Annotation ann_96 (state_offsets). Reads: state.

Original hypothesis (unchanged). Units 8896 (state:19), 10249 (state:3), 11951 and 14942 (state:35), 6730 -> 13827 (state:33, state:24, state:35), 3861 (state:50, identically 0 on train). Each depends only on state one-hot indicators; since the states are mutually exclusive, every such function is exactly an offset per state (least-squares R2 = 1.0 for every exit). Hypothesis: a per-state logit offset table; the largest offsets here are state:33 about -0.25 (switches off 6730 -> 13827), state:35 +0.21 (14942), state:3 +0.30 (10249). Evidence: linear replacement AUC 0.9009 (unchanged); mean ablation 0.9006.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 2.22045e-16).

$$
\begin{aligned}
y_0 &= \mathrm{relu}\big(-0.106 - 1.0 x_{state:19}\big) \\
y_1 &= \mathrm{relu}\big(0.172 + 0.998 x_{state:3_b}\big) \\
y_2 &= \mathrm{relu}\big(-1.74 + 1.0 x_{state:35_bcd}\big) \\
y_3 &= \mathrm{relu}\big(-0.737 + 1.01\,\mathrm{relu}\big(1.26 \\
&\qquad + 0.00244 x_{state:35_bcd} - 0.0113 x_{state:24_b} - 0.000939\,\mathrm{relu}\big(-1.09 \\
&\qquad + 1.0 x_{state:35_bcd}\big) - 0.418 x_{state:33}\big)\big) \\
y_4 &= \mathrm{relu}\big(-0.431 - 0.771 x_{state:50_b}\big)
\end{aligned}
$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C138	direction: state: state other than Massachusetts, Missouri, Arizona, Nevada, Ohio, Wyoming to Arizona (through this part) *shape: categorical; highest to lowest: Arizona, Ohio, Wyoming, state other than Massachusetts, Missouri, Arizona, Nevada, Ohio, Wyoming, Massachusetts, Missouri …*	1% to 1%	0.30	1.35	0.30 to 0.30	3508, 73	disagrees; AUC gain -0.0003 [-0.001, 0.001]	yes
C139	direction: state: state other than Massachusetts, Missouri, Arizona, Nevada, Ohio, Wyoming to Ohio (through this part) *shape: categorical; highest to lowest: Arizona, Ohio, Wyoming, state other than Massachusetts, Missouri, Arizona, Nevada, Ohio, Wyoming, Massachusetts, Missouri …*	1% to 1%	0.21	1.24	0.21 to 0.21	3508, 87	disagrees; AUC gain -0.0003 [-0.001, 0.001]	yes
C140	direction: state: state other than Massachusetts, Missouri, Arizona, Nevada, Ohio, Wyoming to Nevada (through this part) *shape: categorical; highest to lowest: Arizona, Ohio, Wyoming, state other than Massachusetts, Missouri, Arizona, Nevada, Ohio, Wyoming, Massachusetts, Missouri …*	1% to 1%	-0.26	0.77	-0.26 to -0.26	3508, 77	disagrees; AUC gain -0.0003 [-0.001, 0.001]	yes
C141	direction: state: state other than Massachusetts, Missouri, Arizona, Nevada, Ohio, Wyoming to Missouri (through this part) *shape: categorical; highest to lowest: Arizona, Ohio, Wyoming, state other than Massachusetts, Missouri, Arizona, Nevada, Ohio, Wyoming, Massachusetts, Missouri …*	1% to 1%	-0.04	0.97	-0.04 to -0.04	3508, 80	disagrees; AUC gain -0.0003 [-0.001, 0.001]	yes
C142	direction: state: state other than Massachusetts, Missouri, Arizona, Nevada, Ohio, Wyoming to Massachusetts (through this part) *shape: categorical; highest to lowest: Arizona, Ohio, Wyoming, state other than Massachusetts, Missouri, Arizona, Nevada, Ohio, Wyoming, Massachusetts, Missouri …*	1% to 1%	-0.001	1.00	-0.001 to -0.001	3508, 86	disagrees; AUC gain -0.0003 [-0.001, 0.001]	yes
C143	structure: its formula reproduces it on 4000 training patients (largest difference 2.22045e-16)		2e-16			4000	train only	yes

T.9 Daytime calls and Tennessee

Annotation ann_97 (day_calls_state42). Reads: state, total daytime calls.

Original hypothesis (unchanged). 11427 = relu(0.20 + 1.00 state:42 - 0.11 dayCalls), logit weight +0.05. Hypothesis: a tiny hinge that is on for up to about 110 day calls and switches off above about 120 (on for 100% of rows below 110 day calls, 44% at 110-119, 2% at 120+), plus a state:42 offset; contribution at most +0.40 logit. Evidence: linear fit R2 0.999; mean ablation and linear replacement leave AUC at 0.9009.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 5.55112e-17).

$$y_0 = \mathrm{relu}\big(0.198 + 0.995 x_{state:42} - 0.105 x_{totalDayCalls_b}\big)$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C144	direction: state: state other than Tennessee to Tennessee (through this part) *shape: categorical; highest to lowest: Tennessee, state other than Tennessee*	1% to 2%	0.39	1.47	0.38 to 0.39	3932, 68	disagrees; AUC gain -0.0006 [-0.001, -7e-05]	yes
C145	direction: total daytime calls: 75 calls to 125 calls (through this part) *shape: flat; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	-0.01	0.99	-0.01 to -0.01	4000	unclear; AUC gain 4e-05 [-0.0002, 0.0003]	yes
C146	interaction: effect of total daytime calls (75 calls to 125 calls) by state: Tennessee minus state other than Tennessee; stronger in state other than Tennessee *shape: steady; 100% of patients same; verified; risk saturates near 0%*	Tennessee: 2% to 3%; state other than Tennessee: 1% to 1%	-0.003	1.00		68, 3932	agrees; AUC gain -0.0006 [-0.001, -7e-05]	no
C147	threshold: total daytime calls: fold at about 114 calls (stops) *shape: flat; 100% of patients same; verified; risk saturates near 0%*	1% to 1% (typical patient with a fold)	-0.01	0.99	-0.01 to -0.01	3932, 2926	unclear; AUC gain 4e-05 [-0.0002, 0.0003]	no
C148	subgroup: who this part acts on: 2994 of 4000 patients	1% to 1% (typical patient this part acts on (risk without the part, then with it))	0.01	1.01	0.001 to 0.01	2994, 1006	disagrees; AUC gain -0.0005 [-0.001, 9e-06]	no
C149	structure: its formula reproduces it on 4000 training patients (largest difference 5.55112e-17)		6e-17			4000	train only	yes

T.10 Small state unit

Annotation ann_92 (intl_state_unit). Reads: state, total international charge, number of calls to customer service.

Original hypothesis (unchanged). Near-dead unit 252 = relu(-2.22 + 0.76 state:2 + 0.44 state:44 - 0.87 intlCharge + 0.79*161 - 0.38 g(csc)), with 161 = relu(-0.14 - 0.011 state:32) identically 0 on train (dead). 252 is active on 153 of 4000 training rows (78 in state:2, 55 in state:44, mostly with low international charge); its logit contribution is at most +0.10 (weight 0.02). Hypothesis: a negligible state-specific low-international-charge offset. Evidence: mean ablation and linear replacement leave train AUC at 0.9009; restoring it onto the all-linear model changes AUC by -0.0001.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 8.88178e-16).

$$
\begin{aligned}
y_0 = \operatorname{relu}\big(&-2.22 + 0.76 x_{state:2_b} \\
&+ 0.791 \operatorname{relu}\big(-0.14 - 0.0113 x_{state:32}\big) + 0.437 x_{state:44} \\
&- 0.378 \operatorname{relu}\big(1.01 + 0.325 \operatorname{relu}\big(-0.0966 \\
&+ 0.855 x_{numberCustomerServiceCalls_eg}\big) \\
&+ 0.278 x_{numberCustomerServiceCalls_eg}\big) - 0.869 x_{totalIntlCharge_b}\big)
\end{aligned}
$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C150	direction: state: state other than Arkansas, New Mexico, Utah to Arkansas (through this part) *shape: categorical; highest to lowest: Arkansas, Utah, New Mexico, state other than Arkansas, New Mexico, Utah*	1% to 1%	0.05	1.06	0.03 to 0.08	3761, 78	disagrees; AUC gain -5e-05 [-0.0002, 0.0001]	yes
C151	direction: total international charge: 1.84 US dollars to 3.70 US dollars (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1% (typical patient this part acts on)	-0.001	1.00	0.00 to 0.00	4000	unclear; AUC gain 2e-05 [-6e-05, 0.0001]	yes
C152	direction: customer-service calls: 0 calls to 3 calls (through this part) *shape: flat; 100% of patients same; verified; risk saturates near 0%*	1% to 1% (typical patient this part acts on)	-0.0003	1.00	0.00 to 0.00	4000	unclear; AUC gain 0.000 [0.000, 0.000]	yes
C153	direction: state: state other than Arkansas, New Mexico, Utah to Utah (through this part) *shape: categorical; highest to lowest: Arkansas, Utah, New Mexico, state other than Arkansas, New Mexico, Utah*	1% to 1%	0.009	1.01	0.00 to 0.03	3761, 89	disagrees; AUC gain -5e-05 [-0.0002, 0.0001]	yes
C154	interaction: effect of customer-service calls (0 calls to 3 calls) by state: Arkansas minus state other than Arkansas, New Mexico, Utah; stronger in Arkansas *shape: accelerating; 82% of patients same; fallback; risk saturates near 0%*	Arkansas: 2% to 4%; state other than Arkansas, New Mexico, Utah: 1% to 3%	0.28	1.32		78, 3761	disagrees; AUC gain -5e-05 [-0.0002, 0.0001]	no
C155	interaction: effect of total international charge (1.84 US dollars to 3.70 US dollars) by state: Arkansas minus state other than Arkansas, New Mexico, Utah; stronger in state other than Arkansas, New Mexico, Utah *shape: accelerating; 91% of patients same; verified; risk saturates near 0%*	Arkansas: 1% to 3%; state other than Arkansas, New Mexico, Utah: 1% to 2%	-0.07	0.93		78, 3761	disagrees; AUC gain -5e-05 [-0.0002, 0.0001]	no
C156	interaction: effect of customer-service calls (0 calls to 3 calls) by total international charge: total international charge above 2.78 US dollars minus total international charge at or below 2.78 US dollars; stronger in total international charge at or below 2.78 US dollars *shape: accelerating; 78% of patients same; fallback; risk saturates near 0%*	total international charge above 2.78 US dollars: 2% to 3%; total international charge at or below 2.78 US dollars: 1% to 3%	-0.06	0.94		1999, 2001	unclear; AUC gain 2e-05 [-6e-05, 0.0001]	no
C157	threshold: customer-service calls: fold at about 5 calls (stops) *shape: flat; 100% of patients same; verified; risk saturates near 0%*	2% to 2% (typical patient with a fold)	-0.01	0.99	-0.02 to -0.004	66, 63	unclear; AUC gain 0.000 [0.000, 0.000]	no
C158	threshold: total international charge: fold at about 0.43 US dollars (stops) *shape: flat; 97% of patients same; verified; risk saturates near 0%*	1% to 1% (typical patient with a fold)	-0.002	1.00	0.00 to 0.00	3890, 153	unclear; AUC gain 2e-05 [-6e-05, 0.0001]	no
C159	subgroup: who this part acts on: 153 of 4000 patients	1% to 1% (typical patient this part acts on (risk without the part, then with it))	0.04	1.04	0.005 to 0.07	153, 3847	disagrees; AUC gain -2e-05 [-0.0002, 0.0002]	no
C160	null: this part barely changes predictions (switched off)		0.02			4000	train only	no
C161	structure: its formula reproduces it on 4000 training patients (largest difference 8.88178e-16)		9e-16			4000	train only	yes
C164	boundary: boundary		1.00			4000	train only	yes

The network before any explanation

Figure 66 (next page, landscape) is the trained network as it is, with no parts, no splits and no inserted units: inputs at the bottom, hidden units by depth, the output at the top, every connection drawn, every input column its own node (one-hot levels included).

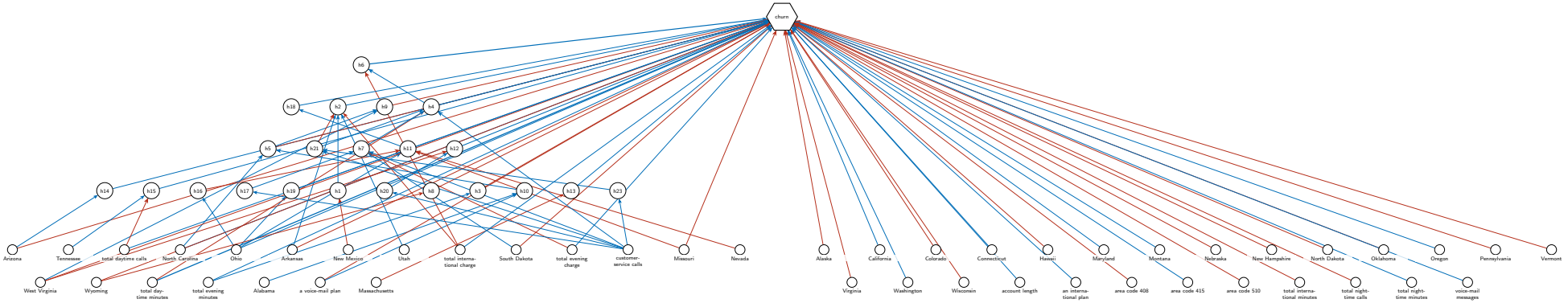


Figure 66: The network as trained, before any explanation, in full. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. Not used (no connection): Georgia, Iowa, Idaho, Illinois, Indiana, Kansas, Kentucky, Louisiana, Maine, Michigan, Minnesota, Mississippi, New Jersey, New York, Rhode Island, South Carolina, Texas, Washington DC, Delaware, Florida, total daytime charge, total evening calls, total night-time charge, total international calls.

2 churn100-s2-c2

Genome `e5394267-5807-44ae-8629-5e157bcfb43a`, explanation `4792056e-ee3e-4fde-a1bb-4c4d809e2d61`. Split: train primary (n=4000), test as a check (n=1000). Claims: 147, verified 138, not supported 9.
AI-assisted draft for review (unverified)

Reference (typical) customer: the typical customer: the median training customer for every number and the most common training category for every category (not a real customer). account length: 100; area code: area code 415; international plan: no international plan; number customer service calls: 1 calls; number vmail messages: 0 messages; state: West Virginia; total day calls: 100 calls; total day charge: 30.69 US dollars; total day minutes: 180 minutes; total eve calls: 100 calls; total eve charge: 17.12 US dollars; total eve minutes: 201 minutes; total intl calls: 4 calls; total intl charge: 2.78 US dollars; total intl minutes: 10 minutes; total night calls: 100 calls; total night charge: 9.02 US dollars; total night minutes: 200 minutes; voice mail plan: no voice-mail plan. Predicted chance that a customer leaves: 1%.

2.1 Summary

For a typical customer (in West Virginia, area code 415, no international or voice-mail plan, 1 call to customer service and median usage) the model predicts a chance of leaving of about 1%; it adds up six parts: fixed-weight inputs, service calls, a voice-mail discount, state units, an area-code unit and a small Utah unit. [C051] Figure 67 shows the model at this level: one box per paragraph below.

How to read the diagrams. Edge colour is the sign of that one connection: blue positive, red negative; thicker is a larger weight. The outline and arrow on an input (and on a part's output box) is its overall effect on the final prediction, read from the verified claims: orange and up arrow raises it, teal and down arrow lowers it, grey and double arrow depends (it goes both ways, or differs between levels). A negative connection into a part that itself raises the prediction means that input lowers the prediction; through a hinge the overall effect can only be read from the claims, not from the edges. Up arrow means towards churn; down arrow, away from it. A double outline marks an input (or a part's output) that acts at a fixed rate everywhere: it adds the same amount to the score per unit, so it is explained by its rate alone (see 'Linear factors'); a dashed double outline, at a fixed rate on the observed data. The normal outline is an input whose effect switches on and off somewhere. For example, total daytime minutes has a negative connection into 'Service calls', a part that raises the predicted chance that a customer leaves, so total daytime minutes lowers it.

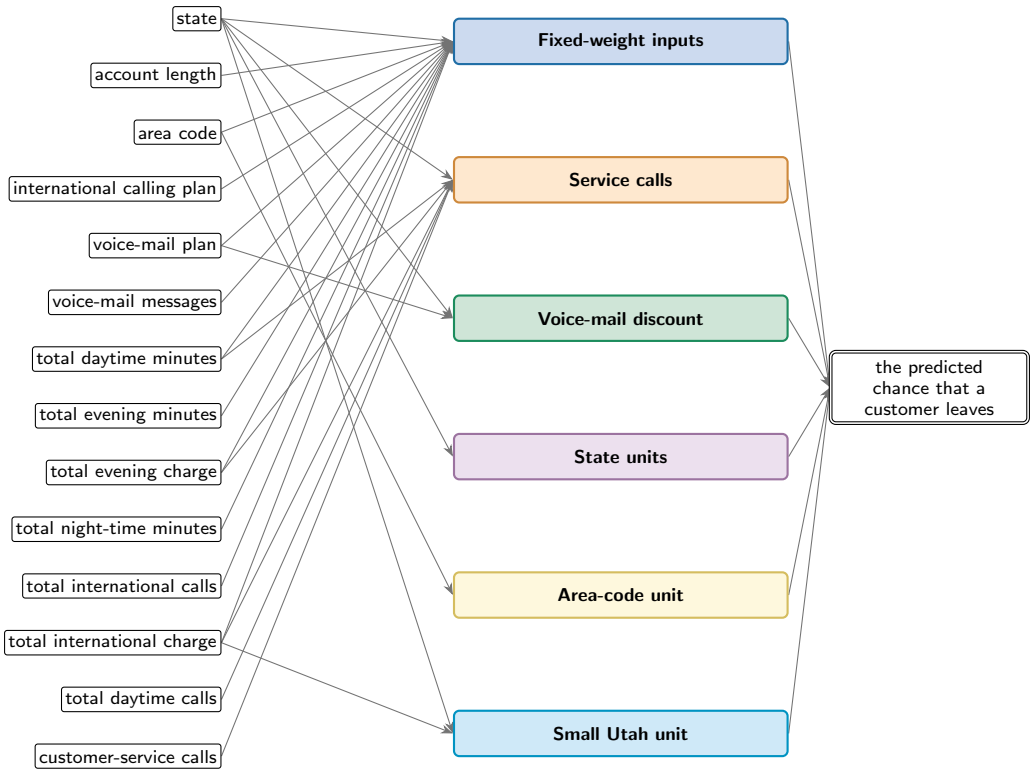


Figure 67: The model at the level of the summary: one box per part.

Fixed-weight inputs. This part adds up most inputs with fixed weights, and daytime minutes count most: they raise a typical customer's chance of leaving steadily, from under 1% at 112 to about 11% at 249 minutes. [C053] An

international plan takes the chance from about 1% to about 14%, a voice-mail plan lowers it to under 1%, and more voice-mail messages raise it steadily, from about 1% at 0 to about 4% at 32 messages. [C052, C061, C056] A higher international charge, more evening minutes, a higher evening charge and more night-time minutes also raise it steadily, each by one to two points. [C057, C059, C060, C065] Through it, the state moves the chance from about 1% in West Virginia to about 5% in Washington and in California, and to under 1% in Hawaii (41 to 122 training customers per state). [C054, C055, C062]

Service calls. This part adds up two parts: customer-service calls combined with low daytime use, and a rise from 4 calls upwards. [C103] Through it, daytime minutes lower the chance only while they stay below about 166 minutes (between 145 and 213 minutes for different customers), in an uneven way: for a typical such customer, from about 28% at 91 minutes to about 1% at 352 minutes. [C098] With an evening charge above 17.12 US dollars, the chance is lowest at about 148 minutes of daytime use and higher either side: from about 1% at 112 minutes to about 13% at 249 minutes, against about 2% to about 9% at or below that charge. [C095] Customer-service calls add to it from 4 calls upwards, slowly at first, then faster: for a typical such customer, from about 3% at 0 calls to about 8% at 4 calls. [C099]

Voice-mail discount. This part reads the voice-mail plan and one state: through it, a voice-mail plan lowers a typical customer's chance from about 1% to under 1%, and customers in North Carolina get under 1% instead of about 1%. [C119, C120]

State units. This part reads only the state: it raises a typical customer's chance from about 1% to about 2% in Washington and in Montana, and by less than a point in Kentucky and Ohio. [C125, C126, C127, C128]

Area-code unit. This part reads the area code and barely changes the predictions (by at most half a point). [C137]

Small Utah unit. This small part reads the international charge and being in Utah, and barely changes the predictions (by at most 0.2 points). [C144]

The whole network. Figure 68 (next page, landscape) draws every unit and connection of the explained model, nested by part: shaded clusters are the parts of the model and darker boxes the parts inside them; circles are hidden units (h1, h2, ...); each cluster's rectangle is its output, and a dashed 'sum' stands for an addition the next unit does (drawn, not a unit of the model); the hexagon is the model's output. An input used by several parts is drawn under each of them (same label, dotted outline, the first copy marked with the number of copies): it is the same input, repeated for readability. Not used by the model (input columns with no connection): state: 25 of 51 levels, total daytime charge, total evening calls, total night-time calls, total night-time charge, total international minutes.

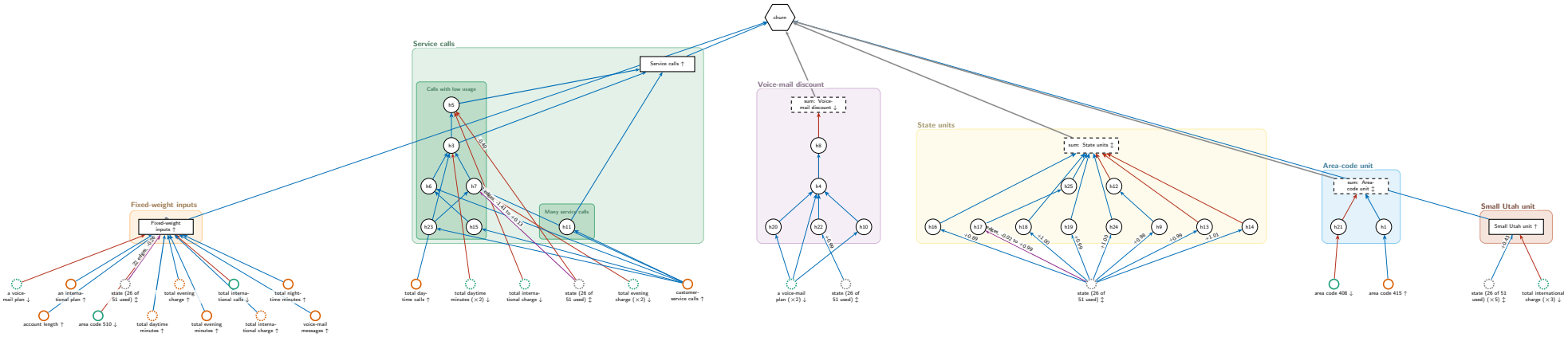


Figure 68: Every unit and connection of the explained model, nested by part. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Worth checking against what is known. Worth checking: the state moves the chance by several points (for example Washington about 9% and California about 5%, Hawaii under 1%), based on 41 to 122 training customers per state. [C003, C004, C012] Worth checking: the model treats a voice-mail plan as lowering the chance of leaving but more voice-mail messages as raising it. [C010, C005] Worth checking: daytime minutes, the evening charge and the international charge raise the chance through the fixed-weight inputs but lower it through calls with low usage. [C053, C104, C060, C107, C057, C105]

2.2 The model, read from the top down

Each part of the explanation in turn, from the whole model down to the smallest parts: what it does for the prediction (hypothesis), what was checked (evidence), a diagram one level down, and where to read on. The technical record of each part is in "Technical detail" at the end.

2.2.1 The whole model

Figure 69 shows what feeds this part; Figure 70 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

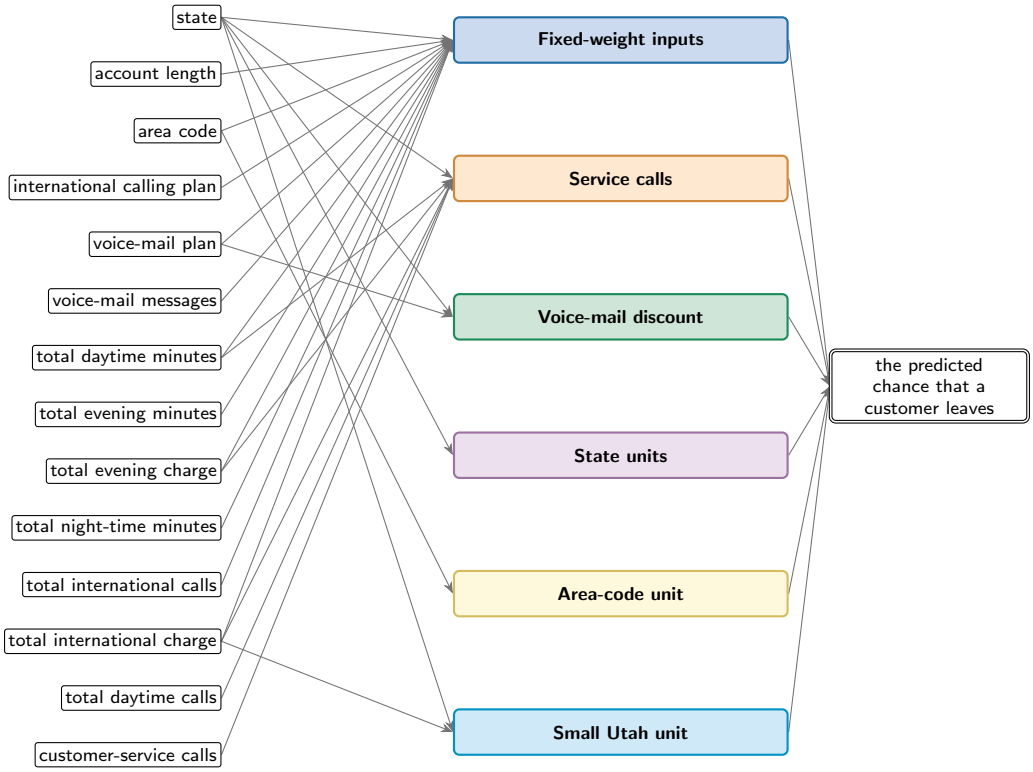


Figure 69: The whole model: what feeds it, one level down.

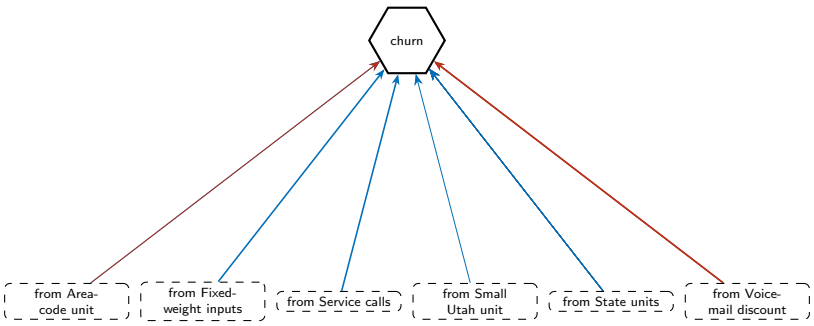


Figure 70: The whole model: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. The model adds up six parts into one score: fixed-weight inputs, service calls, a voice-mail discount, state units, an area-code unit and a small Utah unit. [C051]

Evidence. It is made up of these six parts (each explained below); the score is their sum, which was checked on all 4000 training customers. [C051] Changing one input at a time across the whole model, an international plan takes a typical customer's chance of leaving from about 1% to about 14%, and a voice-mail plan lowers it to under 1%. [C001, C010] In Washington the chance is about 9%, against about 1% in West Virginia. [C003] More customer-service calls raise it slowly at first, then faster, from about 1% at 0 calls to about 2% at 3 calls. [C011] The model ignores the daytime charge, evening calls, international minutes, night-time calls and the night-time charge. [C046, C047, C048, C049, C050]
Evidence plots: Figure 71, Figure 72.

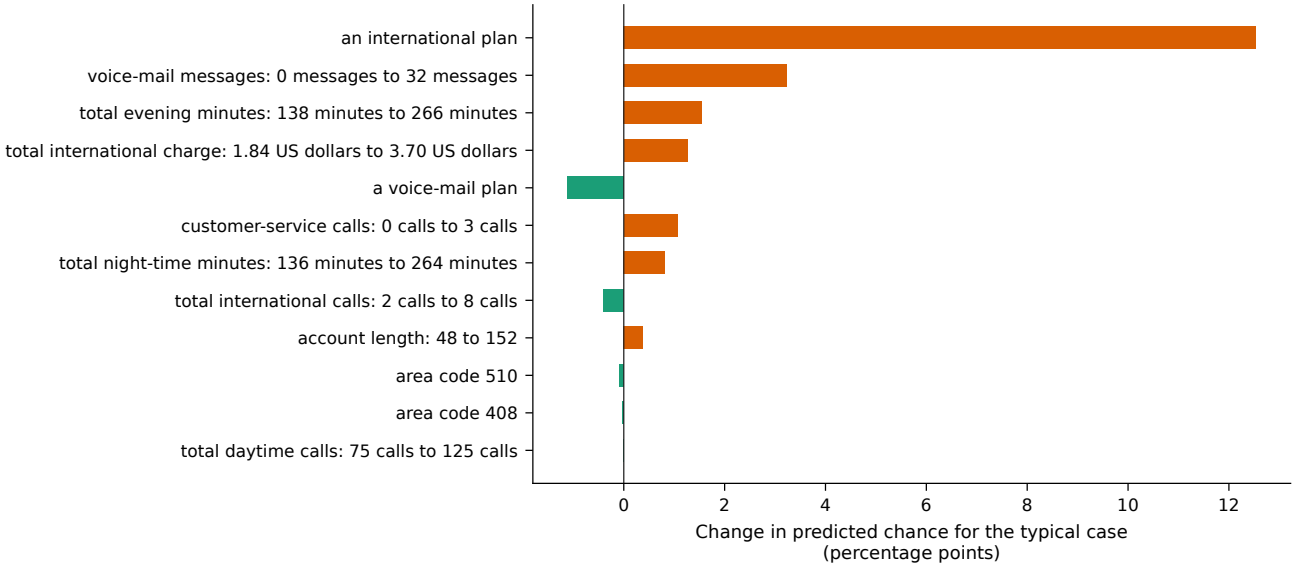


Figure 71: The whole model: Each input's change in the predicted value for the typical case across the whole model, largest first (orange: raises it, teal: lowers it).

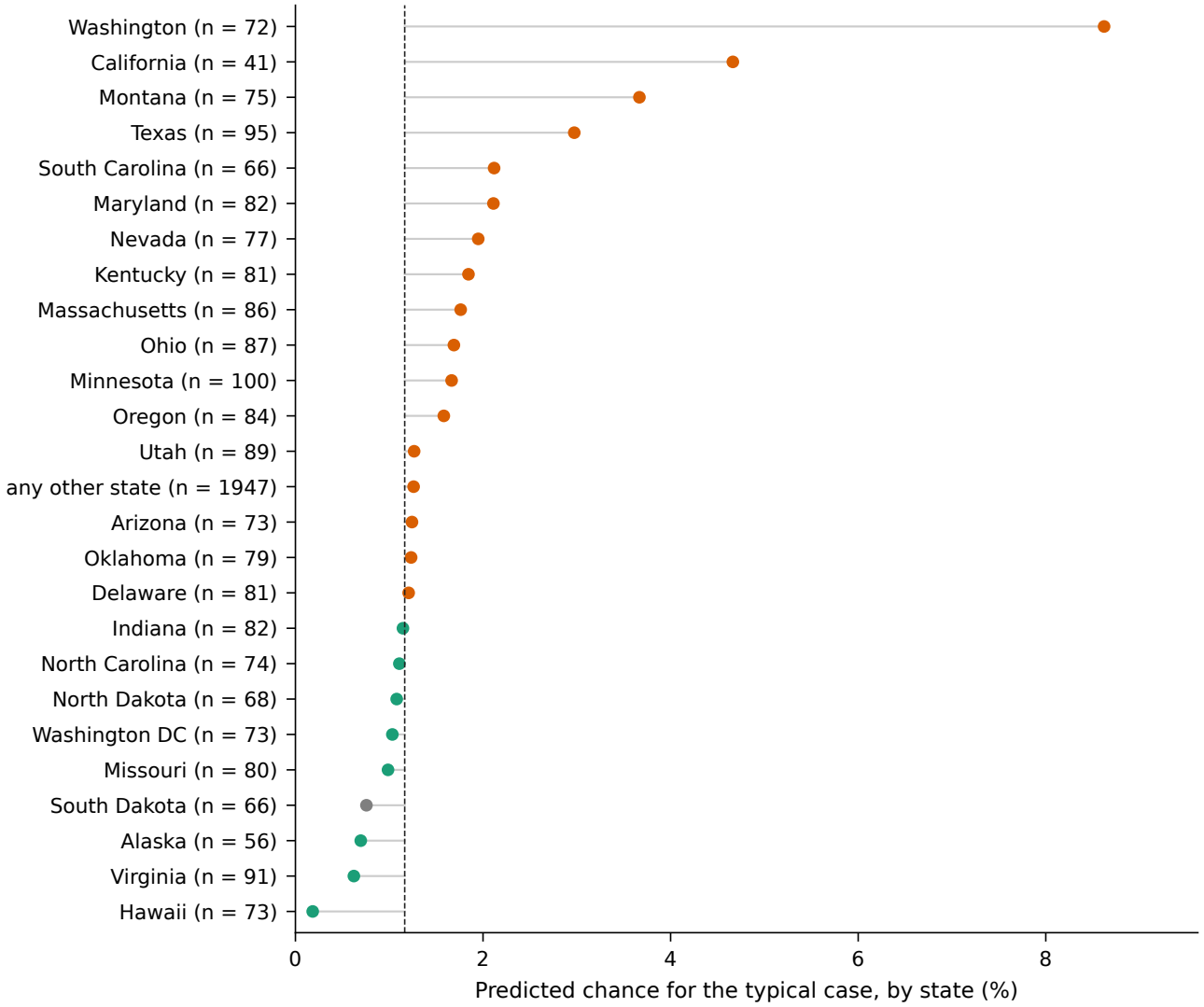


Figure 72: The whole model: Every level of state the model reads: the predicted value for the typical case at that level (dashed: the reference level); n = training customers; grey: few customers or not supported.

Dive deeper: Fixed-weight inputs (2.2.2); Service calls (2.2.3); Voice-mail discount (2.2.6); State units (2.2.7); Area-code unit (2.2.8); Small Utah unit (2.2.9). Technical detail: T.1.

2.2.2 Fixed-weight inputs

Part of: The whole model (2.2.1).
Figure 73 shows what feeds this part; Figure 74 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

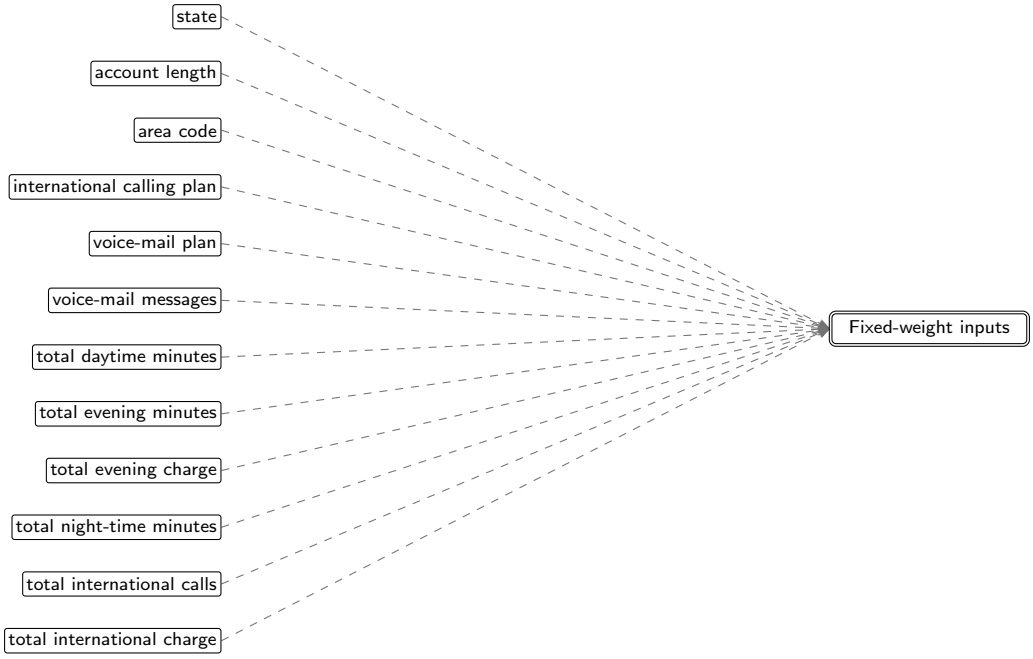


Figure 73: Fixed-weight inputs: what feeds it, one level down.

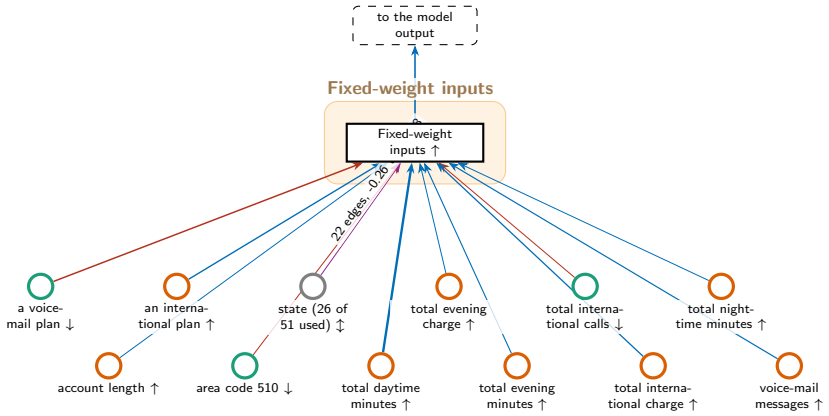


Figure 74: Fixed-weight inputs: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part adds up most inputs with fixed weights: daytime minutes, an international plan and voice-mail messages raise the chance of leaving, steadily for the minutes and messages, and a voice-mail plan lowers it. [C053, C052, C056, C061]

Evidence. Through it, daytime minutes take a typical customer steadily from under 1% at 112 to about 11% at 249 minutes, and an international plan from about 1% to about 14%; a voice-mail plan lowers the chance to under 1%. [C053, C052, C061] Through it, the state moves the chance from about 1% in West Virginia to about 5% in Washington and in California, and to under 1% in Hawaii (41 to 122 training customers per state). [C054, C055, C062] Its formula reproduces the part exactly on all 4000 training customers. [C085]
Evidence plots: Figure 75, Figure 76, Figure 77, Figure 78.

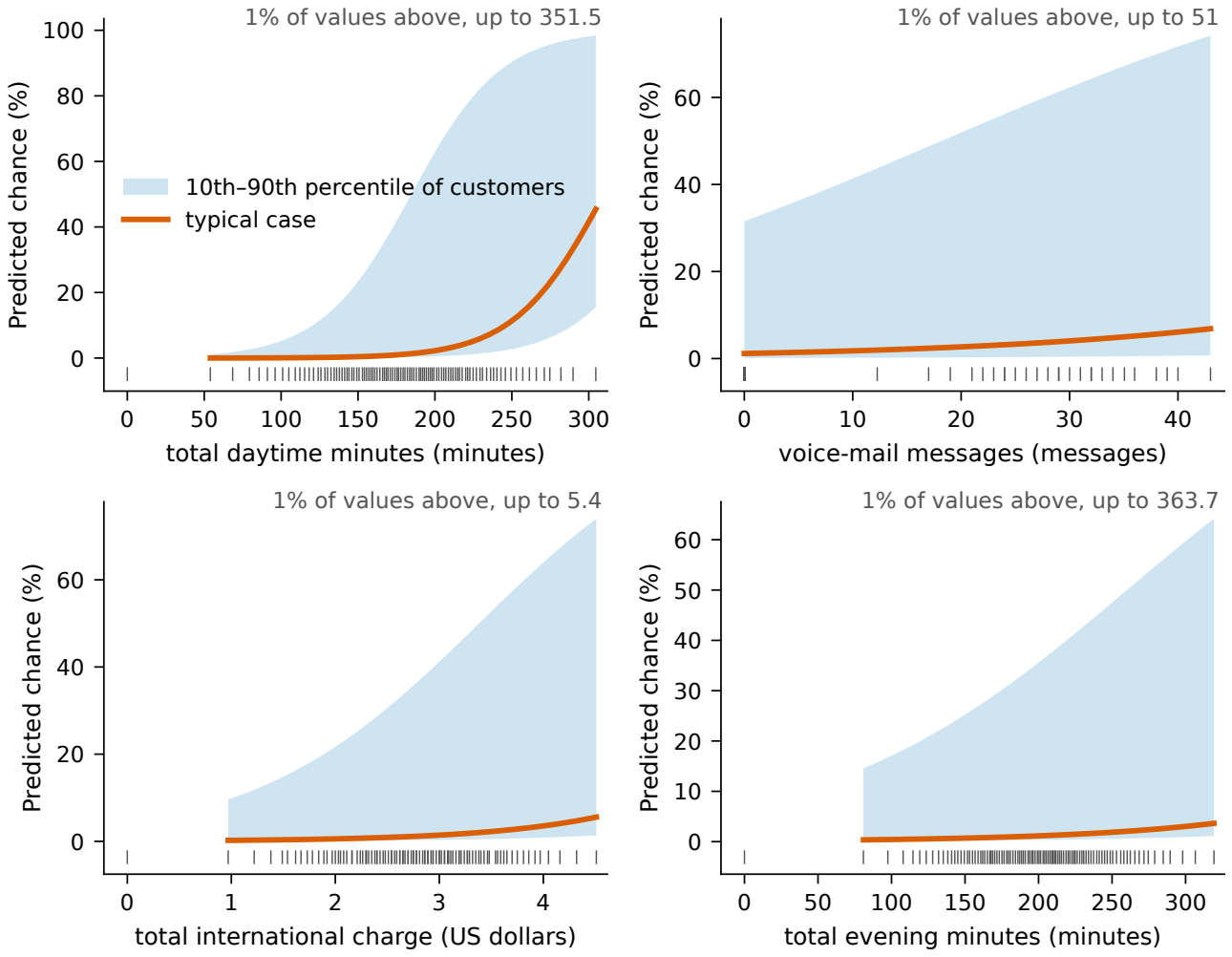


Figure 75: Fixed-weight inputs: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

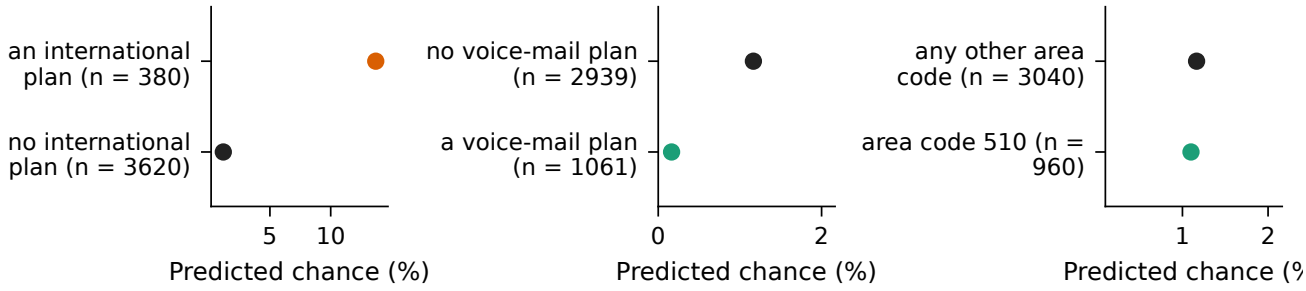


Figure 76: Fixed-weight inputs: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

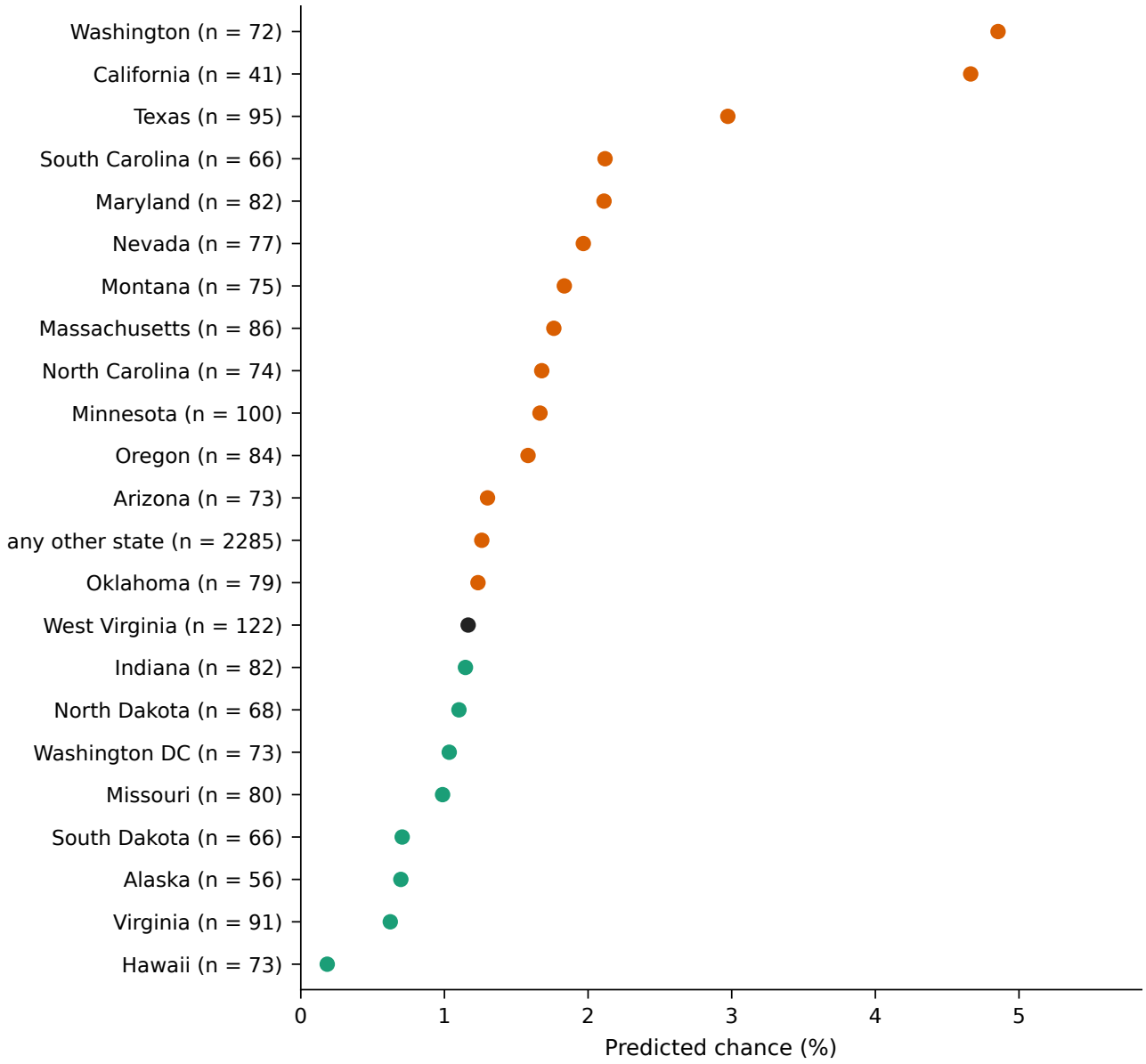


Figure 77: Fixed-weight inputs: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

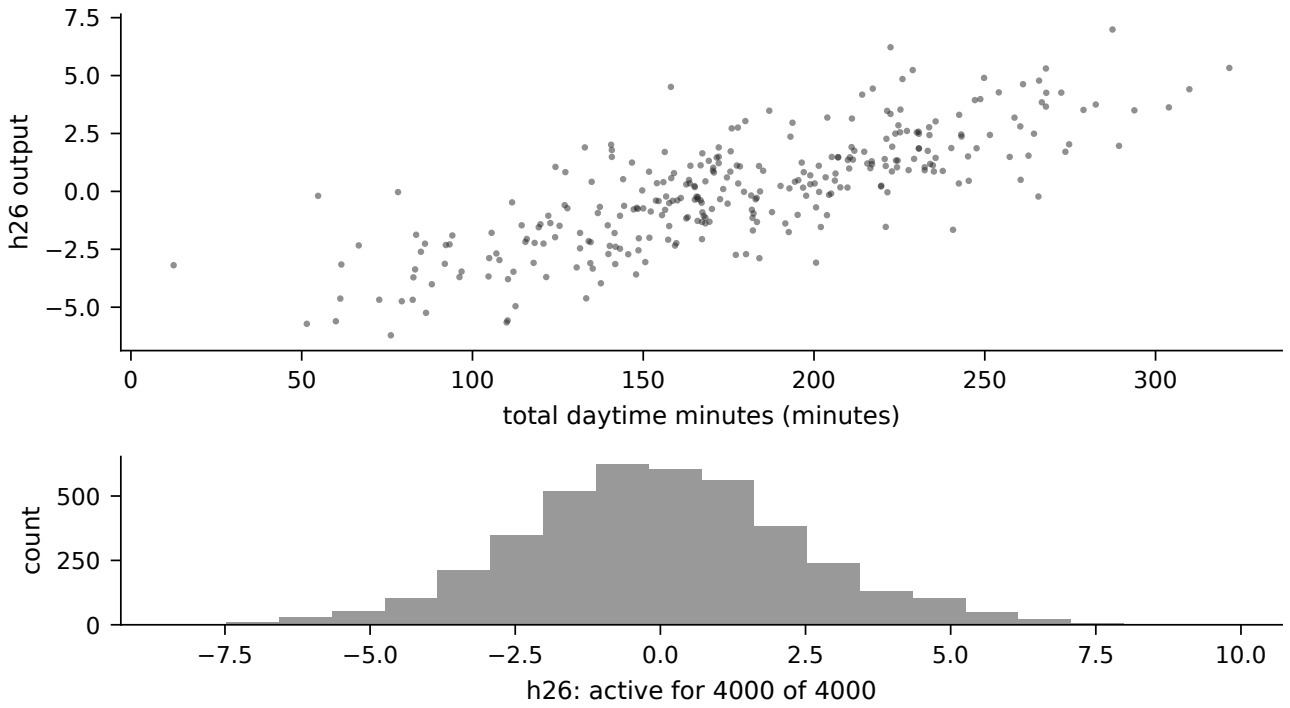


Figure 78: Fixed-weight inputs: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.2.

2.2.3 Service calls

Part of: The whole model (2.2.1).
Figure 79 shows what feeds this part; Figure 80 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

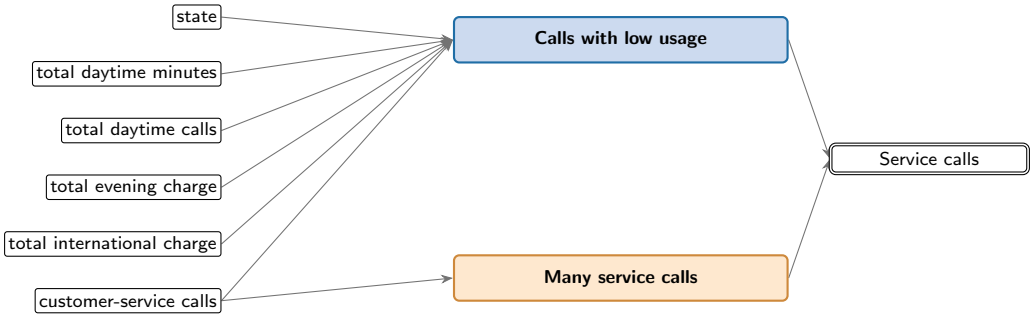


Figure 79: Service calls: what feeds it, one level down.

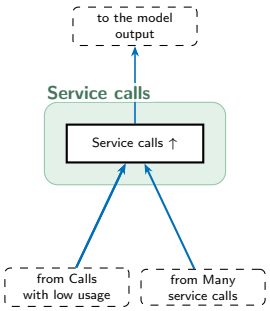


Figure 80: Service calls: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part raises the chance of leaving with calls to customer service, from 4 calls upwards, slowly at first, then faster, and more so for customers with low daytime use, in an uneven way; it adds up two parts. [C103, C099, C098]

Evidence. It is made of two parts that are added: calls with low usage, and many service calls. [C103] Through it, daytime minutes lower the chance only while they stay below about 166 minutes (between 145 and 213 minutes for different customers), in an uneven way: for a typical such customer, from about 28% at 91 minutes to about 1% at 352 minutes. [C098] Customer-service calls add to it from 4 calls upwards, slowly at first, then faster: for a typical such customer, from about 3% at 0 calls to about 8% at 4 calls. [C099] With an evening charge above 17.12 US dollars, the chance is lowest at about 148 minutes of daytime use and higher either side: from about 1% at 112 minutes to about 13% at 249 minutes, against about 2% to about 9% at or below that charge. [C095]
Evidence plots: Figure 81, Figure 82, Figure 83, Figure 84, Figure 85, Figure 86, Figure 87, Figure 88.
Figure 87 shows this part's own output, unit h27: over its two main inputs, with the line where it switches on.

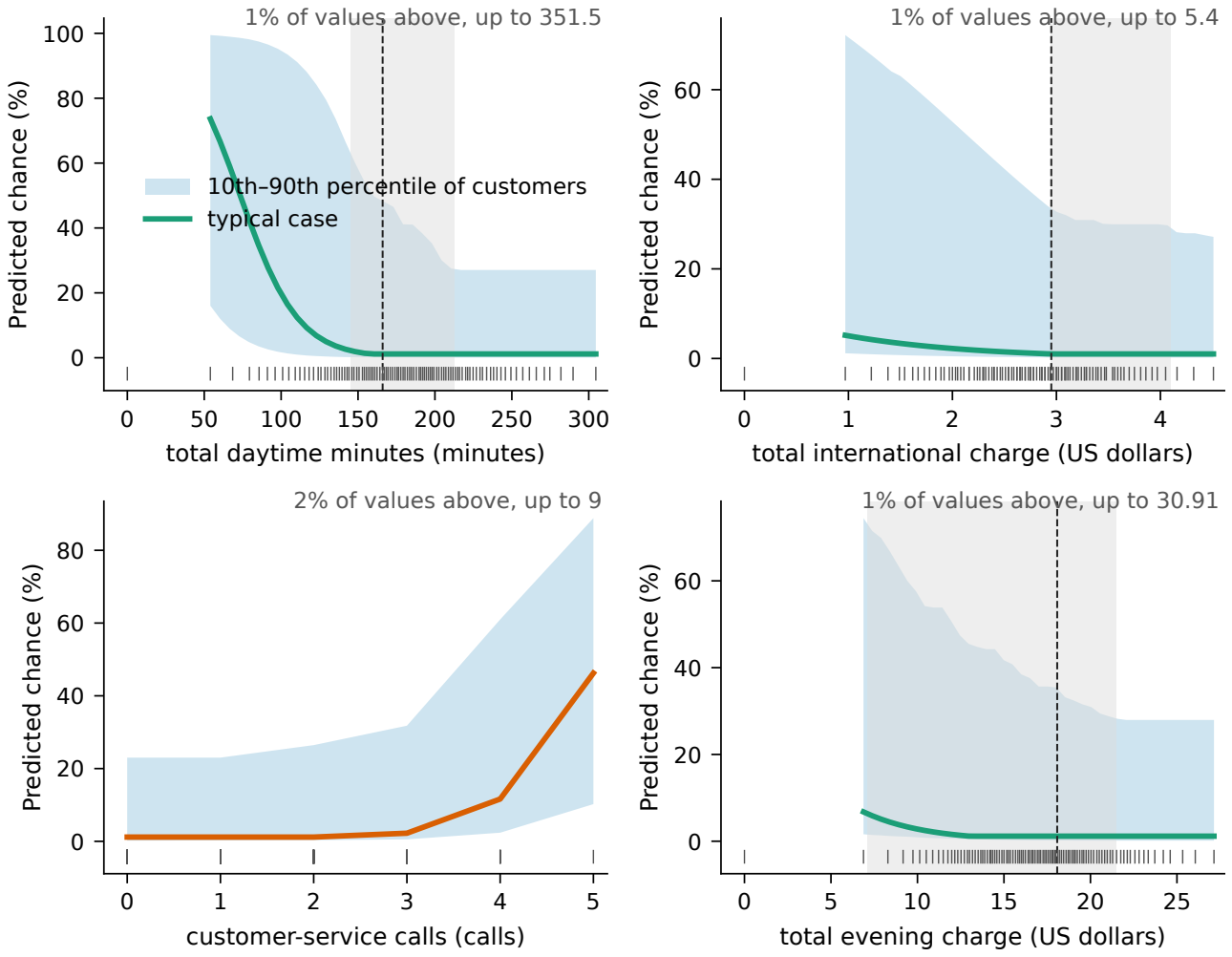


Figure 81: Service calls: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

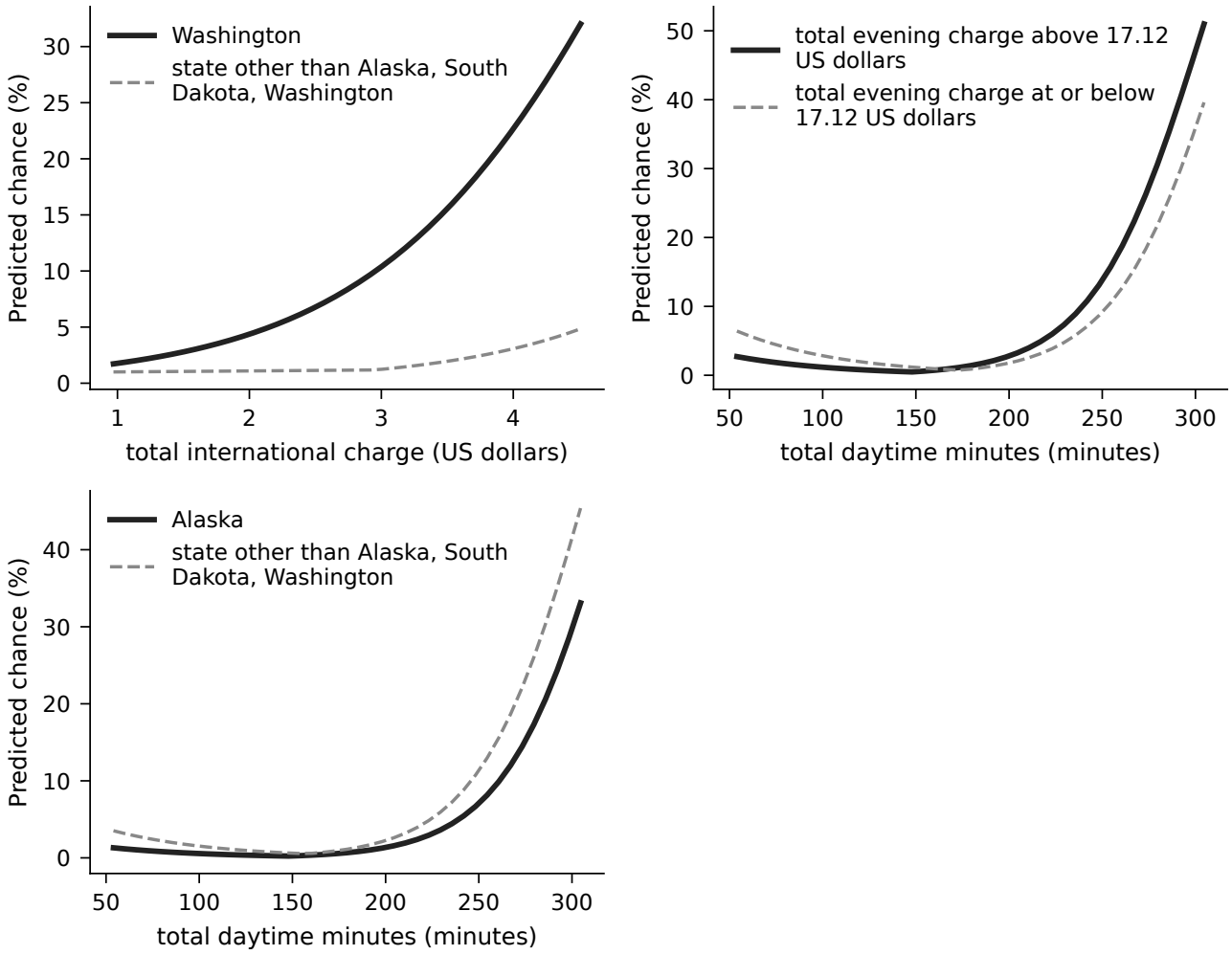


Figure 82: Service calls: How the effect differs between groups: one curve per group, for the typical case of each group.

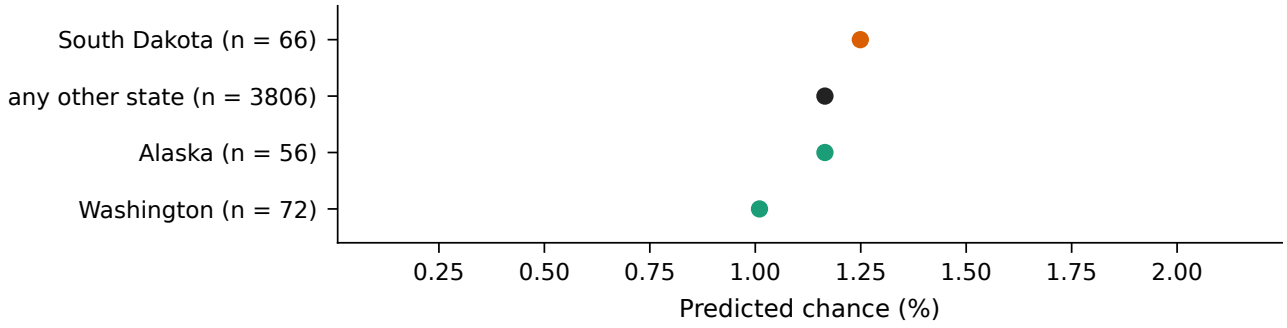


Figure 83: Service calls: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

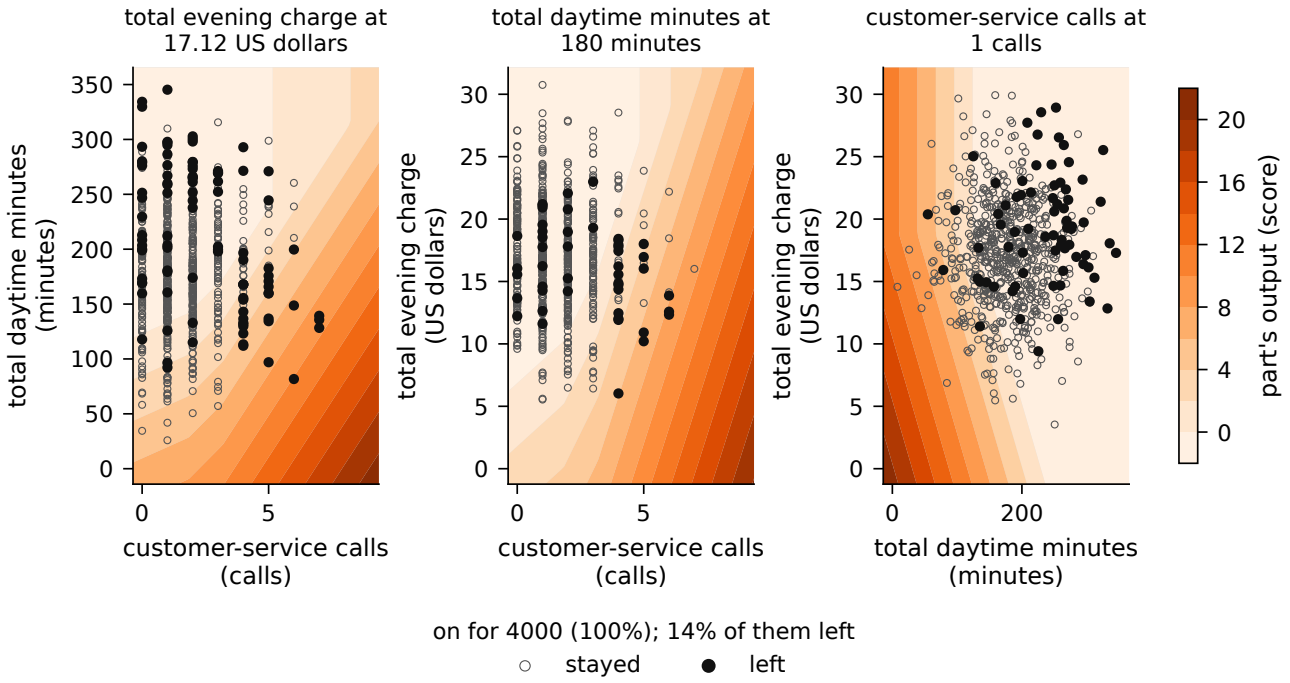


Figure 84: Service calls: The part's own output over each pair of its three main inputs (the third held at its median, named above each panel; the points are the training rows whose third input is between its 40th and 60th percentile), other inputs at the typical case: colour where it is on, flat grey where it is switched off, the black line where it switches on; training customers as hollow (no event) and filled (event) points, so the part can be seen catching the events (the predicted-chance overlay was harder to read). The switch-on line is traced on this map only; the boundary claim (C146) did not pass its check on the data, so no rule is stated.

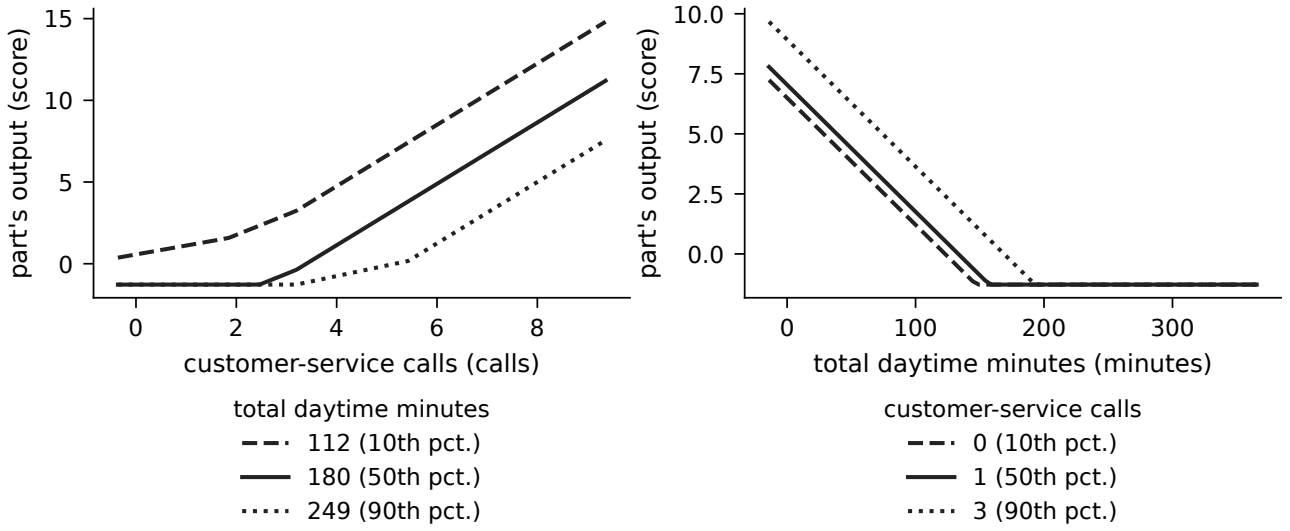


Figure 85: Service calls: Slices of the same map: the part's own output against each input at three values of the other, so the tilted boundary shows as a shifting hinge. The switch-on line is traced on this map only; the boundary claim (C146) did not pass its check on the data, so no rule is stated.

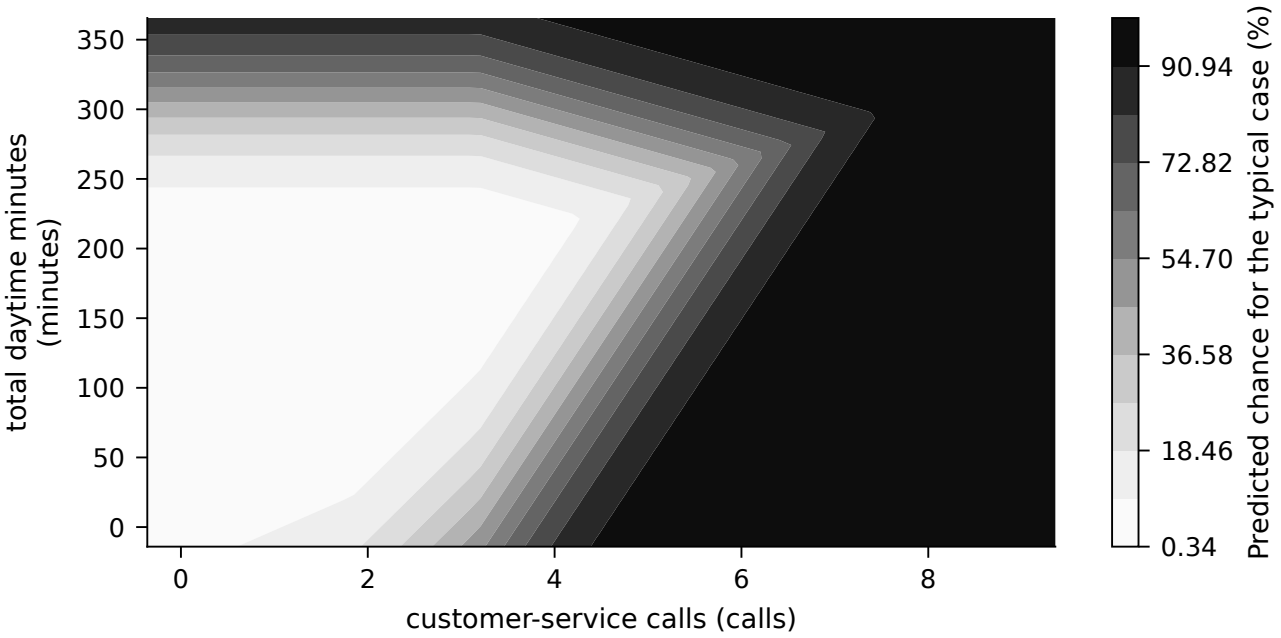


Figure 86: Service calls: The same map on the final scale: the predicted value for the typical case, with the switch-on line (orange). The switch-on line is traced on this map only; the boundary claim (C146) did not pass its check on the data, so no rule is stated.

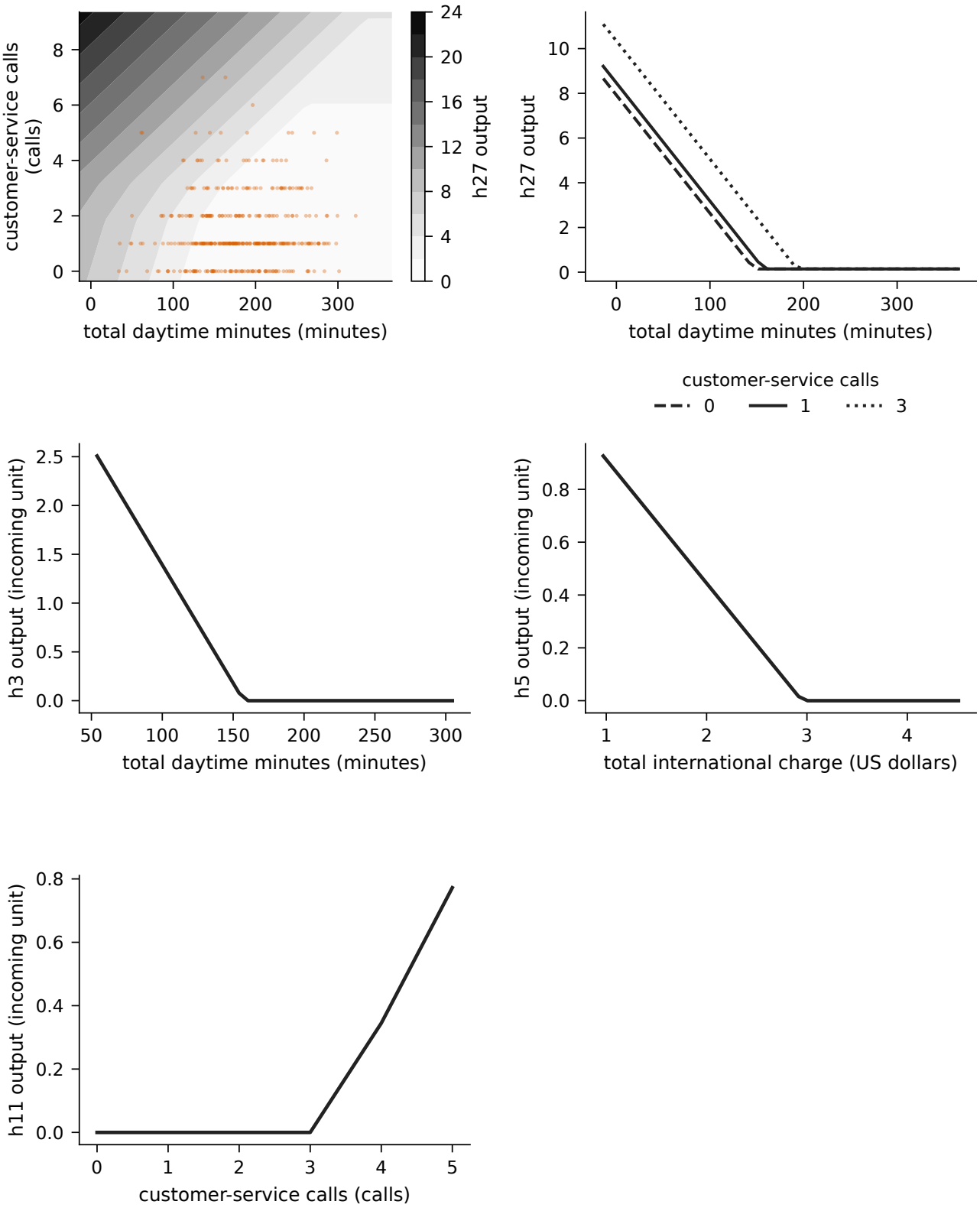


Figure 87: Service calls: This part's own output, unit h27 (combines 3 non-linear units), on its own scale (its activation), with every other input at the typical case; it is active for 3092 of 4000. Left: its output over its two main inputs (orange line: where it switches on; dots: the data); right: slices at three values of the second input; then each incoming non-linear unit against its own main input.

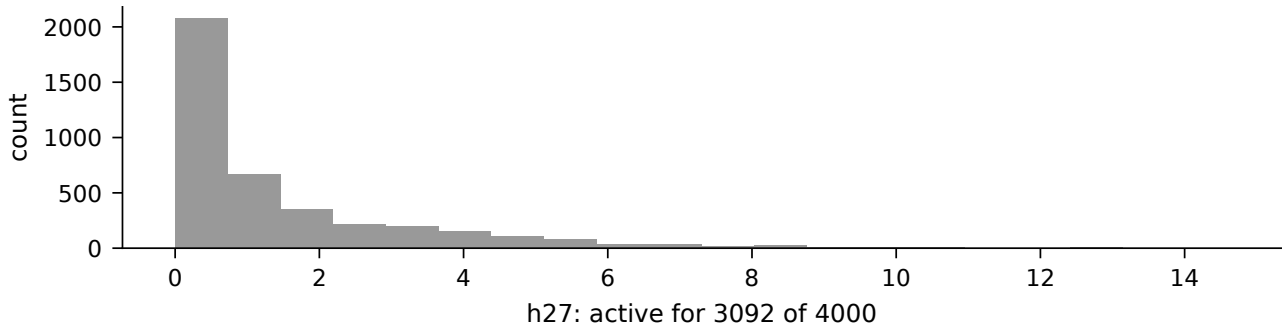


Figure 88: Service calls: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Dive deeper: Calls with low usage (2.2.4); Many service calls (2.2.5). Technical detail: T.3.

2.2.4 Calls with low usage

Part of: Service calls (2.2.3).
Figure 89 shows what feeds this part; Figure 90 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

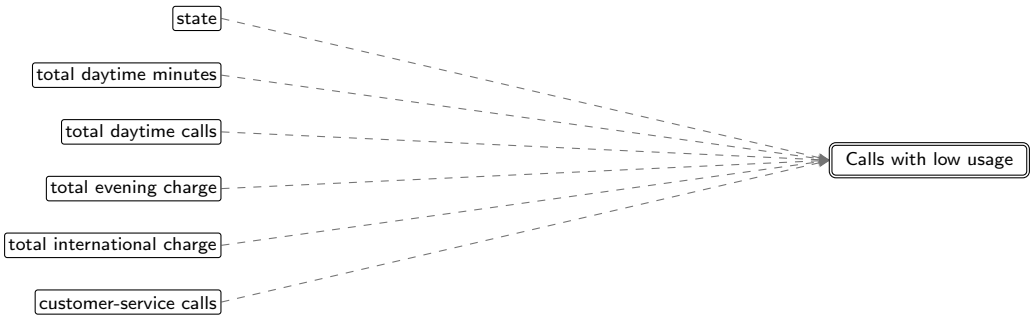


Figure 89: Calls with low usage: what feeds it, one level down.

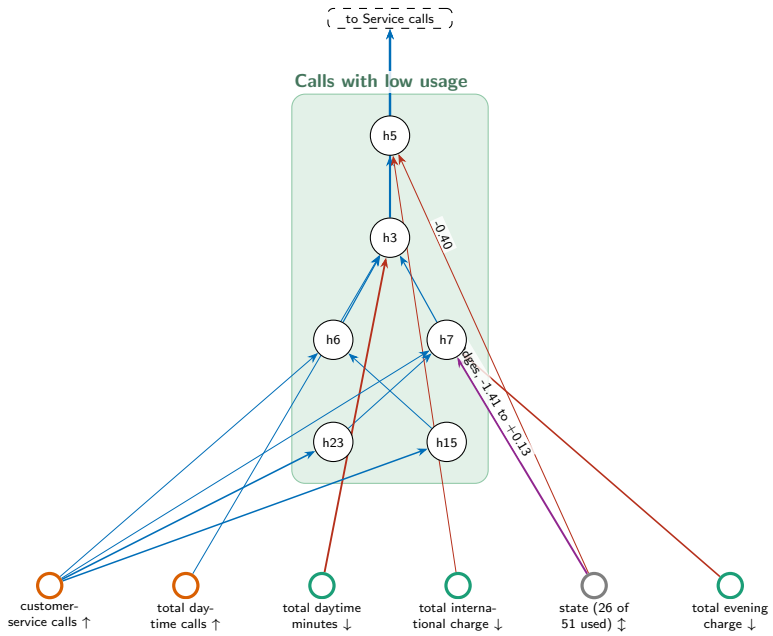


Figure 90: Calls with low usage: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part raises the chance of leaving for customers with low usage, more so when they call customer service: daytime minutes lower it only while they stay below about 166 minutes (between 145 and 213 minutes for different customers), and calls raise it, both in an uneven way. [C112, C106]

Evidence. It acts on 3063 of 4000 training customers, 393 of whom left; for the typical customer it acts on, it changes the chance by less than a point. [C117] Through it, a typical customer's chance falls in an uneven way from about 28% at 91 minutes of daytime use to about 1% at 352 minutes, with no further fall past about 166 minutes (between 145 and 213 minutes for different customers). [C112] A higher international charge lowers it steadily up to about 2.95 US dollars (between 2.95 and 4.10 US dollars for different customers), then no further: for a typical such customer, from about 4% to about 1%. [C115] Customer-service calls raise it in an uneven way, from about 1% at 0 calls to about 2% at 3 calls. [C106] Its formula reproduces the part exactly on all 4000 training customers. [C118]
Evidence plots: Figure 91, Figure 92, Figure 93, Figure 94, Figure 95, Figure 96, Figure 97, Figure 98.
Figure 96 shows this part's own output, unit h3: over its two main inputs, with the line where it switches on.
Figure 97 shows this part's own output, unit h5: over its two main inputs, with the line where it switches on.

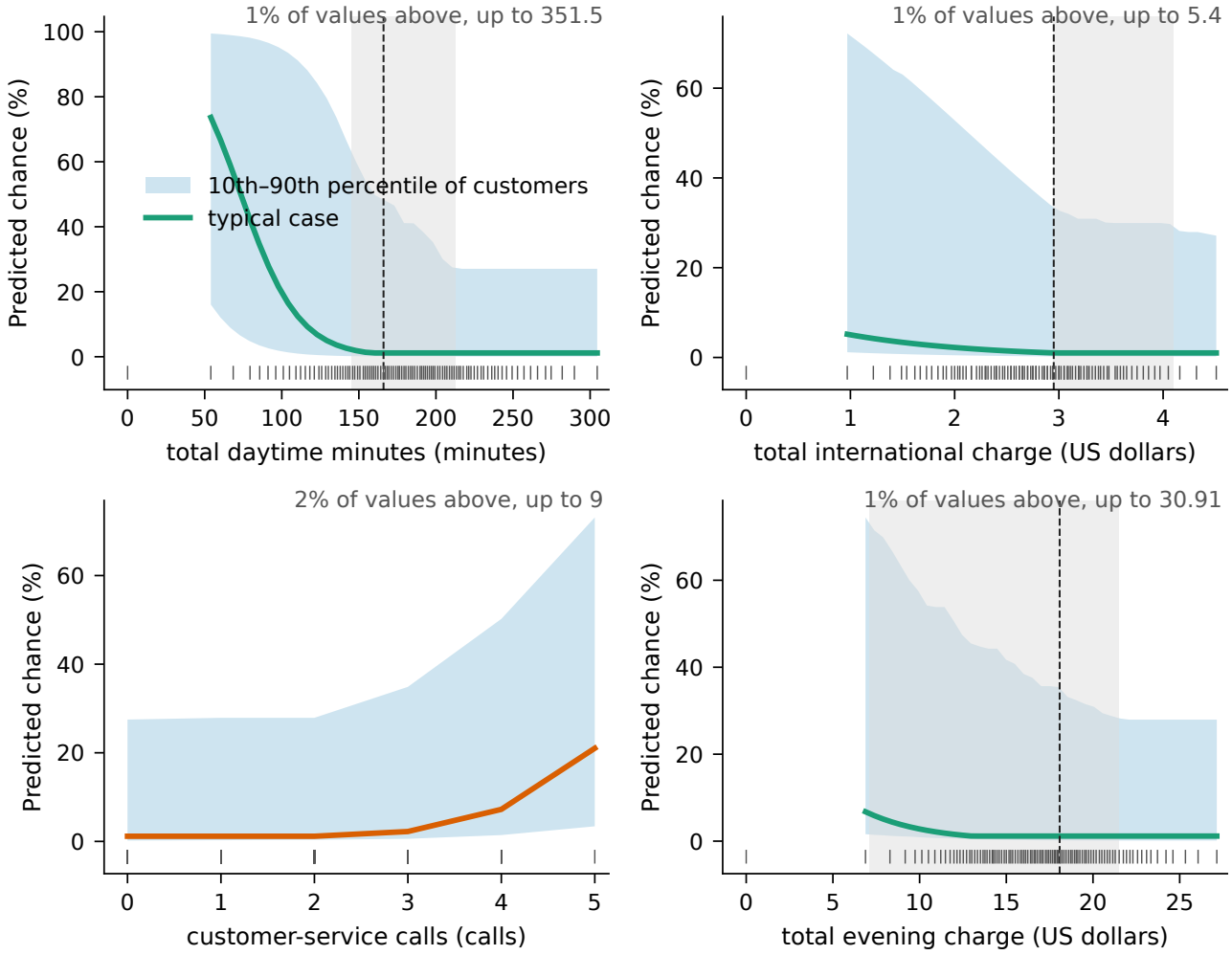


Figure 91: Calls with low usage: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

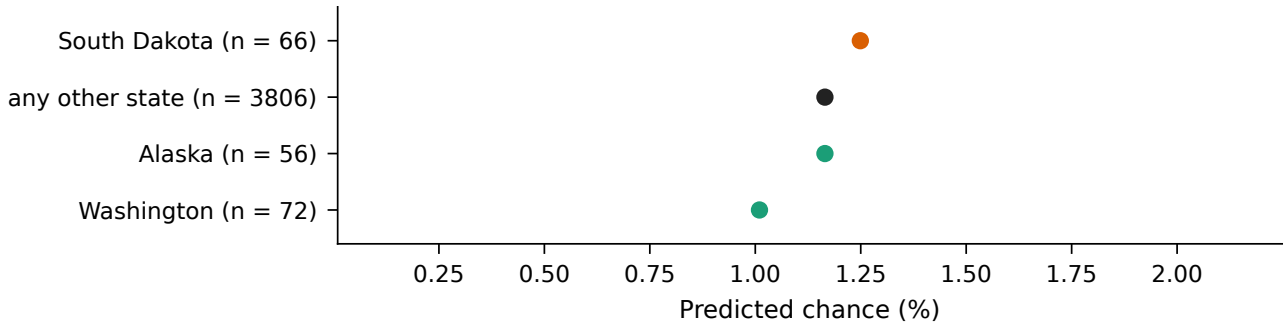


Figure 92: Calls with low usage: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

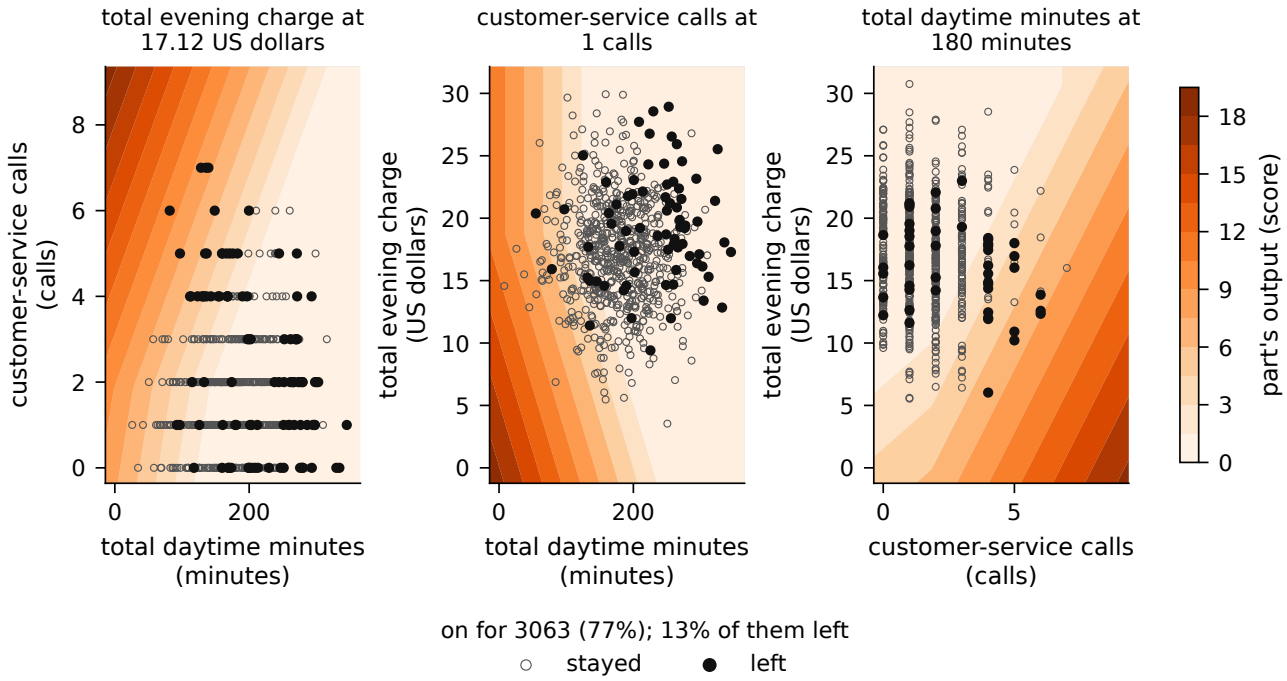


Figure 93: Calls with low usage: The part's own output over each pair of its three main inputs (the third held at its median, named above each panel; the points are the training rows whose third input is between its 40th and 60th percentile), other inputs at the typical case: colour where it is on, flat grey where it is switched off, the black line where it switches on; training customers as hollow (no event) and filled (event) points, so the part can be seen catching the events (the predicted-chance overlay was harder to read). The switch-on line is traced on this map only; the boundary claim (C147) did not pass its check on the data, so no rule is stated.

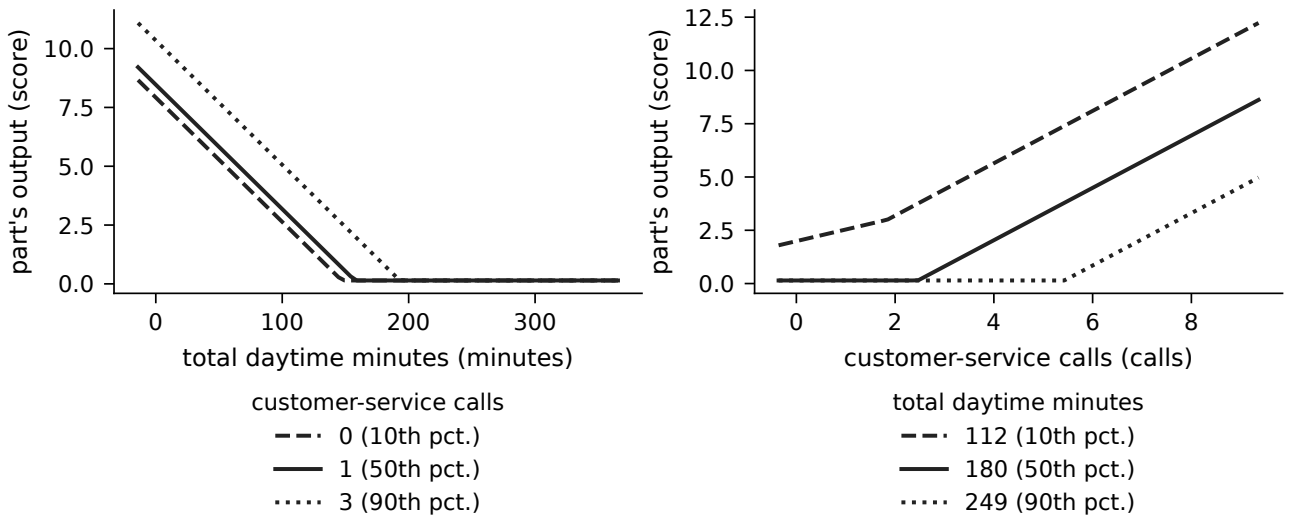


Figure 94: Calls with low usage: Slices of the same map: the part's own output against each input at three values of the other, so the tilted boundary shows as a shifting hinge. The switch-on line is traced on this map only; the boundary claim (C147) did not pass its check on the data, so no rule is stated.

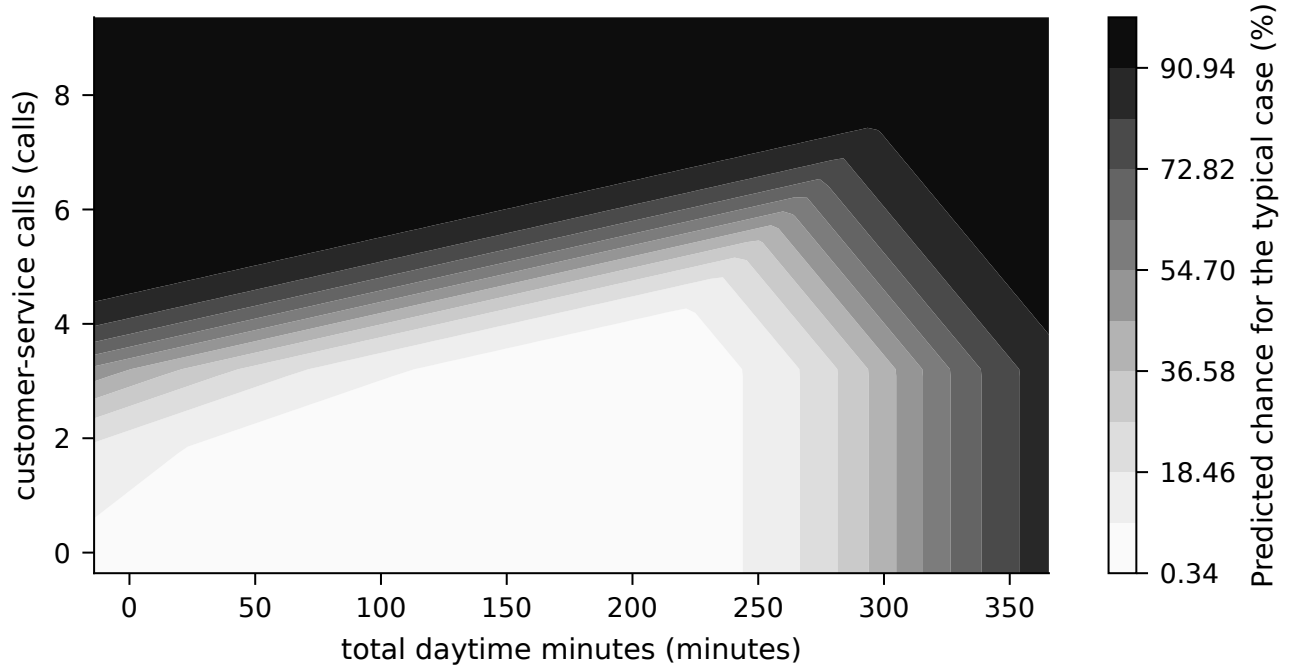


Figure 95: Calls with low usage: The same map on the final scale: the predicted value for the typical case, with the switch-on line (orange). The switch-on line is traced on this map only; the boundary claim (C147) did not pass its check on the data, so no rule is stated.

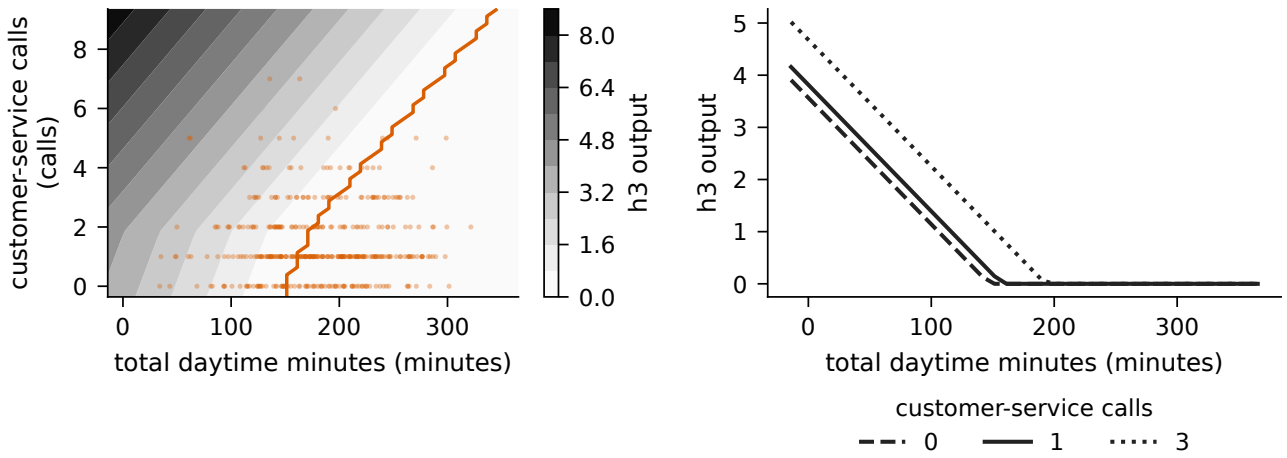


Figure 96: Calls with low usage: This part's own output, unit h3 (the output of 'Calls with low usage'), on its own scale (its activation), with every other input at the typical case; it is active for 1744 of 4000. Left: its output over its two main inputs (orange line: where it switches on; dots: the data); right: slices at three values of the second input.

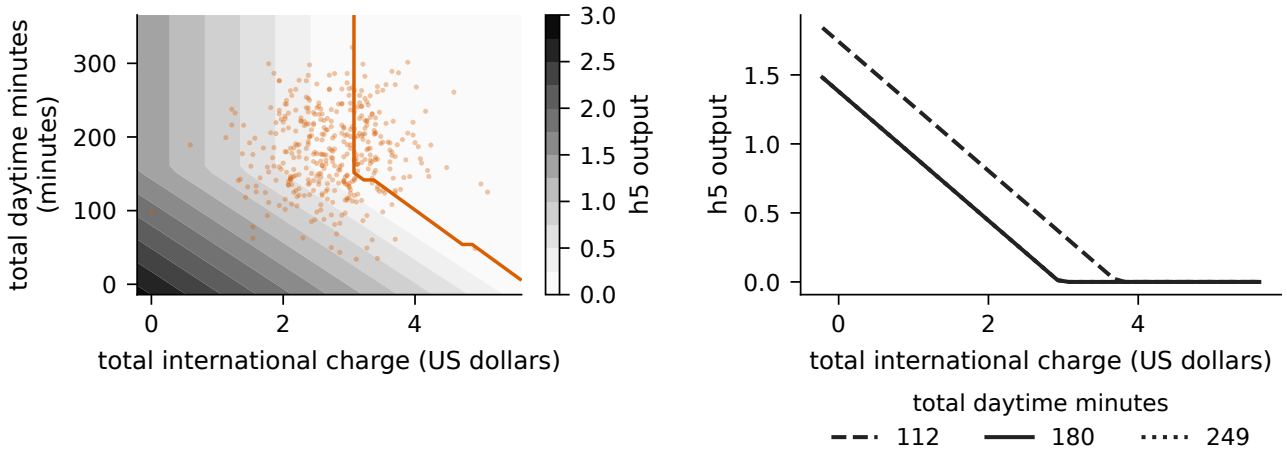


Figure 97: Calls with low usage: This part's own output, unit h5 (the output of 'Calls with low usage'), on its own scale (its activation), with every other input at the typical case; it is active for 2771 of 4000. Left: its output over its two main inputs (orange line: where it switches on; dots: the data); right: slices at three values of the second input.

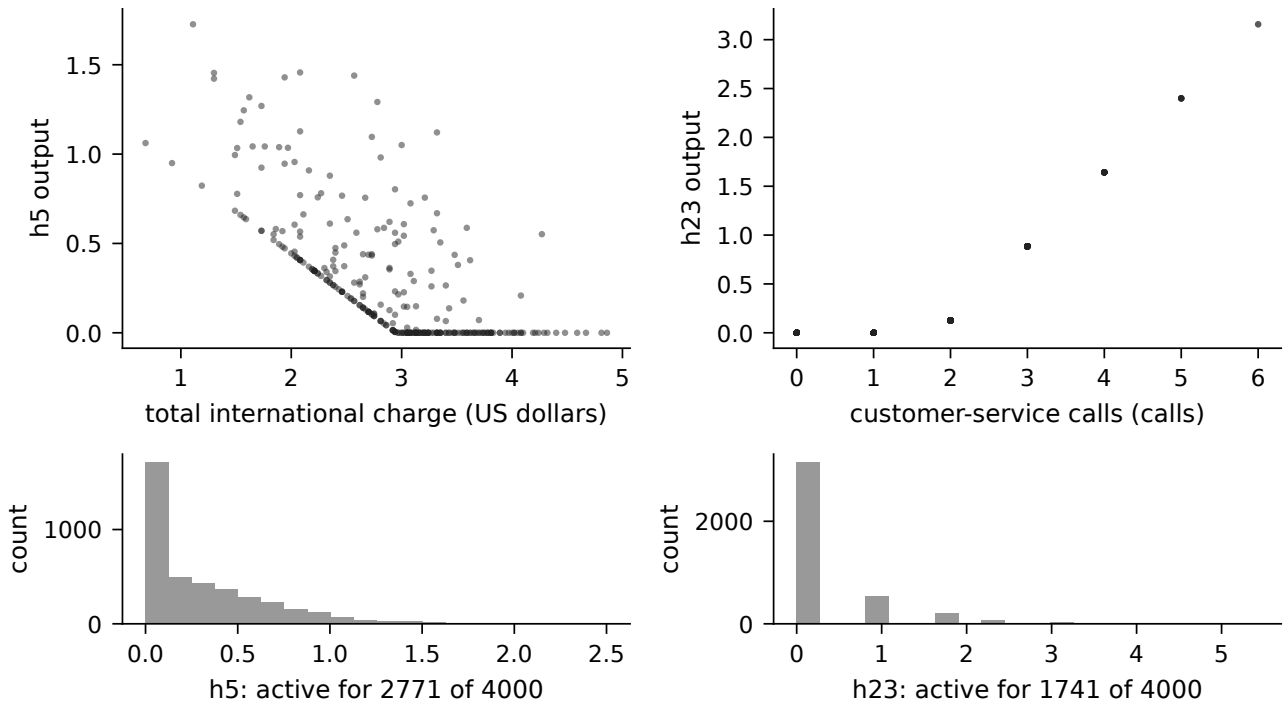


Figure 98: Calls with low usage: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.4.

2.2.5 Many service calls

Part of: Service calls (2.2.3).
Figure 99 shows what feeds this part; Figure 100 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

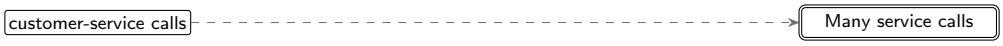


Figure 99: Many service calls: what feeds it, one level down.

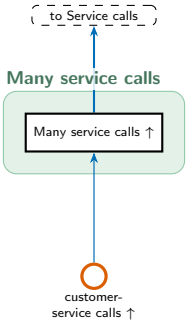


Figure 100: Many service calls: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part raises the chance of leaving from 4 customer-service calls upwards, slowly at first, then faster. [C122]

Evidence. It acts on 318 of 4000 training customers, 159 of whom left; for the typical customer it acts on, the chance is about 7% without it and about 12% with it. [C123] Through it, the chance rises slowly at first, then faster: going from 0 to 4 calls moves a typical such customer's chance from about 1% to about 2%. [C122] Its formula reproduces the part exactly on all 4000 training customers. [C124]

Evidence plots: Figure 101, Figure 102.
Figure 101 shows this part's own output, unit h11: it switches on at about 3.194 calls of customer-service calls, and is zero below.

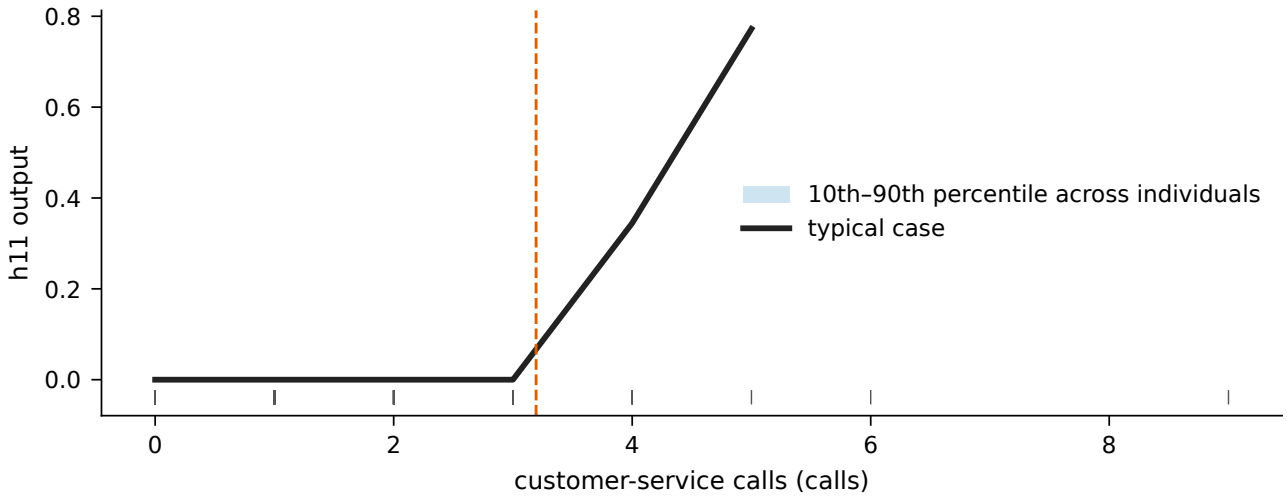


Figure 101: Many service calls: This part's own output, unit h11 (the output of 'Many service calls'), on its own scale (its activation), with every other input at the typical case; it is active for 318 of 4000. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

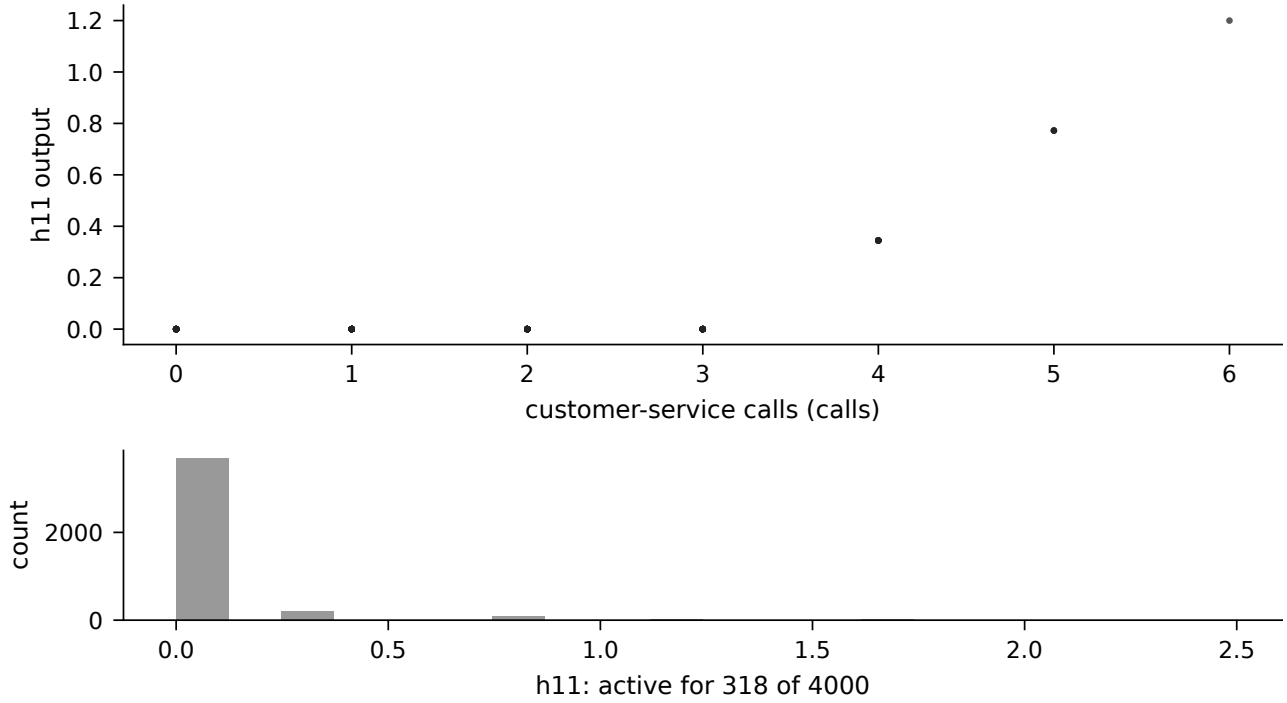


Figure 102: Many service calls: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.5.

2.2.6 Voice-mail discount

Part of: The whole model (2.2.1).
Figure 103 shows what feeds this part; Figure 104 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).



Figure 103: Voice-mail discount: what feeds it, one level down.

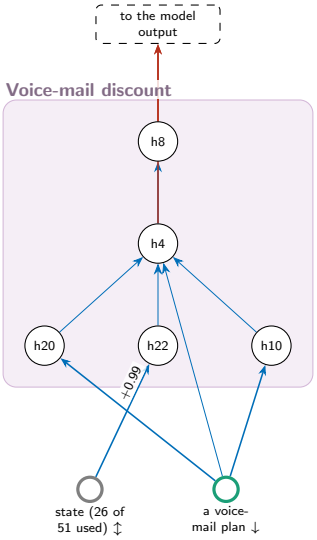


Figure 104: Voice-mail discount: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part gives customers with a voice-mail plan a further discount on the chance of leaving. [C119]

Evidence. Through it, a voice-mail plan lowers a typical customer's chance from about 1% to under 1%, and customers in North Carolina get under 1% instead of about 1%. [C119, C120] Its formula reproduces the part exactly on all 4000 training customers. [C121]
Evidence plots: Figure 105, Figure 106.

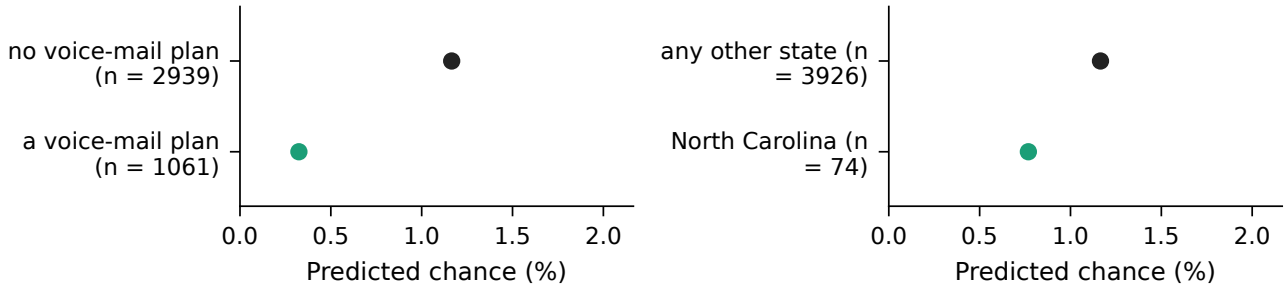


Figure 105: Voice-mail discount: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

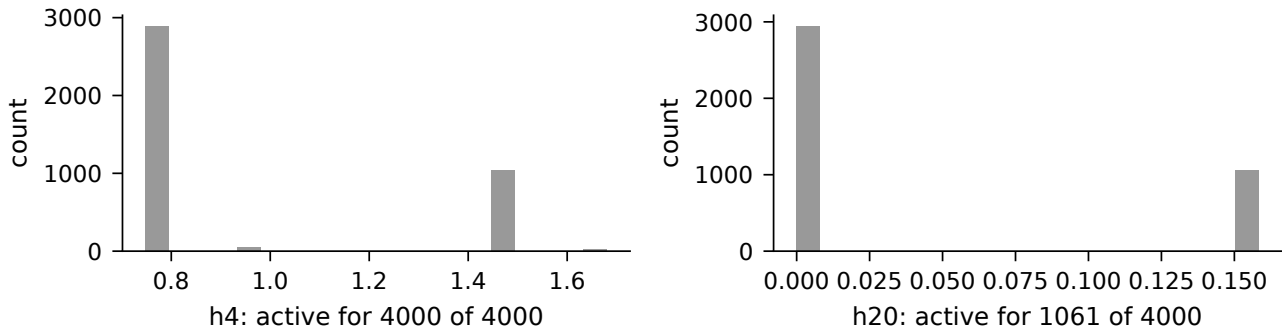


Figure 106: Voice-mail discount: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.6.

2.2.7 State units

Part of: The whole model (2.2.1).
Figure 107 shows what feeds this part; Figure 108 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).



Figure 107: State units: what feeds it, one level down.

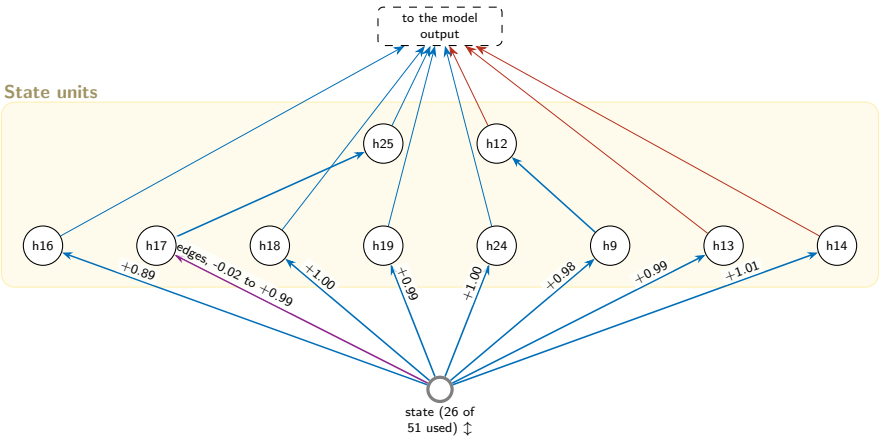


Figure 108: State units: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part reads only the state and raises the chance of leaving in a few states, most in Washington and Montana. [C125, C126]

Evidence. Through it, a typical customer's chance goes from about 1% to about 2% in Washington and in Montana, and up by less than a point in Kentucky and Ohio. [C125, C126, C127, C128] In Arizona, Delaware, North Dakota and Nevada it lowers the chance by less than a tenth of a point. [C129, C130, C131, C132] Its formula reproduces the part exactly on all 4000 training customers. [C133]
Evidence plots: Figure 109, Figure 110.

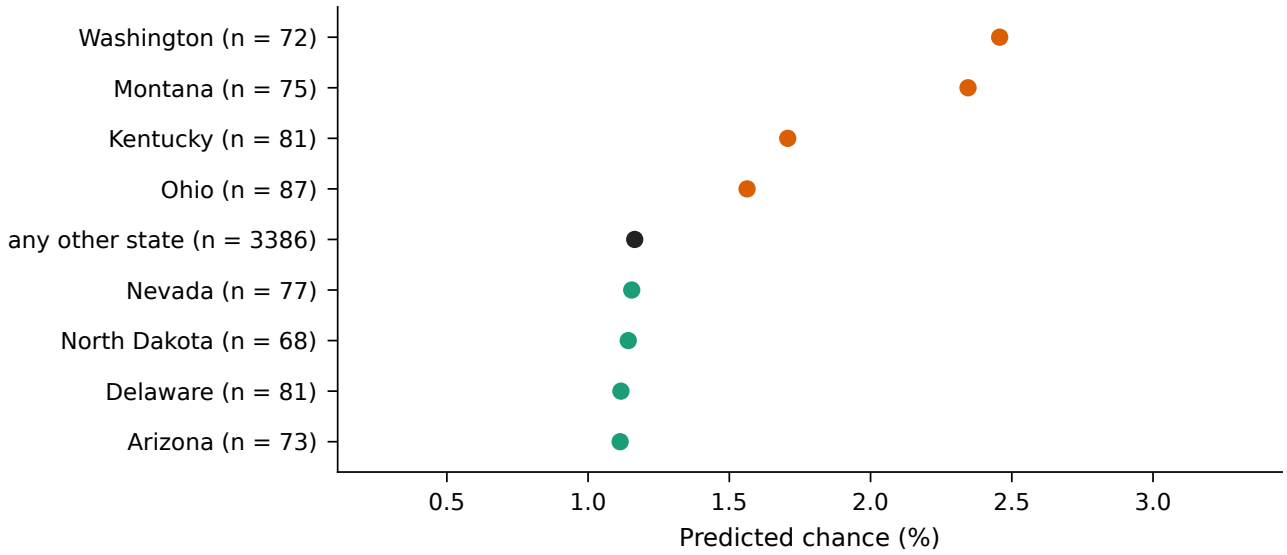


Figure 109: State units: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

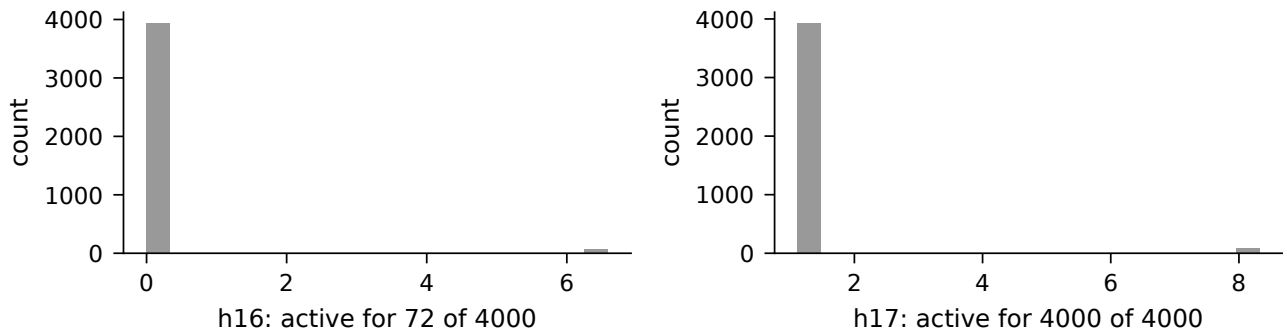


Figure 110: State units: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.7.

2.2.8 Area-code unit

Part of: The whole model (2.2.1).
Figure 111 shows what feeds this part; Figure 112 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).



Figure 111: Area-code unit: what feeds it, one level down.

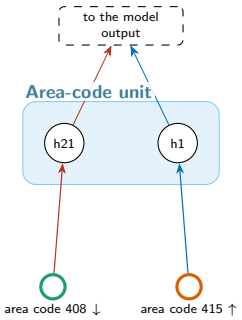


Figure 112: Area-code unit: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part reads the area code and barely changes the predictions. [C137]

Evidence. Through it, a typical customer's chance stays at about 1% in area codes 408 and 510, as in area code 415. [C134, C135] Switching it off changes the predictions by at most half a point. [C137] Its formula reproduces the part exactly on all 4000 training customers. [C138]
Evidence plots: Figure 113, Figure 114.

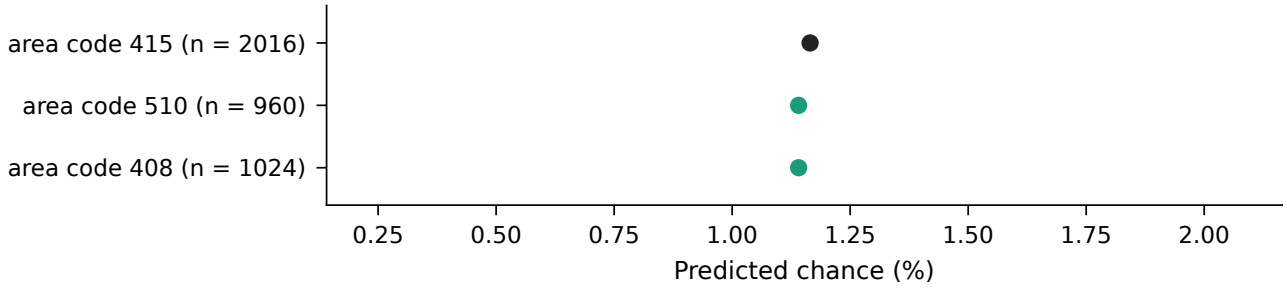


Figure 113: Area-code unit: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

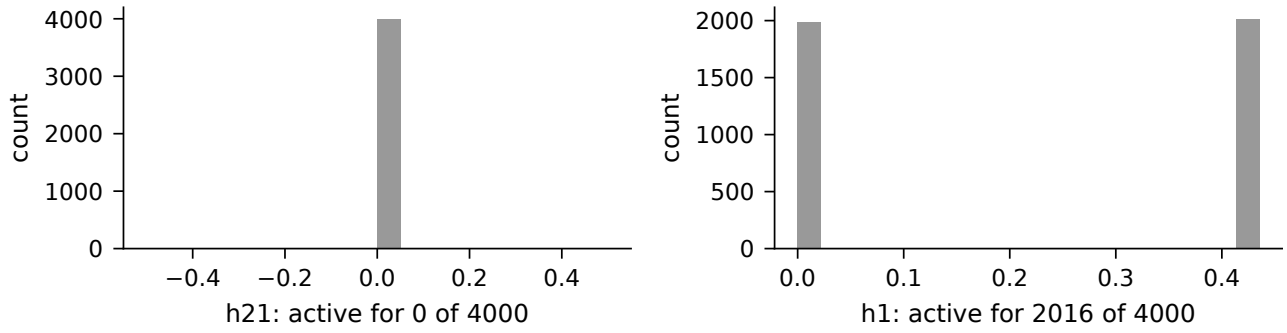


Figure 114: Area-code unit: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.8.

2.2.9 Small Utah unit

Part of: The whole model (2.2.1).
Figure 115 shows what feeds this part; Figure 116 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).



Figure 115: Small Utah unit: what feeds it, one level down.

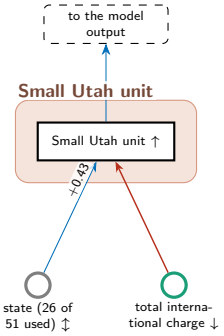


Figure 116: Small Utah unit: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This small part reads the international charge and being in Utah, and barely changes the predictions. [C144]

Evidence. Through it, the international charge barely changes the chance, which stays at about 1% for the typical customer it acts on. [C139] Switching it off changes the predictions by at most 0.2 points. [C144] Its formula reproduces the part exactly on all 4000 training customers. [C145]
Evidence plots: Figure 117, Figure 118, Figure 119, Figure 120.
Figure 119 shows this part's own output, unit h2.

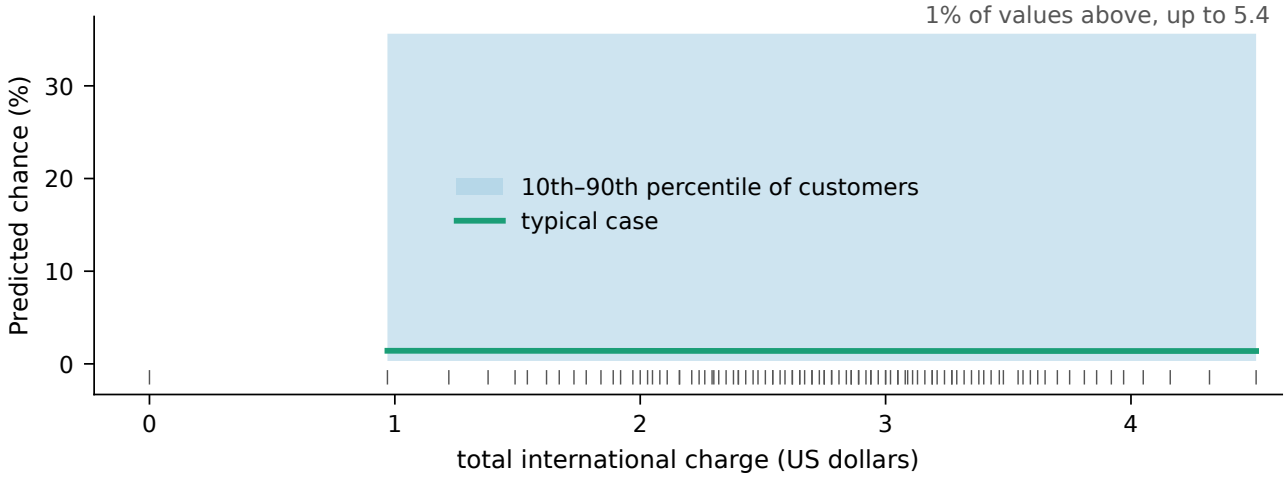


Figure 117: Small Utah unit: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

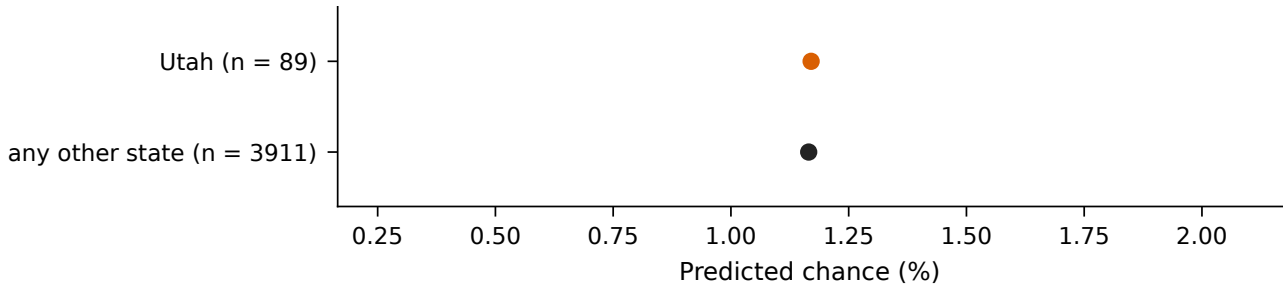


Figure 118: Small Utah unit: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

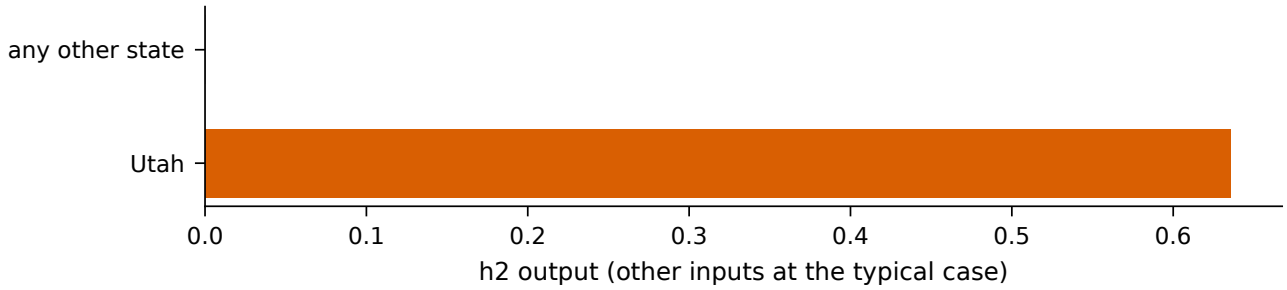


Figure 119: Small Utah unit: This part's own output, unit h2 (the output of 'Small Utah unit'), on its own scale (its activation), with every other input at the typical case; it is active for 100 of 4000. Its output for each combination of the levels it reads.

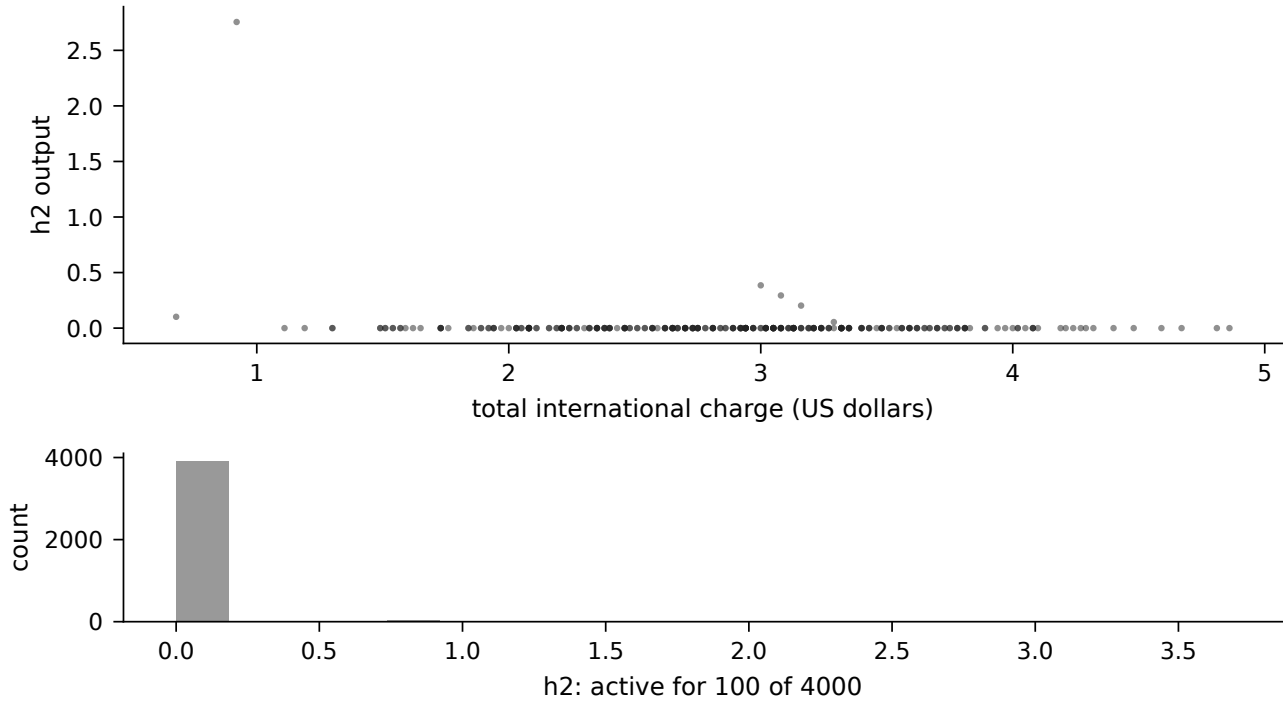


Figure 120: Small Utah unit: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.9.

2.3 Not supported by the model

- C002: direction claim about total daytime minutes in the whole model: not supported by the model (14% of patients have the opposite sign (allowed: 10%)).
- C017: direction claim about total evening charge in the whole model: not supported by the model (52% of patients have the opposite sign (allowed: 10%)).
- C026: direction claim about state in the whole model: not supported by the model (53% of patients have the opposite sign (allowed: 10%)).
- C136: subgroup claim about area code in part ann_94: not supported by the model (its effect is negligible).
- C141: interaction claim about total international charge and state in part ann_93: not supported by the model (the group difference is below 0.1 log-odds; the part's own counterfactual difference has the opposite sign).
- C142: threshold claim about total international charge in part ann_93: not supported by the model (the fold changes the curve by only 0.037 log-odds compared with a straight line (needed 0.1); the effect past the fold is too small or not consistent across patients).
- C143: subgroup claim about state and total international charge in part ann_93: not supported by the model (its effect is negligible).
- C146: boundary claim about number of calls to customer service and total daytime minutes in part ann_96: not supported by the model (no switch-on line could be traced on the first map to check against the data; no switch-on boundary in the observed range).
- C147: boundary claim about total daytime minutes and number of calls to customer service in part ann_90: not supported by the model (no switch-on line could be traced on the first map to check against the data; no switch-on boundary in the observed range).

Technical detail: churn100-s2-c2

The record behind each part: the original hypothesis, the closed form, and every claim with its estimate (log-odds and odds ratio), range across patients, n and test check.

T.1 The whole model

Annotation ann_97 (churn_logit). Reads: state, account length, area code, international calling plan, voice-mail plan, number of voice-mail messages, total daytime minutes, total daytime calls, total daytime charge, total evening minutes, total evening calls, total evening charge, total night-time minutes, total night-time calls, total night-time charge, total international minutes, total international calls, total international charge, number of calls to customer service.

Original hypothesis (unchanged). Root: logit P(churn) = -1.15 + linear_backbone + service_call_nonlinear (identity_svc) - 0.89*1133 - 1.00*4051 (voicemail_chain) + state_offsets + 0.05*161 - 0.25*14653 (area codes, 14653 dead) + 0.007*252, purely additive in the logit; the sigmoid is the only coupling. 30 inputs are unconnected (international minutes, night charge, night calls, evening calls, day charge and 25 state indicators) and have zero effect. Train AUC 0.8944, accuracy 0.896. Performance sources (train): all children replaced by least-squares linear fits gives AUC 0.8314 (an L2 logistic regression refit on all 71 inputs gets 0.8433); restoring csc_usage_gate alone gives 0.8917, csc_hinge alone 0.8462, the others 0.8314 (exactly or nearly linear). So the gain over linear comes from the service-call x low-usage gate. Rare-category switches: state:11 -1.93 logit (73 rows, empirical -1.35); state:47 +1.47 mean (72 rows, empirical +0.92; direct +1.39, +0.76 via 10332, but it switches off 1264, down to -1.38 for some); state:4 +1.34 (41 rows, +0.80); state:0 -1.13 mean, down to -5.0 (56 rows, -0.76; switches off 2796 in the gate); state:26 +1.09 (75 rows, +0.50; via direct + state offsets); state:43 +0.88 (95 rows, +0.48); state:45 -0.71 (91 rows, -0.85).

Closed form. Read through its parts (their formulas are in their own sections).

Split inputs. To annotate the parts, the explanation split these inputs into copies (a refactoring step that does not change what the model computes); the diagrams draw each input once: customer-service calls (2 copies); total daytime minutes (2 copies); total evening charge (2 copies); total international charge (3 copies); a voice-mail plan (2 copies).

State: every level the model reads. Whole-model change in the predicted chance for a typical customer when only the state changes from the reference; n = training customers in that level.

Claim	From	To	n	Chance	OK
C003	West Virginia	Washington	72	1% to 9%	yes
C004	West Virginia	California	41	1% to 5%	yes
C006	West Virginia	Montana	75	1% to 4%	yes
C007	West Virginia	Texas	95	1% to 3%	yes
C013	West Virginia	South Carolina	66	1% to 2%	yes
C014	West Virginia	Maryland	82	1% to 2%	yes
C016	West Virginia	Nevada	77	1% to 2%	yes
C018	West Virginia	Kentucky	81	1% to 2%	yes
C019	West Virginia	Massachusetts	86	1% to 2%	yes
C021	West Virginia	Ohio	87	1% to 2%	yes
C022	West Virginia	Minnesota	100	1% to 2%	yes
C024	West Virginia	Oregon	84	1% to 2%	yes
C030	West Virginia	Utah	89	1% to 1%	yes
C031	West Virginia	state other than Alaska, Hawaii, Indiana, Kentucky, Massa...	1947	1% to 1%	yes
C034	West Virginia	Arizona	73	1% to 1%	yes
C035	West Virginia	Oklahoma	79	1% to 1%	yes
C037	West Virginia	Delaware	81	1% to 1%	yes
C039	West Virginia	Indiana	82	1% to 1%	yes
C036	West Virginia	North Carolina	74	1% to 1%	yes
C033	West Virginia	North Dakota	68	1% to 1%	yes
C029	West Virginia	Washington DC	73	1% to 1%	yes
C028	West Virginia	Missouri	80	1% to 1%	yes
C026	West Virginia	South Dakota	66	1% to 1%	no
C023	West Virginia	Alaska	56	1% to 1%	yes
C020	West Virginia	Virginia	91	1% to 1%	yes
C012	West Virginia	Hawaii	73	1% to <1%	yes

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C001	direction: international calling plan: no international plan to an international plan (whole model) *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 14%	2.60	13.5	2.60 to 2.60	3620, 380	agrees; AUC gain 0.056 [0.032, 0.082]	yes
C002	direction: total daytime minutes: 112 minutes to 249 minutes (whole model) *shape: dip at 154 minutes; 99% of patients same; verified; risk saturates near 0%*	1% to 11%	1.64	5.16	-0.45 to 3.10	4000	agrees; AUC gain 0.081 [0.053, 0.112]	no
C003	direction: state: West Virginia to Washington (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 9%	1.66	5.24	0.76 to 2.22	122, 72	disagrees; AUC gain -0.002 [-0.012, 0.008]	yes
C004	direction: state: West Virginia to California (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 5%	1.42	4.15	1.42 to 1.42	122, 41	disagrees; AUC gain -0.002 [-0.012, 0.008]	yes
C005	direction: voice-mail messages: 0 messages to 32 messages (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 4%	1.36	3.91	1.36 to 1.36	4000	agrees; AUC gain 0.018 [0.006, 0.030]	yes
C006	direction: state: West Virginia to Montana (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 4%	1.17	3.23	1.17 to 1.17	122, 75	disagrees; AUC gain -0.002 [-0.012, 0.008]	yes
C007	direction: state: West Virginia to Texas (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 3%	0.96	2.60	0.96 to 0.96	122, 95	disagrees; AUC gain -0.002 [-0.012, 0.008]	yes
C008	direction: total evening minutes: 138 minutes to 266 minutes (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	1.26	3.54	1.26 to 1.26	4000	agrees; AUC gain 0.014 [0.004, 0.024]	yes
C009	direction: total international charge: 1.84 US dollars to 3.70 US dollars (whole model) *shape: accelerating; 93% of patients same; verified; risk saturates near 0%*	1% to 2%	0.61	1.85	0.15 to 0.79	4000	unclear; AUC gain 0.006 [-4e-05, 0.012]	yes
C010	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (whole model) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	1% to <1%	-3.26	0.04	-3.26 to -3.26	2939, 1061	agrees; AUC gain 0.024 [0.0005, 0.047]	yes
C011	direction: customer-service calls: 0 calls to 3 calls (whole model) *shape: accelerating; 94% of patients same; verified; risk saturates near 0%*	1% to 2%	0.94	2.56	0.00 to 2.43	4000	agrees; AUC gain 0.095 [0.059, 0.134]	yes
C012	direction: state: West Virginia to Hawaii (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to <1%	-1.86	0.16	-1.86 to -1.86	122, 73	disagrees; AUC gain -0.002 [-0.012, 0.008]	yes
C013	direction: state: West Virginia to South Carolina (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 2%	0.61	1.84	0.61 to 0.61	122, 66	disagrees; AUC gain -0.002 [-0.012, 0.008]	yes
C014	direction: state: West Virginia to Maryland (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 2%	0.60	1.83	0.60 to 0.60	122, 82	disagrees; AUC gain -0.002 [-0.012, 0.008]	yes
C015	direction: total night-time minutes: 136 minutes to 264 minutes (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	0.70	2.01	0.70 to 0.70	4000	unclear; AUC gain 0.004 [-0.0008, 0.008]	yes
C016	direction: state: West Virginia to Nevada (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 2%	0.52	1.69	0.52 to 0.52	122, 77	disagrees; AUC gain -0.002 [-0.012, 0.008]	yes
C017	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (whole model) *shape: dip at 12.96 US dollars; 71% of patients same; fallback; risk saturates near 0%*	1% to 2%	-0.20	0.82	-1.79 to 0.87	4000	unclear; AUC gain 0.005 [-0.005, 0.015]	no
C018	direction: state: West Virginia to Kentucky (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 2%	0.47	1.59	0.47 to 0.47	122, 81	disagrees; AUC gain -0.002 [-0.012, 0.008]	yes
C019	direction: state: West Virginia to Massachusetts (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 2%	0.42	1.52	0.42 to 0.42	122, 86	disagrees; AUC gain -0.002 [-0.012, 0.008]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C020	direction: state: West Virginia to Virginia (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	-0.63	0.53	-0.63 to -0.63	122, 91	disagrees; AUC gain -0.002 [-0.012, 0.008]	yes
C021	direction: state: West Virginia to Ohio (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 2%	0.38	1.46	0.38 to 0.38	122, 87	disagrees; AUC gain -0.002 [-0.012, 0.008]	yes
C022	direction: state: West Virginia to Minnesota (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 2%	0.36	1.44	0.36 to 0.36	122, 100	disagrees; AUC gain -0.002 [-0.012, 0.008]	yes
C023	direction: state: West Virginia to Alaska (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	-1.01	0.36	-2.44 to -0.52	122, 56	disagrees; AUC gain -0.002 [-0.012, 0.008]	yes
C024	direction: state: West Virginia to Oregon (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 2%	0.31	1.36	0.31 to 0.31	122, 84	disagrees; AUC gain -0.002 [-0.012, 0.008]	yes
C025	direction: total international calls: 2 calls to 8 calls (whole model) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 0%*	1% to 1%	-0.38	0.68	-0.38 to -0.38	4000	unclear; AUC gain 0.0005 [-0.002, 0.003]	yes
C026	direction: state: West Virginia to South Dakota (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	0.06	1.07	-0.51 to 0.78	122, 66	disagrees; AUC gain -0.002 [-0.012, 0.008]	no
C027	direction: account length: 48 to 152 (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	0.33	1.39	0.33 to 0.33	4000	disagrees; AUC gain -0.001 [-0.004, 0.0006]	yes
C028	direction: state: West Virginia to Missouri (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	-0.17	0.85	-0.17 to -0.17	122, 80	disagrees; AUC gain -0.002 [-0.012, 0.008]	yes
C029	direction: state: West Virginia to Washington DC (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	-0.12	0.89	-0.12 to -0.12	122, 73	disagrees; AUC gain -0.002 [-0.012, 0.008]	yes
C030	direction: state: West Virginia to Utah (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	0.08	1.09	0.08 to 0.09	122, 89	disagrees; AUC gain -0.002 [-0.012, 0.008]	yes
C031	direction: state: West Virginia to state other than Alaska, Hawaii, Indiana, Kentucky, Massachusetts, Maryland, Minnesota, Missouri, Montana, North Carolina, North Dakota, Arizona, Nevada, Ohio, Oklahoma, Oregon, California, South Carolina, South Dakota, Texas, Utah, Virginia, Washington, West Virginia, Washington DC, Delaware (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	0.08	1.08	0.08 to 0.08	122, 1947	disagrees; AUC gain -0.002 [-0.012, 0.008]	yes
C032	direction: area code: area code 415 to area code 510 (whole model) *shape: categorical; highest to lowest: area code 415, area code 408, area code 510*	1% to 1%	-0.08	0.92	-0.08 to -0.08	2016, 960	disagrees; AUC gain -0.0003 [-0.001, 0.0004]	yes
C033	direction: state: West Virginia to North Dakota (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	-0.08	0.93	-0.08 to -0.08	122, 68	disagrees; AUC gain -0.002 [-0.012, 0.008]	yes
C034	direction: state: West Virginia to Arizona (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	0.07	1.07	0.07 to 0.07	122, 73	disagrees; AUC gain -0.002 [-0.012, 0.008]	yes
C035	direction: state: West Virginia to Oklahoma (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	0.06	1.06	0.06 to 0.06	122, 79	disagrees; AUC gain -0.002 [-0.012, 0.008]	yes
C036	direction: state: West Virginia to North Carolina (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	-0.05	0.95	-0.05 to -0.05	122, 74	disagrees; AUC gain -0.002 [-0.012, 0.008]	yes
C037	direction: state: West Virginia to Delaware (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	0.04	1.04	0.04 to 0.04	122, 81	disagrees; AUC gain -0.002 [-0.012, 0.008]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C038	direction: area code: area code 415 to area code 408 (whole model) *shape: categorical; highest to lowest: area code 415, area code 408, area code 510*	1% to 1%	-0.02	0.98	-0.02 to -0.02	2016, 1024	disagrees; AUC gain -0.0003 [-0.001, 0.0004]	yes
C039	direction: state: West Virginia to Indiana (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	-0.02	0.98	-0.02 to -0.02	122, 82	disagrees; AUC gain -0.002 [-0.012, 0.008]	yes
C040	direction: total daytime calls: 75 calls to 125 calls (whole model) *shape: steady; 87% of patients same; fallback; risk saturates near 0%*	1% to 1%	0.13	1.14	0.00 to 0.31	4000	disagrees; AUC gain -0.001 [-0.003, 0.0005]	yes
C041	magnitude: international calling plan: no international plan to an international plan, typical patient *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 14%	2.60	13.5		3620, 380	agrees; AUC gain 0.056 [0.032, 0.082]	yes
C042	magnitude: state: West Virginia to Washington, typical patient *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 9%	2.08	8.00		122, 72	disagrees; AUC gain -0.002 [-0.012, 0.008]	yes
C043	magnitude: voice-mail messages: 0 messages to 32 messages, typical patient *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 4%	1.36	3.91		4000	agrees; AUC gain 0.018 [0.006, 0.030]	yes
C044	magnitude: total evening minutes: 138 minutes to 266 minutes, typical patient *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	1.26	3.54		4000	agrees; AUC gain 0.014 [0.004, 0.024]	yes
C045	magnitude: total international charge: 1.84 US dollars to 3.70 US dollars, typical patient *shape: accelerating; 93% of patients same; verified; risk saturates near 0%*	1% to 2%	0.79	2.20		4000	unclear; AUC gain 0.006 [-4e-05, 0.012]	yes
C046	null: model ignores total daytime charge (exact rule)		0.00			4000	train only	yes
C047	null: model ignores total evening calls (exact rule)		0.00			4000	train only	yes
C048	null: model ignores total international minutes (exact rule)		0.00			4000	train only	yes
C049	null: model ignores total night-time calls (exact rule)		0.00			4000	train only	yes
C050	null: model ignores total night-time charge (exact rule)		0.00			4000	train only	yes
C051	structure: made of linear_backbone, service_call_nonlinear, voicemail_chain, intl_state44_unit, area_code_units, state_offsets; added					4000	train only	yes

T.2 Fixed-weight inputs

Annotation ann_89 (linear_backbone). Reads: state, account length, area code, international calling plan, voice-mail plan, number of voice-mail messages, total daytime minutes, total evening minutes, total evening charge, total night-time minutes, total international calls, total international charge.

Original hypothesis (unchanged). Direct input-to-output edges collected through identity_lin (an identity node). Exactly linear in 33 inputs: day minutes +1.84 per SD, voicemail plan -0.87, international plan +0.76, international charge +0.69, voicemail messages +0.58, evening minutes +0.50, evening charge +0.35, night minutes +0.27, state:11 -0.26 (-1.93 logit for its 73 customers), state:47 +0.18 (+1.39), state:4 +0.14 (+1.34), state:43 +0.13 (+0.88), account length +0.13, international calls -0.16, state:45 -0.11 (-0.71), and smaller state/area-code terms. Hypothesis: the additive backbone; day minutes, voicemail plan and international plan dominate. Evidence: closed form exact (identity); train sensitivities equal the weights (day minutes 1.84, voicemail plan 0.87, international plan 0.76).

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 5.32907e-15).

$$
\begin{aligned}
y_0 = {} & 0.0518x_{state:26_a} + 1.84x_{totalDayMinutes_a} + 0.0673x_{state:40} \\
& + 0.346x_{totalEveCharge_a} + 0.0442x_{state:23} + 0.497x_{totalEveMinutes} \\
& + 0.0622x_{state:33_a} + 0.126x_{accountLength} + 0.0331x_{state:37} + 0.133x_{state:43} \\
& + 0.135x_{state:4} + 0.00424x_{state:3_a} + 0.273x_{totalNightMinutes} \\
& + 0.575x_{numberVmailMessages} + 0.0744x_{state:20} + 0.0391x_{state:27_a} \\
& + 0.694x_{totalIntlCharge_a} + 0.184x_{state:47_a} + 0.763x_{internationalPlan} \\
& + 0.0494x_{state:19} - 0.0703x_{state:0_a} - 0.159x_{totalIntlCalls} \\
& - 0.00299x_{state:36} - 0.0136x_{state:49} - 0.0745x_{state:41_a} - 0.0346x_{state:24} \\
& - 0.0249x_{areaCode:510} - 0.106x_{state:45} - 0.0136x_{state:15} - 0.259x_{state:11} \\
& - 0.0176x_{state:28_a} - 0.0269x_{state:7} - 0.872x_{voiceMailPlan_a}
\end{aligned}
$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C052	direction: international calling plan: no international plan to an international plan (through this part) *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 14%	2.60	13.5	2.60 to 2.60	3620, 380	agrees; AUC gain 0.056 [0.032, 0.082]	yes
C053	direction: total daytime minutes: 112 minutes to 249 minutes (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	<1% to 11%	4.70	110	4.70 to 4.70	4000	agrees; AUC gain 0.157 [0.117, 0.200]	yes
C054	direction: state: West Virginia to Washington (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 5%	1.47	4.33	1.47 to 1.47	122, 72	disagrees; AUC gain -0.004 [-0.013, 0.004]	yes
C055	direction: state: West Virginia to California (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 5%	1.42	4.15	1.42 to 1.42	122, 41	disagrees; AUC gain -0.004 [-0.013, 0.004]	yes
C056	direction: voice-mail messages: 0 messages to 32 messages (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 4%	1.36	3.91	1.36 to 1.36	4000	agrees; AUC gain 0.018 [0.006, 0.030]	yes
C057	direction: total international charge: 1.84 US dollars to 3.70 US dollars (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	<1% to 3%	1.73	5.63	1.73 to 1.73	4000	agrees; AUC gain 0.020 [0.008, 0.032]	yes
C058	direction: state: West Virginia to Texas (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 3%	0.96	2.60	0.96 to 0.96	122, 95	disagrees; AUC gain -0.004 [-0.013, 0.004]	yes
C059	direction: total evening minutes: 138 minutes to 266 minutes (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	1.26	3.54	1.26 to 1.26	4000	agrees; AUC gain 0.014 [0.004, 0.024]	yes
C060	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	0.87	2.40	0.87 to 0.87	4000	agrees; AUC gain 0.007 [0.0006, 0.014]	yes
C061	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	1% to <1%	-1.98	0.14	-1.98 to -1.98	2939, 1061	disagrees; AUC gain -0.0003 [-0.015, 0.015]	yes
C062	direction: state: West Virginia to Hawaii (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to <1%	-1.86	0.16	-1.86 to -1.86	122, 73	disagrees; AUC gain -0.004 [-0.013, 0.004]	yes
C063	direction: state: West Virginia to South Carolina (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 2%	0.61	1.84	0.61 to 0.61	122, 66	disagrees; AUC gain -0.004 [-0.013, 0.004]	yes
C064	direction: state: West Virginia to Maryland (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 2%	0.60	1.83	0.60 to 0.60	122, 82	disagrees; AUC gain -0.004 [-0.013, 0.004]	yes
C065	direction: total night-time minutes: 136 minutes to 264 minutes (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	0.70	2.01	0.70 to 0.70	4000	unclear; AUC gain 0.004 [-0.0008, 0.008]	yes
C066	direction: state: West Virginia to Nevada (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 2%	0.53	1.70	0.53 to 0.53	122, 77	disagrees; AUC gain -0.004 [-0.013, 0.004]	yes
C067	direction: state: West Virginia to Montana (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 2%	0.46	1.59	0.46 to 0.46	122, 75	disagrees; AUC gain -0.004 [-0.013, 0.004]	yes
C068	direction: state: West Virginia to Massachusetts (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 2%	0.42	1.52	0.42 to 0.42	122, 86	disagrees; AUC gain -0.004 [-0.013, 0.004]	yes
C069	direction: state: West Virginia to Virginia (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 1%	-0.63	0.53	-0.63 to -0.63	122, 91	disagrees; AUC gain -0.004 [-0.013, 0.004]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C070	direction: state: West Virginia to North Carolina (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 2%	0.37	1.45	0.37 to 0.37	122, 74	disagrees; AUC gain -0.004 [-0.013, 0.004]	yes
C071	direction: state: West Virginia to Minnesota (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 2%	0.36	1.44	0.36 to 0.36	122, 100	disagrees; AUC gain -0.004 [-0.013, 0.004]	yes
C072	direction: state: West Virginia to Alaska (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 1%	-0.52	0.59	-0.52 to -0.52	122, 56	disagrees; AUC gain -0.004 [-0.013, 0.004]	yes
C073	direction: state: West Virginia to South Dakota (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 1%	-0.51	0.60	-0.51 to -0.51	122, 66	disagrees; AUC gain -0.004 [-0.013, 0.004]	yes
C074	direction: state: West Virginia to Oregon (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 2%	0.31	1.36	0.31 to 0.31	122, 84	disagrees; AUC gain -0.004 [-0.013, 0.004]	yes
C075	direction: total international calls: 2 calls to 8 calls (through this part) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 0%*	1% to 1%	-0.38	0.68	-0.38 to -0.38	4000	unclear; AUC gain 0.0005 [-0.002, 0.003]	yes
C076	direction: account length: 48 to 152 (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	0.33	1.39	0.33 to 0.33	4000	disagrees; AUC gain -0.001 [-0.004, 0.0006]	yes
C077	direction: state: West Virginia to Missouri (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 1%	-0.17	0.85	-0.17 to -0.17	122, 80	disagrees; AUC gain -0.004 [-0.013, 0.004]	yes
C078	direction: state: West Virginia to Arizona (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 1%	0.11	1.12	0.11 to 0.11	122, 73	disagrees; AUC gain -0.004 [-0.013, 0.004]	yes
C079	direction: state: West Virginia to Washington DC (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 1%	-0.12	0.89	-0.12 to -0.12	122, 73	disagrees; AUC gain -0.004 [-0.013, 0.004]	yes
C080	direction: state: West Virginia to state other than Alaska, Hawaii, Indiana, Massachusetts, Maryland, Minnesota, Missouri, Montana, North Carolina, North Dakota, Arizona, Nevada, Oklahoma, Oregon, California, South Carolina, South Dakota, Texas, Virginia, Washington, West Virginia, Washington DC (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 1%	0.08	1.08	0.08 to 0.08	122, 2285	disagrees; AUC gain -0.004 [-0.013, 0.004]	yes
C081	direction: state: West Virginia to Oklahoma (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 1%	0.06	1.06	0.06 to 0.06	122, 79	disagrees; AUC gain -0.004 [-0.013, 0.004]	yes
C082	direction: area code: area code other than area code 510 to area code 510 (through this part) *shape: categorical; highest to lowest: area code other than area code 510, area code 510*	1% to 1%	-0.06	0.94	-0.06 to -0.06	3040, 960	disagrees; AUC gain -6e-05 [-0.0006, 0.0005]	yes
C083	direction: state: West Virginia to North Dakota (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 1%	-0.06	0.94	-0.06 to -0.06	122, 68	disagrees; AUC gain -0.004 [-0.013, 0.004]	yes
C084	direction: state: West Virginia to Indiana (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 1%	-0.02	0.98	-0.02 to -0.02	122, 82	disagrees; AUC gain -0.004 [-0.013, 0.004]	yes
C085	structure: its formula reproduces it on 4000 training patients (largest difference 5.32907e-15)		5e-15			4000	train only	yes

T.3 Service calls

Annotation ann_96 (service_call_nonlinear). Reads: state, total daytime minutes, total daytime calls, total evening charge, total international charge, number of calls to customer service.

Original hypothesis (unchanged). Composition of csc_usage_gate and csc_hinge: their exits are summed by identity_svc = 1.59*807 + 1.81*1264 + 1.52*7208, which enters the logit with weight 1; they share only the customer-service-calls input (split copies). Hypothesis: all of the model's service-call effect is nonlinear (calls have no direct edge): risk jumps from 4 calls, and for frequent callers low day usage raises risk further (the gate). Evidence: mean-ablating both gives AUC 0.7406, linear replacement of both 0.8313 (all-linear model 0.8314); the gate alone restores 0.8917 of 0.8944; sensitivity of identity_svc on train: day minutes 1.18, state:0 0.61, state:47 0.50, service calls 0.46+0.07, international charge 0.44, evening charge 0.43.

Closed form. Read through its parts (their formulas are in their own sections).

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C086	direction: total daytime minutes: 112 minutes to 249 minutes (through this part) *shape: hinge at 166 minutes; 71% of patients same; fallback; risk saturates near 0%; one point with C098 (median fold 166 minutes, typical patient 160 minutes, 10th-90th percentile 145 minutes to 213 minutes, wide)*	12% to 1%	-3.06	0.05	-5.15 to -1.60	4000	agrees; AUC gain 0.059 [0.034, 0.089]	yes
C087	direction: total international charge: 1.84 US dollars to 3.70 US dollars (through this part) *shape: hinge at 2.95 US dollars; 88% of patients same; fallback; risk saturates near 0%; one point with C101 (median fold 2.95 US dollars, typical patient 2.92 US dollars, 10th-90th percentile 2.95 US dollars to 4.10 US dollars, wide)*	3% to 1%	-1.11	0.33	-1.57 to -0.94	4000	unclear; AUC gain 0.004 [-0.004, 0.012]	yes
C088	direction: customer-service calls: 0 calls to 3 calls (through this part) *shape: accelerating; 94% of patients same; verified; risk saturates near 0%*	1% to 2%	0.94	2.56	0.00 to 2.43	4000	agrees; AUC gain 0.095 [0.059, 0.134]	yes
C089	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (through this part) *shape: hinge at 18.08 US dollars; 67% of patients same; fallback; risk saturates near 0%; one point with C100 (median fold 18.08 US dollars, typical patient 12.96 US dollars, 10th-90th percentile 7.14 US dollars to 21.50 US dollars, wide)*	2% to 1%	-1.08	0.34	-2.67 to 0.00	4000	unclear; AUC gain 0.008 [-0.003, 0.019]	yes
C090	direction: state: state other than Alaska, South Dakota, Washington to Washington (through this part) *shape: categorical; highest to lowest: South Dakota, Alaska, state other than Alaska, South Dakota, Washington, Washington*	1% to 1%	-0.57	0.57	-1.46 to 0.00	3806, 72	unclear; AUC gain 0.0004 [-0.0005, 0.001]	yes
C091	direction: state: state other than Alaska, South Dakota, Washington to South Dakota (through this part) *shape: categorical; highest to lowest: South Dakota, Alaska, state other than Alaska, South Dakota, Washington, Washington*	1% to 1%	0.57	1.77	0.00 to 1.29	3806, 66	unclear; AUC gain 0.0004 [-0.0005, 0.001]	yes
C092	direction: state: state other than Alaska, South Dakota, Washington to Alaska (through this part) *shape: categorical; highest to lowest: South Dakota, Alaska, state other than Alaska, South Dakota, Washington, Washington*	1% to 1%	-0.50	0.61	-1.92 to 0.00	3806, 56	unclear; AUC gain 0.0004 [-0.0005, 0.001]	yes
C093	direction: total daytime calls: 75 calls to 125 calls (through this part) *shape: steady; 87% of patients same; fallback; risk saturates near 0%*	1% to 1%	0.13	1.14	0.00 to 0.31	4000	disagrees; AUC gain -0.001 [-0.003, 0.0005]	yes
C094	interaction: effect of total international charge (1.84 US dollars to 3.70 US dollars) by state: Washington minus state other than Alaska, South Dakota, Washington; stronger in Washington *shape: steady; 100% of patients same; verified; risk saturates near 0%*	Washington: 4% to 18%; state other than Alaska, South Dakota, Washington: 1% to 2%	1.14	3.11		72, 3806	agrees; AUC gain 0.0004 [-0.0005, 0.001]	yes
C095	interaction: effect of total daytime minutes (112 minutes to 249 minutes) by total evening charge: total evening charge above 17.12 US dollars minus total evening charge at or below 17.12 US dollars; stronger in total evening charge above 17.12 US dollars *shape: dip at 148 minutes; 98% of patients same; verified; risk saturates near 0%*	total evening charge above 17.12 US dollars: 1% to 13%; total evening charge at or below 17.12 US dollars: 2% to 9%	1.47	4.36		1998, 2002	agrees; AUC gain 0.008 [-0.003, 0.019]	yes
C096	interaction: effect of total daytime minutes (112 minutes to 249 minutes) by state: Alaska minus state other than Alaska, South Dakota, Washington; stronger in Alaska *shape: dip at 148 minutes; 100% of patients same; verified; risk saturates near 0%*	Alaska: <1% to 7%; state other than Alaska, South Dakota, Washington: 1% to 11%	1.07	2.91		56, 3806	agrees; AUC gain 0.0004 [-0.0005, 0.001]	yes
C097	interaction: effect of total evening charge (11.76 US dollars to 22.57 US dollars) by state: Alaska minus state other than Alaska, South Dakota, Washington; stronger in Alaska *shape: steady; 100% of patients same; verified; risk saturates near 0%*	Alaska: <1% to 1%; state other than Alaska, South Dakota, Washington: 1% to 2%	1.08	2.95		56, 3806	agrees; AUC gain 0.0004 [-0.0005, 0.001]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C098	threshold: total daytime minutes: fold at about 166 minutes (stops) *shape: hinge at 166 minutes; 72% of patients same; fallback; risk saturates near 0%; one point with C098 (median fold 166 minutes, typical patient 160 minutes, 10th-90th percentile 145 minutes to 213 minutes, wide)*	28% to 1% (typical patient with a fold)	-4.16	0.02	-6.25 to -2.68	3999, 1743	agrees; AUC gain 0.059 [0.034, 0.089]	yes
C099	threshold: customer-service calls: fold at about 4 calls (starts) *shape: accelerating; 93% of patients same; verified*	3% to 8% (typical patient with a fold)	1.47	4.35	0.52 to 3.17	2700, 528	agrees; AUC gain 0.095 [0.059, 0.134]	yes
C100	threshold: total evening charge: fold at about 18.08 US dollars (stops) for customers in Colorado *shape: hinge at 18.08 US dollars; 68% of patients same; fallback; risk saturates near 0%; one point with C100 (median fold 18.08 US dollars, typical patient 16.50 US dollars, 10th-90th percentile 7.14 US dollars to 21.50 US dollars, wide)*	4% to 1% (typical patient with a fold (customers in Colorado))	-1.95	0.14	-3.27 to 0.00	71, 35	unclear; AUC gain 0.008 [-0.003, 0.019]	yes
C101	threshold: total international charge: fold at about 2.95 US dollars (stops) *shape: hinge at 2.95 US dollars; 93% of patients same; verified; risk saturates near 0%; one point with C101 (median fold 2.95 US dollars, typical patient 2.92 US dollars, 10th-90th percentile 2.95 US dollars to 4.10 US dollars, wide)*	4% to 1% (typical patient with a fold)	-1.47	0.23	-2.17 to -1.19	3886, 2729	unclear; AUC gain 0.004 [-0.004, 0.012]	yes
C102	threshold: total daytime calls: fold at about 75 calls (starts) for customers in Louisiana *shape: hinge at 93 calls; 80% of patients same; fallback; risk saturates near 0%; one point with C102 (median fold 75 calls, typical patient 93 calls, 10th-90th percentile 37 calls to 139 calls, wide)*	1% to 1% (typical patient with a fold (customers in Louisiana))	0.31	1.37	0.02 to 0.45	12, 9 (small n)	disagrees; AUC gain -0.001 [-0.003, 0.0005]	yes
C103	structure: made of csc_usage_gate, csc_hinge; added					4000	train only	yes
C146	boundary: boundary					4000	train only	no

T.4 Calls with low usage

Annotation ann_90 (csc_usage_gate). Reads: state, total daytime minutes, total daytime calls, total evening charge, total international charge, number of calls to customer service.

Original hypothesis (unchanged). Service-call x low-day-usage gate. 807 = relu(-1.30 - 1.30 dayMin + 0.06 dayCalls + 0.61 g1(csc) + 0.58 g2), with g1 = relu(0.95 + 0.24 csc + 0.36 relu(0.92 csc - 0.20)) always active (convex in calls) and g2 = 2796 = relu(0.38 + 0.31 csc + 0.36 relu(0.99 csc - 0.20) - 1.00 eveCharge - 1.41 state:0 + 0.13 state:41); 1264 = relu(0.03 + 0.33*807 - 0.35 intlCharge - 0.40 state:47). The logit gets +1.59*807 + 1.81*1264. Hypothesis: 807 is a hinge that switches on at LOW day minutes, and its kink moves up with each customer-service call (median kink on train: 147 day minutes at 0 calls, 157 at 1, 170 at 2, 194 at 3, 215 at 4, 237 at 5, 270 at 6); high evening charge lowers the kink (2796 is on for 98% of customers in the lowest evening-charge band, 22% in the highest) and state:0 switches 2796 off (all 56 customers). Below the kink the gate's slope is about -2.84 logit per SD of day minutes, more than cancelling the direct +1.84, so the logit is V-shaped in day minutes. 1264 passes 807 on and adds a low-international-charge hinge (on for 1319 rows where 807 is off); state:47 switches 1264 off. Evidence (train, n=4000): CSC>=4 and day minutes < 160: 113 customers, empirical churn 0.88, model mean 0.52 (under-predicted); CSC<4 and day minutes < 120: 476 customers, empirical 0.05, model 0.06. Mean churn by calls 0-6: 0.115/0.110/0.110/0.106/0.437/0.571/0.679, model 0.087/0.078/0.080/0.113/0.246/0.503/0.820; share with 807 on rises 0.31 -> 0.93. Sensitivity (807): day minutes 0.57, state:0 0.29, service calls 0.22, evening charge 0.21. Ablation: mean-ablation AUC 0.7690 (full 0.8944); least-squares linear replacement 0.8461 (R2 0.68/0.74); restored alone onto an all-linearised model 0.8917 (all-linear 0.8314). This is the model's main nonlinearity.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 1.33227e-15).

$$
\begin{aligned}
y_0 = \operatorname{relu}\big(&-1.3 + 0.584\operatorname{relu}\big(0.382 \\
&+ 0.314x_{numberCustomerServiceCalls_abde} + 0.359\operatorname{relu}\big(-0.2 \\
&+ 0.99x_{numberCustomerServiceCalls_abde}\big) + 0.129x_{state{:}41_b} \\
&- 1.41x_{state{:}0_b} - 0.996x_{totalEveCharge_b}\big) \\
&+ 0.608\operatorname{relu}\big(0.951 + 0.362\operatorname{relu}\big(-0.199 \\
&+ 0.917x_{numberCustomerServiceCalls_abde}\big) \\
&+ 0.235x_{numberCustomerServiceCalls_abde}\big) + 0.0575x_{totalDayCalls} \\
&- 1.3x_{totalDayMinutes_b}\big) \\
y_1 = \operatorname{relu}\big(&0.0266 + 0.329\operatorname{relu}\big(-1.3 \\
&+ 0.584\operatorname{relu}\big(0.382 + 0.314x_{numberCustomerServiceCalls_abde} \\
&+ 0.359\operatorname{relu}\big(-0.2 + 0.99x_{numberCustomerServiceCalls_abde}\big) \\
&+ 0.129x_{state{:}41_b} - 1.41x_{state{:}0_b} - 0.996x_{totalEveCharge_b}\big) \\
&+ 0.608\operatorname{relu}\big(0.951 + 0.362\operatorname{relu}\big(-0.199 \\
&+ 0.917x_{numberCustomerServiceCalls_abde}\big) \\
&+ 0.235x_{numberCustomerServiceCalls_abde}\big) + 0.0575x_{totalDayCalls} \\
&- 1.3x_{totalDayMinutes_b}\big) - 0.349x_{totalIntlCharge_c} \\
&- 0.397x_{state{:}47_b}\big)
\end{aligned}
$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C104	direction: total daytime minutes: 112 minutes to 249 minutes (through this part) *shape: hinge at 166 minutes; 71% of patients same; fallback; risk saturates near 0%; one point with C112 (median fold 166 minutes, typical patient 160 minutes, 10th-90th percentile 145 minutes to 213 minutes, wide)*	12% to 1%	-3.06	0.05	-5.15 to -1.60	4000	agrees; AUC gain 0.059 [0.034, 0.089]	yes
C105	direction: total international charge: 1.84 US dollars to 3.70 US dollars (through this part) *shape: hinge at 2.95 US dollars; 88% of patients same; fallback; risk saturates near 0%; one point with C115 (median fold 2.95 US dollars, typical patient 2.92 US dollars, 10th-90th percentile 2.95 US dollars to 4.10 US dollars, wide)*	3% to 1%	-1.11	0.33	-1.57 to -0.94	4000	unclear; AUC gain 0.004 [-0.004, 0.012]	yes
C106	direction: customer-service calls: 0 calls to 3 calls (through this part) *shape: accelerating; 77% of patients same; fallback; risk saturates near 0%*	1% to 2%	0.94	2.56	0.00 to 2.43	4000	agrees; AUC gain 0.062 [0.036, 0.092]	yes
C107	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (through this part) *shape: hinge at 18.08 US dollars; 67% of patients same; fallback; risk saturates near 0%; one point with C113 (median fold 18.08 US dollars, typical patient 12.96 US dollars, 10th-90th percentile 7.14 US dollars to 21.50 US dollars, wide)*	2% to 1%	-1.08	0.34	-2.67 to 0.00	4000	unclear; AUC gain 0.008 [-0.003, 0.019]	yes
C108	direction: state: state other than Alaska, South Dakota, Washington to Washington (through this part) *shape: categorical; highest to lowest: South Dakota, Alaska, state other than Alaska, South Dakota, Washington, Washington*	1% to 1%	-0.57	0.57	-1.46 to 0.00	3806, 72	unclear; AUC gain 0.0004 [-0.0005, 0.001]	yes
C109	direction: state: state other than Alaska, South Dakota, Washington to South Dakota (through this part) *shape: categorical; highest to lowest: South Dakota, Alaska, state other than Alaska, South Dakota, Washington, Washington*	1% to 1%	0.57	1.77	0.00 to 1.29	3806, 66	unclear; AUC gain 0.0004 [-0.0005, 0.001]	yes
C110	direction: state: state other than Alaska, South Dakota, Washington to Alaska (through this part) *shape: categorical; highest to lowest: South Dakota, Alaska, state other than Alaska, South Dakota, Washington, Washington*	1% to 1% (typical patient this part acts on)	-0.50	0.61	-1.92 to 0.00	3806, 56	unclear; AUC gain 0.0004 [-0.0005, 0.001]	yes
C111	direction: total daytime calls: 75 calls to 125 calls (through this part) *shape: steady; 87% of patients same; fallback; risk saturates near 0%*	1% to 1% (typical patient this part acts on)	0.13	1.14	0.00 to 0.31	4000	disagrees; AUC gain -0.001 [-0.003, 0.0005]	yes
C112	threshold: total daytime minutes: fold at about 166 minutes (stops) *shape: hinge at 166 minutes; 72% of patients same; fallback; risk saturates near 0%; one point with C112 (median fold 166 minutes, typical patient 160 minutes, 10th-90th percentile 145 minutes to 213 minutes, wide)*	28% to 1% (typical patient with a fold)	-4.16	0.02	-6.25 to -2.68	3999, 1743	agrees; AUC gain 0.059 [0.034, 0.089]	yes
C113	threshold: total evening charge: fold at about 18.08 US dollars (stops) for customers in Colorado *shape: hinge at 18.08 US dollars; 68% of patients same; fallback; risk saturates near 0%; one point with C113 (median fold 18.08 US dollars, typical patient 16.50 US dollars, 10th-90th percentile 7.14 US dollars to 21.50 US dollars, wide)*	4% to 1% (typical patient with a fold (customers in Colorado))	-1.95	0.14	-3.27 to 0.00	71, 35	unclear; AUC gain 0.008 [-0.003, 0.019]	yes
C114	threshold: customer-service calls: fold at about 4 calls (starts) for customers in Wisconsin *shape: accelerating; 61% of patients same; fallback; risk saturates near 0%*	2% to 5% (typical patient with a fold (customers in Wisconsin))	1.11	3.04	0.00 to 2.88	59, 12 (small n)	agrees; AUC gain 0.062 [0.036, 0.092]	yes
C115	threshold: total international charge: fold at about 2.95 US dollars (stops) *shape: hinge at 2.95 US dollars; 93% of patients same; verified; risk saturates near 0%; one point with C115 (median fold 2.95 US dollars, typical patient 2.92 US dollars, 10th-90th percentile 2.95 US dollars to 4.10 US dollars, wide)*	4% to 1% (typical patient with a fold)	-1.47	0.23	-2.17 to -1.19	3886, 2729	unclear; AUC gain 0.004 [-0.004, 0.012]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C116	threshold: total daytime calls: fold at about 75 calls (starts) for customers in Louisiana *shape: hinge at 93 calls; 80% of patients same; fallback; risk saturates near 0%; one point with C116 (median fold 75 calls, typical patient 93 calls, 10th-90th percentile 37 calls to 139 calls, wide)*	1% to 1% (typical patient with a fold (customers in Louisiana))	0.31	1.37	0.02 to 0.45	12, 9 (small n)	disagrees; AUC gain -0.001 [-0.003, 0.0005]	yes
C117	subgroup: who this part acts on: 3063 of 4000 patients	1% to 1% (typical patient this part acts on (risk without the part, then with it))	1.77	5.84	0.17 to 4.40	3063, 937	agrees; AUC gain 0.134 [0.089, 0.181]	yes
C118	structure: its formula reproduces it on 4000 training patients (largest difference 1.33227e-15)		1e-15			4000	train only	yes
C147	boundary: boundary					4000	train only	no

T.5 Many service calls

Annotation ann_91 (csc_hinge). Reads: number of calls to customer service.

Original hypothesis (unchanged). 7208 = relu(-0.70 + 0.56 csc), logit weight +1.52. Hypothesis: a hinge on customer-service calls that is off for 0-3 calls and switches on from 4 calls (kink at 3.2 calls), adding about +0.65 logit per extra call (contribution +0.52 at 4 calls, +1.17 at 5, +1.82 at 6). Active on exactly the 318 customers with 4+ calls (empirical churn jumps from 0.11 at 3 calls to 0.44 at 4). Evidence: line plot of 7208 on train rises from 0 to 2.5 (correlation with calls 0.67); mean ablation AUC 0.8901, linear replacement 0.8916; restored alone onto the all-linear model 0.8462 (all-linear 0.8314), but largely redundant once the gate is present (gate alone 0.8917).

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 0.0).

$$y_0 = \mathrm{relu}\big(-0.695 + 0.559 x_{numberCustomerServiceCalls_c}\big)$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C122	threshold: customer-service calls: fold at about 4 calls (starts) *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	1% to 2% (typical patient with a fold)	0.52	1.69	0.52 to 0.52	4000, 318	agrees; AUC gain 0.006 [0.0008, 0.012]	yes
C123	subgroup: who this part acts on: 318 of 4000 patients	7% to 12% (typical patient this part acts on (risk without the part, then with it))	0.89	2.44	0.52 to 1.82	318, 3682	agrees; AUC gain 0.006 [0.0008, 0.012]	yes
C124	structure: its formula reproduces it on 4000 training patients (largest difference 0.0)		0.00			4000	train only	yes

T.6 Voice-mail discount

Annotation ann_92 (voicemail_chain). Reads: state, voice-mail plan.

Original hypothesis (unchanged). 5577 = relu(0.99 + 0.96 vmPlan), 13852 = relu(-1.52 + 1.01 vmPlan) (on exactly for the 26.5% with a voicemail plan), 14955 = relu(0.51 + 0.99 state:27), 1133 = relu(0.72 + 0.11 vmPlan + 0.19*5577 + 0.22*13852 + 0.03*14955), 4051 = relu(2.09 + 0.94*1133). All but 13852 are always active, and voicemail plan and state:27 are binary, so the chain is exactly affine in them (least-squares R2 = 1.0). Hypothesis: a further voicemail-plan penalty (chain logit -4.76 with a plan vs -3.47 without, i.e. -1.29 on top of the direct -1.97 for having a plan) plus a small state:27 offset. Evidence: exact linear fit; linear replacement AUC 0.8944 (unchanged). Mean ablation RAISES train AUC to 0.8979: the chain's extra voicemail penalty slightly hurts ranking.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 2.22045e-16).

$$
\begin{aligned}
y_0 = \operatorname{relu}\big(&0.723 + 0.111 x_{voiceMailPlan_bcd} \\
&+ 0.191 \operatorname{relu}\big(0.994 + 0.962 x_{voiceMailPlan_bcd}\big) \\
&+ 0.221 \operatorname{relu}\big(-1.52 + 1.01 x_{voiceMailPlan_bcd}\big) \\
&+ 0.0312 \operatorname{relu}\big(0.513 + 0.993 x_{state:27_b}\big)\big) \\
y_1 = \operatorname{relu}\big(&2.09 + 0.942 \operatorname{relu}\big(0.723 \\
&+ 0.111 x_{voiceMailPlan_bcd} + 0.191 \operatorname{relu}\big(0.994 \\
&+ 0.962 x_{voiceMailPlan_bcd}\big) + 0.221 \operatorname{relu}\big(-1.52 \\
&+ 1.01 x_{voiceMailPlan_bcd}\big) + 0.0312 \operatorname{relu}\big(0.513 \\
&+ 0.993 x_{state:27_b}\big)\big)\big)
\end{aligned}
$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C119	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	1% to <1%	-1.29	0.28	-1.29 to -1.29	2939, 1061	disagrees; AUC gain -0.005 [-0.015, 0.005]	yes
C120	direction: state: state other than North Carolina to North Carolina (through this part) *shape: categorical; highest to lowest: state other than North Carolina, North Carolina*	1% to 1%	-0.42	0.66	-0.42 to -0.42	3926, 74	agrees; AUC gain 0.0004 [0.0001, 0.0008]	yes
C121	structure: its formula reproduces it on 4000 training patients (largest difference 2.22045e-16)		2e-16			4000	train only	yes

T.7 State units

Annotation ann_95 (state_offsets). Reads: state.

Original hypothesis (unchanged). Units 4967 -> 7647 (state:8), 8418 (state:28), 8896 (state:3), 10332 (state:47), 13551 (state:35), 13725 and 16344 (state:26), 11261 -> 16902 (state:17, state:33). Each depends only on state one-hot indicators; since the states are mutually exclusive, every such function is exactly an offset per state (least-squares R2 = 1.0 for every exit). Hypothesis: a per-state logit offset table; the largest offsets here: state:47 +0.76 (10332, adding to its direct +1.39), state:26 +0.71 (13725 + 16344, adding to its direct +0.38), state:17 +0.39 (16902), state:35 +0.30 (13551); state:8/28/3 under 0.05. Evidence: linear replacement AUC 0.8944 (unchanged); mean ablation 0.8935.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 1.77636e-15).

$$
\begin{aligned}
y_0 &= \mathrm{relu}\big(-1.83 + 0.984\,\mathrm{relu}\big(1.76 \\
&\qquad + 0.979 x_{state:8}\big)\big) \\
y_1 &= \mathrm{relu}\big(0.806 + 0.987 x_{state:28_b}\big) \\
y_2 &= \mathrm{relu}\big(-0.144 + 1.01 x_{state:3_b}\big) \\
y_3 &= \mathrm{relu}\big(-0.0137 + 0.894 x_{state:47_c}\big) \\
y_4 &= \mathrm{relu}\big(-0.23 + 0.998 x_{state:35}\big) \\
y_5 &= \mathrm{relu}\big(-0.597 + 0.991 x_{state:26_bc}\big) \\
y_6 &= \mathrm{relu}\big(1.04 + 0.997 x_{state:26_bc}\big) \\
y_7 &= \mathrm{relu}\big(0.0457 + 1.0\,\mathrm{relu}\big(1.42 \\
&\qquad + 0.99 x_{state:17} - 0.0231 x_{state:33_b}\big)\big)
\end{aligned}
$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C125	direction: state: state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware to Washington (through this part) *shape: categorical; highest to lowest: Washington, Montana, Kentucky, Ohio, state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware, Nevada …*	1% to 2%	0.76	2.14	0.76 to 0.76	3386, 72	unclear; AUC gain 0.0005 [-0.004, 0.005]	yes
C126	direction: state: state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware to Montana (through this part) *shape: categorical; highest to lowest: Washington, Montana, Kentucky, Ohio, state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware, Nevada …*	1% to 2%	0.71	2.04	0.71 to 0.71	3386, 75	unclear; AUC gain 0.0005 [-0.004, 0.005]	yes
C127	direction: state: state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware to Kentucky (through this part) *shape: categorical; highest to lowest: Washington, Montana, Kentucky, Ohio, state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware, Nevada …*	1% to 2%	0.39	1.47	0.39 to 0.39	3386, 81	unclear; AUC gain 0.0005 [-0.004, 0.005]	yes
C128	direction: state: state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware to Ohio (through this part) *shape: categorical; highest to lowest: Washington, Montana, Kentucky, Ohio, state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware, Nevada …*	1% to 2%	0.30	1.35	0.30 to 0.30	3386, 87	unclear; AUC gain 0.0005 [-0.004, 0.005]	yes
C129	direction: state: state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware to Arizona (through this part) *shape: categorical; highest to lowest: Washington, Montana, Kentucky, Ohio, state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware, Nevada …*	1% to 1%	-0.05	0.96	-0.05 to -0.05	3386, 73	unclear; AUC gain 0.0005 [-0.004, 0.005]	yes
C130	direction: state: state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware to Delaware (through this part) *shape: categorical; highest to lowest: Washington, Montana, Kentucky, Ohio, state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware, Nevada …*	1% to 1%	-0.04	0.96	-0.04 to -0.04	3386, 81	unclear; AUC gain 0.0005 [-0.004, 0.005]	yes
C131	direction: state: state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware to North Dakota (through this part) *shape: categorical; highest to lowest: Washington, Montana, Kentucky, Ohio, state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware, Nevada …*	1% to 1%	-0.02	0.98	-0.02 to -0.02	3386, 68	unclear; AUC gain 0.0005 [-0.004, 0.005]	yes
C132	direction: state: state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware to Nevada (through this part) *shape: categorical; highest to lowest: Washington, Montana, Kentucky, Ohio, state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware, Nevada …*	1% to 1%	-0.009	0.99	-0.009 to -0.009	3386, 77	unclear; AUC gain 0.0005 [-0.004, 0.005]	yes
C133	structure: its formula reproduces it on 4000 training patients (largest difference 1.77636e-15)		2e-15			4000	train only	yes

T.8 Area-code unit

Annotation ann_94 (area_code_units). Reads: area code.

Original hypothesis (unchanged). 161 = relu(-0.14 + 0.58 areaCode:415), logit weight +0.05: on exactly for area code 415 (binary), so exactly affine (+0.02 logit for area code 415). 14653 = relu(-1.07 - 0.54 areaCode:408) is never active on train (dead). Hypothesis: area code barely matters (direct area code 510 term -0.02 per SD in the backbone). Evidence: sensitivity 0.29 for 161 on its unit scale, 0 for 14653; linear R2 1.0; mean ablation AUC 0.8945.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 0.0).

$$y_0 = \mathrm{relu}\big(-0.137 + 0.577 x_{areaCode:415}\big)$$
$$y_1 = \mathrm{relu}\big(-1.07 - 0.535 x_{areaCode:408}\big)$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C134	direction: area code: area code 415 to area code 408 (through this part) *shape: categorical; highest to lowest: area code 415, area code 408, area code 510*	1% to 1%	-0.02	0.98	-0.02 to -0.02	2016, 1024	disagrees; AUC gain -0.0002 [-0.0005, 6e-05]	yes
C135	direction: area code: area code 415 to area code 510 (through this part) *shape: categorical; highest to lowest: area code 415, area code 408, area code 510*	1% to 1%	-0.02	0.98	-0.02 to -0.02	2016, 960	disagrees; AUC gain -0.0002 [-0.0005, 6e-05]	yes
C136	subgroup: who this part acts on: 2016 of 4000 patients	1% to 1% (typical patient this part acts on (risk without the part, then with it))	0.02	1.02	0.02 to 0.02	2016, 1984	disagrees; AUC gain -0.0002 [-0.0005, 6e-05]	no
C137	null: this part barely changes predictions (switched off)		0.005			4000	train only	yes
C138	structure: its formula reproduces it on 4000 training patients (largest difference 0.0)		0.00			4000	train only	yes

T.9 Small Utah unit

Annotation ann_93 (intl_state44_unit). Reads: state, total international charge.

Original hypothesis (unchanged). 252 = relu(-2.22 + 0.43 state:44 - 0.85 intlCharge), logit weight +0.007. Active on 100 of 4000 training rows (71 in state:44, the rest with very low international charge); logit contribution at most +0.03. Hypothesis: a negligible state:44 / low-international-charge offset. Evidence: sensitivity 0.01/0.02; mean ablation and linear replacement leave train AUC at 0.8944.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 4.44089e-16).

$$y_0 = \mathrm{relu}\big(-2.22 + 0.432 x_{state:44} - 0.851 x_{totalIntlCharge_b}\big)$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C139	direction: total international charge: 1.84 US dollars to 3.70 US dollars (through this part) *shape: flat; 100% of patients same; verified; risk saturates near 0%*	1% to 1% (typical patient this part acts on)	-0.0003	1.00	0.00 to 0.00	4000	unclear; AUC gain 8e-06 [0.000, 5e-05]	yes
C140	direction: state: state other than Utah to Utah (through this part) *shape: categorical; highest to lowest: Utah, state other than Utah*	1% to 1%	0.005	1.01	0.00 to 0.01	3911, 89	unclear; AUC gain 8e-06 [0.000, 5e-05]	yes
C141	interaction: effect of total international charge (1.84 US dollars to 3.70 US dollars) by state: Utah minus state other than Utah; stronger in Utah *shape: accelerating; 96% of patients same; verified; risk saturates near 0%*	Utah: 1% to 3%; state other than Utah: 1% to 2%	0.003	1.00		89, 3911	unclear; AUC gain 8e-06 [0.000, 5e-05]	no
C142	threshold: total international charge: fold at about 0.77 US dollars (stops) *shape: flat; 100% of patients same; verified; risk saturates near 0%*	1% to 1% (typical patient with a fold)	-0.0003	1.00	0.00 to 0.00	4000, 100	unclear; AUC gain 8e-06 [0.000, 5e-05]	no
C143	subgroup: who this part acts on: 100 of 4000 patients	1% to 1% (typical patient this part acts on (risk without the part, then with it))	0.007	1.01	0.001 to 0.01	100, 3900	unclear; AUC gain 0.000 [0.000, 0.000]	no
C144	null: this part barely changes predictions (switched off)		0.002			4000	train only	yes
C145	structure: its formula reproduces it on 4000 training patients (largest difference 4.44089e-16)		4e-16			4000	train only	yes

The network before any explanation

Figure 121 (next page, landscape) is the trained network as it is, with no parts, no splits and no inserted units: inputs at the bottom, hidden units by depth, the output at the top, every connection drawn, every input column its own node (one-hot levels included).

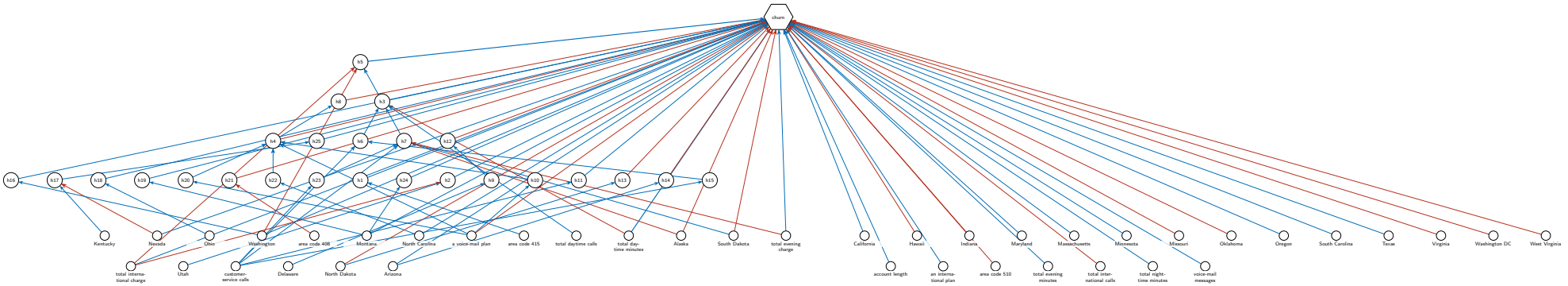


Figure 121: The network as trained, before any explanation, in full. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. Not used (no connection): Alabama, Georgia, Iowa, Idaho, Illinois, Kansas, Louisiana, Arkansas, Maine, Michigan, Mississippi, Nebraska, New Hampshire, New Jersey, New Mexico, New York, Pennsylvania, Rhode Island, Tennessee, Vermont, Wisconsin, Colorado, Wyoming, Connecticut, Florida, total daytime charge, total evening calls, total night-time calls, total night-time charge, total international minutes.

3 churn100-s4-c2

Genome `4206744f-b5ce-4488-bc53-a39aa08fe972`, explanation `33d6db5d-fbdf-41c8-a597-fc997715b92f`. Split: train primary (n=4000), test as a check (n=1000). Claims: 113, verified 110, not supported 3.
AI-assisted draft for review (unverified)

Reference (typical) customer: the typical customer: the median training customer for every number and the most common training category for every category (not a real customer). account length: 100; area code: area code 415; international plan: no international plan; number customer service calls: 1 calls; number vmail messages: 0 messages; state: West Virginia; total day calls: 100 calls; total day charge: 30.69 US dollars; total day minutes: 180 minutes; total eve calls: 100 calls; total eve charge: 17.12 US dollars; total eve minutes: 201 minutes; total intl calls: 4 calls; total intl charge: 2.78 US dollars; total intl minutes: 10 minutes; total night calls: 100 calls; total night charge: 9.02 US dollars; total night minutes: 200 minutes; voice mail plan: no voice-mail plan. Predicted chance that a customer leaves: 3%.

3.1 Summary

For a typical customer (in West Virginia, area code 415, no international or voice-mail plan, no voice-mail messages, 1 call to customer service and median usage) the model predicts a chance of leaving of about 3%; it adds up six parts: fixed-weight inputs, service calls, heavy daytime use, state units, an evening unit and a small voice-mail unit. [C046] Figure 122 shows the model at this level: one box per paragraph below.

How to read the diagrams. Edge colour is the sign of that one connection: blue positive, red negative; thicker is a larger weight. The outline and arrow on an input (and on a part's output box) is its overall effect on the final prediction, read from the verified claims: orange and up arrow raises it, teal and down arrow lowers it, grey and double arrow depends (it goes both ways, or differs between levels). A negative connection into a part that itself raises the prediction means that input lowers the prediction; through a hinge the overall effect can only be read from the claims, not from the edges. Up arrow means towards churn; down arrow, away from it. A double outline marks an input (or a part's output) that acts at a fixed rate everywhere: it adds the same amount to the score per unit, so it is explained by its rate alone (see 'Linear factors'); a dashed double outline, at a fixed rate on the observed data. The normal outline is an input whose effect switches on and off somewhere. For example, a voice-mail plan has a negative connection into 'Fixed-weight inputs', a part that raises the predicted chance that a customer leaves, so a voice-mail plan lowers it.

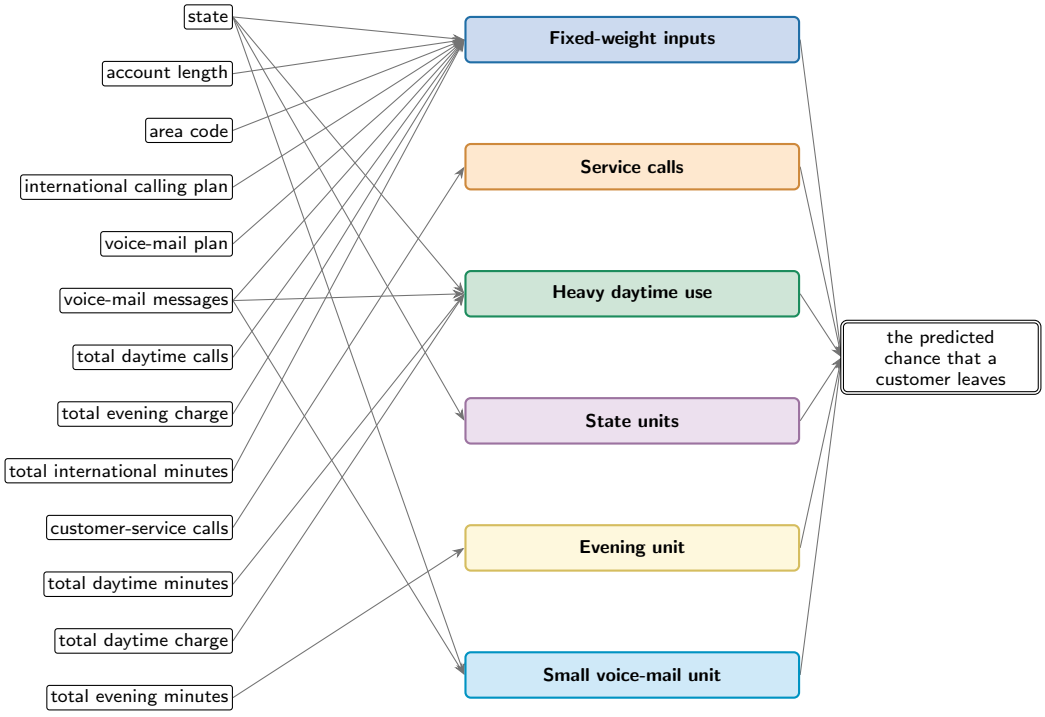


Figure 122: The model at the level of the summary: one box per part.

Fixed-weight inputs. This part adds up most inputs with fixed weights: an international plan raises a typical customer's chance of leaving from about 3% to about 31%, and a voice-mail plan lowers it to under 1%. [C047, C051] More voice-mail messages raise it steadily, from about 3% at 0 to about 10% at 32 messages, flattening out near the bottom because the predicted chance cannot go much lower. [C049] A higher evening charge and more

international minutes also raise it steadily, each from about 2% to about 4% across their usual range. [C055, C056] Through it, the state moves the chance from about 3% in West Virginia to about 11% in Washington and about 7% in New Jersey, and to under 1% in Hawaii and in Rhode Island (72 to 122 training customers per state). [C048, C050, C052, C053]

Service calls. This part adds up two parts: a rise from 3 customer-service calls upwards and a second rise from 4 calls upwards. [C079] From 3 calls upwards it adds to the chance slowly at first, then faster: for a typical such customer, 4 calls take it from about 3% to about 18%. [C078] Up to 3 calls it barely moves a typical customer's chance, which stays at about 3%, rising slowly at first, then faster. [C077]

Heavy daytime use. Daytime minutes add to the chance only once they rise above about 221 minutes (between 186 and 221 minutes for different customers), in an uneven way: for a typical such customer, from about 3% at 0 minutes to about 19% at 271 minutes. [C088] More voice-mail messages lower it in an uneven way: for the typical customer it acts on, from about 12% at 0 to about 6% at 32 messages. [C080] Customers in Washington get less from it: about 7% instead of about 12% for the typical customer it acts on. [C082]

State units. This part reads only the state: it raises a typical customer's chance from about 3% to about 22% in Rhode Island, and by less than a point in Indiana. [C100, C101]

Evening unit. This part reads evening minutes: nothing until about 217 minutes, then steadily, but a typical customer's chance stays at about 3%. [C103, C104]

Small voice-mail unit. This small part reads voice-mail messages and being in Louisiana, and barely changes the predictions (by at most one point). [C111]

The whole network. Figure 123 (next page, landscape) draws every unit and connection of the explained model, nested by part: shaded clusters are the parts of the model and darker boxes the parts inside them; circles are hidden units (h1, h2, ...); each cluster's rectangle is its output, and a dashed 'sum' stands for an addition the next unit does (drawn, not a unit of the model); the hexagon is the model's output. An input used by several parts is drawn under each of them (same label, dotted outline, the first copy marked with the number of copies): it is the same input, repeated for readability. Not used by the model (input columns with no connection): state: 29 of 51 levels, area code 408, area code 510, total evening calls, total night-time minutes, total night-time calls, total night-time charge, total international calls, total international charge.

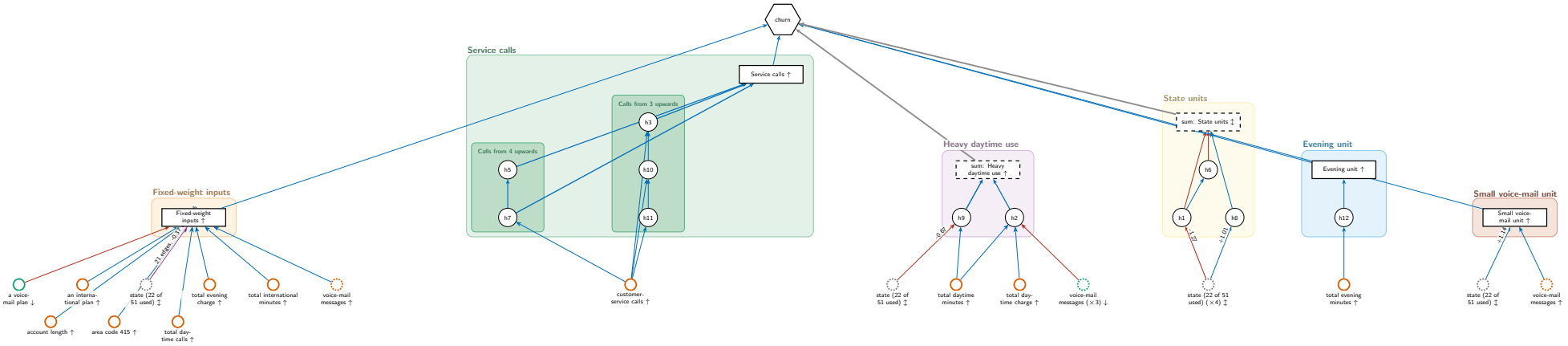


Figure 123: Every unit and connection of the explained model, nested by part. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Worth checking against what is known. Worth checking: for customers in Rhode Island the fixed-weight inputs lower the chance to under 1% while the state units raise it to about 22%, so overall it stays at about 3%. [C053, C100, C033] Worth checking: the state moves the chance by several points (Washington about 11%, New Jersey about 7%, Hawaii under 1%), based on 72 to 122 training customers per state. [C002, C005, C007] Worth checking: the model treats a voice-mail plan as lowering the chance of leaving but more voice-mail messages as raising it. [C006, C003] Worth checking: voice-mail messages raise the chance through the fixed-weight inputs but lower it through heavy daytime use. [C049, C080]

3.2 The model, read from the top down

Each part of the explanation in turn, from the whole model down to the smallest parts: what it does for the prediction (hypothesis), what was checked (evidence), a diagram one level down, and where to read on. The technical record of each part is in "Technical detail" at the end.

3.2.1 The whole model

Figure 124 shows what feeds this part; Figure 125 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

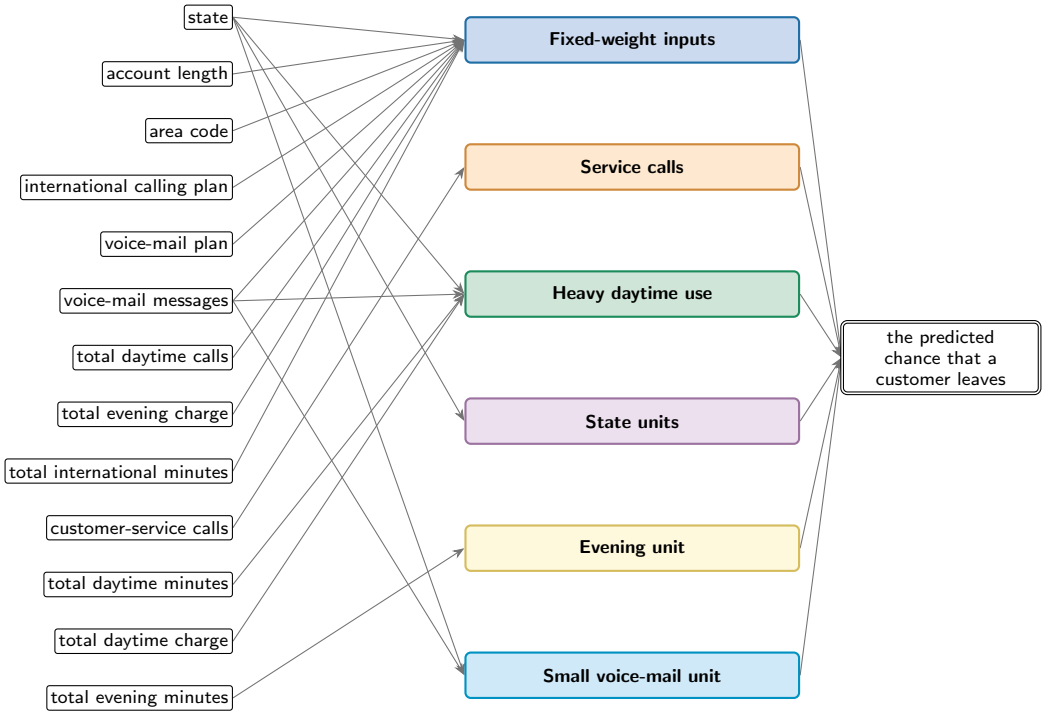


Figure 124: The whole model: what feeds it, one level down.

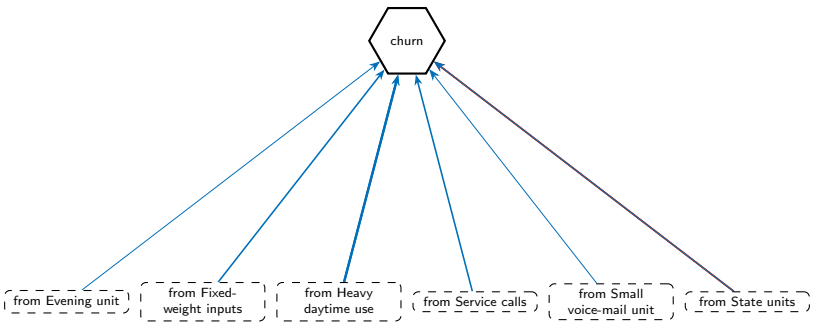


Figure 125: The whole model: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. The model adds up six parts into one score: fixed-weight inputs, service calls, heavy daytime use, state units, an evening unit and a small voice-mail unit. [C046]

Evidence. It is made up of these six parts (each explained below); the score is their sum, which was checked on all 4000 training customers. [C046] Changing one input at a time across the whole model, an international plan takes a typical customer's chance of leaving from about 3% to about 31%, and a voice-mail plan lowers it to under 1%. [C001, C006] More daytime minutes raise it in an uneven way, from about 3% at 112 minutes to about 8% at 249 minutes, flattening out near the bottom because the predicted chance cannot go much lower. [C004] In Washington the chance is about 11%, against about 3% in West Virginia. [C002] The model ignores evening calls, international calls, the international charge, night-time calls, the night-time charge and night-time minutes. [C040, C041, C042, C043, C044, C045]
Evidence plots: Figure 126, Figure 127.

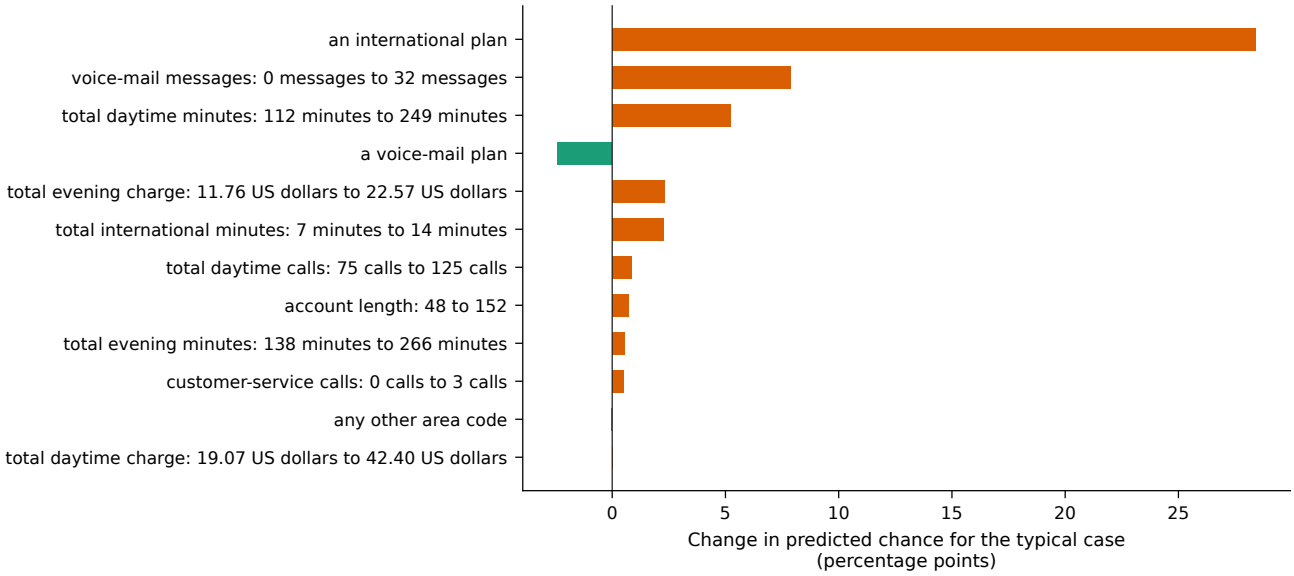


Figure 126: The whole model: Each input's change in the predicted value for the typical case across the whole model, largest first (orange: raises it, teal: lowers it).

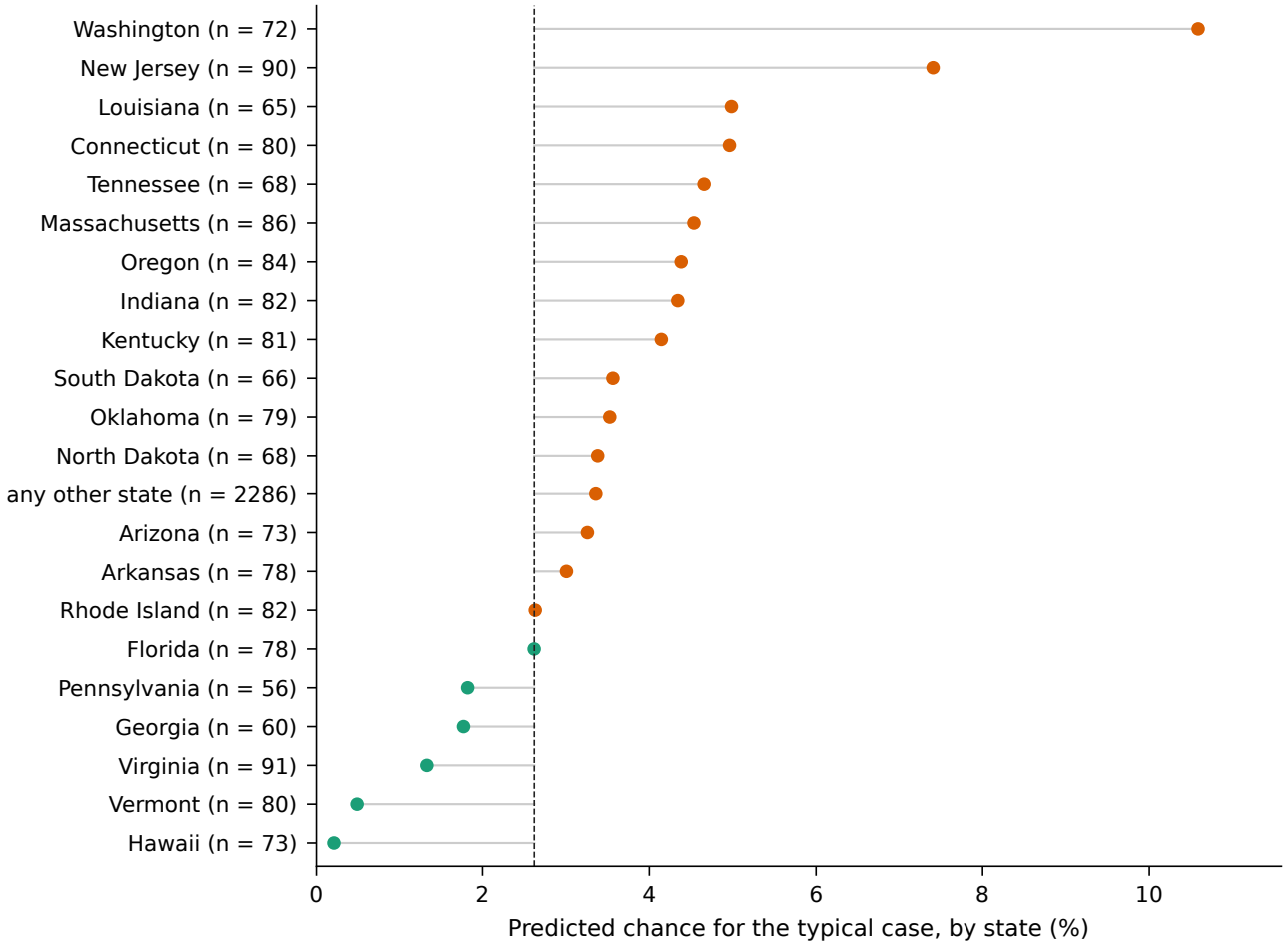


Figure 127: The whole model: Every level of state the model reads: the predicted value for the typical case at that level (dashed: the reference level); n = training customers; grey: few customers or not supported.

Dive deeper: Fixed-weight inputs (5.2.3); Service calls (4.2.7); Heavy daytime use (5.2.7); State units (4.2.6); Evening unit (5.2.2); Small voice-mail unit (5.2.1). Technical detail: T.3.

3.2.2 Fixed-weight inputs

Part of: The whole model (4.2.3).
Figure 128 shows what feeds this part; Figure 129 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

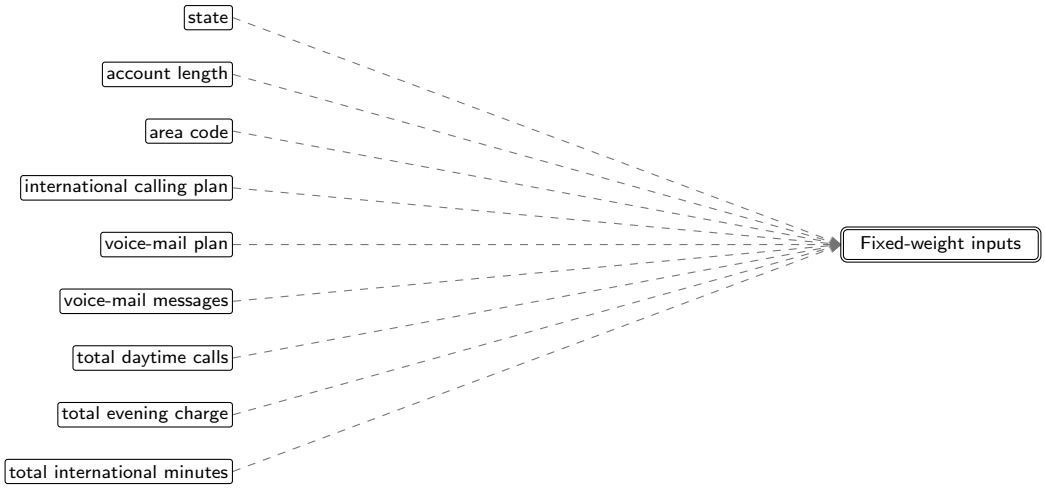


Figure 128: Fixed-weight inputs: what feeds it, one level down.

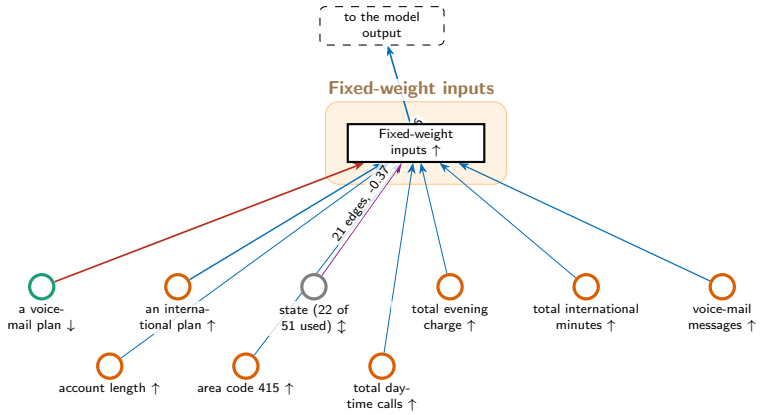


Figure 129: Fixed-weight inputs: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part adds up most inputs with fixed weights: an international plan, voice-mail messages, the evening charge and international minutes raise the chance of leaving, steadily for the numbers, and a voice-mail plan lowers it. [C047, C049, C055, C056, C051]

Evidence. Through it, an international plan takes a typical customer from about 3% to about 31%, and a voice-mail plan lowers the chance to under 1%. [C047, C051] More voice-mail messages raise it steadily, from about 3% at 0 to about 10% at 32 messages, flattening out near the bottom because the predicted chance cannot go much lower. [C049] Through it, the state moves the chance from about 3% in West Virginia to about 11% in Washington and about 7% in New Jersey, and to under 1% in Hawaii and in Rhode Island (72 to 122 training customers per state). [C048, C050, C052, C053] Its formula reproduces the part exactly on all 4000 training customers. [C076]
Evidence plots: Figure 130, Figure 131, Figure 132, Figure 133.

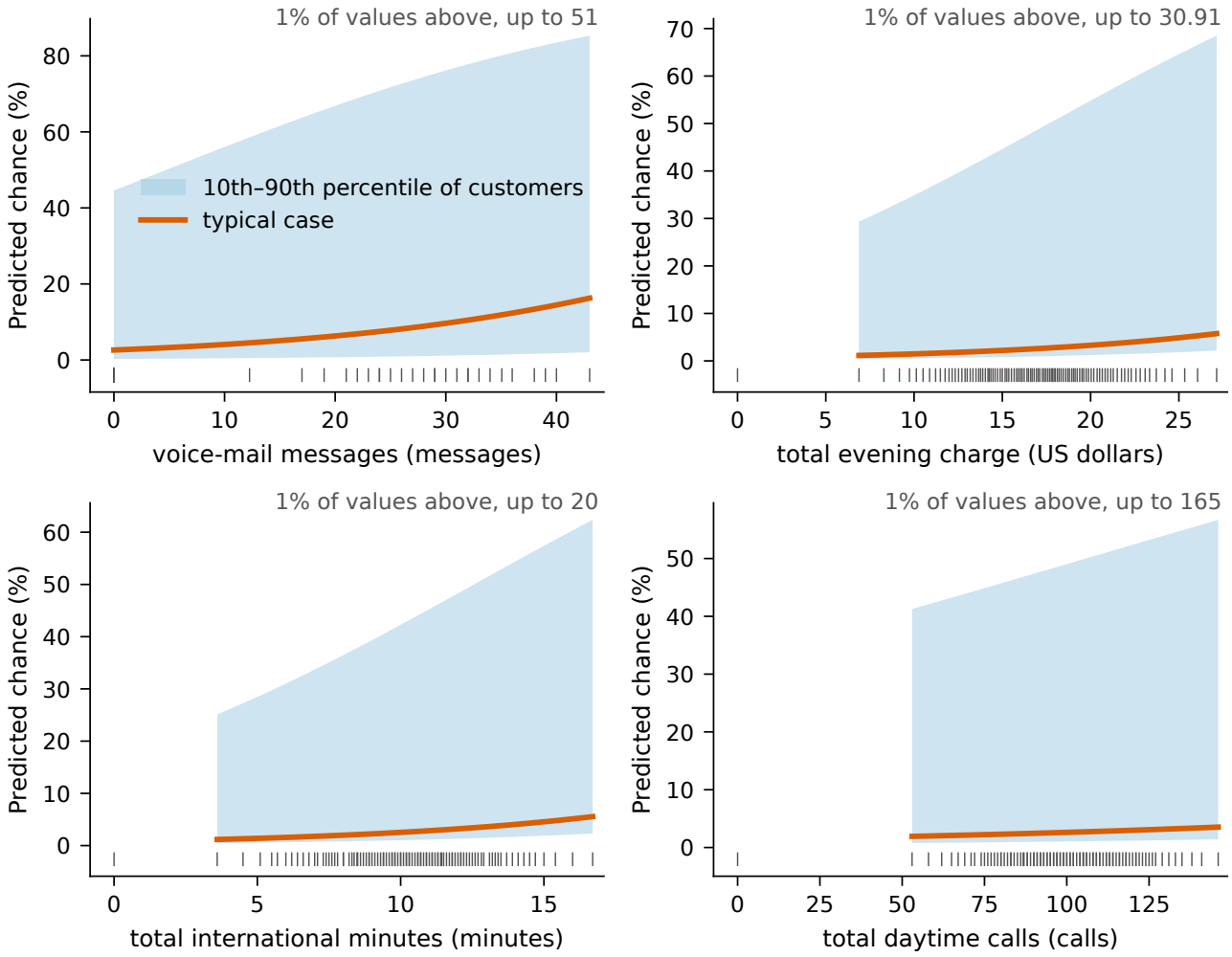


Figure 130: Fixed-weight inputs: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

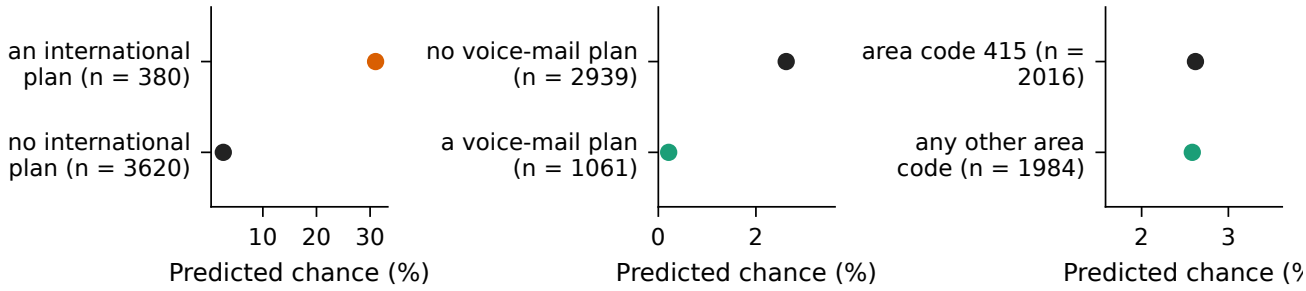


Figure 131: Fixed-weight inputs: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

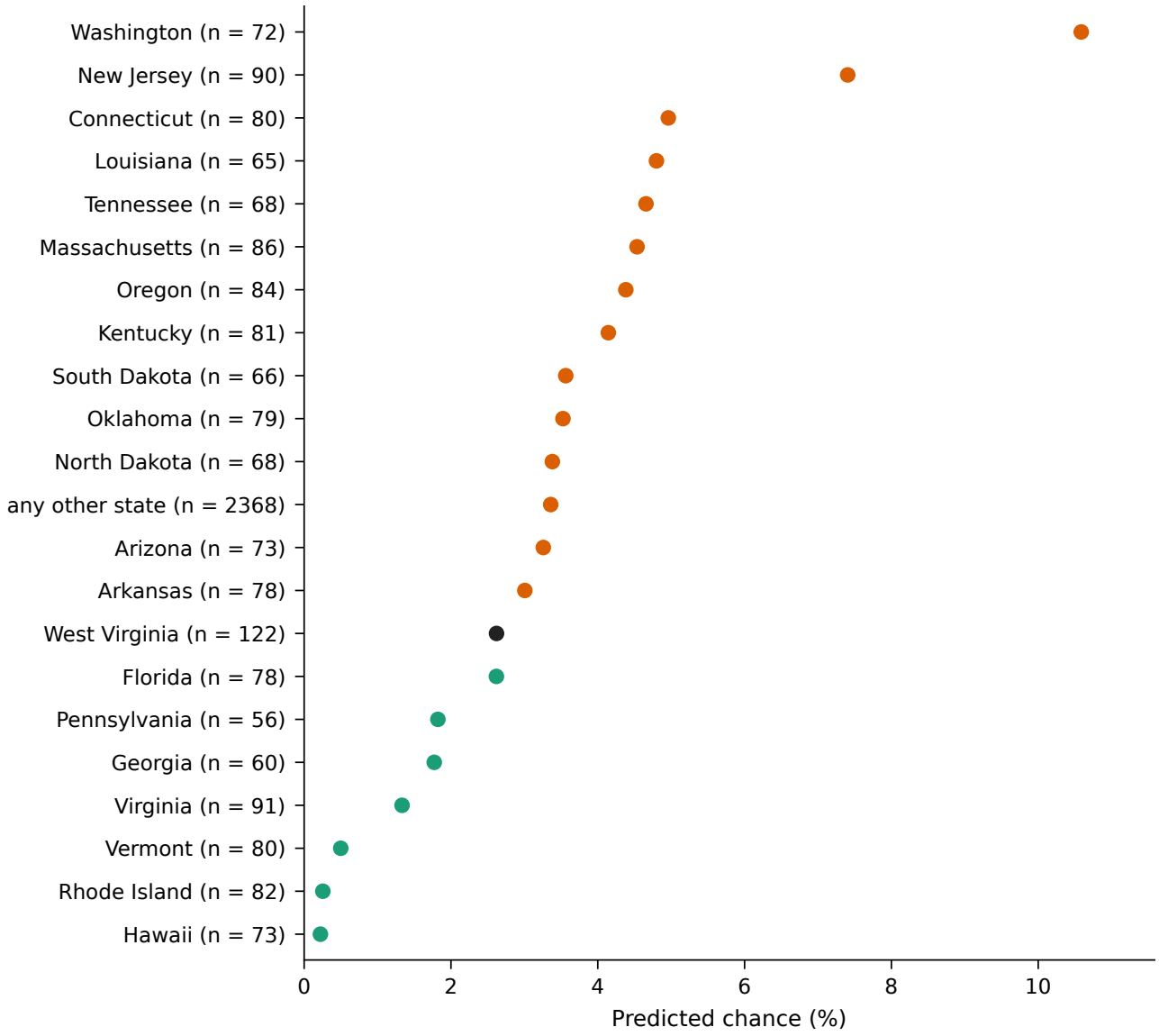


Figure 132: Fixed-weight inputs: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

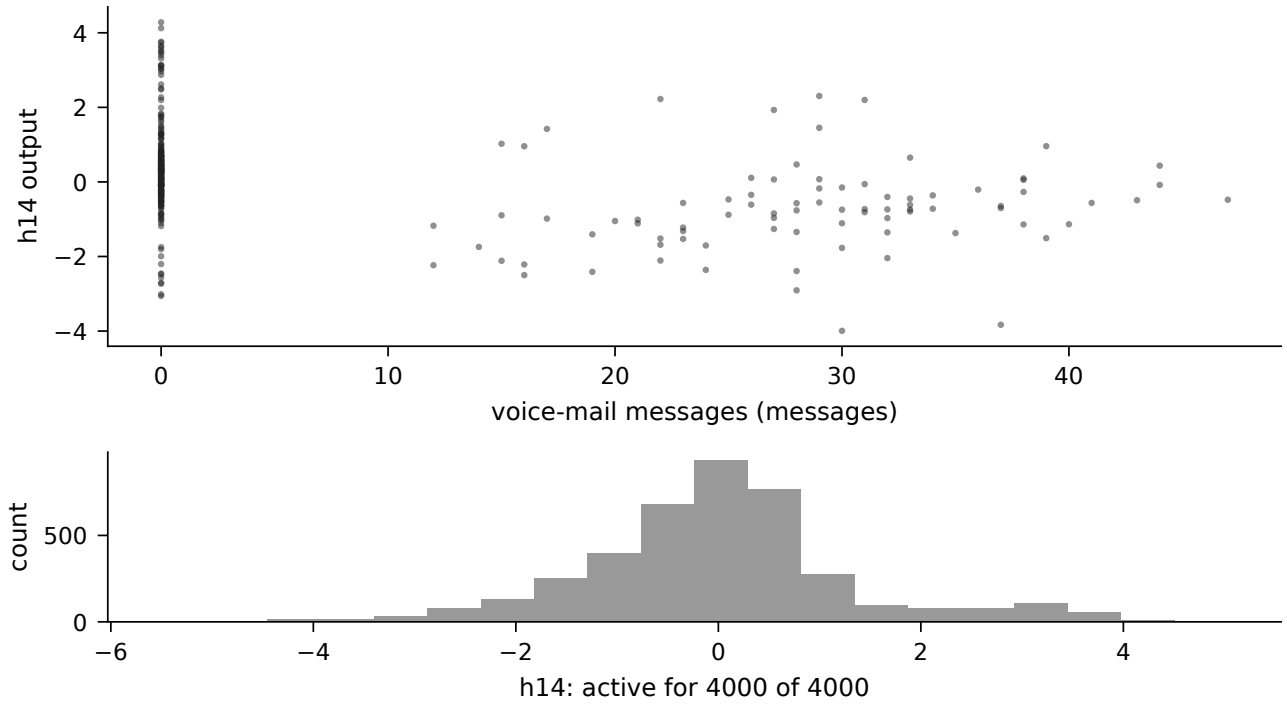


Figure 133: Fixed-weight inputs: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.3.

3.2.3 Service calls

Part of: The whole model (4.2.3).
Figure 134 shows what feeds this part; Figure 135 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).



Figure 134: Service calls: what feeds it, one level down.

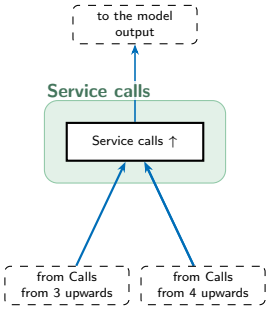


Figure 135: Service calls: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part raises the chance of leaving with calls to customer service from 3 calls upwards, slowly at first, then faster; it adds up two parts. [C079, C078, C077]

Evidence. It is made of two parts that are added: calls from 3 upwards and calls from 4 upwards. [C079] From 3 calls upwards it adds to the chance slowly at first, then faster: for a typical such customer, 4 calls take it from

about 3% to about 18%. [C078] Up to 3 calls it barely moves a typical customer's chance, which stays at about 3%, rising slowly at first, then faster. [C077]
Evidence plots: Figure 136, Figure 137, Figure 138.
Figure 137 shows this part's own output, unit h15: it switches on at about 2.899 calls of customer-service calls, and is zero below.

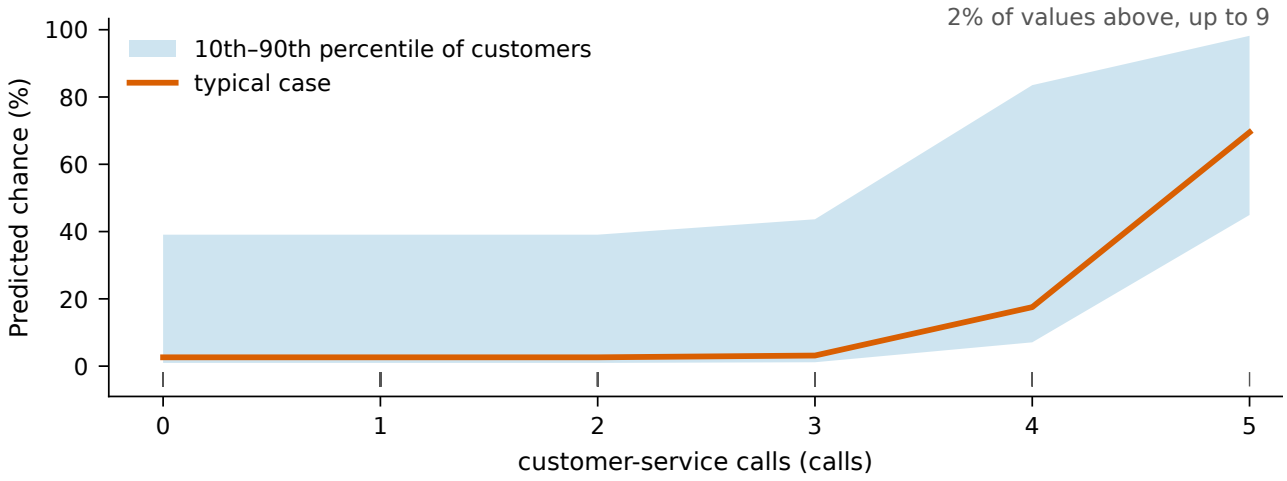


Figure 136: Service calls: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

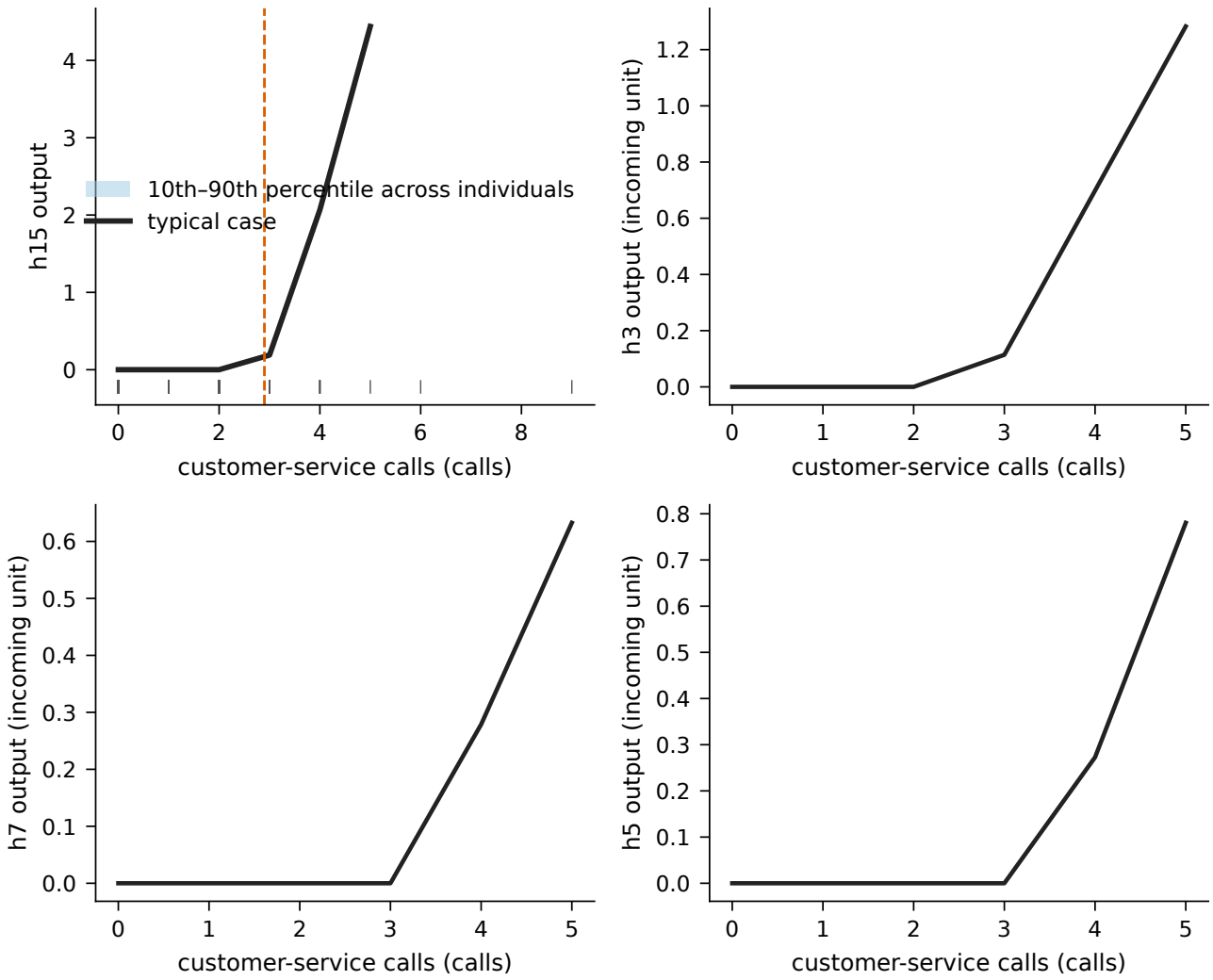


Figure 137: Service calls: This part's own output, unit h15 (combines 3 non-linear units), on its own scale (its activation), with every other input at the typical case; it is active for 848 of 4000. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

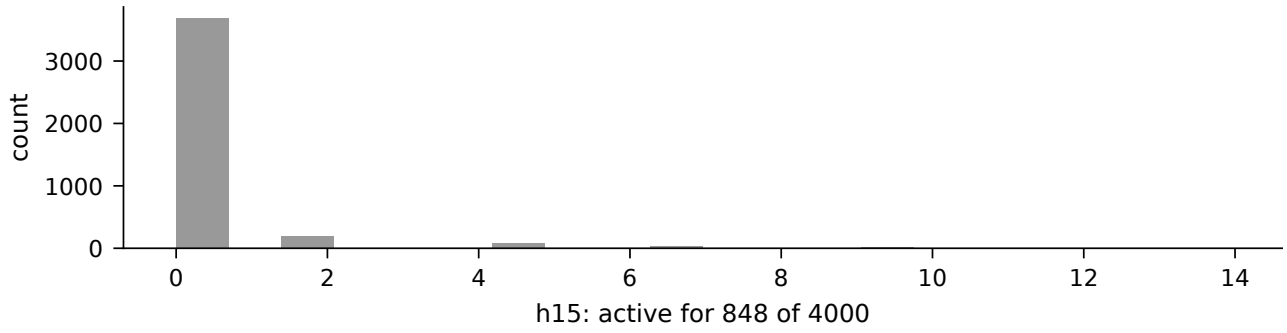


Figure 138: Service calls: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Dive deeper: Calls from 4 upwards (5.2.4); Calls from 3 upwards (5.2.6). Technical detail: T.7.

3.2.4 Calls from 4 upwards

Part of: Service calls (4.2.7).
Figure 139 shows what feeds this part; Figure 140 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).



Figure 139: Calls from 4 upwards: what feeds it, one level down.

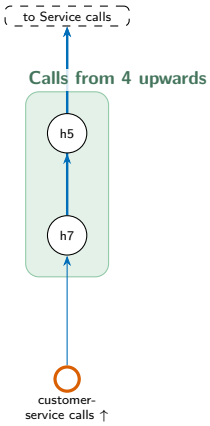


Figure 140: Calls from 4 upwards: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part raises the chance of leaving from 4 customer-service calls upwards, slowly at first, then faster. [C093]

Evidence. It acts on 318 of 4000 training customers, 159 of whom left; for the typical customer it acts on, the chance is about 8% without it and about 18% with it. [C094] Through it, the chance rises slowly at first, then faster: 4 calls take a typical such customer from about 3% to about 6%. [C093] Its formula reproduces the part exactly on all 4000 training customers. [C095]
Evidence plots: Figure 141, Figure 142, Figure 143.
Figure 141 shows this part's own output, unit h7: it switches on at about 3.212 calls of customer-service calls, and is zero below.
Figure 142 shows this part's own output, unit h5: it switches on at about 3.463 calls of customer-service calls, and is zero below.

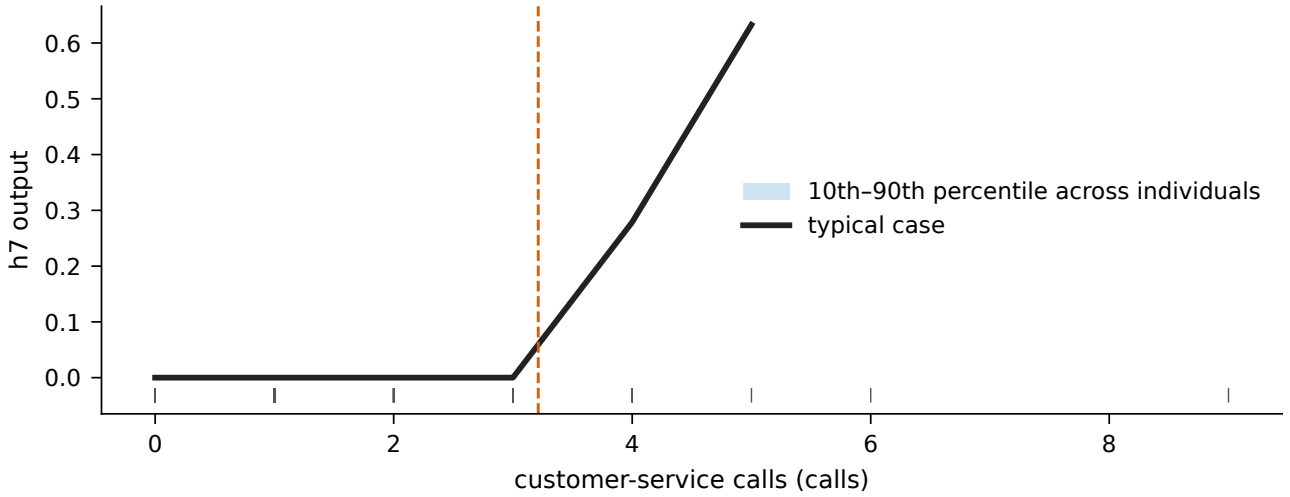


Figure 141: Calls from 4 upwards: This part's own output, unit h7 (the output of 'Calls from 4 upwards'), on its own scale (its activation), with every other input at the typical case; it is active for 318 of 4000. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

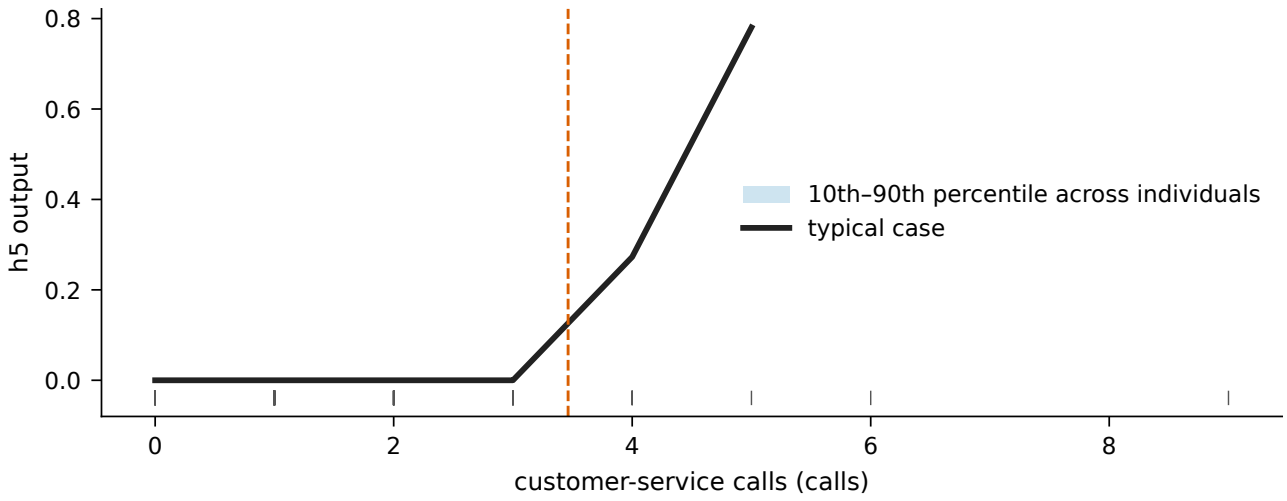


Figure 142: Calls from 4 upwards: This part's own output, unit h5 (the output of 'Calls from 4 upwards'), on its own scale (its activation), with every other input at the typical case; it is active for 318 of 4000. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

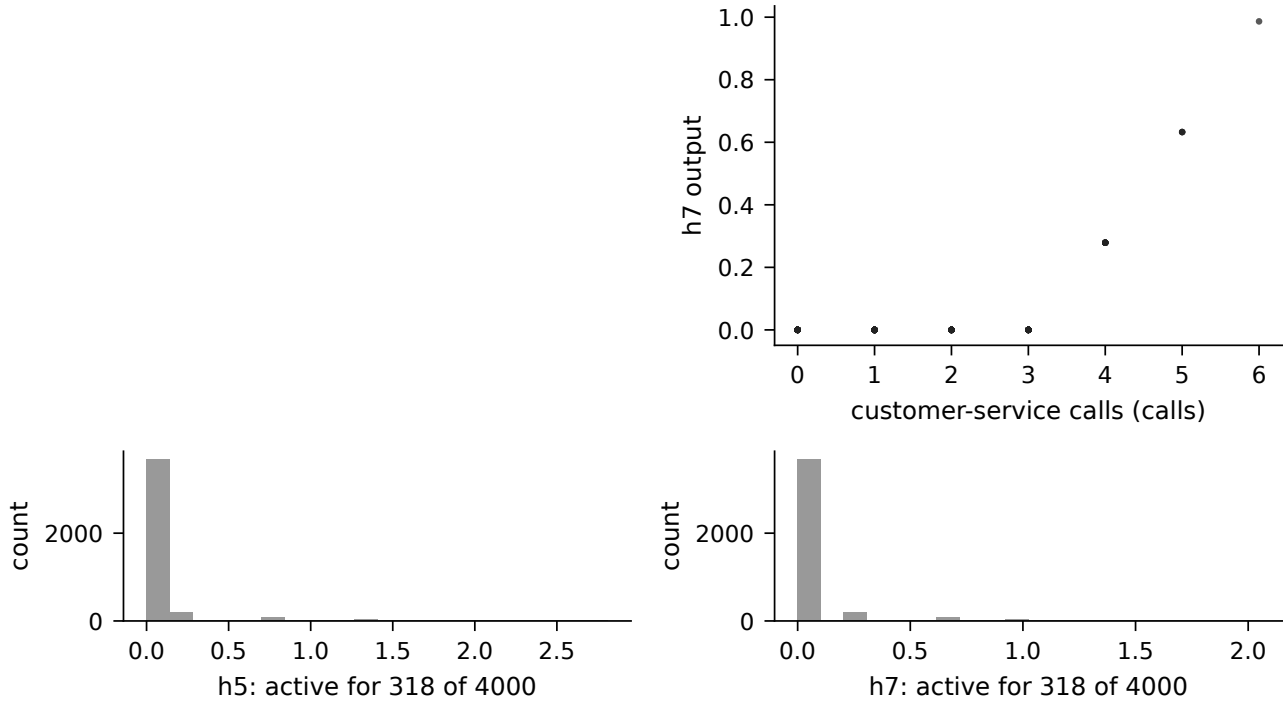


Figure 143: Calls from 4 upwards: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.4.

3.2.5 Calls from 3 upwards

Part of: Service calls (4.2.7).
Figure 144 shows what feeds this part; Figure 145 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

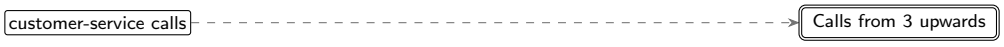


Figure 144: Calls from 3 upwards: what feeds it, one level down.

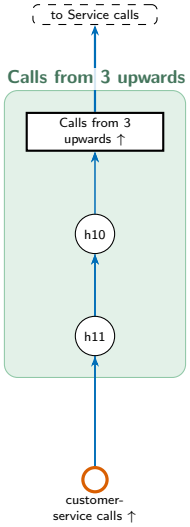


Figure 145: Calls from 3 upwards: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part raises the chance of leaving from 3 customer-service calls upwards, slowly at first, then faster. [C097, C096]

Evidence. It acts on 848 of 4000 training customers, 215 of whom left; for the typical customer it acts on, the chance is about 3% without it and about 4% with it. [C098] Through it, the chance rises slowly at first, then faster: 4 calls take a typical such customer from about 3% to about 8%. [C097] Its formula reproduces the part exactly on all 4000 training customers. [C099]
Evidence plots: Figure 146, Figure 147, Figure 148.
Figure 147 shows this part's own output, unit h3: it switches on at about 2.805 calls of customer-service calls, and is zero below.

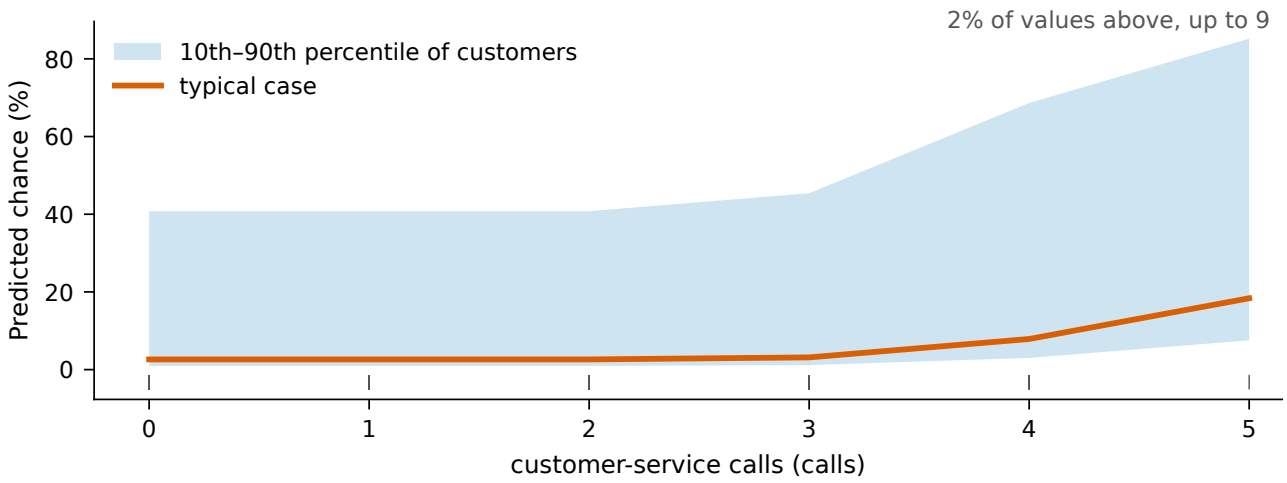


Figure 146: Calls from 3 upwards: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

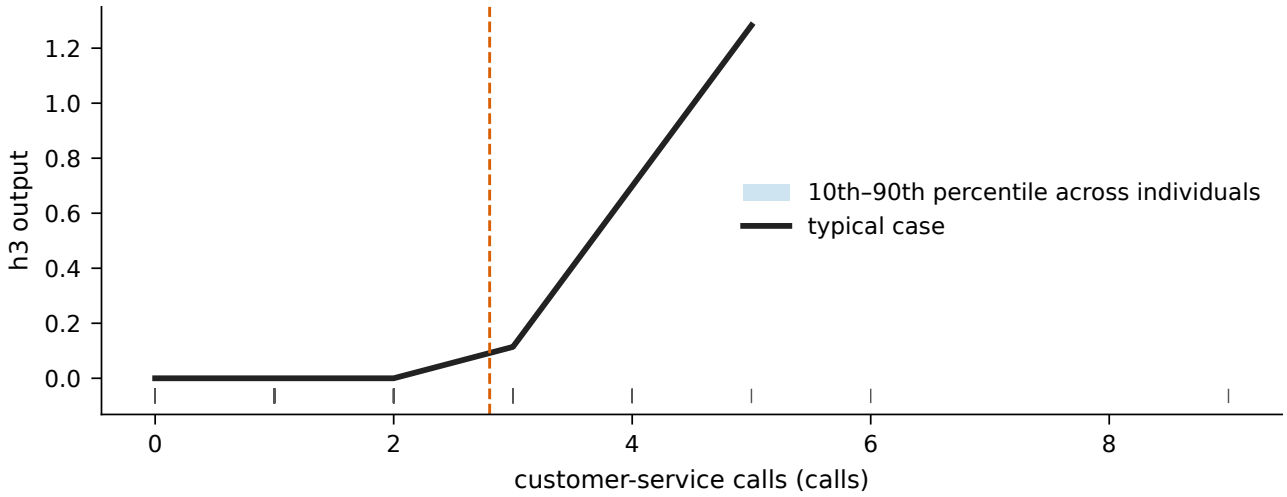


Figure 147: Calls from 3 upwards: This part's own output, unit h3 (the output of 'Calls from 3 upwards'), on its own scale (its activation), with every other input at the typical case; it is active for 848 of 4000. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

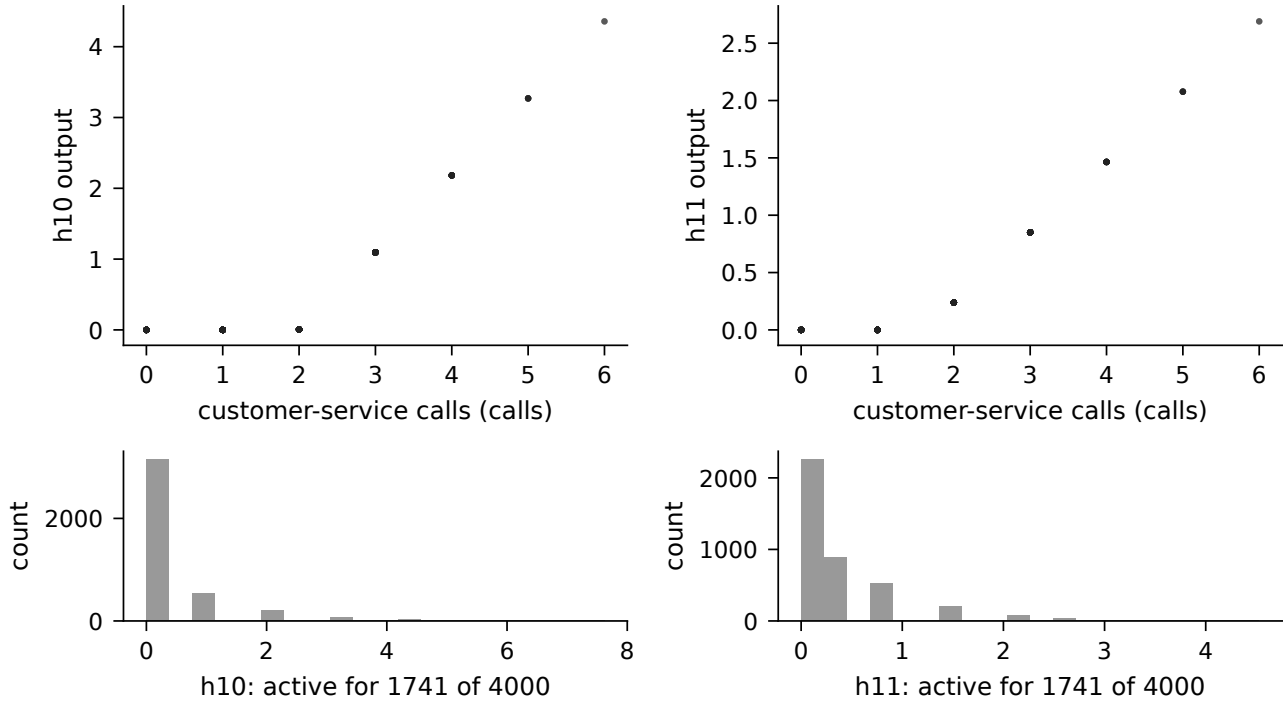


Figure 148: Calls from 3 upwards: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.6.

3.2.6 Heavy daytime use

Part of: The whole model (4.2.3).
Figure 149 shows what feeds this part; Figure 150 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

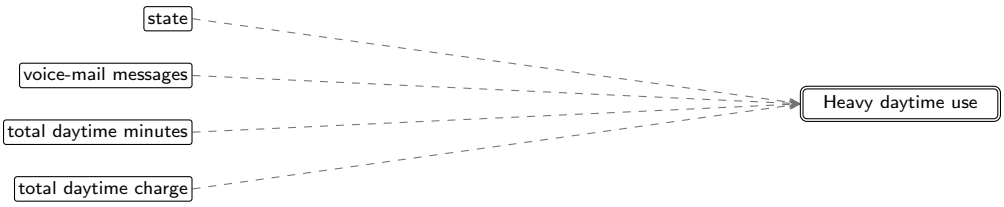


Figure 149: Heavy daytime use: what feeds it, one level down.

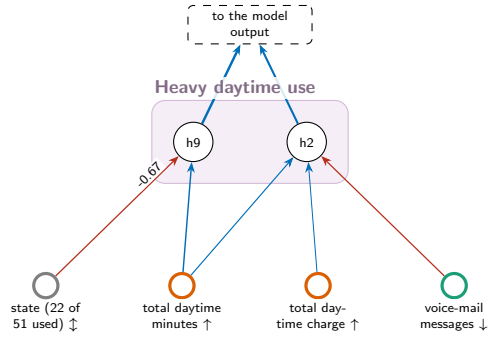


Figure 150: Heavy daytime use: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part raises the chance of leaving for heavy daytime users: daytime minutes add to it only above about 221 minutes (between 186 and 221 minutes for different customers), in an uneven way, and voice-mail messages and being in Washington lower it. [C088, C080, C082]

Evidence. This part acts on 1099 of 4000 training customers, 291 of whom left; for the typical customer it acts on, the chance is about 3% without it and about 12% with it. [C091] Daytime minutes add to the chance only once they rise above about 221 minutes (between 186 and 221 minutes for different customers), in an uneven way: for a typical such customer, from about 3% at 0 minutes to about 19% at 271 minutes. [C088] More voice-mail messages lower it in an uneven way: for the typical customer it acts on, from about 12% at 0 to about 6% at 32 messages. [C080] Customers in Washington get less from it: about 7% instead of about 12% for the typical customer it acts on. [C082] Its formula reproduces the part exactly on all 4000 training customers. [C092]
Evidence plots: Figure 151, Figure 152, Figure 153, Figure 154, Figure 155, Figure 156, Figure 157, Figure 158, Figure 159.
Figure 157 shows this part's own output, unit h2: over its two main inputs, with the line where it switches on.
Figure 158 shows this part's own output, unit h9: it switches on at about 221.1 minutes of total daytime minutes, and is zero below.

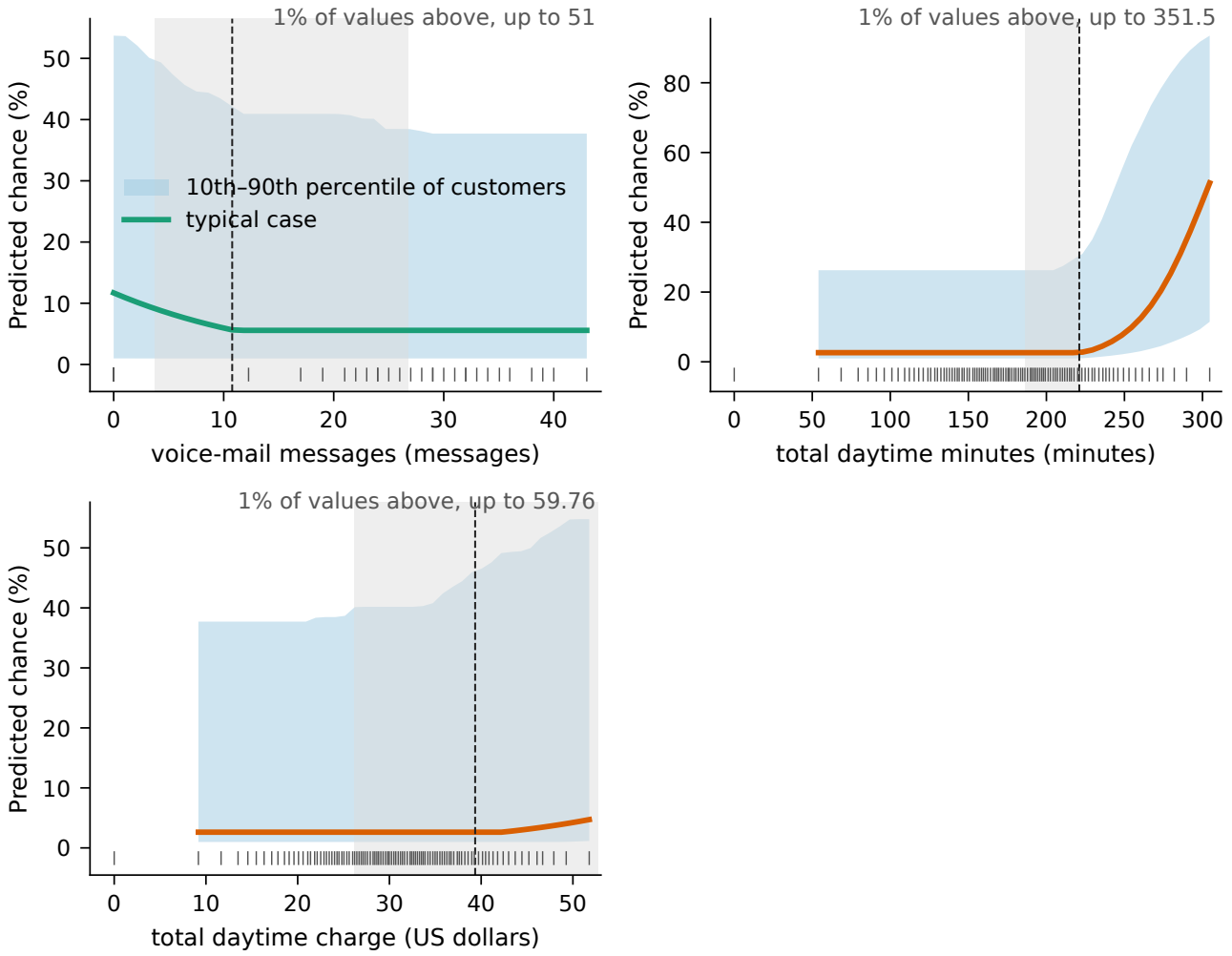


Figure 151: Heavy daytime use: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

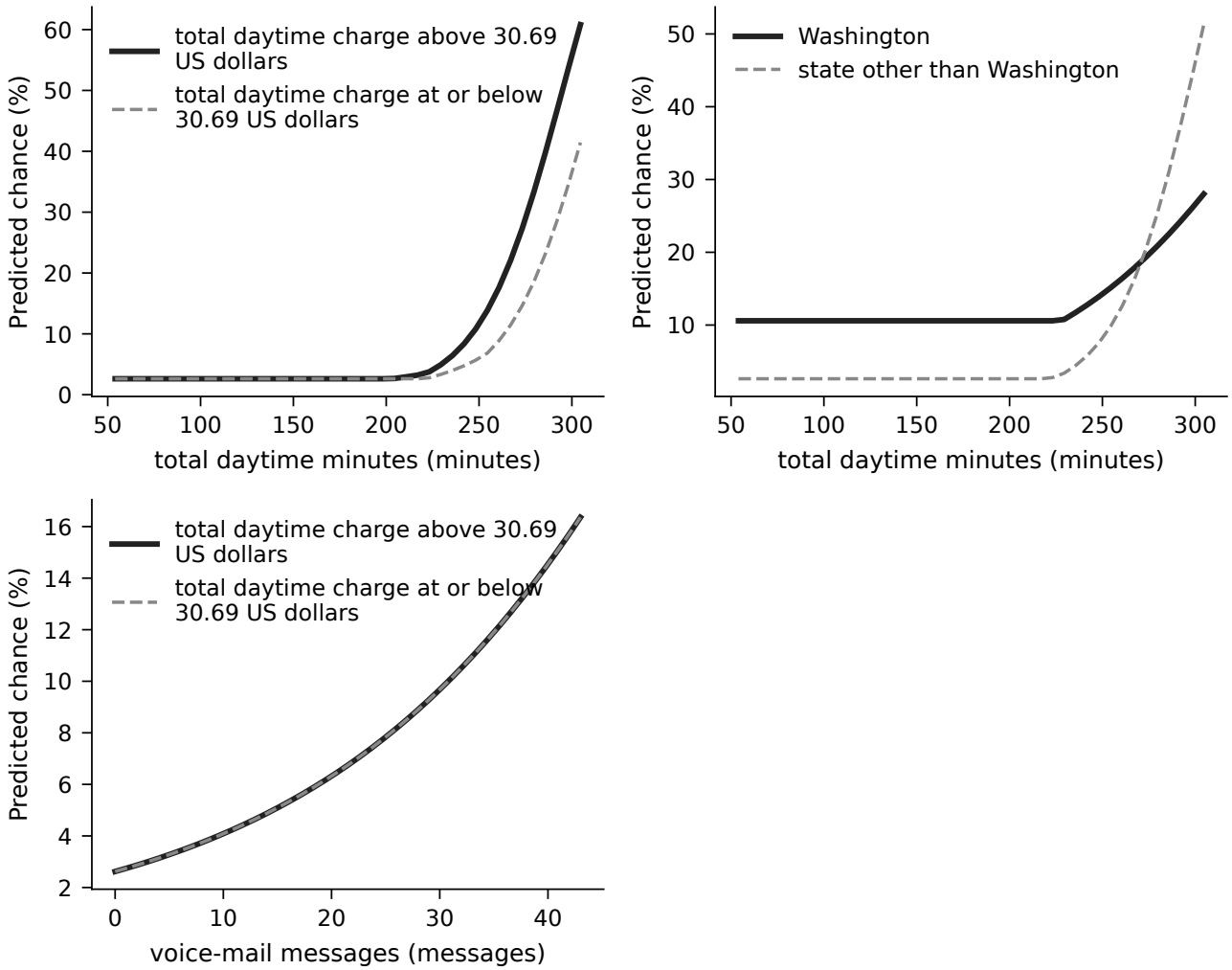


Figure 152: Heavy daytime use: How the effect differs between groups: one curve per group, for the typical case of each group.

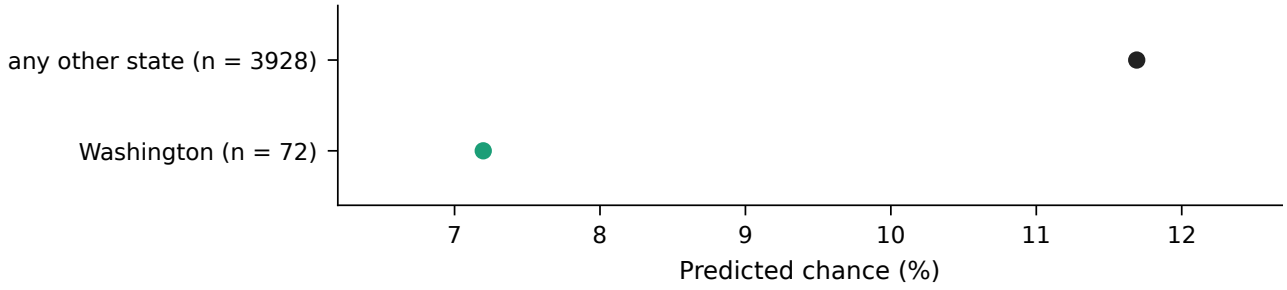


Figure 153: Heavy daytime use: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

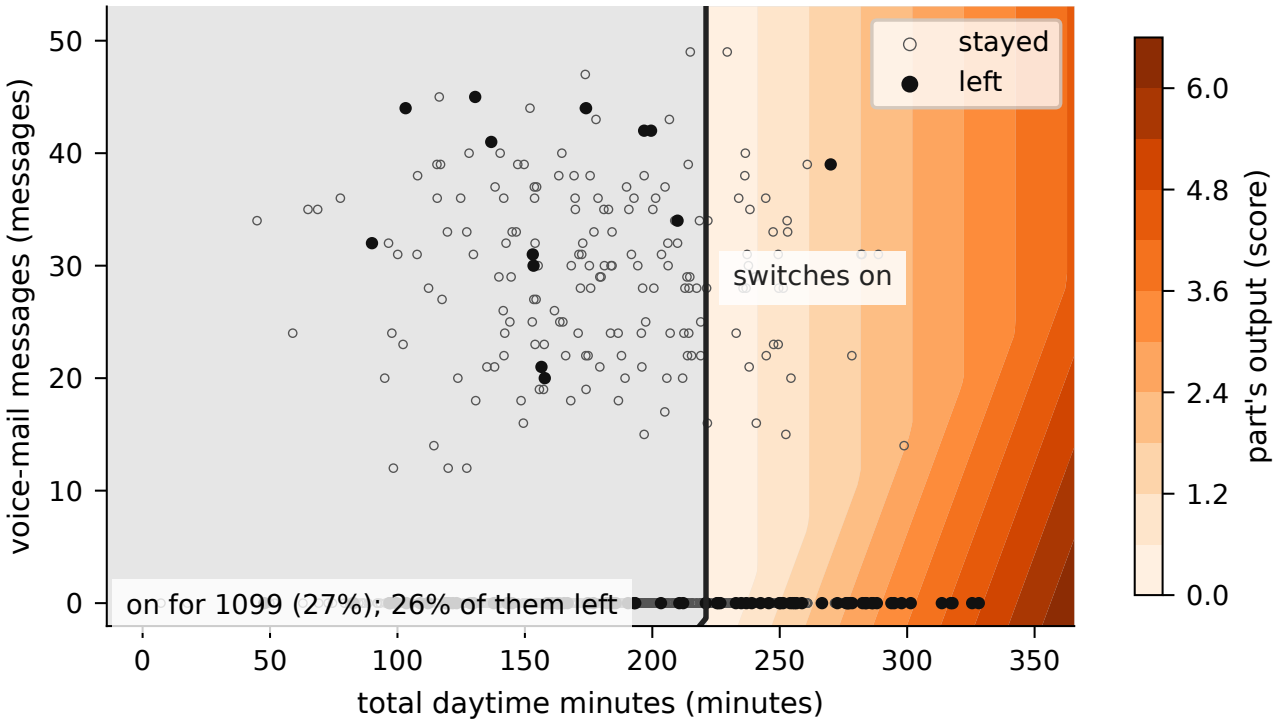


Figure 154: Heavy daytime use: The part's own output over its two numeric inputs, other inputs at the typical case: colour where it is on, flat grey where it is switched off, the black line where it switches on; training customers as hollow (no event) and filled (event) points, so the part can be seen catching the events (the predicted-chance overlay was harder to read). The switch-on line is traced on this map only; the boundary claim (C113) did not pass its check on the data, so no rule is stated.

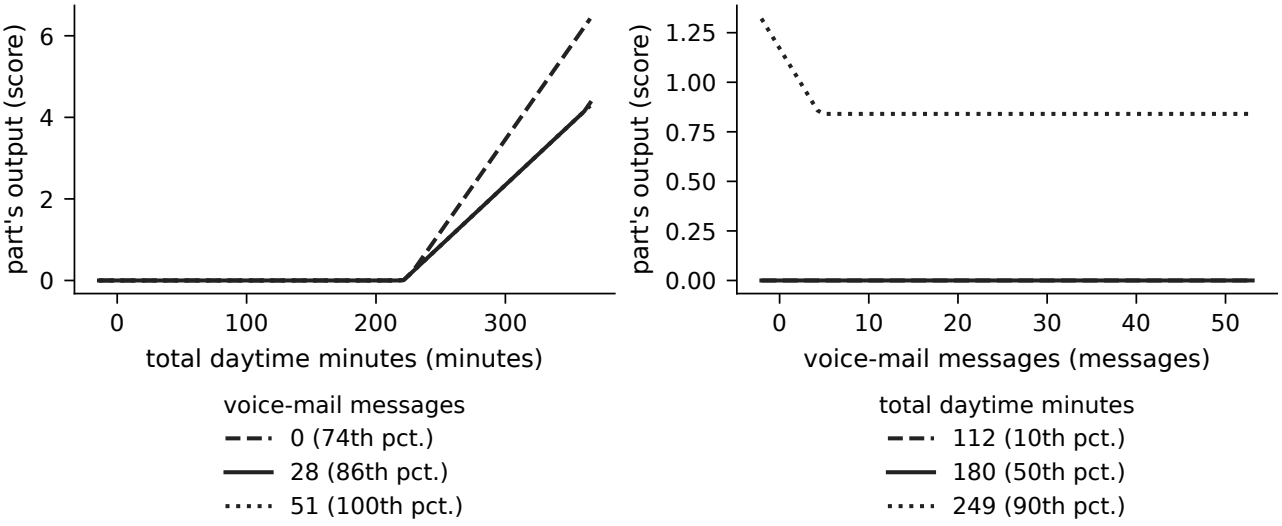


Figure 155: Heavy daytime use: Slices of the same map: the part's own output against each input at three values of the other, so the tilted boundary shows as a shifting hinge. The switch-on line is traced on this map only; the boundary claim (C113) did not pass its check on the data, so no rule is stated.

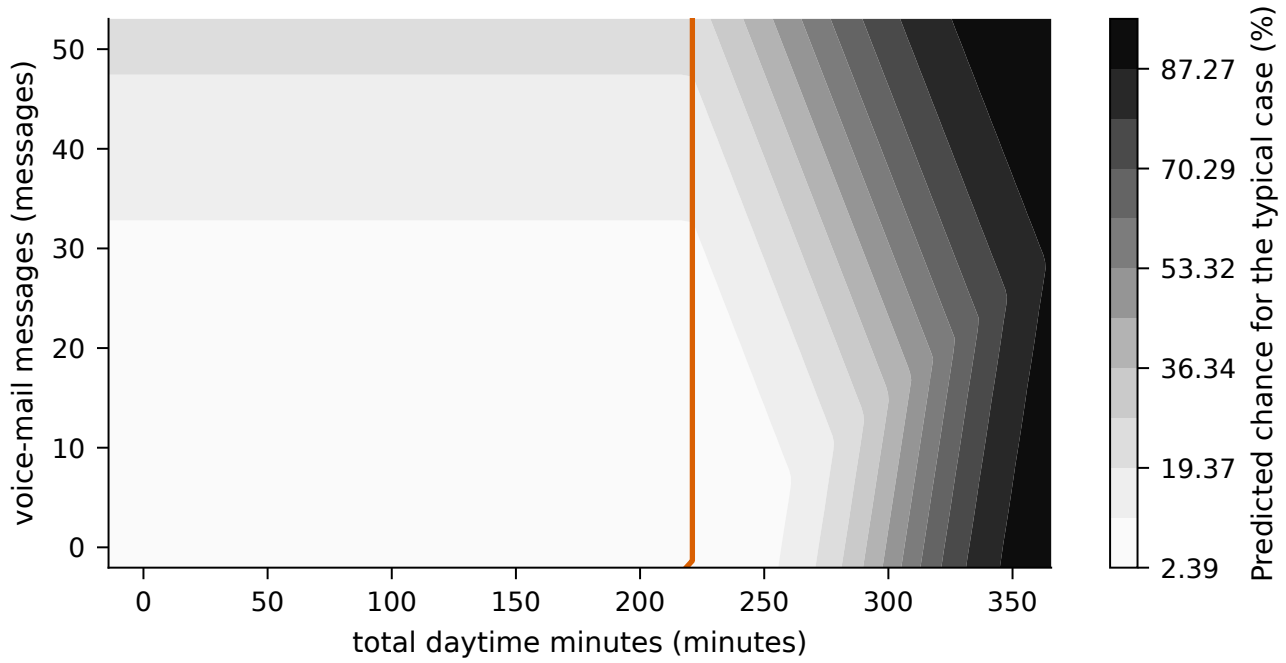


Figure 156: Heavy daytime use: The same map on the final scale: the predicted value for the typical case, with the switch-on line (orange). The switch-on line is traced on this map only; the boundary claim (C113) did not pass its check on the data, so no rule is stated.

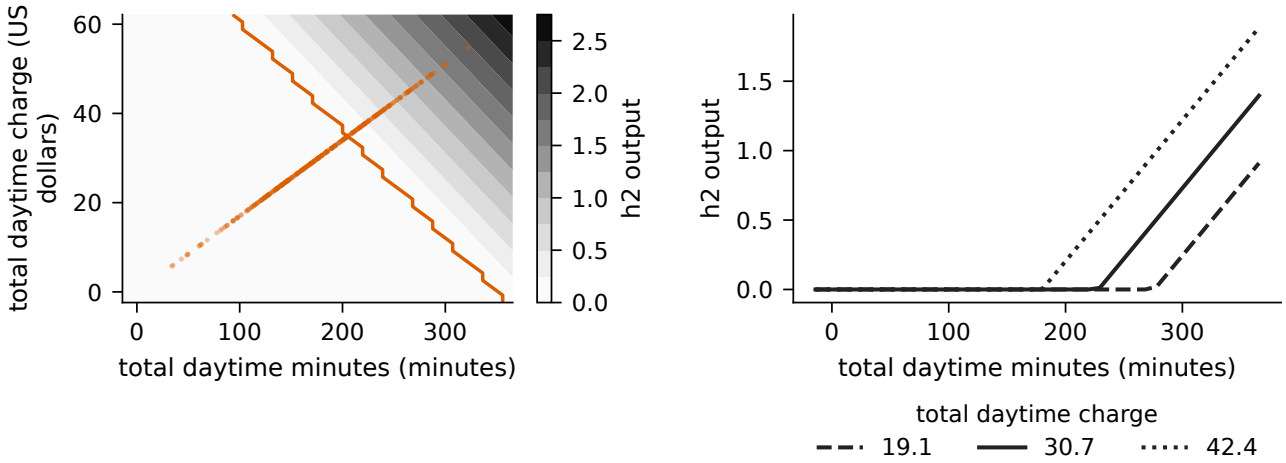


Figure 157: Heavy daytime use: This part's own output, unit h2 (the output of 'Heavy daytime use'), on its own scale (its activation), with every other input at the typical case; it is active for 896 of 4000. Left: its output over its two main inputs (orange line: where it switches on; dots: the data); right: slices at three values of the second input.

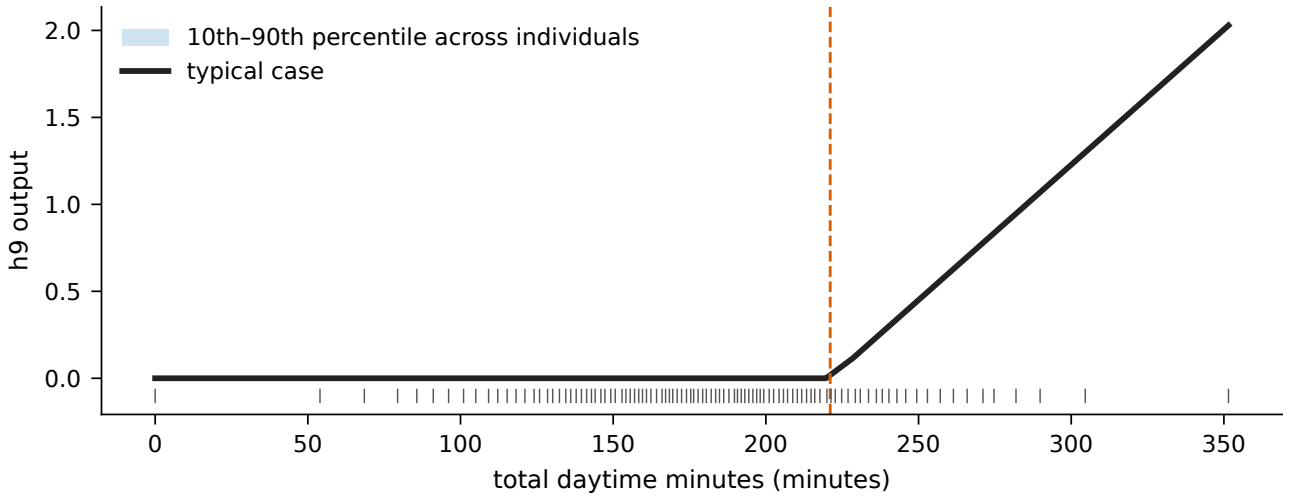


Figure 158: Heavy daytime use: This part's own output, unit h9 (the output of 'Heavy daytime use'), on its own scale (its activation), with every other input at the typical case; it is active for 870 of 4000. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

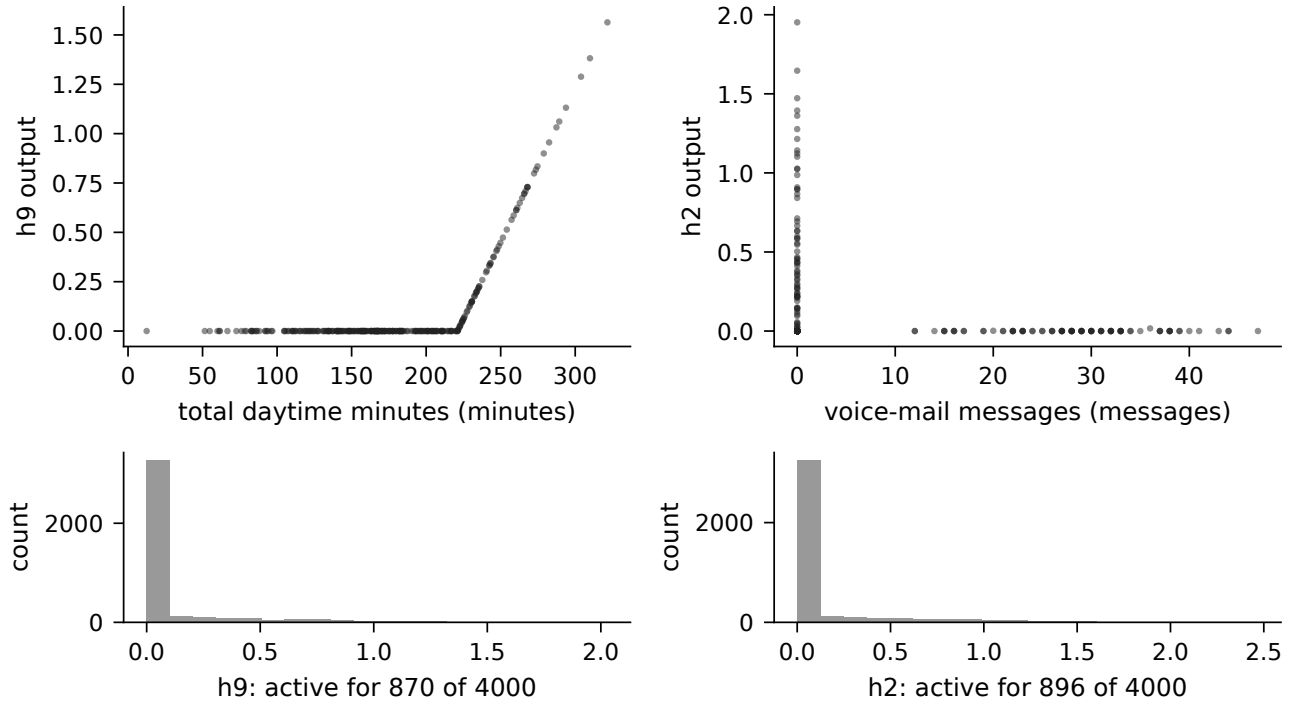


Figure 159: Heavy daytime use: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.7.

3.2.7 State units

Part of: The whole model (4.2.3).
Figure 160 shows what feeds this part; Figure 161 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).



Figure 160: State units: what feeds it, one level down.

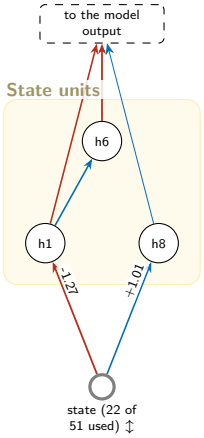


Figure 161: State units: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part reads only the state and raises the chance of leaving in Rhode Island, and a little in Indiana. [C100, C101]

Evidence. Through it, a typical customer's chance goes from about 3% to about 22% in Rhode Island, and up by less than a point in Indiana. [C100, C101] Its formula reproduces the part exactly on all 4000 training customers. [C102] Evidence plots: Figure 162, Figure 163.

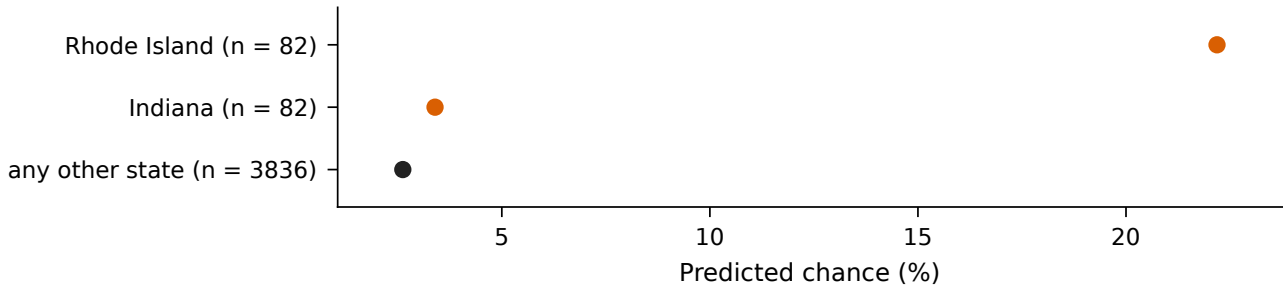


Figure 162: State units: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

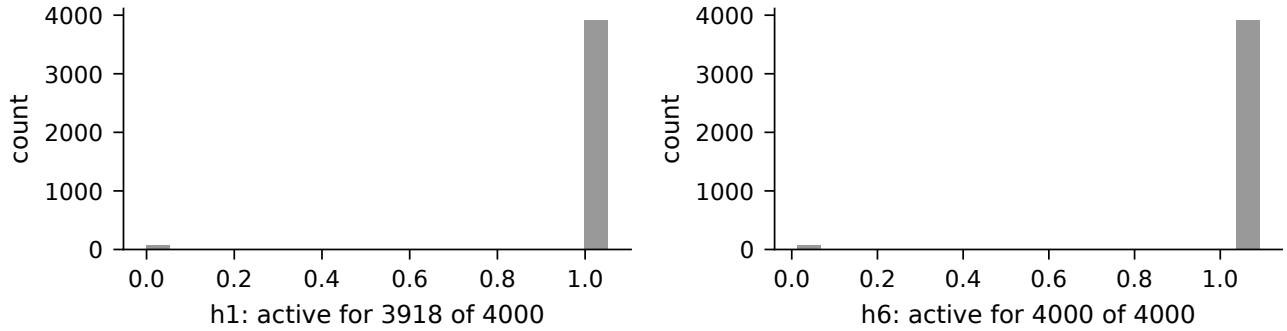


Figure 163: State units: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.6.

3.2.8 Evening unit

Part of: The whole model (4.2.3).
Figure 164 shows what feeds this part; Figure 165 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).



Figure 164: Evening unit: what feeds it, one level down.

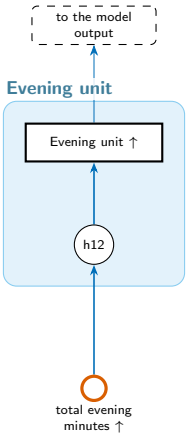


Figure 165: Evening unit: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part reads evening minutes and barely changes the chance of leaving: nothing until about 217 minutes, then steadily. [C103, C104]

Evidence. Through it, evening minutes change nothing until about 217 minutes, then raise the chance steadily, but a typical customer's chance stays at about 3% from 138 to 266 minutes. [C103] Its formula reproduces the part exactly on all 4000 training customers. [C105]
Evidence plots: Figure 166, Figure 167.

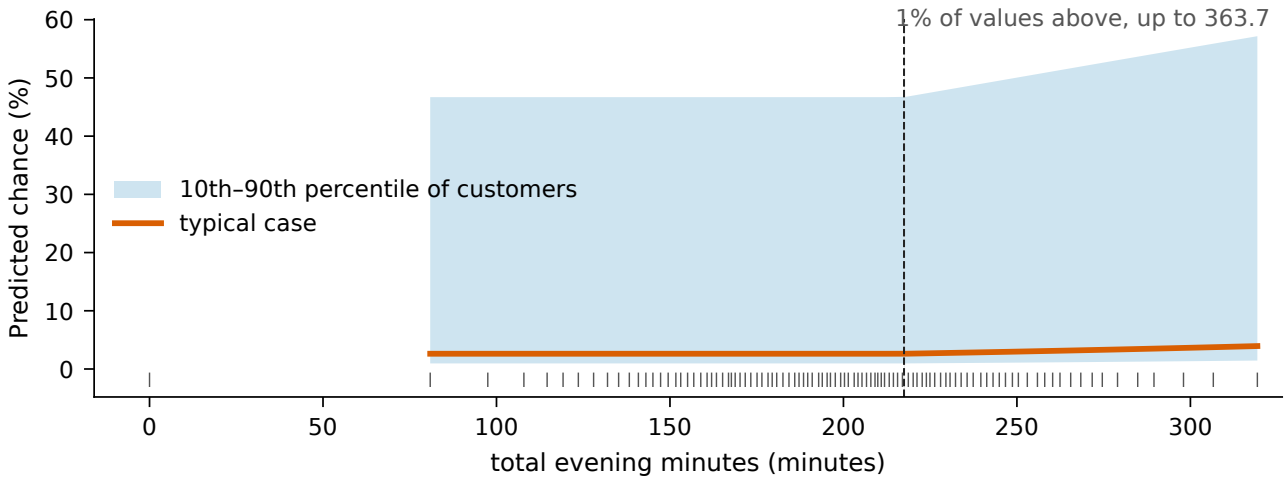


Figure 166: Evening unit: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

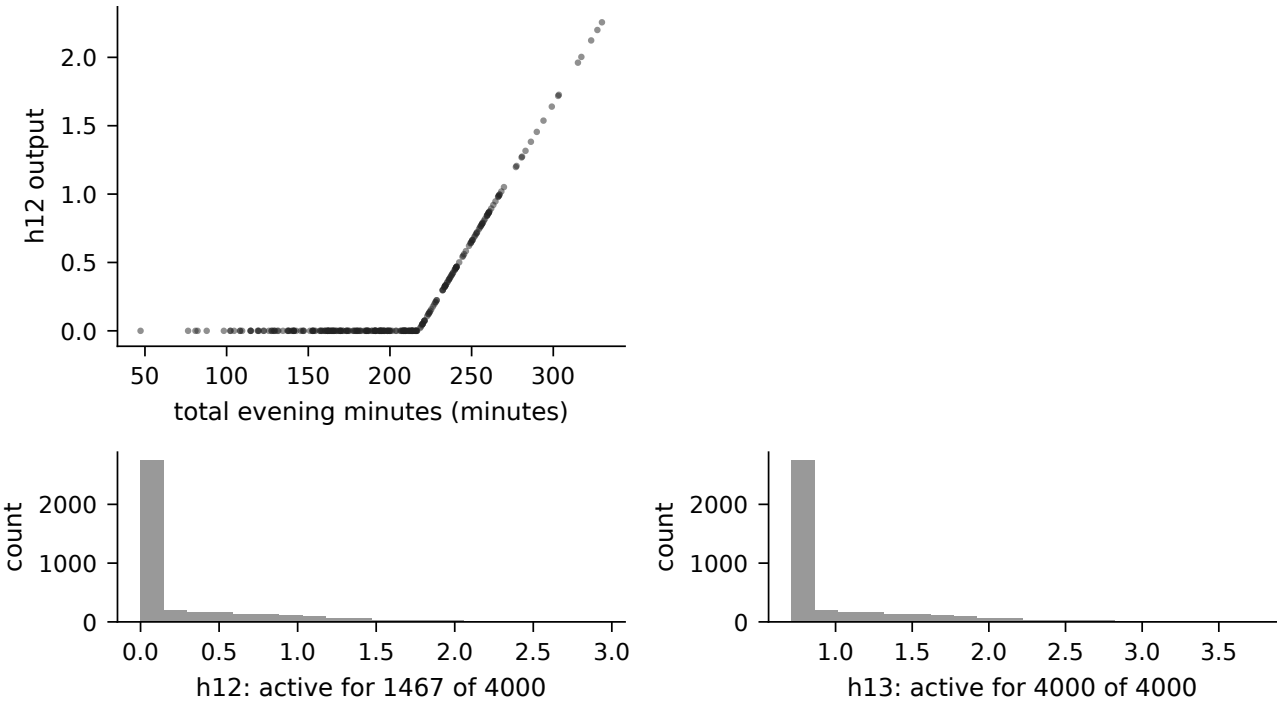


Figure 167: Evening unit: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.2.

3.2.9 Small voice-mail unit

Part of: The whole model (4.2.3).
Figure 168 shows what feeds this part; Figure 169 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).



Figure 168: Small voice-mail unit: what feeds it, one level down.

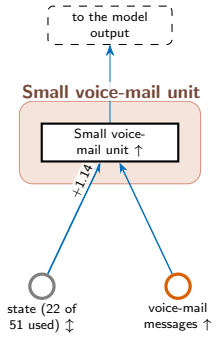


Figure 169: Small voice-mail unit: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This small part reads voice-mail messages and being in Louisiana, and barely changes the predictions. [C111]

Evidence. Through it, a typical customer's chance stays at about 3% in Louisiana as elsewhere, and voice-mail messages barely change it (about 3% at 0 and at 32 messages). [C106, C107] Switching it off changes the predictions by at most one point. [C111] Its formula reproduces the part exactly on all 4000 training customers. [C112]

Evidence plots: Figure 170, Figure 171, Figure 172, Figure 173.

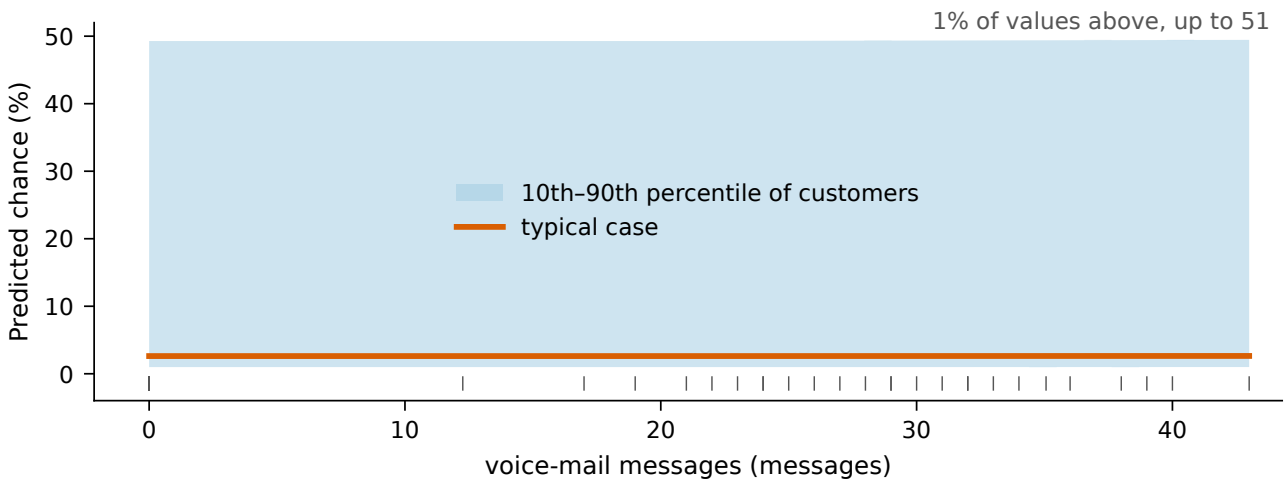


Figure 170: Small voice-mail unit: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

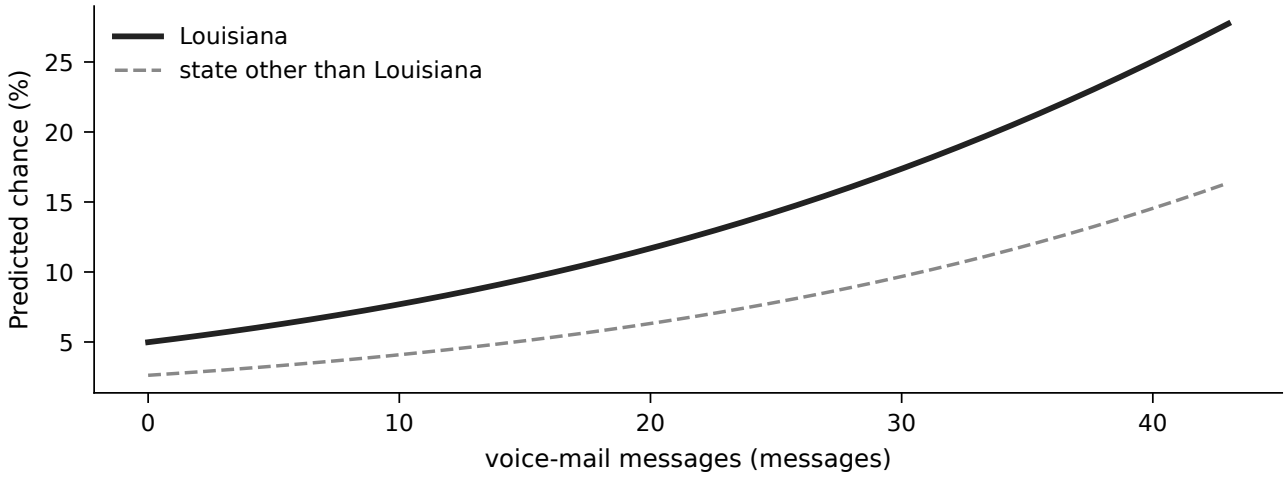


Figure 171: Small voice-mail unit: How the effect differs between groups: one curve per group, for the typical case of each group.

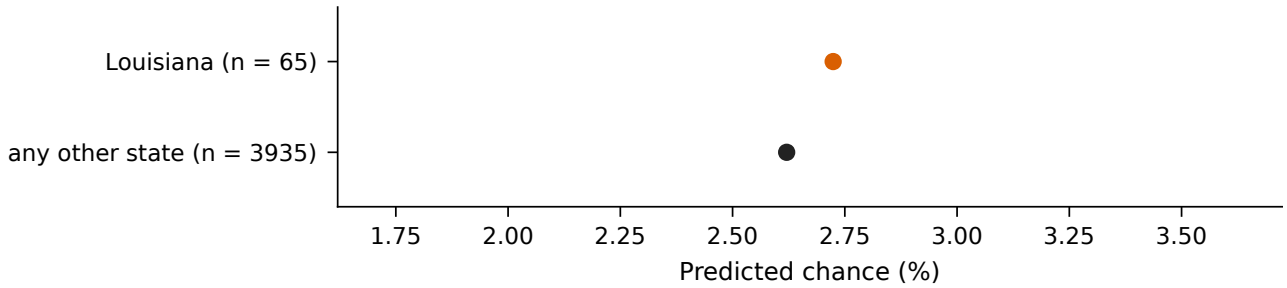


Figure 172: Small voice-mail unit: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

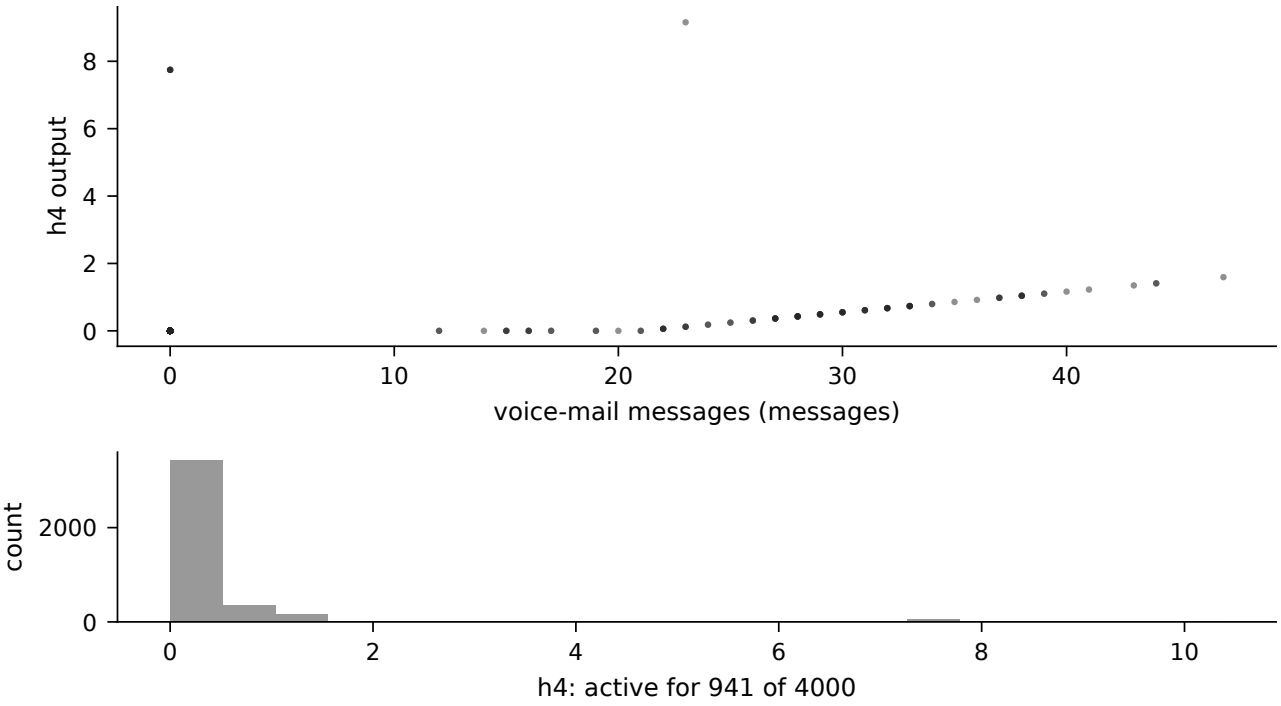


Figure 173: Small voice-mail unit: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.1.

3.3 Not supported by the model

- C109: threshold claim about number of voice-mail messages in part ann_84: not supported by the model (the fold changes the curve by only 0.007 log-odds compared with a straight line (needed 0.1); the effect past the fold is too small or not consistent across patients).
- C110: subgroup claim about state and number of voice-mail messages in part ann_84: not supported by the model (its effect is negligible).
- C113: boundary claim about total daytime minutes and number of voice-mail messages in part ann_82: not supported by the model (the boundary rule matches 0.0 of training rows (needs 0.99)).

Technical detail: churn100-s4-c2

The record behind each part: the original hypothesis, the closed form, and every claim with its estimate (log-odds and odds ratio), range across patients, n and test check.

T.1 The whole model

Annotation ann_87 (churn_logit). Reads: state, account length, area code, international calling plan, voice-mail plan, number of voice-mail messages, total daytime minutes, total daytime calls, total daytime charge, total evening minutes, total evening calls, total evening charge, total night-time minutes, total night-time calls, total night-time charge, total international minutes, total international calls, total international charge, number of calls to customer service.

Original hypothesis (unchanged). Root: logit P(churn) = -1.30 + linear_backbone (identity_lin) + service_calls (identity_svc) + 1.53*3118 + 1.91*10131 (day_usage_hinges) + 0.20*15524 (evening_hinge) - 1.01*1295 - 1.21*7872 + 0.04*9954 (state_offsets) + 0.005*6089, purely additive in the logit; the sigmoid is the only coupling. 37 inputs are unconnected (international charge and calls, night minutes/charge/calls, evening calls, area codes 408/510 and 29 state indicators) and have zero effect. Train AUC 0.8908, accuracy 0.888. Performance sources (train): all children replaced by least-squares linear fits gives AUC 0.8333 (an L2 logistic regression refit on all 71 inputs gets 0.8433); restoring day_usage_hinges alone gives 0.8707, service_calls alone 0.8659, both 0.8907; evening, state and voicemail units restore nothing. So the gain over linear (+0.058) comes from the day-usage and service-call thresholds, which are complementary. Rare-category switches: state:11 -2.75 logit (73 rows, empirical -1.35); state:46 -1.94 (80 rows, -0.91); state:45 -0.94 (91, -0.85); state:47 +0.90 mean (72 rows, empirical +0.92; direct +1.23, but it switches off 10131 so heavy day users lose up to 2.75); state:31 +0.83 (90, +0.67); state:18 +0.41 (65 rows, empirical -0.48: opposite sign); state:39 -0.25 net (82, -0.93: direct -2.61 almost cancelled by the state units).

Closed form. Read through its parts (their formulas are in their own sections).

Split inputs. To annotate the parts, the explanation split these inputs into copies (a refactoring step that does not change what the model computes); the diagrams draw each input once: customer-service calls (2 copies); voice-mail messages (3 copies).

State: every level the model reads. Whole-model change in the predicted chance for a typical customer when only the state changes from the reference; n = training customers in that level.

Claim	From	To	n	Chance	OK
C002	West Virginia	Washington	72	3% to 11%	yes
C005	West Virginia	New Jersey	90	3% to 7%	yes
C008	West Virginia	Louisiana	65	3% to 5%	yes
C009	West Virginia	Connecticut	80	3% to 5%	yes
C013	West Virginia	Tennessee	68	3% to 5%	yes
C014	West Virginia	Massachusetts	86	3% to 5%	yes
C015	West Virginia	Oregon	84	3% to 4%	yes
C016	West Virginia	Indiana	82	3% to 4%	yes
C017	West Virginia	Kentucky	81	3% to 4%	yes
C019	West Virginia	South Dakota	66	3% to 4%	yes
C020	West Virginia	Oklahoma	79	3% to 4%	yes
C024	West Virginia	North Dakota	68	3% to 3%	yes
C025	West Virginia	state other than Georgia, Hawaii, Indiana, Kentucky, Loui...	2286	3% to 3%	yes
C027	West Virginia	Arizona	73	3% to 3%	yes
C030	West Virginia	Arkansas	78	3% to 3%	yes
C033	West Virginia	Rhode Island	82	3% to 3%	yes
C034	West Virginia	Florida	78	3% to 3%	yes
C023	West Virginia	Pennsylvania	56	3% to 2%	yes
C022	West Virginia	Georgia	60	3% to 2%	yes
C018	West Virginia	Virginia	91	3% to 1%	yes
C012	West Virginia	Vermont	80	3% to <1%	yes
C007	West Virginia	Hawaii	73	3% to <1%	yes

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C001	direction: international calling plan: no international plan to an international plan (whole model) *shape: categorical; highest to lowest: an international plan, no international plan*	3% to 31%	2.82	16.7	2.82 to 2.82	3620, 380	agrees; AUC gain 0.060 [0.032, 0.090]	yes
C002	direction: state: West Virginia to Washington (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 11%	1.27	3.58	0.64 to 1.48	122, 72	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C003	direction: voice-mail messages: 0 messages to 32 messages (whole model) *shape: steady; 72% of patients same; fallback; risk saturates near 0%*	3% to 11%	1.21	3.35	0.39 to 1.47	4000	agrees; AUC gain 0.017 [0.005, 0.031]	yes
C004	direction: total daytime minutes: 112 minutes to 249 minutes (whole model) *shape: hinge at 223 minutes; 50% of patients same; fallback; risk saturates near 0%; matches the fold at 221 minutes*	3% to 8%	1.13	3.10	0.83 to 1.79	4000	agrees; AUC gain 0.036 [0.015, 0.059]	yes
C005	direction: state: West Virginia to New Jersey (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 7%	1.09	2.97	1.09 to 1.09	122, 90	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C006	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (whole model) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	3% to <1%	-2.54	0.08	-2.54 to -2.54	2939, 1061	agrees; AUC gain 0.033 [0.014, 0.053]	yes
C007	direction: state: West Virginia to Hawaii (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to <1%	-2.50	0.08	-2.50 to -2.50	122, 73	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C008	direction: state: West Virginia to Louisiana (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 5%	0.67	1.95	0.67 to 0.67	122, 65	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C009	direction: state: West Virginia to Connecticut (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 5%	0.66	1.94	0.66 to 0.66	122, 80	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C010	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	2% to 4%	0.88	2.42	0.88 to 0.88	4000	agrees; AUC gain 0.013 [0.004, 0.022]	yes
C011	direction: total international minutes: 7 minutes to 14 minutes (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	2% to 4%	0.85	2.35	0.85 to 0.85	4000	unclear; AUC gain 0.0008 [-0.007, 0.008]	yes
C012	direction: state: West Virginia to Vermont (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to <1%	-1.68	0.19	-1.68 to -1.68	122, 80	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C013	direction: state: West Virginia to Tennessee (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 5%	0.60	1.82	0.60 to 0.60	122, 68	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C014	direction: state: West Virginia to Massachusetts (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 5%	0.57	1.77	0.57 to 0.57	122, 86	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C015	direction: state: West Virginia to Oregon (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 4%	0.53	1.70	0.53 to 0.53	122, 84	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C016	direction: state: West Virginia to Indiana (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 4%	0.52	1.69	0.52 to 0.52	122, 82	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C017	direction: state: West Virginia to Kentucky (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 4%	0.47	1.61	0.47 to 0.47	122, 81	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C018	direction: state: West Virginia to Virginia (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 1%	-0.69	0.50	-0.69 to -0.69	122, 91	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C019	direction: state: West Virginia to South Dakota (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 4%	0.32	1.37	0.32 to 0.32	122, 66	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C020	direction: state: West Virginia to Oklahoma (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 4%	0.31	1.36	0.31 to 0.31	122, 79	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C021	direction: total daytime calls: 75 calls to 125 calls (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	2% to 3%	0.33	1.40	0.33 to 0.33	4000	disagrees; AUC gain -0.002 [-0.005, 0.0008]	yes
C022	direction: state: West Virginia to Georgia (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 2%	-0.40	0.67	-0.40 to -0.40	122, 60	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C023	direction: state: West Virginia to Pennsylvania (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 2%	-0.37	0.69	-0.37 to -0.37	122, 56	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C024	direction: state: West Virginia to North Dakota (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 3%	0.26	1.30	0.26 to 0.26	122, 68	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C025	direction: state: West Virginia to state other than Georgia, Hawaii, Indiana, Kentucky, Louisiana, Massachusetts, Arkansas, North Dakota, Arizona, New Jersey, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Dakota, Tennessee, Virginia, Vermont, Washington, West Virginia, Connecticut, Florida (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 3%	0.26	1.29	0.26 to 0.26	122, 2286	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C026	direction: account length: 48 to 152 (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	2% to 3%	0.28	1.32	0.28 to 0.28	4000	disagrees; AUC gain -0.001 [-0.004, 0.0006]	yes
C027	direction: state: West Virginia to Arizona (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 3%	0.22	1.25	0.22 to 0.22	122, 73	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C028	direction: total evening minutes: 138 minutes to 266 minutes (whole model) *shape: hinge at 218 minutes; 100% of patients same; verified; risk saturates near 0%; matches the fold at 217 minutes*	3% to 3%	0.20	1.22	0.20 to 0.20	4000	unclear; AUC gain 0.003 [-8e-05, 0.005]	yes
C029	direction: customer-service calls: 0 calls to 3 calls (whole model) *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	3% to 3%	0.19	1.21	0.19 to 0.19	4000	agrees; AUC gain 0.111 [0.070, 0.152]	yes
C030	direction: state: West Virginia to Arkansas (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 3%	0.14	1.15	0.14 to 0.14	122, 78	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C031	direction: area code: area code 415 to area code other than area code 415 (whole model) *shape: categorical; highest to lowest: area code 415, area code other than area code 415*	3% to 3%	-0.01	0.99	-0.01 to -0.01	2016, 1984	disagrees; AUC gain -0.0002 [-0.0004, 5e-05]	yes
C032	direction: total daytime charge: 19.07 US dollars to 42.40 US dollars (whole model) *shape: hinge at 42.20 US dollars; 77% of patients same; fallback; risk saturates near 0%; matches the fold at 39.34 US dollars*	3% to 3%	0.24	1.27	0.00 to 0.94	4000	unclear; AUC gain 0.005 [-0.0004, 0.012]	yes
C033	direction: state: West Virginia to Rhode Island (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 3%	0.004	1.00	0.004 to 0.004	122, 82	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C034	direction: state: West Virginia to Florida (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 3%	-0.0006	1.00	-0.0006 to -0.0006	122, 78	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C035	magnitude: international calling plan: no international plan to an international plan, typical patient *shape: categorical; highest to lowest: an international plan, no international plan*	3% to 31%	2.82	16.7		3620, 380	agrees; AUC gain 0.060 [0.032, 0.090]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C036	magnitude: state: West Virginia to Washington, typical patient *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 11%	1.48	4.40		122, 72	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C037	magnitude: voice-mail messages: 0 messages to 32 messages, typical patient *shape: steady; 72% of patients same; fallback; risk saturates near 0%*	3% to 11%	1.47	4.37		4000	agrees; AUC gain 0.017 [0.005, 0.031]	yes
C038	magnitude: total daytime minutes: 112 minutes to 249 minutes, typical patient *shape: hinge at 223 minutes; 50% of patients same; fallback; risk saturates near 0%; matches the fold at 221 minutes*	3% to 8%	1.15	3.17		4000	agrees; AUC gain 0.036 [0.015, 0.059]	yes
C039	magnitude: voice-mail plan: no voice-mail plan to a voice-mail plan, typical patient *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	3% to <1%	-2.54	0.08		2939, 1061	agrees; AUC gain 0.033 [0.014, 0.053]	yes
C040	null: model ignores total evening calls (exact rule)		0.00			4000	train only	yes
C041	null: model ignores total international calls (exact rule)		0.00			4000	train only	yes
C042	null: model ignores total international charge (exact rule)		0.00			4000	train only	yes
C043	null: model ignores total night-time calls (exact rule)		0.00			4000	train only	yes
C044	null: model ignores total night-time charge (exact rule)		0.00			4000	train only	yes
C045	null: model ignores total night-time minutes (exact rule)		0.00			4000	train only	yes
C046	structure: made of linear_backbone, service_calls, day_usage_hinges, evening_hinge, vmail_state18_unit, state_offsets; added					4000	train only	yes

T.2 Fixed-weight inputs

Annotation ann_79 (linear_backbone). Reads: state, account length, area code, international calling plan, voice-mail plan, number of voice-mail messages, total daytime calls, total evening charge, total international minutes.

Original hypothesis (unchanged). Direct input-to-output edges collected through identity_lin (an identity node). Exactly linear in 29 inputs: voicemail plan -1.12 per SD (-2.54 logit for having a plan, mostly offset for heavy message users by voicemail messages +0.62 per SD), international plan +0.83 (+2.82 logit for having one), evening charge +0.35, international minutes +0.34, day calls +0.13, account length +0.11, and state offsets: state:11 -0.37 (-2.75 logit for its 73 customers), state:39 -0.37 (-2.61, but the state_offsets units give back +2.36), state:46 -0.27 (-1.94), state:47 +0.16 (+1.23), state:31 +0.12 (+0.83), state:45 -0.14 (-0.94), then smaller states and area code 415 (+0.01). Day minutes, day charge and customer-service calls have NO direct edge: they act only through hinges. Evidence: closed form exact (identity); train sensitivities equal the weights (voicemail plan 1.12, voicemail messages 0.62, international plan 0.83).

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 1.33227e-15).

$$
\begin{aligned}
y_0 = {} & 0.00719x_{areaCode:415} + 0.106x_{accountLength} + 0.0569x_{state:6} \\
& + 0.163x_{state:47_a} + 0.133x_{totalDayCalls} + 0.337x_{totalIntlMinutes} \\
& + 0.044x_{state:42} + 0.007x_{state:36} + 0.00781x_{state:41} \\
& + 0.621x_{numberVmailMessages_a} + 0.0397x_{state:37} + 0.35x_{totalEveCharge} \\
& + 0.0453x_{state:19} + 0.047x_{state:18_a} + 0.826x_{internationalPlan} \\
& + 0.000897x_{state:28} + 0.0307x_{state:17} + 0.124x_{state:31} - 0.37x_{state:39_a} \\
& - 0.0355x_{state:9} - 0.0158x_{state:2} - 0.00422x_{state:3} - 0.271x_{state:46} \\
& - 1.12x_{voiceMailPlan} - 0.141x_{state:45} - 0.044x_{state:49} - 0.0737x_{state:38} \\
& - 0.0797x_{state:10} - 0.368x_{state:11}
\end{aligned}
$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C047	direction: international calling plan: no international plan to an international plan (through this part) *shape: categorical; highest to lowest: an international plan, no international plan*	3% to 31%	2.82	16.7	2.82 to 2.82	3620, 380	agrees; AUC gain 0.060 [0.032, 0.090]	yes
C048	direction: state: West Virginia to Washington (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 11%	1.48	4.40	1.48 to 1.48	122, 72	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C049	direction: voice-mail messages: 0 messages to 32 messages (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	3% to 10%	1.47	4.35	1.47 to 1.47	4000	agrees; AUC gain 0.016 [0.003, 0.029]	yes
C050	direction: state: West Virginia to New Jersey (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 7%	1.09	2.97	1.09 to 1.09	122, 90	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C051	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	3% to <1%	-2.54	0.08	-2.54 to -2.54	2939, 1061	agrees; AUC gain 0.033 [0.014, 0.053]	yes
C052	direction: state: West Virginia to Hawaii (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to <1%	-2.50	0.08	-2.50 to -2.50	122, 73	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C053	direction: state: West Virginia to Rhode Island (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to <1%	-2.36	0.09	-2.36 to -2.36	122, 82	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C054	direction: state: West Virginia to Connecticut (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 5%	0.66	1.94	0.66 to 0.66	122, 80	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C055	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	2% to 4%	0.88	2.42	0.88 to 0.88	4000	agrees; AUC gain 0.013 [0.004, 0.022]	yes
C056	direction: total international minutes: 7 minutes to 14 minutes (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	2% to 4%	0.85	2.35	0.85 to 0.85	4000	unclear; AUC gain 0.0008 [-0.007, 0.008]	yes
C057	direction: state: West Virginia to Louisiana (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 5%	0.63	1.87	0.63 to 0.63	122, 65	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C058	direction: state: West Virginia to Vermont (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to <1%	-1.68	0.19	-1.68 to -1.68	122, 80	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C059	direction: state: West Virginia to Tennessee (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 5%	0.60	1.82	0.60 to 0.60	122, 68	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C060	direction: state: West Virginia to Massachusetts (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 5%	0.57	1.77	0.57 to 0.57	122, 86	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C061	direction: state: West Virginia to Oregon (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 4%	0.53	1.70	0.53 to 0.53	122, 84	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C062	direction: state: West Virginia to Kentucky (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 4%	0.47	1.61	0.47 to 0.47	122, 81	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C063	direction: state: West Virginia to Virginia (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 1%	-0.69	0.50	-0.69 to -0.69	122, 91	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C064	direction: state: West Virginia to South Dakota (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 4%	0.32	1.37	0.32 to 0.32	122, 66	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C065	direction: state: West Virginia to Oklahoma (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts ...*	3% to 4%	0.31	1.36	0.31 to 0.31	122, 79	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C066	direction: total daytime calls: 75 calls to 125 calls (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	2% to 3%	0.33	1.40	0.33 to 0.33	4000	disagrees; AUC gain -0.002 [-0.005, 0.0008]	yes
C067	direction: state: West Virginia to Georgia (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts ...*	3% to 2%	-0.40	0.67	-0.40 to -0.40	122, 60	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C068	direction: state: West Virginia to Pennsylvania (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts ...*	3% to 2%	-0.37	0.69	-0.37 to -0.37	122, 56	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C069	direction: state: West Virginia to North Dakota (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts ...*	3% to 3%	0.26	1.30	0.26 to 0.26	122, 68	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C070	direction: state: West Virginia to state other than Georgia, Hawaii, Kentucky, Louisiana, Massachusetts, Arkansas, North Dakota, Arizona, New Jersey, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Dakota, Tennessee, Virginia, Vermont, Washington, West Virginia, Connecticut, Florida (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts ...*	3% to 3%	0.26	1.29	0.26 to 0.26	122, 2368	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C071	direction: account length: 48 to 152 (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	2% to 3%	0.28	1.32	0.28 to 0.28	4000	disagrees; AUC gain -0.001 [-0.004, 0.0006]	yes
C072	direction: state: West Virginia to Arizona (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts ...*	3% to 3%	0.22	1.25	0.22 to 0.22	122, 73	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C073	direction: state: West Virginia to Arkansas (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts ...*	3% to 3%	0.14	1.15	0.14 to 0.14	122, 78	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C074	direction: area code: area code 415 to area code other than area code 415 (through this part) *shape: categorical; highest to lowest: area code 415, area code other than area code 415*	3% to 3%	-0.01	0.99	-0.01 to -0.01	2016, 1984	disagrees; AUC gain -0.0002 [-0.0004, 5e-05]	yes
C075	direction: state: West Virginia to Florida (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts ...*	3% to 3%	-0.0006	1.00	-0.0006 to -0.0006	122, 78	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C076	structure: its formula reproduces it on 4000 training patients (largest difference 1.33227e-15)		1e-15			4000	train only	yes

T.3 Service calls

Annotation ann_86 (service_calls). Reads: number of calls to customer service.

Original hypothesis (unchanged). Composition of csc_moderate_chain and csc_high_hinge: their exits are summed by identity_svc = 1.66*5268 + 1.76*9436 + 1.54*6713, which enters the logit with weight 1; they read the same input (two split copies of customer-service calls). Hypothesis: all of the model's service-call effect is a threshold: nothing for 0-2 calls, +0.19 logit at 3 calls, then about +2.4 logit per extra call (+2.07 at 4, +4.44 at 5, +6.81 at 6, +9.18 at 7). Evidence (train): churn by calls 0-7 is 0.115/0.110/0.110/0.106/0.437/0.571/0.679/0.545 against model 0.118/0.108/0.105/0.115/0.327/0.749/0.956/0.986: the threshold is right, but the model under-predicts at 4 calls and is over-confident from 5 calls (116 customers). Mean-ablating both children gives AUC 0.7976, linear replacement 0.8711; restoring service_calls alone onto the all-linear model gives 0.8659 (all-linear 0.8333).

Closed form. Read through its parts (their formulas are in their own sections).

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C077	direction: customer-service calls: 0 calls to 3 calls (through this part) *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	3% to 3%	0.19	1.21	0.19 to 0.19	4000	agrees; AUC gain 0.111 [0.070, 0.152]	yes
C078	threshold: customer-service calls: fold at about 3 calls (starts) *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	3% to 18% (typical patient with a fold)	2.07	7.90	2.07 to 2.07	4000, 848	agrees; AUC gain 0.111 [0.070, 0.152]	yes
C079	structure: made of csc_moderate_chain, csc_high_hinge; added					4000	train only	yes

T.4 Calls from 4 upwards

Annotation ann_81 (csc_high_hinge). Reads: number of calls to customer service.

Original hypothesis (unchanged). 9436 = relu(-0.58 + 0.46 csc), 6713 = relu(-0.13 + 1.43*9436); logit weights +1.76 and +1.54 (via identity_svc). Hypothesis: a second hinge on customer-service calls, off for 0-3 calls and switching on from 4 calls, adding about +1.4 logit per extra call (+0.91 at 4, +2.32 at 5, +3.72 at 6). Active on exactly the 8% of customers with 4+ calls (318 rows; empirical churn jumps from 0.11 at 3 calls to 0.44 at 4). Evidence: line plot correlation 0.66; mean ablation AUC 0.8761; linear replacement 0.8857 (R2 0.44/0.41); restored alone onto all-linear 0.8571.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 0.0).

$$
\begin{aligned}
y_0 &= \operatorname{relu}\big(-0.582 + 0.462 x_{numberCustomerServiceCalls_b}\big) \\
y_1 &= \operatorname{relu}\big(-0.127 + 1.43 \operatorname{relu}\big(-0.582 \\
&\qquad + 0.462 x_{numberCustomerServiceCalls_b}\big)\big)
\end{aligned}
$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C093	threshold: customer-service calls: fold at about 4 calls (starts) *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	3% to 6% (typical patient with a fold)	0.91	2.49	0.91 to 0.91	4000, 318	agrees; AUC gain 0.023 [0.012, 0.035]	yes
C094	subgroup: who this part acts on: 318 of 4000 patients	8% to 18% (typical patient this part acts on (risk without the part, then with it))	1.71	5.51	0.91 to 3.72	318, 3682	agrees; AUC gain 0.023 [0.012, 0.035]	yes
C095	structure: its formula reproduces it on 4000 training patients (largest difference 0.0)		0.00			4000	train only	yes

T.5 Calls from 3 upwards

Annotation ann_80 (csc_moderate_chain). Reads: number of calls to customer service.

Original hypothesis (unchanged). 12730 = relu(-0.03 + 0.80 csc), 12171 = relu(-0.44 + 0.84 csc + 0.72*12730), 5268 = relu(-0.56 + 0.25 csc + 0.36*12171); logit weight +1.66 (via identity_svc). Hypothesis: a convex hinge on customer-service calls that is zero for 0-2 calls, just switches on at 3 calls (mean logit +0.19) and then adds about +1.0 logit per extra call (+1.16 at 4, +2.12 at 5, +3.09 at 6). Evidence (train, n=4000): line plot of 5268 against calls rises from 0 to 3.6 (correlation 0.74); 5268 active on 21% of customers (all with 3+ calls). Mean ablation AUC 0.8733 (full 0.8908); linear replacement 0.8852 (R2 0.55); restored alone onto the all-linearised model 0.8565 (all-linear 0.8333).

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 4.44089e-16).

$$\begin{aligned} y_0 = \operatorname{relu}\big(&-0.557 + 0.358 \operatorname{relu}\big(-0.443 \\ &+ 0.719 \operatorname{relu}\big(-0.0259 + 0.801 x_{numberCustomerServiceCalls_acd}\big) \\ &+ 0.845 x_{numberCustomerServiceCalls_acd}\big) \\ &+ 0.254 x_{numberCustomerServiceCalls_acd}\big) \end{aligned}$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C096	direction: customer-service calls: 0 calls to 3 calls (through this part) *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	3% to 3%	0.19	1.21	0.19 to 0.19	4000	agrees; AUC gain 0.028 [0.014, 0.042]	yes
C097	threshold: customer-service calls: fold at about 3 calls (starts) *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	3% to 8% (typical patient with a fold)	1.16	3.18	1.16 to 1.16	4000, 848	agrees; AUC gain 0.028 [0.014, 0.042]	yes
C098	subgroup: who this part acts on: 848 of 4000 patients	3% to 4% (typical patient this part acts on (risk without the part, then with it))	0.76	2.13	0.19 to 2.12	848, 3152	agrees; AUC gain 0.028 [0.014, 0.042]	yes
C099	structure: its formula reproduces it on 4000 training patients (largest difference 4.44089e-16)		4e-16			4000	train only	yes

T.6 Heavy daytime use

Annotation ann_82 (day_usage_hinges). Reads: state, number of voice-mail messages, total daytime minutes, total daytime charge.

Original hypothesis (unchanged). 3118 = relu(-0.86 + 0.55 dayMin + 0.38 dayCharge - 0.65 vmailMessages), logit weight +1.53; 10131 = relu(-0.72 + 0.84 dayMin - 0.67 state:47), logit weight +1.91. Day minutes and day charge are collinear (r = 1.0000), so 3118 is one hinge in day usage. Hypothesis: day usage has no effect below about 200 minutes, then raises churn risk steeply. 3118 switches on at about 209 day minutes for customers with no voicemail messages (its kink moves up about 25 minutes per 10 messages: 267 at 20, 292 at 30), so it is on for only 2% of message users; 10131 switches on at about 221 day minutes for everyone except state:47 (off for all 72 state:47 customers). Evidence (train): combined logit contribution 0 below 200 minutes, +0.62 at 220-240, +1.61 at 240-260, +2.59 at 260-280, +4.26 above 280; churn by band 0.06 (200-220), 0.14, 0.26, 0.46, 0.67 vs model 0.08, 0.15, 0.28, 0.43, 0.68 (well calibrated). Sensitivity: 3118 vmail messages 0.15, day minutes 0.12, day charge 0.09; 10131 day minutes 0.18, state:47 0.14. Mean ablation AUC 0.8107 (the largest single drop); linear replacement 0.8657 (R2 0.46/0.45); restored alone onto all-linear 0.8707 (all-linear 0.8333).

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 4.44089e-16).

$$
\begin{aligned}
y_0 = \mathrm{relu}\big(&-0.856 + 0.382 x_{totalDayCharge} + 0.547 x_{totalDayMinutes} \\
&- 0.65 x_{numberVmailMessages_b}\big) \\
y_1 = \mathrm{relu}\big(&-0.721 + 0.837 x_{totalDayMinutes} \\
&- 0.672 x_{state:47_b}\big)
\end{aligned}
$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C080	direction: voice-mail messages: 0 messages to 32 messages (through this part) *shape: hinge at 11 messages; 64% of patients same; fallback; one point with C089 (median fold 11 messages, typical patient 11 messages, 10th-90th percentile 4 messages to 27 messages, wide)*	12% to 6% (typical patient this part acts on)	-0.26	0.77	-1.08 to 0.00	4000	agrees; AUC gain 0.006 [0.003, 0.009]	yes
C081	direction: total daytime minutes: 112 minutes to 249 minutes (through this part) *shape: hinge at 221 minutes; 50% of patients same; fallback; risk saturates near 0%; one point with C088 (median fold 221 minutes, typical patient 223 minutes, 10th-90th percentile 186 minutes to 221 minutes, wide)*	3% to 8%	1.13	3.10	0.83 to 1.79	4000	agrees; AUC gain 0.036 [0.015, 0.059]	yes
C082	direction: state: state other than Washington to Washington (through this part) *shape: categorical; highest to lowest: state other than Washington, Washington*	12% to 7% (typical patient this part acts on)	-0.21	0.81	-0.84 to 0.00	3928, 72	agrees; AUC gain 0.0004 [2e-05, 0.0008]	yes
C083	direction: total daytime charge: 19.07 US dollars to 42.40 US dollars (through this part) *shape: hinge at 39.34 US dollars; 77% of patients same; fallback; risk saturates near 0%; one point with C090 (median fold 39.34 US dollars, typical patient 42.20 US dollars, 10th-90th percentile 26.14 US dollars to 52.75 US dollars, wide)*	3% to 3%	0.24	1.27	0.00 to 0.94	4000	unclear; AUC gain 0.005 [-0.0004, 0.012]	yes
C084	interaction: effect of total daytime minutes (112 minutes to 249 minutes) by total daytime charge: total daytime charge above 30.69 US dollars minus total daytime charge at or below 30.69 US dollars; stronger in total daytime charge above 30.69 US dollars *shape: accelerating; 74% of patients same; fallback; risk saturates near 0%*	total daytime charge above 30.69 US dollars: 3% to 11%; total daytime charge at or below 30.69 US dollars: 3% to 6%	0.54	1.71		1997, 2003	agrees; AUC gain 0.005 [-0.0004, 0.012]	yes
C085	interaction: effect of total daytime minutes (112 minutes to 249 minutes) by state: Washington minus state other than Washington; stronger in state other than Washington *shape: hinge at 223 minutes; 51% of patients same; fallback; risk saturates near 0%*	Washington: 11% to 14%; state other than Washington: 3% to 8%	-0.75	0.47		72, 3928	agrees; AUC gain 0.0004 [2e-05, 0.0008]	yes
C086	interaction: effect of voice-mail messages (0 messages to 32 messages) by total daytime charge: total daytime charge above 30.69 US dollars minus total daytime charge at or below 30.69 US dollars; stronger in total daytime charge at or below 30.69 US dollars *shape: steady; 100% of patients same; verified; risk saturates near 0%*	total daytime charge above 30.69 US dollars: 3% to 11%; total daytime charge at or below 30.69 US dollars: 3% to 11%	-0.53	0.59		1997, 2003	agrees; AUC gain 0.005 [-0.0004, 0.012]	yes
C087	interaction: effect of voice-mail messages (0 messages to 32 messages) by total daytime minutes: total daytime minutes above 180 minutes minus total daytime minutes at or below 180 minutes; stronger in total daytime minutes at or below 180 minutes *shape: steady; 100% of patients same; verified; risk saturates near 0%*	total daytime minutes above 180 minutes: 3% to 11%; total daytime minutes at or below 180 minutes: 3% to 11%	-0.53	0.59		1997, 2003	agrees; AUC gain 0.036 [0.015, 0.059]	yes
C088	threshold: total daytime minutes: fold at about 221 minutes (starts) *shape: hinge at 221 minutes; 48% of patients same; fallback; risk saturates near 0%; one point with C088 (median fold 221 minutes, typical patient 223 minutes, 10th-90th percentile 186 minutes to 221 minutes, wide)*	3% to 19% (typical patient with a fold)	1.98	7.25	1.48 to 2.79	3995, 1099	agrees; AUC gain 0.036 [0.015, 0.059]	yes
C089	threshold: voice-mail messages: fold at about 11 messages (stops) for customers in New Jersey *shape: hinge at 11 messages; 74% of patients same; fallback; one point with C089 (median fold 11 messages, typical patient 11 messages, 10th-90th percentile 4 messages to 27 messages, wide)*	24% to 13% (typical patient with a fold (customers in New Jersey))	-0.93	0.40	-1.97 to -0.27	36, 26	agrees; AUC gain 0.006 [0.003, 0.009]	yes
C090	threshold: total daytime charge: fold at about 39.34 US dollars (starts) for customers in North Carolina *shape: hinge at 39.34 US dollars; 88% of patients same; fallback; one point with C090 (median fold 39.34 US dollars, typical patient 39.00 US dollars, 10th-90th percentile 26.14 US dollars to 52.75 US dollars, wide)*	4% to 5% (typical patient with a fold (customers in North Carolina))	0.51	1.67	0.00 to 1.28	50, 17 (small n)	unclear; AUC gain 0.005 [-0.0004, 0.012]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C091	subgroup: who this part acts on: 1099 of 4000 patients	3% to 12% (typical patient this part acts on (risk without the part, then with it))	1.49	4.42	0.12 to 3.47	1099, 2901	agrees; AUC gain 0.080 [0.048, 0.112]	yes
C092	structure: its formula reproduces it on 4000 training patients (largest difference 4.44089e-16)		4e-16			4000	train only	yes
C113	boundary: boundary		0.00			4000	train only	no

T.7 State units

Annotation ann_85 (state_offsets). Reads: state.

Original hypothesis (unchanged). 1295 = relu(0.87 - 1.27 state:39) -> 7872 = relu(0.01 + 1.02*1295) (logit weights -1.01, -1.21); 9954 = relu(-0.38 + 1.01 state:15) (weight +0.04). Each depends only on one binary state indicator, so each is exactly an offset (least-squares R2 = 1.0). Hypothesis: the pair is a constant -2.33 logit for everyone except state:39 customers, for whom it is switched off (+2.36 logit), almost cancelling state:39's direct -2.61 (net -0.25, against an empirical log-odds offset of -0.93 for its 82 customers); 9954 adds +0.27 for state:15. Evidence: sensitivity 1.24/1.27 for state:39, 0.02 for state:15; linear replacement AUC 0.8908 (unchanged); mean ablation 0.8898.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 0.0).

$$
\begin{aligned}
y_0 &= \operatorname{relu}\big(0.868 - 1.27 x_{state:39_b}\big) \\
y_1 &= \operatorname{relu}\big(0.0138 + 1.02 \operatorname{relu}\big(0.868 \\
&\qquad - 1.27 x_{state:39_b}\big)\big) \\
y_2 &= \operatorname{relu}\big(-0.38 + 1.01 x_{state:15}\big)
\end{aligned}
$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C100	direction: state: state other than Indiana, Rhode Island to Rhode Island (through this part) *shape: categorical; highest to lowest: Rhode Island, Indiana, state other than Indiana, Rhode Island*	3% to 22%	2.36	10.6	2.36 to 2.36	3836, 82	unclear; AUC gain 0.005 [-0.002, 0.016]	yes
C101	direction: state: state other than Indiana, Rhode Island to Indiana (through this part) *shape: categorical; highest to lowest: Rhode Island, Indiana, state other than Indiana, Rhode Island*	3% to 3%	0.27	1.31	0.27 to 0.27	3836, 82	unclear; AUC gain 0.005 [-0.002, 0.016]	yes
C102	structure: its formula reproduces it on 4000 training patients (largest difference 0.0)		0.00			4000	train only	yes

T.8 Evening unit

Annotation ann_83 (evening_hinge). Reads: total evening minutes.

Original hypothesis (unchanged). 14484 = relu(-0.33 + 1.01 eveMinutes), 15524 = relu(0.71 + 1.02*14484) (always active); logit weight +0.20. Hypothesis: a small hinge in evening minutes, constant (+0.14 logit) below about 217 evening minutes and rising steadily above (+0.21 at 220-250, +0.36 at 250-300, +0.56 above 300); it adds to the direct evening-charge term (+0.35 per SD). Evidence: line plot correlation 0.79; mean ablation AUC 0.8907, linear replacement 0.8907 (R2 0.62): negligible for ranking.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 0.0).

$$y_0 = \mathrm{relu}\big(0.714 + 1.02\,\mathrm{relu}\big(-0.33 + 1.01 x_{totalEveMinutes}\big)\big)$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C103	direction: total evening minutes: 138 minutes to 266 minutes (through this part) *shape: hinge at 217 minutes; 100% of patients same; verified; risk saturates near 0%; one point with C104 (median fold 217 minutes, typical patient 218 minutes, 10th-90th percentile 217 minutes to 217 minutes)*	3% to 3%	0.20	1.22	0.20 to 0.20	4000	unclear; AUC gain 0.003 [-8e-05, 0.005]	yes
C104	threshold: total evening minutes: fold at about 217 minutes (starts) *shape: hinge at 217 minutes; 100% of patients same; verified; risk saturates near 0%; one point with C104 (median fold 217 minutes, typical patient 218 minutes, 10th-90th percentile 217 minutes to 217 minutes)*	3% to 3% (typical patient with a fold)	0.28	1.32	0.28 to 0.28	4000, 1467	unclear; AUC gain 0.003 [-8e-05, 0.005]	yes
C105	structure: its formula reproduces it on 4000 training patients (largest difference 0.0)		0.00			4000	train only	yes

T.9 Small voice-mail unit

Annotation ann_84 (vmail_state18_unit). Reads: state, number of voice-mail messages.

Original hypothesis (unchanged). 6089 = relu(-0.67 + 1.14 state:18 + 0.83 vmailMessages), logit weight +0.005. On for customers with more than about 21 voicemail messages and for all 65 state:18 customers; logit contribution at most +0.05. Hypothesis: negligible. Evidence: mean ablation and linear replacement both leave train AUC at 0.8908.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 1.77636e-15).

$$y_0 = \mathrm{relu}\big(-0.668 + 1.14 x_{state:18_b} + 0.828 x_{numberVmailMessages_c}\big)$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C106	direction: state: state other than Louisiana to Louisiana (through this part) *shape: categorical; highest to lowest: Louisiana, state other than Louisiana*	3% to 3%	0.04	1.04	0.04 to 0.05	3935, 65	disagrees; AUC gain -6e-05 [-0.0002, 5e-05]	yes
C107	direction: voice-mail messages: 0 messages to 32 messages (through this part) *shape: flat; 100% of patients same; verified; risk saturates near 0%*	3% to 3%	0.004	1.00	0.003 to 0.003	4000	unclear; AUC gain 2e-05 [-9e-05, 0.0001]	yes
C108	interaction: effect of voice-mail messages (0 messages to 32 messages) by state: Louisiana minus state other than Louisiana; stronger in Louisiana *shape: steady; 78% of patients same; fallback*	Louisiana: 5% to 19%; state other than Louisiana: 3% to 11%	0.14	1.15		65, 3935	unclear; AUC gain -6e-05 [-0.0002, 5e-05]	yes
C109	threshold: voice-mail messages: fold at about 22 messages (starts) *shape: flat; 100% of patients same; verified; risk saturates near 0%*	3% to 3% (typical patient with a fold)	0.005	1.00	0.005 to 0.005	3935, 876	unclear; AUC gain 2e-05 [-9e-05, 0.0001]	no
C110	subgroup: who this part acts on: 941 of 4000 patients	2% to 2% (typical patient this part acts on (risk without the part, then with it))	0.006	1.01	0.0009 to 0.007	941, 3059	disagrees; AUC gain -5e-05 [-0.0002, 9e-05]	no
C111	null: this part barely changes predictions (switched off)		0.01			4000	train only	yes
C112	structure: its formula reproduces it on 4000 training patients (largest difference 1.77636e-15)		2e-15			4000	train only	yes

The network before any explanation

Figure 174 (next page, landscape) is the trained network as it is, with no parts, no splits and no inserted units: inputs at the bottom, hidden units by depth, the output at the top, every connection drawn, every input column its own node (one-hot levels included).

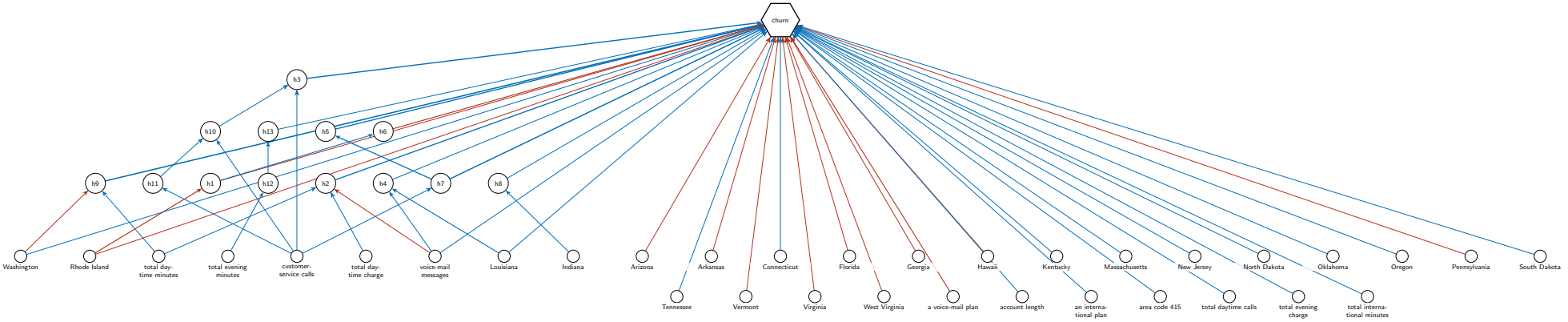


Figure 174: The network as trained, before any explanation, in full. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. Not used (no connection): Alaska, Alabama, Iowa, Idaho, Illinois, Kansas, Maryland, Maine, Michigan, Minnesota, Missouri, Mississippi, Montana, North Carolina, Nebraska, New Hampshire, New Mexico, Nevada, New York, Ohio, California, South Carolina, Texas, Utah, Wisconsin, Colorado, Wyoming, Washington DC, Delaware, area code 408, area code 510, total evening calls, total night-time minutes, total night-time calls, total night-time charge, total international calls, total international charge.

4 churn100-s4-c3

Genome `27f540ac-6acb-4d58-9972-cd71fae31cf0`, explanation `ffdb207f-1e0f-4d2e-a330-f47b1ca75c4a`. Split: train primary (n=4000), test as a check (n=1000). Claims: 123, verified 121, not supported 2.
AI-assisted draft for review (unverified)

Reference (typical) customer: the typical customer: the median training customer for every number and the most common training category for every category (not a real customer). account length: 100; area code: area code 415; international plan: no international plan; number customer service calls: 1 calls; number vmail messages: 0 messages; state: West Virginia; total day calls: 100 calls; total day charge: 30.69 US dollars; total day minutes: 180 minutes; total eve calls: 100 calls; total eve charge: 17.12 US dollars; total eve minutes: 201 minutes; total intl calls: 4 calls; total intl charge: 2.78 US dollars; total intl minutes: 10 minutes; total night calls: 100 calls; total night charge: 9.02 US dollars; total night minutes: 200 minutes; voice mail plan: no voice-mail plan. Predicted chance that a customer leaves: 1%.

4.1 Summary

For a typical customer (in West Virginia, area code 415, no international or voice-mail plan, no voice-mail messages, 1 call to customer service and median usage) the model predicts a chance of leaving of about 1%; it adds up four parts: fixed-weight inputs, service calls, heavy daytime use and state units. [C054] Figure 175 shows the model at this level: one box per paragraph below.

How to read the diagrams. Edge colour is the sign of that one connection: blue positive, red negative; thicker is a larger weight. The outline and arrow on an input (and on a part's output box) is its overall effect on the final prediction, read from the verified claims: orange and up arrow raises it, teal and down arrow lowers it, grey and double arrow depends (it goes both ways, or differs between levels). A negative connection into a part that itself raises the prediction means that input lowers the prediction; through a hinge the overall effect can only be read from the claims, not from the edges. Up arrow means towards churn; down arrow, away from it. A double outline marks an input (or a part's output) that acts at a fixed rate everywhere: it adds the same amount to the score per unit, so it is explained by its rate alone (see 'Linear factors'); a dashed double outline, at a fixed rate on the observed data. The normal outline is an input whose effect switches on and off somewhere. For example, a voice-mail plan has a negative connection into 'Fixed-weight inputs', a part that raises the predicted chance that a customer leaves, so a voice-mail plan lowers it.

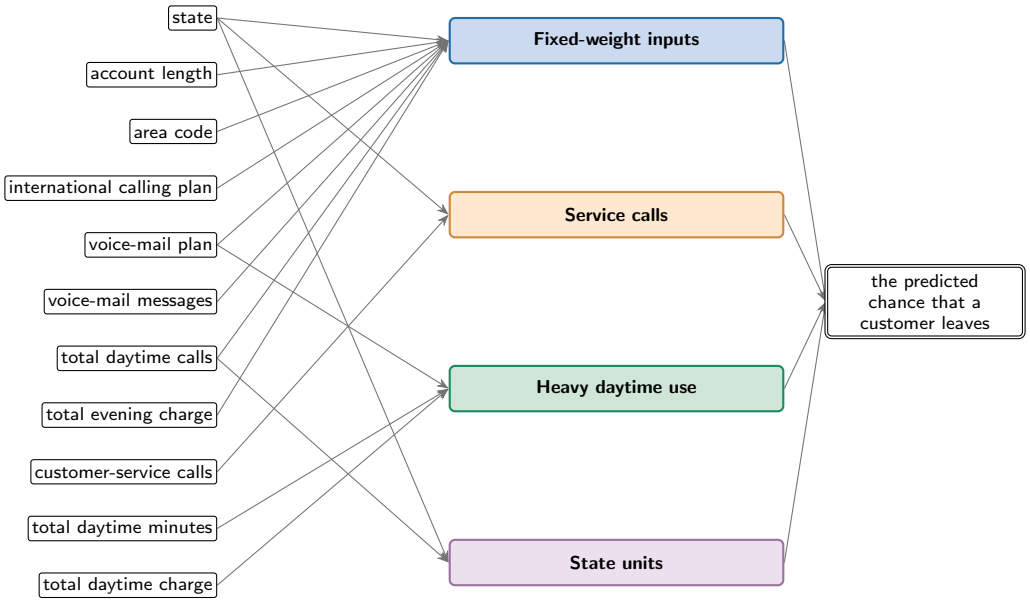


Figure 175: The model at the level of the summary: one box per part.

Fixed-weight inputs. This part adds up most inputs with fixed weights: an international plan raises a typical customer's chance of leaving from about 1% to about 18%, and a voice-mail plan lowers it to below 1%. [C055, C063] More voice-mail messages raise it steadily, from about 1% at 0 to about 6% at 32 messages, flattening out near the bottom because the predicted chance cannot go much lower. [C056] A higher evening charge also raises it steadily, from about 1% at 11.76 US dollars to about 2% at 22.57 US dollars, flattening out near the bottom because the predicted chance cannot go much lower. [C059] Through it, the state moves the chance from about 1% in West Virginia to about 5% in Washington and about 4% in Nevada, and to below 1% in Rhode Island, Hawaii and Vermont (72 to 82 training customers per state). [C057, C058, C061, C062, C064]

Service calls. This part adds up two parts: an early rise and a late rise in calls to customer service. [C090] From about 3 calls upwards it adds to the chance slowly at first, then faster: for a typical such customer, 4 calls take it from about 1% to about 10%, flattening out near the bottom because the predicted chance cannot go much lower. [C089] Up to 3 calls the chance rises slowly at first, then faster, from about 1% to about 2% for a typical customer, flattening out near the bottom because the predicted chance cannot go much lower. [C084] That early rise happens only outside Nebraska, California and Colorado: in Nebraska the chance stays at about 1% from 0 to 3 calls, against a rise from about 1% to about 2% elsewhere, slowly at first, then faster, flattening out near the bottom because the predicted chance cannot go much lower. [C088]

Heavy daytime use. Daytime minutes add to the chance only once they rise above about 209 minutes, in an uneven way: for a typical such customer, from about 1% at 0 minutes to about 16% at 271 minutes, flattening out near the bottom because the predicted chance cannot go much lower. [C100] The rise is much larger without a voice-mail plan: from 112 to 249 daytime minutes the chance goes from about 1% to about 6% without one, but stays below 1% with one, in an uneven way, flattening out near the bottom because the predicted chance cannot go much lower. [C099] This part acts on 1074 of 4000 training customers, 290 of whom left; for the typical customer it acts on, the chance is about 1% without it and about 4% with it. [C102]

State units. This part reads only the state: it raises a typical customer's chance from about 1% to about 12% in Rhode Island, to about 3% in Vermont and to about 2% in Tennessee. [C104, C105, C106] It lowers the chance to below 1% in Virginia and in Nevada. [C107, C108]

The whole network. Figure 176 (next page, landscape) draws every unit and connection of the explained model, nested by part: shaded clusters are the parts of the model and darker boxes the parts inside them; circles are hidden units (h1, h2, ...); each cluster's rectangle is its output, and a dashed 'sum' stands for an addition the next unit does (drawn, not a unit of the model); the hexagon is the model's output. An input used by several parts is drawn under each of them (same label, dotted outline, the first copy marked with the number of copies): it is the same input, repeated for readability. Not used by the model (input columns with no connection): state: 20 of 51 levels, area code 408, area code 510, total evening minutes, total evening calls, total night-time minutes, total night-time calls, total night-time charge, total international minutes, total international calls, total international charge.

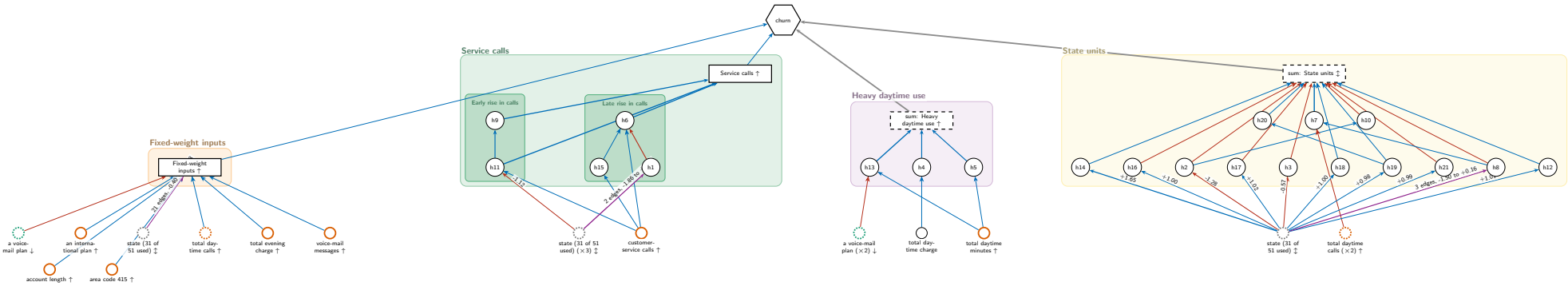


Figure 176: Every unit and connection of the explained model, nested by part. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Worth checking against what is known. Worth checking: for customers in Rhode Island the fixed-weight inputs lower the chance to below 1% while the state units raise it to about 12%, so overall it is about 1%. [C061, C104, C025] Worth checking: for customers in Nevada the fixed-weight inputs raise the chance to about 4% while the state units lower it to below 1%, so overall it is about 2%. [C058, C108, C018] Worth checking: the state moves the chance by several points (Washington about 5%, Maine about 3%, Hawaii below 1%), based on 72 to 86 training customers per state. [C004, C006, C007] Worth checking: the model treats a voice-mail plan as lowering the chance of leaving but more voice-mail messages as raising it. [C008, C002] Worth checking: the early rise with customer-service calls does not happen for customers in Nebraska. [C093]

4.2 The model, read from the top down

Each part of the explanation in turn, from the whole model down to the smallest parts: what it does for the prediction (hypothesis), what was checked (evidence), a diagram one level down, and where to read on. The technical record of each part is in "Technical detail" at the end.

4.2.1 The whole model

Figure 177 shows what feeds this part; Figure 178 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

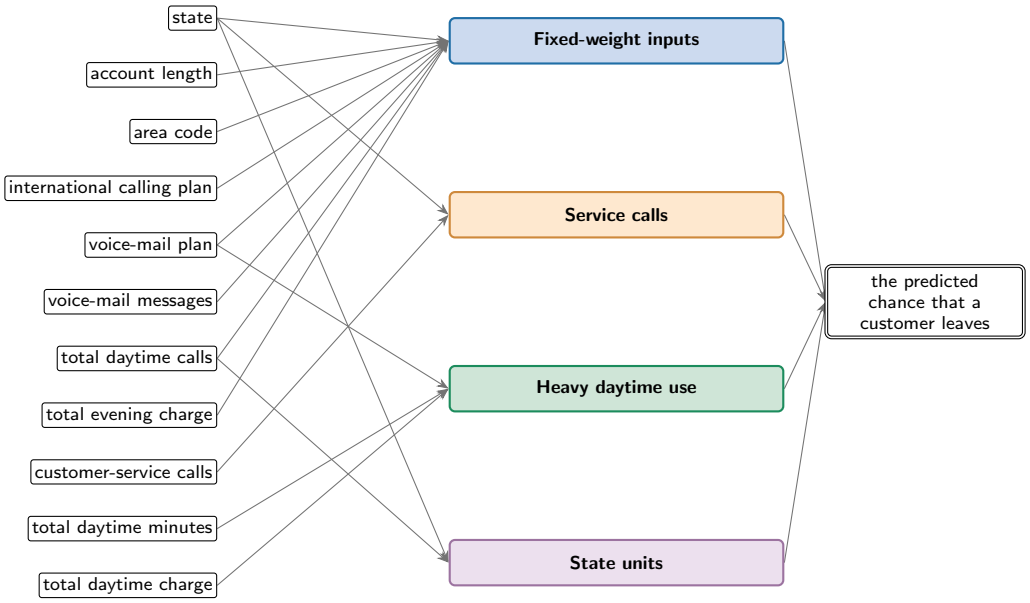


Figure 177: The whole model: what feeds it, one level down.

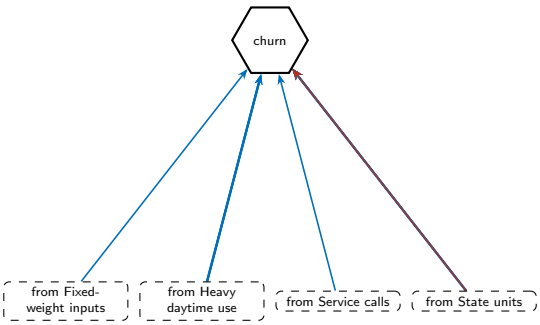


Figure 178: The whole model: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. The model adds up four parts into one score: fixed-weight inputs, service calls, heavy daytime use and state units. [C054]

Evidence. It is made up of these four parts (each explained below); the score is their sum, which was checked on all 4000 training customers. [C054] Changing one input at a time across the whole model, an international

plan takes a typical customer's chance of leaving from about 1% to about 18%, and a voice-mail plan lowers it to below 1%. [C001, C008] More voice-mail messages raise it steadily, from about 1% at 0 to about 6% at 32 messages, flattening out near the bottom because the predicted chance cannot go much lower. [C002] More daytime minutes raise it in an uneven way, from about 1% at 112 minutes to about 6% at 249 minutes, flattening out near the bottom because the predicted chance cannot go much lower. [C003] The model ignores evening calls, evening minutes, international calls, the international charge, international minutes, night-time calls, the night-time charge and night-time minutes. [C046, C047, C048, C049, C050, C051, C052, C053]
Evidence plots: Figure 179, Figure 180.

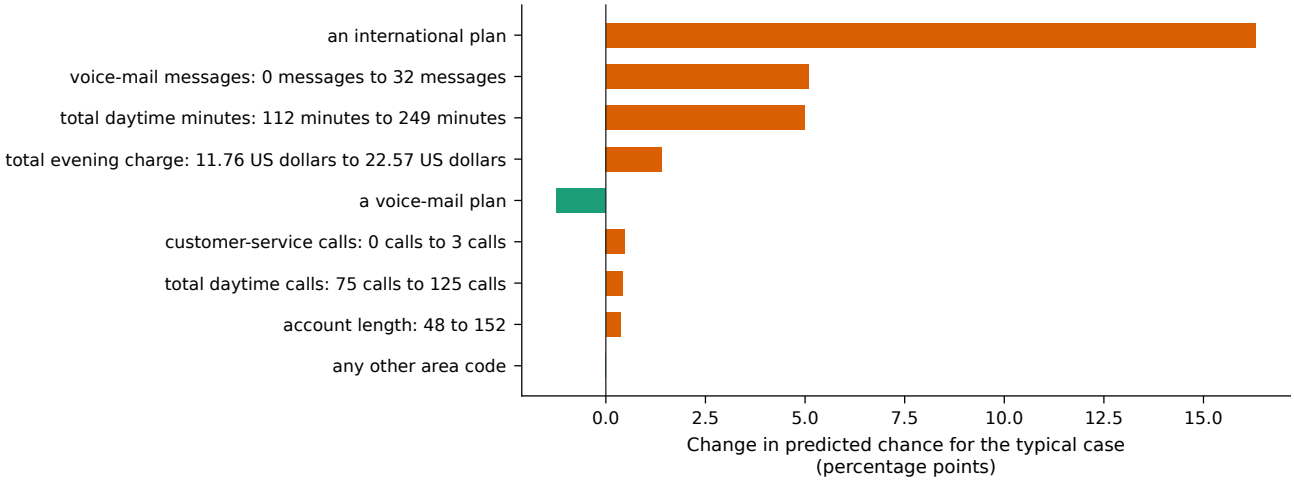


Figure 179: The whole model: Each input's change in the predicted value for the typical case across the whole model, largest first (orange: raises it, teal: lowers it).

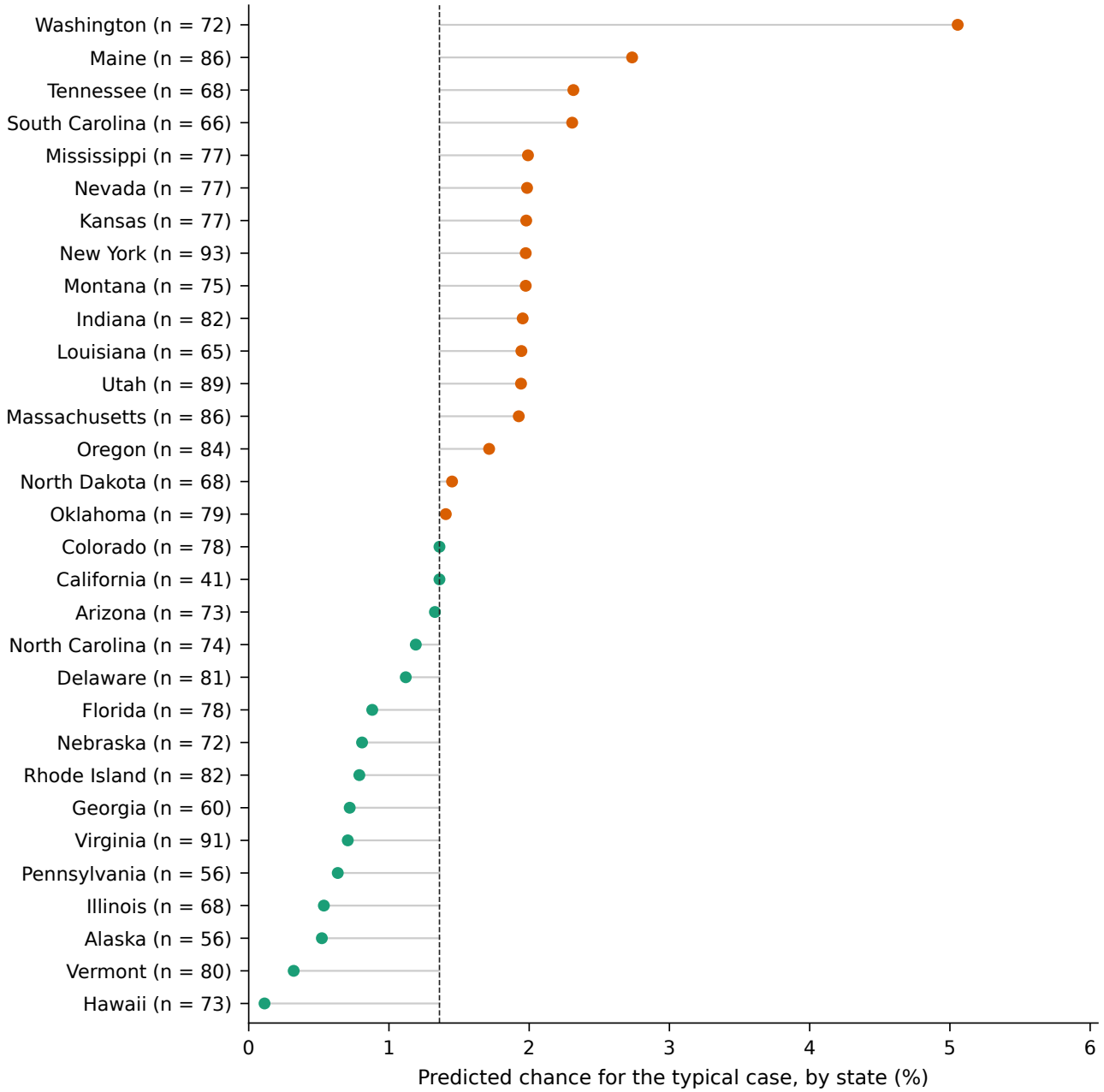


Figure 180: The whole model: Every level of state the model reads: the predicted value for the typical case at that level (dashed: the reference level); n = training customers; grey: few customers or not supported.

Dive deeper: Fixed-weight inputs (5.2.7); Service calls (4.2.3); Heavy daytime use (4.2.6); State units (4.2.7). Technical detail: T.1.

4.2.2 Fixed-weight inputs

Part of: The whole model (4.2.1).
Figure 181 shows what feeds this part; Figure 182 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

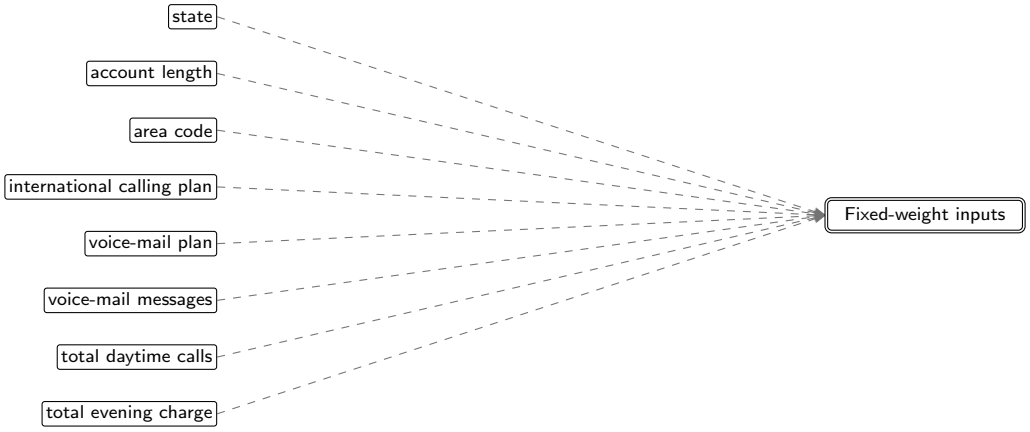


Figure 181: Fixed-weight inputs: what feeds it, one level down.

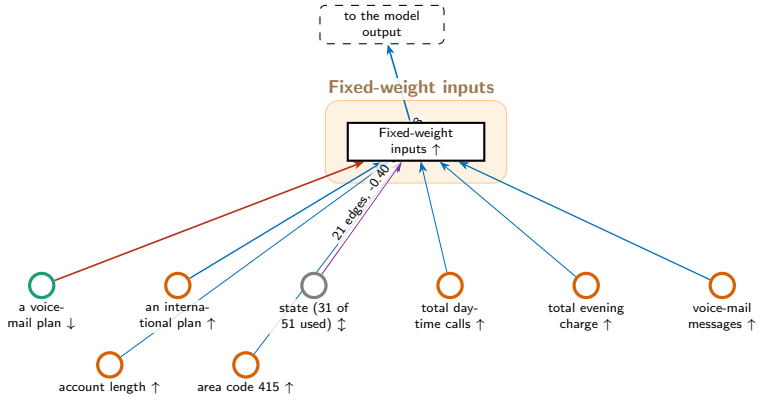


Figure 182: Fixed-weight inputs: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part adds up most inputs with fixed weights: an international plan, voice-mail messages and the evening charge raise the chance of leaving, steadily for the numbers, and a voice-mail plan lowers it. [C055, C056, C059, C063]

Evidence. Through it, an international plan takes a typical customer from about 1% to about 18%, and a voice-mail plan lowers the chance to below 1%. [C055, C063] More voice-mail messages raise it steadily, from about 1% at 0 to about 6% at 32 messages, flattening out near the bottom because the predicted chance cannot go much lower. [C056] A higher evening charge also raises it steadily, from about 1% at 11.76 US dollars to about 2% at 22.57 US dollars, flattening out near the bottom because the predicted chance cannot go much lower. [C059] Through it, the state moves the chance from about 1% in West Virginia to about 5% in Washington and about 4% in Nevada, and to below 1% in Rhode Island, Hawaii and Vermont (72 to 82 training customers per state). [C057, C058, C061, C062, C064] Its formula reproduces the part exactly on all 4000 training customers. [C083]
Evidence plots: Figure 183, Figure 184, Figure 185, Figure 186.

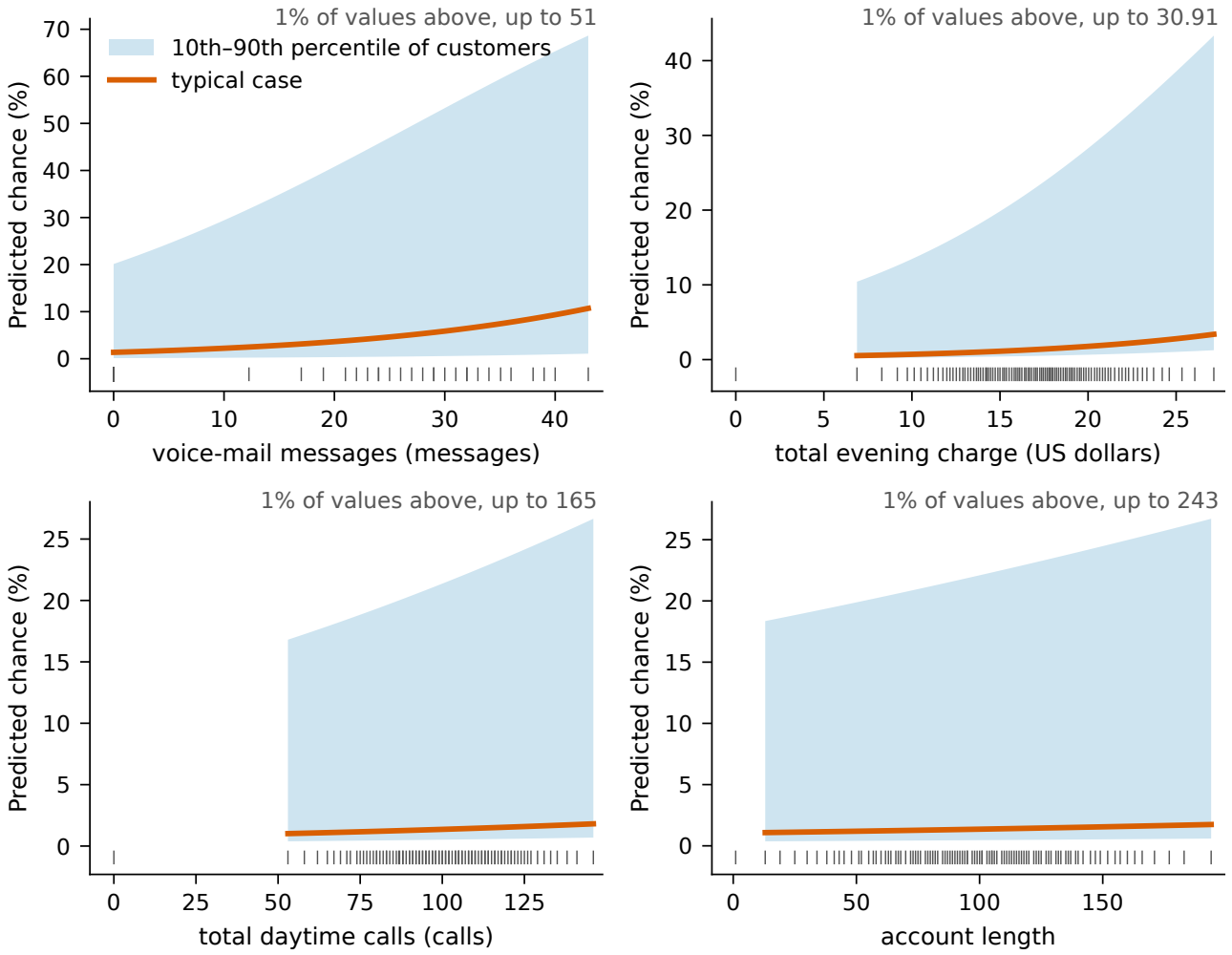


Figure 183: Fixed-weight inputs: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

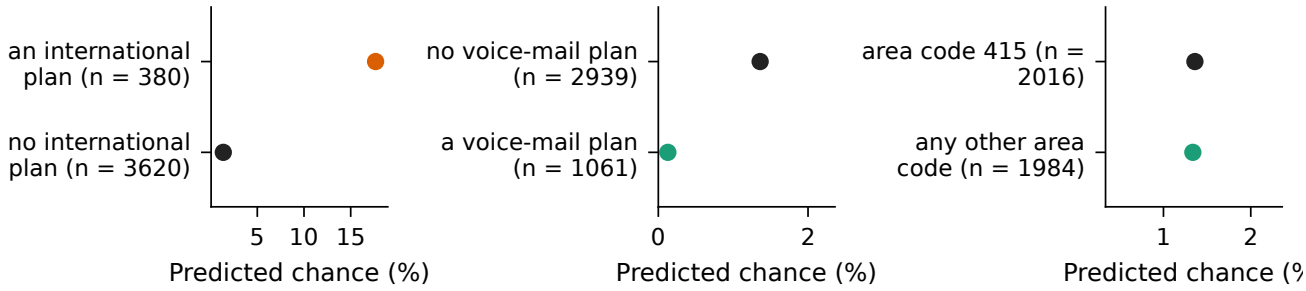


Figure 184: Fixed-weight inputs: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

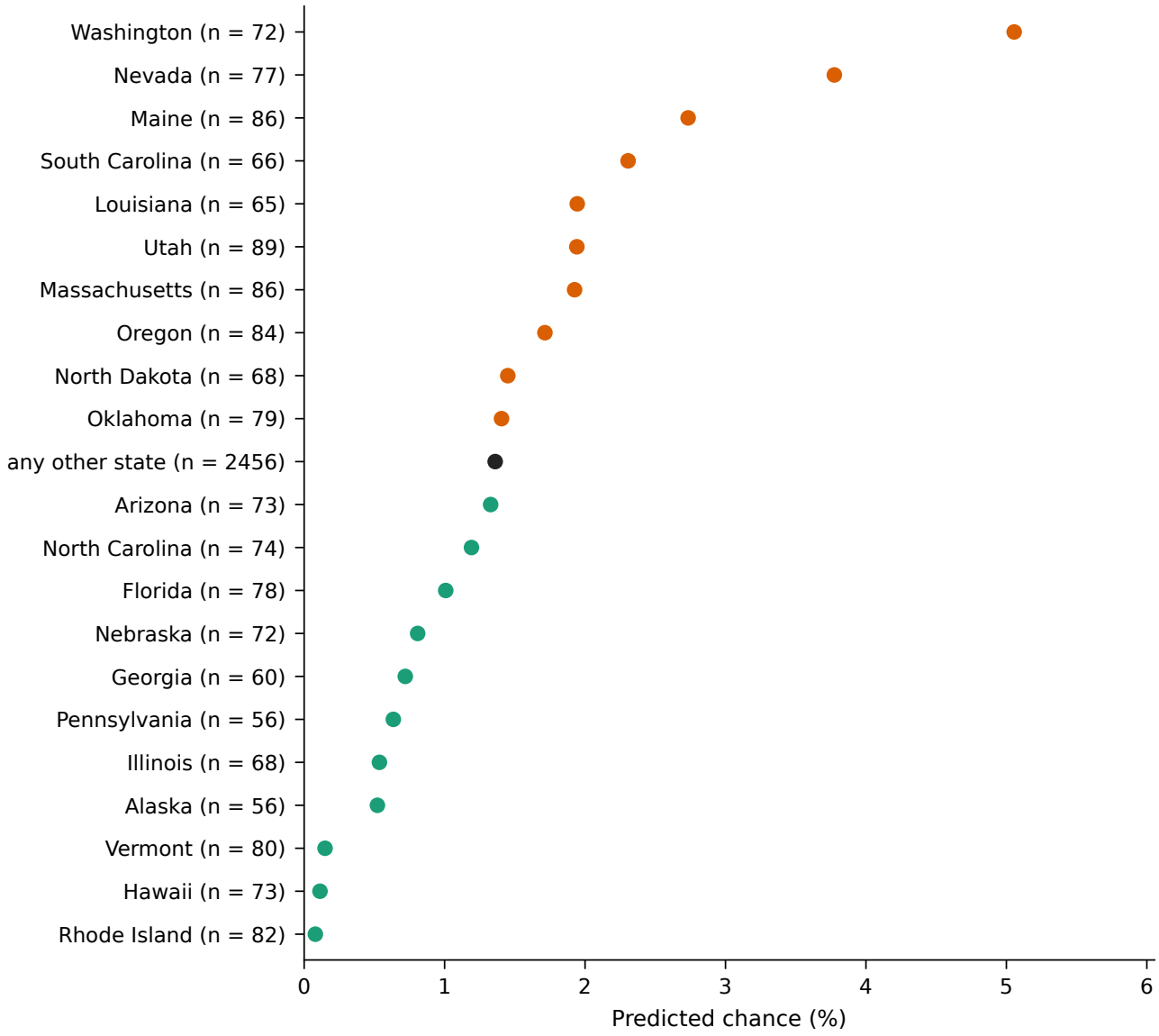


Figure 185: Fixed-weight inputs: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

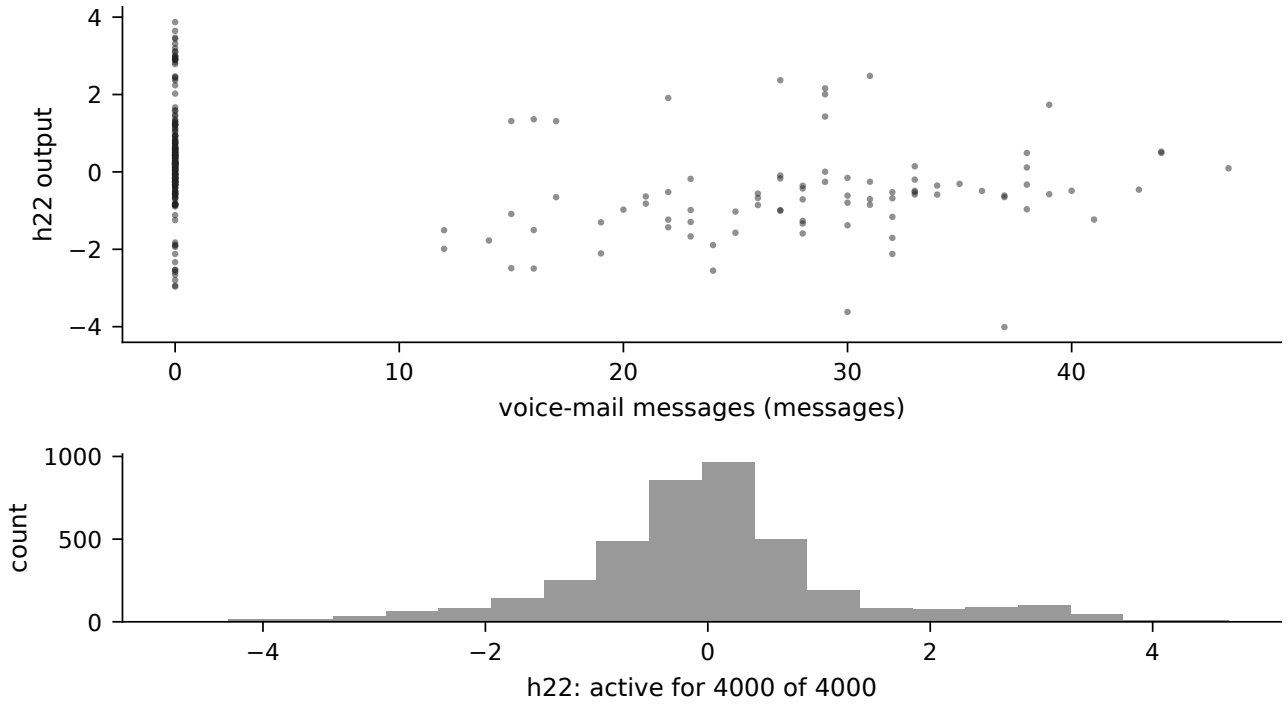


Figure 186: Fixed-weight inputs: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.7.

4.2.3 Service calls

Part of: The whole model (4.2.1).
Figure 187 shows what feeds this part; Figure 188 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

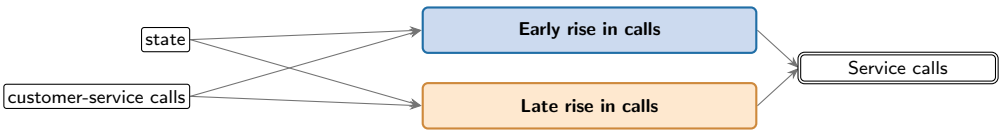


Figure 187: Service calls: what feeds it, one level down.

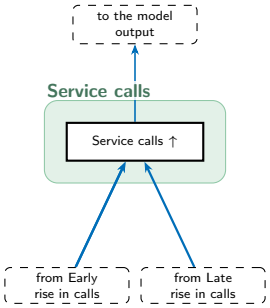


Figure 188: Service calls: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part raises the chance of leaving with calls to customer service, slowly at first, then faster from about 3 calls upwards; it adds up two parts. [C090, C089]

Evidence. It is made of two parts that are added: an early rise and a late rise in calls to customer service. [C090] From about 3 calls upwards it adds to the chance slowly at first, then faster: for a typical such customer, 4 calls

take it from about 1% to about 10%, flattening out near the bottom because the predicted chance cannot go much lower. [C089] Up to 3 calls the chance rises slowly at first, then faster, from about 1% to about 2% for a typical customer, flattening out near the bottom because the predicted chance cannot go much lower. [C084] That early rise happens only outside Nebraska, California and Colorado: in Nebraska the chance stays at about 1% from 0 to 3 calls, against a rise from about 1% to about 2% elsewhere, slowly at first, then faster, flattening out near the bottom because the predicted chance cannot go much lower. [C088]
Evidence plots: Figure 189, Figure 190, Figure 191, Figure 192, Figure 193.
Figure 192 shows this part's own output, unit h23: it switches on at about 2.827 calls of customer-service calls, and is zero below.

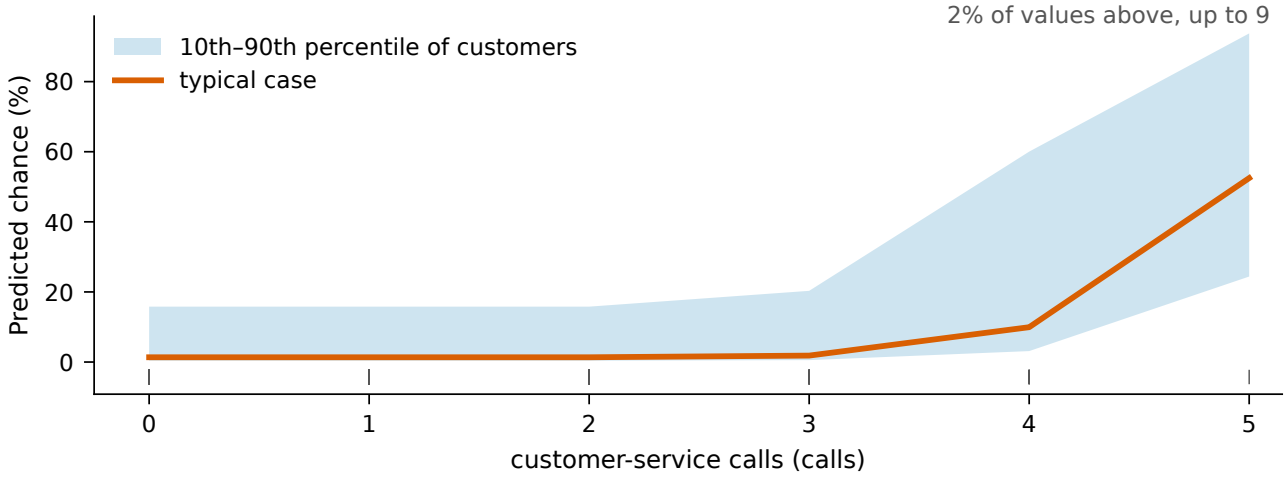


Figure 189: Service calls: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

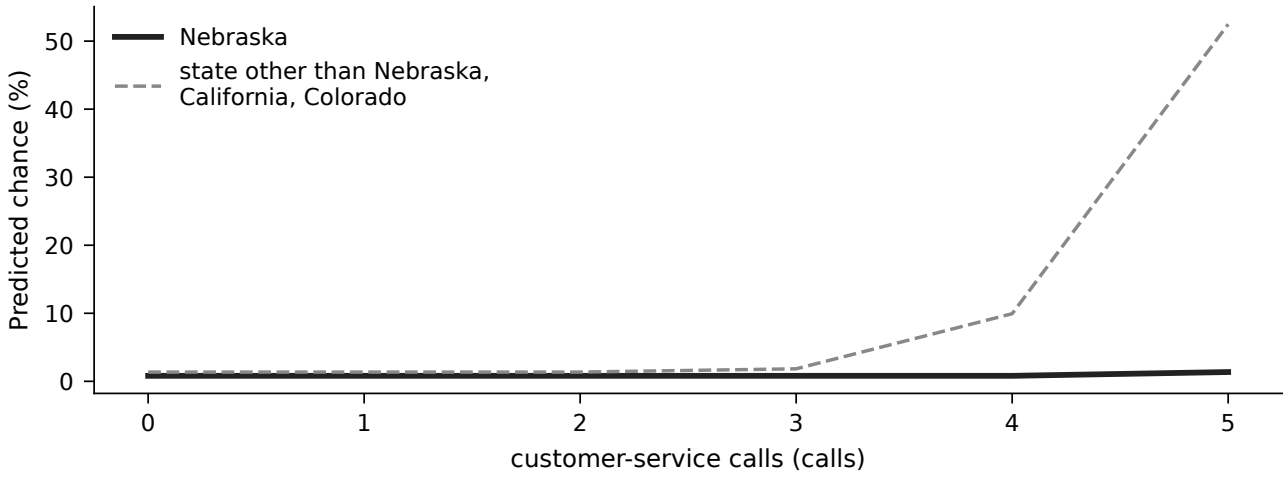


Figure 190: Service calls: How the effect differs between groups: one curve per group, for the typical case of each group.

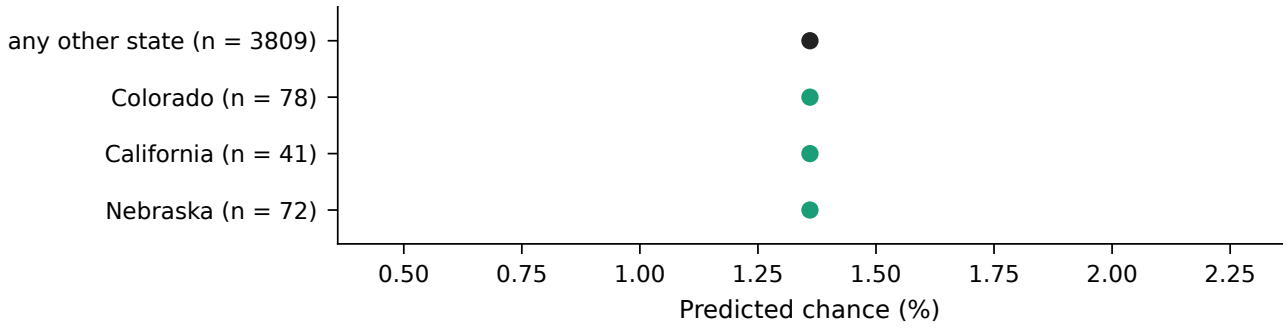


Figure 191: Service calls: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

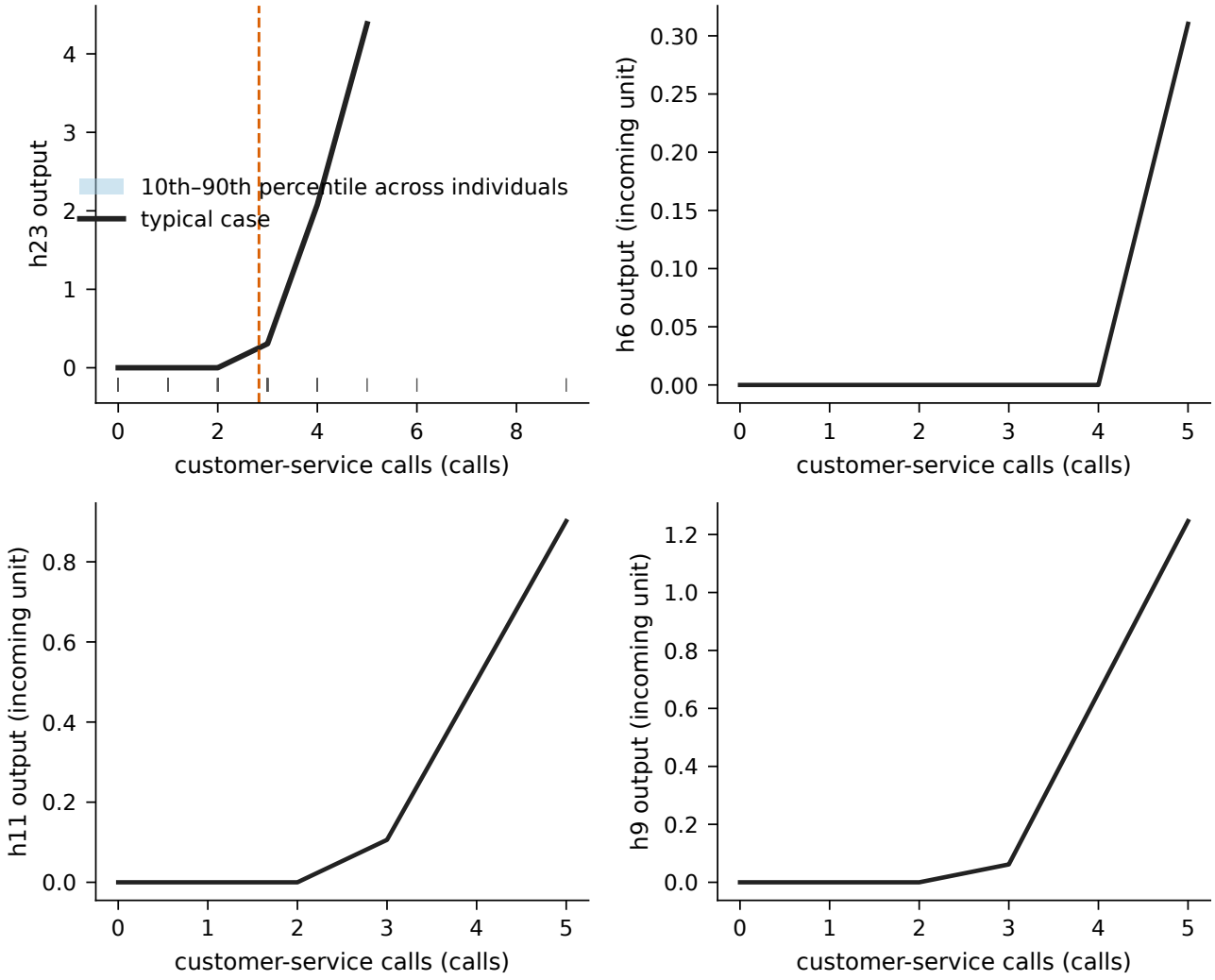


Figure 192: Service calls: This part's own output, unit h23 (combines 3 non-linear units), on its own scale (its activation), with every other input at the typical case; it is active for 831 of 4000. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

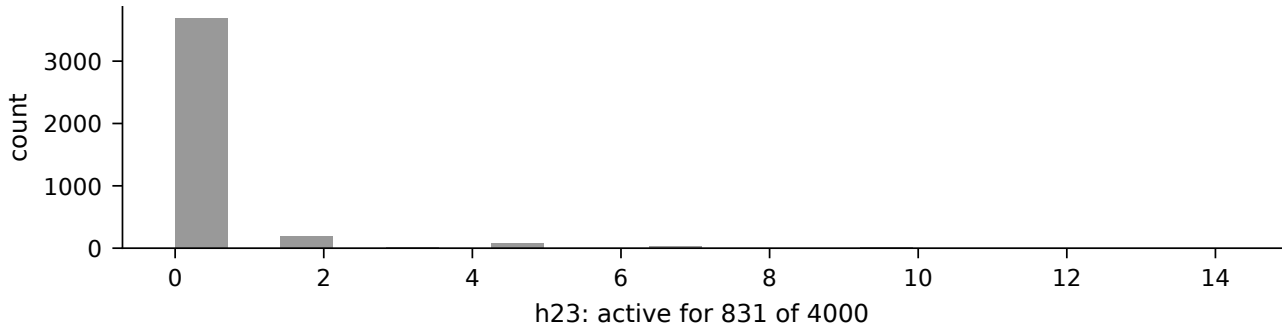


Figure 193: Service calls: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Dive deeper: Early rise in calls (5.2.1); Late rise in calls (5.2.2). Technical detail: T.3.

4.2.4 Early rise in calls

Part of: Service calls (4.2.3).
Figure 194 shows what feeds this part; Figure 195 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

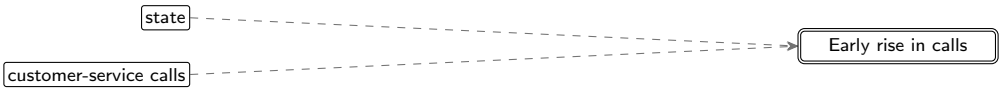


Figure 194: Early rise in calls: what feeds it, one level down.

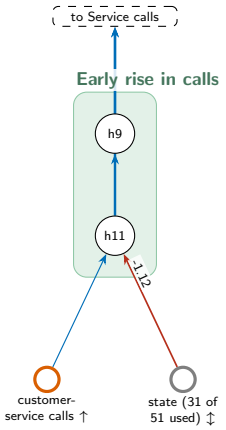


Figure 195: Early rise in calls: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part raises the chance of leaving from about 3 customer-service calls upwards, in an uneven way. [C094]

Evidence. It acts on 830 of 4000 training customers, 213 of whom left; for the typical customer it acts on, the chance is about 1% without it and about 2% with it. [C095] Through it, calls add to the chance only from about 3 calls upwards, in an uneven way: for a typical such customer, 4 calls take it from about 1% to about 10%, flattening out near the bottom because the predicted chance cannot go much lower. [C094] Up to 3 calls it raises the chance slowly at first, then faster, but only outside Nebraska: there from about 1% to about 2%, while in Nebraska it stays at about 1%, flattening out near the bottom because the predicted chance cannot go much lower. [C093] Its formula reproduces the part exactly on all 4000 training customers. [C096]
Evidence plots: Figure 196, Figure 197, Figure 198, Figure 199, Figure 200, Figure 201.
Figure 199 shows this part's own output, unit h11: it switches on at about 2.734 calls of customer-service calls, and is zero below.
Figure 200 shows this part's own output, unit h9: it switches on at about 2.896 calls of customer-service calls, and is zero below.

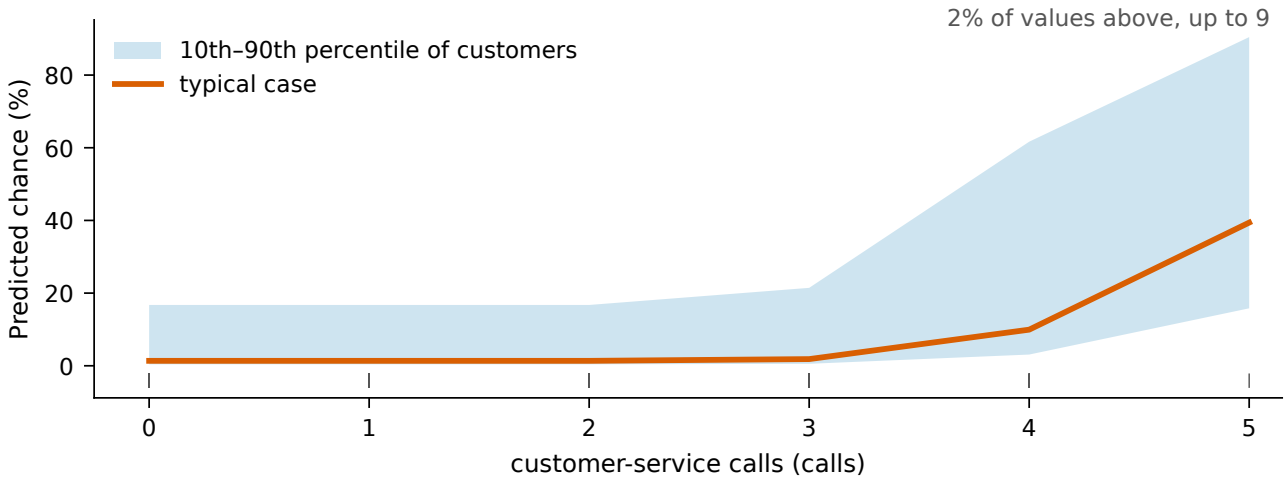


Figure 196: Early rise in calls: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

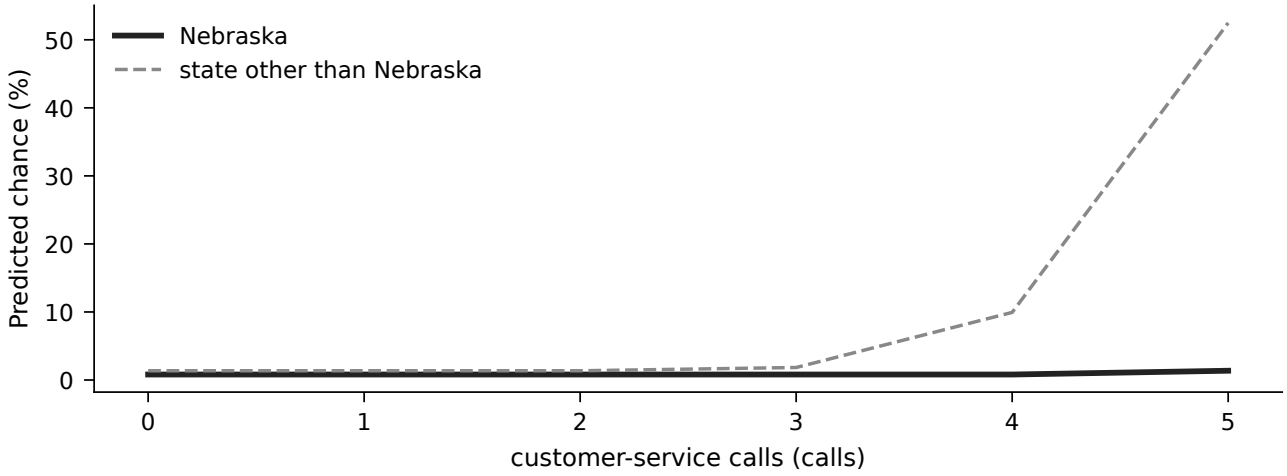


Figure 197: Early rise in calls: How the effect differs between groups: one curve per group, for the typical case of each group.

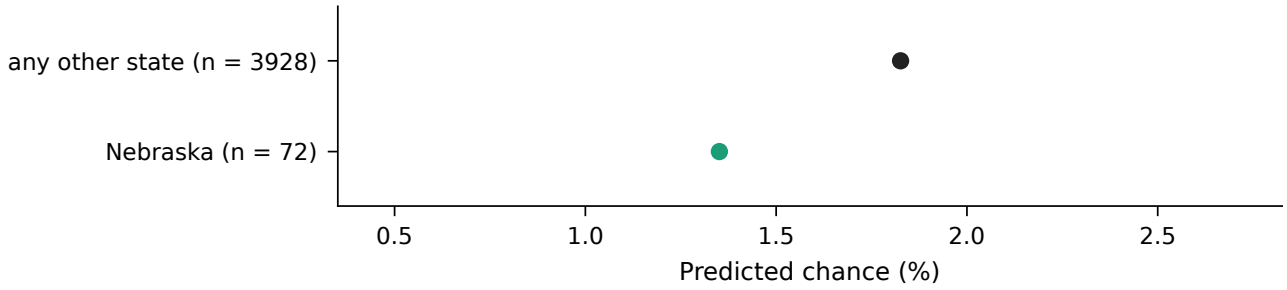


Figure 198: Early rise in calls: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

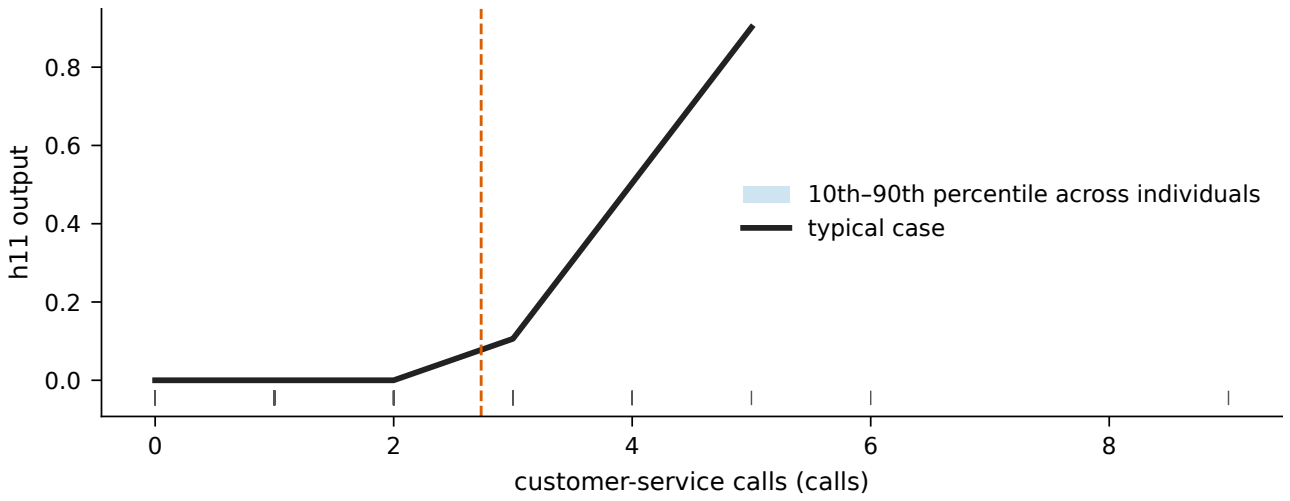


Figure 199: Early rise in calls: This part's own output, unit h11 (the output of 'Early rise in calls'), on its own scale (its activation), with every other input at the typical case; it is active for 830 of 4000. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

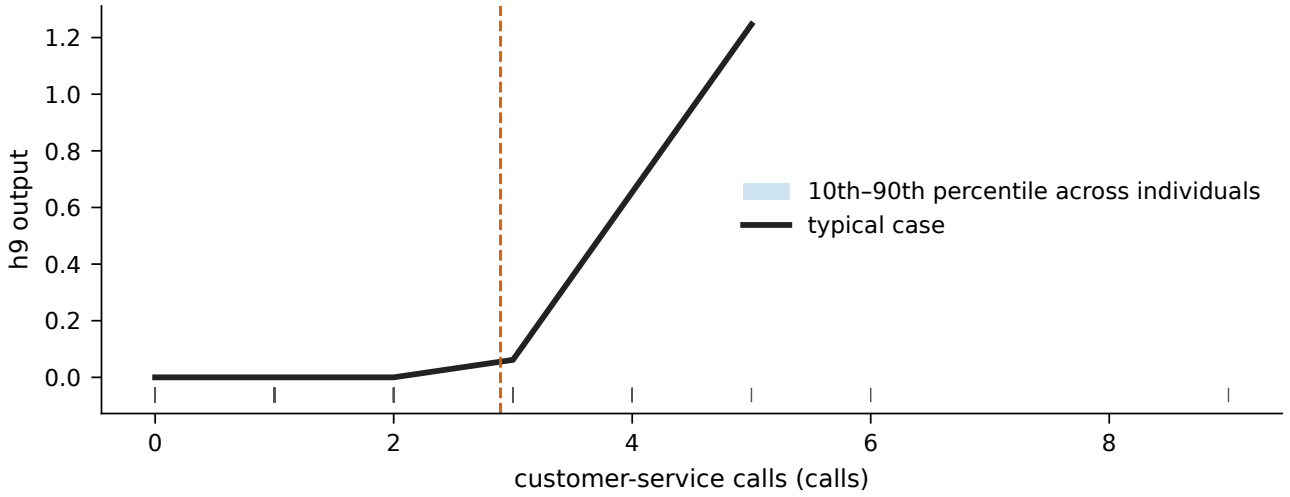


Figure 200: Early rise in calls: This part's own output, unit h9 (the output of 'Early rise in calls'), on its own scale (its activation), with every other input at the typical case; it is active for 830 of 4000. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

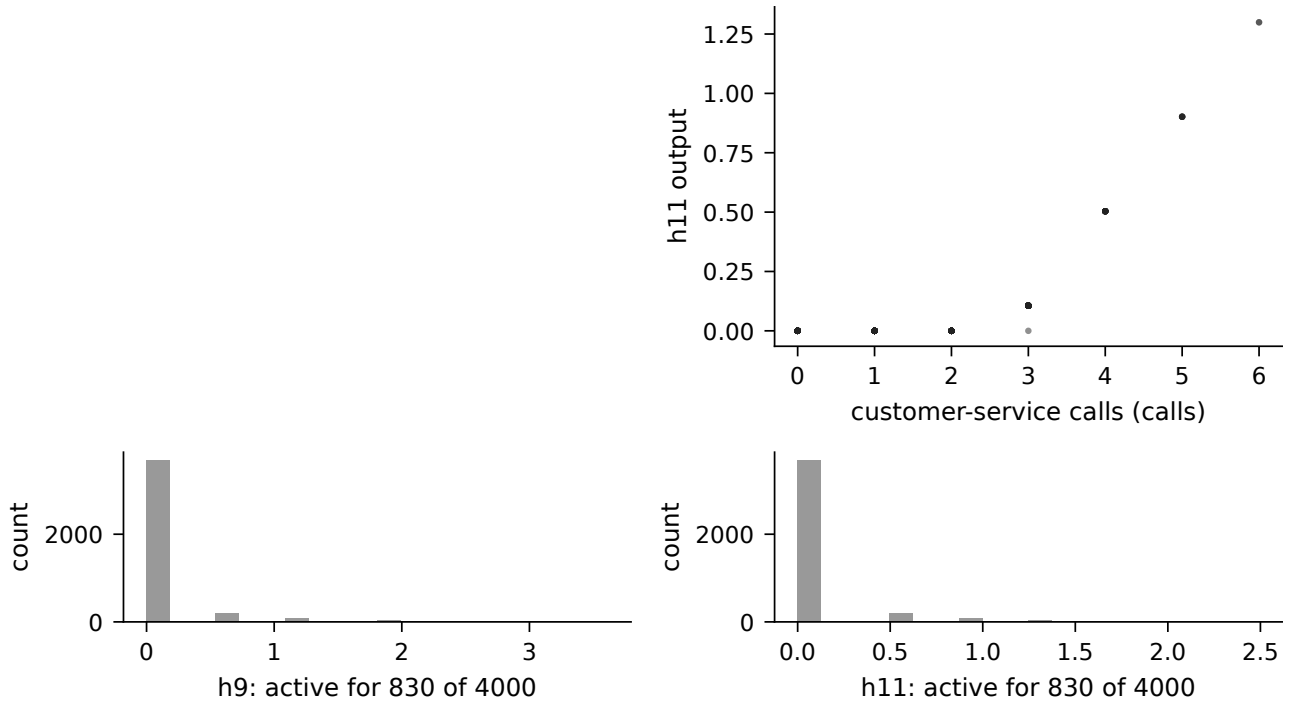


Figure 201: Early rise in calls: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.1.

4.2.5 Late rise in calls

Part of: Service calls (4.2.3).
Figure 202 shows what feeds this part; Figure 203 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

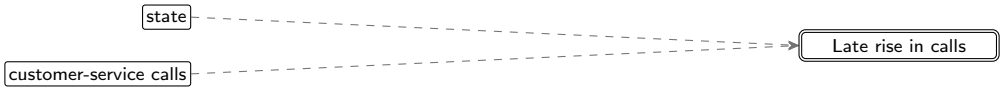


Figure 202: Late rise in calls: what feeds it, one level down.

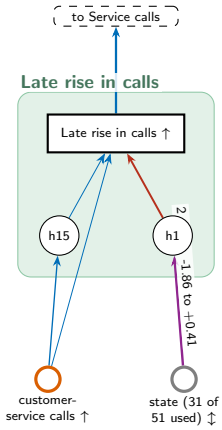


Figure 203: Late rise in calls: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part raises the chance of leaving for a small group of customers, more in Colorado and less in California. [C122, C117, C118]

Evidence. It acts on 134 of 4000 training customers, 74 of whom left; for the typical customer it acts on, the chance is about 39% without it and about 75% with it. [C122] For the typical customer it acts on, the chance through it is about 75% in Colorado and about 39% in California, against about 52% in other states. [C117, C118] Its formula reproduces the part exactly on all 4000 training customers. [C123]
Evidence plots: Figure 204, Figure 205, Figure 206, Figure 207, Figure 208.
Figure 207 shows this part's own output, unit h6: it switches on at about 5 calls of customer-service calls, and is zero below.

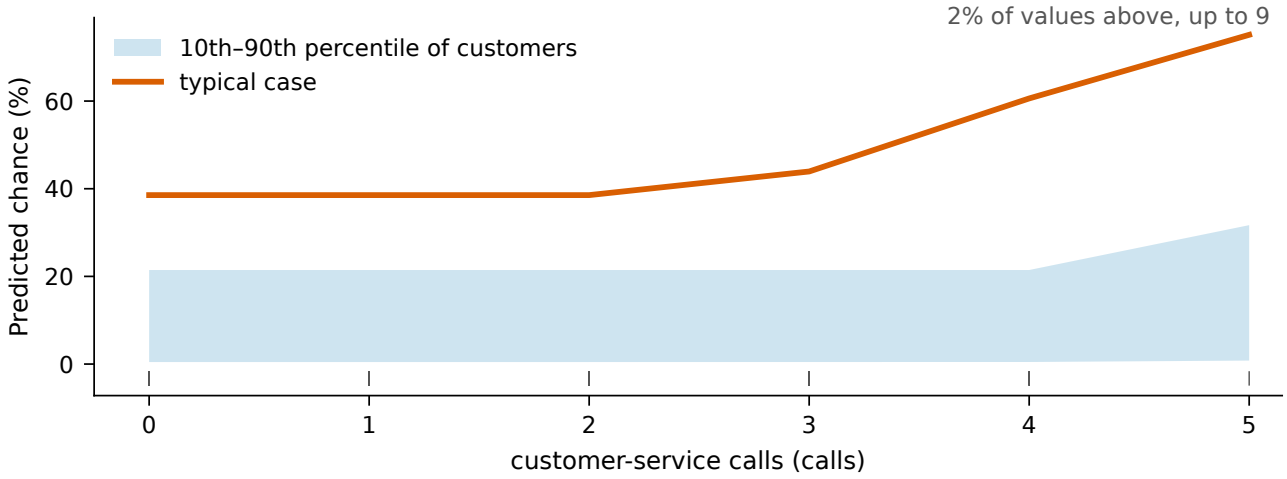


Figure 204: Late rise in calls: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

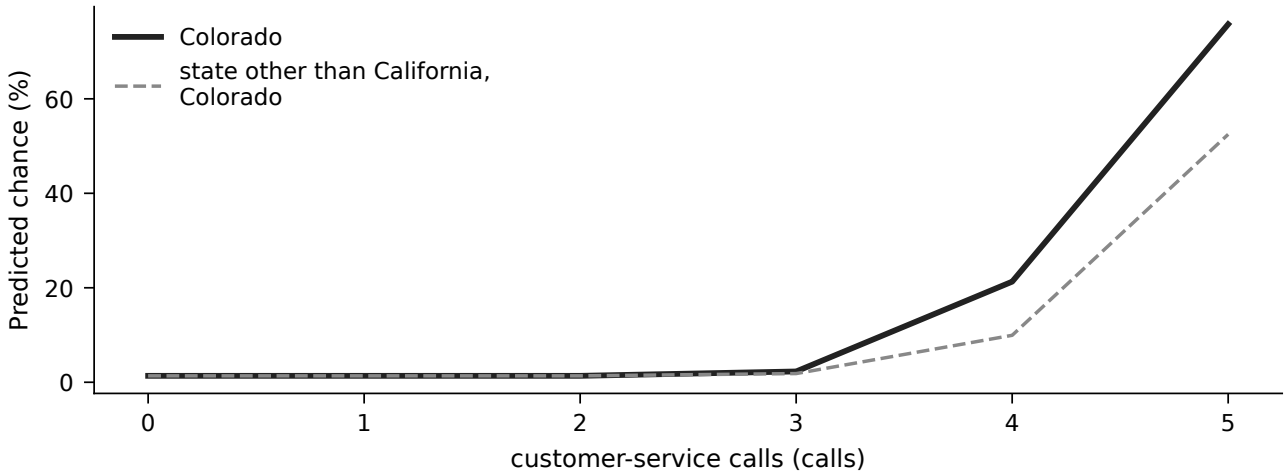


Figure 205: Late rise in calls: How the effect differs between groups: one curve per group, for the typical case of each group.

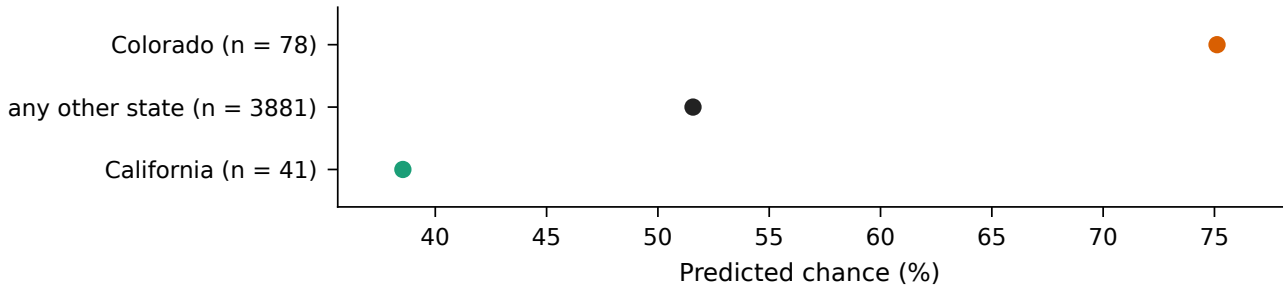


Figure 206: Late rise in calls: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

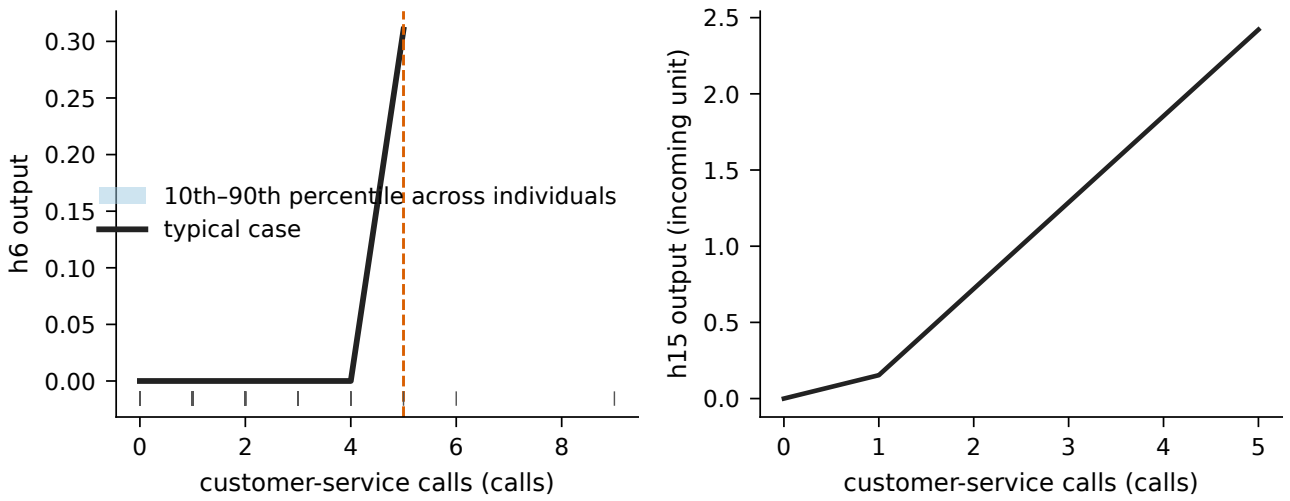


Figure 207: Late rise in calls: This part's own output, unit h6 (combines 2 non-linear units), on its own scale (its activation), with every other input at the typical case; it is active for 134 of 4000. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

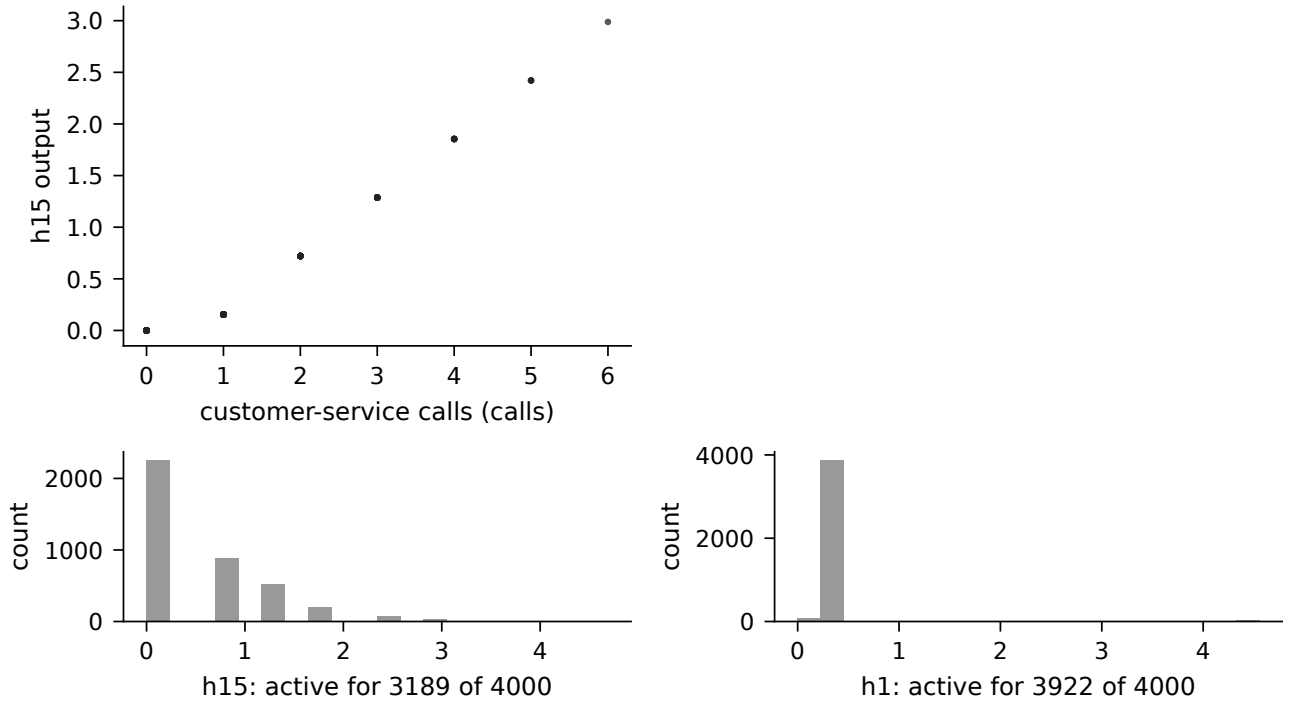


Figure 208: Late rise in calls: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.2.

4.2.6 Heavy daytime use

Part of: The whole model (4.2.1).
Figure 209 shows what feeds this part; Figure 210 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

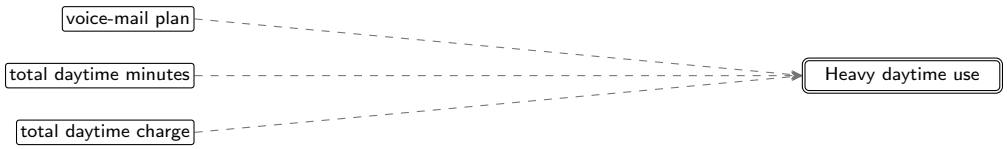


Figure 209: Heavy daytime use: what feeds it, one level down.

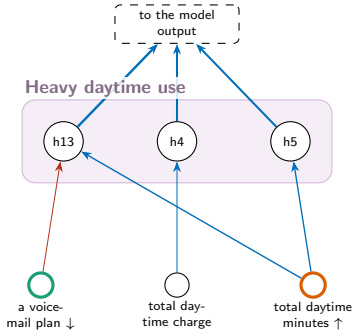


Figure 210: Heavy daytime use: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part raises the chance of leaving for heavy daytime users: daytime minutes add to it only above about 209 minutes, in an uneven way, and much less with a voice-mail plan. [C100, C099]

Evidence. This part acts on 1074 of 4000 training customers, 290 of whom left; for the typical customer it acts on, the chance is about 1% without it and about 4% with it. [C102] Daytime minutes add to the chance only once they rise above about 209 minutes, in an uneven way: for a typical such customer, from about 1% at 0 minutes to about 16% at 271 minutes, flattening out near the bottom because the predicted chance cannot go much lower. [C100] The rise is much larger without a voice-mail plan: from 112 to 249 daytime minutes the chance goes from about 1% to about 6% without one, but stays below 1% with one, in an uneven way, flattening out near the bottom because the predicted chance cannot go much lower. [C099] A voice-mail plan lowers it: for the typical customer it acts on, from about 4% to about 2%. [C098] Its formula reproduces the part exactly on all 4000 training customers. [C103]
Evidence plots: Figure 211, Figure 212, Figure 213, Figure 214, Figure 215, Figure 216.
Figure 214 shows this part's own output, unit h13: it switches on at about 209 minutes of total daytime minutes, and is zero below.
Figure 215 shows this part's own output, unit h5: it switches on at about 223.9 minutes of total daytime minutes, and is zero below.

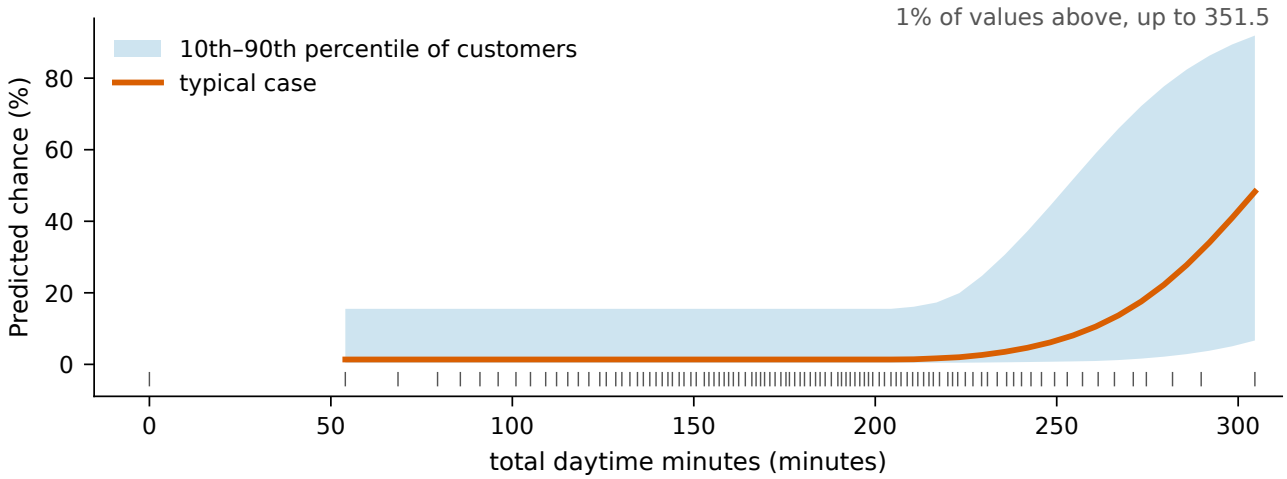


Figure 211: Heavy daytime use: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

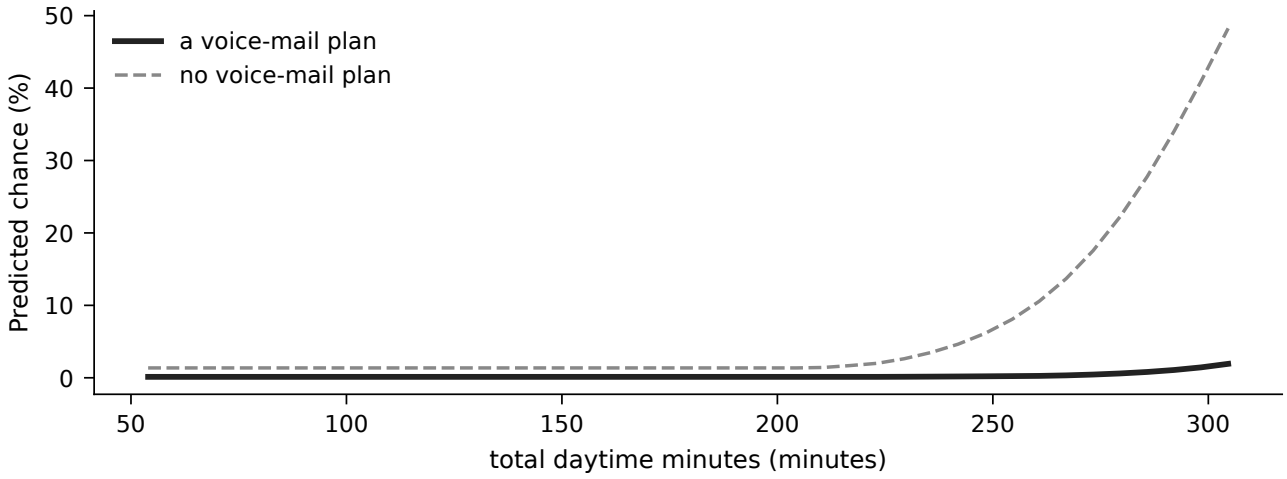


Figure 212: Heavy daytime use: How the effect differs between groups: one curve per group, for the typical case of each group.

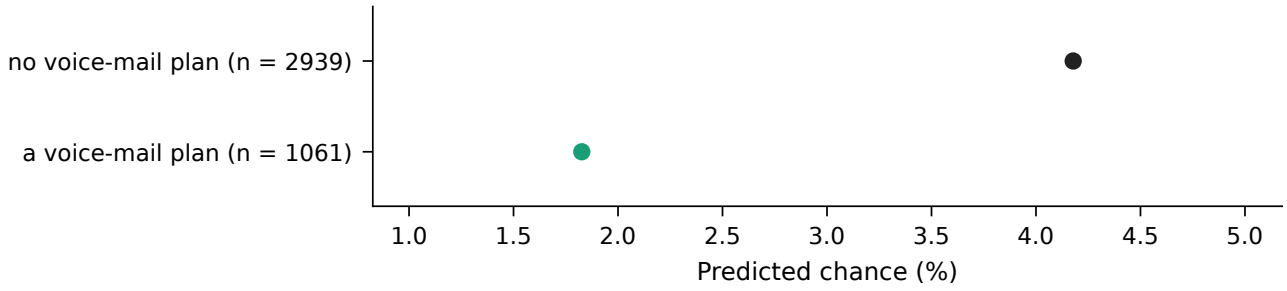


Figure 213: Heavy daytime use: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

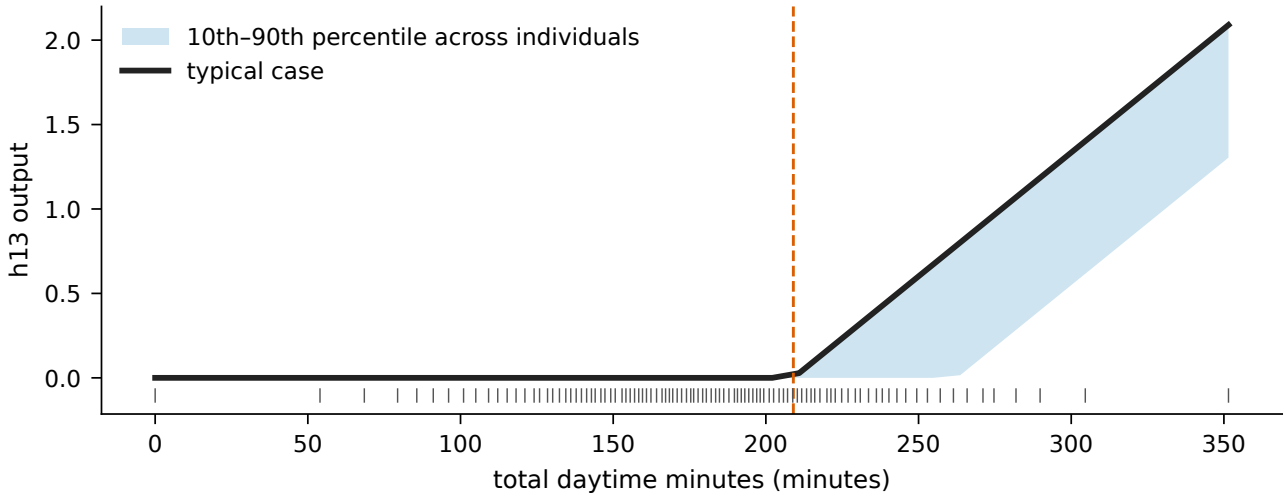


Figure 214: Heavy daytime use: This part's own output, unit h13 (the output of 'Heavy daytime use'), on its own scale (its activation), with every other input at the typical case; it is active for 929 of 4000. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

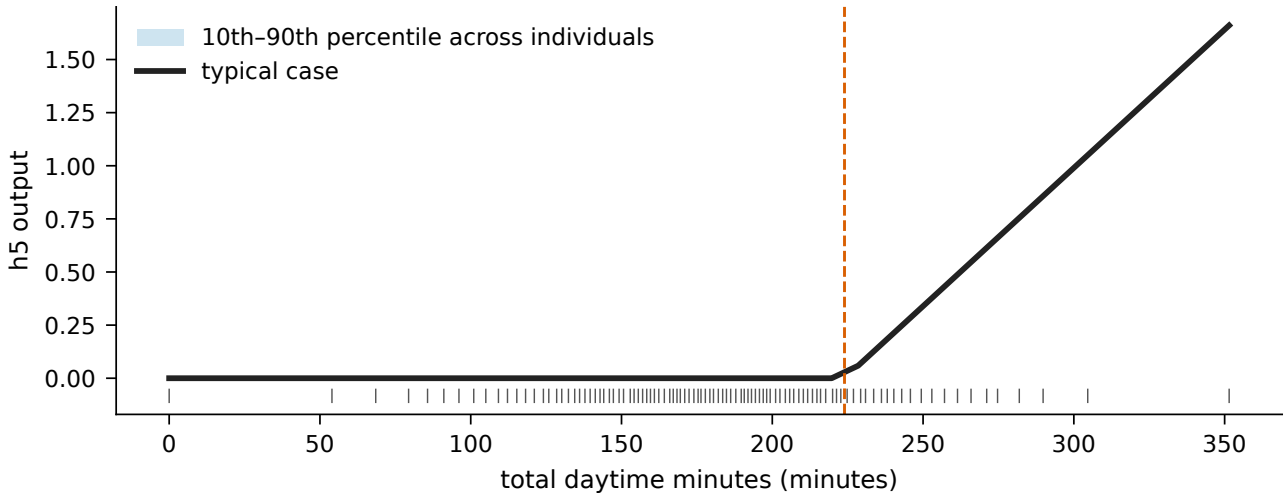


Figure 215: Heavy daytime use: This part's own output, unit h5 (the output of 'Heavy daytime use'), on its own scale (its activation), with every other input at the typical case; it is active for 818 of 4000. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

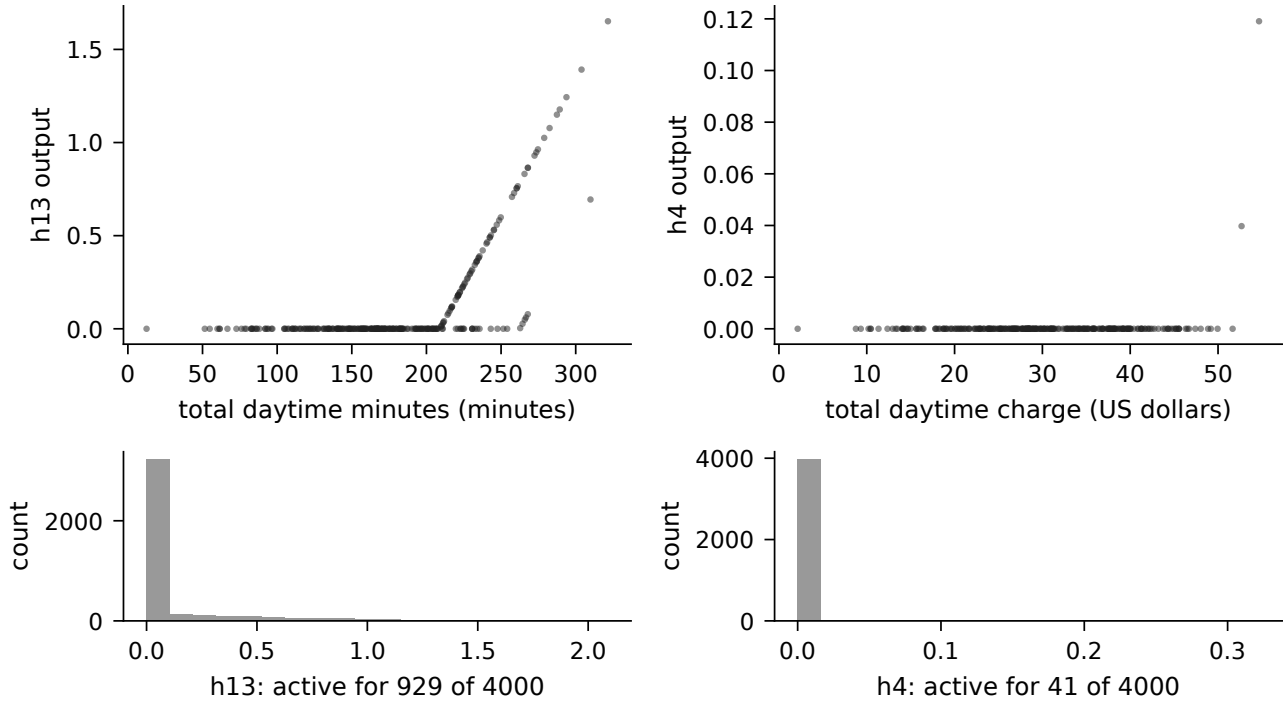


Figure 216: Heavy daytime use: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.6.

4.2.7 State units

Part of: The whole model (4.2.1).
Figure 217 shows what feeds this part; Figure 218 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).



Figure 217: State units: what feeds it, one level down.

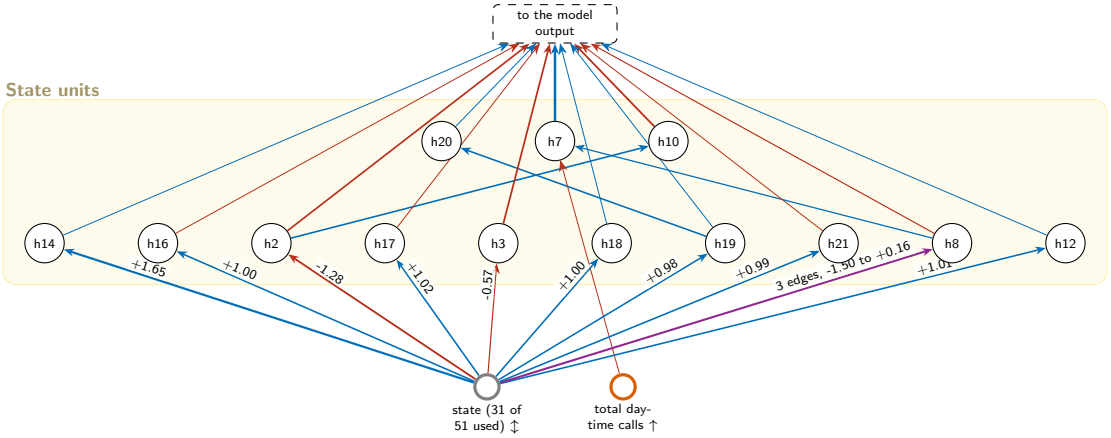


Figure 218: State units: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part reads only the state: it raises the chance of leaving in Rhode Island, Vermont and Tennessee, and lowers it in Virginia and Nevada. [C104, C105, C106, C107, C108]

Evidence. This part reads only the state: it raises a typical customer's chance from about 1% to about 12% in Rhode Island, to about 3% in Vermont and to about 2% in Tennessee. [C104, C105, C106] It lowers the chance to below 1% in Virginia and in Nevada. [C107, C108] Its formula reproduces the part exactly on all 4000 training customers. [C116]
Evidence plots: Figure 219, Figure 220.

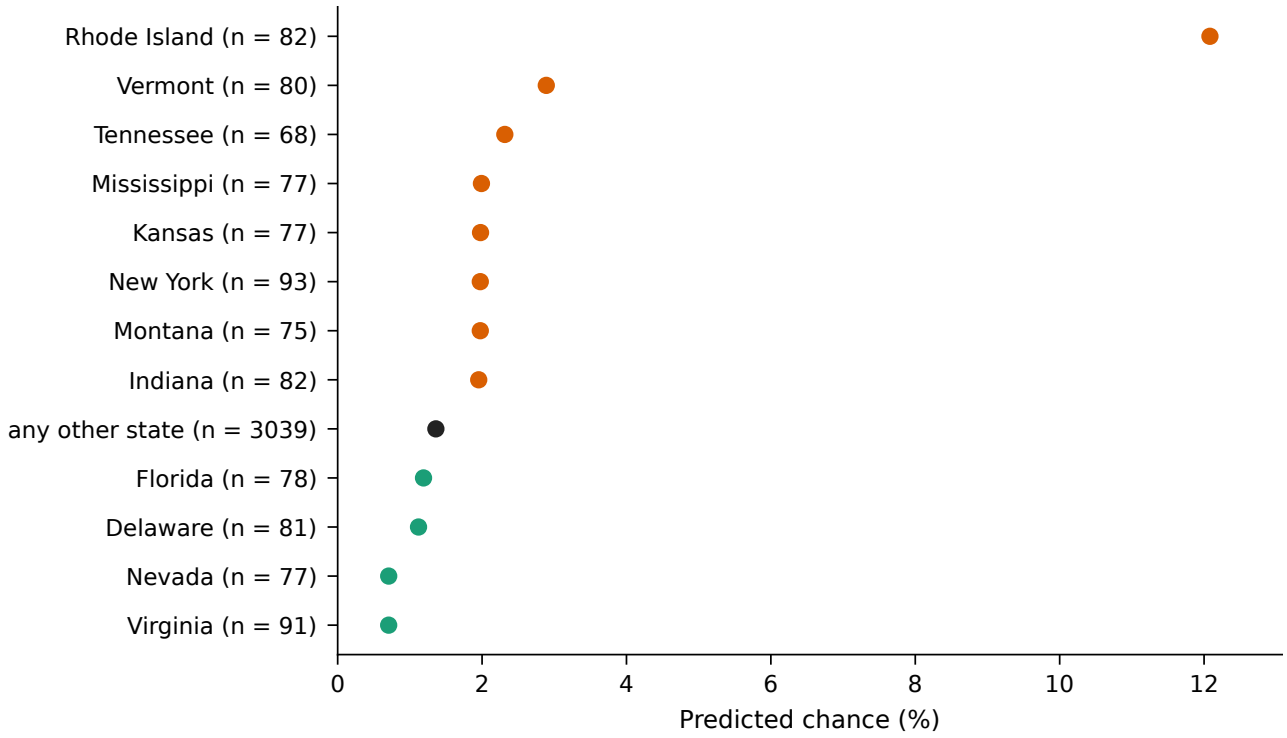


Figure 219: State units: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

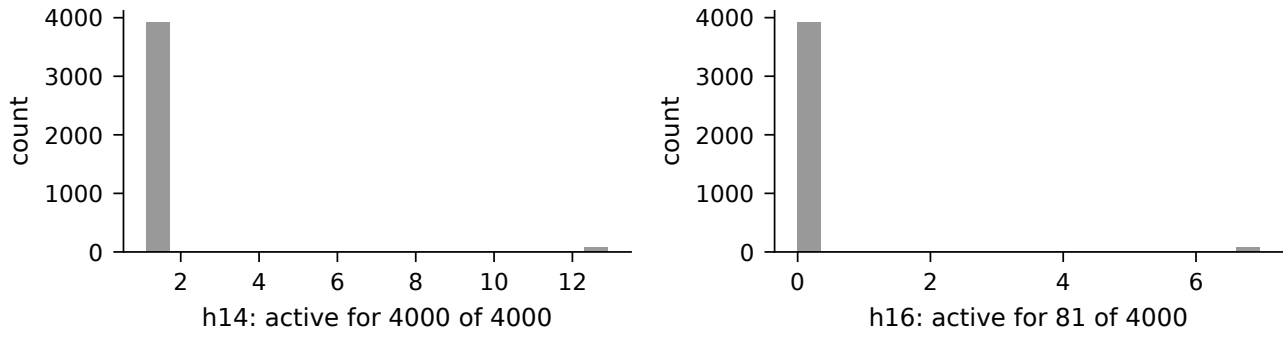


Figure 220: State units: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.7.

4.3 Not supported by the model

- C101: threshold claim about total daytime charge in part ann_85: not supported by the model (the effect past the fold is too small or not consistent across patients).
- C121: threshold claim about number of calls to customer service in part ann_83: not supported by the model (the effect past the fold is too small or not consistent across patients).

Technical detail: churn100-s4-c3

The record behind each part: the original hypothesis, the closed form, and every claim with its estimate (log-odds and odds ratio), range across patients, n and test check.

T.1 The whole model

Annotation ann_88 (churn_logit). Reads: state, account length, area code, international calling plan, voice-mail plan, number of voice-mail messages, total daytime minutes, total daytime calls, total daytime charge, total evening minutes, total evening calls, total evening charge, total night-time minutes, total night-time calls, total night-time charge, total international minutes, total international calls, total international charge, number of calls to customer service.

Original hypothesis (unchanged). Root: logit P(churn) = -1.35 + linear_backbone (identity_lin) + service_calls (identity_svc) + 1.48*3118 + 1.50*4000 + 1.89*10131 (day_usage_hinges) + the state_offsets units (-1.11*1295 - 1.25*7872 - 1.17*1390 - 0.56*5786 + small terms; 5533 is never active), purely additive in the logit; the sigmoid is the only coupling. 30 inputs are unconnected (international minutes, calls and charge, night minutes/charge/calls, evening minutes and calls, area codes 408/510 and 20 state indicators) and have zero effect. Train AUC 0.8884, accuracy 0.8795; the mean predicted churn is 0.082 against a base rate of 0.1415 (the model ranks well but under-predicts). Performance sources (train): all children replaced by least-squares linear fits gives AUC 0.8352 (an L2 logistic regression refit on all 71 inputs gets 0.8433); restoring day_usage_hinges alone gives 0.8713, service_calls alone 0.8657, both 0.8884; the state units restore nothing (exactly linear). So the gain over linear (+0.053) comes from the day-usage and service-call thresholds, which are complementary. Rare-category switches: state:11 -2.50 logit (73 rows, empirical -1.35); state:46 -1.46 net (80 rows, -0.91); state:47 +1.35 (72 rows, +0.92); state:0 -0.97 (56, -0.76); state:14 -0.94 (68, -0.53); state:29 -0.53 direct plus switching off the early service-call hinge (mean -0.78, down to -6.15 for many-call customers; 72 rows, -0.79); state:39 -0.55 net (82, -0.93: direct -2.85 mostly cancelled by the state units); state:33 +0.38 net (direct +1.05, unit -0.68); state:18 +0.36 (65 rows, empirical -0.48: opposite sign); state:4 switches off the late chain (41 rows, empirical +0.80).

Closed form. Read through its parts (their formulas are in their own sections).

Split inputs. To annotate the parts, the explanation split these inputs into copies (a refactoring step that does not change what the model computes); the diagrams draw each input once: customer-service calls (2 copies); total daytime calls (2 copies); a voice-mail plan (2 copies).

State: every level the model reads. Whole-model change in the predicted chance for a typical customer when only the state changes from the reference; n = training customers in that level.

Claim	From
C004	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C006	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C010	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C011	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C017	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C018	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C019	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C020	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C021	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C022	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C023	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C024	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C026	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C032	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C035	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C036	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C039	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C040	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C037	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C034	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C033	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C028	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C027	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C025	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C016	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C015	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C014	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C013	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C012	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C009	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C007	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C001	direction: international calling plan: no international plan to an international plan (whole model) *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 18%	2.74	15.6	2.74 to 2.74	3620, 380	agrees; AUC gain 0.066 [0.035, 0.098]	yes
C002	direction: voice-mail messages: 0 messages to 32 messages (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 6%	1.61	5.00	1.61 to 1.61	4000	agrees; AUC gain 0.014 [0.0002, 0.029]	yes
C003	direction: total daytime minutes: 112 minutes to 249 minutes (whole model) *shape: accelerating; 100% of patients same; fallback; risk saturates near 0%*	1% to 6%	1.30	3.67	0.49 to 1.59	4000	agrees; AUC gain 0.074 [0.044, 0.105]	yes
C004	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Washington (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 5%	1.35	3.86	1.35 to 1.35	1693, 72	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C005	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	1.01	2.73	1.01 to 1.01	4000	agrees; AUC gain 0.014 [0.003, 0.026]	yes
C006	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Maine (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 3%	0.71	2.04	0.71 to 0.71	1693, 86	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C007	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Hawaii (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to <1%	-2.50	0.08	-2.50 to -2.50	1693, 73	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C008	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (whole model) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	1% to <1%	-2.62	0.07	-3.50 to -2.38	2939, 1061	agrees; AUC gain 0.049 [0.028, 0.072]	yes
C009	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Vermont (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to <1%	-1.46	0.23	-1.46 to -1.46	1693, 80	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C010	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Tennessee (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 2%	0.54	1.72	0.54 to 0.54	1693, 68	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C011	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to South Carolina (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 2%	0.54	1.71	0.54 to 0.54	1693, 66	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C012	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Alaska (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	-0.97	0.38	-0.97 to -0.97	1693, 56	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C013	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Illinois (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	-0.94	0.39	-0.94 to -0.94	1693, 68	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C014	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Pennsylvania (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	-0.77	0.46	-0.77 to -0.77	1693, 56	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C015	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Virginia (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	-0.66	0.52	-0.66 to -0.66	1693, 91	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C016	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Georgia (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	-0.64	0.53	-0.64 to -0.64	1693, 60	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C017	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Mississippi (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 2%	0.39	1.47	0.39 to 0.39	1693, 77	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C018	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Nevada (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 2%	0.38	1.47	0.38 to 0.38	1693, 77	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C019	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Kansas (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 2%	0.38	1.46	0.38 to 0.38	1693, 77	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C020	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Montana (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 2%	0.38	1.46	0.38 to 0.38	1693, 75	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C021	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to New York (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 2%	0.38	1.46	0.38 to 0.38	1693, 93	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C022	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Indiana (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 2%	0.37	1.45	0.37 to 0.37	1693, 82	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C023	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Louisiana (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 2%	0.36	1.44	0.36 to 0.36	1693, 65	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C024	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Utah (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 2%	0.36	1.44	0.36 to 0.36	1693, 89	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C025	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Rhode Island (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	-0.55	0.58	-0.55 to -0.55	1693, 82	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C026	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Massachusetts (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 2%	0.35	1.42	0.35 to 0.35	1693, 86	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C027	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Nebraska (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	-0.81	0.44	-0.83 to -0.53	1693, 72	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C028	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Florida (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	-0.44	0.64	-0.44 to -0.44	1693, 78	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C029	direction: customer-service calls: 0 calls to 3 calls (whole model) *shape: accelerating; 99% of patients same; verified; risk saturates near 0%*	1% to 2%	0.30	1.36	0.31 to 0.31	4000	agrees; AUC gain 0.106 [0.065, 0.145]	yes
C030	direction: total daytime calls: 75 calls to 125 calls (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	0.32	1.37	0.32 to 0.32	4000	disagrees; AUC gain -0.002 [-0.005, 0.001]	yes
C031	direction: account length: 48 to 152 (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	0.28	1.32	0.28 to 0.28	4000	disagrees; AUC gain -0.001 [-0.004, 0.001]	yes
C032	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Oregon (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 2%	0.23	1.26	0.23 to 0.23	1693, 84	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C033	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Delaware (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	-0.20	0.82	-0.20 to -0.20	1693, 81	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C034	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to North Carolina (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	-0.13	0.87	-0.13 to -0.13	1693, 74	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C035	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to North Dakota (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	0.06	1.07	0.06 to 0.06	1693, 68	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C036	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Oklahoma (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	0.03	1.03	0.03 to 0.03	1693, 79	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C037	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Arizona (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	-0.02	0.98	-0.02 to -0.02	1693, 73	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C038	direction: area code: area code 415 to area code other than area code 415 (whole model) *shape: categorical; highest to lowest: area code 415, area code other than area code 415*	1% to 1%	-0.02	0.98	-0.02 to -0.02	2016, 1984	disagrees; AUC gain -7e-05 [-0.0004, 0.0003]	yes
C039	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to California (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	-0.03	0.97	0.00 to 0.00	1693, 41	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C040	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Colorado (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	0.11	1.11	0.00 to 0.22	1693, 78	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C041	magnitude: international calling plan: no international plan to an international plan, typical patient *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 18%	2.74	15.6		3620, 380	agrees; AUC gain 0.066 [0.035, 0.098]	yes
C042	magnitude: voice-mail messages: 0 messages to 32 messages, typical patient *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 6%	1.61	5.00		4000	agrees; AUC gain 0.014 [0.0002, 0.029]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C043	magnitude: total daytime minutes: 112 minutes to 249 minutes, typical patient *shape: accelerating; 100% of patients same; fallback; risk saturates near 0%*	1% to 6%	1.59	4.92		4000	agrees; AUC gain 0.074 [0.044, 0.105]	yes
C044	magnitude: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Washington, typical patient *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 5%	1.35	3.86		1693, 72	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C045	magnitude: total evening charge: 11.76 US dollars to 22.57 US dollars, typical patient *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	1.01	2.73		4000	agrees; AUC gain 0.014 [0.003, 0.026]	yes
C046	null: model ignores total evening calls (exact rule)		0.00			4000	train only	yes
C047	null: model ignores total evening minutes (exact rule)		0.00			4000	train only	yes
C048	null: model ignores total international calls (exact rule)		0.00			4000	train only	yes
C049	null: model ignores total international charge (exact rule)		0.00			4000	train only	yes
C050	null: model ignores total international minutes (exact rule)		0.00			4000	train only	yes
C051	null: model ignores total night-time calls (exact rule)		0.00			4000	train only	yes
C052	null: model ignores total night-time charge (exact rule)		0.00			4000	train only	yes
C053	null: model ignores total night-time minutes (exact rule)		0.00			4000	train only	yes
C054	structure: made of linear_backbone, service_calls, day_usage_hinges, state_offsets; added					4000	train only	yes

T.2 Fixed-weight inputs

Annotation ann_82 (linear_backbone). Reads: state, account length, area code, international calling plan, voice-mail plan, number of voice-mail messages, total daytime calls, total evening charge.

Original hypothesis (unchanged). Direct input-to-output edges collected through identity_lin (an identity node). Exactly linear in 28 inputs: voicemail plan -1.05 per SD (-2.61 logit for having a plan), international plan +0.80 (+2.75 logit for having one), voicemail messages +0.68 per SD (offsets most of the plan effect for heavy message users), evening charge +0.40, day calls +0.13, account length +0.11, and state offsets: state:39 -0.40 per SD (-2.85 logit, but state_offsets gives back +2.30), state:11 -0.33 (-2.50), state:46 -0.31 (-2.22, of which state_offsets gives back +0.77), state:47 +0.18 (+1.35), state:33 +0.14 (+1.05, of which state_offsets takes back -0.68), state:0 -0.97, state:14 -0.94, state:38 -0.77, state:21 +0.71, state:10 -0.64, state:40 +0.54, state:29 -0.53, then smaller states and area code 415 (+0.01). Day minutes, day charge and customer-service calls have NO direct edge: they act only through hinges. Evidence: closed form exact (identity); train sensitivities equal the weights (voicemail plan 1.05, international plan 0.80, voicemail messages 0.68, state:39 0.40, evening charge 0.40).

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 1.77636e-15).

$$
\begin{aligned}
y_0 = {} & 0.0337x_{state:37} + 0.144x_{state:33_a} + 0.679x_{numberVmailMessages} \\
& + 0.0685x_{state:40} + 0.00461x_{state:36} + 0.18x_{state:47} + 0.398x_{totalEveCharge} \\
& + 0.103x_{state:21} + 0.126x_{totalDayCalls_a} + 0.00838x_{state:28} \\
& + 0.00889x_{areaCode:415} + 0.0459x_{state:18} + 0.805x_{internationalPlan} \\
& + 0.0513x_{state:19} + 0.106x_{accountLength} + 0.0534x_{state:44} \\
& - 1.05x_{voiceMailPlan_a} - 0.335x_{state:11} - 0.404x_{state:39_a} \\
& - 0.311x_{state:46_a} - 0.0781x_{state:10} - 0.0904x_{state:38} - 0.00327x_{state:3} \\
& - 0.114x_{state:0} - 0.07x_{state:29_a} - 0.0181x_{state:27} - 0.0419x_{state:9_a} \\
& - 0.122x_{state:14}
\end{aligned}
$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C055	direction: international calling plan: no international plan to an international plan (through this part) *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 18%	2.74	15.6	2.74 to 2.74	3620, 380	agrees; AUC gain 0.066 [0.035, 0.098]	yes
C056	direction: voice-mail messages: 0 messages to 32 messages (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 6%	1.61	5.00	1.61 to 1.61	4000	agrees; AUC gain 0.014 [0.0002, 0.029]	yes
C057	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Washington (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 5%	1.35	3.86	1.35 to 1.35	2456, 72	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C058	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Nevada (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 4%	1.05	2.85	1.05 to 1.05	2456, 77	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C059	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	1.01	2.73	1.01 to 1.01	4000	agrees; AUC gain 0.014 [0.003, 0.026]	yes
C060	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Maine (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 3%	0.71	2.04	0.71 to 0.71	2456, 86	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C061	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Rhode Island (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to <1%	-2.85	0.06	-2.85 to -2.85	2456, 82	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C062	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Hawaii (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to <1%	-2.50	0.08	-2.50 to -2.50	2456, 73	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C063	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	1% to <1%	-2.38	0.09	-2.38 to -2.38	2939, 1061	agrees; AUC gain 0.038 [0.017, 0.061]	yes
C064	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Vermont (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to <1%	-2.22	0.11	-2.22 to -2.22	2456, 80	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C065	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to South Carolina (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 2%	0.54	1.71	0.54 to 0.54	2456, 66	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C066	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Alaska (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 1%	-0.97	0.38	-0.97 to -0.97	2456, 56	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C067	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Illinois (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 1%	-0.94	0.39	-0.94 to -0.94	2456, 68	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C068	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Pennsylvania (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 1%	-0.77	0.46	-0.77 to -0.77	2456, 56	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C069	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Georgia (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 1%	-0.64	0.53	-0.64 to -0.64	2456, 60	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C070	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Louisiana (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 2%	0.36	1.44	0.36 to 0.36	2456, 65	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C071	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Utah (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 2%	0.36	1.44	0.36 to 0.36	2456, 89	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C072	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Massachusetts (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 2%	0.35	1.42	0.35 to 0.35	2456, 86	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C073	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Nebraska (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 1%	-0.53	0.59	-0.53 to -0.53	2456, 72	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C074	direction: total daytime calls: 75 calls to 125 calls (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	0.32	1.37	0.32 to 0.32	4000	disagrees; AUC gain -0.002 [-0.005, 0.001]	yes
C075	direction: account length: 48 to 152 (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	0.28	1.32	0.28 to 0.28	4000	disagrees; AUC gain -0.001 [-0.004, 0.001]	yes
C076	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Oregon (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 2%	0.23	1.26	0.23 to 0.23	2456, 84	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C077	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Florida (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 1%	-0.30	0.74	-0.30 to -0.30	2456, 78	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C078	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to North Carolina (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 1%	-0.13	0.87	-0.13 to -0.13	2456, 74	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C079	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to North Dakota (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 1%	0.06	1.07	0.06 to 0.06	2456, 68	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C080	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Oklahoma (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 1%	0.03	1.03	0.03 to 0.03	2456, 79	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C081	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Arizona (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 1%	-0.02	0.98	-0.02 to -0.02	2456, 73	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C082	direction: area code: area code 415 to area code other than area code 415 (through this part) *shape: categorical; highest to lowest: area code 415, area code other than area code 415*	1% to 1%	-0.02	0.98	-0.02 to -0.02	2016, 1984	disagrees; AUC gain -7e-05 [-0.0004, 0.0003]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C083	structure: its formula reproduces it on 4000 training patients (largest difference 1.77636e-15)		2e-15			4000	train only	yes

T.3 Service calls

Annotation ann_87 (service_calls). Reads: state, number of calls to customer service.

Original hypothesis (unchanged). Composition of csc_early_hinge and csc_late_chain: their exits are summed by identity_svc = 1.88*9436 + 1.73*6713 + 1.71*5268, which enters the logit with weight 1; they read the same input (two copies of customer-service calls) plus state:29 (early hinge) and state:4/state:5 (late chain). Hypothesis: all of the model's service-call effect is a threshold: nothing for 0-2 calls, +0.30 logit at 3 calls, then about +2.3 logit per extra call (+2.06 at 4, +4.40 at 5, +6.67 at 6, +9.28 at 7); the early hinge carries most of it, the late chain adds an extra bend from 5 calls. For state:29 customers only the late chain remains. Evidence (train): churn by calls 0-7 is 0.115/0.110/0.110/0.106/0.437/0.571/0.679/0.545 against model 0.058/0.055/0.050/0.062/0.209/0.573/0.890/0.979: the threshold is right, but the model under-predicts up to 4 calls and is over-confident from 6 calls (42 customers). Sensitivity of identity_svc: state:29 1.04, calls (early copy) 0.48, state:4 0.13. Mean-ablating both children gives AUC 0.8016, linear replacement 0.8713; restoring service_calls alone onto the all-linear model gives 0.8657 (all-linear 0.8352).

Closed form. Read through its parts (their formulas are in their own sections).

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C084	direction: customer-service calls: 0 calls to 3 calls (through this part) *shape: accelerating; 99% of patients same; verified; risk saturates near 0%*	1% to 2%	0.30	1.36	0.31 to 0.31	4000	agrees; AUC gain 0.106 [0.065, 0.145]	yes
C085	direction: state: state other than Nebraska, California, Colorado to Nebraska (through this part) *shape: categorical; highest to lowest: Nebraska, California, Colorado, state other than Nebraska, California, Colorado*	1% to 1%	-0.29	0.75	-0.31 to 0.00	3809, 72	unclear; AUC gain 0.0001 [-0.0007, 0.002]	yes
C086	direction: state: state other than Nebraska, California, Colorado to California (through this part) *shape: categorical; highest to lowest: Nebraska, California, Colorado, state other than Nebraska, California, Colorado*	1% to 1%	-0.03	0.97	0.00 to 0.00	3809, 41	unclear; AUC gain 0.0001 [-0.0007, 0.002]	yes
C087	direction: state: state other than Nebraska, California, Colorado to Colorado (through this part) *shape: categorical; highest to lowest: Nebraska, California, Colorado, state other than Nebraska, California, Colorado*	1% to 1%	0.11	1.11	0.00 to 0.22	3809, 78	unclear; AUC gain 0.0001 [-0.0007, 0.002]	yes
C088	interaction: effect of customer-service calls (0 calls to 3 calls) by state: Nebraska minus state other than Nebraska, California, Colorado; stronger in state other than Nebraska, California, Colorado *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	Nebraska: 1% to 1%; state other than Nebraska, California, Colorado: 1% to 2%	-0.31	0.74		72, 3809	agrees; AUC gain 0.0001 [-0.0007, 0.002]	yes
C089	threshold: customer-service calls: fold at about 3 calls (starts) *shape: accelerating; 99% of patients same; verified; risk saturates near 0%*	1% to 10% (typical patient with a fold)	2.06	7.84	2.08 to 2.08	4000, 831	agrees; AUC gain 0.106 [0.065, 0.145]	yes
C090	structure: made of csc_late_chain, csc_early_hinge; added					4000	train only	yes

T.4 Early rise in calls

Annotation ann_84 (csc_early_hinge). Reads: state, number of calls to customer service.

Original hypothesis (unchanged). 9436 = relu(-0.62 + 0.52 csc - 1.12 state:29), 6713 = relu(-0.10 + 1.49*9436); logit weights +1.88 and +1.73 (via identity_svc). Hypothesis: the main hinge on customer-service calls: off for 0-2 calls, just on at 3 calls (+0.30 logit), then about +1.8 logit per extra call (+2.04 at 4, +3.85 at 5, +5.43 at 6, +7.40 at 7); never on for the 72 state:29 customers. Active on 20.7% of customers (all with 3+ calls, outside state:29); empirical churn jumps from 0.11 at 3 calls to 0.44 at 4. Evidence: sensitivity 0.35 (6713) / 0.23 (9436) to state:29 and 0.16 / 0.11 to calls; the line_plot sweeps state:29 (first entry), so it is flat at 0 (tool quirk: line plots vary only the first entry). Mean ablation AUC 0.8281; linear replacement 0.8736 (R2 0.57/0.50); restored alone onto all-linear 0.8654.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 8.88178e-16).

$$
\begin{aligned}
y_0 &= \operatorname{relu}\big(-0.615 + 0.52 x_{numberCustomerServiceCalls_b} \\
&\qquad - 1.12 x_{state:29_b}\big) \\
y_1 &= \operatorname{relu}\big(-0.0961 + 1.49 \operatorname{relu}\big(-0.615 \\
&\qquad + 0.52 x_{numberCustomerServiceCalls_b} - 1.12 x_{state:29_b}\big)\big)
\end{aligned}
$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C091	direction: customer-service calls: 0 calls to 3 calls (through this part) *shape: accelerating; 100% of patients same; fallback; risk saturates near 0%*	1% to 2%	0.30	1.35	0.31 to 0.31	4000	agrees; AUC gain 0.082 [0.050, 0.114]	yes
C092	direction: state: state other than Nebraska to Nebraska (through this part) *shape: categorical; highest to lowest: state other than Nebraska, Nebraska*	2% to 1% (typical patient this part acts on)	-0.29	0.75	-0.31 to 0.00	3928, 72	unclear; AUC gain 8e-05 [-0.0002, 0.0003]	yes
C093	interaction: effect of customer-service calls (0 calls to 3 calls) by state: Nebraska minus state other than Nebraska; stronger in state other than Nebraska *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	Nebraska: 1% to 1%; state other than Nebraska: 1% to 2%	-0.31	0.73		72, 3928	agrees; AUC gain 8e-05 [-0.0002, 0.0003]	yes
C094	threshold: customer-service calls: fold at about 3 calls (starts) *shape: accelerating; 100% of patients same; fallback; risk saturates near 0%*	1% to 10% (typical patient with a fold)	2.08	8.00	2.08 to 2.08	3928, 830	agrees; AUC gain 0.082 [0.050, 0.114]	yes
C095	subgroup: who this part acts on: 830 of 4000 patients	1% to 2% (typical patient this part acts on (risk without the part, then with it))	1.36	3.88	0.31 to 3.85	830, 3170	agrees; AUC gain 0.083 [0.051, 0.116]	yes
C096	structure: its formula reproduces it on 4000 training patients (largest difference 8.88178e-16)		9e-16			4000	train only	yes

T.5 Late rise in calls

Annotation ann_83 (csc_late_chain). Reads: state, number of calls to customer service.

Original hypothesis (unchanged). 12374 = relu(0.48 + 0.74 csc) (always active except at 0 calls, where it is just below its kink), 349 = relu(0.22 + 0.41 state:4 - 1.86 state:5), 5268 = relu(-0.56 + 0.33 csc + 0.26*12374 - 1.39*349); logit weight +1.71 (via identity_svc). 349 takes three values: 0.44 for most customers, 0 in state:5 (78 customers) and 4.55 in state:4 (41 customers). Hypothesis: a late hinge on customer-service calls that is off for 0-4 calls for most customers and adds about +0.65 logit per call from 5 calls (+0.55 at 5, +1.24 at 6, +1.88 at 7); in state:5 it starts one call earlier (on from 3 calls) and in state:4 it is always off. Evidence (train, n=4000): 5268 is active on 3.4% of customers (on for 3% at 4 calls, 99% at 5); sensitivity of 5268 is 0.074 to state:4, 0.017 to calls, 0.017 to state:5; line plot correlation 0.51. Mean ablation AUC 0.8889 (full 0.8884): it adds nothing to ranking on its own; linear replacement 0.8887 (R2 0.27); restored alone onto the all-linearised model 0.8401 (all-linear 0.8352).

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 2.22045e-16).

$$\begin{aligned} y_0 = \operatorname{relu}\big(&-0.557 + 0.256\operatorname{relu}\big(0.476 \\ &+ 0.74x_{numberCustomerServiceCalls_ac}\big) + 0.327x_{numberCustomerServiceCalls_ac} \\ &- 1.39\operatorname{relu}\big(0.219 + 0.414x_{state:4} - 1.86x_{state:5}\big)\big) \end{aligned}$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C117	direction: state: state other than California, Colorado to Colorado (through this part) *shape: categorical; highest to lowest: Colorado, state other than California, Colorado, California*	52% to 75% (typical patient this part acts on)	0.11	1.11	0.00 to 0.22	3881, 78	unclear; AUC gain 2e-05 [-0.0009, 0.001]	yes
C118	direction: state: state other than California, Colorado to California (through this part) *shape: categorical; highest to lowest: Colorado, state other than California, Colorado, California*	52% to 39% (typical patient this part acts on)	-0.03	0.97	0.00 to 0.00	3881, 41	unclear; AUC gain 2e-05 [-0.0009, 0.001]	yes
C119	direction: customer-service calls: 0 calls to 3 calls (through this part) *shape: accelerating; 2% of patients same; fallback*	39% to 44% (typical patient this part acts on)	0.004	1.00	0.00 to 0.00	4000	unclear; AUC gain 0.0003 [-0.0007, 0.002]	yes
C120	interaction: effect of customer-service calls (0 calls to 3 calls) by state: Colorado minus state other than California, Colorado; stronger in Colorado *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	Colorado: 1% to 2%; state other than California, Colorado: 1% to 2%	0.23	1.26		78, 3881	agrees; AUC gain 2e-05 [-0.0009, 0.001]	yes
C121	threshold: customer-service calls: fold at about 5 calls (starts) *shape: hinge at 4 calls; 98% of patients same; verified; risk saturates near 0%*	1% to 1% (typical patient with a fold)	0.02	1.02	0.00 to 0.00	3959, 134	unclear; AUC gain 0.0003 [-0.0007, 0.002]	no
C122	subgroup: who this part acts on: 134 of 4000 patients	39% to 75% (typical patient this part acts on (risk without the part, then with it))	0.85	2.35	0.53 to 1.88	134, 3866	unclear; AUC gain 0.0003 [-0.0008, 0.002]	yes
C123	structure: its formula reproduces it on 4000 training patients (largest difference 2.22045e-16)		2e-16			4000	train only	yes

T.6 Heavy daytime use

Annotation ann_85 (day_usage_hinges). Reads: voice-mail plan, total daytime minutes, total daytime charge.

Original hypothesis (unchanged). 10131 = relu(-0.63 + 0.79 dayMin - 0.35 voiceMailPlan), logit weight +1.89; 4000 = relu(-0.56 + 0.70 dayMin), weight +1.50; 3118 = relu(-0.84 + 0.36 dayCharge), weight +1.48. Day minutes and day charge are collinear (r = 1.0000), so all three are hinges in day usage. Hypothesis: day usage has no effect below about 200 minutes, then raises churn risk steeply in three stages: 10131 switches on at about 209 day minutes without a voicemail plan (about 263 with one), 4000 at about 224 minutes, and 3118 at a day charge of about 51.7 (about 304 minutes; on for only 1% of customers). Evidence (train): combined logit contribution 0 below 200 minutes, +0.05 at 200-220, +0.51 at 220-240, +1.31 at 240-260, +2.19 at 260-280, +3.10 at 280-300, +4.69 above 300 (voicemail-plan holders about 1.1-1.7 lower from 240 minutes up); churn by band 0.06/0.14/0.26/0.46/0.58/0.79 against model 0.04/0.08/0.15/0.24/0.37/0.64 (right shape, under-predicted, as is the whole model). Sensitivity: 10131 day minutes 0.18, voicemail plan 0.08; 4000 day minutes 0.14; 3118 day charge 0.004. Mean ablation AUC 0.8083 (the largest single drop); linear replacement 0.8657 (R2 0.47/0.44/0.05); restored alone onto all-linear 0.8713 (all-linear 0.8352).

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 2.22045e-16).

$$
\begin{aligned}
y_0 &= \mathrm{relu}\big(-0.836 + 0.365 x_{totalDayCharge}\big) \\
y_1 &= \mathrm{relu}\big(-0.565 + 0.7 x_{totalDayMinutes}\big) \\
y_2 &= \mathrm{relu}\big(-0.625 + 0.789 x_{totalDayMinutes} \\
&\qquad - 0.347 x_{voiceMailPlan_b}\big)
\end{aligned}
$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C097	direction: total daytime minutes: 112 minutes to 249 minutes (through this part) *shape: accelerating; 100% of patients same; fallback; risk saturates near 0%*	1% to 6%	1.30	3.67	0.49 to 1.59	4000	agrees; AUC gain 0.074 [0.044, 0.105]	yes
C098	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	4% to 2% (typical patient this part acts on)	-0.24	0.79	-1.12 to 0.00	2939, 1061	agrees; AUC gain 0.007 [0.003, 0.011]	yes
C099	interaction: effect of total daytime minutes (112 minutes to 249 minutes) by voice-mail plan: a voice-mail plan minus no voice-mail plan; stronger in no voice-mail plan *shape: accelerating; 100% of patients same; fallback; risk saturates near 0%*	a voice-mail plan: <1% to <1%; no voice-mail plan: 1% to 6%	-1.11	0.33		1061, 2939	agrees; AUC gain 0.007 [0.003, 0.011]	yes
C100	threshold: total daytime minutes: fold at about 209 minutes (starts) *shape: accelerating; 100% of patients same; fallback; risk saturates near 0%*	1% to 16% (typical patient with a fold)	2.24	9.41	1.15 to 2.63	4000, 1074	agrees; AUC gain 0.074 [0.044, 0.105]	yes
C101	threshold: total daytime charge: fold at about 51.68 US dollars (starts) *shape: flat; 100% of patients same; verified; risk saturates near 0%*	1% to 1% (typical patient with a fold)	0.00	1.00	0.00 to 0.00	4000, 41	unclear; AUC gain 3e-05 [-3e-05, 0.0001]	no
C102	subgroup: who this part acts on: 1074 of 4000 patients	1% to 4% (typical patient this part acts on (risk without the part, then with it))	1.28	3.61	0.12 to 3.02	1074, 2926	agrees; AUC gain 0.080 [0.049, 0.113]	yes
C103	structure: its formula reproduces it on 4000 training patients (largest difference 2.22045e-16)		2e-16			4000	train only	yes

T.7 State units

Annotation ann_86 (state_offsets). Reads: state, total daytime calls.

Original hypothesis (unchanged). Thirteen units, each reading only one-hot state indicators, so each is exactly an offset per state (least-squares R2 = 1.0): 1295 -> 7872 (state:39; weights -1.11, -1.25: a constant -2.32 logit for everyone except state:39, where it is switched off, +2.30, against state:39's direct -2.85; net -0.55 vs empirical log-odds offset -0.93, 82 customers), 11361 (state:46, +0.77 against direct -2.22), 5786 (state:26/33/34: +0.38 for state:26 and state:34, -0.68 for state:33 against its direct +1.05), 1390 (state:25 +0.39), 17102 (state:45 -0.66), 15490 -> 15904 (state:42 +0.54), 15362 (state:16 +0.38), 9954 (state:15 +0.37), 12690 (state:8 -0.20), 13068 (state:9 -0.14, plus direct -0.30), and 5533, which also reads day calls but is never active on train (pre-activation at most -0.30), so it contributes exactly 0. Evidence: sensitivities are on the matching state only (state:39 1.25/1.28, state:46 1.65, state:42 0.98); linear replacement AUC 0.8884 (unchanged); mean ablation 0.8856.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 1.11022e-16).

$$
\begin{aligned}
y_{0} &= \mathrm{relu}\big(0.78 - 1.28 x_{state:39_b}\big) \\
y_{1} &= \mathrm{relu}\big(0.0571 + 1.02\,\mathrm{relu}\big(0.78 \\
&\qquad - 1.28 x_{state:39_b}\big)\big) \\
y_{2} &= \mathrm{relu}\big(0.252 - 0.574 x_{state:25}\big) \\
y_{3} &= \mathrm{relu}\big(-0.38 + 1.01 x_{state:15}\big) \\
y_{4} &= \mathrm{relu}\big(1.36 + 1.65 x_{state:46_b}\big) \\
y_{5} &= \mathrm{relu}\big(-0.0337 + 1.0 x_{state:8}\big) \\
y_{6} &= \mathrm{relu}\big(-0.372 + 1.02 x_{state:9_b}\big) \\
y_{7} &= \mathrm{relu}\big(-0.581 + 0.999 x_{state:16}\big) \\
y_{8} &= \mathrm{relu}\big(0.361 + 0.984 x_{state:42}\big) \\
y_{9} &= \mathrm{relu}\big(-0.0356 + 0.993\,\mathrm{relu}\big(0.361 \\
&\qquad + 0.984 x_{state:42}\big)\big) \\
y_{10} &= \mathrm{relu}\big(-1.42 + 0.995 x_{state:45}\big) \\
y_{11} &= \mathrm{relu}\big(0.383 + 0.163 x_{state:33_b} - 1.5 x_{state:26} \\
&\qquad - 0.732 x_{state:34}\big) \\
y_{12} &= \mathrm{relu}\big(-1.54 + 0.655\,\mathrm{relu}\big(0.383 \\
&\qquad + 0.163 x_{state:33_b} - 1.5 x_{state:26} - 0.732 x_{state:34}\big) \\
&\qquad - 0.00543 x_{totalDayCalls_b}\big)
\end{aligned}
$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C104	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Rhode Island (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 12%	2.30	9.97	2.30 to 2.30	3039, 82	agrees; AUC gain 0.015 [0.003, 0.029]	yes
C105	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Vermont (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 3%	0.77	2.16	0.77 to 0.77	3039, 80	agrees; AUC gain 0.015 [0.003, 0.029]	yes
C106	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Tennessee (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 2%	0.54	1.72	0.54 to 0.54	3039, 68	agrees; AUC gain 0.015 [0.003, 0.029]	yes
C107	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Virginia (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 1%	-0.66	0.52	-0.66 to -0.66	3039, 91	agrees; AUC gain 0.015 [0.003, 0.029]	yes
C108	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Nevada (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 1%	-0.66	0.52	-0.66 to -0.66	3039, 77	agrees; AUC gain 0.015 [0.003, 0.029]	yes
C109	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Mississippi (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 2%	0.39	1.47	0.39 to 0.39	3039, 77	agrees; AUC gain 0.015 [0.003, 0.029]	yes
C110	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Kansas (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 2%	0.38	1.46	0.38 to 0.38	3039, 77	agrees; AUC gain 0.015 [0.003, 0.029]	yes
C111	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Montana (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 2%	0.38	1.46	0.38 to 0.38	3039, 75	agrees; AUC gain 0.015 [0.003, 0.029]	yes
C112	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to New York (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 2%	0.38	1.46	0.38 to 0.38	3039, 93	agrees; AUC gain 0.015 [0.003, 0.029]	yes
C113	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Indiana (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 2%	0.37	1.45	0.37 to 0.37	3039, 82	agrees; AUC gain 0.015 [0.003, 0.029]	yes
C114	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Delaware (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 1%	-0.20	0.82	-0.20 to -0.20	3039, 81	agrees; AUC gain 0.015 [0.003, 0.029]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C115	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Florida (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 1%	-0.14	0.87	-0.14 to -0.14	3039, 78	agrees; AUC gain 0.015 [0.003, 0.029]	yes
C116	structure: its formula reproduces it on 4000 training patients (largest difference 1.11022e-16)		1e-16			4000	train only	yes

The network before any explanation

Figure 221 (next page, landscape) is the trained network as it is, with no parts, no splits and no inserted units: inputs at the bottom, hidden units by depth, the output at the top, every connection drawn, every input column its own node (one-hot levels included).

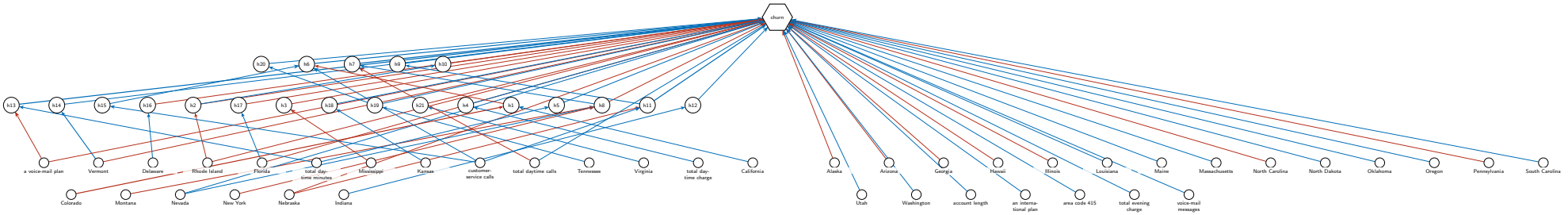


Figure 221: The network as trained, before any explanation, in full. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. Not used (no connection): Alabama, Iowa, Idaho, Kentucky, Arkansas, Maryland, Michigan, Minnesota, Missouri, New Hampshire, New Jersey, New Mexico, Ohio, South Dakota, Texas, Wisconsin, West Virginia, Wyoming, Connecticut, Washington DC, area code 408, area code 510, total evening minutes, total evening calls, total night-time minutes, total night-time calls, total night-time charge, total international minutes, total international calls, total international charge.

5 churn100-s7-c1

Genome `32d24e98-266e-450c-88c8-5bdc1abbf5dd`, explanation `b370983e-3d7d-40c8-8b8f-f37c41a96cc1`. Split: train primary (n=4000), test as a check (n=1000). Claims: 119, verified 109, not supported 10.
AI-assisted draft for review (unverified)

Reference (typical) customer: the typical customer: the median training customer for every number and the most common training category for every category (not a real customer). account length: 100; area code: area code 415; international plan: no international plan; number customer service calls: 1 calls; number vmail messages: 0 messages; state: West Virginia; total day calls: 100 calls; total day charge: 30.69 US dollars; total day minutes: 180 minutes; total eve calls: 100 calls; total eve charge: 17.12 US dollars; total eve minutes: 201 minutes; total intl calls: 4 calls; total intl charge: 2.78 US dollars; total intl minutes: 10 minutes; total night calls: 100 calls; total night charge: 9.02 US dollars; total night minutes: 200 minutes; voice mail plan: no voice-mail plan. Predicted chance that a customer leaves: 1%.

5.1 Summary

For a typical customer (in West Virginia, area code 415, no international or voice-mail plan, 1 call to customer service and median usage) the model predicts a chance of leaving of about 1%; it adds up five parts: usage and service calls, the international plan, fixed-weight inputs, a voice-mail and South Dakota adjustment and state adjustments. [C047] Figure 222 shows the model at this level: one box per paragraph below.

How to read the diagrams. Edge colour is the sign of that one connection: blue positive, red negative; thicker is a larger weight. The outline and arrow on an input (and on a part's output box) is its overall effect on the final prediction, read from the verified claims: orange and up arrow raises it, teal and down arrow lowers it, grey and double arrow depends (it goes both ways, or differs between levels). A negative connection into a part that itself raises the prediction means that input lowers the prediction; through a hinge the overall effect can only be read from the claims, not from the edges. Up arrow means towards churn; down arrow, away from it. A double outline marks an input (or a part's output) that acts at a fixed rate everywhere: it adds the same amount to the score per unit, so it is explained by its rate alone (see 'Linear factors'); a dashed double outline, at a fixed rate on the observed data. The normal outline is an input whose effect switches on and off somewhere. For example, a voice-mail plan has a negative connection into 'Usage and service calls', a part that raises the predicted chance that a customer leaves, so a voice-mail plan lowers it.

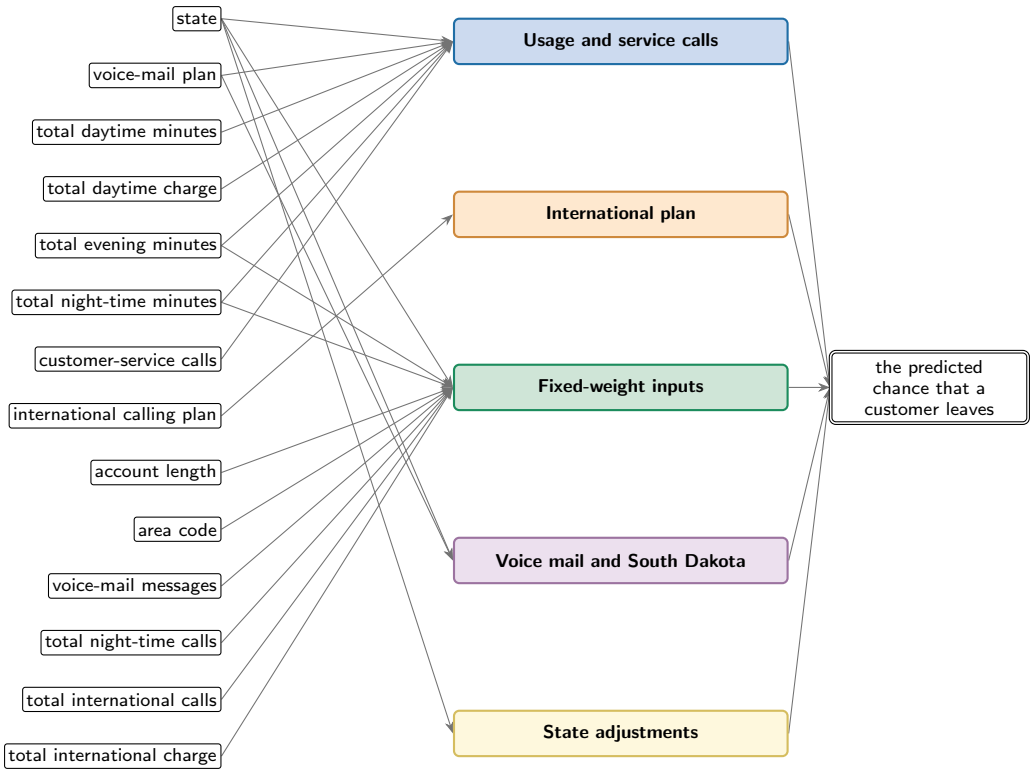


Figure 222: The model at the level of the summary: one box per part.

Usage and service calls. This part combines a usage score, built from daytime, evening and night-time minutes and the voice-mail plan, with calls to customer service, not as a simple sum. [C060] Customer-service calls add only

from about 3 calls upwards, for most customers slowly at first, then faster, but for some customers without a voice-mail plan they add nothing until a point, then steadily: going from 0 to 4 calls takes a typical customer from about 1% to about 15%. [C055] Through it, daytime minutes add to the chance only above about 187 minutes for customers without a voice-mail plan, in an uneven way: for a typical customer, going from 0 to 271 minutes takes it from about 1% to about 11%. [C056] Daytime minutes count more with more than 201 evening minutes, in an uneven way: from about 1% to about 11% between 112 and 249 minutes, against about 1% to about 2% with fewer. [C067]

International plan. This part reads the international plan alone and acts on the 380 of 4000 training customers who have one, 161 of whom left; for the typical customer it acts on, the chance is about 2% without it and about 26% with it. [C078] For a typical customer, an international plan takes the chance of leaving from about 1% to about 15%. [C077]

Fixed-weight inputs. This part adds up many inputs with fixed weights, each moving a typical customer's chance by a few points at most: compared with most other states, Washington raises it to about 3% and Montana to about 2%, and Vermont, Rhode Island and Colorado lower it to under 1%. [C080, C081, C082, C084, C085] A higher international charge raises it steadily, from under 1% at 1.84 to about 1% at 3.70 US dollars, and more voice-mail messages and a longer account also raise it steadily, by less than a point. [C083, C086, C098] More international calls lower it, each extra call taking away about the same, from about 1% at 2 to under 1% at 8 calls, and more evening and night-time minutes lower it steadily through this part, by less than a point. [C087, C092, C096]

Voice mail and South Dakota. This part reads the voice-mail plan and the state: through it, a customer in South Dakota has under 1% instead of about 1%, and a voice-mail plan also lowers a typical customer's chance to under 1%. [C108, C109]

State adjustments. This part reads only the state: it moves a typical customer's chance from about 1% in West Virginia to about 5% in South Dakota and about 2% in Indiana and in most other states, and keeps it under 1% in Virginia (66 to 122 training customers per named state; the full list is in the technical detail). [C111, C112, C114, C115]

The whole network. Figure 223 (next page, landscape) draws every unit and connection of the explained model, nested by part: shaded clusters are the parts of the model and darker boxes the parts inside them; circles are hidden units (h1, h2, ...); each cluster's rectangle is its output, and a dashed 'sum' stands for an addition the next unit does (drawn, not a unit of the model); the hexagon is the model's output. An input used by several parts is drawn under each of them (same label, dotted outline, the first copy marked with the number of copies): it is the same input, repeated for readability. Not used by the model (input columns with no connection): state: 28 of 51 levels, area code 408, area code 415, total daytime calls, total evening calls, total evening charge, total night-time charge, total international minutes.

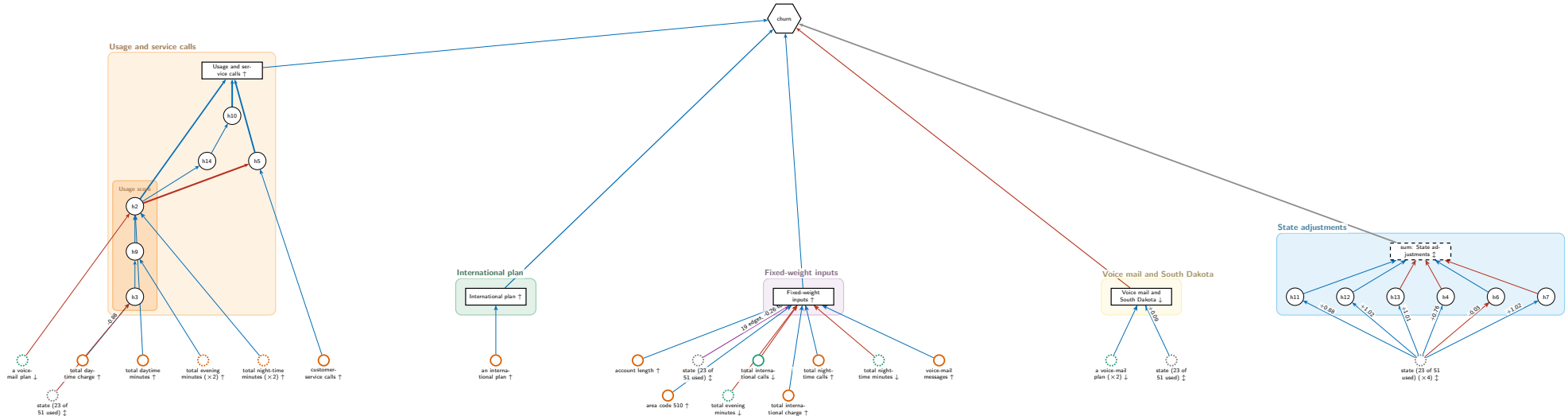


Figure 223: Every unit and connection of the explained model, nested by part. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Worth checking against what is known. Worth checking: the state moves the chance by several points (for example Washington about 5% and Vermont under 1% across the whole model), based on 72 to 122 training customers per state. [C002, C022] Worth checking: South Dakota gets a rise through the state adjustments (to about 5%) and a fall through the voice-mail and South Dakota adjustment (to under 1%), so the two pull against each other. [C111, C108] Worth checking: more evening and night-time minutes raise the chance through the usage score but lower it through the fixed-weight inputs. [C063, C064, C092, C096] Worth checking: the model treats a voice-mail plan as lowering the chance of leaving but more voice-mail messages as raising it. [C027, C023]

5.2 The model, read from the top down

Each part of the explanation in turn, from the whole model down to the smallest parts: what it does for the prediction (hypothesis), what was checked (evidence), a diagram one level down, and where to read on. The technical record of each part is in "Technical detail" at the end.

5.2.1 The whole model

Figure 224 shows what feeds this part; Figure 225 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

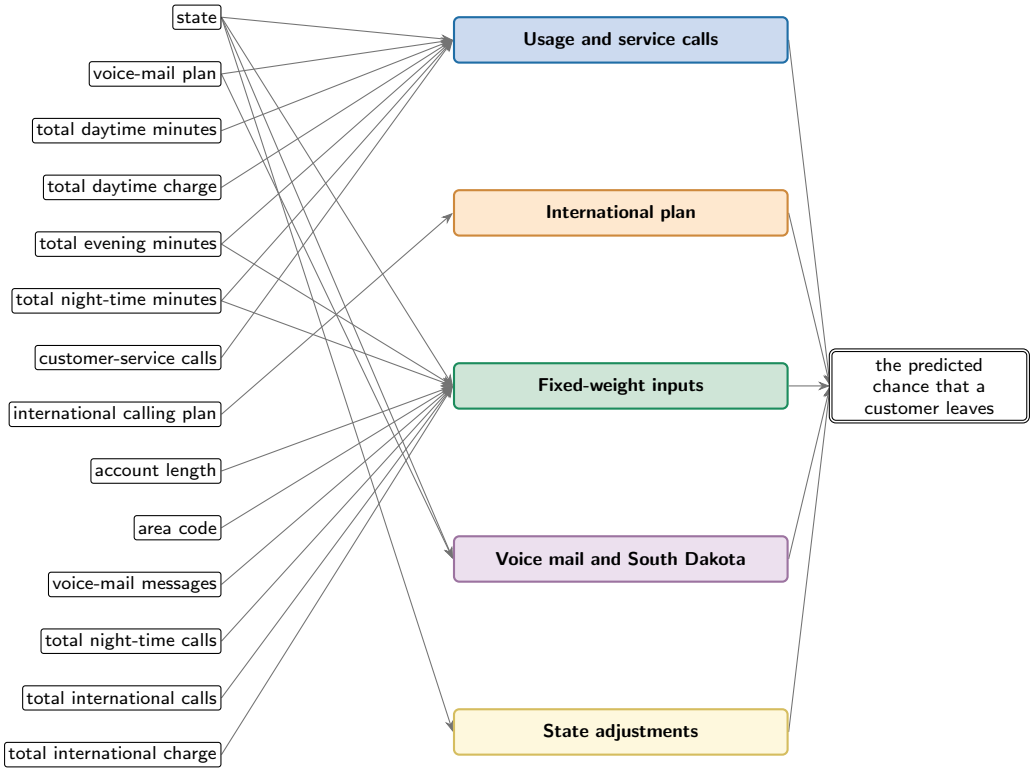


Figure 224: The whole model: what feeds it, one level down.

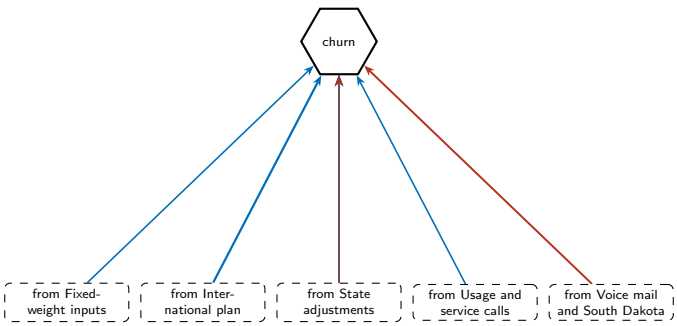


Figure 225: The whole model: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. The model adds up five parts into one score: usage and service calls, the international plan, fixed-weight inputs, a voice-mail and South Dakota adjustment, and state adjustments. [C047]

Evidence. It is made up of these five parts (each explained below); the score is their sum, which was checked on all 4000 training customers. [C047] Changing one input at a time across the whole model, an international plan takes a typical customer's chance of leaving from about 1% to about 15%, and more daytime minutes raise it in an uneven way, from about 1% at 112 to about 4% at 249 minutes. [C001, C004] In Washington the chance is about 5% and in Montana about 4%, against about 1% in West Virginia and under 1% in Vermont. [C002, C003, C022] The model ignores daytime calls, evening calls, the evening charge, international minutes and the night-time charge. [C042, C043, C044, C045, C046]
Evidence plots: Figure 226, Figure 227.

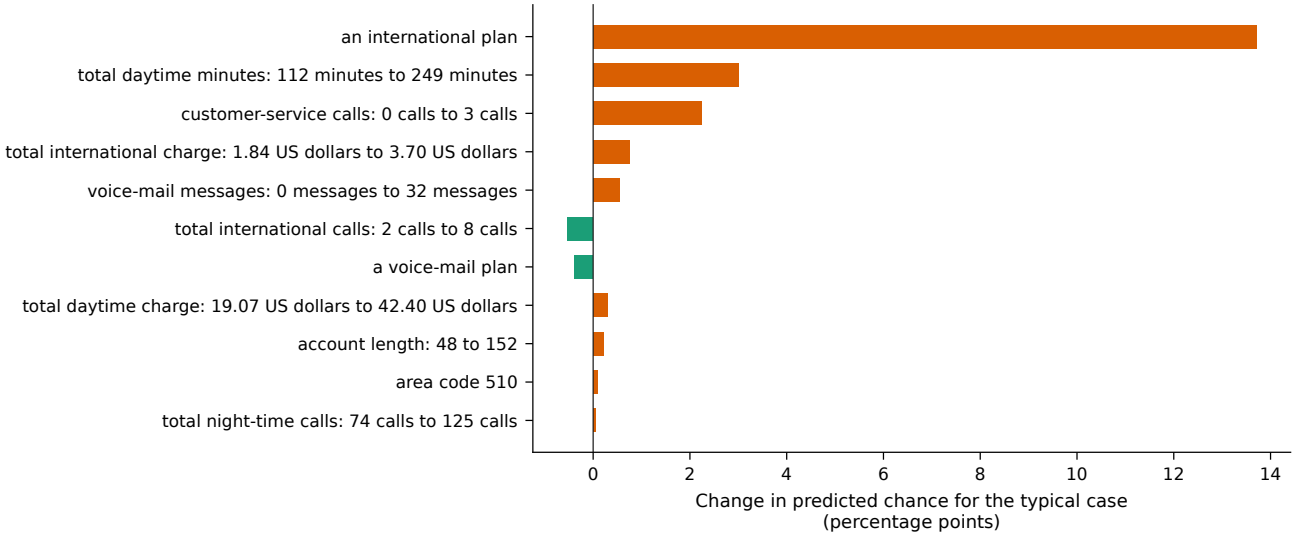


Figure 226: The whole model: Each input's change in the predicted value for the typical case across the whole model, largest first (orange: raises it, teal: lowers it).

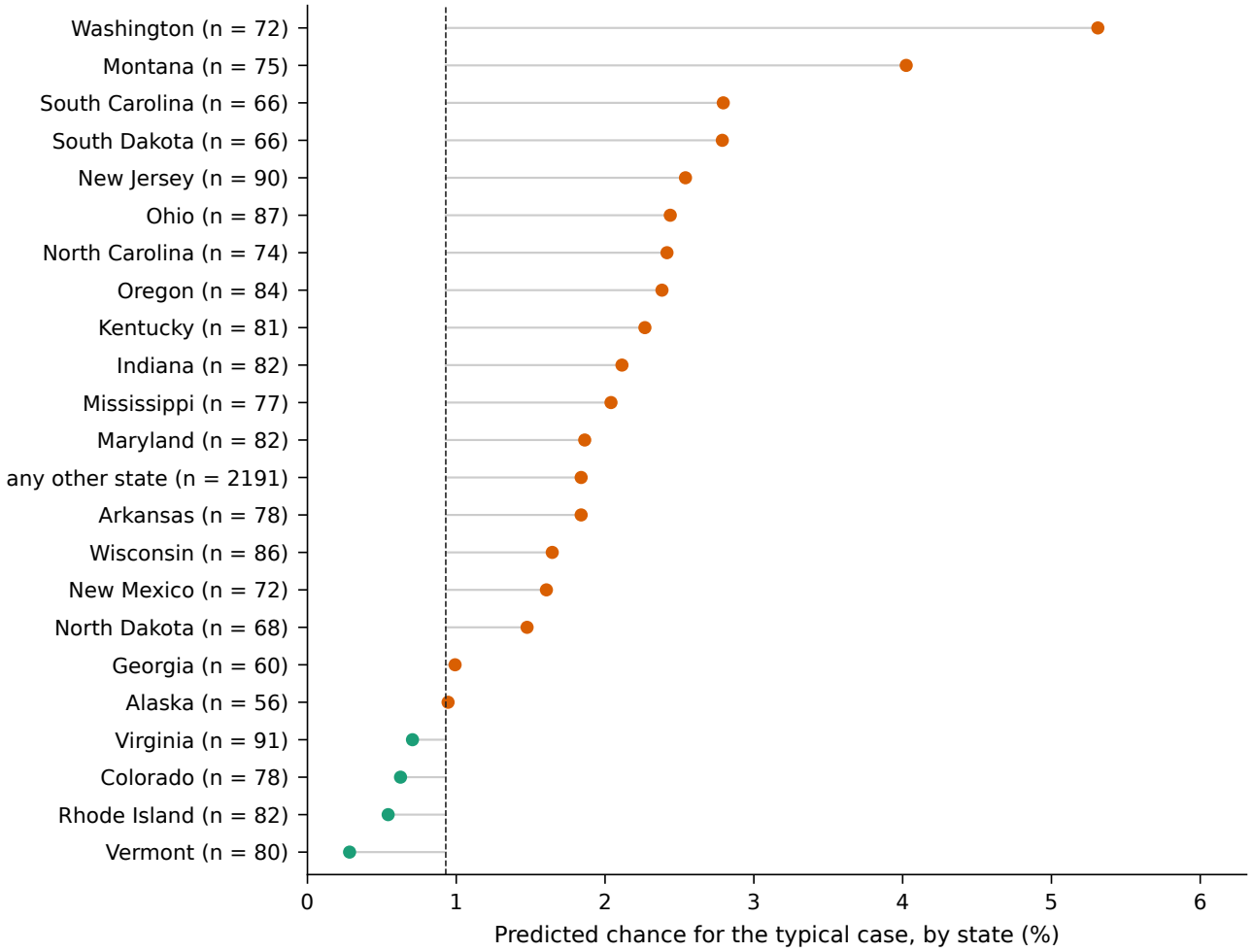


Figure 227: The whole model: Every level of state the model reads: the predicted value for the typical case at that level (dashed: the reference level); n = training customers; grey: few customers or not supported.

Dive deeper: Usage and service calls (5.2.2); International plan (5.2.4); Fixed-weight inputs (5.2.5); Voice mail and South Dakota (5.2.6); State adjustments (5.2.7). Technical detail: T.1.

5.2.2 Usage and service calls

Part of: The whole model (5.2.1).
Figure 228 shows what feeds this part; Figure 229 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

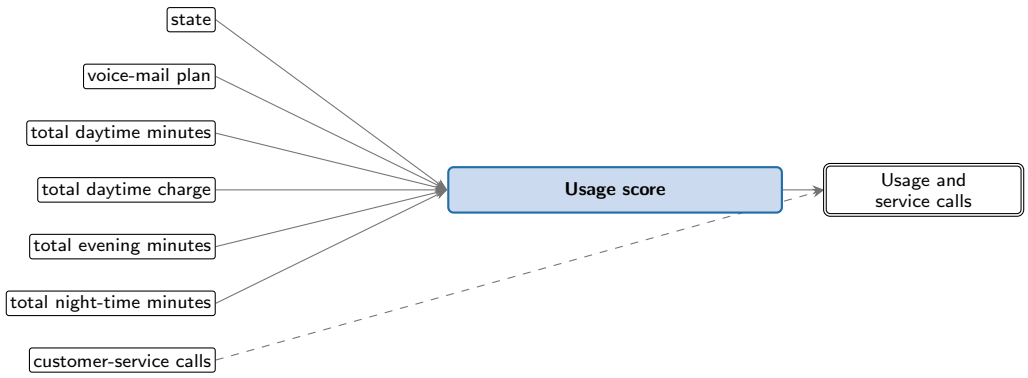


Figure 228: Usage and service calls: what feeds it, one level down.

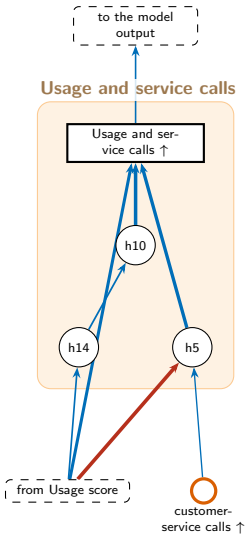


Figure 229: Usage and service calls: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part raises the chance of leaving for heavy daytime users without a voice-mail plan and for customers who call customer service 3 times or more: daytime minutes in an uneven way, and calls for most customers slowly at first, then faster, but for some customers without a voice-mail plan nothing until a point, then steadily. [C056, C055]

Evidence. It is made up of the usage score (explained below) together with customer-service calls, combined in a way that is not a simple sum. [C060] Through it, daytime minutes add only above about 187 minutes for customers without a voice-mail plan, in an uneven way, from about 1% to about 11% between 0 and 271 minutes for a typical customer. [C056] Customer-service calls add only from about 3 calls upwards, for most customers slowly at first, then faster, but for some customers without a voice-mail plan nothing until a point, then steadily, from about 1% to about 15% between 0 and 4 calls. [C055]
Evidence plots: Figure 230, Figure 231, Figure 232, Figure 233, Figure 234, Figure 235, Figure 236.
Figure 235 shows this part's own output, unit h16: over its two main inputs, with the line where it switches on.

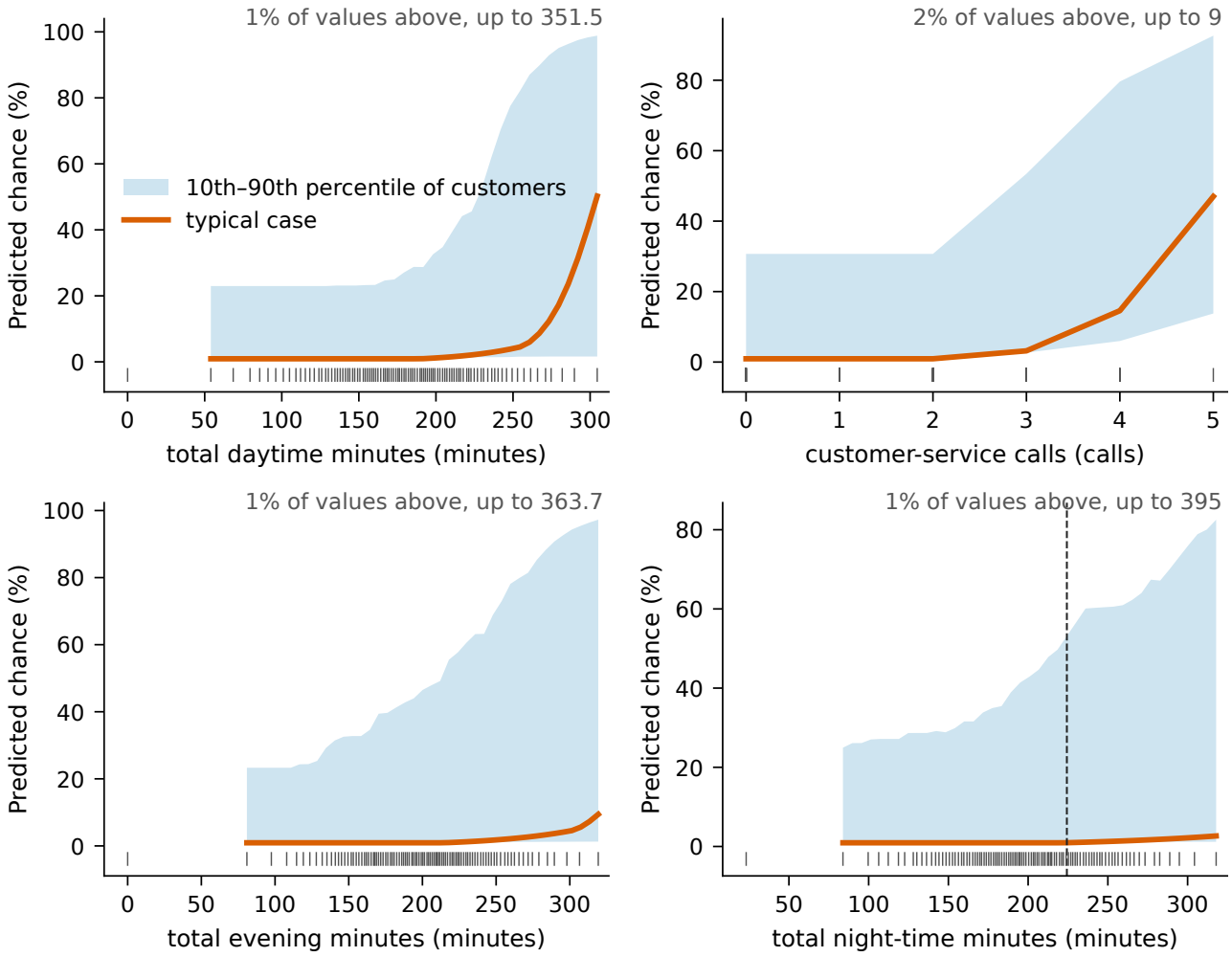


Figure 230: Usage and service calls: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

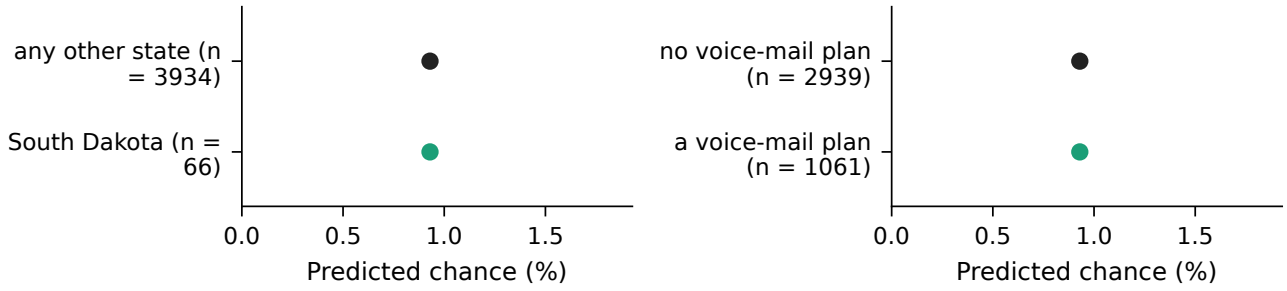


Figure 231: Usage and service calls: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

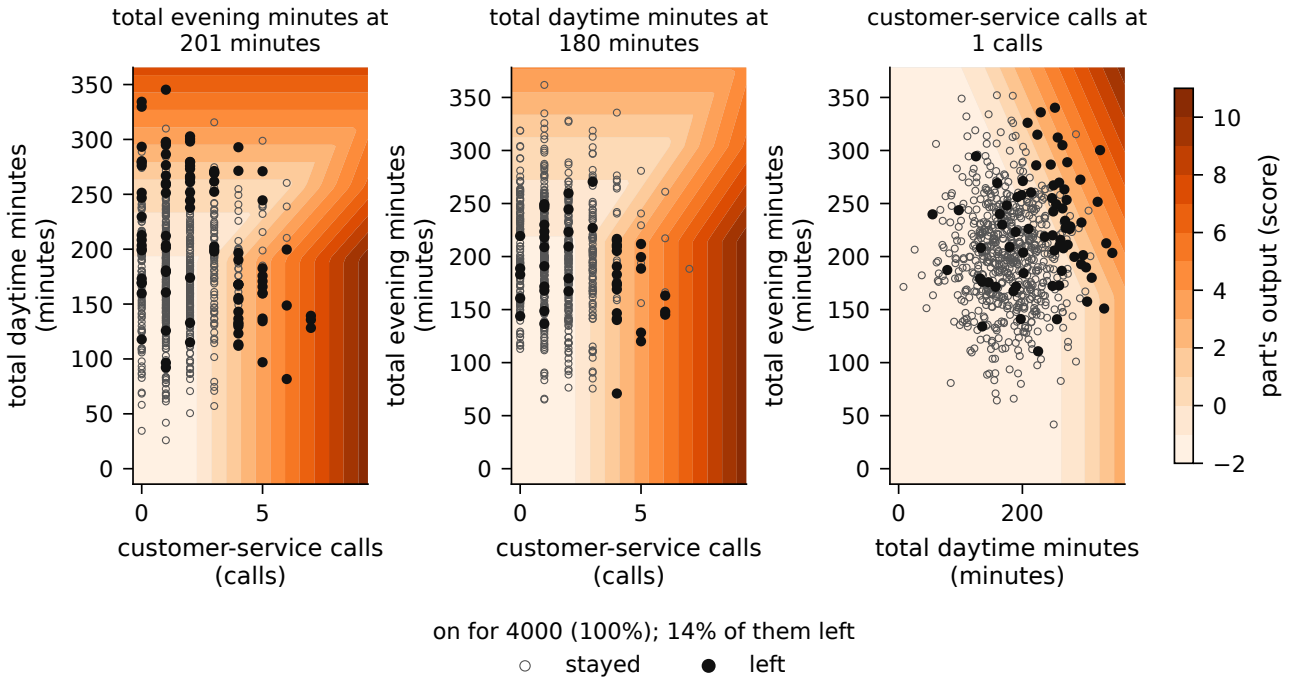


Figure 232: Usage and service calls: The part's own output over each pair of its three main inputs (the third held at its median, named above each panel; the points are the training rows whose third input is between its 40th and 60th percentile), other inputs at the typical case: colour where it is on, flat grey where it is switched off, the black line where it switches on; training customers as hollow (no event) and filled (event) points, so the part can be seen catching the events (the predicted-chance overlay was harder to read). The switch-on line is traced on this map only; the boundary claim (C118) did not pass its check on the data, so no rule is stated.

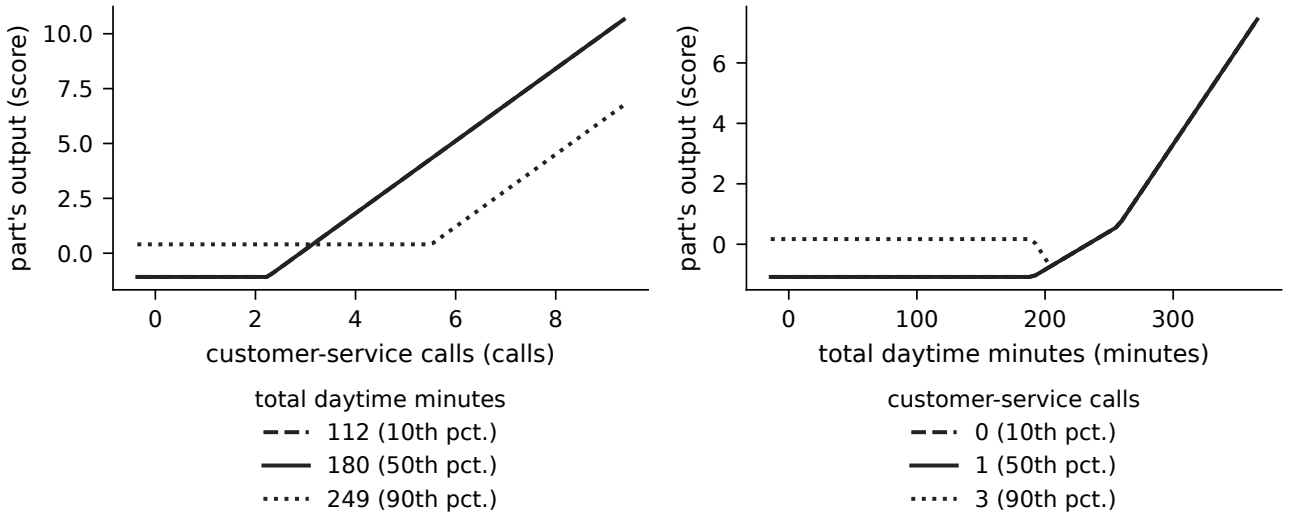


Figure 233: Usage and service calls: Slices of the same map: the part's own output against each input at three values of the other, so the tilted boundary shows as a shifting hinge. The switch-on line is traced on this map only; the boundary claim (C118) did not pass its check on the data, so no rule is stated.

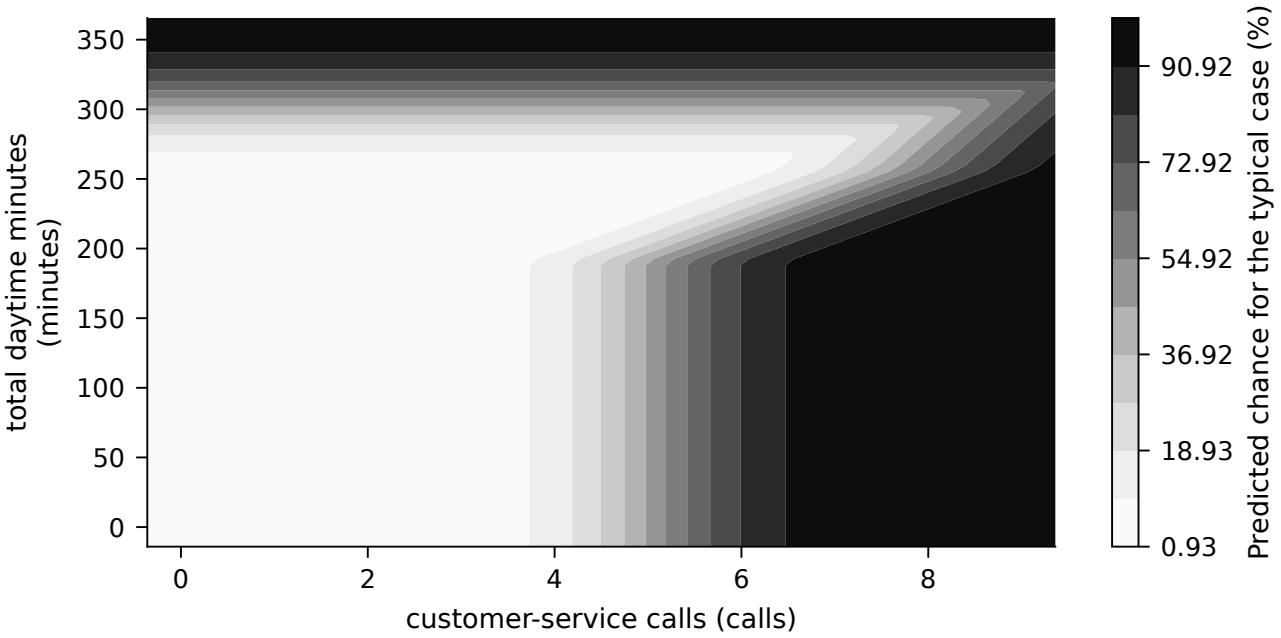


Figure 234: Usage and service calls: The same map on the final scale: the predicted value for the typical case, with the switch-on line (orange). The switch-on line is traced on this map only; the boundary claim (C118) did not pass its check on the data, so no rule is stated.

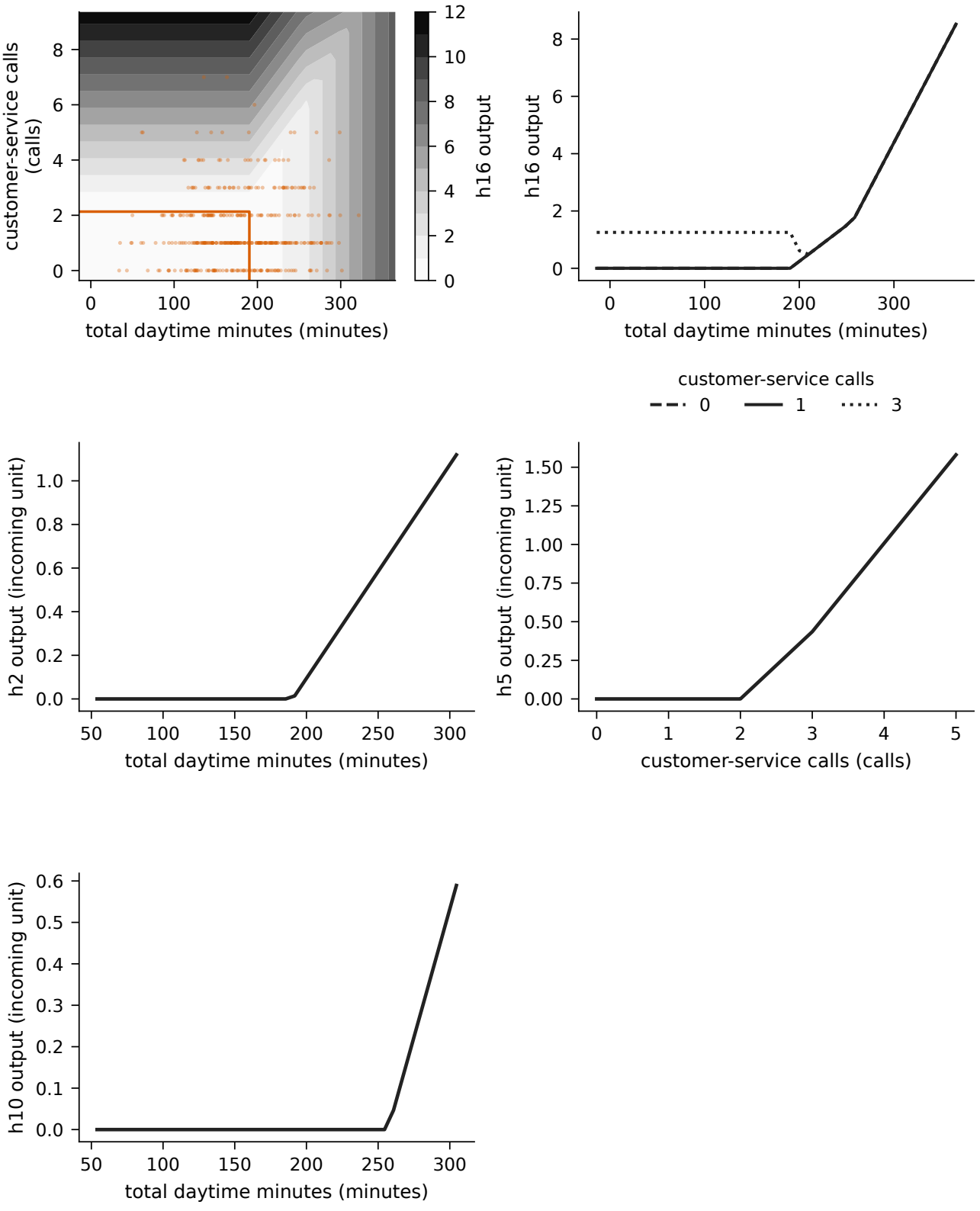


Figure 235: Usage and service calls: This part's own output, unit h16 (combines 3 non-linear units), on its own scale (its activation), with every other input at the typical case; it is active for 1927 of 4000. Left: its output over its two main inputs (orange line: where it switches on; dots: the data); right: slices at three values of the second input; then each incoming non-linear unit against its own main input.

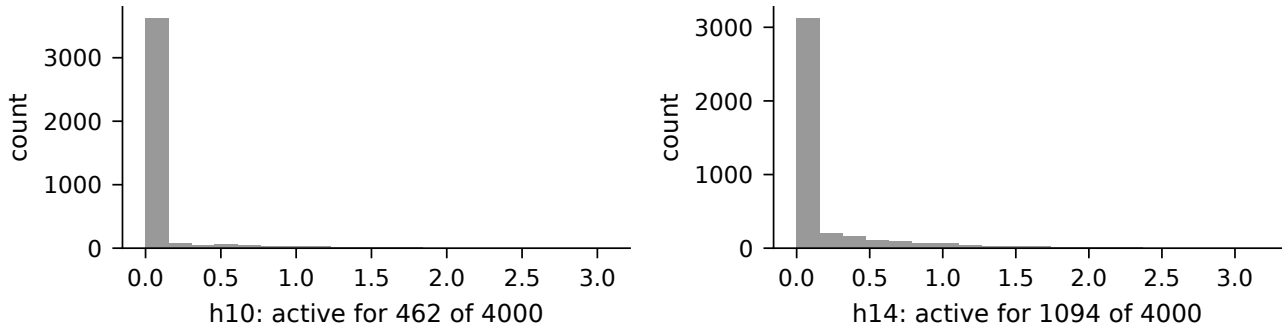


Figure 236: Usage and service calls: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Dive deeper: Usage score (5.2.3). Technical detail: T.2.

5.2.3 Usage score

Part of: Usage and service calls (5.2.2).
Figure 237 shows what feeds this part; Figure 238 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

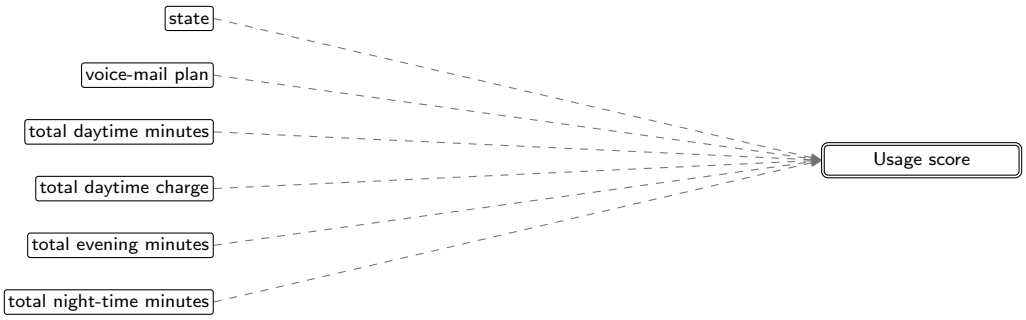


Figure 237: Usage score: what feeds it, one level down.

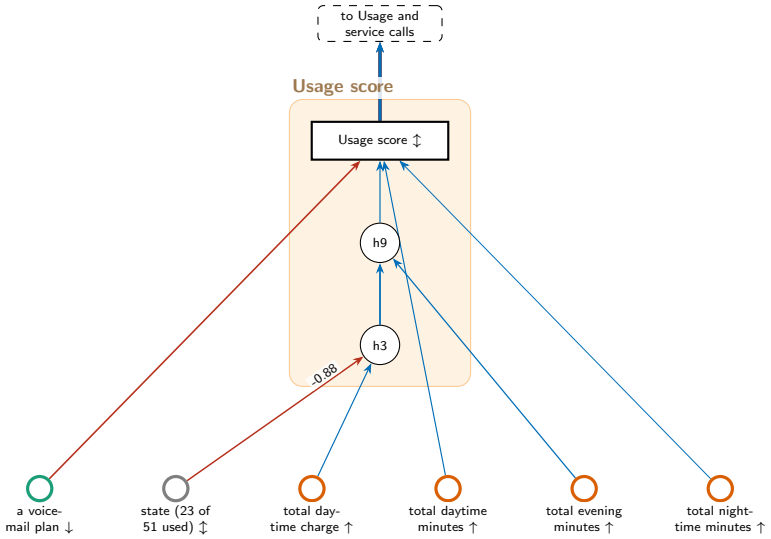


Figure 238: Usage score: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part builds a usage score: heavy daytime use raises the chance of leaving, in an uneven way and more so with high evening minutes, and a voice-mail plan lowers it. [C071, C067, C062]

Evidence. Through it, daytime minutes add only above about 187 minutes for customers without a voice-mail plan, in an uneven way, from about 1% to about 11% between 0 and 271 minutes. [C071] Daytime minutes count more with more than 201 evening minutes (about 1% to about 11%, against about 1% to about 2%) and with more than 200 night-time minutes (about 1% to about 7%, against about 1% to about 3%), in an uneven way. [C067, C068] A voice-mail plan lowers the chance through it from about 3% to about 1% for the typical customer it acts on. [C062] Its formula reproduces the part exactly on all 4000 training customers. [C076]
Evidence plots: Figure 239, Figure 240, Figure 241, Figure 242, Figure 243, Figure 244, Figure 245, Figure 246.

Worked example: reading this part. Figure 242 maps the part's own output over its main inputs. In the grey region it is switched off and adds nothing; past the black line it switches on, and the colour shows how much it adds to the score. The line is the rule in the claim (C119): it switches on when total daytime minutes above about 220 (for total evening minutes from about 0.00 to 164); total daytime minutes plus about 0.75 × total evening minutes above about 341 (for total evening minutes from about 164 to 364). 1382 of 4000 training customers lie on the 'on' side, and 23% of them left, against 10% of the others: the filled points crowd on the 'on' side. Figure 243 cuts the map into slices: the hinge moves as the other input changes, which is what a tilted line means. Figure 244 shows what this does to the final prediction for the typical case.
Figure 245 shows this part's own output, unit h2: over its two main inputs, with the line where it switches on.

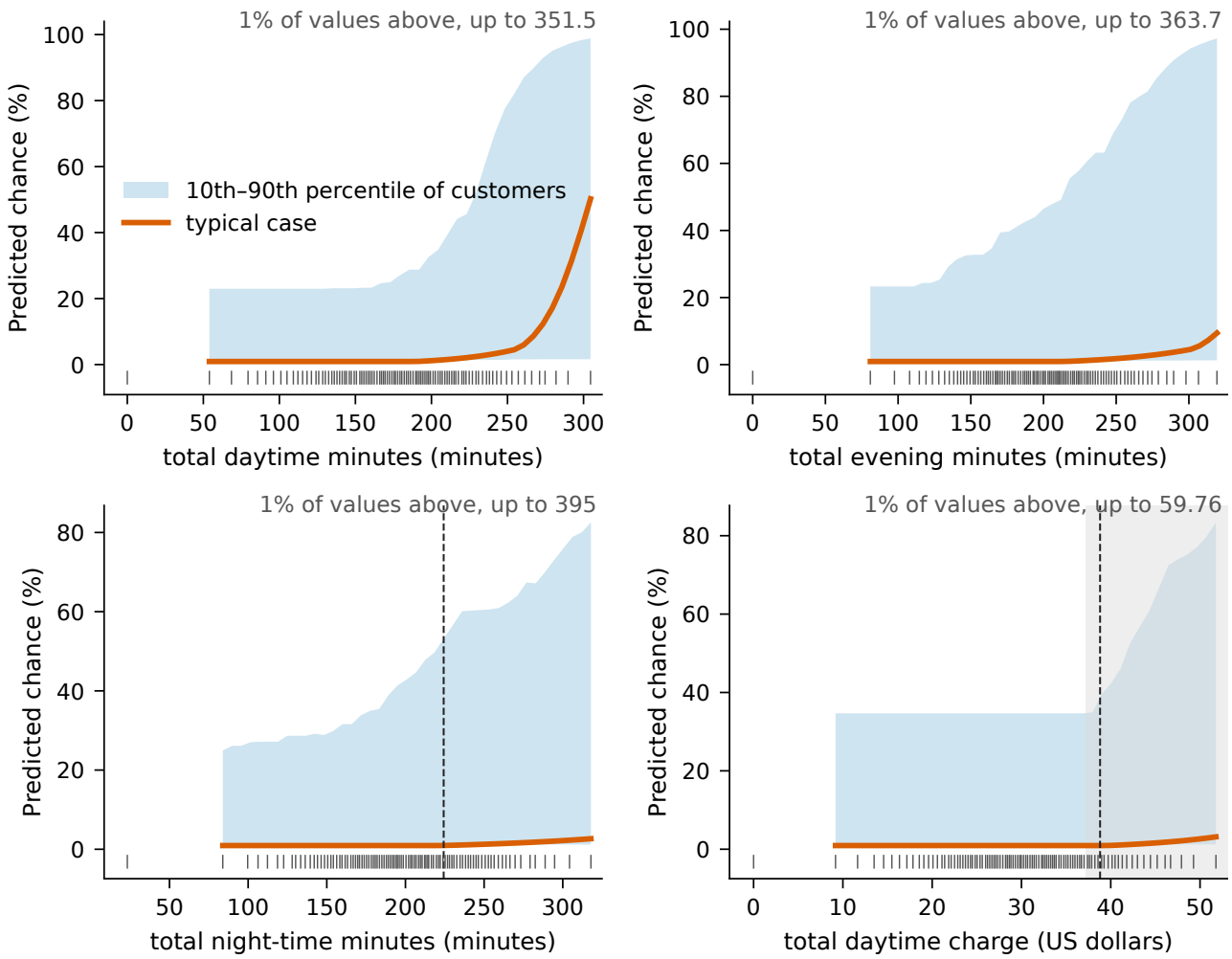


Figure 239: Usage score: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

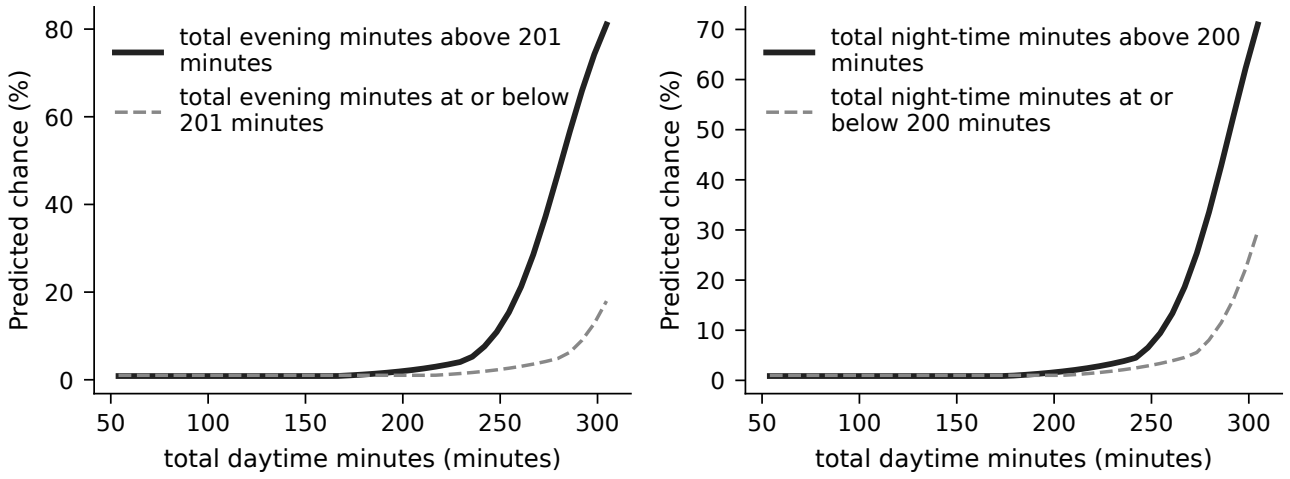


Figure 240: Usage score: How the effect differs between groups: one curve per group, for the typical case of each group.

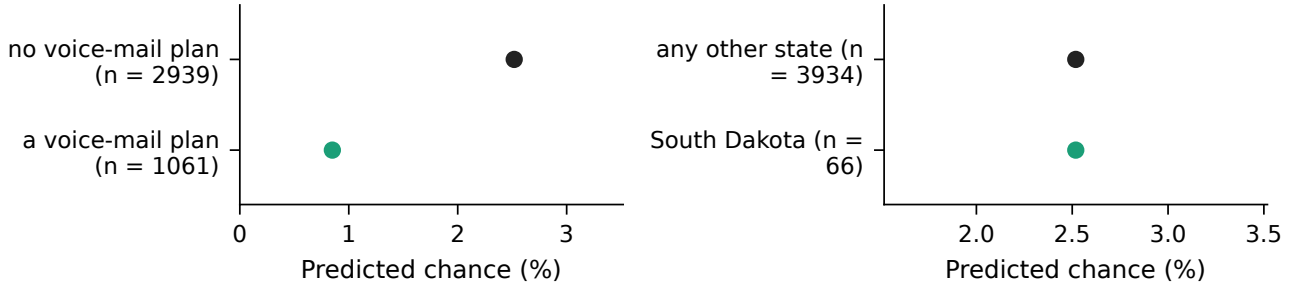


Figure 241: Usage score: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

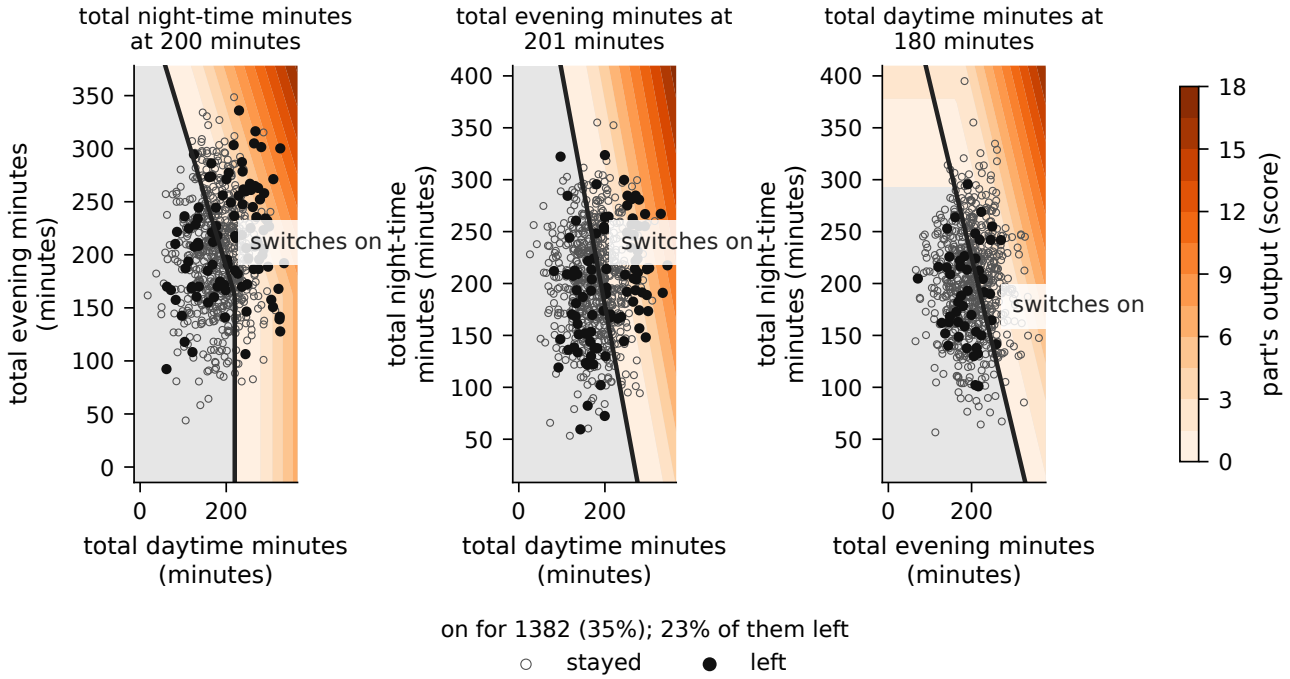


Figure 242: Usage score: The part's own output over each pair of its three main inputs (the third held at its median, named above each panel; the points are the training rows whose third input is between its 40th and 60th percentile), other inputs at the typical case: colour where it is on, flat grey where it is switched off, the black line where it switches on; training customers as hollow (no event) and filled (event) points, so the part can be seen catching the events (the predicted-chance overlay was harder to read).

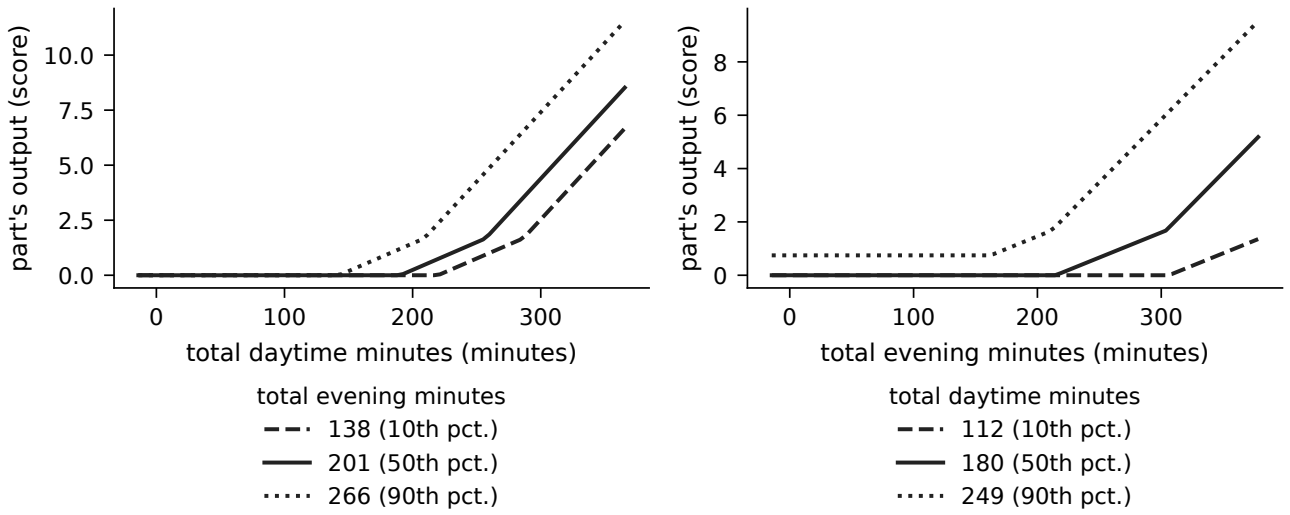


Figure 243: Usage score: Slices of the same map: the part's own output against each input at three values of the other, so the tilted boundary shows as a shifting hinge.

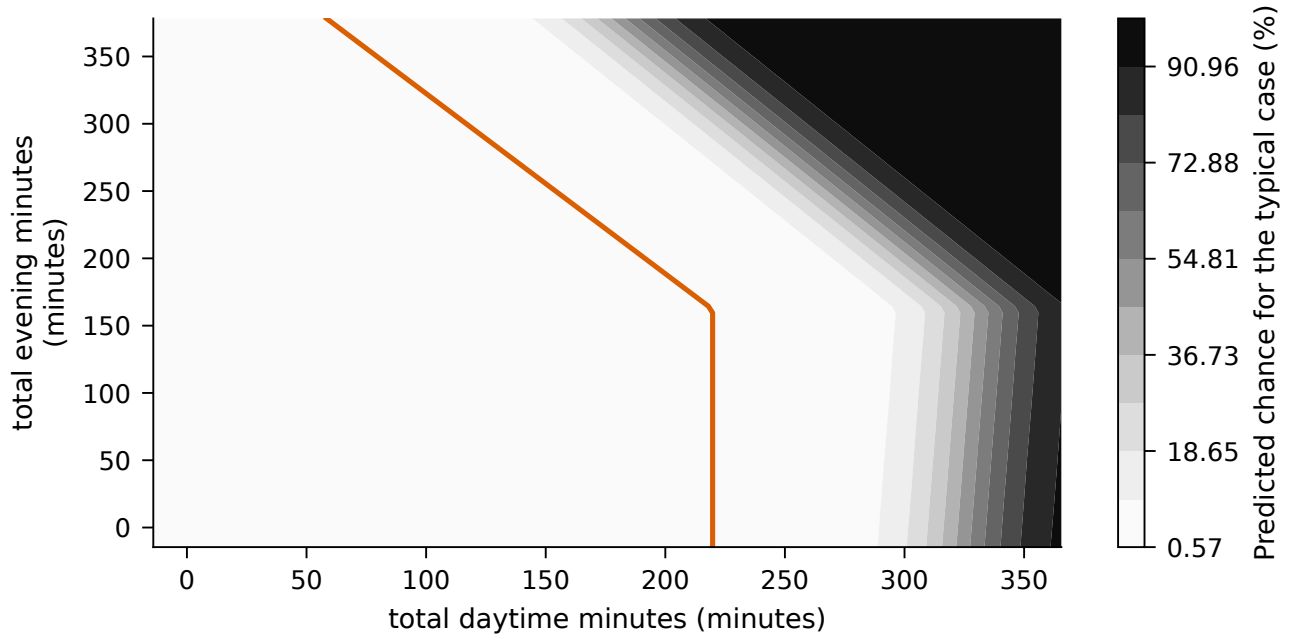


Figure 244: Usage score: The same map on the final scale: the predicted value for the typical case, with the switch-on line (orange).

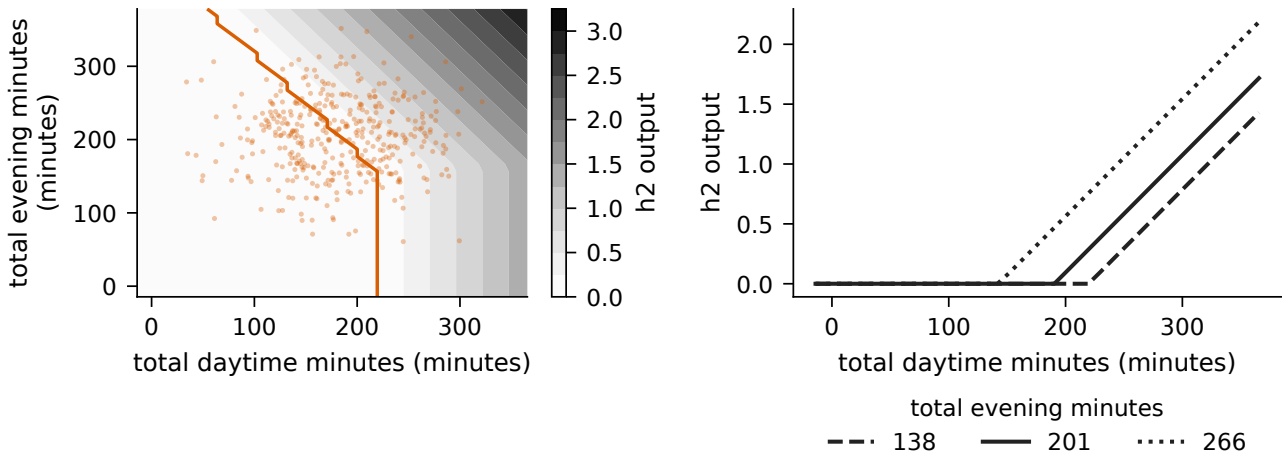


Figure 245: Usage score: This part's own output, unit h2 (the output of 'Usage score'), on its own scale (its activation), with every other input at the typical case; it is active for 1382 of 4000. Left: its output over its two main inputs (orange line: where it switches on; dots: the data); right: slices at three values of the second input.

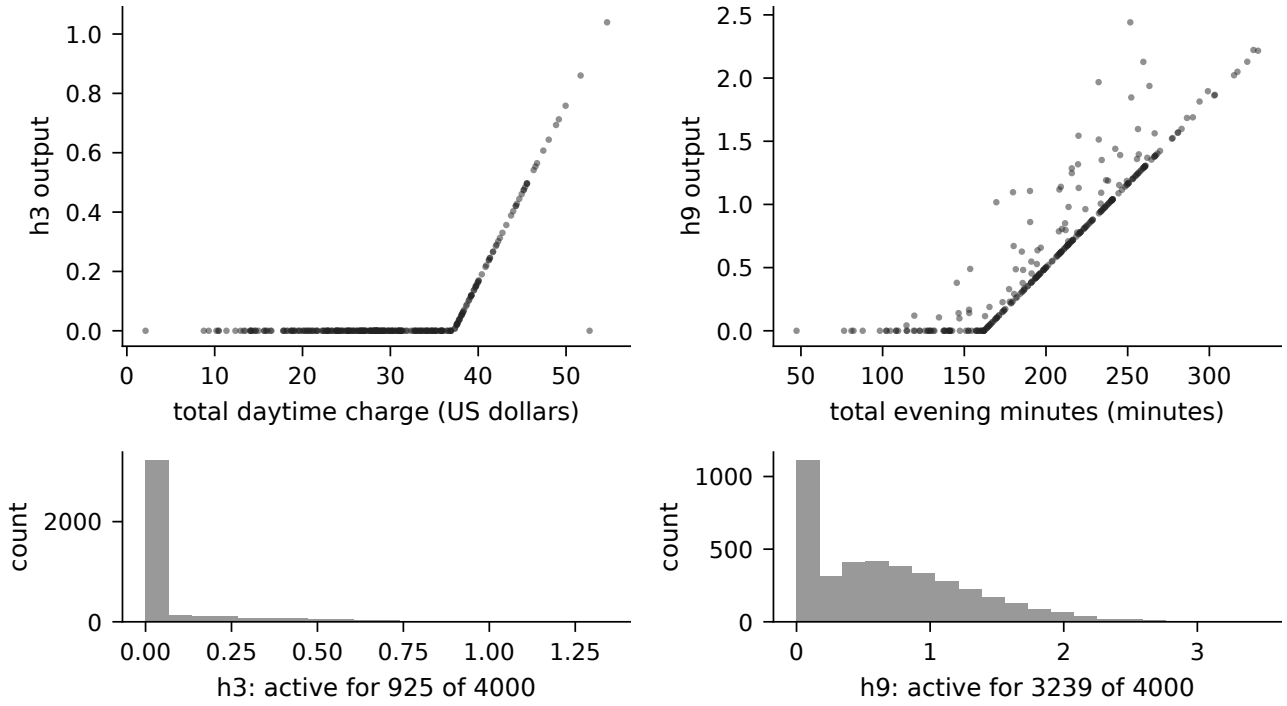


Figure 246: Usage score: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.3.

5.2.4 International plan

Part of: The whole model (5.2.1).
Figure 247 shows what feeds this part; Figure 248 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

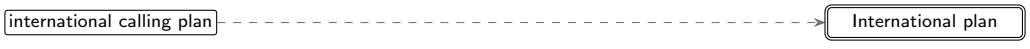


Figure 247: International plan: what feeds it, one level down.



Figure 248: International plan: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part raises the chance of leaving for customers with an international plan. [C077]

Evidence. It acts on the 380 of 4000 training customers with an international plan, 161 of whom left; for the typical customer it acts on, the chance is about 2% without it and about 26% with it. [C078] For a typical customer, an international plan takes the chance from about 1% to about 15%. [C077] Its formula reproduces the part exactly on all 4000 training customers. [C079]
Evidence plots: Figure 249, Figure 250.

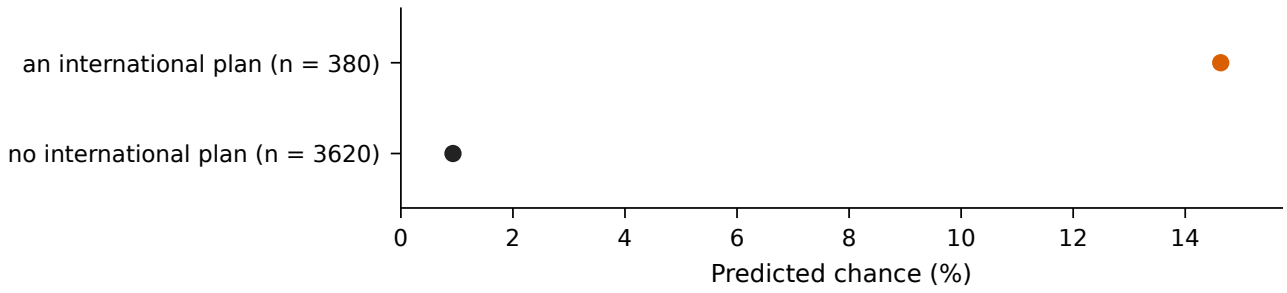


Figure 249: International plan: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

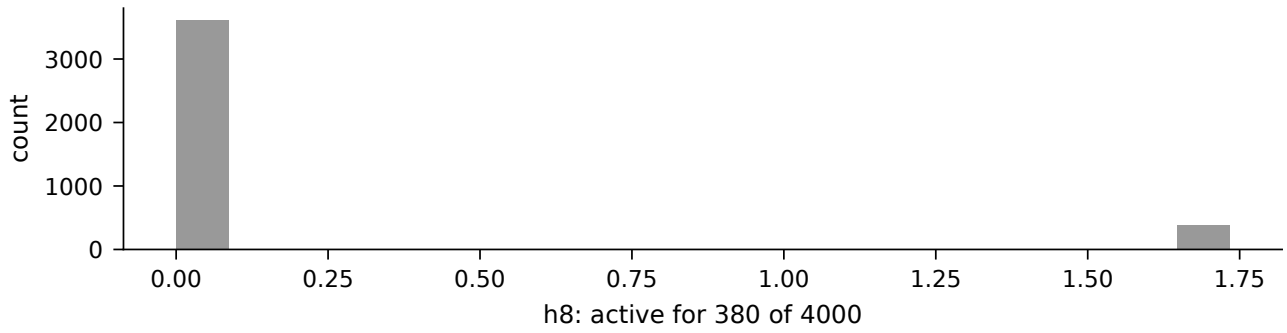


Figure 250: International plan: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.4.

5.2.5 Fixed-weight inputs

Part of: The whole model (5.2.1).
Figure 251 shows what feeds this part; Figure 252 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

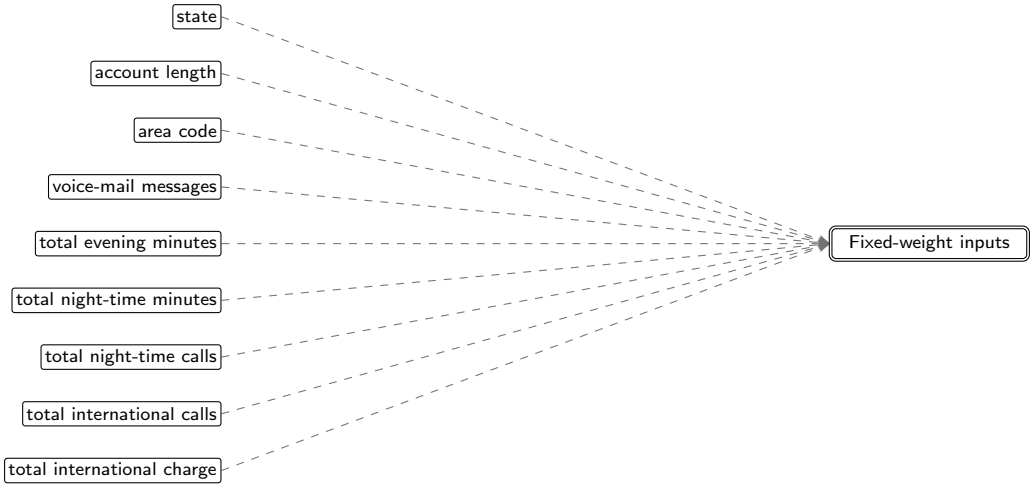


Figure 251: Fixed-weight inputs: what feeds it, one level down.

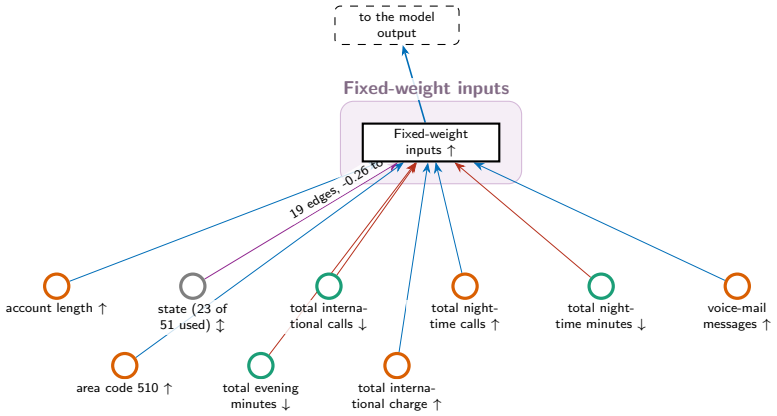


Figure 252: Fixed-weight inputs: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part adds small fixed-weight effects: a higher international charge, more voice-mail messages and a longer account raise the chance steadily, more international calls lower it, each extra call taking away about the same, and some states shift it up or down. [C083, C086, C098, C087, C080, C082]

Evidence. Through it, compared with most other states, Washington raises a typical customer's chance to about 3% and Montana to about 2%, and Vermont, Rhode Island and Colorado lower it to under 1%. [C080, C081, C082, C084, C085] The international charge raises it steadily, from under 1% at 1.84 to about 1% at 3.70 US dollars, and international calls lower it from about 1% at 2 to under 1% at 8 calls, each extra call taking away about the same. [C083, C087] More evening and night-time minutes lower it steadily through this part, by less than a point. [C092, C096] Its formula reproduces the part exactly on all 4000 training customers. [C107]
Evidence plots: Figure 253, Figure 254, Figure 255, Figure 256.

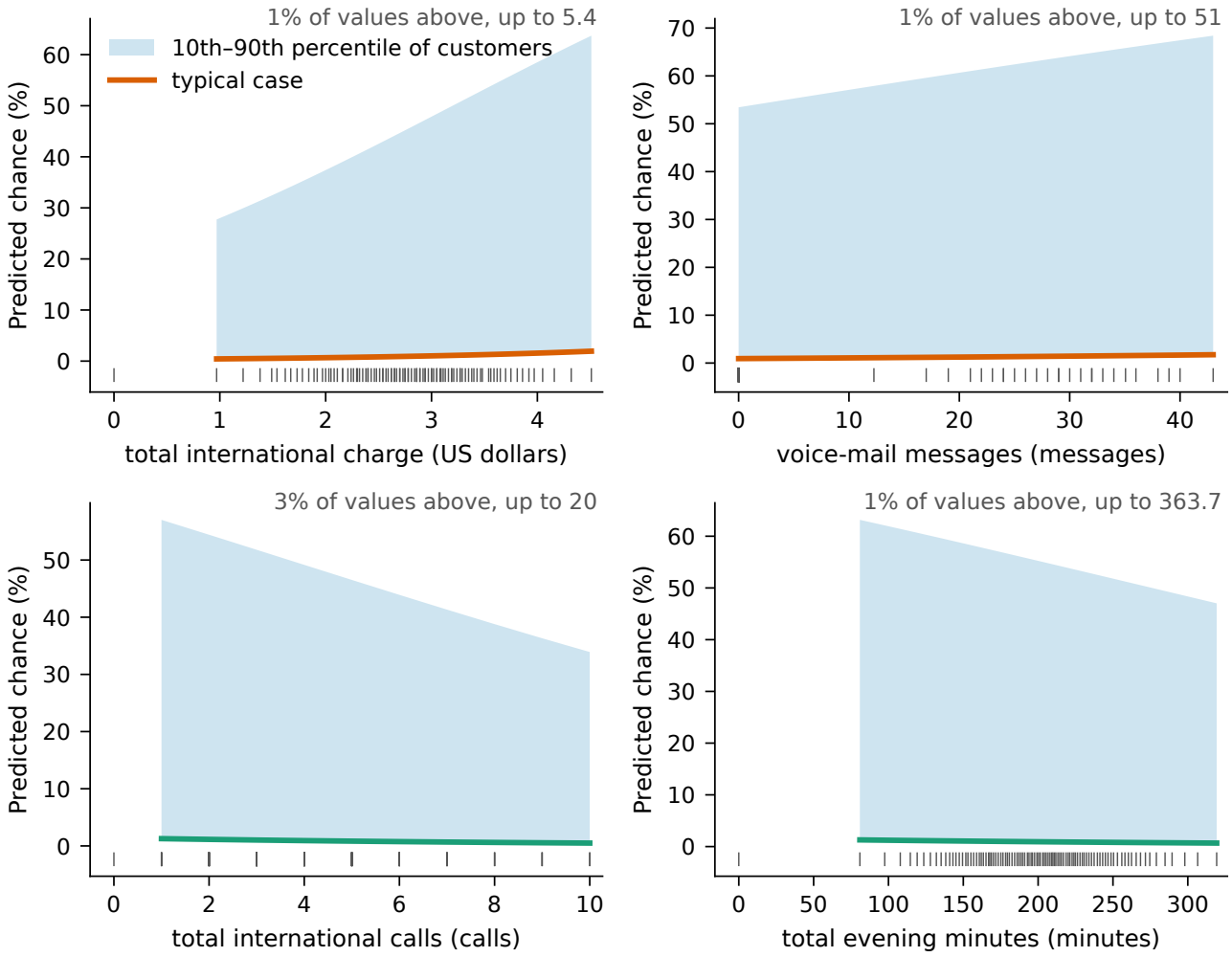


Figure 253: Fixed-weight inputs: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

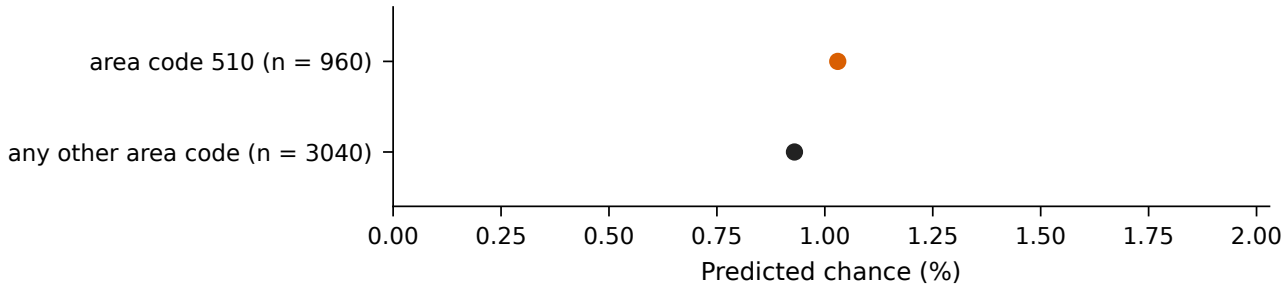


Figure 254: Fixed-weight inputs: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

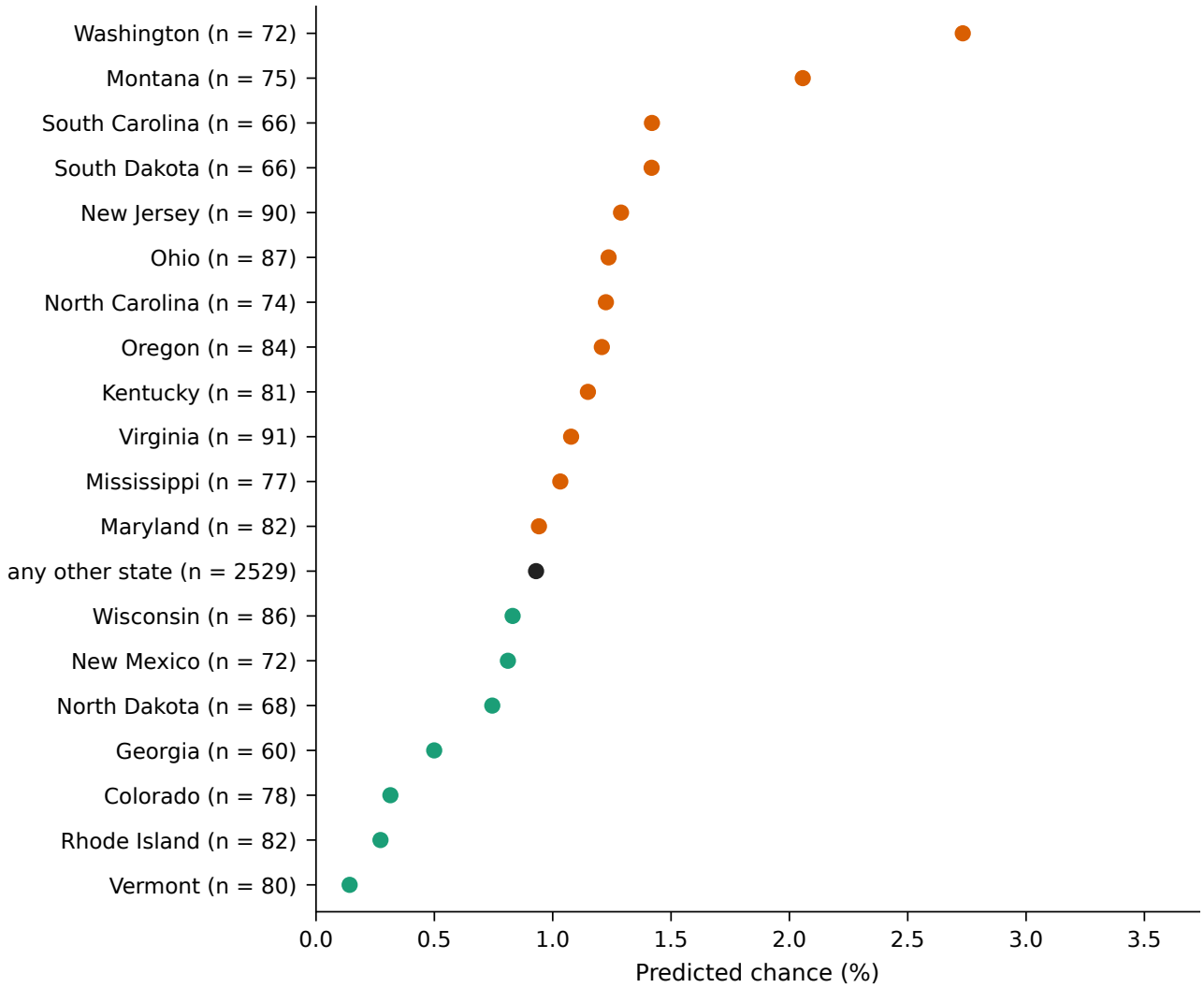


Figure 255: Fixed-weight inputs: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

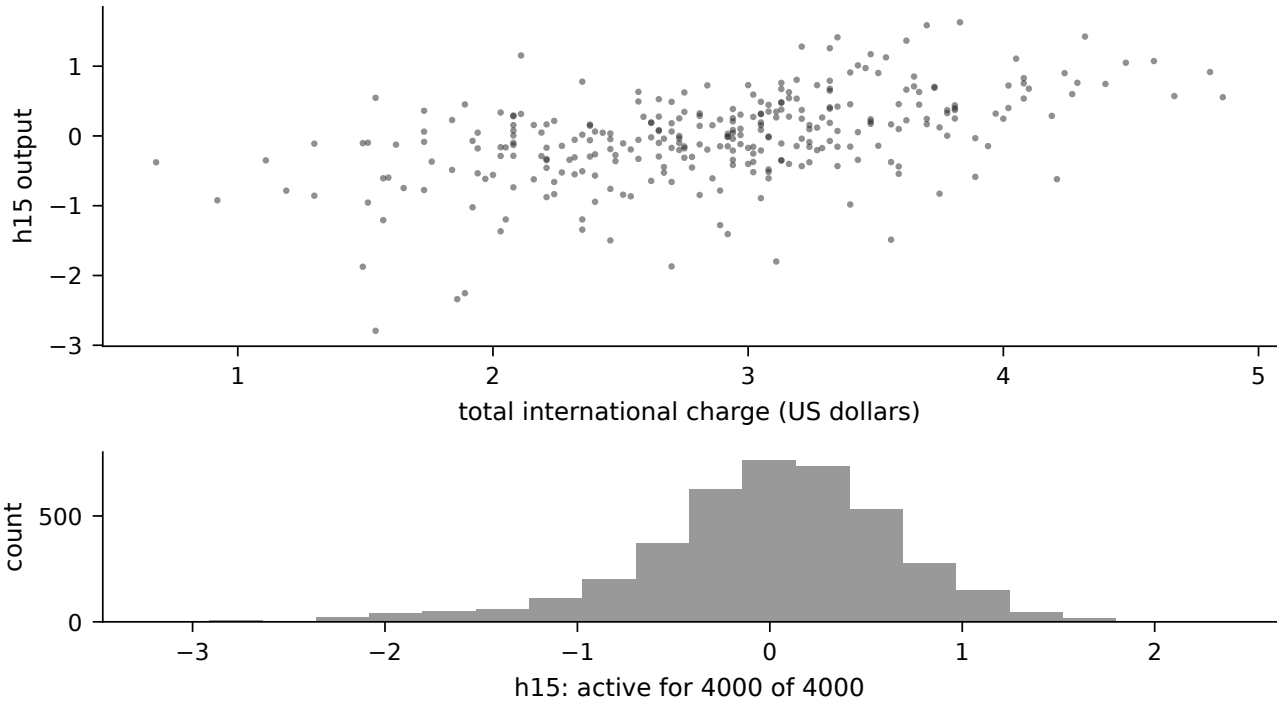


Figure 256: Fixed-weight inputs: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.5.

5.2.6 Voice mail and South Dakota

Part of: The whole model (5.2.1).
Figure 257 shows what feeds this part; Figure 258 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).



Figure 257: Voice mail and South Dakota: what feeds it, one level down.

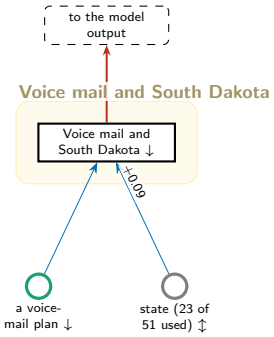


Figure 258: Voice mail and South Dakota: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part lowers the chance of leaving for customers in South Dakota and for customers with a voice-mail plan. [C108, C109]

Evidence. Through it, a customer in South Dakota has under 1% instead of about 1%, and a voice-mail plan also takes a typical customer to under 1%. [C108, C109] Its formula reproduces the part exactly on all 4000 training customers. [C110]

Evidence plots: Figure 259, Figure 260.

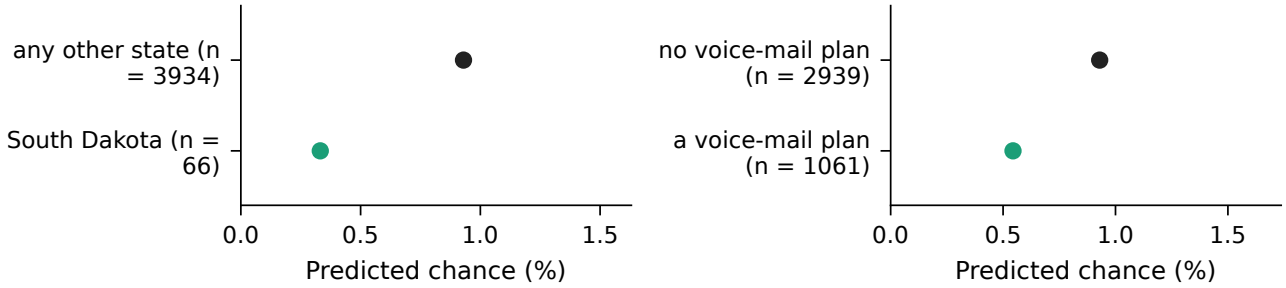


Figure 259: Voice mail and South Dakota: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

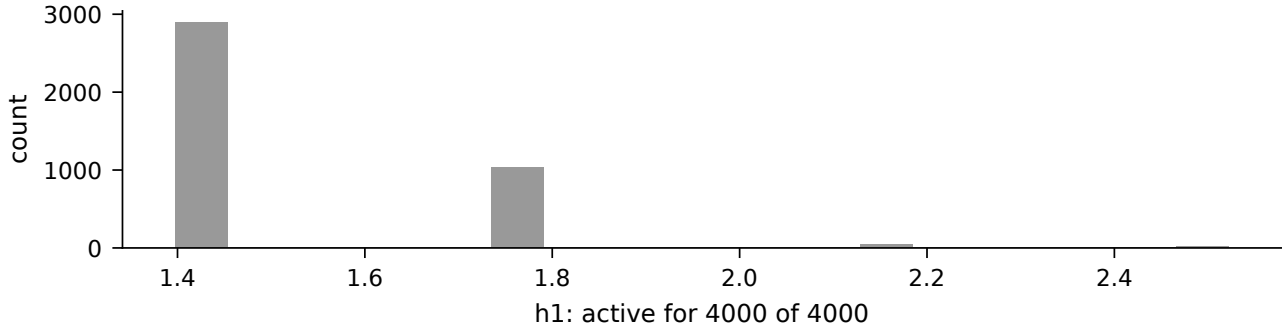


Figure 260: Voice mail and South Dakota: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.6.

5.2.7 State adjustments

Part of: The whole model (5.2.1).
Figure 261 shows what feeds this part; Figure 262 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).



Figure 261: State adjustments: what feeds it, one level down.

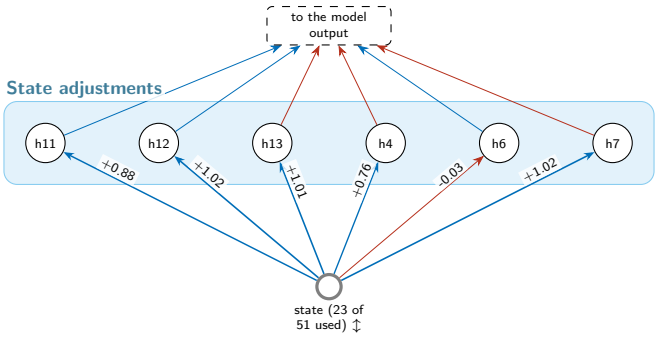


Figure 262: State adjustments: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part shifts the chance of leaving by state: it is lowest in West Virginia and Virginia and highest in South Dakota. [C111, C114, C115]

Evidence. Through it, a typical customer's chance is about 5% in South Dakota and about 2% in Indiana and in most other states, against about 1% in West Virginia and under 1% in Virginia. [C111, C112, C114, C115] Its formula reproduces the part exactly on all 4000 training customers. [C117]
Evidence plots: Figure 263, Figure 264.

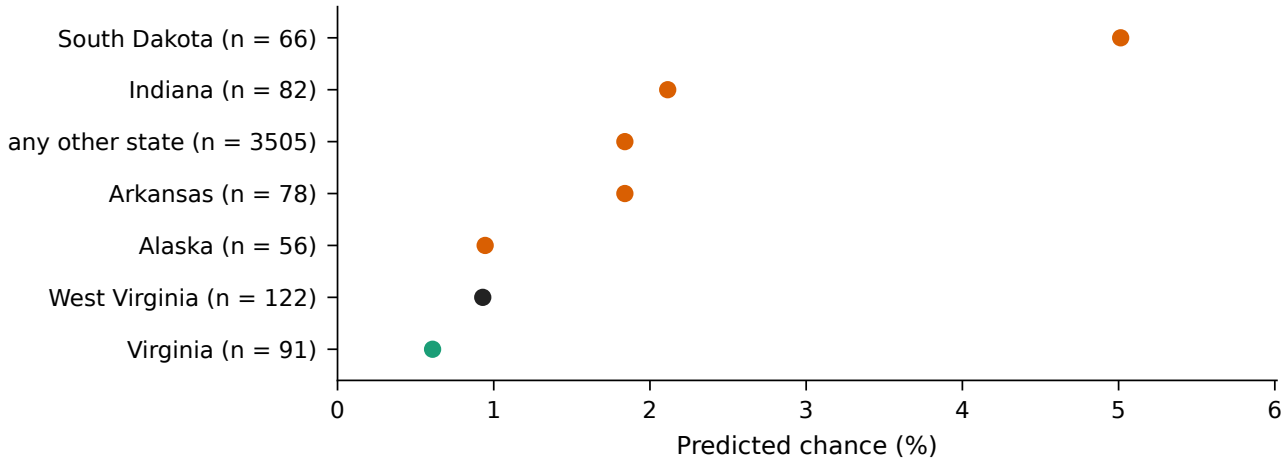


Figure 263: State adjustments: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

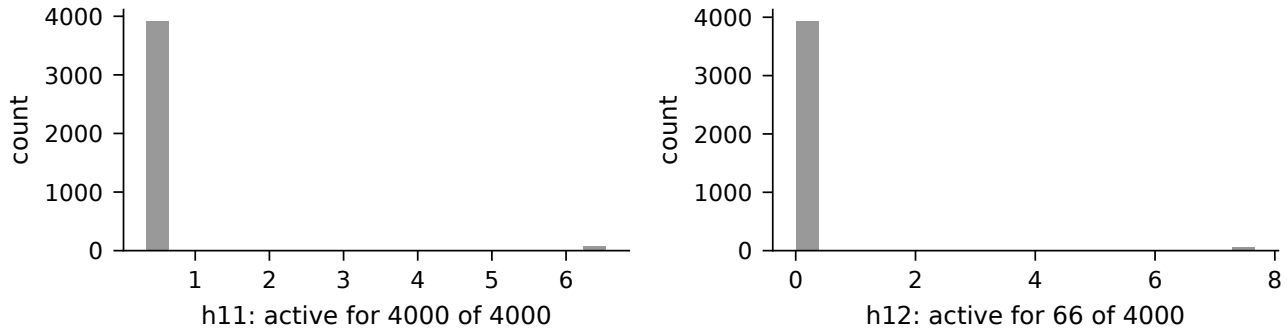


Figure 264: State adjustments: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.7.

5.3 Not supported by the model

- C016: direction claim about total evening minutes in the whole model: not supported by the model (58% of patients have the opposite sign (allowed: 10%)).
- C032: direction claim about total night-time minutes in the whole model: not supported by the model (63% of patients have the opposite sign (allowed: 10%)).
- C058: threshold claim about total night-time minutes in part ann_83: not supported by the model (the folds are spread over 39% of the observed range (allowed 20%), also within each category group: no single threshold; the effect past the fold is too small or not consistent across patients).
- C059: threshold claim about total evening minutes in part ann_83: not supported by the model (the effect past the fold is too small or not consistent across patients; no training patient is past the fold).
- C069: interaction claim about total daytime minutes and voice-mail plan in part ann_79: not supported by the model (the part's counterfactual difference has the same sign for only 89% of affected patients).
- C070: interaction claim about total evening minutes and voice-mail plan in part ann_79: not supported by the model (the part's counterfactual difference has the same sign for only 89% of affected patients).
- C073: threshold claim about total night-time minutes in part ann_79: not supported by the model (the folds are spread over 39% of the observed range (allowed 20%), also within each category group: no single threshold; the effect past the fold is too small or not consistent across patients).

- C074: threshold claim about total evening minutes in part ann_79: not supported by the model (the effect past the fold is too small or not consistent across patients; no training patient is past the fold).
- C075: subgroup claim about state and voice-mail plan in part ann_79: not supported by the model (its effect goes both ways across the patients it acts on).
- C118: boundary claim about number of calls to customer service and total daytime minutes in part ann_83: not supported by the model (no switch-on line could be traced on the first map to check against the data; no switch-on boundary in the observed range).

Technical detail: churn100-s7-c1

The record behind each part: the original hypothesis, the closed form, and every claim with its estimate (log-odds and odds ratio), range across patients, n and test check.

T.1 The whole model

Annotation ann_84 (churn_logit). Reads: state, account length, area code, international calling plan, voice-mail plan, number of voice-mail messages, total daytime minutes, total daytime calls, total daytime charge, total evening minutes, total evening calls, total evening charge, total night-time minutes, total night-time calls, total night-time charge, total international minutes, total international calls, total international charge, number of calls to customer service.

Original hypothesis (unchanged). Root: logit P(churn) = -1.96 + linear_backbone (identity_lin) + usage_response (identity_usage) + 1.68*6711 (intl_plan_switch) - 1.40*582 (vmail_state41_unit) + state_offsets (-0.26*2067 - 0.10*5821 + 0.02*12726 + 0.13*13164 - 0.12*13420 + 0.03*5667), purely additive in the logit; the sigmoid is the only coupling. 35 inputs are unconnected (international minutes, night charge, evening charge and calls, day calls, area codes 408/415 and 29 state indicators) and have zero effect. Train AUC 0.9189, accuracy 0.921. Performance sources (train): all units replaced by least-squares linear fits gives AUC 0.8336 (an L2 logistic regression refit on all 71 inputs gets 0.8433); restoring usage_response alone gives 0.9189 (all the gain); its parts alone: usage hinge 0.8803, heavy-use hinge 0.8870, service-calls gate 0.8731. The international-plan, voicemail and state units are exact switches/offsets and restore nothing. Rare-category switches: state:46 -1.89 logit (80 rows, empirical log-odds offset -0.91); state:39 -1.23 (82, -0.93); state:47 +1.10 (72, +0.92); state:5 -1.09 (78, -0.51); state:45 -0.97 (91, -0.85); state:26 +0.81 (75, +0.50); state:49 -0.69 (122, -0.32); state:0 -0.68 (56, -0.76); state:10 -0.63 (60, -0.22); state:41 +0.25 mean, range -3.85..+0.43 (66, -0.04: it switches off the 1406 day-charge hinge, so heavy users lose up to 3.85).

Closed form. Read through its parts (their formulas are in their own sections).

Split inputs. To annotate the parts, the explanation split these inputs into copies (a refactoring step that does not change what the model computes); the diagrams draw each input once: total evening minutes (2 copies); total night-time minutes (2 copies); a voice-mail plan (2 copies).

State: every level the model reads. Whole-model change in the predicted chance for a typical customer when only the state changes from the reference; n = training customers in that level.

Claim	From	To	n	Chance	OK
C002	West Virginia	Washington	72	1% to 5%	yes
C003	West Virginia	Montana	75	1% to 4%	yes
C006	West Virginia	South Carolina	66	1% to 3%	yes
C007	West Virginia	South Dakota	66	1% to 3%	yes
C008	West Virginia	New Jersey	90	1% to 3%	yes
C009	West Virginia	Ohio	87	1% to 2%	yes
C010	West Virginia	North Carolina	74	1% to 2%	yes
C011	West Virginia	Oregon	84	1% to 2%	yes
C012	West Virginia	Kentucky	81	1% to 2%	yes
C013	West Virginia	Indiana	82	1% to 2%	yes
C014	West Virginia	Mississippi	77	1% to 2%	yes
C015	West Virginia	Maryland	82	1% to 2%	yes
C017	West Virginia	Arkansas	78	1% to 2%	yes
C018	West Virginia	state other than Alaska, Georgia, Indiana, Kentucky, Arka...	2191	1% to 2%	yes
C020	West Virginia	Wisconsin	86	1% to 2%	yes
C021	West Virginia	New Mexico	72	1% to 2%	yes
C024	West Virginia	North Dakota	68	1% to 1%	yes
C034	West Virginia	Georgia	60	1% to 1%	yes
C036	West Virginia	Alaska	56	1% to 1%	yes
C030	West Virginia	Virginia	91	1% to 1%	yes
C028	West Virginia	Colorado	78	1% to 1%	yes
C026	West Virginia	Rhode Island	82	1% to 1%	yes
C022	West Virginia	Vermont	80	1% to <1%	yes

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C001	direction: international calling plan: no international plan to an international plan (whole model) *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 15%	2.91	18.3	2.91 to 2.91	3620, 380	agrees; AUC gain 0.060 [0.033, 0.090]	yes
C002	direction: state: West Virginia to Washington (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 5%	1.79	5.98	1.79 to 1.79	122, 72	disagrees; AUC gain -0.004 [-0.017, 0.006]	yes
C003	direction: state: West Virginia to Montana (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 4%	1.50	4.47	1.50 to 1.50	122, 75	disagrees; AUC gain -0.004 [-0.017, 0.006]	yes
C004	direction: total daytime minutes: 112 minutes to 249 minutes (whole model) *shape: accelerating; 67% of patients same; fallback; risk saturates near 0%*	1% to 4%	1.37	3.93	0.00 to 4.32	4000	agrees; AUC gain 0.035 [0.016, 0.052]	yes
C005	direction: customer-service calls: 0 calls to 3 calls (whole model) *shape: accelerating; 89% of patients same; mixed (but hinge for customers without a voice-mail plan); risk saturates near 0%*	1% to 3%	0.86	2.37	0.00 to 1.25	4000	agrees; AUC gain 0.097 [0.061, 0.136]	yes
C006	direction: state: West Virginia to South Carolina (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 3%	1.12	3.06	1.12 to 1.12	122, 66	disagrees; AUC gain -0.004 [-0.017, 0.006]	yes
C007	direction: state: West Virginia to South Dakota (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 3%	0.94	2.55	0.79 to 1.12	122, 66	disagrees; AUC gain -0.004 [-0.017, 0.006]	yes
C008	direction: state: West Virginia to New Jersey (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 3%	1.02	2.78	1.02 to 1.02	122, 90	disagrees; AUC gain -0.004 [-0.017, 0.006]	yes
C009	direction: state: West Virginia to Ohio (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 2%	0.98	2.66	0.98 to 0.98	122, 87	disagrees; AUC gain -0.004 [-0.017, 0.006]	yes
C010	direction: state: West Virginia to North Carolina (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 2%	0.97	2.64	0.97 to 0.97	122, 74	disagrees; AUC gain -0.004 [-0.017, 0.006]	yes
C011	direction: state: West Virginia to Oregon (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 2%	0.96	2.60	0.96 to 0.96	122, 84	disagrees; AUC gain -0.004 [-0.017, 0.006]	yes
C012	direction: state: West Virginia to Kentucky (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 2%	0.91	2.47	0.91 to 0.91	122, 81	disagrees; AUC gain -0.004 [-0.017, 0.006]	yes
C013	direction: state: West Virginia to Indiana (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 2%	0.83	2.30	0.83 to 0.83	122, 82	disagrees; AUC gain -0.004 [-0.017, 0.006]	yes
C014	direction: state: West Virginia to Mississippi (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 2%	0.80	2.22	0.80 to 0.80	122, 77	disagrees; AUC gain -0.004 [-0.017, 0.006]	yes
C015	direction: state: West Virginia to Maryland (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 2%	0.70	2.02	0.70 to 0.70	122, 82	disagrees; AUC gain -0.004 [-0.017, 0.006]	yes
C016	direction: total evening minutes: 138 minutes to 266 minutes (whole model) *shape: dip at 212 minutes; 36% of patients same; fallback; risk saturates near 0%*	1% to 2%	0.67	1.96	-0.35 to 3.65	4000	agrees; AUC gain 0.030 [0.010, 0.050]	no
C017	direction: state: West Virginia to Arkansas (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 2%	0.69	2.00	0.69 to 0.69	122, 78	disagrees; AUC gain -0.004 [-0.017, 0.006]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C018	direction: state: West Virginia to state other than Alaska, Georgia, Indiana, Kentucky, Arkansas, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, West Virginia, Colorado (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 2%	0.69	2.00	0.69 to 0.69	122, 2191	disagrees; AUC gain -0.004 [-0.017, 0.006]	yes
C019	direction: total international charge: 1.84 US dollars to 3.70 US dollars (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	0.80	2.23	0.80 to 0.80	4000	unclear; AUC gain 0.003 [-0.005, 0.011]	yes
C020	direction: state: West Virginia to Wisconsin (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 2%	0.58	1.78	0.58 to 0.58	122, 86	disagrees; AUC gain -0.004 [-0.017, 0.006]	yes
C021	direction: state: West Virginia to New Mexico (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 2%	0.55	1.74	0.55 to 0.55	122, 72	disagrees; AUC gain -0.004 [-0.017, 0.006]	yes
C022	direction: state: West Virginia to Vermont (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to <1%	-1.20	0.30	-1.20 to -1.20	122, 80	disagrees; AUC gain -0.004 [-0.017, 0.006]	yes
C023	direction: voice-mail messages: 0 messages to 32 messages (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	0.47	1.60	0.47 to 0.47	4000	unclear; AUC gain 0.0008 [-0.004, 0.006]	yes
C024	direction: state: West Virginia to North Dakota (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 1%	0.47	1.60	0.47 to 0.47	122, 68	disagrees; AUC gain -0.004 [-0.017, 0.006]	yes
C025	direction: total international calls: 2 calls to 8 calls (whole model) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 0%*	1% to 1%	-0.63	0.53	-0.63 to -0.63	4000	unclear; AUC gain 0.002 [-0.004, 0.008]	yes
C026	direction: state: West Virginia to Rhode Island (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 1%	-0.54	0.58	-0.54 to -0.54	122, 82	disagrees; AUC gain -0.004 [-0.017, 0.006]	yes
C027	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (whole model) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	1% to 1%	-1.37	0.25	-3.46 to -0.54	2939, 1061	agrees; AUC gain 0.019 [0.005, 0.031]	yes
C028	direction: state: West Virginia to Colorado (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 1%	-0.40	0.67	-0.40 to -0.40	122, 78	disagrees; AUC gain -0.004 [-0.017, 0.006]	yes
C029	direction: total daytime charge: 19.07 US dollars to 42.40 US dollars (whole model) *shape: hinge at 40.06 US dollars; 50% of patients same; fallback; risk saturates near 0%; matches the fold at 38.81 US dollars*	1% to 1%	0.24	1.27	0.00 to 1.11	4000	unclear; AUC gain 0.0003 [-0.007, 0.008]	yes
C030	direction: state: West Virginia to Virginia (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 1%	-0.28	0.76	-0.28 to -0.28	122, 91	disagrees; AUC gain -0.004 [-0.017, 0.006]	yes
C031	direction: account length: 48 to 152 (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	0.23	1.26	0.23 to 0.23	4000	disagrees; AUC gain -0.001 [-0.003, 0.001]	yes
C032	direction: total night-time minutes: 136 minutes to 264 minutes (whole model) *shape: dip at 218 minutes; 30% of patients same; fallback; risk saturates near 0%*	1% to 1%	0.34	1.40	-0.29 to 2.38	4000	agrees; AUC gain 0.007 [0.0006, 0.013]	no
C033	direction: area code: area code other than area code 510 to area code 510 (whole model) *shape: categorical; highest to lowest: area code 510, area code other than area code 510*	1% to 1%	0.10	1.11	0.10 to 0.10	3040, 960	unclear; AUC gain 8e-05 [-0.001, 0.002]	yes
C034	direction: state: West Virginia to Georgia (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 1%	0.07	1.07	0.07 to 0.07	122, 60	disagrees; AUC gain -0.004 [-0.017, 0.006]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C035	direction: total night-time calls: 74 calls to 125 calls (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	0.06	1.07	0.06 to 0.06	4000	disagrees; AUC gain -9e-05 [-0.0008, 0.0006]	yes
C036	direction: state: West Virginia to Alaska (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 1%	0.02	1.02	0.02 to 0.02	122, 56	disagrees; AUC gain -0.004 [-0.017, 0.006]	yes
C037	magnitude: international calling plan: no international plan to an international plan, typical patient *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 15%	2.91	18.3		3620, 380	agrees; AUC gain 0.060 [0.033, 0.090]	yes
C038	magnitude: state: West Virginia to Washington, typical patient *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 5%	1.79	5.98		122, 72	disagrees; AUC gain -0.004 [-0.017, 0.006]	yes
C039	magnitude: total daytime minutes: 112 minutes to 249 minutes, typical patient *shape: accelerating; 67% of patients same; fallback; risk saturates near 0%*	1% to 4%	1.47	4.37		4000	agrees; AUC gain 0.035 [0.016, 0.052]	yes
C040	magnitude: customer-service calls: 0 calls to 3 calls, typical patient *shape: accelerating; 89% of patients same; mixed (but hinge for customers without a voice-mail plan); risk saturates near 0%*	1% to 3%	1.25	3.50		4000	agrees; AUC gain 0.097 [0.061, 0.136]	yes
C041	magnitude: total international charge: 1.84 US dollars to 3.70 US dollars, typical patient *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	0.80	2.23		4000	unclear; AUC gain 0.003 [-0.005, 0.011]	yes
C042	null: model ignores total daytime calls (exact rule)		0.00			4000	train only	yes
C043	null: model ignores total evening calls (exact rule)		0.00			4000	train only	yes
C044	null: model ignores total evening charge (exact rule)		0.00			4000	train only	yes
C045	null: model ignores total international minutes (exact rule)		0.00			4000	train only	yes
C046	null: model ignores total night-time charge (exact rule)		0.00			4000	train only	yes
C047	structure: made of linear_backbone, usage_response, intl_plan_switch, vmail_state41_unit, state_offsets; added					4000	train only	yes

T.2 Usage and service calls

Annotation ann_83 (usage_response). Reads: state, voice-mail plan, total daytime minutes, total daytime charge, total evening minutes, total night-time minutes, number of calls to customer service.

Original hypothesis (unchanged). Composition on usage_hinge: identity_usage = 2.57*879 + 3.06*10552 + 2.88*4344 enters the logit with weight 1. 10552 = relu(-0.64 + 1.17*relu(-0.16 + 1.08*879)) is a second, steeper hinge that opens once 879 exceeds 0.65 (about 240 day minutes at median other use; active for 12% of customers), so the usage effect accelerates. 4344 = relu(-0.38 + 0.75 calls - 3.24*879) is a service-calls gate: shut for 0-2 calls, open from 3 calls, and closed again by usage (at 3 calls once 879 > 0.14, at 5 calls once 879 > 0.49). Hypothesis: usage and service calls interact non-additively: service calls raise churn only for low-usage customers (or voicemail-plan customers, for whom usage is off), heavy users get the usage hinges instead. Evidence (train): usage logit (879+10552) by day minutes 0.14 (<200), 0.58 (200-220), 1.08 (220-240), 2.02 (240-260), 3.43 (260-280), 6.37 (>=280); churn 0.10/0.06/0.14/0.26/0.46/0.67 vs model 0.09/0.09/0.17/0.30/0.49/0.69. Without a voicemail plan churn reaches 0.84 above 280 minutes (model 0.86), with a plan 0.04 (model 0.09). Gate logit at 0-7 calls: 0/0/0/0.80/2.11/3.59/5.30/7.08 (1.25/2.90/4.55/6.20 at 3-6 calls when usage is off, 0.10/0.50/1.05/2.01 when it is on). Churn by calls 0.115/0.110/0.110/0.106/0.437/0.571/0.679/0.545 vs model 0.116/0.111/0.109/0.150/0.319/0.598/0.849/0.940: the gate opens one call too early (3 calls: +0.80 logit, empirical churn unchanged) and is over-confident from 6 calls (39 customers). Mean ablation of all three terms AUC 0.6557 (the largest drop: full 0.9189); gate alone 0.8202, heavy-use hinge alone 0.9093; linear replacement 0.8336 (R2 0.53/0.29/0.53); restoring usage_response alone onto the all-linear model recovers 0.9189 (all-linear 0.8336), i.e. all of the gain.

Closed form. Read through its parts (their formulas are in their own sections).

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C048	direction: total daytime minutes: 112 minutes to 249 minutes (through this part) *shape: accelerating; 67% of patients same; fallback; risk saturates near 0%*	1% to 4%	1.37	3.93	0.00 to 4.32	4000	agrees; AUC gain 0.035 [0.016, 0.052]	yes
C049	direction: customer-service calls: 0 calls to 3 calls (through this part) *shape: accelerating; 89% of patients same; mixed (but hinge for customers without a voice-mail plan); risk saturates near 0%*	1% to 3%	0.86	2.37	0.00 to 1.25	4000	agrees; AUC gain 0.097 [0.061, 0.136]	yes
C050	direction: total evening minutes: 138 minutes to 266 minutes (through this part) *shape: accelerating; 40% of patients same; fallback; risk saturates near 0%*	1% to 2%	1.03	2.79	0.00 to 4.01	4000	agrees; AUC gain 0.034 [0.013, 0.055]	yes
C051	direction: total night-time minutes: 136 minutes to 264 minutes (through this part) *shape: hinge at 224 minutes; 35% of patients same; fallback; risk saturates near 0%*	1% to 1%	0.63	1.87	0.00 to 2.67	4000	agrees; AUC gain 0.01 [0.003, 0.016]	yes
C052	direction: total daytime charge: 19.07 US dollars to 42.40 US dollars (through this part) *shape: hinge at 38.81 US dollars; 50% of patients same; fallback; risk saturates near 0%; one point with C057 (median fold 38.81 US dollars, typical patient 40.06 US dollars, 10th-90th percentile 37.22 US dollars to 53.94 US dollars, wide)*	1% to 1%	0.24	1.27	0.00 to 1.11	4000	unclear; AUC gain 0.0003 [-0.007, 0.008]	yes
C053	direction: state: state other than South Dakota to South Dakota (through this part) *shape: categorical; highest to lowest: South Dakota, state other than South Dakota*	1% to 1%	-0.18	0.83	-0.32 to 0.00	3934, 66	unclear; AUC gain 0.0002 [-0.0002, 0.0007]	yes
C054	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: a voice-mail plan, no voice-mail plan*	1% to 1%	-0.83	0.44	-2.92 to 0.00	2939, 1061	agrees; AUC gain 0.013 [0.0009, 0.023]	yes
C055	threshold: customer-service calls: fold at about 3 calls (starts) *shape: accelerating; 86% of patients same; mixed (but hinge for customers without a voice-mail plan); risk saturates near 0%*	1% to 15% (typical patient with a fold)	2.20	9.02	0.00 to 2.90	3833, 614	agrees; AUC gain 0.097 [0.061, 0.136]	yes
C056	threshold: total daytime minutes: fold at about 187 minutes (starts) for customers without a voice-mail plan *shape: accelerating; 67% of patients same; fallback; risk saturates near 0%*	1% to 11% (typical patient with a fold)	2.80	16.4	0.00 to 6.18	3014, 1382	agrees; AUC gain 0.035 [0.016, 0.052]	yes
C057	threshold: total daytime charge: fold at about 38.81 US dollars (starts) for customers without a voice-mail plan *shape: hinge at 38.81 US dollars; 45% of patients same; fallback; risk saturates near 0%; one point with C057 (median fold 38.81 US dollars, typical patient 36.87 US dollars, 10th-90th percentile 37.22 US dollars to 53.94 US dollars, wide)*	1% to 2% (typical patient with a fold)	0.73	2.07	0.00 to 2.28	2578, 579	unclear; AUC gain 0.0003 [-0.007, 0.008]	yes
C058	threshold: total night-time minutes: fold at about 220 minutes (starts) for customers without a voice-mail plan *shape: hinge at 230 minutes; 45% of patients same; fallback; risk saturates near 0%*	1% to 2% (typical patient with a fold)	0.75	2.11	-0.06 to 2.45	2340, 1023	agrees; AUC gain 0.01 [0.003, 0.016]	no
C059	threshold: total evening minutes: fold at about 318 minutes (starts) for customers with a voice-mail plan *shape: step at 307 minutes; 50% of patients same; fallback; risk saturates near 0%*	2% to 2% (typical patient with a fold (customers with a voice-mail plan))	-0.01	0.99	0.00 to 0.00	43, 0 (small n)	agrees; AUC gain 0.034 [0.013, 0.055]	no
C060	structure: made of usage_hinge and inputs number customer service calls; combined non-additively					4000	train only	yes
C118	boundary: boundary					4000	train only	no

T.3 Usage score

Annotation ann_79 (usage_hinge). Reads: state, voice-mail plan, total daytime minutes, total daytime charge, total evening minutes, total night-time minutes.

Original hypothesis (unchanged). 879 = relu(-0.99 + 0.53 dayMin + 0.22 nightMin - 1.01 vmailPlan + 0.55*7558), 7558 = relu(0.51 + 0.67 eveMin + 1.21*1406), 1406 = relu(-0.50 + 0.54 dayCharge - 0.88 state:41). Day minutes and day charge are collinear (r = 1.0000), so 879 is a hinge in total usage, dominated by day use. Hypothesis: a usage hinge that is never active for voicemail-plan customers (0 of 1061) and active for 47% of the rest; at median evening/night use it switches on at about 190 day minutes and rises steadily, with a second slope from the 1406 day-charge hinge (on for 24% of customers, never in state:41). Its exit feeds the logit directly (weight 2.57) and drives the heavy-use hinge and the service-calls gate in usage_response. Evidence (train, n=4000): sensitivity of 879 to voicemail plan 0.35, day minutes 0.18, evening minutes 0.12, state:41 0.09, night minutes 0.08, day charge 0.05. Mean ablation of its direct logit term AUC 0.9017; linear replacement 0.9107 (R2 0.53); restored alone onto the all-linearised model 0.8803 (all-linear 0.8336).

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 8.88178e-16).

$$
\begin{aligned}
y_0 = \operatorname{relu}\big(&-0.993 + 0.219 x_{totalNightMinutes_b} \\
&+ 0.527 x_{totalDayMinutes} + 0.554 \operatorname{relu}\big(0.515 \\
&+ 1.21 \operatorname{relu}\big(-0.503 + 0.545 x_{totalDayCharge} \\
&- 0.885 x_{state:41_c}\big) + 0.665 x_{totalEveMinutes_b}\big) \\
&- 1.01 x_{voiceMailPlan_b}\big)
\end{aligned}
$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C061	direction: total daytime minutes: 112 minutes to 249 minutes (through this part) *shape: accelerating; 67% of patients same; fallback; risk saturates near 0%*	1% to 4%	1.37	3.93	0.00 to 4.32	4000	agrees; AUC gain 0.035 [0.016, 0.052]	yes
C062	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	3% to 1% (typical patient this part acts on)	-0.83	0.44	-2.92 to 0.00	2939, 1061	agrees; AUC gain 0.013 [0.0009, 0.023]	yes
C063	direction: total evening minutes: 138 minutes to 266 minutes (through this part) *shape: accelerating; 40% of patients same; fallback; risk saturates near 0%*	1% to 2%	1.03	2.79	0.00 to 4.01	4000	agrees; AUC gain 0.034 [0.013, 0.055]	yes
C064	direction: total night-time minutes: 136 minutes to 264 minutes (through this part) *shape: hinge at 224 minutes; 35% of patients same; fallback; risk saturates near 0%*	1% to 1%	0.63	1.87	0.00 to 2.67	4000	agrees; AUC gain 0.01 [0.003, 0.016]	yes
C065	direction: total daytime charge: 19.07 US dollars to 42.40 US dollars (through this part) *shape: hinge at 38.81 US dollars; 50% of patients same; fallback; risk saturates near 0%; one point with C072 (median fold 38.81 US dollars, typical patient 40.06 US dollars, 10th-90th percentile 37.22 US dollars to 53.94 US dollars, wide)*	1% to 1%	0.24	1.27	0.00 to 1.11	4000	unclear; AUC gain 0.0003 [-0.007, 0.008]	yes
C066	direction: state: state other than South Dakota to South Dakota (through this part) *shape: categorical; highest to lowest: South Dakota, state other than South Dakota*	3% to 3% (typical patient this part acts on)	-0.18	0.83	-0.32 to 0.00	3934, 66	unclear; AUC gain 0.0002 [-0.0002, 0.0007]	yes
C067	interaction: effect of total daytime minutes (112 minutes to 249 minutes) by total evening minutes: total evening minutes above 201 minutes minus total evening minutes at or below 201 minutes; stronger in total evening minutes above 201 minutes *shape: accelerating; 75% of patients same; fallback; risk saturates near 0%*	total evening minutes above 201 minutes: 1% to 11%; total evening minutes at or below 201 minutes: 1% to 2%	1.54	4.66		1998, 2002	agrees; AUC gain 0.034 [0.013, 0.055]	yes
C068	interaction: effect of total daytime minutes (112 minutes to 249 minutes) by total night-time minutes: total night-time minutes above 200 minutes minus total night-time minutes at or below 200 minutes; stronger in total night-time minutes above 200 minutes *shape: accelerating; 74% of patients same; fallback; risk saturates near 0%*	total night-time minutes above 200 minutes: 1% to 7%; total night-time minutes at or below 200 minutes: 1% to 3%	1.00	2.71		2000, 2000	agrees; AUC gain 0.01 [0.003, 0.016]	yes
C069	interaction: effect of total daytime minutes (112 minutes to 249 minutes) by voice-mail plan: a voice-mail plan minus no voice-mail plan; stronger in no voice-mail plan *shape: accelerating; 70% of patients same; fallback; risk saturates near 0%*	a voice-mail plan: 1% to 1%; no voice-mail plan: 1% to 4%	-1.86	0.16		1061, 2939	agrees; AUC gain 0.013 [0.0009, 0.023]	no
C070	interaction: effect of total evening minutes (138 minutes to 266 minutes) by voice-mail plan: a voice-mail plan minus no voice-mail plan; stronger in no voice-mail plan *shape: dip at 212 minutes; 48% of patients same; fallback; risk saturates near 0%*	a voice-mail plan: 1% to <1%; no voice-mail plan: 1% to 2%	-1.40	0.25		1061, 2939	agrees; AUC gain 0.013 [0.0009, 0.023]	no
C071	threshold: total daytime minutes: fold at about 187 minutes (starts) for customers without a voice-mail plan *shape: accelerating; 67% of patients same; fallback; risk saturates near 0%*	1% to 11% (typical patient with a fold)	2.80	16.4	0.00 to 6.18	3014, 1382	agrees; AUC gain 0.035 [0.016, 0.052]	yes
C072	threshold: total daytime charge: fold at about 38.81 US dollars (starts) for customers without a voice-mail plan *shape: hinge at 38.81 US dollars; 45% of patients same; fallback; risk saturates near 0%; one point with C072 (median fold 38.81 US dollars, typical patient 36.87 US dollars, 10th-90th percentile 37.22 US dollars to 53.94 US dollars, wide)*	1% to 2% (typical patient with a fold)	0.73	2.07	0.00 to 2.28	2578, 579	unclear; AUC gain 0.0003 [-0.007, 0.008]	yes
C073	threshold: total night-time minutes: fold at about 220 minutes (starts) for customers without a voice-mail plan *shape: hinge at 230 minutes; 45% of patients same; fallback; risk saturates near 0%*	1% to 2% (typical patient with a fold)	0.75	2.11	-0.06 to 2.45	2340, 1023	agrees; AUC gain 0.01 [0.003, 0.016]	no

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C074	threshold: total evening minutes: fold at about 318 minutes (starts) for customers with a voice-mail plan *shape: step at 307 minutes; 50% of patients same; fallback; risk saturates near 0%*	2% to 2% (typical patient with a fold (customers with a voice-mail plan))	-0.01	0.99	0.00 to 0.00	43, 0 (small n)	agrees; AUC gain 0.034 [0.013, 0.055]	no
C075	subgroup: who this part acts on: 1382 of 4000 patients	1% to 3% (typical patient this part acts on (risk without the part, then with it))	1.77	5.86	-0.34 to 5.50	1382, 2618	agrees; AUC gain 0.214 [0.160, 0.273]	no
C076	structure: its formula reproduces it on 4000 training patients (largest difference 8.88178e-16)		9e-16			4000	train only	yes
C119	boundary: boundary		1.00			4000	train only	yes

T.4 International plan

Annotation ann_81 (intl_plan_switch). Reads: international calling plan.

Original hypothesis (unchanged). 6711 = relu(-0.23 + 0.64 intlPlan), logit weight +1.68: exactly 0 for customers without an international plan and +2.91 logit for the 380 with one (R2 1.0). Hypothesis: an international-plan switch, the model's only route for the plan (no direct edge). Empirical churn 0.42 with the plan vs 0.11 without; model 0.45 vs 0.11. No interaction with international usage (international minutes are unconnected; international charge and calls act linearly in the backbone). Evidence: line plot correlation 1.00; mean ablation AUC 0.8611; linear replacement 0.9189 (identical, exact switch).

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 0.0).

$$y_0 = \mathrm{relu}\big(-0.228 + 0.635 x_{internationalPlan}\big)$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C077	direction: international calling plan: no international plan to an international plan (through this part) *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 15%	2.91	18.3	2.91 to 2.91	3620, 380	agrees; AUC gain 0.060 [0.033, 0.090]	yes
C078	subgroup: who this part acts on: 380 of 4000 patients	2% to 26% (typical patient this part acts on (risk without the part, then with it))	2.91	18.3	2.91 to 2.91	380, 3620	agrees; AUC gain 0.060 [0.033, 0.090]	yes
C079	structure: its formula reproduces it on 4000 training patients (largest difference 0.0)		0.00			4000	train only	yes

T.5 Fixed-weight inputs

Annotation ann_78 (linear_backbone). Reads: state, account length, area code, number of voice-mail messages, total evening minutes, total night-time minutes, total night-time calls, total international calls, total international charge.

Original hypothesis (unchanged). Direct input-to-output edges collected through identity_lin (an identity node). Exactly linear in 27 inputs, all with small weights (at most 0.32 per SD): international charge +0.32, international calls -0.26, voicemail messages +0.20, evening minutes -0.14 and night minutes -0.11 (these two also feed the usage hinge with positive weights), account length +0.09, area code 510 +0.04 (+0.10 logit), and state offsets: state:46 -1.89 logit (80 customers), state:39 -1.23 (82), state:47 +1.10 (72), state:5 -1.09 (78), state:26 +0.81, state:10 -0.63, state:40 +0.43, state:41 +0.43, state:31 +0.33, then smaller. International plan, voicemail plan, day minutes and customer-service calls have NO direct edge. Evidence: closed form exact (identity, R2 1.0); train sensitivities equal the weights (international charge 0.32, international calls 0.26, state:46 0.26, voicemail messages 0.20). Mean ablation AUC 0.9076 (full 0.9189).

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 6.66134e-16).

$$
\begin{aligned}
y_0 = {} & 0.321 x_{totalIntlCharge} + 0.199 x_{numberVmailMessages} + 0.042 x_{state:35} \\
& + 0.0442 x_{areaCode:510} + 0.0249 x_{totalNightCalls} + 0.0544 x_{state:41_a} \\
& + 0.0545 x_{state:40} + 0.0145 x_{state:25} + 0.00187 x_{state:20} + 0.0301 x_{state:17} \\
& + 0.146 x_{state:47} + 0.0375 x_{state:27} + 0.0379 x_{state:37} + 0.0871 x_{accountLength} \\
& + 0.0223 x_{state:45_a} + 0.049 x_{state:31} + 0.109 x_{state:26} - 0.0289 x_{state:28} \\
& - 0.139 x_{totalEveMinutes_a} - 0.175 x_{state:39} - 0.264 x_{totalIntlCalls} \\
& - 0.0184 x_{state:32} - 0.0761 x_{state:10} - 0.114 x_{totalNightMinutes_a} \\
& - 0.0165 x_{state:48} - 0.264 x_{state:46} - 0.151 x_{state:5}
\end{aligned}
$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C080	direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to Washington (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 3%	1.10	2.99	1.10 to 1.10	2529, 72	disagrees; AUC gain -0.004 [-0.016, 0.006]	yes
C081	direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to Montana (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 2%	0.81	2.24	0.81 to 0.81	2529, 75	disagrees; AUC gain -0.004 [-0.016, 0.006]	yes
C082	direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to Vermont (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to <1%	-1.89	0.15	-1.89 to -1.89	2529, 80	disagrees; AUC gain -0.004 [-0.016, 0.006]	yes
C083	direction: total international charge: 1.84 US dollars to 3.70 US dollars (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	0.80	2.23	0.80 to 0.80	4000	unclear; AUC gain 0.003 [-0.005, 0.011]	yes
C084	direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to Rhode Island (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to <1%	-1.23	0.29	-1.23 to -1.23	2529, 82	disagrees; AUC gain -0.004 [-0.016, 0.006]	yes
C085	direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to Colorado (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to <1%	-1.09	0.34	-1.09 to -1.09	2529, 78	disagrees; AUC gain -0.004 [-0.016, 0.006]	yes
C086	direction: voice-mail messages: 0 messages to 32 messages (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	0.47	1.60	0.47 to 0.47	4000	unclear; AUC gain 0.0008 [-0.004, 0.006]	yes
C087	direction: total international calls: 2 calls to 8 calls (through this part) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 0%*	1% to 1%	-0.63	0.53	-0.63 to -0.63	4000	unclear; AUC gain 0.002 [-0.004, 0.008]	yes
C088	direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to South Carolina (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 1%	0.43	1.53	0.43 to 0.43	2529, 66	disagrees; AUC gain -0.004 [-0.016, 0.006]	yes
C089	direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to South Dakota (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 1%	0.43	1.53	0.43 to 0.43	2529, 66	disagrees; AUC gain -0.004 [-0.016, 0.006]	yes
C090	direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to Georgia (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to <1%	-0.63	0.53	-0.63 to -0.63	2529, 60	disagrees; AUC gain -0.004 [-0.016, 0.006]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C091	direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to New Jersey (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 1%	0.33	1.39	0.33 to 0.33	2529, 90	disagrees; AUC gain -0.004 [-0.016, 0.006]	yes
C092	direction: total evening minutes: 138 minutes to 266 minutes (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	-0.35	0.70	-0.35 to -0.35	4000	disagrees; AUC gain -0.002 [-0.005, 0.0005]	yes
C093	direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to Ohio (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 1%	0.29	1.33	0.29 to 0.29	2529, 87	disagrees; AUC gain -0.004 [-0.016, 0.006]	yes
C094	direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to North Carolina (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 1%	0.28	1.32	0.28 to 0.28	2529, 74	disagrees; AUC gain -0.004 [-0.016, 0.006]	yes
C095	direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to Oregon (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 1%	0.26	1.30	0.26 to 0.26	2529, 84	disagrees; AUC gain -0.004 [-0.016, 0.006]	yes
C096	direction: total night-time minutes: 136 minutes to 264 minutes (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	-0.29	0.75	-0.29 to -0.29	4000	disagrees; AUC gain -0.002 [-0.006, 0.0006]	yes
C097	direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to Kentucky (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 1%	0.21	1.24	0.21 to 0.21	2529, 81	disagrees; AUC gain -0.004 [-0.016, 0.006]	yes
C098	direction: account length: 48 to 152 (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	0.23	1.26	0.23 to 0.23	4000	disagrees; AUC gain -0.001 [-0.003, 0.001]	yes
C099	direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to North Dakota (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 1%	-0.22	0.80	-0.22 to -0.22	2529, 68	disagrees; AUC gain -0.004 [-0.016, 0.006]	yes
C100	direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to Virginia (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 1%	0.15	1.16	0.15 to 0.15	2529, 91	disagrees; AUC gain -0.004 [-0.016, 0.006]	yes
C101	direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to New Mexico (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 1%	-0.14	0.87	-0.14 to -0.14	2529, 72	disagrees; AUC gain -0.004 [-0.016, 0.006]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C102	direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to Mississippi (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 1%	0.11	1.11	0.11 to 0.11	2529, 77	disagrees; AUC gain -0.004 [-0.016, 0.006]	yes
C103	direction: area code: area code other than area code 510 to area code 510 (through this part) *shape: categorical; highest to lowest: area code 510, area code other than area code 510*	1% to 1%	0.10	1.11	0.10 to 0.10	3040, 960	unclear; AUC gain 8e-05 [-0.001, 0.002]	yes
C104	direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to Wisconsin (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 1%	-0.11	0.89	-0.11 to -0.11	2529, 86	disagrees; AUC gain -0.004 [-0.016, 0.006]	yes
C105	direction: total night-time calls: 74 calls to 125 calls (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	0.06	1.07	0.06 to 0.06	4000	disagrees; AUC gain -9e-05 [-0.0008, 0.0006]	yes
C106	direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to Maryland (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 1%	0.01	1.01	0.01 to 0.01	2529, 82	disagrees; AUC gain -0.004 [-0.016, 0.006]	yes
C107	structure: its formula reproduces it on 4000 training patients (largest difference 6.66134e-16)		7e-16			4000	train only	yes

T.6 Voice mail and South Dakota

Annotation ann_80 (vmail_state41_unit). Reads: state, voice-mail plan.

Original hypothesis (unchanged). 582 = relu(1.51 + 0.17 vmailPlan + 0.09 state:41), logit weight -1.40. The pre-activation stays between 1.40 and 2.52 on train, so the unit is always active and exactly affine (R2 1.0): a constant -1.96 logit for customers without a voicemail plan outside state:41, -0.54 extra for a voicemail plan and -1.04 extra for state:41. Hypothesis: a fixed offset plus a voicemail-plan and state:41 discount. Evidence: sensitivity 0.17 (voicemail plan), 0.09 (state:41); linear replacement AUC 0.9189 (unchanged), mean ablation 0.9184.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 4.44089e-16).

$$y_0 = \mathrm{relu}\big(1.51 + 0.17 x_{voiceMailPlan_a} + 0.0942 x_{state:41_b}\big)$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C108	direction: state: state other than South Dakota to South Dakota (through this part) *shape: categorical; highest to lowest: state other than South Dakota, South Dakota*	1% to <1%	-1.04	0.35	-1.04 to -1.04	3934, 66	agrees; AUC gain 0.001 [0.0004, 0.002]	yes
C109	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	1% to 1%	-0.54	0.58	-0.54 to -0.54	2939, 1061	unclear; AUC gain 0.002 [-0.003, 0.008]	yes
C110	structure: its formula reproduces it on 4000 training patients (largest difference 4.44089e-16)		4e-16			4000	train only	yes

T.7 State adjustments

Annotation ann_82 (state_offsets). Reads: state.

Original hypothesis (unchanged). Six single-state units, each a function of one binary indicator and so exactly an offset (R2 1.0): 2067 state:45 -1.12 logit (91 customers; with its direct +0.15, net -0.97), 13164 state:41 +1.04 (66; it offsets most of state:41's -1.04 in vmail_state41_unit), 13420 state:49 -0.69 (122), 5821 state:0 -0.68 (56), 12726 state:15 +0.14 (82; always active, constant +0.008), 5667 state:2 is never active (dead unit, zero effect). Hypothesis: per-state offsets. Evidence: sensitivities as listed; linear replacement AUC 0.9189 (unchanged); mean ablation 0.9167.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 0.0).

$$
\begin{aligned}
y_0 &= \mathrm{relu}\big(-0.691 + 0.757 x_{state:45_b}\big) \\
y_1 &= \mathrm{relu}\big(-1.7 + 1.02 x_{state:0}\big) \\
y_2 &= \mathrm{relu}\big(0.47 + 0.878 x_{state:15}\big) \\
y_3 &= \mathrm{relu}\big(-0.236 + 1.02 x_{state:41_d}\big) \\
y_4 &= \mathrm{relu}\big(0.00575 + 1.01 x_{state:49}\big) \\
y_5 &= \mathrm{relu}\big(-0.325 - 0.0317 x_{state:2}\big)
\end{aligned}
$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C111	direction: state: West Virginia to South Dakota (through this part) *shape: categorical; highest to lowest: South Dakota, Indiana, Arkansas, state other than Alaska, Indiana, Arkansas, South Dakota, Virginia, West Virginia, Alaska, West Virginia …*	1% to 5%	1.73	5.62	1.73 to 1.73	122, 66	disagrees; AUC gain -0.0004 [-0.006, 0.004]	yes
C112	direction: state: West Virginia to Indiana (through this part) *shape: categorical; highest to lowest: South Dakota, Indiana, Arkansas, state other than Alaska, Indiana, Arkansas, South Dakota, Virginia, West Virginia, Alaska, West Virginia …*	1% to 2%	0.83	2.30	0.83 to 0.83	122, 82	disagrees; AUC gain -0.0004 [-0.006, 0.004]	yes
C113	direction: state: West Virginia to Arkansas (through this part) *shape: categorical; highest to lowest: South Dakota, Indiana, Arkansas, state other than Alaska, Indiana, Arkansas, South Dakota, Virginia, West Virginia, Alaska, West Virginia …*	1% to 2%	0.69	2.00	0.69 to 0.69	122, 78	disagrees; AUC gain -0.0004 [-0.006, 0.004]	yes
C114	direction: state: West Virginia to state other than Alaska, Indiana, Arkansas, South Dakota, Virginia, West Virginia (through this part) *shape: categorical; highest to lowest: South Dakota, Indiana, Arkansas, state other than Alaska, Indiana, Arkansas, South Dakota, Virginia, West Virginia, Alaska, West Virginia …*	1% to 2%	0.69	2.00	0.69 to 0.69	122, 3505	disagrees; AUC gain -0.0004 [-0.006, 0.004]	yes
C115	direction: state: West Virginia to Virginia (through this part) *shape: categorical; highest to lowest: South Dakota, Indiana, Arkansas, state other than Alaska, Indiana, Arkansas, South Dakota, Virginia, West Virginia, Alaska, West Virginia …*	1% to 1%	-0.43	0.65	-0.43 to -0.43	122, 91	disagrees; AUC gain -0.0004 [-0.006, 0.004]	yes
C116	direction: state: West Virginia to Alaska (through this part) *shape: categorical; highest to lowest: South Dakota, Indiana, Arkansas, state other than Alaska, Indiana, Arkansas, South Dakota, Virginia, West Virginia, Alaska, West Virginia …*	1% to 1%	0.02	1.02	0.02 to 0.02	122, 56	disagrees; AUC gain -0.0004 [-0.006, 0.004]	yes
C117	structure: its formula reproduces it on 4000 training patients (largest difference 0.0)		0.00			4000	train only	yes

The network before any explanation

Figure 265 (next page, landscape) is the trained network as it is, with no parts, no splits and no inserted units: inputs at the bottom, hidden units by depth, the output at the top, every connection drawn, every input column its own node (one-hot levels included).

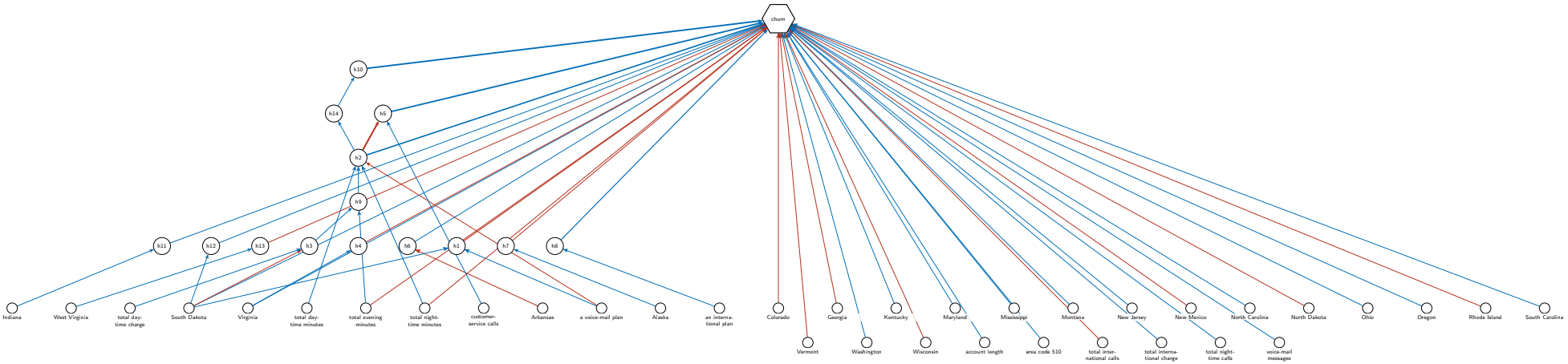


Figure 265: The network as trained, before any explanation, in full. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. Not used (no connection): Alabama, Hawaii, Iowa, Idaho, Illinois, Kansas, Louisiana, Massachusetts, Maine, Michigan, Minnesota, Missouri, Nebraska, Arizona, New Hampshire, Nevada, New York, Oklahoma, Pennsylvania, California, Tennessee, Texas, Utah, Wyoming, Connecticut, Washington DC, Delaware, Florida, area code 408, area code 415, total daytime calls, total evening calls, total evening charge, total night-time charge, total international minutes.

6 heart-s1-c1

Genome `a2df22ce-983d-47b6-bbd6-3176d38ed5ff`, explanation `492866e6-a73b-4465-92a8-ecad82a2adcc`. Split: train primary (n=242), test as a check (n=61). Claims: 47, verified 47, not supported 0.
AI-assisted draft for review (unverified)

Reference (typical) patient: the typical patient: the median training patient for every number and the most common training category for every category (not a real patient). age: 56 years; ca: 0 vessels; chol: 240 mg/dL (6.2 mmol/L); cp: no chest pain (asymptomatic); exang: no chest pain on exercise; fbs: fasting blood sugar of 120 mg/dL or less; oldpeak: 0.8 mm; restecg: a normal resting ECG; sex: male; slope: an upsloping ST segment; thal: a normal thallium scan; thalach: 152 beats per minute; trestbps: 130 mm Hg. Predicted risk of heart disease: 18%.

6.1 Summary

For a typical patient (a 56-year-old man with no chest pain symptoms and median test results) the model predicts a risk of heart disease of about 18%; it adds up most of its inputs independently, and five parts add to that sum: two add more for each extra vessel coloured, one lowers the risk for typical angina, one lowers it for non-anginal chest pain, one acts only on the 42 of 242 training patients with a normal resting ECG and marked ST depression, and one lowers it for a higher maximum heart rate, with nothing until about 145 beats per minute, then steadily. [C29, C30, C40, C31, C43, C38, C46] Figure 266 shows the model at this level: one box per paragraph below.

How to read the diagrams. Edge colour is the sign of that one connection: blue positive, red negative; thicker is a larger weight. The outline and arrow on an input (and on a part's output box) is its overall effect on the final prediction, read from the verified claims: orange and up arrow raises it, teal and down arrow lowers it, grey and double arrow depends (it goes both ways, or differs between levels). A negative connection into a part that itself raises the prediction means that input lowers the prediction; through a hinge the overall effect can only be read from the claims, not from the edges. Up arrow means towards heart disease; down arrow, away from it. A double outline marks an input (or a part's output) that acts at a fixed rate everywhere: it adds the same amount to the score per unit, so it is explained by its rate alone (see 'Linear factors'); a dashed double outline, at a fixed rate on the observed data. The normal outline is an input whose effect switches on and off somewhere. For example, normal ECG has a positive connection into 'ST depression with a normal ECG', a part that raises the predicted risk of heart disease, so normal ECG raises it.

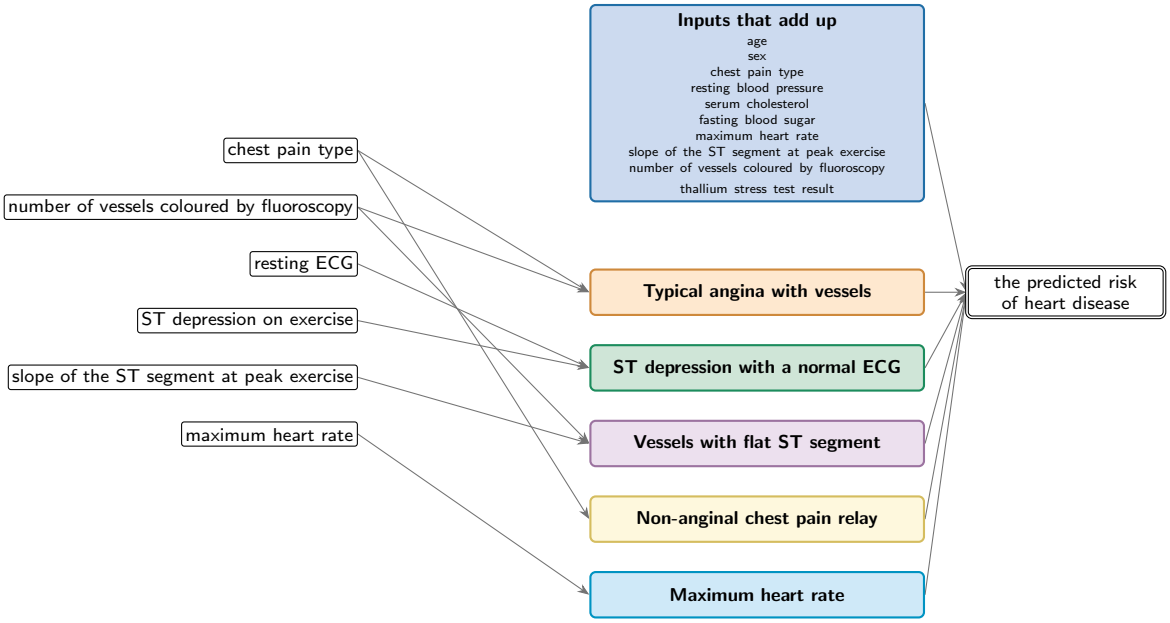


Figure 266: The model at the level of the summary: one box per part.

Inputs that add up. *Lowering the risk.* These add up independently, and these lower the risk: being a woman takes a typical patient's predicted risk from about 18% to below 1% and fasting blood sugar above 120 mg/dL to about 8%; through this sum alone, compared with no chest pain symptoms, typical angina or non-anginal chest pain lowers it to about 2% and atypical angina to about 5% (chest pain type also acts through the parts below). [C10, C19, C12, C17]

Raising the risk. A reversible defect (or the undefined code 4) on the thallium scan raises it to about 55%; higher cholesterol (from about 8% at 193 mg/dL to about 45% at 308 mg/dL), higher resting blood pressure (from about 9% at 110 mm Hg to about 35% at 155 mm Hg) and older age (from about 10% at 42 years to about 25% at 66 years) raise it steadily. [C05, C07, C08, C13] Through this sum alone, each extra vessel coloured by fluoroscopy adds about the same (from about 18% at 0 to about 80% at 2 vessels) and a flat ST segment raises it to about 87%, while the maximum heart rate barely changes it; all three also act through the parts below. [C04, C03, C21]
Rare results. Rarer results also count (the model has little data on this, n = 15 and 17): a fixed defect on the thallium scan lowers the risk to about 3%, and through this sum alone a downsloping ST segment raises it to about 32%; the model ignores exercise-induced angina. [C14, C15, C28]

Typical angina with vessels. This part combines chest pain type with the number of vessels coloured: through it, typical angina lowers a typical patient's predicted risk from about 18% to about 2%, and each extra vessel adds about the same, from about 18% at 0 to about 61% at 2 vessels. [C31, C30] Across the whole model, more vessels raise the predicted risk, and even more so with typical angina, each extra vessel adding about the same: going from 0 to 2 vessels moves a typical patient from below 1% to about 55% with typical angina, and from about 18% to about 99% with any other chest pain type. [C32] Across the whole model, typical angina takes a typical patient's predicted risk to below 1%. [C09]

ST depression with a normal ECG. This part combines the resting ECG with ST depression on exercise and acts on 42 of 242 training patients, nearly all with a normal resting ECG, 26 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 98% without it and about 99% with it. [C38] Through it, more ST depression raises the predicted risk only once it rises above about 1.0 mm, for patients with a normal resting ECG, in an uneven way: for a typical such patient from about 13% at 0 mm to about 63% at 3.5 mm. [C37] More ST depression therefore raises the predicted risk, but only if the resting ECG is normal: going from 0 to 2.8 mm moves a typical patient from about 18% to about 55% with a normal resting ECG, in an uneven way, and leaves it at about 18% otherwise. [C36]

Vessels with flat ST segment. This part combines the number of vessels coloured with a flat ST segment: through it, each extra vessel adds about the same, from about 18% at 0 to about 36% at 2 vessels for a typical patient, and a flat ST segment raises the risk from about 18% to about 23%. [C40, C41] Across the whole model, each extra vessel takes a typical patient from about 18% at 0 to about 99% at 2 vessels, about the same for each, flattening out near the top because the predicted risk cannot go much higher, and a flat ST segment takes it to about 91%. [C01, C02]

Non-anginal chest pain relay. This part reads only non-anginal chest pain: through it, non-anginal chest pain lowers a typical patient's predicted risk from about 18% to about 8%. [C43] Across the whole model, non-anginal chest pain takes it to about 1%. [C11]

Maximum heart rate. This part reads only the maximum heart rate: a higher maximum heart rate lowers the predicted risk with nothing until about 145 beats per minute, then steadily, for a typical patient from about 19% at 118 to about 14% at 175 beats per minute. [C45, C46] Across the whole model the same holds, from about 19% at 118 to about 14% at 175 beats per minute. [C20]

The whole network. Figure 267 (next page, landscape) draws every unit and connection of the explained model, nested by part: shaded clusters are the parts of the model and darker boxes the parts inside them; circles are hidden units (h1, h2, ...); each cluster's rectangle is its output, and a dashed 'sum' stands for an addition the next unit does (drawn, not a unit of the model); the hexagon is the model's output. An input used by several parts is drawn under each of them (same label, dotted outline, the first copy marked with the number of copies): it is the same input, repeated for readability. Not used by the model (input columns with no connection): male, sugar 120 or less, ST-T abnormal ECG, LVH on ECG, no exercise pain, exercise pain, upsloping ST, thallium code 4, reversible defect.

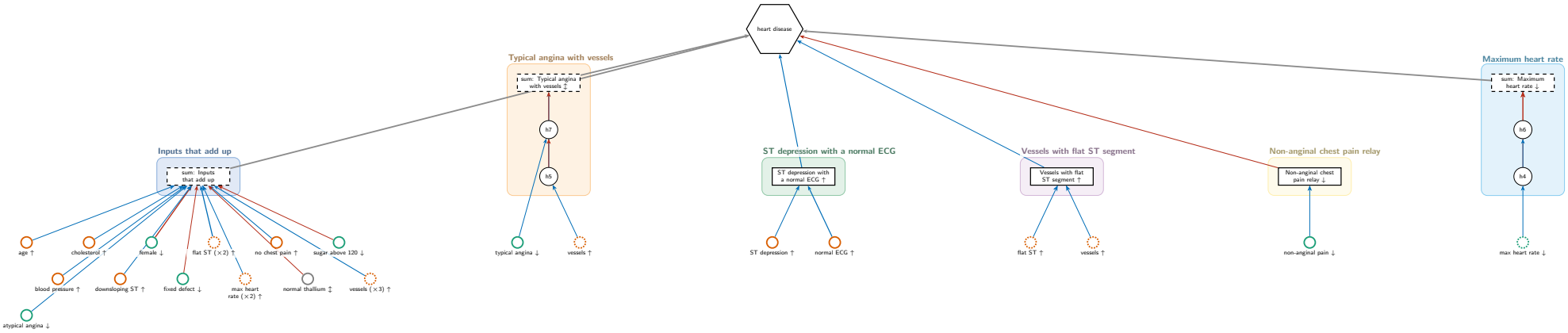


Figure 267: Every unit and connection of the explained model, nested by part. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Worth checking against what is known. Worth checking: patients with typical angina or non-anginal chest pain get the lowest predicted risk of all chest pain types (below 1% and about 1% for a typical patient), below patients with no chest pain symptoms. [C09, C11] Worth checking: the model treats a fixed defect on the thallium scan as lowering the risk compared with a normal scan, from only 15 training patients. [C14] Worth checking: the model treats fasting blood sugar above 120 mg/dL as lowering the risk of heart disease (for a typical patient from about 18% to about 8%). [C19] Worth checking: going from 0 to 2 vessels coloured moves a typical patient's predicted risk from about 18% to about 99%. [C01] Worth checking: ST depression on exercise raises the predicted risk only when the resting ECG is normal. [C36] Worth checking: the model ignores exercise-induced angina. [C28]

6.2 The model, read from the top down

Each part of the explanation in turn, from the whole model down to the smallest parts: what it does for the prediction (hypothesis), what was checked (evidence), a diagram one level down, and where to read on. The technical record of each part is in "Technical detail" at the end.

6.2.1 The whole model

Figure 268 shows what feeds this part; Figure 269 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

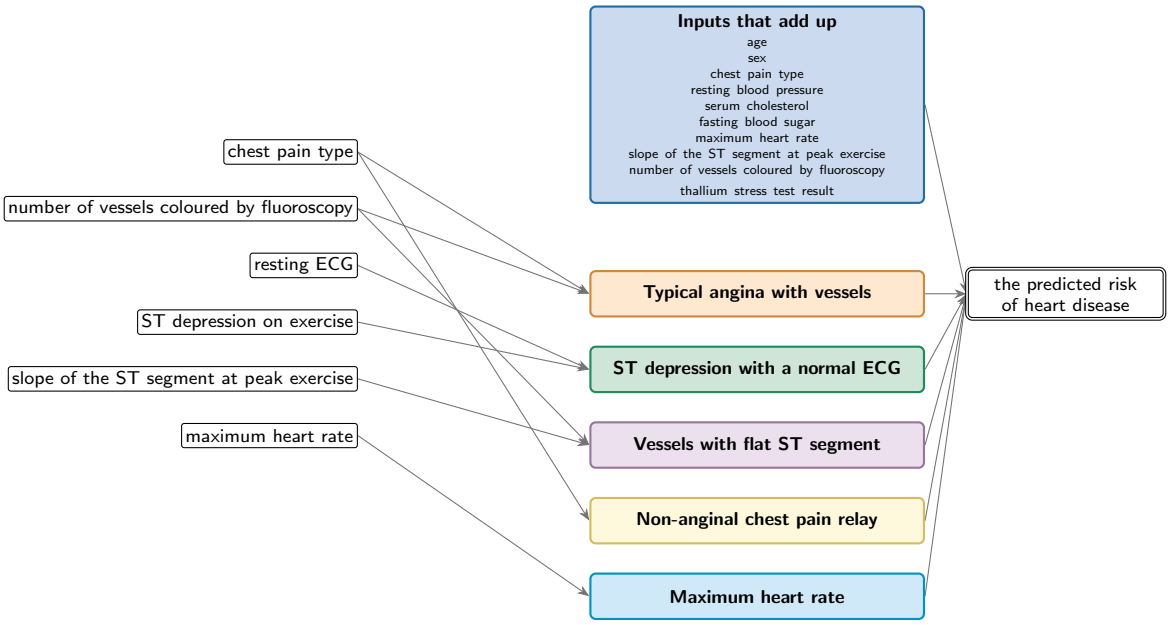


Figure 268: The whole model: what feeds it, one level down.

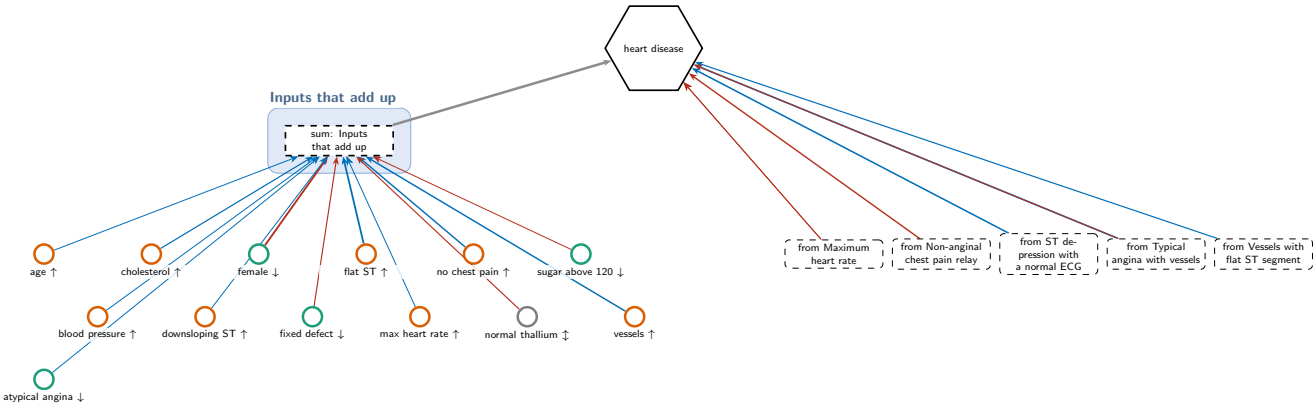


Figure 269: The whole model: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. The model adds up its inputs into one score: the vessels coloured (each extra vessel adding about the same), the ST segment, chest pain type and sex each move the predicted risk on their own, five parts add to

that sum, and exercise-induced angina is ignored. [C29, C01, C02, C10, C28]

Evidence. It is made up of the inputs that feed the score directly and five parts: vessels with a flat ST segment, the non-anginal chest pain relay, ST depression with a normal ECG, the maximum heart rate part and typical angina with vessels (each explained below); the score is their sum, which was checked on all 242 training patients. [C29] Changing one input at a time across the whole model, each extra vessel adds about the same, taking a typical patient's predicted risk from about 18% at 0 to about 99% at 2 vessels, flattening out near the top because the predicted risk cannot go much higher, against about 80% through the sum alone. [C01, C04] More ST depression raises it in an uneven way across the whole model, from about 18% at 0 mm to about 55% at 2.8 mm; it enters the score only through the part with a normal ECG. [C06] The model ignores exercise-induced angina. [C28]
Evidence plots: Figure 270.

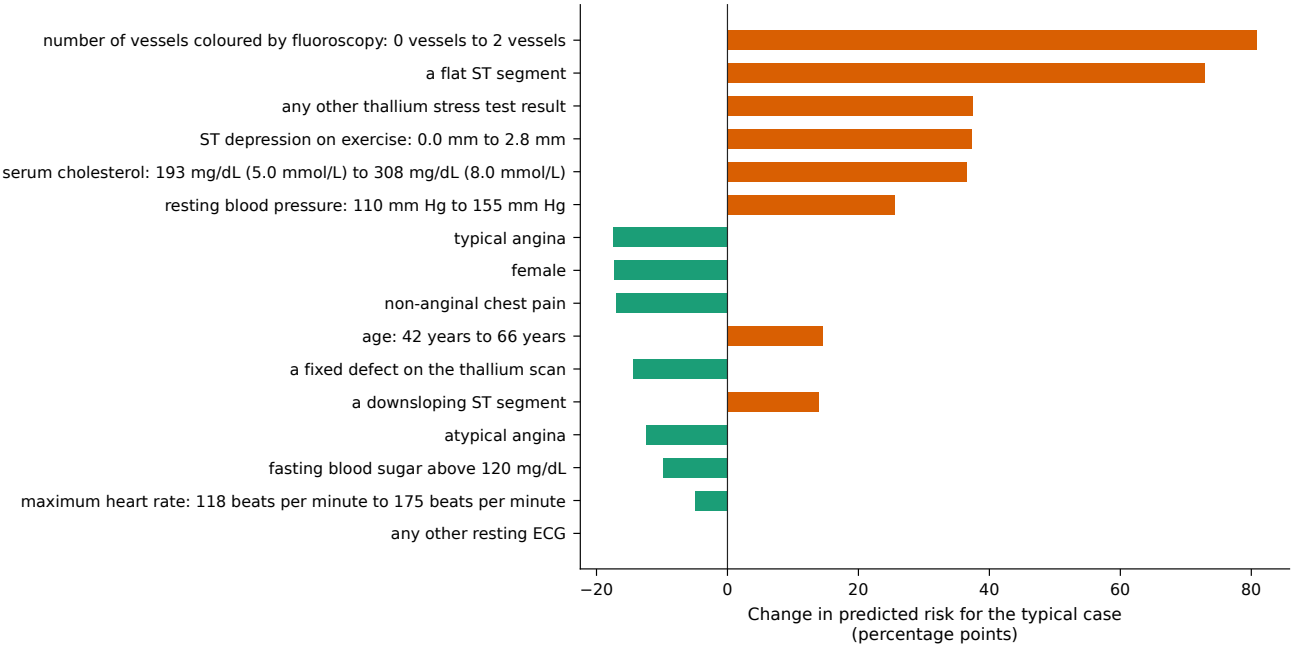


Figure 270: The whole model: Each input's change in the predicted value for the typical case across the whole model, largest first (orange: raises it, teal: lowers it).

Dive deeper: Typical angina with vessels (12.2.2); ST depression with a normal ECG (10.2.2); Vessels with flat ST segment (13.2.1); Non-anginal chest pain relay (9.2.2); Maximum heart rate (11.2.2). Technical detail: T.3.

6.2.2 Typical angina with vessels

Part of: The whole model (12.2.3).
Figure 271 shows what feeds this part; Figure 272 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).



Figure 271: Typical angina with vessels: what feeds it, one level down.

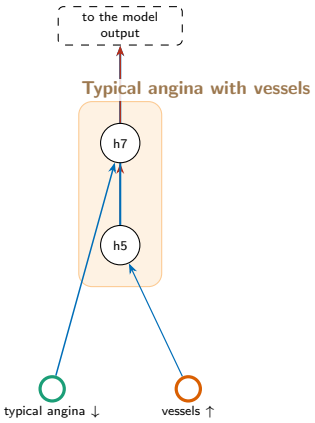


Figure 272: Typical angina with vessels: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part lowers the predicted risk for typical angina and adds risk for each extra vessel coloured, about the same for each, with the vessels counting more for patients with typical angina. [C31, C30, C32]

Evidence. Through it, typical angina takes a typical patient's predicted risk from about 18% to about 2%, and each extra vessel adds about the same, from about 18% at 0 to about 61% at 2 vessels. [C31, C30] Across the whole model, going from 0 to 2 vessels moves a typical patient from below 1% to about 55% with typical angina, each extra vessel adding about the same, and from about 18% to about 99% with any other chest pain type. [C32] Its formula reproduces the part exactly on all 242 training patients. [C33]
Evidence plots: Figure 273, Figure 274, Figure 275, Figure 276, Figure 277, Figure 278.
Figure 276 shows this part's own output, unit h5.
Figure 277 shows this part's own output, unit h7.

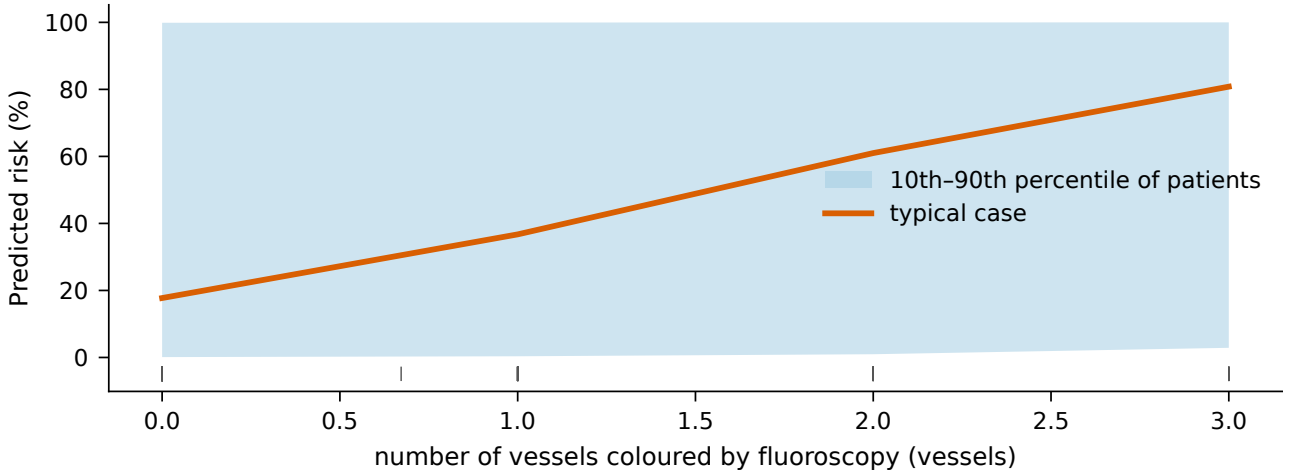


Figure 273: Typical angina with vessels: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual patients (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

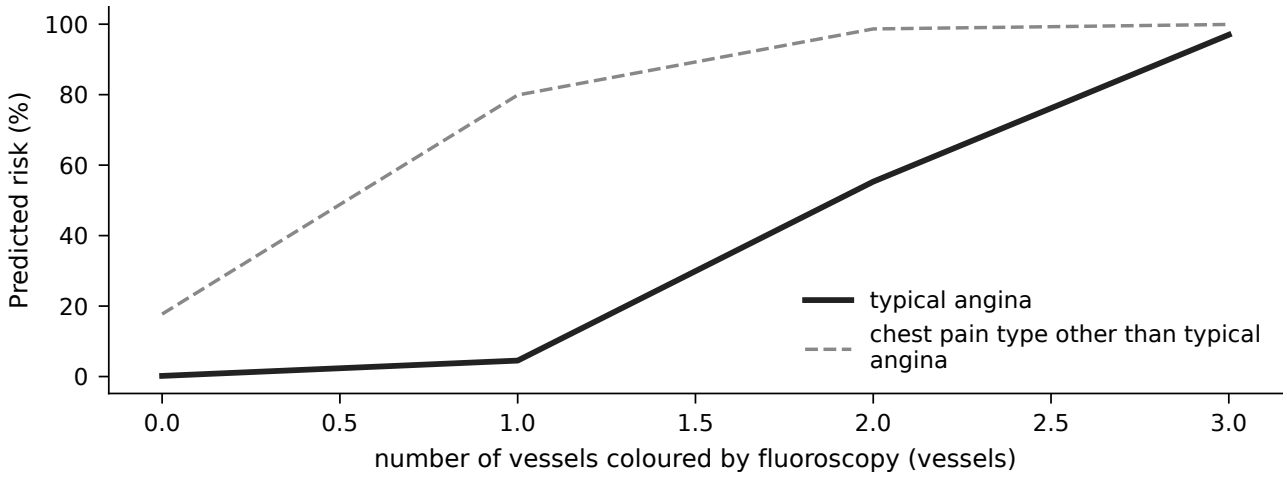


Figure 274: Typical angina with vessels: How the effect differs between groups: one curve per group, for the typical case of each group.

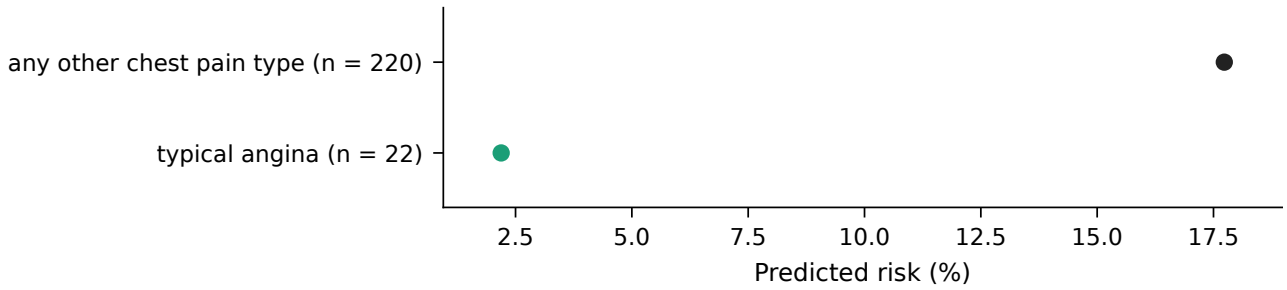


Figure 275: Typical angina with vessels: The predicted value for the typical case at each level, through this part (n = training patients; grey: fewer than 20).

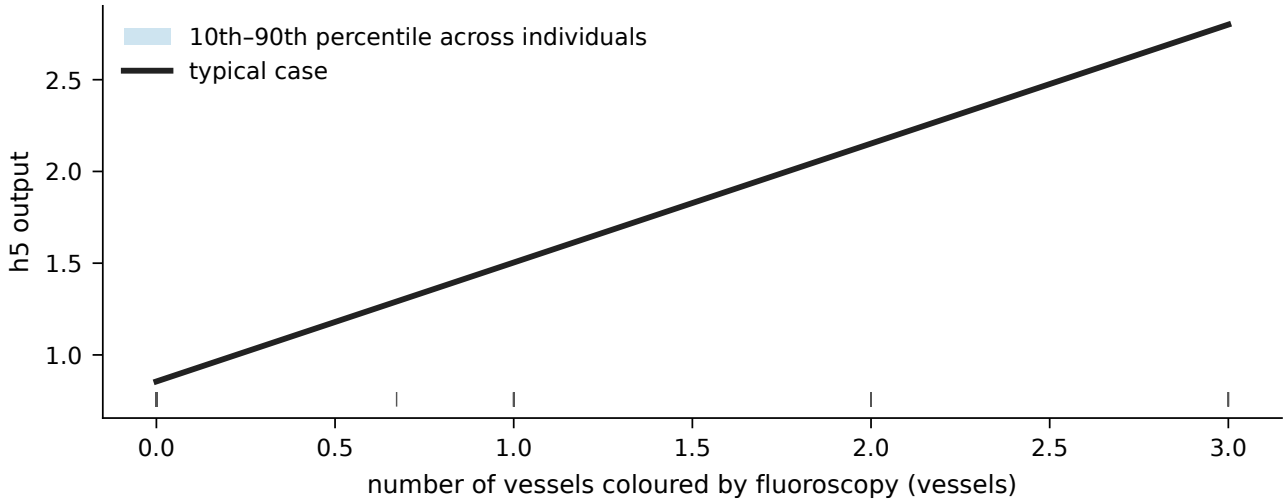


Figure 276: Typical angina with vessels: This part's own output, unit h5 (the output of 'Typical angina with vessels'), on its own scale (its activation), with every other input at the typical case; it is active for 242 of 242. Bold: the typical case; band: 10th to 90th percentile across individuals; ticks: the data.

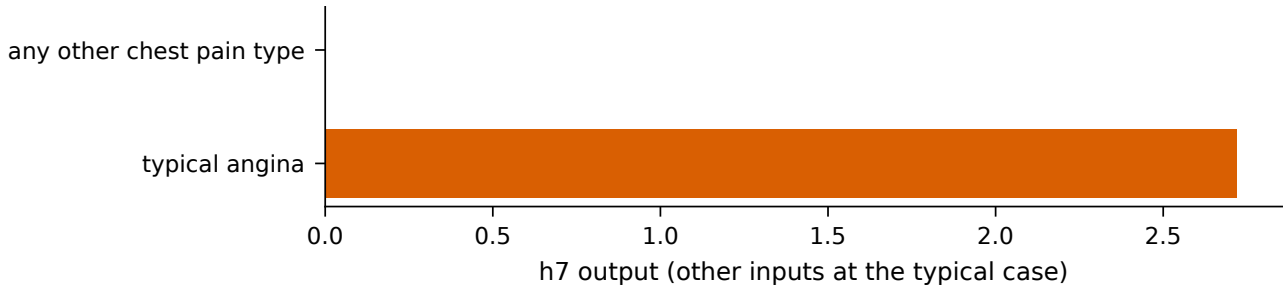


Figure 277: Typical angina with vessels: This part's own output, unit h7 (the output of 'Typical angina with vessels'), on its own scale (its activation), with every other input at the typical case; it is active for 22 of 242. Its output for each combination of the levels it reads.

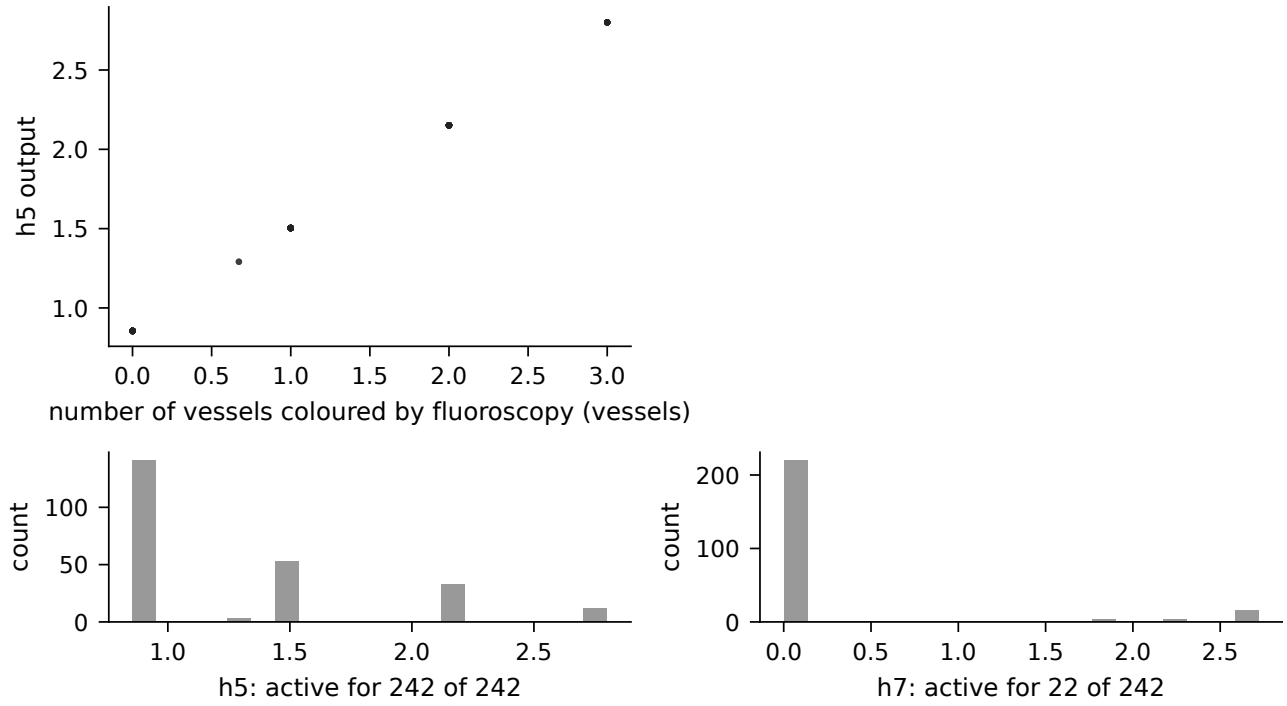


Figure 278: Typical angina with vessels: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.2.

6.2.3 ST depression with a normal ECG

Part of: The whole model (12.2.3).
Figure 279 shows what feeds this part; Figure 280 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

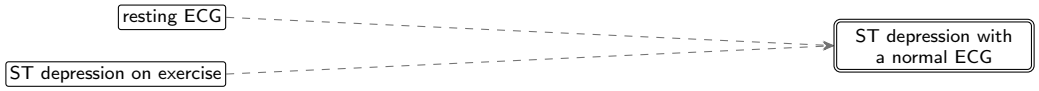


Figure 279: ST depression with a normal ECG: what feeds it, one level down.

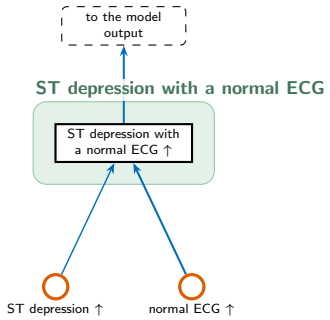


Figure 280: ST depression with a normal ECG: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part raises the predicted risk with ST depression on exercise, in an uneven way, only for patients with a normal resting ECG, once it rises above about 1.0 mm. [C37, C36]

Evidence. It acts on 42 of 242 training patients, nearly all with a normal resting ECG, 26 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 98% without it and about 99% with it. [C38] Through it, more ST depression raises the predicted risk only once it rises above about 1.0 mm, in an uneven way: for a typical patient with a normal resting ECG from about 13% at 0 mm to about 63% at 3.5 mm. [C37] Going from 0 to 2.8 mm of ST depression moves a typical patient from about 18% to about 55% with a normal resting ECG, in an uneven way, and leaves it at about 18% otherwise. [C36] Its formula reproduces the part exactly on all 242 training patients. [C39]
Evidence plots: Figure 281, Figure 282, Figure 283, Figure 284, Figure 285.
Figure 284 shows this part's own output, unit h3: it switches on at about 1.006 mm of ST depression on exercise, and is zero below.

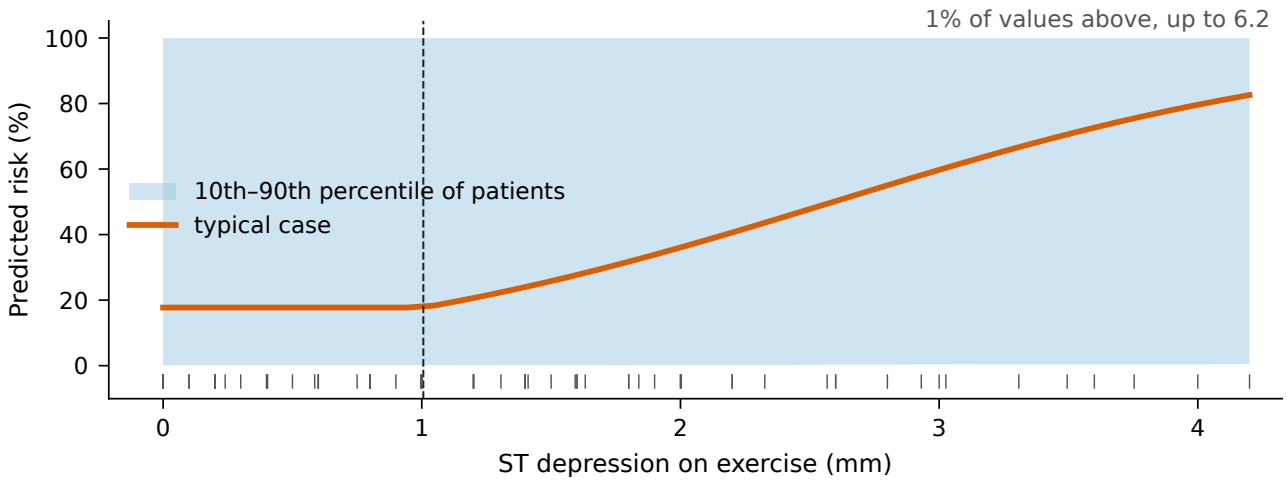


Figure 281: ST depression with a normal ECG: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual patients (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

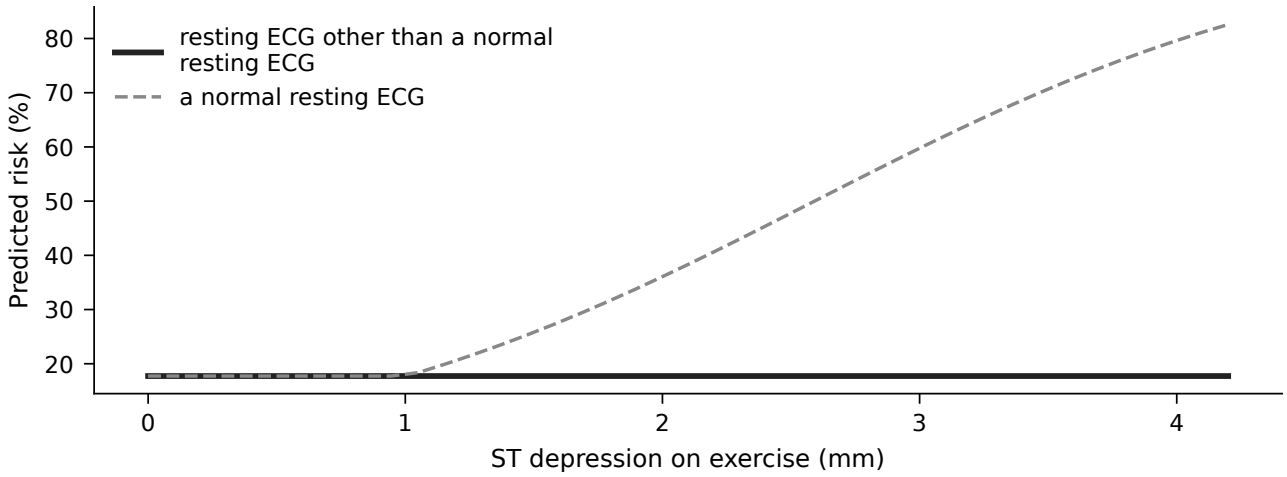


Figure 282: ST depression with a normal ECG: How the effect differs between groups: one curve per group, for the typical case of each group.

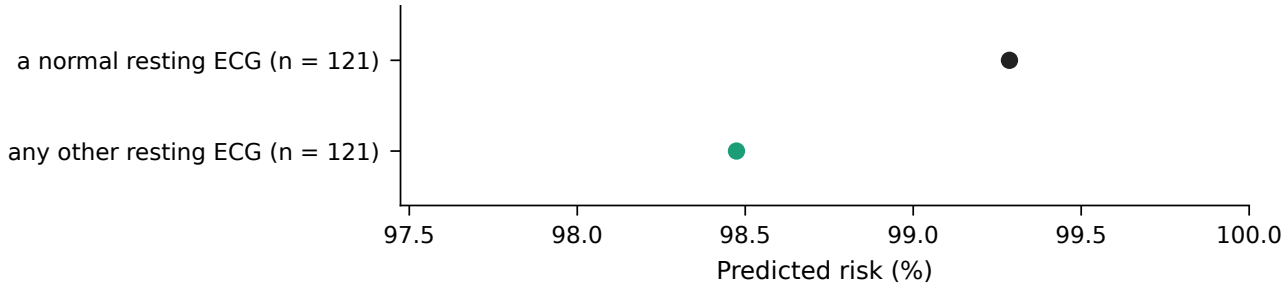


Figure 283: ST depression with a normal ECG: The predicted value for the typical case at each level, through this part (n = training patients; grey: fewer than 20).

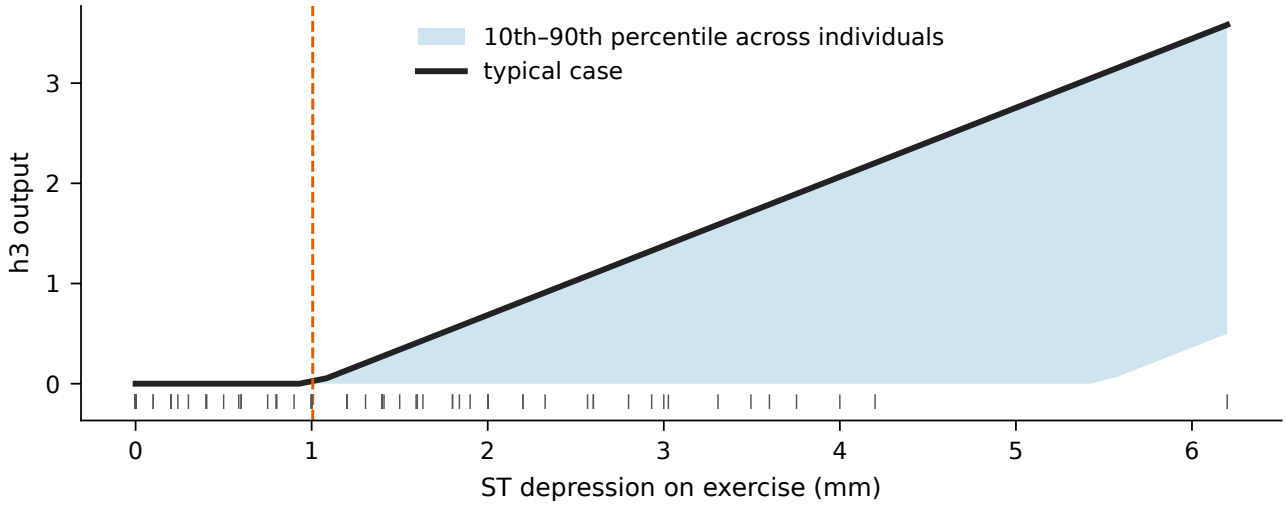


Figure 284: ST depression with a normal ECG: This part's own output, unit h3 (the output of 'ST depression with a normal ECG'), on its own scale (its activation), with every other input at the typical case; it is active for 42 of 242. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

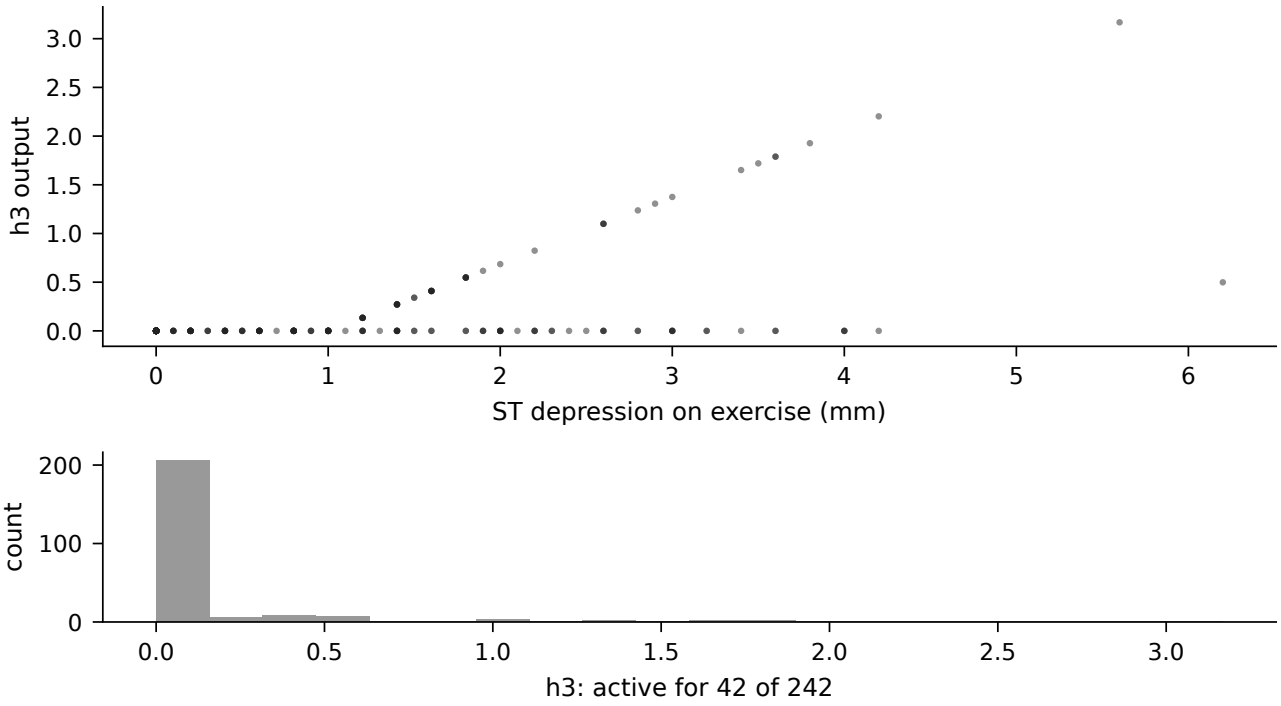


Figure 285: ST depression with a normal ECG: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.2.

6.2.4 Vessels with flat ST segment

Part of: The whole model (12.2.3).
Figure 286 shows what feeds this part; Figure 287 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

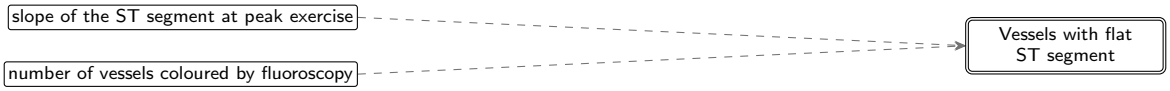


Figure 286: Vessels with flat ST segment: what feeds it, one level down.

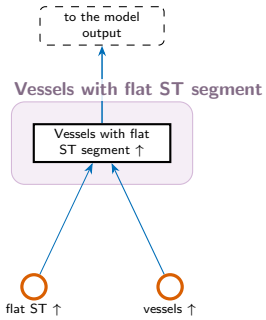


Figure 287: Vessels with flat ST segment: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part adds risk for each extra vessel coloured, about the same for each, and for a flat ST segment. [C40, C41]

Evidence. Through it, each extra vessel adds about the same, from about 18% at 0 to about 36% at 2 vessels for a typical patient, and a flat ST segment raises the risk from about 18% to about 23%. [C40, C41] Its formula reproduces the part exactly on all 242 training patients. [C42]

Evidence plots: Figure 288, Figure 289, Figure 290.

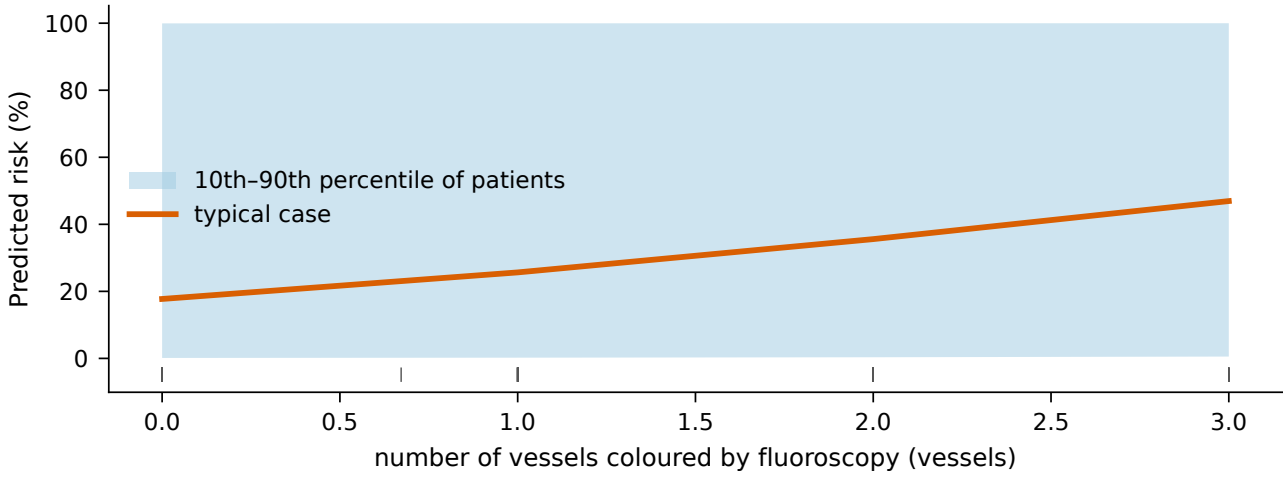


Figure 288: Vessels with flat ST segment: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual patients (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

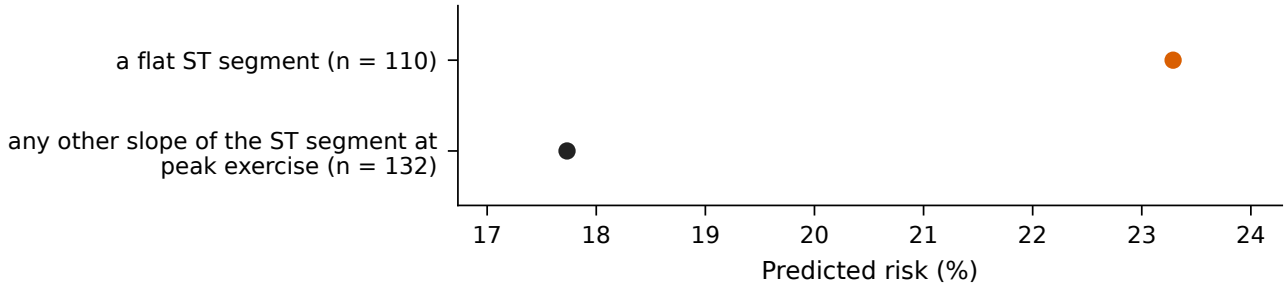


Figure 289: Vessels with flat ST segment: The predicted value for the typical case at each level, through this part (n = training patients; grey: fewer than 20).

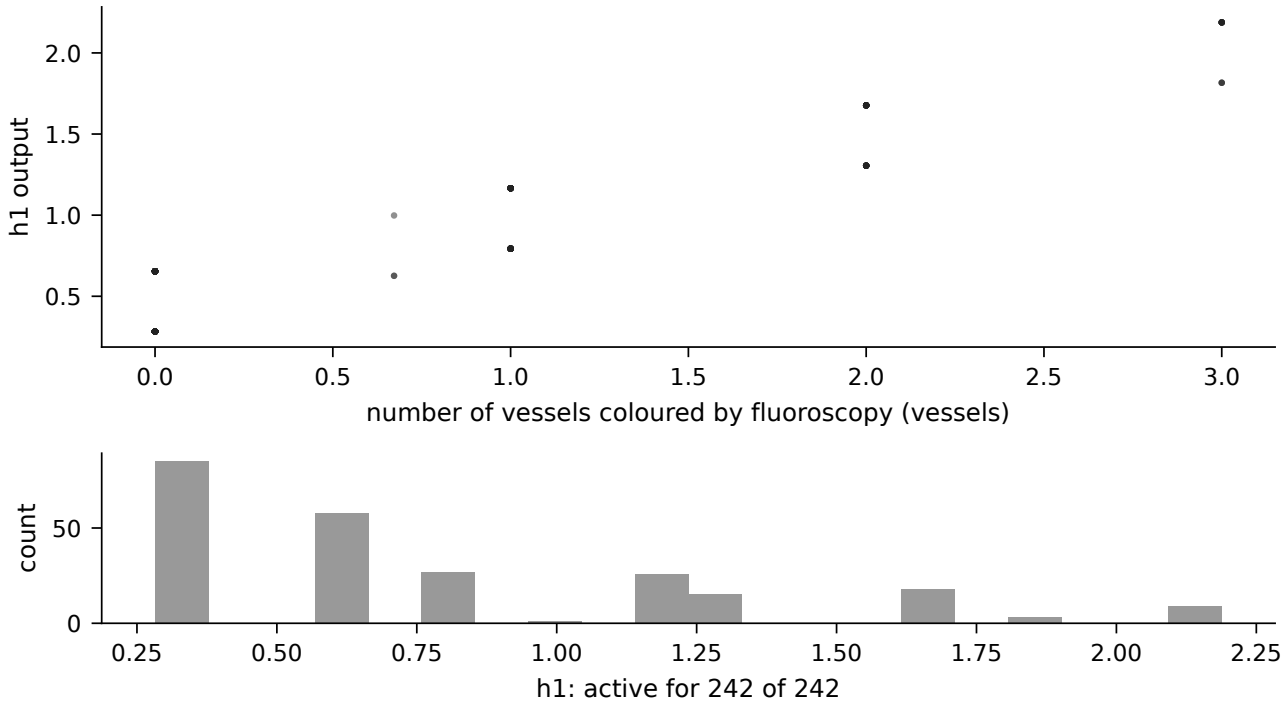


Figure 290: Vessels with flat ST segment: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.1.

6.2.5 Non-anginal chest pain relay

Part of: The whole model (12.2.3).
Figure 291 shows what feeds this part; Figure 292 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).



Figure 291: Non-anginal chest pain relay: what feeds it, one level down.

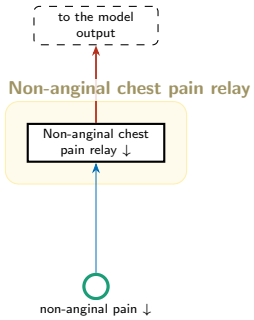


Figure 292: Non-anginal chest pain relay: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part lowers the predicted risk for patients with non-anginal chest pain. [C43]

Evidence. Through it, non-anginal chest pain takes a typical patient's predicted risk from about 18% to about 8%. [C43] Its formula reproduces the part exactly on all 242 training patients. [C44]
Evidence plots: Figure 293, Figure 294.

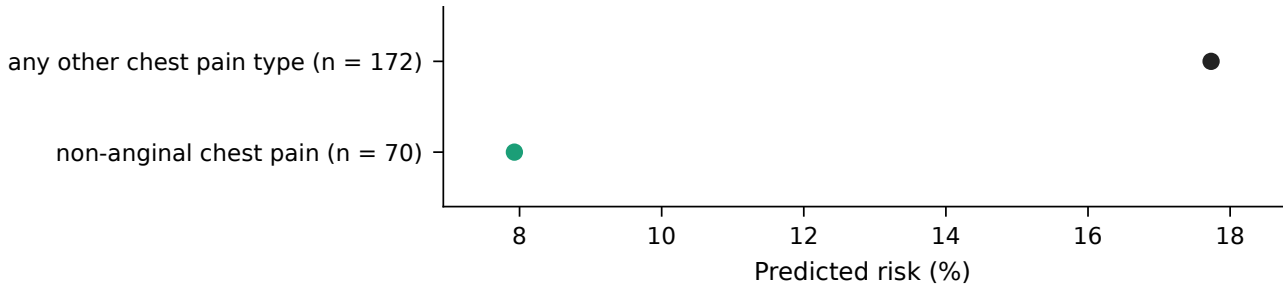


Figure 293: Non-anginal chest pain relay: The predicted value for the typical case at each level, through this part (n = training patients; grey: fewer than 20).

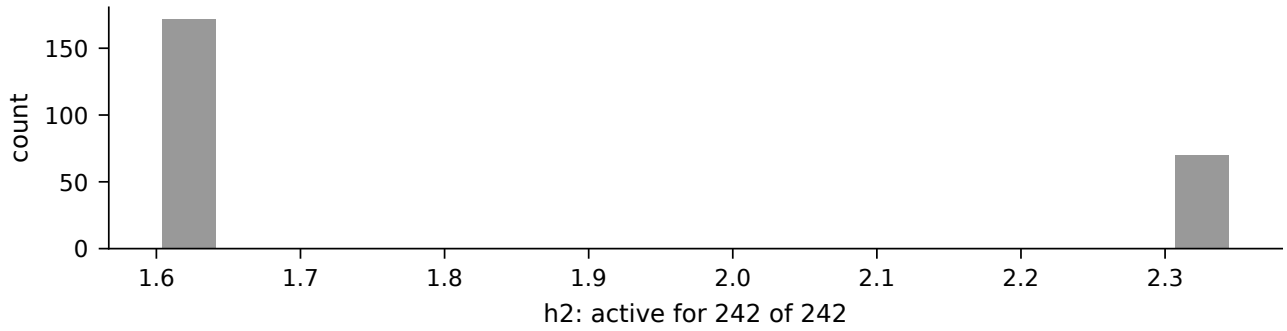


Figure 294: Non-anginal chest pain relay: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.2.

6.2.6 Maximum heart rate

Part of: The whole model (12.2.3).
Figure 295 shows what feeds this part; Figure 296 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).



Figure 295: Maximum heart rate: what feeds it, one level down.

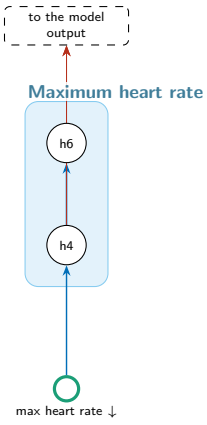


Figure 296: Maximum heart rate: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part lowers the predicted risk for a higher maximum heart rate: nothing until about 145 beats per minute, then steadily. [C45, C46]

Evidence. Through it, a higher maximum heart rate takes a typical patient from about 19% at 71 to about 13% at 182 beats per minute, with nothing until about 145 beats per minute, then steadily. [C46] Its formula reproduces the part exactly on all 242 training patients. [C47]
Evidence plots: Figure 297, Figure 298, Figure 299, Figure 300.
Figure 298 shows this part's own output, unit h4: it switches on at about 144.7 beats per minute of maximum heart rate, and is zero below.
Figure 299 shows this part's own output, unit h6.

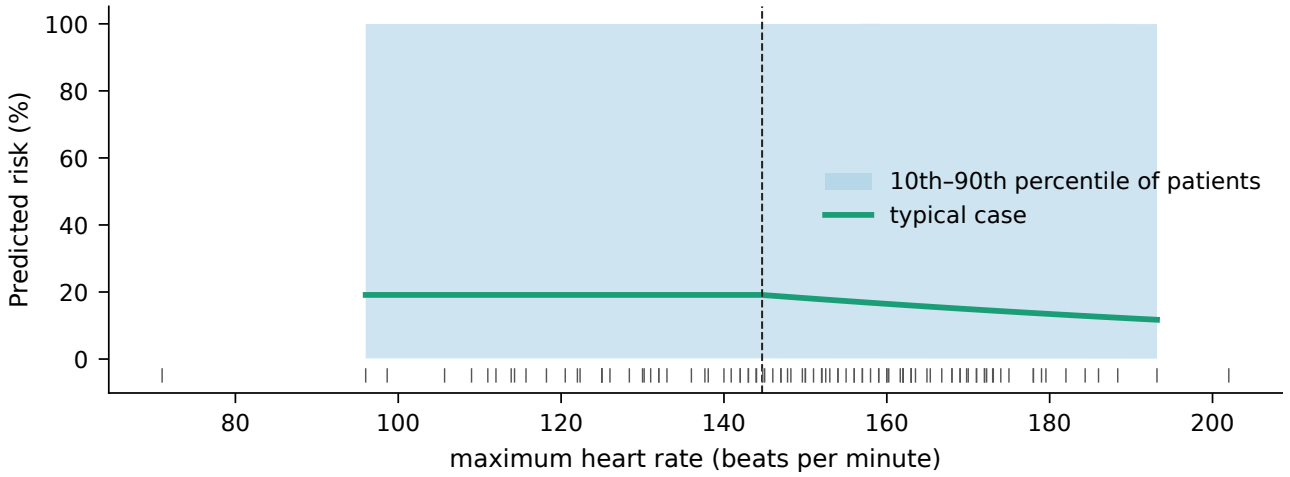


Figure 297: Maximum heart rate: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual patients (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

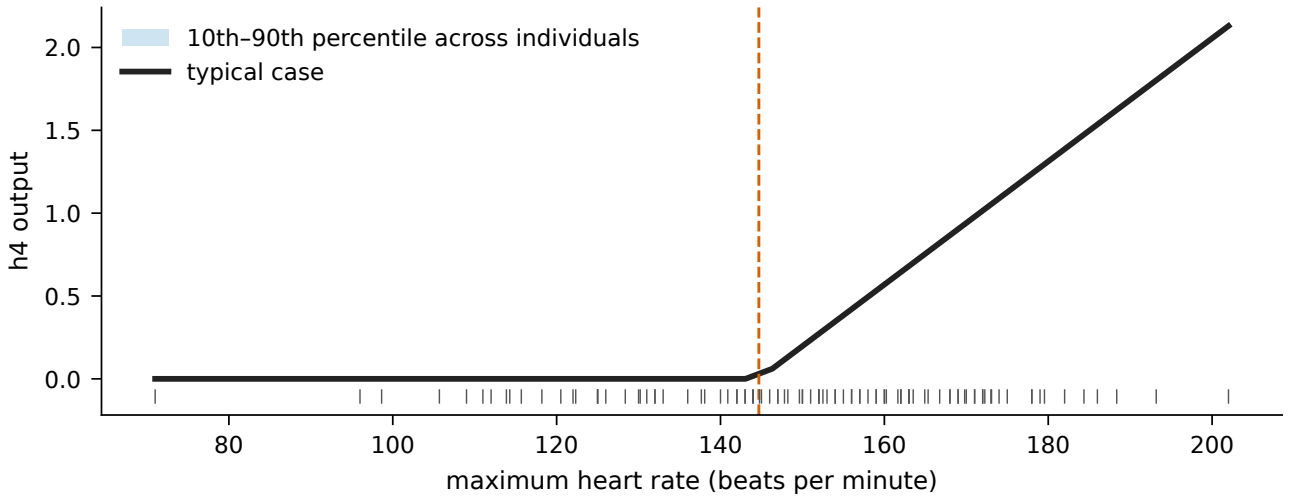


Figure 298: Maximum heart rate: This part's own output, unit h4 (the output of 'Maximum heart rate'), on its own scale (its activation), with every other input at the typical case; it is active for 155 of 242. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

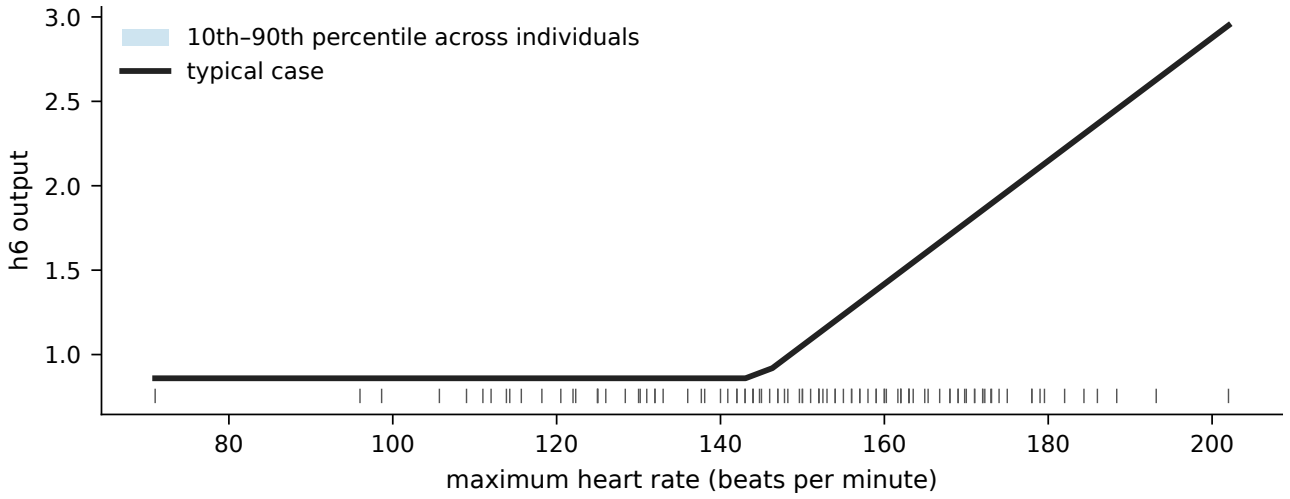


Figure 299: Maximum heart rate: This part's own output, unit h6 (the output of 'Maximum heart rate'), on its own scale (its activation), with every other input at the typical case; it is active for 242 of 242. Bold: the typical case; band: 10th to 90th percentile across individuals; ticks: the data.

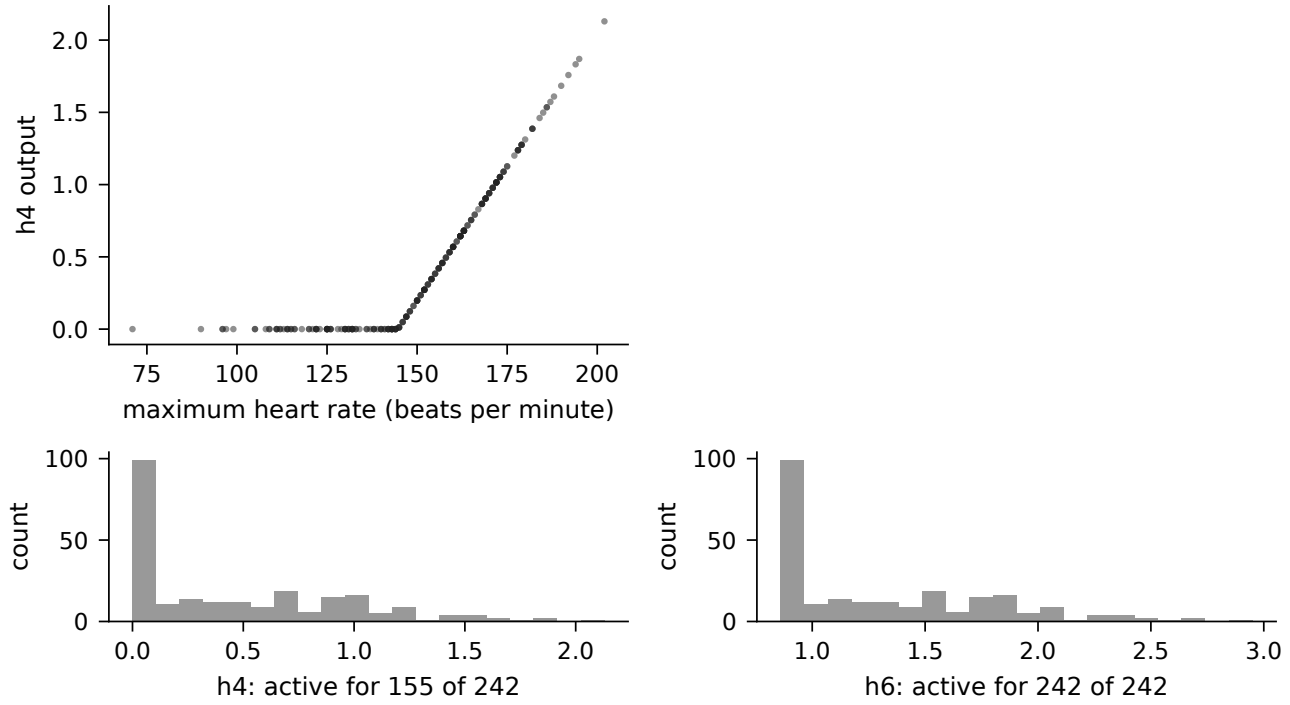


Figure 300: Maximum heart rate: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.2.

6.3 Not supported by the model

Every claim passed its checks.

Technical detail: heart-s1-c1

The record behind each part: the original hypothesis, the closed form, and every claim with its estimate (log-odds and odds ratio), range across patients, n and test check.

T.1 The whole model

Annotation ann_34 (heart_disease_logit). Reads: age, sex, chest pain type, resting blood pressure, serum cholesterol, fasting blood sugar, resting ECG, maximum heart rate reached during the exercise test, exercise-induced angina, ST depression during exercise (compared with rest), slope of the ST segment at peak exercise, number of major vessels coloured by fluoroscopy, thallium stress test result.

Original hypothesis (unchanged). Hypothesis: the heart-disease logit is essentially a logistic regression. 13 direct input->output edges (largest: sex:0 -1.87, slope:2 +1.72, ca +1.30, cp:4 +1.25, chol +0.93, thal:3 -0.87, thal:6 -0.86, trestbps +0.67, cp:2 +0.44, age +0.40, fbs:1 -0.32, slope:3 +0.20, thalach +0.002) are added to five children: ca_slope2_relay and the 2245 relay (raise the effective ca weight to 2.60 per SD), cp3_relay (non-anginal pain lowers risk), st_depression_normal_ecg_gate (ST depression raises risk only with a normal resting ECG), typical_angina_ca_switch's 3210 (typical angina lowers risk, less with more vessels) and max_heart_rate_hinge (small). Nine inputs are unconnected (sex:1, fbs:0, restecg:1, restecg:2, exang:0, exang:1, slope:1, thal:4, thal:7): exercise-induced angina is ignored. Evidence (train, n=242): AUC 0.9392; all hidden units replaced by their best linear fits 0.9355 (= L2 logistic regression 0.9355); all ReLUs as identity 0.9078; all hidden mean-ablated 0.9241. The gain over linear (+0.0037) comes from the ECG/ST gate (+0.0034 restored alone), the heart-rate hinge (+0.0008) and the typical-angina switch (+0.0003); the relays contribute none. Direct edges carry logit SD 4.25 of 5.63.

Closed form. Read through its parts (their formulas are in their own sections).

Split inputs. To annotate the parts, the explanation split these inputs into copies (a refactoring step that does not change what the model computes); the diagrams draw each input once: vessels (3 copies); flat ST (2 copies); max heart rate (2 copies).

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C01	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (whole model) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	18% to 99%	5.89	362	5.83 to 5.83	242	unclear; AUC gain 0.041 [-0.032, 0.123]	yes
C02	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	18% to 91%	3.81	44.9	3.81 to 3.81	115, 110	disagrees; AUC gain -0.011 [-0.065, 0.036]	yes
C03	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment (direct contribution only, not through any part) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	18% to 87%	3.46	31.9	3.46 to 3.46	115, 110	disagrees; AUC gain -0.012 [-0.059, 0.030]	yes
C04	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (direct contribution only, not through any part) *shape: steady (equal steps); 100% of patients same; verified*	18% to 80%	2.91	18.3	2.91 to 2.91	242	disagrees; AUC gain -0.001 [-0.039, 0.032]	yes
C05	direction: thallium stress test result: a normal thallium scan to thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan (whole model) *shape: categorical; highest to lowest: thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	18% to 55%	1.74	5.71	1.74 to 1.74	135, 92	unclear; AUC gain 0.012 [-0.017, 0.044]	yes
C06	direction: ST depression on exercise: 0.0 mm to 2.8 mm (whole model) *shape: hinge at 1.0 mm; 100% of patients same; fallback; matches the fold at 1.0 mm*	18% to 55%	0.87	2.38	0.00 to 1.74	242	unclear; AUC gain 0.005 [-0.005, 0.022]	yes
C07	direction: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L) (whole model) *shape: steady; 100% of patients same; verified*	8% to 45%	2.22	9.23	2.22 to 2.22	242	disagrees; AUC gain -0.022 [-0.066, 0.011]	yes
C08	direction: resting blood pressure: 110 mm Hg to 155 mm Hg (whole model) *shape: steady; 100% of patients same; verified*	9% to 35%	1.65	5.20	1.65 to 1.65	242	unclear; AUC gain 0.000 [-0.022, 0.020]	yes
C09	direction: chest pain type: no chest pain (asymptomatic) to typical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	18% to <1%	-4.55	0.01	-4.78 to -4.08	108, 22	agrees; AUC gain 0.041 [0.003, 0.083]	yes
C10	direction: sex: male to female (whole model) *shape: categorical; highest to lowest: male, female*	18% to <1%	-3.94	0.02	-3.94 to -3.94	159, 83	unclear; AUC gain 0.023 [-0.014, 0.068]	yes
C11	direction: chest pain type: no chest pain (asymptomatic) to non-anginal chest pain (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	18% to 1%	-3.43	0.03	-3.43 to -3.43	108, 70	agrees; AUC gain 0.041 [0.003, 0.083]	yes
C12	direction: chest pain type: no chest pain (asymptomatic) to chest pain type other than atypical angina, no chest pain (asymptomatic) (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, chest pain type other than atypical angina, no chest pain (asymptomatic)*	18% to 2%	-2.51	0.08	-2.51 to -2.51	108, 92	agrees; AUC gain 0.037 [0.006, 0.074]	yes
C13	direction: age: 42 years to 66 years (whole model) *shape: steady; 100% of patients same; verified*	10% to 25%	1.05	2.87	1.05 to 1.05	242	disagrees; AUC gain -0.008 [-0.018, 0.001]	yes
C14	direction: thallium stress test result: a normal thallium scan to a fixed defect on the thallium scan (whole model) *shape: categorical; highest to lowest: thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	18% to 3%	-1.83	0.16	-1.83 to -1.83	135, 15 (small n)	unclear; AUC gain 0.012 [-0.017, 0.044]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C15	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a downsloping ST segment (direct contribution only, not through any part) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	18% to 32%	0.77	2.16	0.77 to 0.77	115, 17 (small n)	disagrees; AUC gain -0.012 [-0.059, 0.030]	yes
C16	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a downsloping ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	18% to 32%	0.77	2.16	0.77 to 0.77	115, 17 (small n)	disagrees; AUC gain -0.011 [-0.065, 0.036]	yes
C17	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, chest pain type other than atypical angina, no chest pain (asymptomatic)*	18% to 5%	-1.34	0.26	-1.34 to -1.34	108, 42	agrees; AUC gain 0.037 [0.006, 0.074]	yes
C18	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	18% to 5%	-1.34	0.26	-1.34 to -1.34	108, 42	agrees; AUC gain 0.041 [0.003, 0.083]	yes
C19	direction: fasting blood sugar: fasting blood sugar of 120 mg/dL or less to fasting blood sugar above 120 mg/dL (whole model) *shape: categorical; highest to lowest: fasting blood sugar of 120 mg/dL or less, fasting blood sugar above 120 mg/dL*	18% to 8%	-0.92	0.40	-0.92 to -0.92	209, 33	disagrees; AUC gain -0.001 [-0.010, 0.009]	yes
C20	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (whole model) *shape: hinge at 145 beats per minute; 100% of patients same; verified; matches the fold at 145 beats per minute*	19% to 14%	-0.36	0.70	-0.36 to -0.36	242	unclear; AUC gain 0.002 [0.000, 0.009]	yes
C21	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (direct contribution only, not through any part) *shape: flat; 100% of patients same; verified*	18% to 18%	0.005	1.00	0.005 to 0.005	242	unclear; AUC gain 0.000 [0.000, 0.000]	yes
C22	direction: resting ECG: a normal resting ECG to resting ECG other than a normal resting ECG (whole model) *shape: categorical; highest to lowest: a normal resting ECG, resting ECG other than a normal resting ECG*	18% to 18%	-0.46	0.63	-1.74 to 0.00	121, 121	disagrees; AUC gain -0.008 [-0.021, 0.000]	yes
C23	magnitude: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels, typical patient *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	18% to 99%	5.83	340		242	unclear; AUC gain 0.041 [-0.032, 0.123]	yes
C24	magnitude: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment, typical patient *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	18% to 91%	3.81	44.9		115, 110	disagrees; AUC gain -0.011 [-0.065, 0.036]	yes
C25	magnitude: thallium stress test result: a normal thallium scan to thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan, typical patient *shape: categorical; highest to lowest: thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	18% to 55%	1.74	5.71		135, 92	unclear; AUC gain 0.012 [-0.017, 0.044]	yes
C26	magnitude: ST depression on exercise: 0.0 mm to 2.8 mm, typical patient *shape: hinge at 1.0 mm; 100% of patients same; fallback; matches the fold at 1.0 mm*	18% to 55%	1.74	5.68		242	unclear; AUC gain 0.005 [-0.005, 0.022]	yes
C27	magnitude: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L), typical patient *shape: steady; 100% of patients same; verified*	8% to 45%	2.22	9.23		242	disagrees; AUC gain -0.022 [-0.066, 0.011]	yes
C28	null: model ignores exercise-induced angina (exact rule)		0.00			242	train only	yes

C29	structure: made of ca_slope2_relay, cp3_relay, st_depression_normal_ecg_gate, max_heart_rate_hinge, typical_angina_ca_switch and inputs age, sex, cp, trestbps, chol, fbs, thalach, slope, ca, thal; added	242	train only	yes

T.2 Typical angina with vessels

Annotation ann_33 (typical_angina_ca_switch). Reads: chest pain type, number of major vessels coloured by fluoroscopy.

Original hypothesis (unchanged). Hypothesis: (i) node 2245 = ReLU(0.578 ca + 1.276) is an always-on ca relay adding +0.88 logit per SD of ca (output weight 1.528); (ii) node 3210 = ReLU(0.901 cp:1 - 0.644*2245 + 0.420) is a typical-angina switch: it is on only for cp:1 patients and then lowers the logit by 0.833 x its activation, from about -2.26 at the lowest ca (-0.73 SD) to about -1.22 at the highest ca (2.64 SD). So typical angina lowers predicted risk, and the reduction shrinks with more narrowed vessels. Evidence (train, n=242): 2245 never clips (0.86..2.80); 3210 active for all 22 cp:1 patients and none of the 220 others. Mean ablation AUC 2245 0.9339, 3210 0.9354; linear replacement 0.9392 / 0.9391 (3210 is a cp:1 switch whose within-group response is linear in ca).

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 0.0).

$$
\begin{aligned}
y_0 &= \operatorname{relu}\big(1.28 + 0.578 x_{ca_c}\big) \\
y_1 &= \operatorname{relu}\big(0.42 + 0.901 x_{cp:1} - 0.644 \operatorname{relu}\big(1.28 \\
&\qquad + 0.578 x_{ca_c}\big)\big)
\end{aligned}
$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C30	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (through this part) *shape: steady (equal steps); 100% of patients same; verified*	18% to 61%	2.04	7.73	1.98 to 1.98	242	disagrees; AUC gain -0.003 [-0.032, 0.019]	yes
C31	direction: chest pain type: chest pain type other than typical angina to typical angina (through this part) *shape: categorical; highest to lowest: chest pain type other than typical angina, typical angina*	18% to 2%	-2.04	0.13	-2.26 to -1.57	220, 22	unclear; AUC gain 0.001 [-0.005, 0.008]	yes
C32	interaction: effect of number of vessels coloured by fluoroscopy (0 vessels to 2 vessels) by chest pain type: typical angina minus chest pain type other than typical angina; stronger in typical angina *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 0%*	typical angina: <1% to 55%; chest pain type other than typical angina: 18% to 99%	0.70	2.00		22, 220	agrees; AUC gain 0.001 [-0.005, 0.008]	yes
C33	structure: its formula reproduces it on 242 training patients (largest difference 0.0)		0.00			242	train only	yes

T.3 ST depression with a normal ECG

Annotation ann_31 (st_depression_normal_ecg_gate). Reads: resting ECG, ST depression during exercise (compared with rest).

Original hypothesis (unchanged). Hypothesis: node 1801 = ReLU(1.542 restecg:0 + 0.817 oldpeak - 1.521) is a gate in which ST depression (oldpeak) raises predicted risk only for patients with a normal resting ECG (restecg:0), once oldpeak is above about its mean (threshold -0.03 SD); with an abnormal resting ECG the threshold is 3.75 SD and the gate is effectively closed. Evidence (train, n=242): active for 41 of 121 normal-ECG patients and 1 of 121 others; adds 1.40 x activation = +1.14 logit per SD oldpeak when open. It is the model's main nonlinearity: mean ablation AUC 0.9339, linear replacement 0.9362, and restoring it onto the all-linear model gains +0.0034 AUC. oldpeak has no direct edge, so the model ignores ST depression for patients with an abnormal resting ECG.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 2.22045e-16).

$$y_0 = \mathrm{relu}\big(-1.52 + 1.54 x_{restecg:0} + 0.818 x_{oldpeak}\big)$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C34	direction: ST depression on exercise: 0.0 mm to 2.8 mm (through this part) *shape: hinge at 1.0 mm; 100% of patients same; fallback; one point with C37 (median fold 1.0 mm, typical patient 1.0 mm, 10th-90th percentile 1.0 mm to 1.0 mm)*	18% to 55%	0.87	2.38	0.00 to 1.74	242	unclear; AUC gain 0.005 [-0.005, 0.022]	yes
C35	direction: resting ECG: a normal resting ECG to resting ECG other than a normal resting ECG (through this part) *shape: categorical; highest to lowest: a normal resting ECG, resting ECG other than a normal resting ECG*	99% to 98% (typical patient this part acts on)	-0.46	0.63	-1.74 to 0.00	121, 121	disagrees; AUC gain -0.008 [-0.021, 0.000]	yes
C36	interaction: effect of ST depression on exercise (0.0 mm to 2.8 mm) by resting ECG: resting ECG other than a normal resting ECG minus a normal resting ECG; stronger in a normal resting ECG *shape: hinge at 1.0 mm; 100% of patients same; fallback*	resting ECG other than a normal resting ECG: 18% to 18%; a normal resting ECG: 18% to 55%	-1.74	0.18		121, 121	agrees; AUC gain -0.008 [-0.021, 0.000]	yes
C37	threshold: ST depression on exercise: fold at about 1.0 mm (starts) for patients with a normal resting ECG *shape: hinge at 1.0 mm; 100% of patients same; fallback; one point with C37 (median fold 1.0 mm, typical patient 1.0 mm, 10th-90th percentile 1.0 mm to 1.0 mm)*	13% to 63% (typical patient with a fold (patients with a normal resting ECG))	2.41	11.1	2.41 to 2.41	121, 41	unclear; AUC gain 0.005 [-0.005, 0.022]	yes
C38	subgroup: who this part acts on: 42 of 242 patients	98% to 99% (typical patient this part acts on (risk without the part, then with it))	1.08	2.95	0.19 to 2.50	42, 200	unclear; AUC gain 0.005 [-0.006, 0.025]	yes
C39	structure: its formula reproduces it on 242 training patients (largest difference 2.22045e-16)		2e-16			242	train only	yes

T.4 Vessels with flat ST segment

Annotation ann_29 (ca_slope2_relay). Reads: slope of the ST segment at peak exercise, number of major vessels coloured by fluoroscopy.

Original hypothesis (unchanged). Hypothesis: node 664 = ReLU(0.456 ca + 0.185 slope:2 + 0.783) is an affine relay that strengthens the direct ca and flat-ST-slope (slope:2) effects. Evidence (train, n=242): pre-activation 0.28..2.19, never clips, so the part is exactly affine on the data; it adds +0.42 logit per SD of ca and +0.17 per SD of slope:2 (output weight 0.92). Mean ablation AUC 0.9392 -> 0.9375; linear replacement leaves AUC at 0.9392. No concept beyond reweighting.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 2.22045e-16).

$$y_0 = \mathrm{relu}\big(0.783 + 0.185 x_{slope:2_b} + 0.456 x_{ca_b}\big)$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C40	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (through this part) *shape: steady (equal steps); 100% of patients same; verified*	18% to 36%	0.94	2.56	0.94 to 0.94	242	disagrees; AUC gain -0.002 [-0.019, 0.011]	yes
C41	direction: slope of the ST segment at peak exercise: slope of the ST segment at peak exercise other than a flat ST segment to a flat ST segment (through this part) *shape: categorical; highest to lowest: a flat ST segment, slope of the ST segment at peak exercise other than a flat ST segment*	18% to 23%	0.34	1.41	0.34 to 0.34	132, 110	disagrees; AUC gain -0.003 [-0.016, 0.005]	yes
C42	structure: its formula reproduces it on 242 training patients (largest difference 2.22045e-16)		2e-16			242	train only	yes

T.5 Non-anginal chest pain relay

Annotation ann_30 (cp3_relay). Reads: chest pain type.

Original hypothesis (unchanged). Hypothesis: node 1173 = ReLU(0.336 cp:3 + 1.818) is an always-on relay through which non-anginal chest pain (cp:3) lowers predicted risk. Evidence (train, n=242): pre-activation 1.60..2.34, never clips (entry-exit correlation 1.0); with output weight -1.24 it contributes -0.417 logit per SD of cp:3, about -0.92 logit for non-anginal pain vs other types. cp:3 has no direct edge, so this is its only route. Mean ablation AUC 0.9373; linear replacement 0.9392.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 0.0).

$$y_0 = \mathrm{relu}\big(1.82 + 0.336 x_{cp:3}\big)$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C43	direction: chest pain type: chest pain type other than non-anginal chest pain to non-anginal chest pain (through this part) *shape: categorical; highest to lowest: chest pain type other than non-anginal chest pain, non-anginal chest pain*	18% to 8%	-0.92	0.40	-0.92 to -0.92	172, 70	unclear; AUC gain 0.001 [-0.008, 0.013]	yes
C44	structure: its formula reproduces it on 242 training patients (largest difference 0.0)		0.00			242	train only	yes

T.6 Maximum heart rate

Annotation ann_32 (max_heart_rate_hinge). Reads: maximum heart rate reached during the exercise test.

Original hypothesis (unchanged). Hypothesis: a small hinge on maximum heart rate (thalach). Node 2009 = ReLU(0.833 thalach + 0.201) feeds the output (-0.151) and the always-on relay 2650 = ReLU(0.982*2009 + 0.859) (-0.172), a net -0.32 x 2009: higher maximum heart rate lowers predicted risk, but only above about -0.24 SD; below that it has no effect (the direct thalach edge, +0.002, is negligible). Evidence (train, n=242): 2009 active for 64% of patients; logit contribution SD 0.16 (small); mean ablation AUC 0.9391, linear replacement 0.9392, restore-onto-linear gain +0.0008.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 0.0).

$$\begin{aligned} y_0 &= \mathrm{relu}\big(0.201 + 0.833 x_{thalach_b}\big) \\ y_1 &= \mathrm{relu}\big(0.859 + 0.982\,\mathrm{relu}\big(0.201 \\ &\qquad + 0.833 x_{thalach_b}\big)\big) \end{aligned}$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C45	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (through this part) *shape: hinge at 145 beats per minute; 100% of patients same; verified; one point with C46 (median fold 145 beats per minute, typical patient 145 beats per minute, 10th-90th percentile 145 beats per minute to 145 beats per minute)*	19% to 14%	-0.36	0.70	-0.36 to -0.36	242	unclear; AUC gain 0.002 [0.000, 0.009]	yes
C46	threshold: maximum heart rate: fold at about 145 beats per minute (starts) *shape: hinge at 145 beats per minute; 100% of patients same; verified; one point with C46 (median fold 145 beats per minute, typical patient 145 beats per minute, 10th-90th percentile 145 beats per minute to 145 beats per minute)*	19% to 13% (typical patient with a fold)	-0.44	0.64	-0.44 to -0.44	242, 155	unclear; AUC gain 0.002 [0.000, 0.009]	yes
C47	structure: its formula reproduces it on 242 training patients (largest difference 0.0)		0.00			242	train only	yes

The network before any explanation

Figure 301 (next page, landscape) is the trained network as it is, with no parts, no splits and no inserted units: inputs at the bottom, hidden units by depth, the output at the top, every connection drawn, every input column its own node (one-hot levels included).

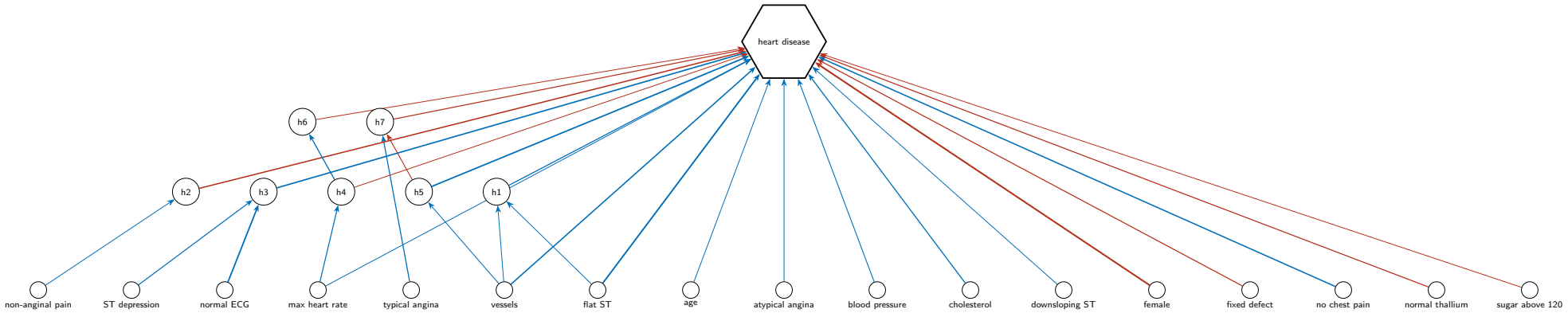


Figure 301: The network as trained, before any explanation, in full. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. Not used (no connection): male, sugar 120 or less, ST-T abnormal ECG, LVH on ECG, no exercise pain, exercise pain, upsloping ST, thallium code 4, reversible defect.

7 heart-s2-c1

Genome `22757c22-3eac-4248-961b-ed0cd500a680`, explanation `eac2ad3d-7d0e-4112-9d34-7f0f217a2c31`. Split: train primary (n=242), test as a check (n=61). Claims: 55, verified 54, not supported 1.
AI-assisted draft for review (unverified)

Reference (typical) patient: the typical patient: the median training patient for every number and the most common training category for every category (not a real patient). age: 56 years; ca: 0 vessels; chol: 240 mg/dL (6.2 mmol/L); cp: no chest pain (asymptomatic); exang: no chest pain on exercise; fbs: fasting blood sugar of 120 mg/dL or less; oldpeak: 0.8 mm; restecg: a normal resting ECG; sex: male; slope: an upsloping ST segment; thal: a normal thallium scan; thalach: 152 beats per minute; trestbps: 130 mm Hg. Predicted risk of heart disease: 29%.

7.1 Summary

For a typical patient (a 56-year-old man with no chest pain symptoms and median test results) the model predicts a risk of heart disease of about 29%; it adds up most of its inputs independently, and three parts add to that sum: one acts on 80 of 242 training patients through age and maximum heart rate, one on 121 through ST depression on exercise, and one on 117 through the ST segment, ST depression and cholesterol. [C29, C38, C42, C52] Figure 302 shows the model at this level: one box per paragraph below.

How to read the diagrams. Edge colour is the sign of that one connection: blue positive, red negative; thicker is a larger weight. The outline and arrow on an input (and on a part's output box) is its overall effect on the final prediction, read from the verified claims: orange and up arrow raises it, teal and down arrow lowers it, grey and double arrow depends (it goes both ways, or differs between levels). A negative connection into a part that itself raises the prediction means that input lowers the prediction; through a hinge the overall effect can only be read from the claims, not from the edges. Up arrow means towards heart disease; down arrow, away from it. A double outline marks an input (or a part's output) that acts at a fixed rate everywhere: it adds the same amount to the score per unit, so it is explained by its rate alone (see 'Linear factors'); a dashed double outline, at a fixed rate on the observed data. The normal outline is an input whose effect switches on and off somewhere. For example, age has a negative connection into 'Age and heart rate gate', a part that lowers the predicted risk of heart disease, so age raises it.

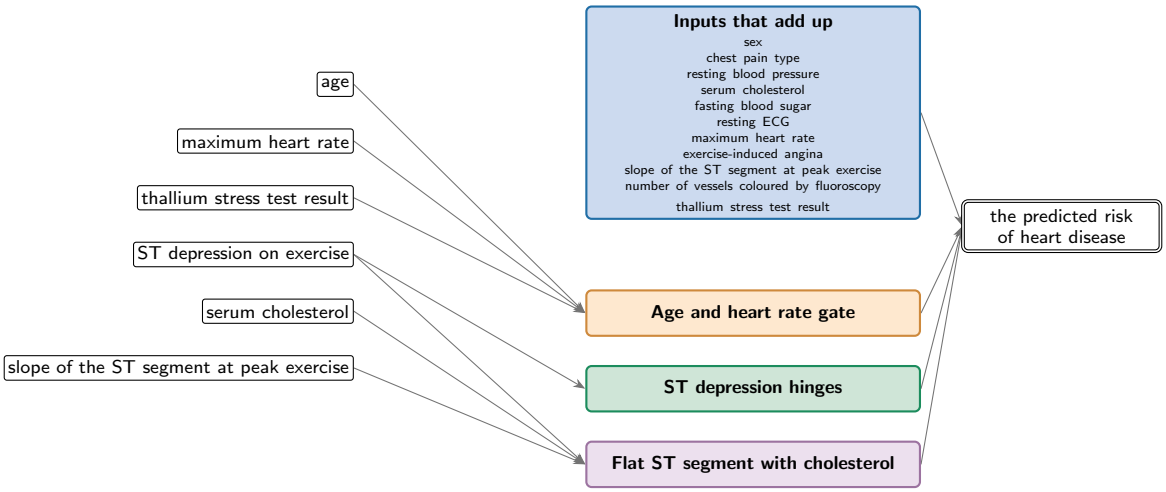


Figure 302: The model at the level of the summary: one box per part.

Inputs that add up. *Lowering the risk.* These add up independently, and these lower the risk: compared with no chest pain symptoms, typical angina takes a typical patient's predicted risk from about 29% to about 3%, non-anginal chest pain to about 5% and atypical angina to about 14%; being a woman lowers it to about 3% and fasting blood sugar above 120 mg/dL to about 14%; and through this sum alone a higher maximum heart rate lowers it steadily, from about 32% at 118 to about 28% at 175 beats per minute (maximum heart rate also acts through the age and heart rate gate, below). [C06, C09, C17, C07, C18, C23]
Raising the risk. Each extra vessel coloured by fluoroscopy adds about the same, from about 29% at 0 to about 94% at 2 vessels, flattening out near the top because the predicted risk cannot go much higher; higher resting blood pressure raises it steadily, from about 20% at 110 mm Hg to about 44% at 155 mm Hg, and chest pain brought on by exercise raises it to about 43%. [C01, C08, C19] Through this sum alone, a flat rather than upsloping ST segment raises it to about 73%, higher cholesterol raises it steadily from about 18% at 193 mg/dL to about

49% at 308 mg/dL, and a reversible defect on the thallium scan raises it to about 42%; all three also act through the parts below. [C03, C04, C20]
Rare results. Some rarer results also count (the model has little data on this, n = 2 to 17): through this sum alone a downsloping ST segment raises the risk to about 50% and a fixed defect on the thallium scan lowers it to about 11%, and an ST-T wave abnormality on the resting ECG raises it to about 46%. [C10, C13, C16]

Age and heart rate gate. This part reads age, maximum heart rate and the thallium scan, and acts on 80 of 242 training patients, 9 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 4% without it and about 1% with it. [C38] For patients without a reversible defect on the thallium scan, it switches on when age in years minus about 0.28 × the maximum heart rate in beats per minute is below about 11.4 (for maximum heart rates from about 71 to 202 beats per minute); 80 of 242 training patients are on that side, and 11% of them had heart disease, against 60% of the others. [C54] Through it, older age raises the predicted risk steadily up to about 56 years (between 45 and 61 years for different patients), then no further for most patients, but for patients with a reversible defect on the thallium scan it rises in a jump: for a typical patient from about 12% at 42 to about 29% at 66 years. [C30, C37] Across the whole model, a higher maximum heart rate lowers the risk in an uneven way, more without a reversible defect on the thallium scan (from about 32% at 118 to about 21% at 175 beats per minute, against about 45% to about 40% with one). [C35]

ST depression hinges. This part reads only ST depression on exercise and acts on 121 of 242 training patients, 77 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 89% without it and about 91% with it. [C42] Through it, more ST depression raises the predicted risk slowly at first, then faster, adding risk only once it rises above about 0.8 mm: for a typical patient from about 29% at 0 mm to about 48% at 2.8 mm. [C40, C41]

Flat ST segment with cholesterol. This part reads the slope of the ST segment, ST depression on exercise and cholesterol, and acts on 117 of 242 training patients, 78 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 86% without it and about 90% with it. [C52] For patients whose ST segment is not flat, it switches on when ST depression on exercise is above about 2.23 mm (for cholesterol from about 126 to 186 mg/dL), or when ST depression minus about 0.012 × the cholesterol in mg/dL is above about 0.011 (for cholesterol from about 186 to 417 mg/dL); 117 of 242 training patients are on, and 67% of them had heart disease, against 23% of the others. [C55] Across the whole model, more ST depression raises the risk slowly at first, then faster, and even more so with a flat ST segment: from about 75% to about 91% then, against about 29% to about 48% otherwise. [C48] Higher cholesterol raises it steadily across the whole model, from about 67% to about 88% with a flat ST segment and from about 18% to about 49% otherwise, while through this part it lowers the risk in an uneven way, for the typical patient the part acts on from about 91% to about 88%. [C47, C46]

The whole network. Figure 303 (next page, landscape) draws every unit and connection of the explained model, nested by part: shaded clusters are the parts of the model and darker boxes the parts inside them; circles are hidden units (h1, h2, . . .); each cluster's rectangle is its output, and a dashed 'sum' stands for an addition the next unit does (drawn, not a unit of the model); the hexagon is the model's output. An input used by several parts is drawn under each of them (same label, dotted outline, the first copy marked with the number of copies): it is the same input, repeated for readability. Not used by the model (input columns with no connection): non-anginal pain, sugar 120 or less, normal ECG, LVH on ECG, no exercise pain, downsloping ST, normal thallium, thallium code 4.

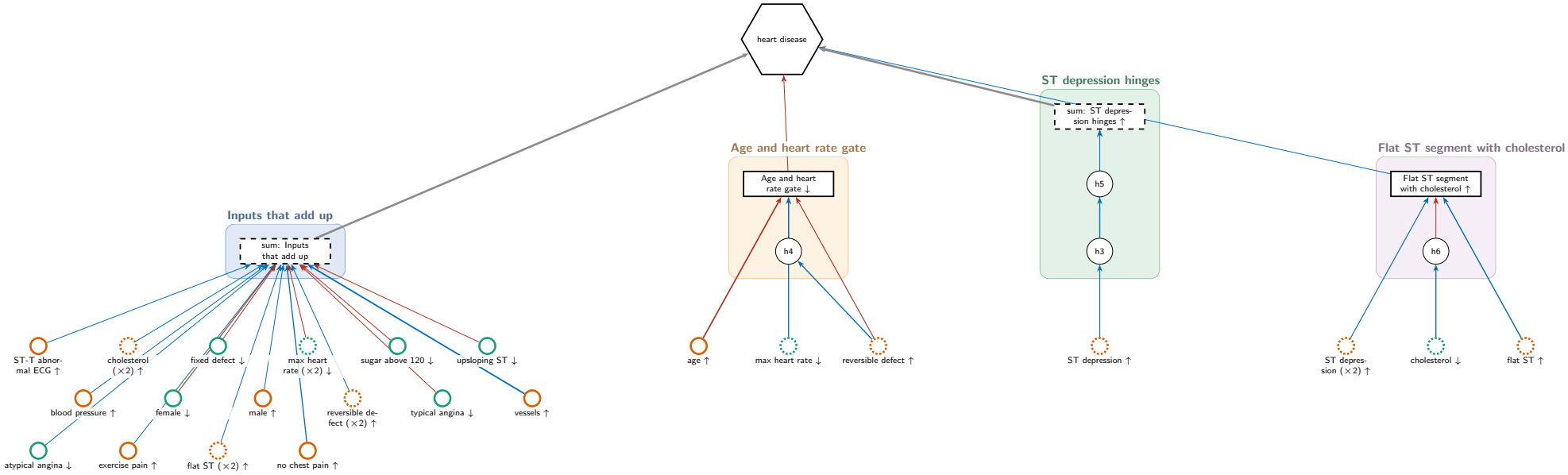


Figure 303: Every unit and connection of the explained model, nested by part. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Worth checking against what is known. Worth checking: going from 0 to 2 vessels coloured moves a typical patient's predicted risk from about 29% to about 94%. [C01] Worth checking: patients with typical angina get the lowest predicted risk of all chest pain types, below patients with no chest pain symptoms. [C06, C09, C17] Worth checking: the model treats fasting blood sugar above 120 mg/dL as lowering the risk of heart disease (for a typical patient from about 29% to about 14%). [C18] Worth checking: the model treats a fixed defect on the thallium scan as lowering the risk (for a typical patient from about 29% to about 11%), from only 15 training patients. [C14] Worth checking: across the whole model, older age raises the predicted risk only for patients without a reversible defect on the thallium scan, and through one part only up to about 56 years. [C33, C37]

7.2 The model, read from the top down

Each part of the explanation in turn, from the whole model down to the smallest parts: what it does for the prediction (hypothesis), what was checked (evidence), a diagram one level down, and where to read on. The technical record of each part is in "Technical detail" at the end.

7.2.1 The whole model

Figure 304 shows what feeds this part; Figure 305 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

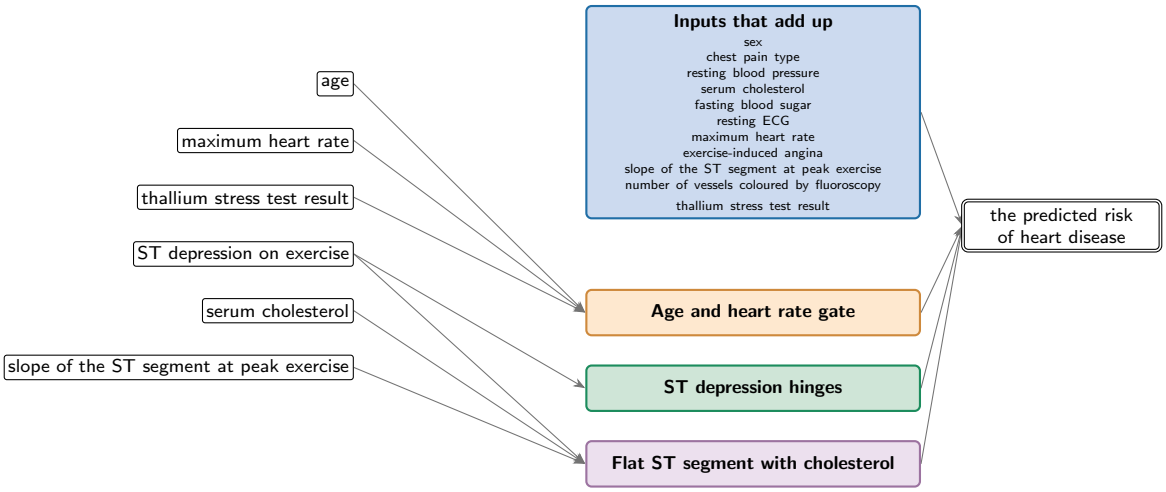


Figure 304: The whole model: what feeds it, one level down.

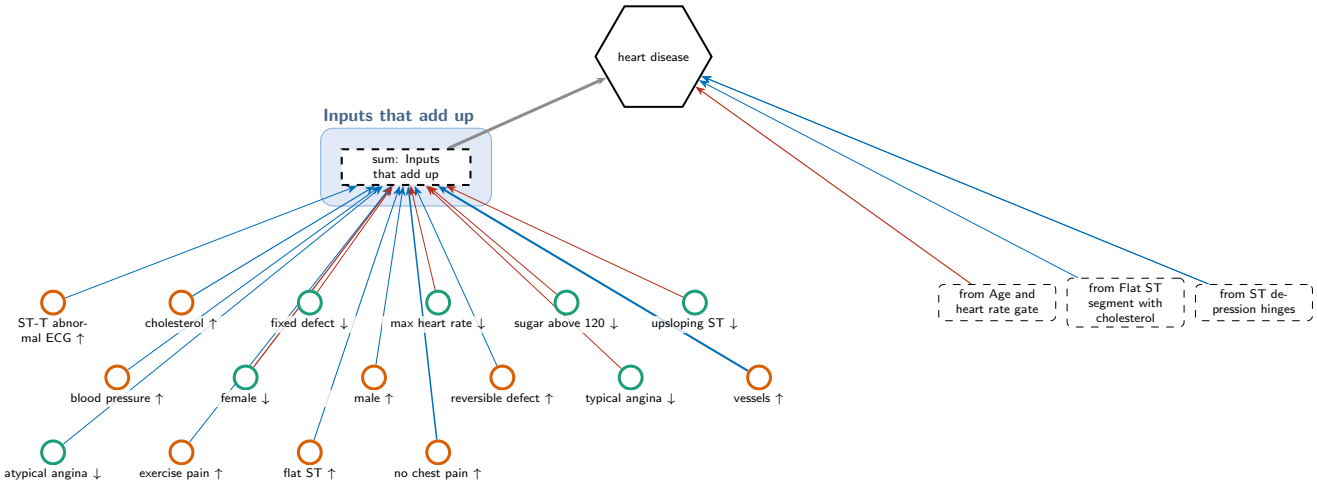


Figure 305: The whole model: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. The model adds up its inputs into one score: the vessels coloured (each extra vessel adding about the same), the ST segment, chest pain type and sex each move the predicted risk on their own, and three parts add to that sum: one for age and maximum heart rate, one for ST depression and one for the ST segment with cholesterol. [C29, C01, C02, C06, C07]

Evidence. It is made up of the inputs that feed the score directly and three parts, the age and heart rate gate, the ST depression hinges and the flat ST segment with cholesterol part (each explained below); the score is their sum, which was checked on all 242 training patients. [C29] Age and ST depression on exercise do not feed the score directly, so they act only through these parts. [C29, C30, C40] Changing one input at a time across the whole model, a flat rather than upsloping ST segment takes a typical patient's predicted risk from about 29% to about 77%, against about 73% through the sum alone. [C02, C03] Across the whole model, a higher maximum heart rate lowers it in an uneven way, from about 32% at 118 to about 21% at 175 beats per minute, against about 28% through the sum alone, where it falls steadily. [C22, C23] Across the whole model, a fixed defect on the thallium scan lowers it to about 11% (the model has little data on this, n = 15), and a reversible defect raises it to about 42%. [C14, C21] Evidence plots: Figure 306.

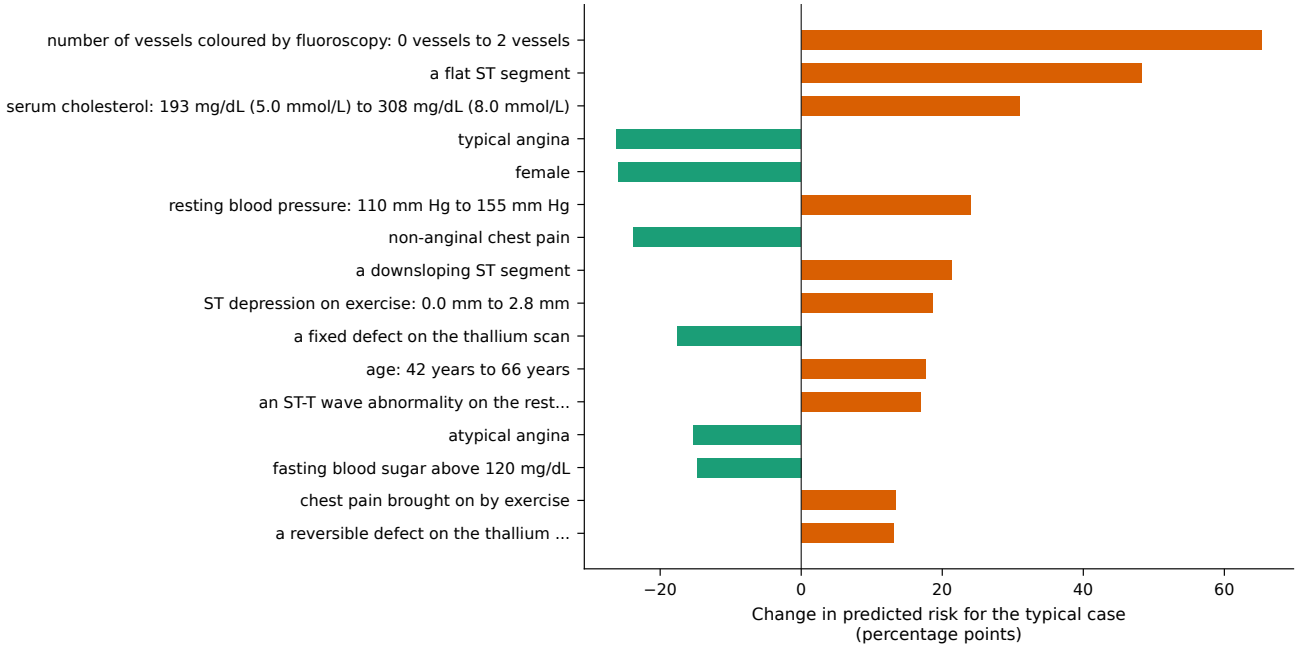


Figure 306: The whole model: Each input's change in the predicted value for the typical case across the whole model, largest first (orange: raises it, teal: lowers it).

Dive deeper: Age and heart rate gate (10.2.2); ST depression hinges (12.2.2); Flat ST segment with cholesterol (11.2.2). Technical detail: T.3.

7.2.2 Age and heart rate gate

Part of: The whole model (12.2.3).
Figure 307 shows what feeds this part; Figure 308 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

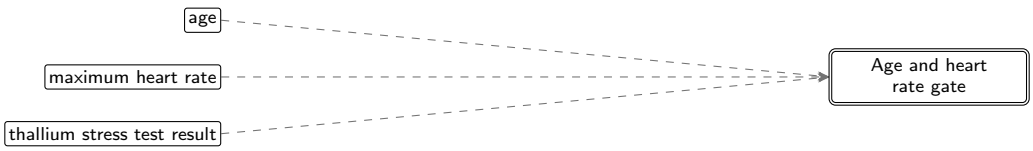


Figure 307: Age and heart rate gate: what feeds it, one level down.

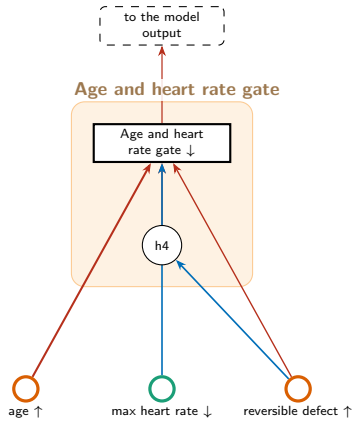


Figure 308: Age and heart rate gate: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part lowers the predicted risk for patients whose age is low relative to their maximum heart rate; for the typical patient it acts on, a reversible defect on the thallium scan raises the risk through it from about 1% to about 4%. [C54, C32]

Evidence. For patients without a reversible defect on the thallium scan, it switches on when age in years minus about 0.28 × the maximum heart rate in beats per minute is below about 11.4 (for maximum heart rates from about 71 to 202 beats per minute); 80 of 242 training patients are on that side, and 11% of them had heart disease, against 60% of the others. [C54] Through it, older age raises the predicted risk steadily up to about 56 years (between 45 and 61 years for different patients), then no further for most patients, but for patients with a reversible defect on the thallium scan it rises in a jump: for a typical patient from about 12% at 42 to about 29% at 66 years. [C30, C37] Across the whole model, older age raises the risk, then no further, only if the thallium scan shows no reversible defect: from about 12% to about 29%, while with one it stays at about 42%. [C33] A higher maximum heart rate lowers the risk through it in an uneven way, from about 29% at 118 to about 22% at 175 beats per minute, and across the whole model it lowers it more without a reversible defect (from about 32% to about 21%, against about 45% to about 40%). [C31, C35] Its formula reproduces the part exactly on all 242 training patients. [C39]
Evidence plots: Figure 309, Figure 310, Figure 311, Figure 312, Figure 313, Figure 314, Figure 315, Figure 316.

Worked example: reading this part. Figure 312 maps the part's own output over its main inputs. In the grey region it is switched off and adds nothing; past the black line it switches on, and the colour shows how much it adds to the score. The line is the rule in the claim (C54): it switches on when age minus about 0.28 × maximum heart rate below about 11.4 (for maximum heart rate from about 71.0 to 202). 80 of 242 training patients lie on the 'on' side, and 11% of them had heart disease, against 60% of the others: the 'on' side holds mostly hollow points, so the part picks out low-risk cases. Figure 313 cuts the map into slices: the hinge moves as the other input changes, which is what a tilted line means. Figure 314 shows what this does to the final prediction for the typical case.
Figure 315 shows this part's own output, unit h2: over its two main inputs, with the line where it switches on.

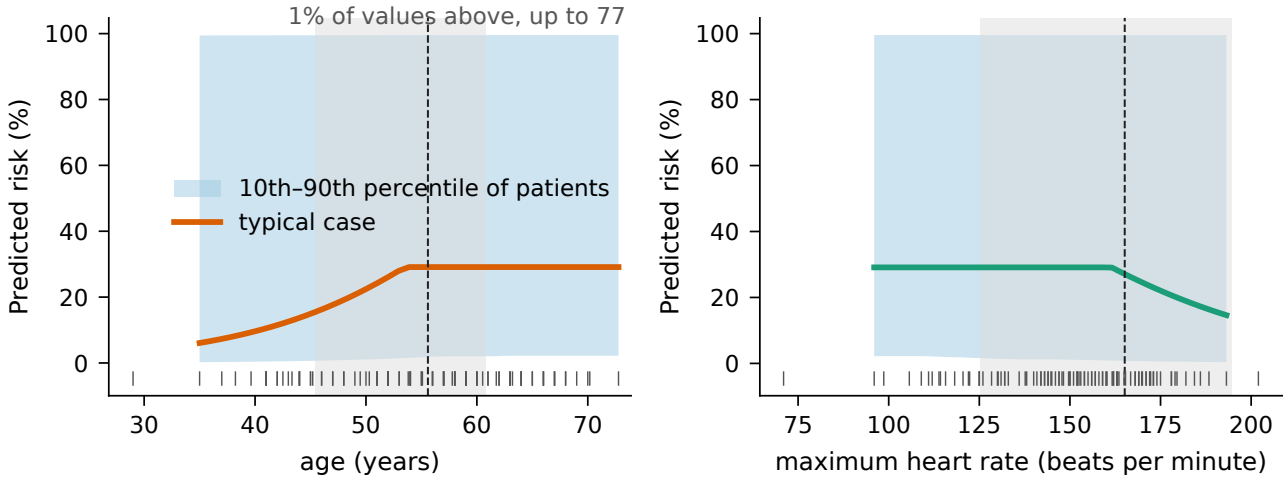


Figure 309: Age and heart rate gate: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual patients (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

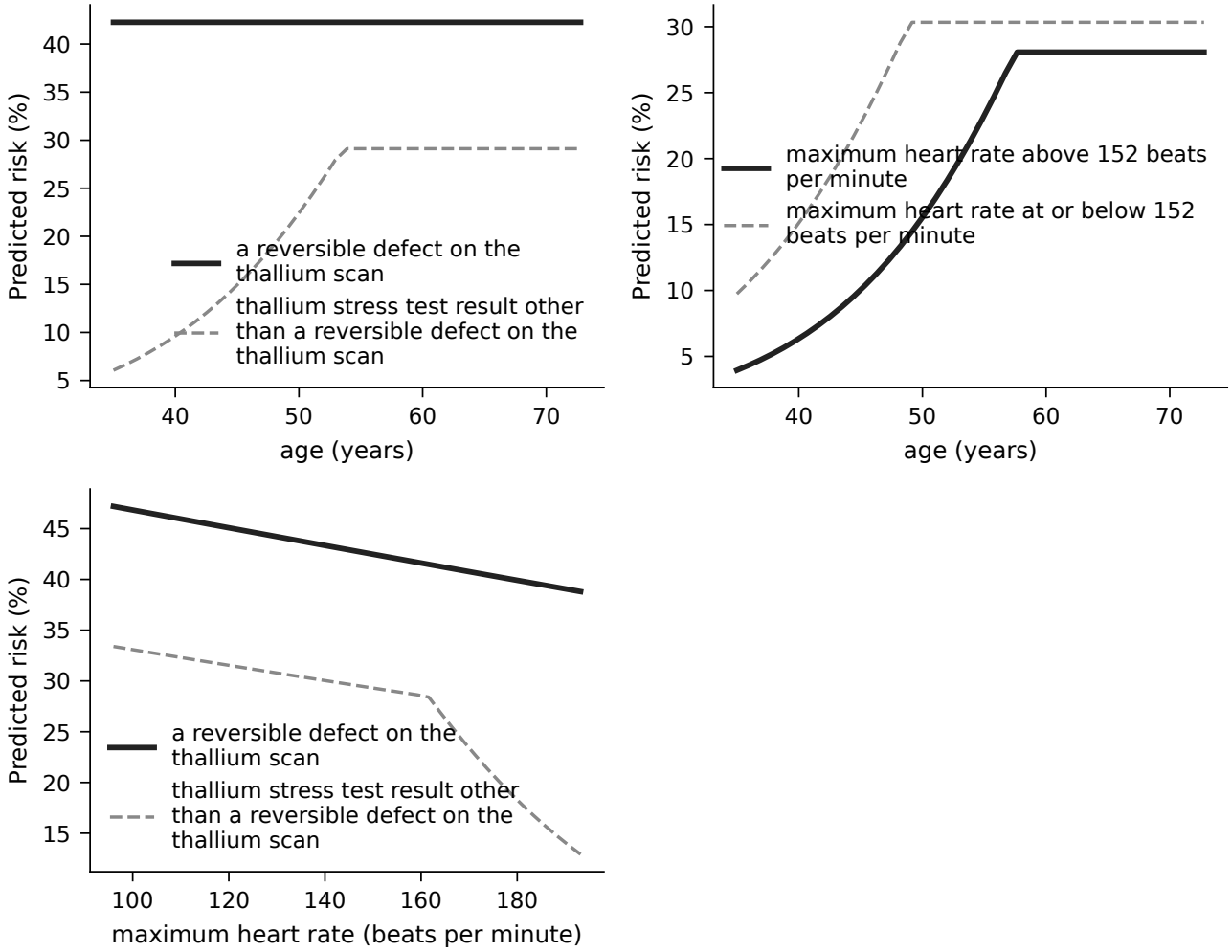


Figure 310: Age and heart rate gate: How the effect differs between groups: one curve per group, for the typical case of each group.

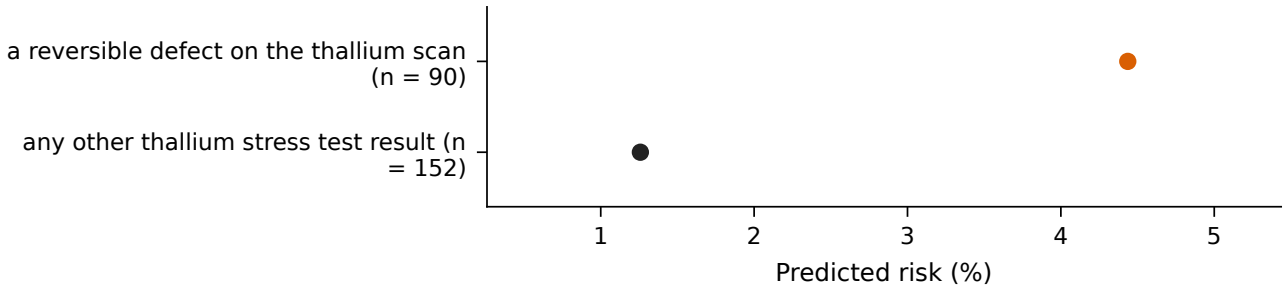


Figure 311: Age and heart rate gate: The predicted value for the typical case at each level, through this part (n = training patients; grey: fewer than 20).

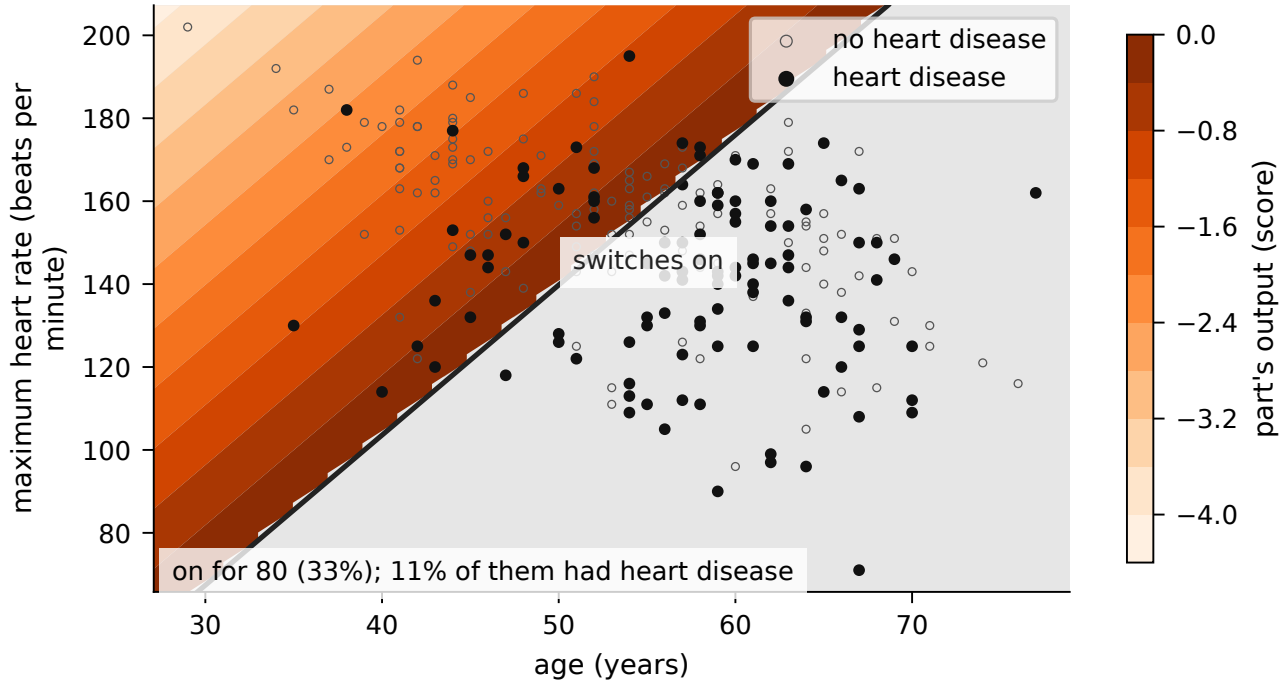


Figure 312: Age and heart rate gate: The part's own output over its two numeric inputs, other inputs at the typical case: colour where it is on, flat grey where it is switched off, the black line where it switches on; training patients as hollow (no event) and filled (event) points, so the part can be seen catching the events (the predicted-chance overlay was harder to read).

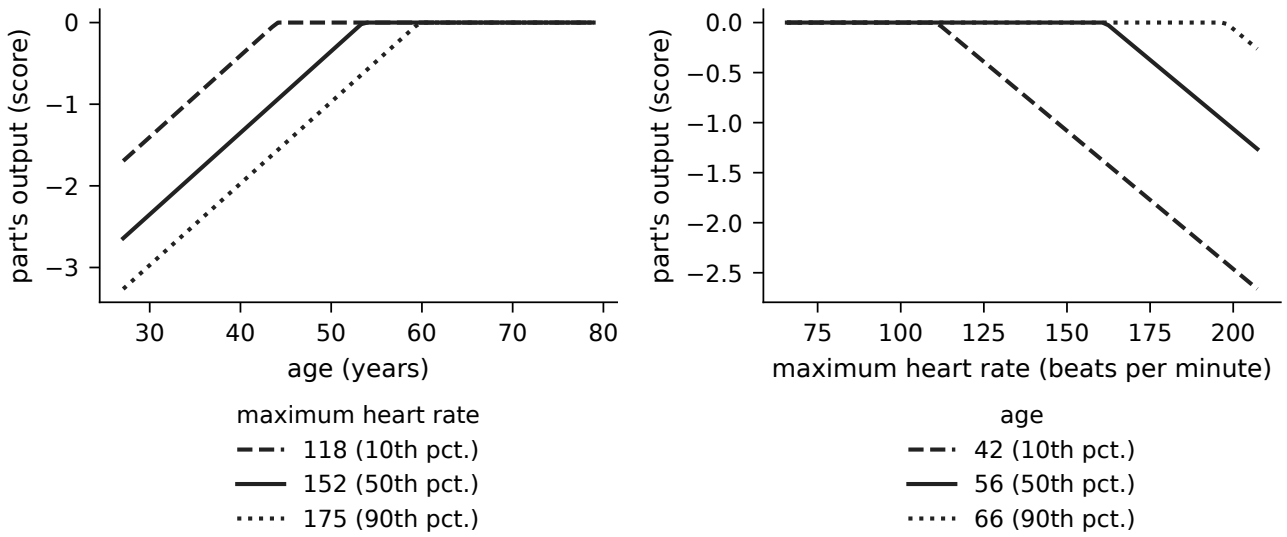


Figure 313: Age and heart rate gate: Slices of the same map: the part's own output against each input at three values of the other, so the tilted boundary shows as a shifting hinge.

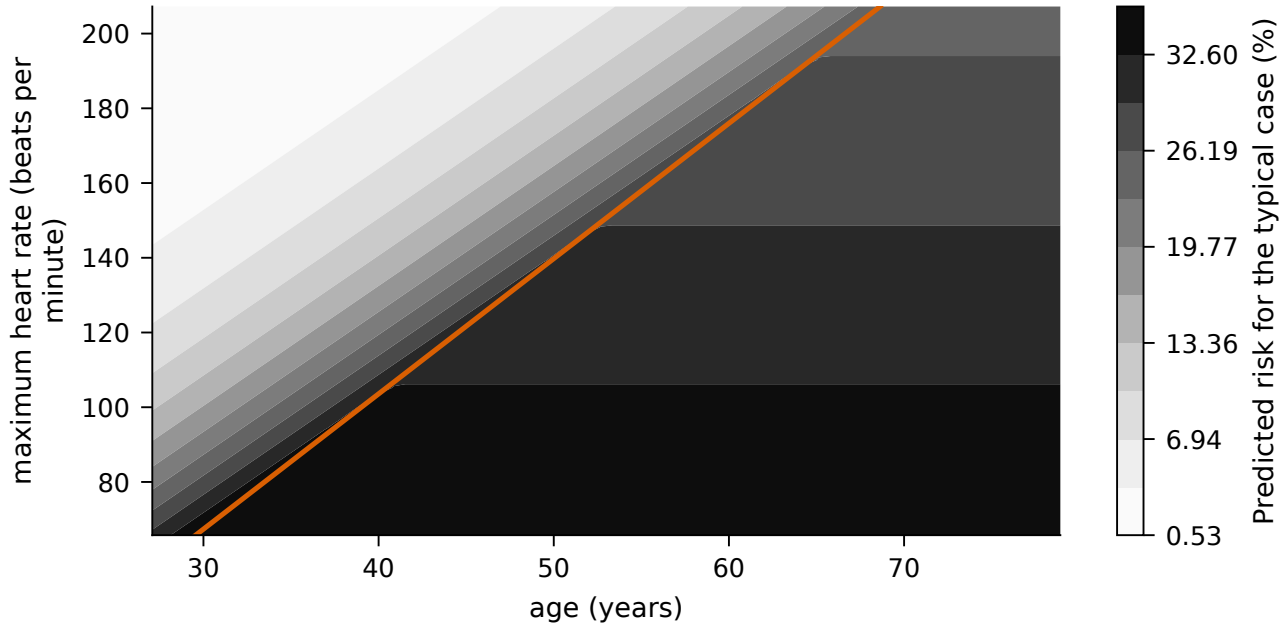


Figure 314: Age and heart rate gate: The same map on the final scale: the predicted value for the typical case, with the switch-on line (orange).

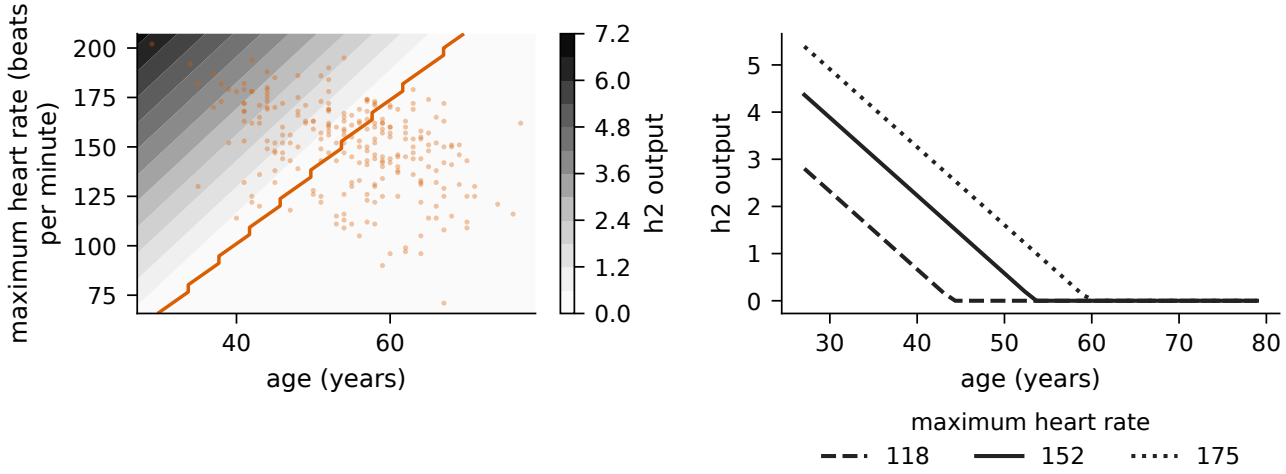


Figure 315: Age and heart rate gate: This part's own output, unit h2 (the output of 'Age and heart rate gate'), on its own scale (its activation), with every other input at the typical case; it is active for 80 of 242. Left: its output over its two main inputs (orange line: where it switches on; dots: the data); right: slices at three values of the second input.

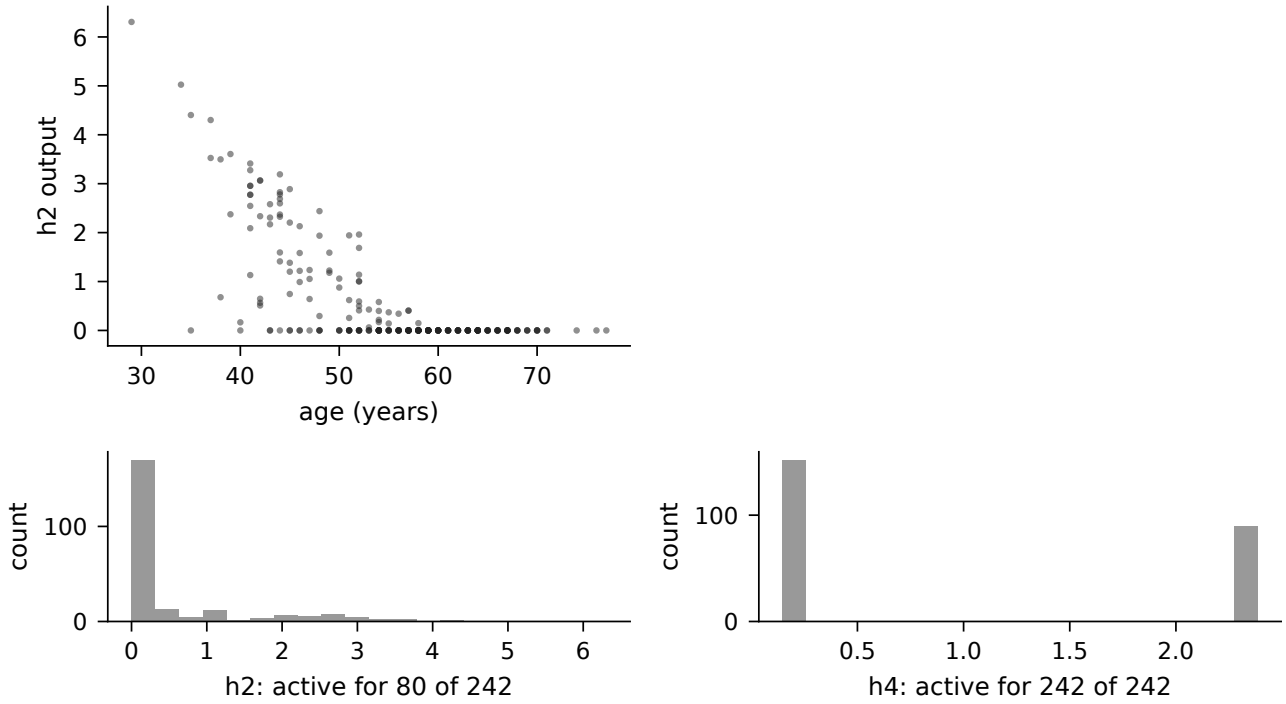


Figure 316: Age and heart rate gate: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.2.

7.2.3 ST depression hinges

Part of: The whole model (12.2.3).
Figure 317 shows what feeds this part; Figure 318 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).



Figure 317: ST depression hinges: what feeds it, one level down.

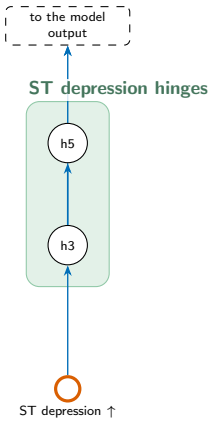


Figure 318: ST depression hinges: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part raises the predicted risk with ST depression on exercise, slowly at first, then faster, adding risk only once it rises above about 0.8 mm. [C40, C41]

Evidence. It acts on 121 of 242 training patients, 77 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 89% without it and about 91% with it. [C42] Through it, more ST depression raises the risk slowly at first, then faster, adding risk only once it rises above about 0.8 mm: for a typical patient from about 29% at 0 mm to about 48% at 2.8 mm. [C40, C41] Its formula reproduces the part exactly on all 242 training patients. [C43]
Evidence plots: Figure 319, Figure 320, Figure 321, Figure 322.
Figure 320 shows this part's own output, unit h3: it switches on at about 0.7582 mm of ST depression on exercise, and is zero below.
Figure 321 shows this part's own output, unit h5: it switches on at about 1.666 mm of ST depression on exercise, and is zero below.

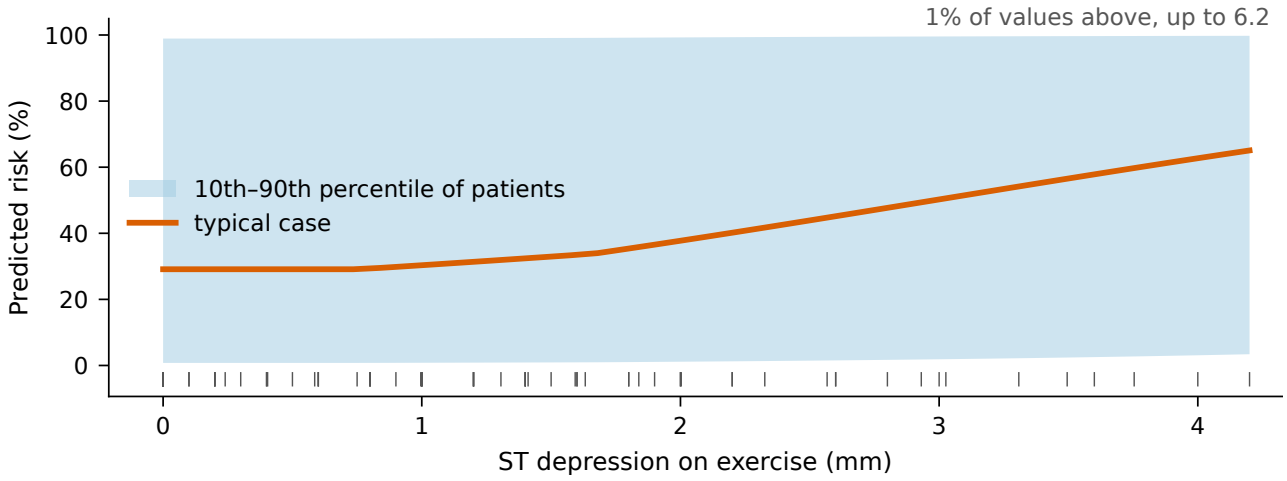


Figure 319: ST depression hinges: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual patients (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

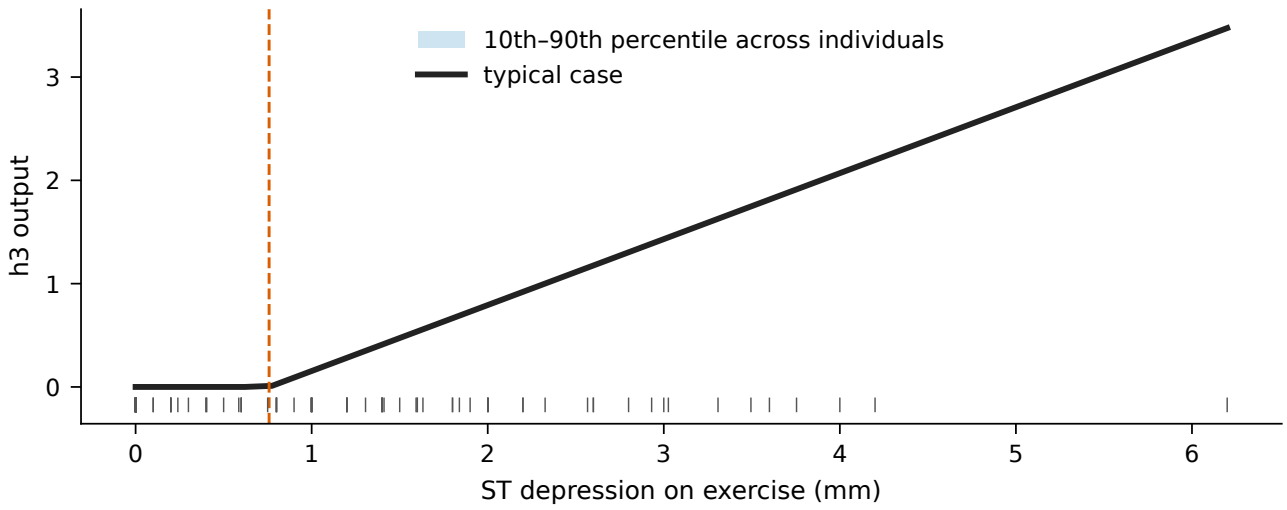


Figure 320: ST depression hinges: This part's own output, unit h3 (the output of 'ST depression hinges'), on its own scale (its activation), with every other input at the typical case; it is active for 121 of 242. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

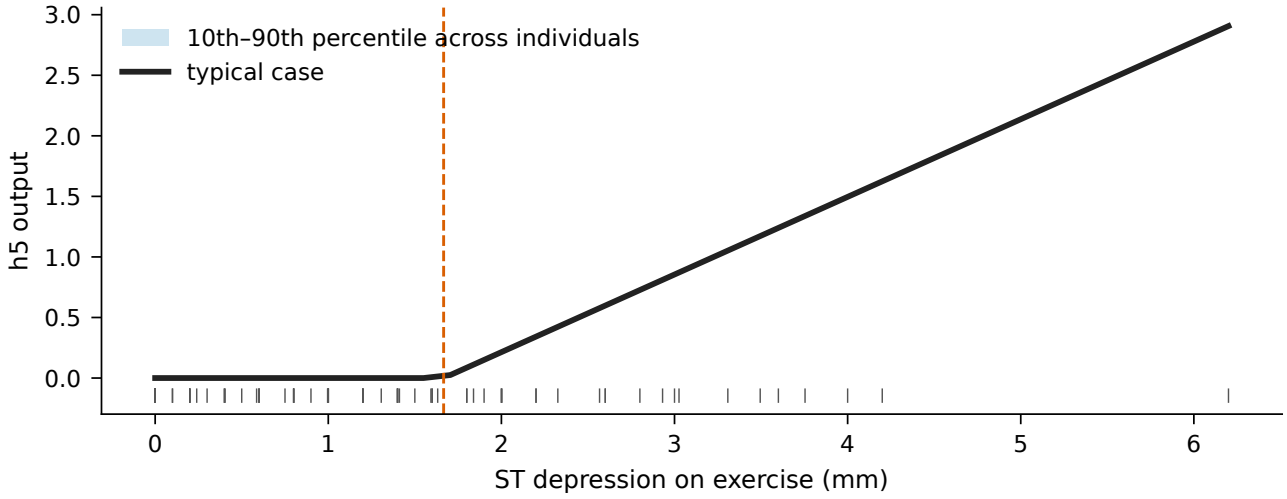


Figure 321: ST depression hinges: This part's own output, unit h5 (the output of 'ST depression hinges'), on its own scale (its activation), with every other input at the typical case; it is active for 58 of 242. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

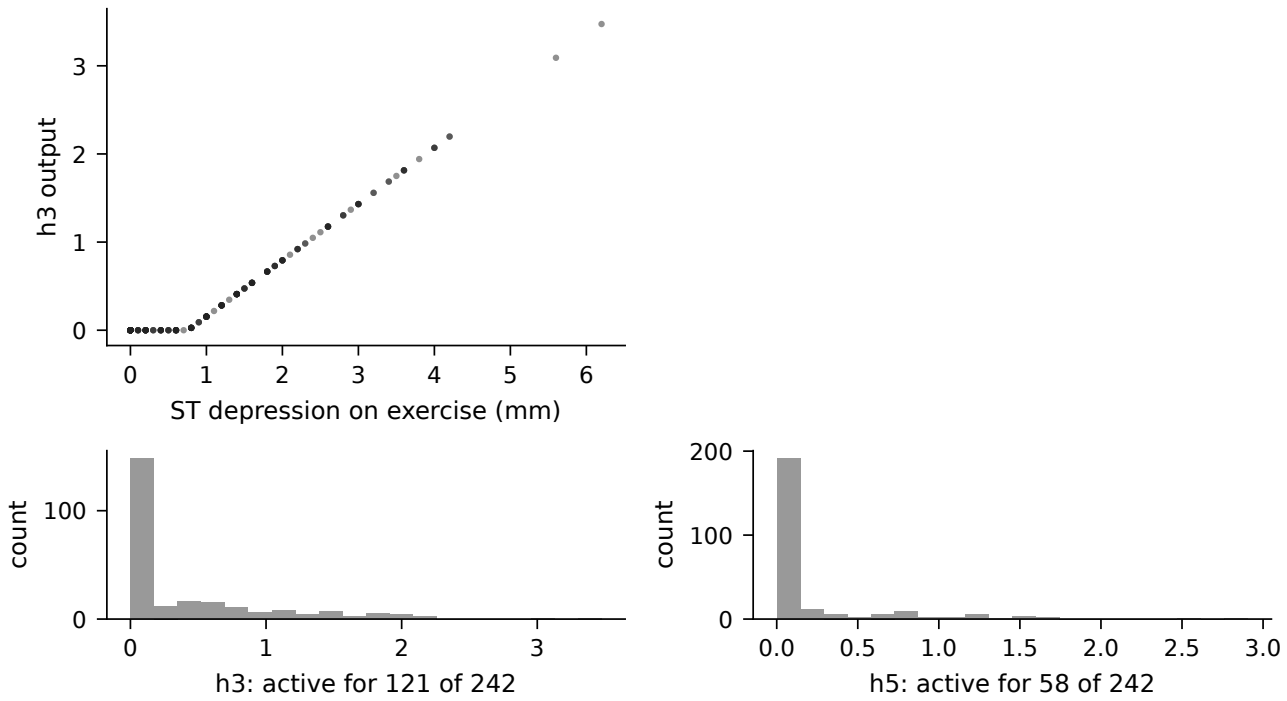


Figure 322: ST depression hinges: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.2.

7.2.4 Flat ST segment with cholesterol

Part of: The whole model (12.2.3).
Figure 323 shows what feeds this part; Figure 324 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

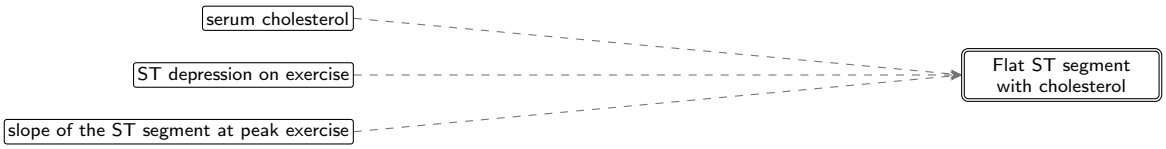


Figure 323: Flat ST segment with cholesterol: what feeds it, one level down.

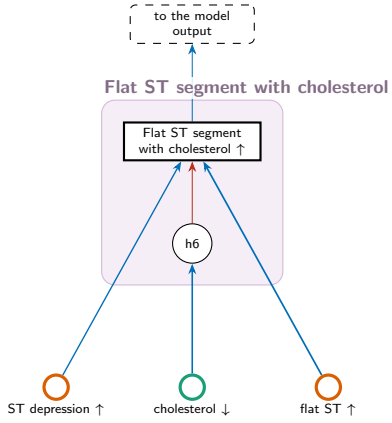


Figure 324: Flat ST segment with cholesterol: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part raises the predicted risk mainly for patients with a flat ST segment, and makes ST depression on exercise count more for them, rising slowly at first, then faster. [C44, C48]

Evidence. It acts on 117 of 242 training patients, 78 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 86% without it and about 90% with it. [C52] For patients whose ST segment is not flat, it switches on when ST depression on exercise is above about 2.23 mm (for cholesterol from about 126 to 186 mg/dL), or when ST depression minus about 0.012 × the cholesterol in mg/dL is above about 0.011 (for cholesterol from about 186 to 417 mg/dL); 117 of 242 training patients are on, and 67% of them had heart disease, against 23% of the others. [C55] Through it, a flat ST segment raises a typical patient's predicted risk from about 29% to about 34%; for the typical patient it acts on, more ST depression raises it in an uneven way from about 87% to about 92%, and higher cholesterol lowers it in an uneven way from about 91% to about 88%. [C44, C45, C46] Across the whole model, more ST depression raises the risk slowly at first, then faster, from about 75% to about 91% with a flat ST segment, against about 29% to about 48% otherwise. [C48] Its formula reproduces the part exactly on all 242 training patients. [C53]
Evidence plots: Figure 325, Figure 326, Figure 327, Figure 328, Figure 329, Figure 330, Figure 331, Figure 332.

Worked example: reading this part. Figure 328 maps the part's own output over its main inputs. In the grey region it is switched off and adds nothing; past the black line it switches on, and the colour shows how much it adds to the score. The line is the rule in the claim (C55): it switches on when ST depression on exercise above about 2.23 (for serum cholesterol from about 126 to 186); ST depression on exercise minus about 0.012 × serum cholesterol above about 0.011 (for serum cholesterol from about 186 to 417). 117 of 242 training patients lie on the 'on' side, and 67% of them had heart disease, against 23% of the others: the filled points crowd on the 'on' side. Figure 329 cuts the map into slices: the hinge moves as the other input changes, which is what a tilted line means. Figure 330 shows what this does to the final prediction for the typical case.
Figure 331 shows this part's own output, unit h1: it switches on at about 2.935 mm of ST depression on exercise, and is zero below.

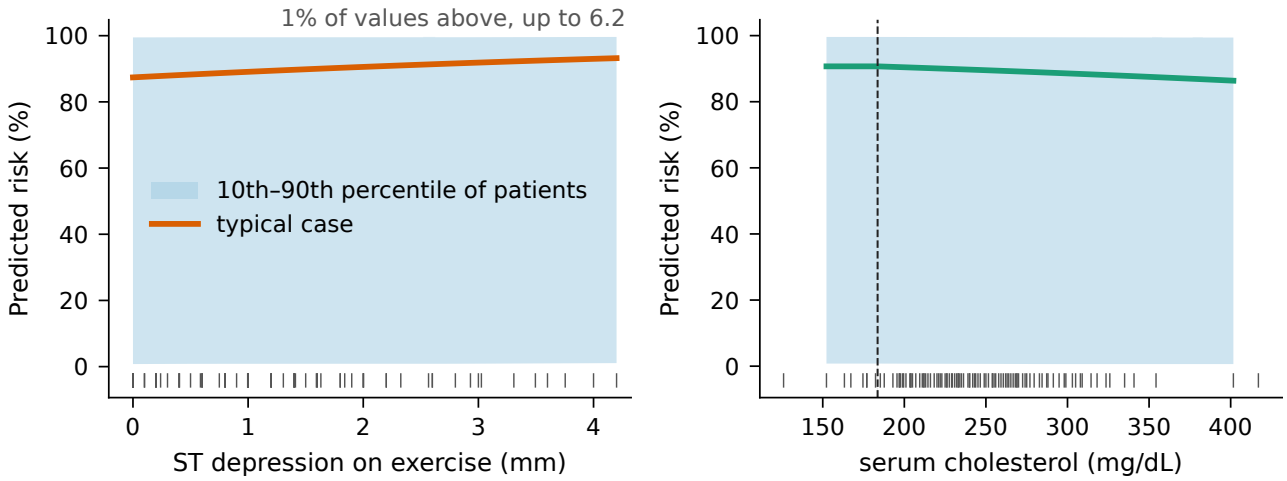


Figure 325: Flat ST segment with cholesterol: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual patients (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

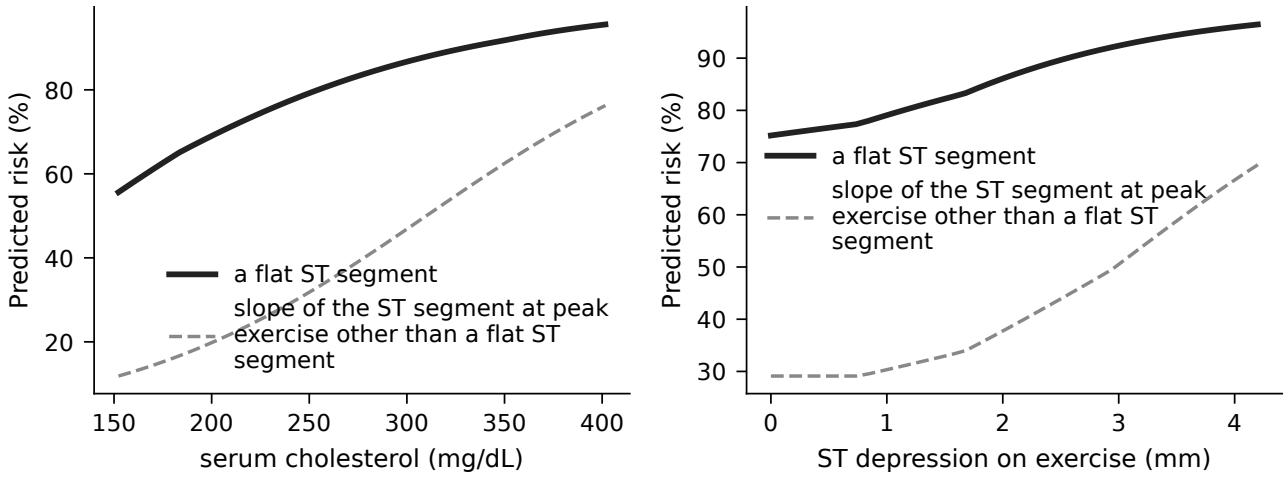


Figure 326: Flat ST segment with cholesterol: How the effect differs between groups: one curve per group, for the typical case of each group.

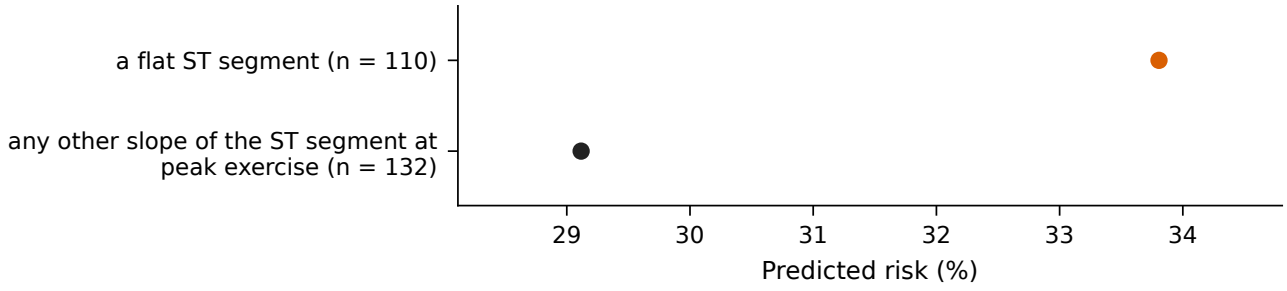


Figure 327: Flat ST segment with cholesterol: The predicted value for the typical case at each level, through this part (n = training patients; grey: fewer than 20).

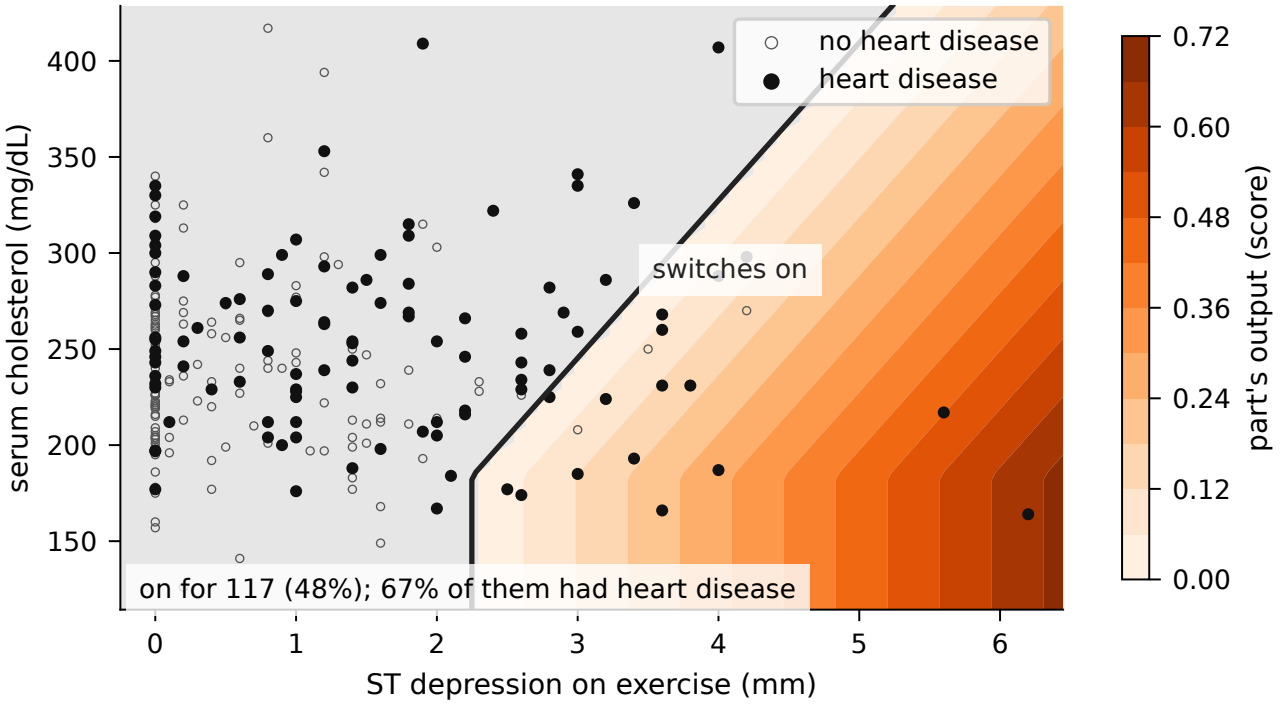


Figure 328: Flat ST segment with cholesterol: The part's own output over its two numeric inputs, other inputs at the typical case: colour where it is on, flat grey where it is switched off, the black line where it switches on; training patients as hollow (no event) and filled (event) points, so the part can be seen catching the events (the predicted-chance overlay was harder to read).

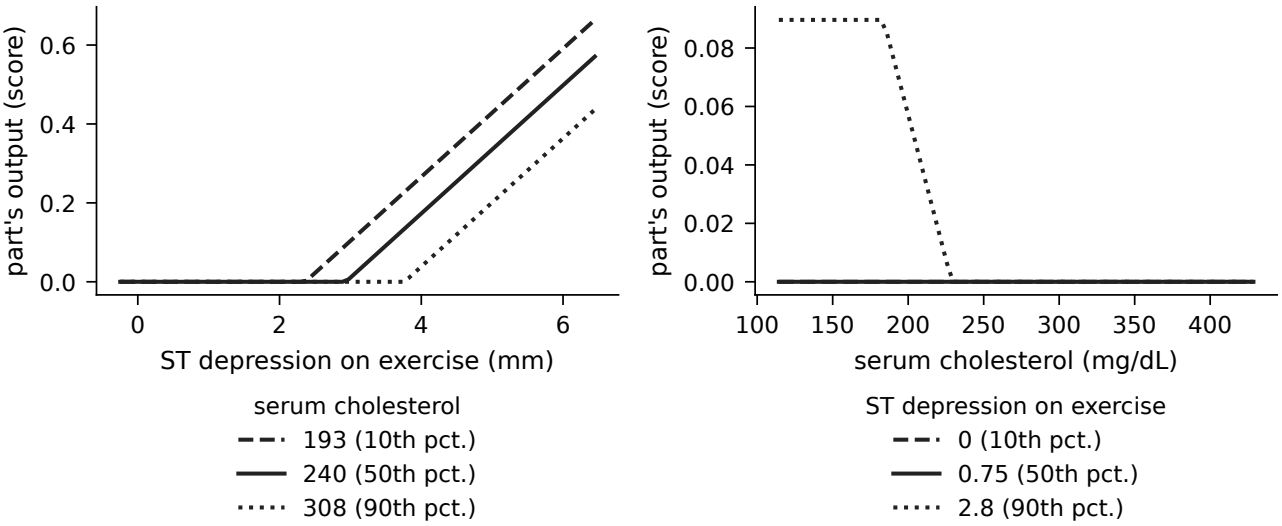


Figure 329: Flat ST segment with cholesterol: Slices of the same map: the part's own output against each input at three values of the other, so the tilted boundary shows as a shifting hinge.

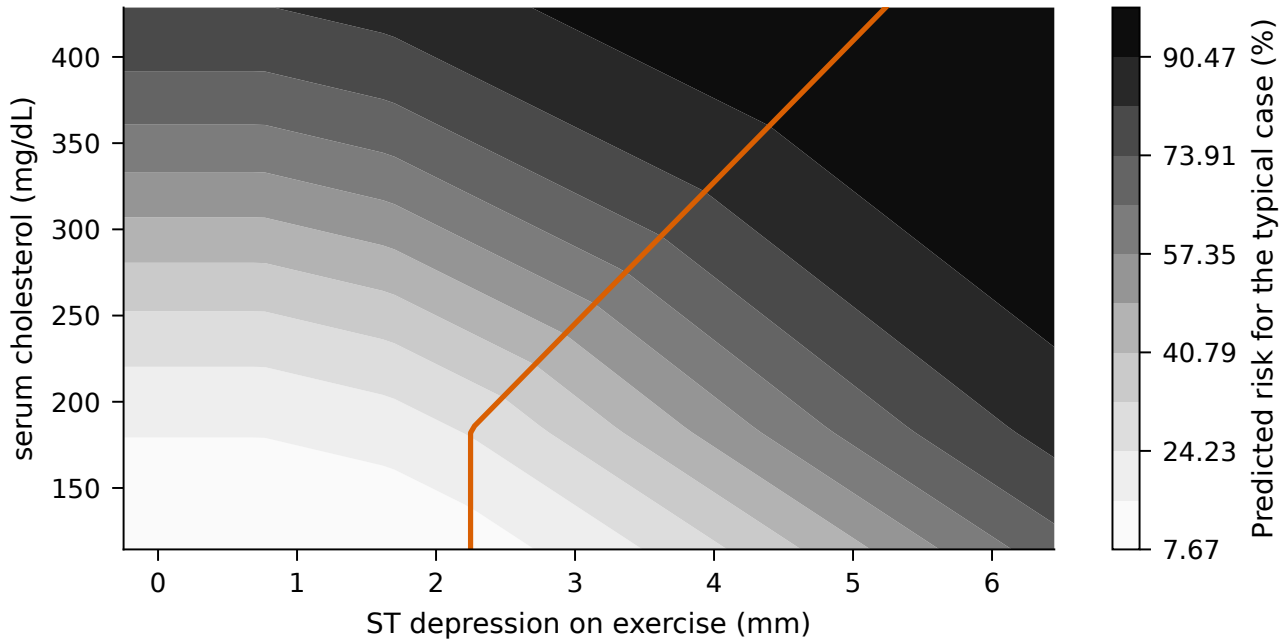


Figure 330: Flat ST segment with cholesterol: The same map on the final scale: the predicted value for the typical case, with the switch-on line (orange).

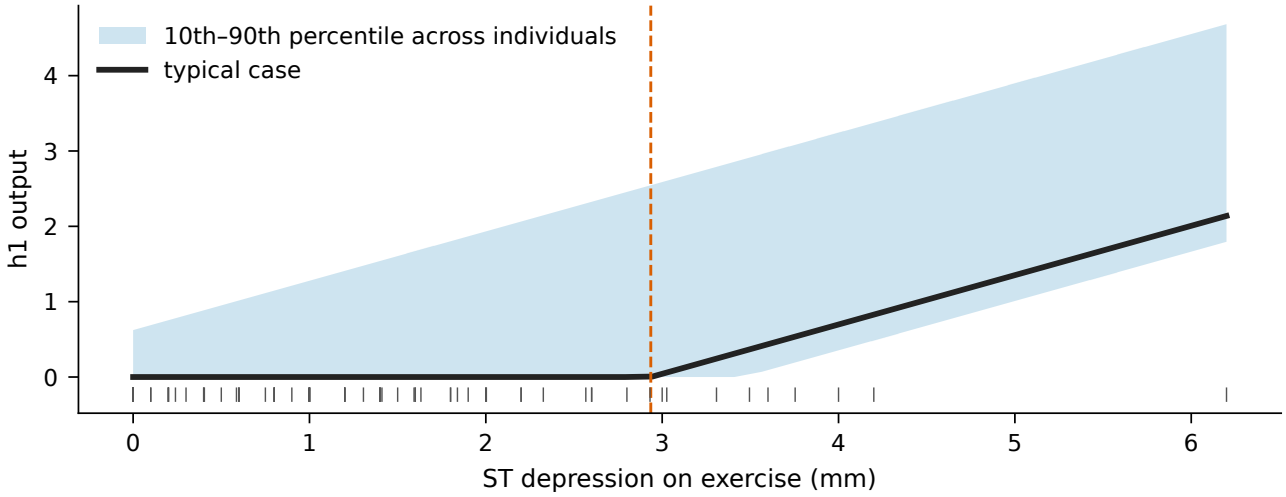


Figure 331: Flat ST segment with cholesterol: This part's own output, unit h1 (the output of 'Flat ST segment with cholesterol'), on its own scale (its activation), with every other input at the typical case; it is active for 117 of 242. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

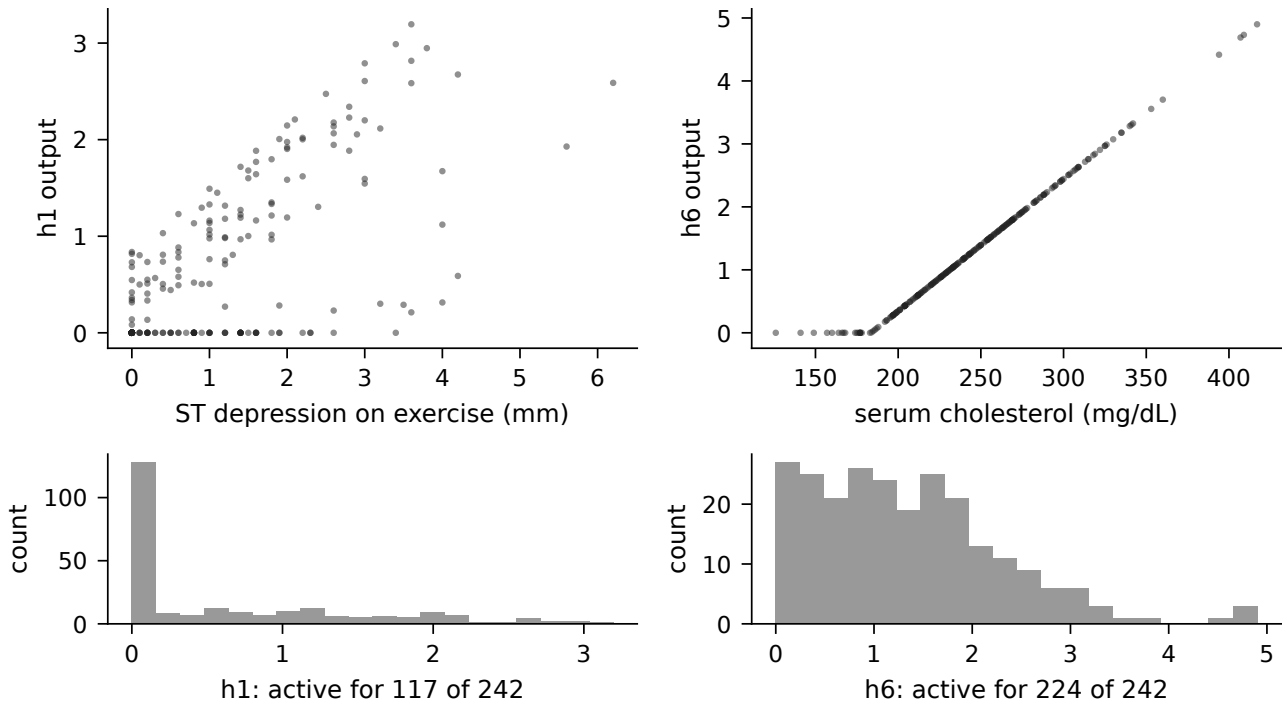


Figure 332: Flat ST segment with cholesterol: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.2.

7.3 Not supported by the model

- C49: interaction claim about serum cholesterol and ST depression during exercise (compared with rest) in part ann_32: not supported by the model (the group difference is below 0.1 log-odds).

Technical detail: heart-s2-c1

The record behind each part: the original hypothesis, the closed form, and every claim with its estimate (log-odds and odds ratio), range across patients, n and test check.

T.1 The whole model

Annotation ann_34 (heart_disease_logit). Reads: age, sex, chest pain type, resting blood pressure, serum cholesterol, fasting blood sugar, resting ECG, maximum heart rate reached during the exercise test, exercise-induced angina, ST depression during exercise (compared with rest), slope of the ST segment at peak exercise, number of major vessels coloured by fluoroscopy, thallium stress test result.

Original hypothesis (unchanged). Root: p = sigmoid(z), z = -0.227 + LINEAR CORE - 0.605*age_hr_reserve_gate + 0.248*st_slope_chol_gate + 0.378*oldpeak_hinges[1946] + 0.419*oldpeak_hinges[3284]. LINEAR CORE (16 direct input->output edges, z-scored inputs): ca +1.653, cp:4 +0.986, sex:0 -0.742, chol +0.613, slope:2 +0.495, trestbps +0.465, slope:1 -0.453, sex:1 +0.447, cp:2 +0.397, fbs:1 -0.308, thal:7 +0.279, exang:1 +0.278, thal:6 -0.278, cp:1 -0.180, thalach -0.079, restecg:1 +0.066. Eight inputs are unused (cp:3, fbs:0, restecg:0, restecg:2, exang:0, slope:3, thal:3, thal:4; zero-effect entries of this root). Age and oldpeak have no direct edge; they act only through the children. COMPOSITION: purely additive in the logit. The three children add nonlinear corrections to the linear core: (1) age_hr_reserve_gate, a protective ReLU on when max heart rate is high for age and there is no reversible thallium defect; (2) st_slope_chol_gate, a flat-ST-slope indicator scaled up by oldpeak and down by cholesterol; (3) oldpeak_hinges, a convex two-kink ST-depression dose-response. The children share no hidden nodes; they are coupled only through shared inputs (thal:7, thalach, chol and slope:2 also have direct edges, oldpeak feeds (2) and (3)). EVIDENCE (train, n=242): full model AUC 0.9432, acc 0.880 (compute_performance). Linear core alone AUC 0.930; logit SD 3.24 (core) vs 0.95 (all nonlinear readouts). Zero-ablating one child at a time: hr gate 0.937, flat-slope gate 0.942, oldpeak hinges 0.942. Replacing all hidden readouts with one least-squares linear fit in their six inputs gives 0.937; an L2 logistic regression on all 26 inputs gives 0.936. Mean |SHAP| (train, all 242): ca 0.152, cp:4 0.111, sex:0 0.071, slope:2 0.066, slope:1 0.049, thal:7 0.048, chol 0.044. Interpretation: close to logistic regression plus a protective heart-rate-reserve-for-age gate (blocked by a reversible defect) and two small oldpeak/flat-slope corrections. Counter-directional linear signs (cp:2 +0.40, fbs:1 -0.31, negative chol inside the flat-slope gate) are reported, not explained. Retrained copy of the 10-seed heart-s2 genome (same NEAT key 9083): same mechanisms, weights within about 0.03.

Closed form. Read through its parts (their formulas are in their own sections).

Split inputs. To annotate the parts, the explanation split these inputs into copies (a refactoring step that does not change what the model computes); the diagrams draw each input once: cholesterol (2 copies); ST depression (2 copies); flat ST (2 copies); reversible defect (3 copies); max heart rate (2 copies).

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C01	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (whole model) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	29% to 94%	3.71	40.7	3.71 to 3.71	242	unclear; AUC gain 0.032 [-0.044, 0.111]	yes
C02	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	29% to 77%	2.14	8.54	1.93 to 2.43	115, 110	disagrees; AUC gain -0.013 [-0.060, 0.026]	yes
C03	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment (direct contribution only, not through any part) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	29% to 73%	1.90	6.70	1.90 to 1.90	115, 110	disagrees; AUC gain -0.014 [-0.055, 0.021]	yes
C04	direction: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L) (direct contribution only, not through any part) *shape: steady; 100% of patients same; verified*	18% to 49%	1.46	4.32	1.46 to 1.46	242	disagrees; AUC gain -0.030 [-0.068, -0.004]	yes
C05	direction: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L) (whole model) *shape: steady; 100% of patients same; verified*	18% to 49%	1.35	3.87	1.24 to 1.46	242	disagrees; AUC gain -0.030 [-0.068, -0.004]	yes
C06	direction: chest pain type: no chest pain (asymptomatic) to typical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	29% to 3%	-2.61	0.07	-2.61 to -2.61	108, 22	agrees; AUC gain 0.033 [0.005, 0.072]	yes
C07	direction: sex: male to female (whole model) *shape: categorical; highest to lowest: male, female*	29% to 3%	-2.51	0.08	-2.51 to -2.51	159, 83	unclear; AUC gain 0.030 [-0.011, 0.079]	yes
C08	direction: resting blood pressure: 110 mm Hg to 155 mm Hg (whole model) *shape: steady; 100% of patients same; verified*	20% to 44%	1.15	3.16	1.15 to 1.15	242	disagrees; AUC gain -0.003 [-0.022, 0.014]	yes
C09	direction: chest pain type: no chest pain (asymptomatic) to non-anginal chest pain (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	29% to 5%	-1.98	0.14	-1.98 to -1.98	108, 70	agrees; AUC gain 0.033 [0.005, 0.072]	yes
C10	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a downsloping ST segment (direct contribution only, not through any part) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	29% to 50%	0.91	2.48	0.91 to 0.91	115, 17 (small n)	disagrees; AUC gain -0.014 [-0.055, 0.021]	yes
C11	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a downsloping ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	29% to 50%	0.91	2.48	0.91 to 0.91	115, 17 (small n)	disagrees; AUC gain -0.013 [-0.060, 0.026]	yes
C12	direction: ST depression on exercise: 0.0 mm to 2.8 mm (whole model) *shape: accelerating; 100% of patients same; verified*	29% to 48%	1.01	2.74	0.80 to 1.25	242	unclear; AUC gain 0.005 [-0.009, 0.022]	yes
C13	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan to a fixed defect on the thallium scan (direct contribution only, not through any part) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan, a fixed defect on the thallium scan*	29% to 11%	-1.15	0.32	-1.15 to -1.15	137, 15 (small n)	unclear; AUC gain 0.002 [-0.016, 0.019]	yes
C14	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan to a fixed defect on the thallium scan (whole model) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan, a fixed defect on the thallium scan*	29% to 11%	-1.15	0.32	-1.15 to -1.15	137, 15 (small n)	unclear; AUC gain 0.015 [-0.01, 0.044]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C15	direction: age: 42 years to 66 years (whole model) *shape: hinge at 54 years; 86% of patients same; mixed (but step for men and patients with a reversible defect on the thallium scan); matches the fold at 56 years*	12% to 29%	0.74	2.11	0.00 to 1.71	242	disagrees; AUC gain -0.006 [-0.022, 0.008]	yes
C16	direction: resting ECG: resting ECG other than an ST-T wave abnormality on the resting ECG to an ST-T wave abnormality on the resting ECG (whole model) *shape: categorical; highest to lowest: an ST-T wave abnormality on the resting ECG, resting ECG other than an ST-T wave abnormality on the resting ECG*	29% to 46%	0.73	2.08	0.73 to 0.73	240, 2 (small n)	unclear; AUC gain 0.000 [0.000, 0.000]	yes
C17	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	29% to 14%	-0.94	0.39	-0.94 to -0.94	108, 42	agrees; AUC gain 0.033 [0.005, 0.072]	yes
C18	direction: fasting blood sugar: fasting blood sugar of 120 mg/dL or less to fasting blood sugar above 120 mg/dL (whole model) *shape: categorical; highest to lowest: fasting blood sugar of 120 mg/dL or less, fasting blood sugar above 120 mg/dL*	29% to 14%	-0.90	0.41	-0.90 to -0.90	209, 33	unclear; AUC gain 0.002 [-0.012, 0.020]	yes
C19	direction: exercise-induced angina: no chest pain on exercise to chest pain brought on by exercise (whole model) *shape: categorical; highest to lowest: chest pain brought on by exercise, no chest pain on exercise*	29% to 43%	0.59	1.80	0.59 to 0.59	160, 82	unclear; AUC gain 0.005 [-0.006, 0.022]	yes
C20	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan to a reversible defect on the thallium scan (direct contribution only, not through any part) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan, a fixed defect on the thallium scan*	29% to 42%	0.58	1.78	0.58 to 0.58	137, 90	unclear; AUC gain 0.002 [-0.016, 0.019]	yes
C21	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan to a reversible defect on the thallium scan (whole model) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan, a fixed defect on the thallium scan*	29% to 42%	0.99	2.68	0.58 to 2.14	137, 90	unclear; AUC gain 0.015 [-0.01, 0.044]	yes
C22	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (whole model) *shape: accelerating; 54% of patients same; fallback*	32% to 21%	-0.63	0.53	-1.77 to -0.20	242	unclear; AUC gain 0.01 [-0.001, 0.032]	yes
C23	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (direct contribution only, not through any part) *shape: steady; 100% of patients same; verified*	32% to 28%	-0.20	0.82	-0.20 to -0.20	242	unclear; AUC gain 0.003 [-0.004, 0.016]	yes
C24	magnitude: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels, typical patient *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	29% to 94%	3.71	40.7		242	unclear; AUC gain 0.032 [-0.044, 0.111]	yes
C25	magnitude: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment, typical patient *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	29% to 77%	2.12	8.33		115, 110	disagrees; AUC gain -0.013 [-0.060, 0.026]	yes
C26	magnitude: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L), typical patient *shape: steady; 100% of patients same; verified*	18% to 49%	1.46	4.32		242	disagrees; AUC gain -0.030 [-0.068, -0.004]	yes
C27	magnitude: chest pain type: no chest pain (asymptomatic) to typical angina, typical patient *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	29% to 3%	-2.61	0.07		108, 22	agrees; AUC gain 0.033 [0.005, 0.072]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C28	magnitude: sex: male to female, typical patient *shape: categorical; highest to lowest: male, female*	29% to 3%	-2.51	0.08		159, 83	unclear; AUC gain 0.030 [-0.011, 0.079]	yes
C29	structure: made of age_hr_reserve_gate, st_slope_chol_gate, oldpeak_hinges and inputs sex, cp, trestbps, chol, fbs, restecg, thalach, exang, slope, ca, thal; added					242	train only	yes

T.2 Age and heart rate gate

Annotation ann_31 (age_hr_reserve_gate). Reads: age, maximum heart rate reached during the exercise test, thallium stress test result.

Original hypothesis (unchanged). Heart-rate-reserve gate (protective). 1718 = ReLU(-0.809 - 1.502*age + 1.023*thalach - 0.772*thal:7 - 0.730*R3195), relay R3195 = ReLU(0.979 + 1.081*thal:7); read out -0.605 into the logit. Hypothesis: R3195 is an always-on affine relay of thal:7, so the gate is a single ReLU, 1718 = ReLU(-1.524 - 1.502*age + 1.023*thalach - 1.561*thal:7), that opens when the achieved max heart rate is high for the patient's age and there is no reversible thallium defect, and then lowers the predicted risk. EVIDENCE (train, n=242): R3195 is active for 100% (pre-activation >= 0.148), so the reduction is exact on the data. The gate is active for 33% (51% without a reversible defect, 3% with one). Disease rate 11% when active vs 60% when inactive. Mean |sensitivity| age 0.50 > thalach 0.34 > thal:7 copies 0.26/0.26. Zero-ablating its readout lowers train AUC 0.943 -> 0.937, the largest single nonlinear effect; a linear refit in its inputs recovers only to 0.939. Supported. Same mechanism and nearly the same weights as the 10-seed heart-s2 record (its generation-38 parent), re-derived here from this genome's own weights.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 8.88178e-16).

$$\begin{aligned} y_0 = \operatorname{relu}\big(&-0.809 + 1.02 x_{thalach_b} - 1.5 x_{age} \\ &- 0.772 x_{thal:7_b} - 0.73 \operatorname{relu}\big(0.979 + 1.08 x_{thal:7_c}\big)\big) \end{aligned}$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C30	direction: age: 42 years to 66 years (through this part) *shape: hinge at 56 years; 86% of patients same; mixed (but step for patients with a reversible defect on the thallium scan); one point with C37 (median fold 56 years, typical patient 54 years, 10th-90th percentile 45 years to 61 years, wide)*	12% to 29%	0.74	2.11	0.00 to 1.71	242	disagrees; AUC gain -0.006 [-0.022, 0.008]	yes
C31	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (through this part) *shape: hinge at 165 beats per minute; 68% of patients same; fallback; one point with C36 (median fold 165 beats per minute, typical patient 162 beats per minute, 10th-90th percentile 125 beats per minute to 195 beats per minute, wide)*	29% to 22%	-0.43	0.65	-1.57 to 0.00	242	unclear; AUC gain 0.009 [-0.002, 0.028]	yes
C32	direction: thallium stress test result: thallium stress test result other than a reversible defect on the thallium scan to a reversible defect on the thallium scan (through this part) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result other than a reversible defect on the thallium scan*	1% to 4% (typical patient this part acts on)	0.41	1.50	0.00 to 1.56	152, 90	unclear; AUC gain 0.01 [-0.003, 0.029]	yes
C33	interaction: effect of age (42 years to 66 years) by thallium stress test result: a reversible defect on the thallium scan minus thallium stress test result other than a reversible defect on the thallium scan; stronger in thallium stress test result other than a reversible defect on the thallium scan *shape: hinge at 54 years; 97% of patients same; verified*	a reversible defect on the thallium scan: 42% to 42%; thallium stress test result other than a reversible defect on the thallium scan: 12% to 29%	-1.17	0.31		90, 152	agrees; AUC gain 0.01 [-0.003, 0.029]	yes
C34	interaction: effect of age (42 years to 66 years) by maximum heart rate: maximum heart rate above 152 beats per minute minus maximum heart rate at or below 152 beats per minute; stronger in maximum heart rate above 152 beats per minute *shape: hinge at 58 years; 73% of patients same; mixed (but step for patients with a reversible defect on the thallium scan)*	maximum heart rate above 152 beats per minute: 8% to 28%; maximum heart rate at or below 152 beats per minute: 18% to 30%	0.83	2.30		121, 121	agrees; AUC gain 0.009 [-0.002, 0.028]	yes
C35	interaction: effect of maximum heart rate (118 beats per minute to 175 beats per minute) by thallium stress test result: a reversible defect on the thallium scan minus thallium stress test result other than a reversible defect on the thallium scan; stronger in thallium stress test result other than a reversible defect on the thallium scan *shape: accelerating; 79% of patients same; fallback*	a reversible defect on the thallium scan: 45% to 40%; thallium stress test result other than a reversible defect on the thallium scan: 32% to 21%	0.67	1.95		90, 152	agrees; AUC gain 0.01 [-0.003, 0.029]	yes
C36	threshold: maximum heart rate: fold at about 165 beats per minute (starts) for patients with a fixed defect on the thallium scan *shape: hinge at 165 beats per minute; 77% of patients same; fallback; one point with C36 (median fold 165 beats per minute, typical patient 164 beats per minute, 10th-90th percentile 125 beats per minute to 195 beats per minute, wide)*	91% to 87% (typical patient with a fold (patients with a fixed defect on the thallium scan))	-0.56	0.57	-1.57 to 0.00	15, 3 (small n)	unclear; AUC gain 0.009 [-0.002, 0.028]	yes
C37	threshold: age: fold at about 56 years (stops) for patients with a normal thallium scan *shape: hinge at 56 years; 99% of patients same; verified; risk saturates near 0%; one point with C37 (median fold 56 years, typical patient 56 years, 10th-90th percentile 45 years to 61 years, wide)*	<1% to <1% (typical patient with a fold (patients with a normal thallium scan))	1.35	3.84	0.44 to 1.98	135, 73	disagrees; AUC gain -0.006 [-0.022, 0.008]	yes
C38	subgroup: who this part acts on: 80 of 242 patients	4% to 1% (typical patient this part acts on (risk without the part, then with it))	-1.06	0.35	-1.99 to -0.20	80, 162	unclear; AUC gain 0.017 [-0.001, 0.052]	yes
C39	structure: its formula reproduces it on 242 training patients (largest difference 8.88178e-16)		9e-16			242	train only	yes
C54	boundary: boundary		1.00			242	train only	yes

T.3 ST depression hinges

Annotation ann_33 (oldpeak_hinges). Reads: ST depression during exercise (compared with rest).

Original hypothesis (unchanged). Oldpeak hinge pair (raises risk). 1946 = ReLU(0.177 + 0.757*oldpeak), 3284 = ReLU(-0.582 + 1.004*1946); read out +0.378 and +0.419. Hypothesis: a convex piecewise-linear dose-response in ST depression alone: no effect for the lowest oldpeak values, a moderate rise from z-oldpeak -0.23, and a steeper rise (slope about doubles) above z-oldpeak 0.53. EVIDENCE (train, n=242): 1946 is active for 50% (disease rate 64%), 3284 for 24% (disease rate 79%). The readout 0.378*1946 + 0.419*3284 has slope 0, 0.286 and 0.605 per z-unit in the three pieces. Removing both readouts lowers train AUC 0.943 -> 0.942; a straight-line refit in oldpeak gives 0.9425. Supported: a small convex ST-depression term. Oldpeak has no direct edge; it acts only here and in the flat-slope gate.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 0.0).

$$y_0 = \mathrm{relu}\big(0.177 + 0.757 x_{oldpeak_b}\big)$$
$$y_1 = \mathrm{relu}\big(-0.582 + 1.0\,\mathrm{relu}\big(0.177 + 0.757 x_{oldpeak_b}\big)\big)$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C40	direction: ST depression on exercise: 0.0 mm to 2.8 mm (through this part) *shape: accelerating; 100% of patients same; verified*	29% to 48%	0.80	2.22	0.80 to 0.80	242	unclear; AUC gain 0.005 [-0.008, 0.020]	yes
C41	threshold: ST depression on exercise: fold at about 0.8 mm (starts) *shape: accelerating; 100% of patients same; verified*	29% to 57% (typical patient with a fold)	1.15	3.16	1.15 to 1.15	242, 121	unclear; AUC gain 0.005 [-0.008, 0.020]	yes
C42	subgroup: who this part acts on: 121 of 242 patients	89% to 91% (typical patient this part acts on (risk without the part, then with it))	0.43	1.54	0.03 to 1.15	121, 121	unclear; AUC gain 0.005 [-0.008, 0.020]	yes
C43	structure: its formula reproduces it on 242 training patients (largest difference 0.0)		0.00			242	train only	yes

T.4 Flat ST segment with cholesterol

Annotation ann_32 (st_slope_chol_gate). Reads: serum cholesterol, ST depression during exercise (compared with rest), slope of the ST segment at peak exercise.

Original hypothesis (unchanged). Flat-ST-slope gate (raises risk). 1476 = ReLU(0.255 + 1.150*slope:2 + 0.776*oldpeak - 0.380*C3464), C3464 = ReLU(1.302 + 1.011*chol); read out +0.248. Hypothesis: a one-sided detector that is on for patients with a flat peak-exercise ST segment, its height raised by ST depression and lowered by cholesterol through an almost-always-active chol relay. EVIDENCE (train, n=242): C3464 is active for 93% (off only for z-chol < -1.29), so 1476 is nearly ReLU(-0.239 + 0.776*oldpeak + 1.150*slope:2 - 0.384*chol). Active for 98% of the 110 flat-slope patients and 7% of the others; corr with oldpeak 0.69. Disease rate 67% when active vs 23% when inactive. Mean |sensitivity| slope:2 0.56 > oldpeak 0.38 > chol 0.17. Zero-ablating its readout lowers train AUC 0.943 -> 0.942 (linear refit 0.943). Supported but small. The negative chol term partly offsets chol's direct +0.613 edge for flat-slope patients; no clinical reading is offered.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 4.44089e-16).

$$y_0 = \operatorname{relu}\big(0.255 + 0.776x_{oldpeak_a} + 1.15x_{slope:2_b} - 0.38\operatorname{relu}\big(1.3 + 1.01x_{chol_b}\big)\big)$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C44	direction: slope of the ST segment at peak exercise: slope of the ST segment at peak exercise other than a flat ST segment to a flat ST segment (through this part) *shape: categorical; highest to lowest: a flat ST segment, slope of the ST segment at peak exercise other than a flat ST segment*	29% to 34%	0.24	1.27	0.03 to 0.53	132, 110	disagrees; AUC gain -0.005 [-0.019, 0.003]	yes
C45	direction: ST depression on exercise: 0.0 mm to 2.8 mm (through this part) *shape: steady; 43% of patients same; fallback*	87% to 92% (typical patient this part acts on)	0.21	1.24	0.00 to 0.45	242	disagrees; AUC gain -0.001 [-0.009, 0.006]	yes
C46	direction: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L) (through this part) *shape: hinge at 184 mg/dL (4.8 mmol/L); 46% of patients same; fallback; one point with C50 (median fold 184 mg/dL (4.8 mmol/L), typical patient 183 mg/dL (4.7 mmol/L), 10th-90th percentile 184 mg/dL (4.8 mmol/L) to 184 mg/dL (4.8 mmol/L))*	91% to 88% (typical patient this part acts on)	-0.11	0.90	-0.23 to 0.00	242	unclear; AUC gain 0.003 [-0.007, 0.019]	yes
C47	interaction: effect of serum cholesterol (193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L)) by slope of the ST segment at peak exercise: a flat ST segment minus slope of the ST segment at peak exercise other than a flat ST segment; stronger in slope of the ST segment at peak exercise other than a flat ST segment *shape: steady; 100% of patients same; verified*	a flat ST segment: 67% to 88%; slope of the ST segment at peak exercise other than a flat ST segment: 18% to 49%	-0.21	0.81		110, 132	agrees; AUC gain -0.005 [-0.019, 0.003]	yes
C48	interaction: effect of ST depression on exercise (0.0 mm to 2.8 mm) by slope of the ST segment at peak exercise: a flat ST segment minus slope of the ST segment at peak exercise other than a flat ST segment; stronger in a flat ST segment *shape: accelerating; 100% of patients same; verified*	a flat ST segment: 75% to 91%; slope of the ST segment at peak exercise other than a flat ST segment: 29% to 48%	0.42	1.53		110, 132	agrees; AUC gain -0.005 [-0.019, 0.003]	yes
C49	interaction: effect of serum cholesterol (193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L)) by ST depression on exercise: ST depression on exercise above 0.8 mm minus ST depression on exercise at or below 0.8 mm; stronger in ST depression on exercise at or below 0.8 mm *shape: steady; 100% of patients same; verified*	ST depression on exercise above 0.8 mm: 22% to 54%; ST depression on exercise at or below 0.8 mm: 18% to 49%	-0.10	0.91		121, 121	agrees; AUC gain -0.001 [-0.009, 0.006]	no
C50	threshold: serum cholesterol: fold at about 184 mg/dL (4.8 mmol/L) (starts) *shape: hinge at 184 mg/dL (4.8 mmol/L); 46% of patients same; fallback; one point with C50 (median fold 184 mg/dL (4.8 mmol/L), typical patient 183 mg/dL (4.7 mmol/L), 10th-90th percentile 184 mg/dL (4.8 mmol/L) to 184 mg/dL (4.8 mmol/L))*	91% to 89% (typical patient with a fold)	-0.26	0.77	-0.28 to -0.21	123, 114	unclear; AUC gain 0.003 [-0.007, 0.019]	yes
C51	threshold: ST depression on exercise: fold at about 2.8 mm (starts) *shape: hinge at 2.8 mm; 88% of patients same; fallback; one point with C51 (median fold 2.8 mm, typical patient 3.0 mm, 10th-90th percentile 1.5 mm to 3.7 mm, wide)*	27% to 29% (typical patient with a fold)	0.13	1.14	0.00 to 0.33	148, 23	disagrees; AUC gain -0.001 [-0.009, 0.006]	yes
C52	subgroup: who this part acts on: 117 of 242 patients	86% to 90% (typical patient this part acts on (risk without the part, then with it))	0.31	1.36	0.08 to 0.55	117, 125	disagrees; AUC gain -0.001 [-0.016, 0.011]	yes
C53	structure: its formula reproduces it on 242 training patients (largest difference 4.44089e-16)		4e-16			242	train only	yes
C55	boundary: boundary		1.00			242	train only	yes

The network before any explanation

Figure 333 (next page, landscape) is the trained network as it is, with no parts, no splits and no inserted units: inputs at the bottom, hidden units by depth, the output at the top, every connection drawn, every input column its own node (one-hot levels included).

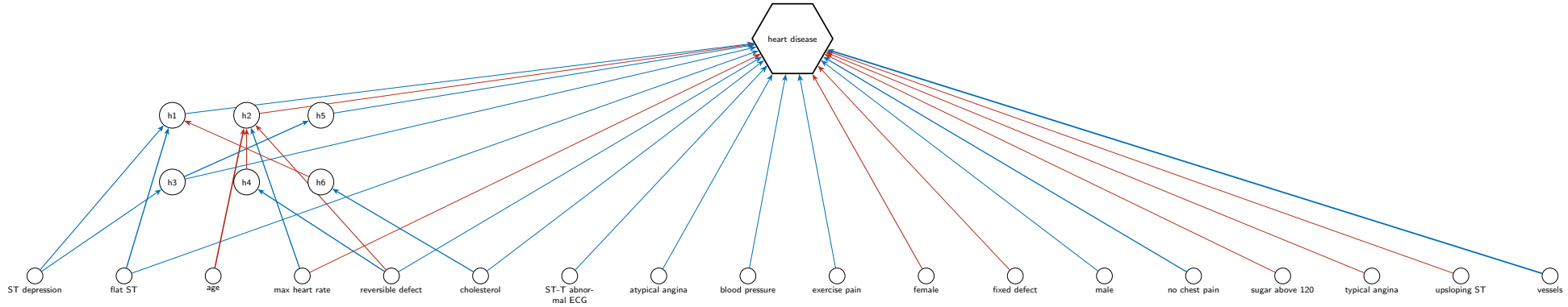


Figure 333: The network as trained, before any explanation, in full. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. Not used (no connection): non-anginal pain, sugar 120 or less, normal ECG, LVH on ECG, no exercise pain, downsloping ST, normal thallium, thallium code 4.

8 heart-s2-c2

Genome `d4bbc960-816a-4781-8979-fdd4165debf2`, explanation `3531d96c-aa00-4141-863a-2c85155afe8f`. Split: train primary (n=242), test as a check (n=61). Claims: 42, verified 41, not supported 1.
AI-assisted draft for review (unverified)

Reference (typical) patient: the typical patient: the median training patient for every number and the most common training category for every category (not a real patient). age: 56 years; ca: 0 vessels; chol: 240 mg/dL (6.2 mmol/L); cp: no chest pain (asymptomatic); exang: no chest pain on exercise; fbs: fasting blood sugar of 120 mg/dL or less; oldpeak: 0.8 mm; restecg: a normal resting ECG; sex: male; slope: an upsloping ST segment; thal: a normal thallium scan; thalach: 152 beats per minute; trestbps: 130 mm Hg. Predicted risk of heart disease: 32%.

8.1 Summary

For a typical patient (a 56-year-old man with no chest pain symptoms and median test results) the model predicts a risk of heart disease of about 32%; it adds up most of its inputs independently, and two parts add to that sum: one acts on 104 of 242 training patients through chest pain type, maximum heart rate and the thallium scan, and one through the resting ECG barely changes the predictions. [C29, C37, C41] Figure 334 shows the model at this level: one box per paragraph below.

How to read the diagrams. Edge colour is the sign of that one connection: blue positive, red negative; thicker is a larger weight. The outline and arrow on an input (and on a part's output box) is its overall effect on the final prediction, read from the verified claims: orange and up arrow raises it, teal and down arrow lowers it, grey and double arrow depends (it goes both ways, or differs between levels). A negative connection into a part that itself raises the prediction means that input lowers the prediction; through a hinge the overall effect can only be read from the claims, not from the edges. Up arrow means towards heart disease; down arrow, away from it. A double outline marks an input (or a part's output) that acts at a fixed rate everywhere: it adds the same amount to the score per unit, so it is explained by its rate alone (see 'Linear factors'); a dashed double outline, at a fixed rate on the observed data. The normal outline is an input whose effect switches on and off somewhere. For example, fixed defect has a negative connection into 'Heart rate gate', a part that lowers the predicted risk of heart disease, so fixed defect raises it.

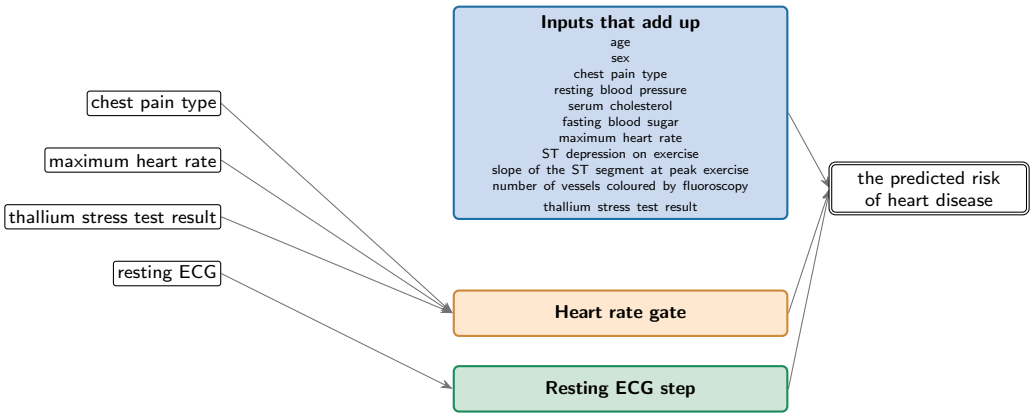


Figure 334: The model at the level of the summary: one box per part.

Inputs that add up. *Lowering the risk.* These add up independently, and these lower the risk: through this sum alone, compared with no chest pain symptoms, typical angina takes a typical patient's predicted risk from about 32% to about 2%, non-anginal chest pain to about 7% and atypical angina to about 14%; being a woman lowers it to about 4% and fasting blood sugar above 120 mg/dL to about 20%; and a higher maximum heart rate lowers it steadily, from about 37% at 118 to about 28% at 175 beats per minute (chest pain type and maximum heart rate also act through the heart rate gate, below). [C04, C12, C16, C08, C18, C20]
Raising the risk. Each extra vessel coloured by fluoroscopy adds about the same, from about 32% at 0 to about 96% at 2 vessels, flattening out near the top because the predicted risk cannot go much higher; a flat rather than upsloping ST segment raises it to about 81%; and ST depression on exercise, cholesterol and resting blood pressure each raise it steadily, from about 25% at 0 mm to about 55% at 2.8 mm, from about 22% at 193 mg/dL to about 50% at 308 mg/dL, and from about 21% at 110 mm Hg to about 48% at 155 mm Hg. [C01, C02, C03, C07, C09] Through this sum alone a reversible defect on the thallium scan raises it to about 57% (the thallium scan also acts through the heart rate gate, below); older age raises it steadily but only from about 31% at 42 to about 32% at 66 years; and the model ignores chest pain brought on by exercise. [C10, C21, C28]

Rare results. Some rarer results also count (the model has little data on this, n = 15 to 17): a downsloping ST segment raises the risk to about 54%, and through this sum alone a fixed defect on the thallium scan lowers it to about 12%. [C13, C14]

Heart rate gate. This part reads chest pain type, maximum heart rate and the thallium scan, and acts on 104 of 242 training patients, 24 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 5% without it and about 2% with it. [C37] Across the whole model, non-anginal chest pain lowers the predicted risk, and even more so without a fixed defect on the thallium scan: from about 32% to about 3% then, against about 12% to about 2% with one (the model has little data on this, n = 15). [C33] Across the whole model, a higher maximum heart rate lowers the risk, and even more so with non-anginal chest pain, where it falls steadily from about 7% at 118 to about 2% at 175 beats per minute, flattening out near the bottom because the predicted risk cannot go much lower; for other chest pain types it falls from about 37% to about 26%. [C35] Through this part alone, for most patients a higher maximum heart rate does nothing until about 168 beats per minute, then lowers the risk steadily, but for patients with non-anginal chest pain it lowers it steadily all along: for a typical patient from about 32% at 118 to about 29% at 175 beats per minute (the model has little data on this, n = 14). [C31, C36]

Resting ECG step. This part reads only the resting ECG: signs of left ventricular hypertrophy take a typical patient's predicted risk from about 32% to about 31%, and switching the part off changes any prediction by at most about half a percentage point. [C39, C41]

The whole network. Figure 335 (next page, landscape) draws every unit and connection of the explained model, nested by part: shaded clusters are the parts of the model and darker boxes the parts inside them; circles are hidden units (h1, h2, ...); each cluster's rectangle is its output, and a dashed 'sum' stands for an addition the next unit does (drawn, not a unit of the model); the hexagon is the model's output. An input used by several parts is drawn under each of them (same label, dotted outline, the first copy marked with the number of copies): it is the same input, repeated for readability. Not used by the model (input columns with no connection): sugar 120 or less, normal ECG, ST-T abnormal ECG, no exercise pain, exercise pain, downsloping ST, normal thallium, thallium code 4.

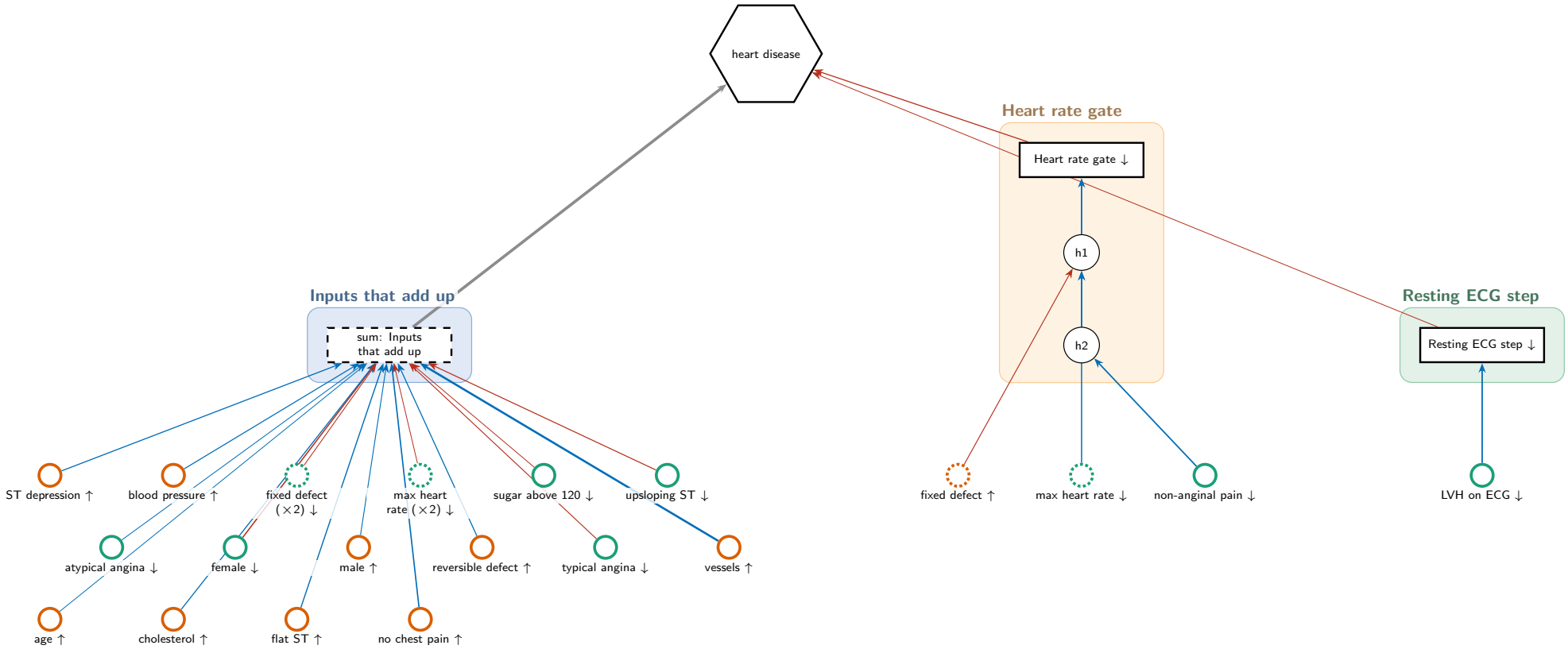


Figure 335: Every unit and connection of the explained model, nested by part. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Worth checking against what is known. Worth checking: going from 0 to 2 vessels coloured moves a typical patient's predicted risk from about 32% to about 96%. [C01] Worth checking: patients with typical angina get the lowest predicted risk of all chest pain types, below patients with no chest pain symptoms. [C05, C06, C17] Worth checking: the model treats fasting blood sugar above 120 mg/dL as lowering the risk of heart disease (for a typical patient from about 32% to about 20%). [C18] Worth checking: the model treats a fixed defect on the thallium scan as lowering the risk (for a typical patient from about 32% to about 12%), from only 15 training patients. [C15] Worth checking: the model ignores whether exercise brings on chest pain. [C28]

8.2 The model, read from the top down

Each part of the explanation in turn, from the whole model down to the smallest parts: what it does for the prediction (hypothesis), what was checked (evidence), a diagram one level down, and where to read on. The technical record of each part is in "Technical detail" at the end.

8.2.1 The whole model

Figure 336 shows what feeds this part; Figure 337 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

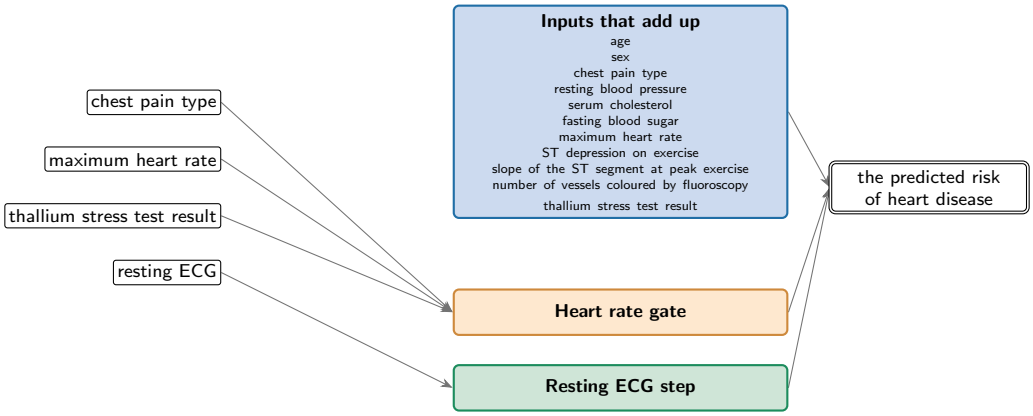


Figure 336: The whole model: what feeds it, one level down.

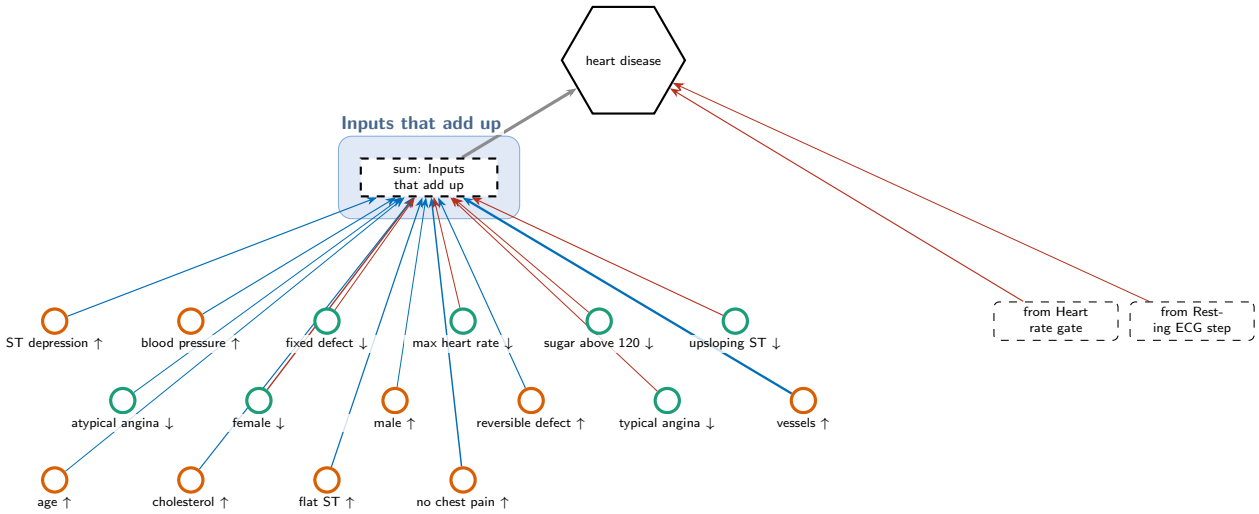


Figure 337: The whole model: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. The model adds up its inputs into one score: the vessels coloured (each extra vessel adding about the same), the ST segment, chest pain type and sex each move the predicted risk on their own, and two parts add to that sum: a heart rate gate that lowers the risk, and a small step for the resting ECG. [C29, C01, C02, C05, C08]

Evidence. It is made up of the inputs that feed the score directly and two parts, the heart rate gate and the resting ECG step (each explained below); the score is their sum, which was checked on all 242 training patients. [C29]

Chest pain brought on by exercise feeds nothing, so the model ignores it. [C28] Across the whole model, compared with no chest pain symptoms, typical angina takes a typical patient's predicted risk from about 32% to about 2%, non-anginal chest pain to about 3% and atypical angina to about 14%. [C05, C06, C17] Across the whole model, a higher maximum heart rate lowers the risk, for most patients slowly at first, then faster, but for patients with non-anginal chest pain steadily: from about 37% at 118 to about 26% at 175 beats per minute, against about 28% through the sum alone. [C19, C20] Across the whole model, a reversible defect on the thallium scan raises it to about 57%, and a fixed defect lowers it to about 12% (the model has little data on this, n = 15). [C11, C15]
Evidence plots: Figure 338.

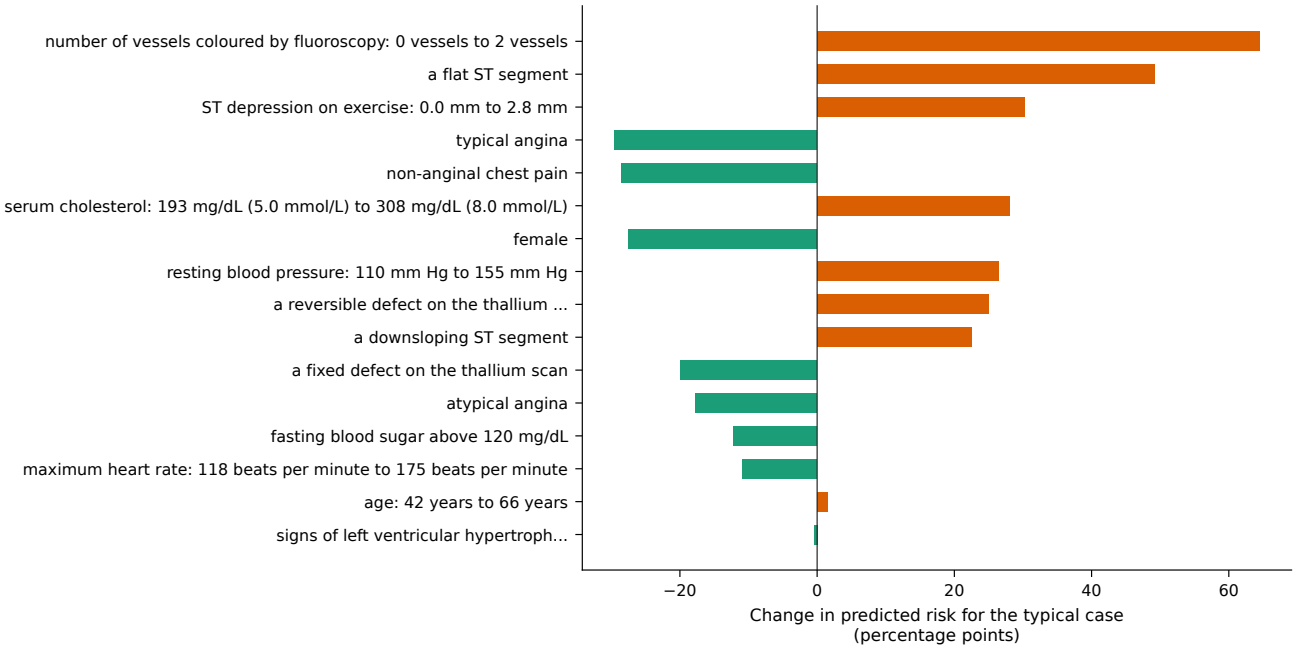


Figure 338: The whole model: Each input's change in the predicted value for the typical case across the whole model, largest first (orange: raises it, teal: lowers it).

Dive deeper: Heart rate gate (13.2.3); Resting ECG step (13.2.1). Technical detail: T.2.

8.2.2 Heart rate gate

Part of: The whole model (9.2.2).
Figure 339 shows what feeds this part; Figure 340 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

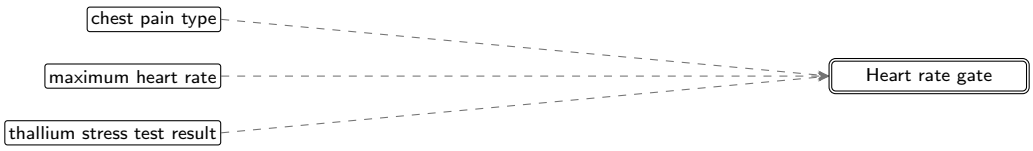


Figure 339: Heart rate gate: what feeds it, one level down.

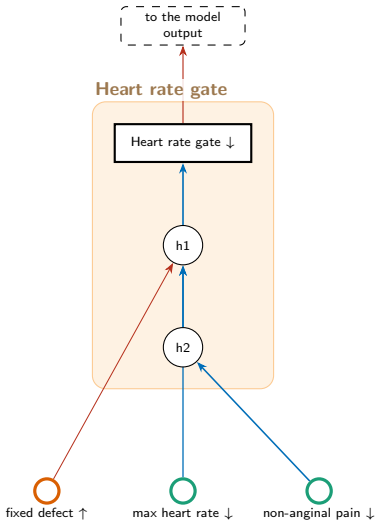


Figure 340: Heart rate gate: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part lowers the predicted risk for patients with non-anginal chest pain or a high maximum heart rate, unless the thallium scan shows a fixed defect (the model has little data on this, n = 15). [C37, C33]

Evidence. It acts on 104 of 242 training patients, 24 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 5% without it and about 2% with it. [C37] Through it, non-anginal chest pain takes a typical patient's predicted risk from about 32% to about 16%, and for the typical patient it acts on a fixed defect on the thallium scan raises it from about 2% to about 4% (the model has little data on this, n = 15). [C30, C32] Through it, for most patients a higher maximum heart rate does nothing until about 168 beats per minute, then lowers the risk steadily, but for patients with non-anginal chest pain it lowers it steadily all along: for a typical patient from about 32% at 118 to about 29% at 175 beats per minute (the model has little data on this, n = 14). [C31, C36] Across the whole model, a higher maximum heart rate lowers the risk more without a fixed defect on the thallium scan, for most patients slowly at first, then faster, but for patients with non-anginal chest pain steadily: from about 37% to about 26%, against about 14% to about 10% with one (the model has little data on this, n = 15). [C34] Its formula reproduces the part exactly on all 242 training patients. [C38]
Evidence plots: Figure 341, Figure 342, Figure 343, Figure 344, Figure 345.
Figure 344 shows this part's own output, unit h4: it switches on at about 167.7 beats per minute of maximum heart rate, and is zero below.

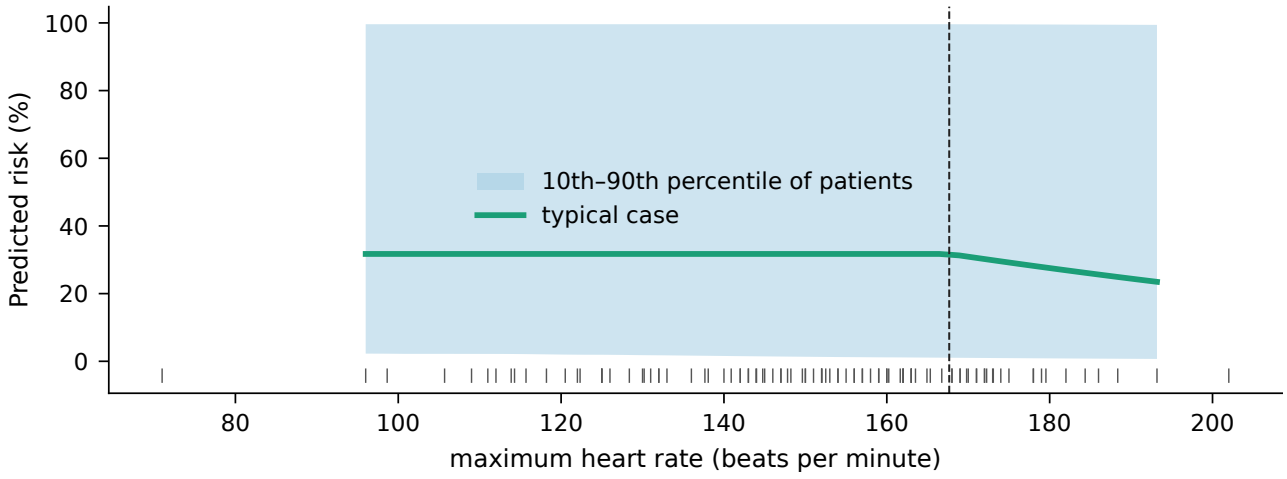


Figure 341: Heart rate gate: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual patients (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

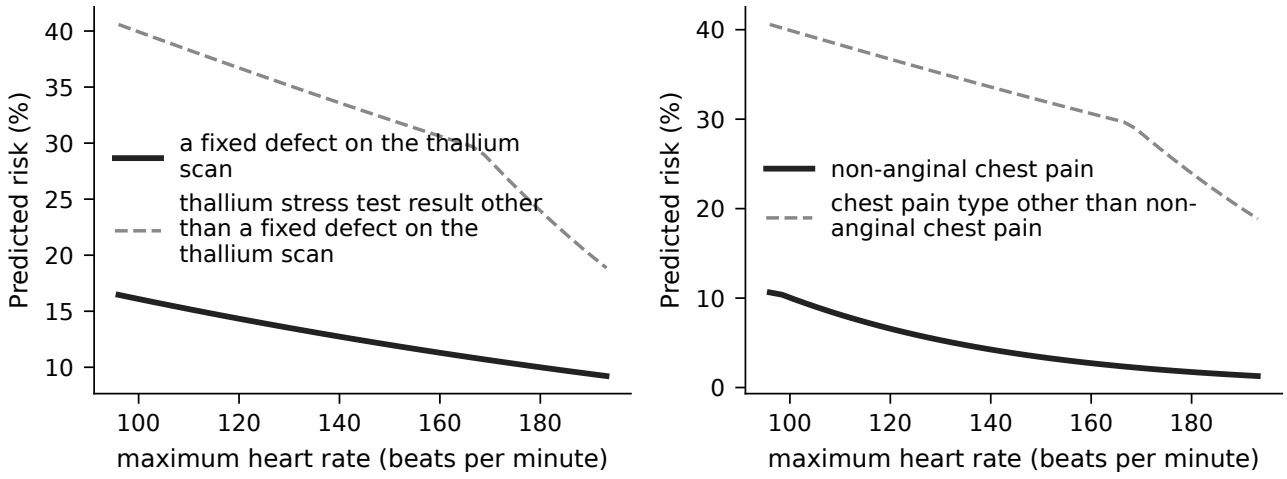


Figure 342: Heart rate gate: How the effect differs between groups: one curve per group, for the typical case of each group.

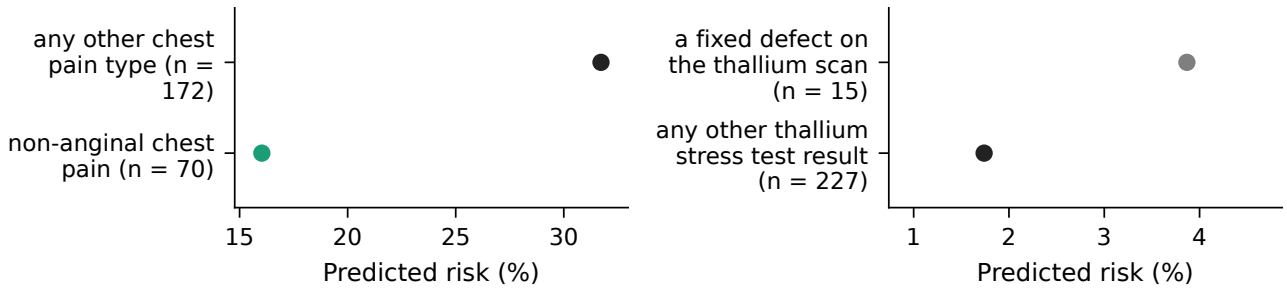


Figure 343: Heart rate gate: The predicted value for the typical case at each level, through this part (n = training patients; grey: fewer than 20).

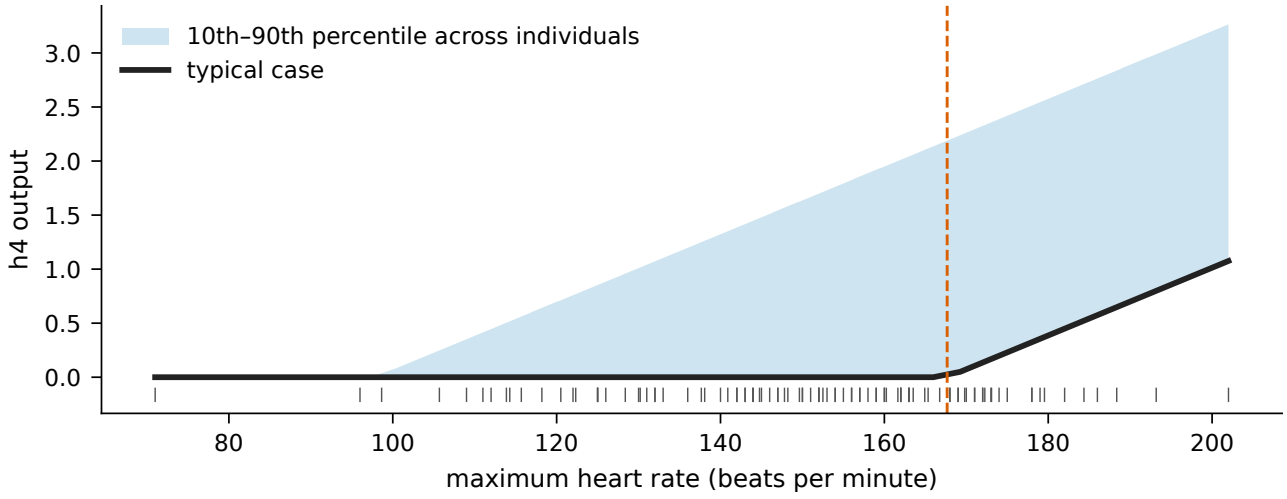


Figure 344: Heart rate gate: This part's own output, unit h4 (the output of 'Heart rate gate'), on its own scale (its activation), with every other input at the typical case; it is active for 104 of 242. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

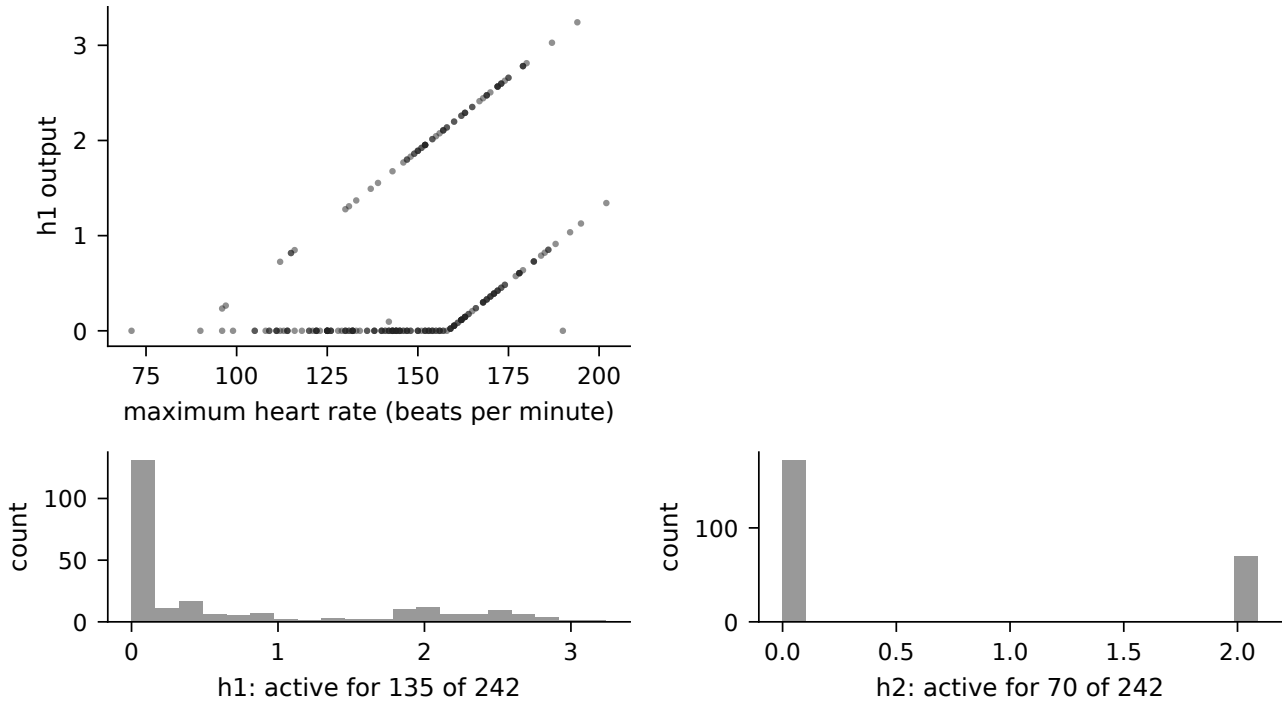


Figure 345: Heart rate gate: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.3.

8.2.3 Resting ECG step

Part of: The whole model (9.2.2).
Figure 346 shows what feeds this part; Figure 347 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

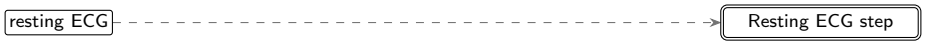


Figure 346: Resting ECG step: what feeds it, one level down.

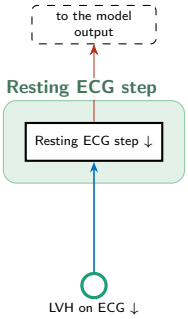


Figure 347: Resting ECG step: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part barely changes the predictions: signs of left ventricular hypertrophy on the resting ECG take a typical patient's predicted risk from about 32% to about 31%. [C39, C41]

Evidence. Switching it off changes any training patient's predicted risk by at most about half a percentage point. [C41] Its formula reproduces the part exactly on all 242 training patients. [C42]
Evidence plots: Figure 348, Figure 349.

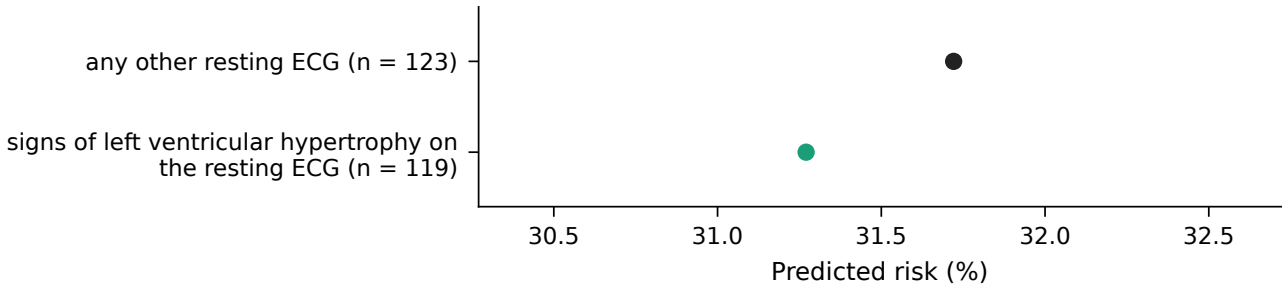


Figure 348: Resting ECG step: The predicted value for the typical case at each level, through this part (n = training patients; grey: fewer than 20).

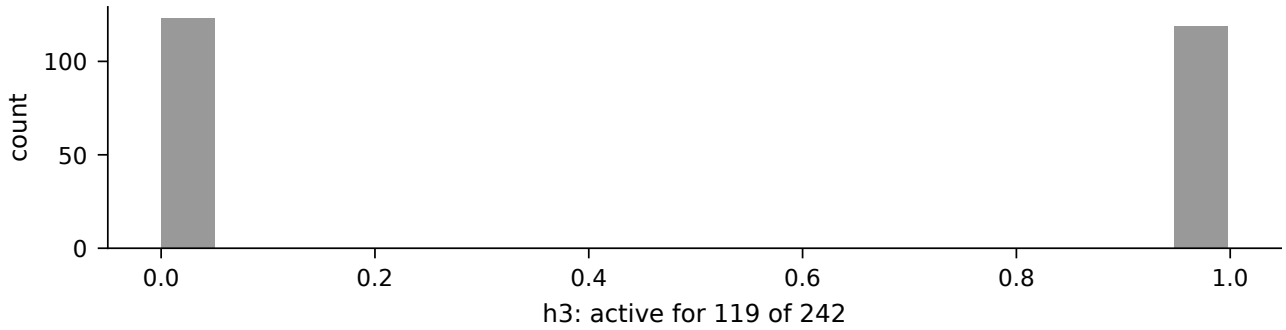


Figure 349: Resting ECG step: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.1.

8.3 Not supported by the model

- C40: subgroup claim about resting ECG in part ann_29: not supported by the model (its effect is negligible).

Technical detail: heart-s2-c2

The record behind each part: the original hypothesis, the closed form, and every claim with its estimate (log-odds and odds ratio), range across patients, n and test check.

T.1 The whole model

Annotation ann_30 (heart_disease_logit). Reads: age, sex, chest pain type, resting blood pressure, serum cholesterol, fasting blood sugar, resting ECG, maximum heart rate reached during the exercise test, exercise-induced angina, ST depression during exercise (compared with rest), slope of the ST segment at peak exercise, number of major vessels coloured by fluoroscopy, thallium stress test result.

Original hypothesis (unchanged). Root: heart-disease logit at output 0 (sigmoid). Essentially logistic regression: 16 direct edges (ca +1.78, cp:4 +0.87, sex:0 -0.77, slope:2 +0.63, oldpeak +0.56, chol +0.54, thal:7 +0.50, trestbps +0.49, slope:1 -0.47, cp:1 -0.37, sex:1 +0.34, thal:6 -0.30, cp:2 +0.26, fbs:1 -0.22, thalach -0.15, age +0.03) carry logit SD 3.66 and train AUC 0.9343 alone (L2 logistic regression on all 26 inputs: 0.9355). The two children are added to the logit: heart_rate_nonanginal_gate (x -0.519; +0.006 AUC, the only consequential nonlinearity) and restecg2_relay (x -0.021, negligible). Eight inputs are unconnected, zero effect: thal:3, thal:4, slope:3, exang:0, exang:1, restecg:0, restecg:1, fbs:0 (exercise angina is ignored). Evidence (train, n=242): composed closed form; sensitivity and mean |SHAP| both rank ca first (0.16), then cp:4, sex:0, slope:2; full-model train AUC 0.9403, accuracy 0.897.

Closed form. Read through its parts (their formulas are in their own sections).

Split inputs. To annotate the parts, the explanation split these inputs into copies (a refactoring step that does not change what the model computes); the diagrams draw each input once: fixed defect (2 copies); max heart rate (2 copies).

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C01	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (whole model) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	32% to 96%	3.99	54.0	3.99 to 3.99	242	unclear; AUC gain 0.037 [-0.038, 0.119]	yes
C02	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	32% to 81%	2.21	9.14	2.21 to 2.21	115, 110	disagrees; AUC gain -0.004 [-0.044, 0.034]	yes
C03	direction: ST depression on exercise: 0.0 mm to 2.8 mm (whole model) *shape: steady; 100% of patients same; verified*	25% to 55%	1.31	3.72	1.31 to 1.31	242	unclear; AUC gain 0.011 [-0.005, 0.034]	yes
C04	direction: chest pain type: no chest pain (asymptomatic) to typical angina (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	32% to 2%	-3.03	0.05	-3.03 to -3.03	108, 22	agrees; AUC gain 0.039 [0.009, 0.076]	yes
C05	direction: chest pain type: no chest pain (asymptomatic) to typical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	32% to 2%	-3.03	0.05	-3.03 to -3.03	108, 22	agrees; AUC gain 0.057 [0.017, 0.104]	yes
C06	direction: chest pain type: no chest pain (asymptomatic) to non-anginal chest pain (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	32% to 3%	-2.53	0.08	-2.88 to -1.96	108, 70	agrees; AUC gain 0.057 [0.017, 0.104]	yes
C07	direction: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L) (whole model) *shape: steady; 100% of patients same; verified*	22% to 50%	1.28	3.58	1.28 to 1.28	242	disagrees; AUC gain -0.011 [-0.039, 0.012]	yes
C08	direction: sex: male to female (whole model) *shape: categorical; highest to lowest: male, female*	32% to 4%	-2.34	0.10	-2.34 to -2.34	159, 83	unclear; AUC gain 0.027 [-0.01, 0.074]	yes
C09	direction: resting blood pressure: 110 mm Hg to 155 mm Hg (whole model) *shape: steady; 100% of patients same; verified*	21% to 48%	1.22	3.38	1.22 to 1.22	242	unclear; AUC gain 0.009 [-0.013, 0.034]	yes
C10	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan to a reversible defect on the thallium scan (direct contribution only, not through any part) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan, a fixed defect on the thallium scan*	32% to 57%	1.04	2.82	1.04 to 1.04	137, 90	unclear; AUC gain 0.018 [-0.012, 0.053]	yes
C11	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan to a reversible defect on the thallium scan (whole model) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan, a fixed defect on the thallium scan*	32% to 57%	1.04	2.82	1.04 to 1.04	137, 90	unclear; AUC gain 0.018 [-0.012, 0.053]	yes
C12	direction: chest pain type: no chest pain (asymptomatic) to non-anginal chest pain (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	32% to 7%	-1.75	0.17	-1.75 to -1.75	108, 70	agrees; AUC gain 0.039 [0.009, 0.076]	yes
C13	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a downsloping ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	32% to 54%	0.94	2.56	0.94 to 0.94	115, 17 (small n)	disagrees; AUC gain -0.004 [-0.044, 0.034]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C14	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan to a fixed defect on the thallium scan (direct contribution only, not through any part) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan, a fixed defect on the thallium scan*	32% to 12%	-1.24	0.29	-1.24 to -1.24	137, 15 (small n)	unclear; AUC gain 0.018 [-0.012, 0.053]	yes
C15	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan to a fixed defect on the thallium scan (whole model) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan, a fixed defect on the thallium scan*	32% to 12%	-1.00	0.37	-1.24 to -0.42	137, 15 (small n)	unclear; AUC gain 0.018 [-0.012, 0.053]	yes
C16	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	32% to 14%	-1.05	0.35	-1.05 to -1.05	108, 42	agrees; AUC gain 0.039 [0.009, 0.076]	yes
C17	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	32% to 14%	-1.05	0.35	-1.05 to -1.05	108, 42	agrees; AUC gain 0.057 [0.017, 0.104]	yes
C18	direction: fasting blood sugar: fasting blood sugar of 120 mg/dL or less to fasting blood sugar above 120 mg/dL (whole model) *shape: categorical; highest to lowest: fasting blood sugar of 120 mg/dL or less, fasting blood sugar above 120 mg/dL*	32% to 20%	-0.65	0.52	-0.65 to -0.65	209, 33	unclear; AUC gain 0.000 [-0.011, 0.013]	yes
C19	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (whole model) *shape: accelerating; 67% of patients same; mixed (but steady for patients with non-anginal chest pain)*	37% to 26%	-0.73	0.48	-1.32 to -0.51	242	agrees; AUC gain 0.014 [0.002, 0.037]	yes
C20	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (direct contribution only, not through any part) *shape: steady; 100% of patients same; verified*	37% to 28%	-0.39	0.68	-0.39 to -0.39	242	unclear; AUC gain 0.006 [0.000, 0.020]	yes
C21	direction: age: 42 years to 66 years (whole model) *shape: steady; 100% of patients same; verified*	31% to 32%	0.07	1.08	0.07 to 0.07	242	disagrees; AUC gain -0.001 [-0.005, 0.000]	yes
C22	direction: resting ECG: resting ECG other than signs of left ventricular hypertrophy on the resting ECG to signs of left ventricular hypertrophy on the resting ECG (whole model) *shape: categorical; highest to lowest: resting ECG other than signs of left ventricular hypertrophy on the resting ECG, signs of left ventricular hypertrophy on the resting ECG*	32% to 31%	-0.02	0.98	-0.02 to -0.02	123, 119	unclear; AUC gain 0.000 [0.000, 0.000]	yes
C23	magnitude: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels, typical patient *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	32% to 96%	3.99	54.0		242	unclear; AUC gain 0.037 [-0.038, 0.119]	yes
C24	magnitude: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment, typical patient *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	32% to 81%	2.21	9.14		115, 110	disagrees; AUC gain -0.004 [-0.044, 0.034]	yes
C25	magnitude: ST depression on exercise: 0.0 mm to 2.8 mm, typical patient *shape: steady; 100% of patients same; verified*	25% to 55%	1.31	3.72		242	unclear; AUC gain 0.011 [-0.005, 0.034]	yes
C26	magnitude: chest pain type: no chest pain (asymptomatic) to typical angina, typical patient *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	32% to 2%	-3.03	0.05		108, 22	agrees; AUC gain 0.057 [0.017, 0.104]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C27	magnitude: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L), typical patient *shape: steady; 100% of patients same; verified*	22% to 50%	1.28	3.58		242	disagrees; AUC gain -0.011 [-0.039, 0.012]	yes
C28	null: model ignores exercise-induced angina (exact rule)		0.00			242	train only	yes
C29	structure: made of heart_rate_nonanginal_gate, restecg2_relay and inputs age, sex, cp, trestbps, chol, fbs, thalach, oldpeak, slope, ca, thal; added					242	train only	yes

T.2 Heart rate gate

Annotation ann_28 (heart_rate_nonanginal_gate). Reads: chest pain type, maximum heart rate reached during the exercise test, thallium stress test result.

Original hypothesis (unchanged). Protective heart-rate gate, gated by chest-pain type and thallium. 2942 = ReLU(0.968 cp:3 + 0.571) is a non-anginal-pain indicator (2.09 for the 70 cp:3 patients, 0 otherwise). 3842 = ReLU(1.021*ReLU(0.688 thalach - 0.374 thal:6 + 1.028*2942 - 0.347) - 0.295) is read out -0.519 into the logit. The gate opens once max heart rate passes a threshold set by the other two inputs: about -2.3 SD with non-anginal pain (open for 66 of 68), about +0.78 SD for other pain types without a fixed defect (38 of 159), and never with a fixed thallium defect (0 of 15). Once open, each SD of thalach lowers the logit by 0.36 more (total -0.52 vs -0.15 closed), and non-anginal pain lowers it by about 1.1. Evidence (train, n=242): closed form from get_formula; gate open for 104 patients (disease rate 23% vs 60% when closed); logit contribution SD 0.45; removing it drops train AUC 0.9403 -> 0.9344, a linear refit on cp:3, thalach, thal:6 recovers 0.9378 (R2 0.87), so most of its value is the gating rather than the linear trend. It is the model's only consequential nonlinearity.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 4.44089e-16).

$$\begin{aligned} y_0 = \operatorname{relu}\big(&-0.295 + 1.02\operatorname{relu}\big(-0.347 \\ &+ 0.688x_{thalach_b} + 1.03\operatorname{relu}\big(0.571 + 0.968x_{cp:3}\big) \\ &- 0.374x_{thal:6_b}\big)\big) \end{aligned}$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C30	direction: chest pain type: chest pain type other than non-anginal chest pain to non-anginal chest pain (through this part) *shape: categorical; highest to lowest: chest pain type other than non-anginal chest pain, non-anginal chest pain*	32% to 16%	-0.79	0.46	-1.14 to -0.21	172, 70	unclear; AUC gain 0.008 [-0.012, 0.030]	yes
C31	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (through this part) *shape: hinge at 168 beats per minute; 70% of patients same; mixed (but steady for patients with non-anginal chest pain); one point with C36 (median fold 168 beats per minute, typical patient 166 beats per minute, 10th-90th percentile 168 beats per minute to 168 beats per minute)*	32% to 29%	-0.34	0.71	-0.93 to -0.12	242	unclear; AUC gain 0.008 [0.000, 0.022]	yes
C32	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan to a fixed defect on the thallium scan (through this part) *shape: categorical; highest to lowest: a fixed defect on the thallium scan, thallium stress test result other than a fixed defect on the thallium scan*	2% to 4% (typical patient this part acts on)	0.24	1.27	0.00 to 0.82	227, 15 (small n)	unclear; AUC gain 0.001 [0.000, 0.006]	yes
C33	interaction: effect of chest pain type (chest pain type other than non-anginal chest pain to non-anginal chest pain) by thallium stress test result: a fixed defect on the thallium scan minus thallium stress test result other than a fixed defect on the thallium scan; stronger in thallium stress test result other than a fixed defect on the thallium scan *shape: categorical; highest to lowest: chest pain type other than non-anginal chest pain, non-anginal chest pain*	a fixed defect on the thallium scan: 12% to 2%; thallium stress test result other than a fixed defect on the thallium scan: 32% to 3%	0.79	2.20		15, 227 (small n)	agrees; AUC gain 0.001 [0.000, 0.006]	yes
C34	interaction: effect of maximum heart rate (118 beats per minute to 175 beats per minute) by thallium stress test result: a fixed defect on the thallium scan minus thallium stress test result other than a fixed defect on the thallium scan; stronger in thallium stress test result other than a fixed defect on the thallium scan *shape: accelerating; 70% of patients same; mixed (but steady for patients with non-anginal chest pain)*	a fixed defect on the thallium scan: 14% to 10%; thallium stress test result other than a fixed defect on the thallium scan: 37% to 26%	0.30	1.35		15, 227 (small n)	agrees; AUC gain 0.001 [0.000, 0.006]	yes
C35	interaction: effect of maximum heart rate (118 beats per minute to 175 beats per minute) by chest pain type: non-anginal chest pain minus chest pain type other than non-anginal chest pain; stronger in non-anginal chest pain *shape: steady; 97% of patients same; verified; risk saturates near 0%*	non-anginal chest pain: 7% to 2%; chest pain type other than non-anginal chest pain: 37% to 26%	-0.80	0.45		70, 172	agrees; AUC gain 0.008 [-0.012, 0.030]	yes
C36	threshold: maximum heart rate: fold at about 168 beats per minute (starts) for patients with no chest pain symptoms (asymptomatic) *shape: hinge at 168 beats per minute; 100% of patients same; verified; risk saturates near 100%; one point with C36 (median fold 168 beats per minute, typical patient 166 beats per minute, 10th-90th percentile 168 beats per minute to 168 beats per minute)*	99% to 99% (typical patient with a fold (patients with no chest pain symptoms (asymptomatic)))	-0.23	0.79	-0.23 to -0.23	99, 14 (small n)	unclear; AUC gain 0.008 [0.000, 0.022]	yes
C37	subgroup: who this part acts on: 104 of 242 patients	5% to 2% (typical patient this part acts on (risk without the part, then with it))	-0.67	0.51	-1.22 to -0.05	104, 138	unclear; AUC gain 0.018 [0.000, 0.044]	yes
C38	structure: its formula reproduces it on 242 training patients (largest difference 4.44089e-16)		4e-16			242	train only	yes

T.3 Resting ECG step

Annotation ann_29 (restecg2_relay). Reads: resting ECG.

Original hypothesis (unchanged). restecg:2 relay: 3607 = ReLU(0.997 restecg:2 - 0.016) is 0.998 for the 119 patients with left ventricular hypertrophy on the resting ECG and 0 for the other 123, i.e. exactly affine in the binary input on the data. Read out -0.021, it lowers the logit by 0.02 for LVH: negligible. Evidence (train, n=242): closed form; entry-exit correlation 1.0; logit SD 0.01; removing it changes train AUC 0.9403 -> 0.9401.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 0.0).

$$y_0 = \mathrm{relu}\big(-0.016 + 0.997 x_{restecg:2}\big)$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C39	direction: resting ECG: resting ECG other than signs of left ventricular hypertrophy on the resting ECG to signs of left ventricular hypertrophy on the resting ECG (through this part) *shape: categorical; highest to lowest: resting ECG other than signs of left ventricular hypertrophy on the resting ECG, signs of left ventricular hypertrophy on the resting ECG*	32% to 31%	-0.02	0.98	-0.02 to -0.02	123, 119	unclear; AUC gain 0.000 [0.000, 0.000]	yes
C40	subgroup: who this part acts on: 119 of 242 patients	85% to 84% (typical patient this part acts on (risk without the part, then with it))	-0.02	0.98	-0.02 to -0.02	119, 123	unclear; AUC gain 0.000 [0.000, 0.000]	no
C41	null: this part barely changes predictions (switched off)		0.005			242	train only	yes
C42	structure: its formula reproduces it on 242 training patients (largest difference 0.0)		0.00			242	train only	yes

The network before any explanation

Figure 350 (next page, landscape) is the trained network as it is, with no parts, no splits and no inserted units: inputs at the bottom, hidden units by depth, the output at the top, every connection drawn, every input column its own node (one-hot levels included).

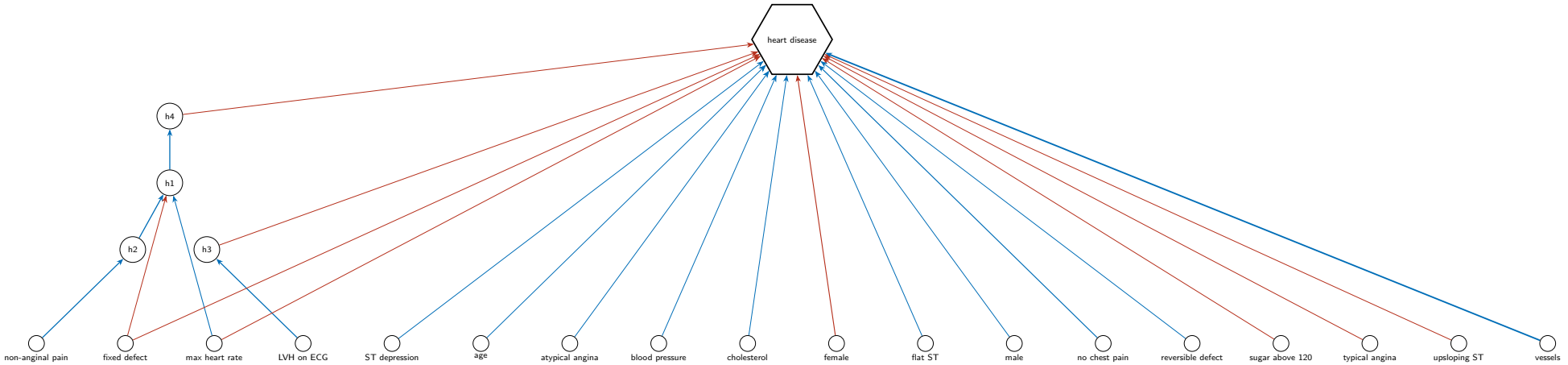


Figure 350: The network as trained, before any explanation, in full. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. Not used (no connection): sugar 120 or less, normal ECG, ST-T abnormal ECG, no exercise pain, exercise pain, downsloping ST, normal thallium, thallium code 4.

9 heart-s2-c3

Genome `1a5255ec-4f4d-4ff4-916a-1b5c4622ccc9`, explanation `6565cffd-9c19-4ae2-b635-a2e35f3d485a`. Split: train primary (n=242), test as a check (n=61). Claims: 52, verified 46, not supported 6.
AI-assisted draft for review (unverified)

Reference (typical) patient: the typical patient: the median training patient for every number and the most common training category for every category (not a real patient). age: 56 years; ca: 0 vessels; chol: 240 mg/dL (6.2 mmol/L); cp: no chest pain (asymptomatic); exang: no chest pain on exercise; fbs: fasting blood sugar of 120 mg/dL or less; oldpeak: 0.8 mm; restecg: a normal resting ECG; sex: male; slope: an upsloping ST segment; thal: a normal thallium scan; thalach: 152 beats per minute; trestbps: 130 mm Hg. Predicted risk of heart disease: 30%.

9.1 Summary

For a typical patient (a 56-year-old man with no chest pain symptoms and median test results) the model predicts a risk of heart disease of about 30%; it adds up most of its inputs independently, and three parts that all read chest pain type add to that sum: one acts on 76 of 242 training patients through maximum heart rate, one on 61 through cholesterol, and one barely changes the predictions. [C29, C34, C40, C50] Figure 351 shows the model at this level: one box per paragraph below.

How to read the diagrams. Edge colour is the sign of that one connection: blue positive, red negative; thicker is a larger weight. The outline and arrow on an input (and on a part's output box) is its overall effect on the final prediction, read from the verified claims: orange and up arrow raises it, teal and down arrow lowers it, grey and double arrow depends (it goes both ways, or differs between levels). A negative connection into a part that itself raises the prediction means that input lowers the prediction; through a hinge the overall effect can only be read from the claims, not from the edges. Up arrow means towards heart disease; down arrow, away from it. A double outline marks an input (or a part's output) that acts at a fixed rate everywhere: it adds the same amount to the score per unit, so it is explained by its rate alone (see 'Linear factors'); a dashed double outline, at a fixed rate on the observed data. The normal outline is an input whose effect switches on and off somewhere. For example, non-anginal pain has a negative connection into 'ST depression gate with little effect', a part that lowers the predicted risk of heart disease, so non-anginal pain raises it.

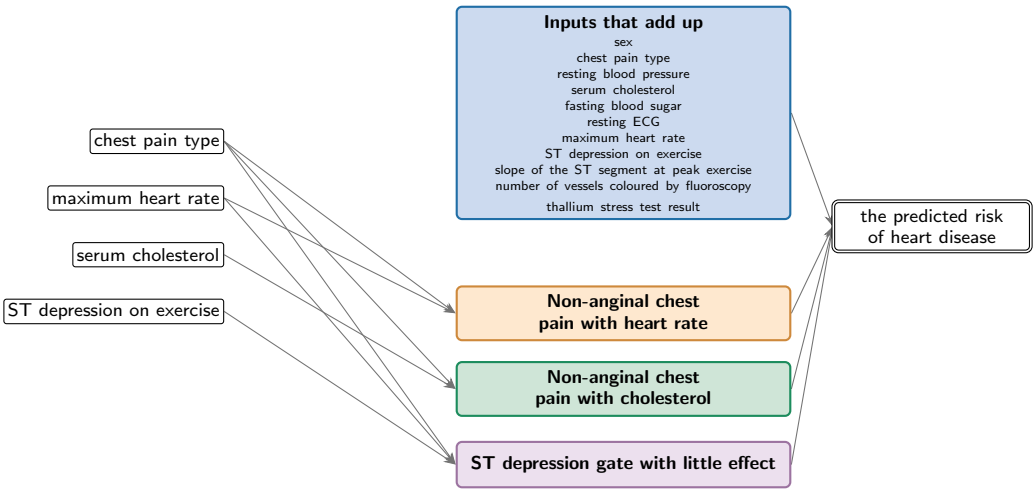


Figure 351: The model at the level of the summary: one box per part.

Inputs that add up. *Lowering the risk.* These add up independently, and these lower the risk: through this sum alone, compared with no chest pain symptoms, typical angina takes a typical patient's predicted risk from about 30% to about 2%, non-anginal chest pain to about 7% and atypical angina to about 13% (chest pain type also acts through the parts below); being a woman lowers it to about 4% and fasting blood sugar above 120 mg/dL to about 19%; and through this sum alone a higher maximum heart rate lowers it steadily, from about 36% at 118 to about 27% at 175 beats per minute (it also acts through the parts below). [C05, C14, C16, C10, C18, C19]
Raising the risk. Each extra vessel coloured by fluoroscopy adds about the same, from about 30% at 0 to about 96% at 2 vessels, flattening out near the top because the predicted risk cannot go much higher; a flat rather than upsloping ST segment raises it to about 80%, higher resting blood pressure raises it steadily from about 20% at 110 mm Hg to about 46% at 155 mm Hg, and a reversible defect on the thallium scan raises it to about 55%. [C01,

C02, C11, C12] Through this sum alone, more ST depression on exercise raises it steadily from about 23% at 0 mm to about 53% at 2.8 mm, and higher cholesterol raises it steadily from about 21% at 193 mg/dL to about 48% at 308 mg/dL; both also act through the parts below. [C03, C07]
Rare results. Some rarer results also count (the model has little data on this, n = 15 to 17): a downsloping ST segment raises the risk to about 54% and a fixed defect on the thallium scan lowers it to about 12%, while signs of left ventricular hypertrophy on the resting ECG raise it from about 30% to about 33%. [C13, C15, C21]

Non-anginal chest pain with heart rate. This part reads chest pain type and maximum heart rate, and acts on 76 of 242 training patients, mostly patients with non-anginal chest pain, 15 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 5% without it and about 2% with it. [C34] Through it, non-anginal chest pain rather than any other chest pain type takes a typical patient's predicted risk from about 30% to about 17%. [C30] Across the whole model, a higher maximum heart rate lowers the risk slowly at first, then faster, and more with non-anginal chest pain: from about 7% at 118 to about 2% at 175 beats per minute then, flattening out near the bottom because the predicted risk cannot go much lower, against about 36% to about 27% for other chest pain types. [C32]

Non-anginal chest pain with cholesterol. This part reads chest pain type and cholesterol, and acts on 61 of 242 training patients, all of them with non-anginal chest pain, 13 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 3% without it and about 2% with it. [C40] Through it, for patients with non-anginal chest pain, higher cholesterol raises the predicted risk steadily up to about 282 mg/dL, where the part switches off, then no further: for the typical patient it acts on, from about 2% at 193 mg/dL to about 3% at 308 mg/dL. [C37, C39] Across the whole model, higher cholesterol raises the risk in an uneven way from about 2% to about 7% with non-anginal chest pain, against about 21% to about 48% for other chest pain types. [C38]

ST depression gate with little effect. This part reads chest pain type, maximum heart rate and ST depression on exercise, and barely changes the predictions (by at most 0.5 percentage points). [C50] It switches on when ST depression on exercise minus about 0.019 × the maximum heart rate exceeds about -1.71, which never happens with non-anginal chest pain; 73 of 242 training patients are on, and 79% of them had heart disease, against 29% of the others. [C52]

The whole network. Figure 352 (next page, landscape) draws every unit and connection of the explained model, nested by part: shaded clusters are the parts of the model and darker boxes the parts inside them; circles are hidden units (h1, h2, ...); each cluster's rectangle is its output, and a dashed 'sum' stands for an addition the next unit does (drawn, not a unit of the model); the hexagon is the model's output. An input used by several parts is drawn under each of them (same label, dotted outline, the first copy marked with the number of copies): it is the same input, repeated for readability. Not used by the model (input columns with no connection): age, sugar 120 or less, normal ECG, ST-T abnormal ECG, no exercise pain, exercise pain, downsloping ST, normal thallium, thallium code 4.

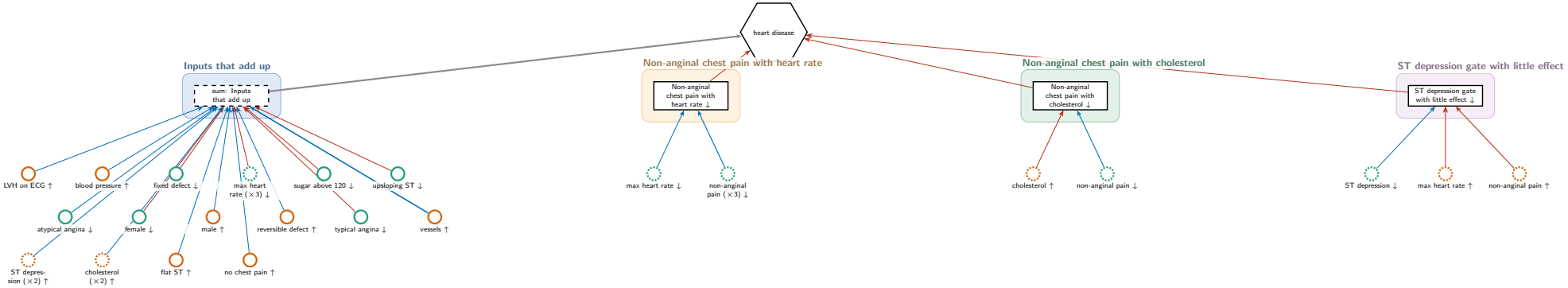


Figure 352: Every unit and connection of the explained model, nested by part. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Worth checking against what is known. Worth checking: going from 0 to 2 vessels coloured moves a typical patient's predicted risk from about 30% to about 96%. [C01] Worth checking: patients with typical angina get the lowest predicted risk of all chest pain types (about 2%), below non-anginal chest pain (about 3%), atypical angina (about 13%) and no chest pain symptoms (about 30%). [C06, C09, C17] Worth checking: the model treats fasting blood sugar above 120 mg/dL as lowering the risk of heart disease (for a typical patient from about 30% to about 19%). [C18] Worth checking: the model treats a fixed defect on the thallium scan as lowering the risk (for a typical patient from about 30% to about 12%), from only 15 training patients. [C15] Worth checking: the model ignores age. [C27] Worth checking: with non-anginal chest pain, one part lowers the predicted risk for cholesterol below about 282 mg/dL, so cholesterol moves the risk from about 2% to about 7% for these patients, against about 21% to about 48% for others. [C38, C39]

9.2 The model, read from the top down

Each part of the explanation in turn, from the whole model down to the smallest parts: what it does for the prediction (hypothesis), what was checked (evidence), a diagram one level down, and where to read on. The technical record of each part is in "Technical detail" at the end.

9.2.1 The whole model

Figure 353 shows what feeds this part; Figure 354 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

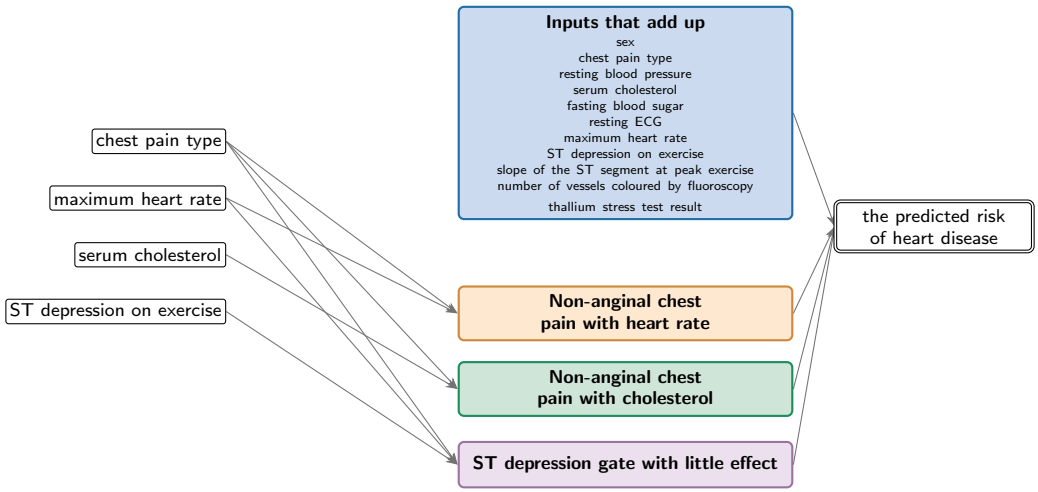


Figure 353: The whole model: what feeds it, one level down.

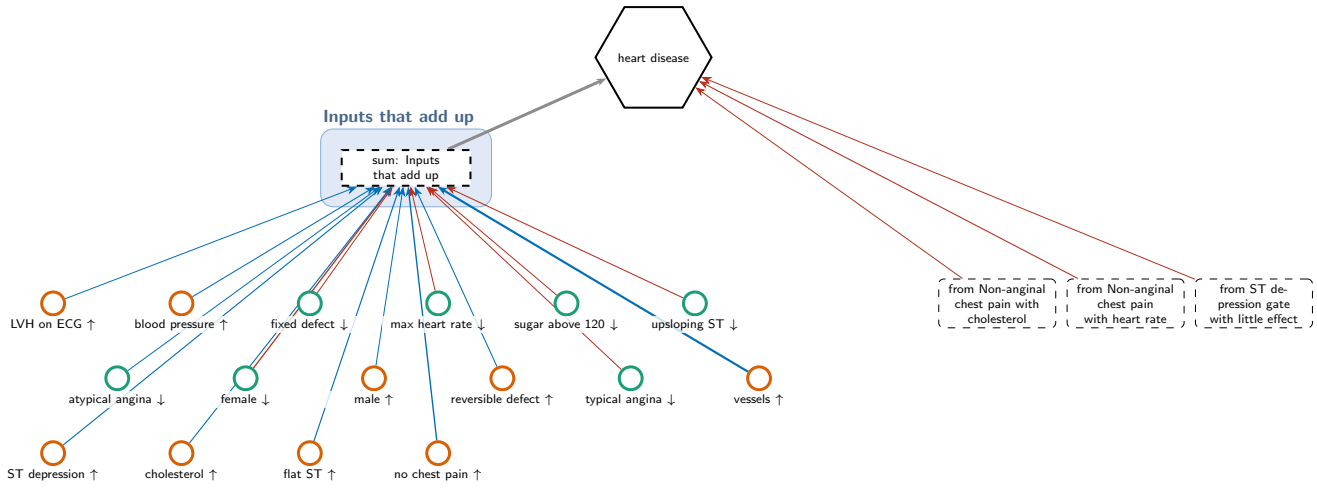


Figure 354: The whole model: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. The model adds up its inputs into one score: the vessels coloured (each extra vessel adding about the same), the ST segment, sex and chest pain type each move the predicted risk on their own, and three parts

add to that sum, two of which lower the risk further for patients with non-anginal chest pain. [C29, C01, C02, C10, C09]

Evidence. It is made up of the inputs that feed the score directly and three parts, the non-anginal chest pain parts with heart rate and with cholesterol and an ST depression gate (each explained below); the score is their sum, which was checked on all 242 training patients. [C29] Changing one input at a time across the whole model, non-anginal chest pain rather than no chest pain symptoms takes a typical patient's predicted risk from about 30% to about 3%, against about 7% through the sum alone. [C09, C14] Across the whole model, a higher maximum heart rate lowers it slowly at first, then faster, from about 36% at 118 to about 27% at 175 beats per minute, where through the sum alone it falls steadily over the same range. [C20, C19] Across the whole model, more ST depression raises it steadily from about 23% to about 53%, and higher cholesterol steadily from about 21% to about 48%. [C04, C08] It ignores age and chest pain brought on by exercise. [C27, C28]
Evidence plots: Figure 355.

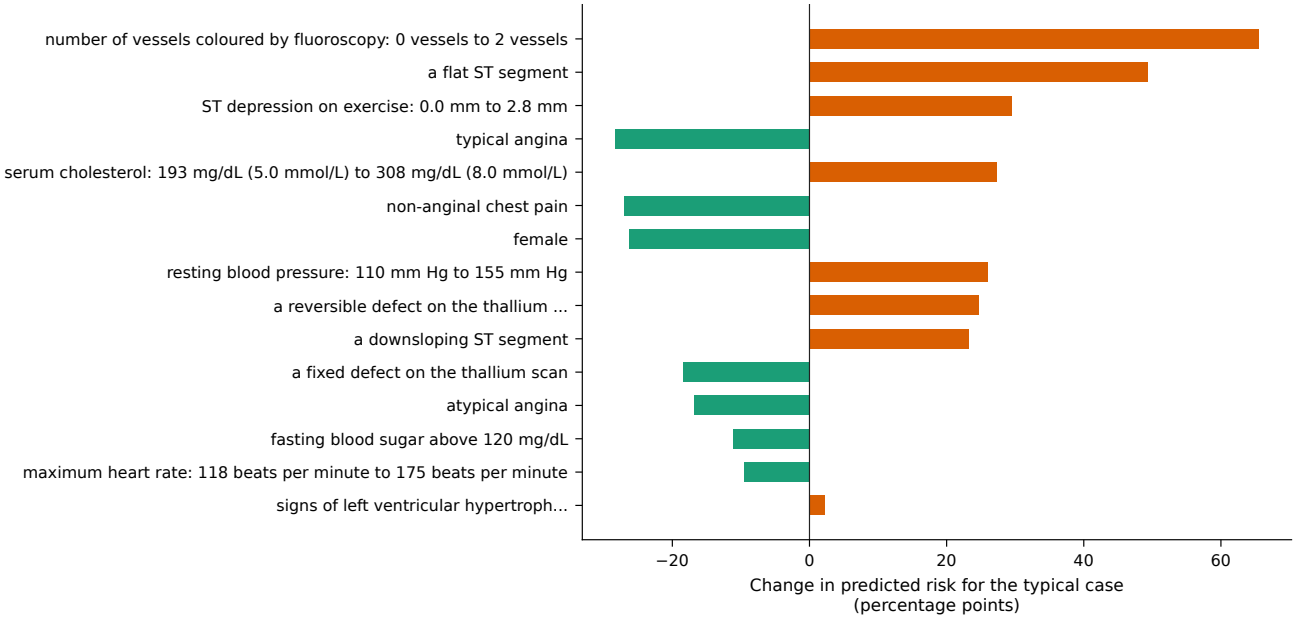


Figure 355: The whole model: Each input's change in the predicted value for the typical case across the whole model, largest first (orange: raises it, teal: lowers it).

Dive deeper: Non-anginal chest pain with heart rate (9.2.2); Non-anginal chest pain with cholesterol (11.2.2); ST depression gate with little effect (10.2.2). Technical detail: T.2.

9.2.2 Non-anginal chest pain with heart rate

Part of: The whole model (12.2.2).
Figure 356 shows what feeds this part; Figure 357 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

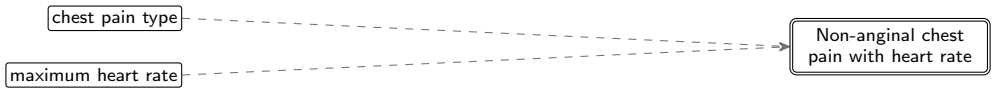


Figure 356: Non-anginal chest pain with heart rate: what feeds it, one level down.

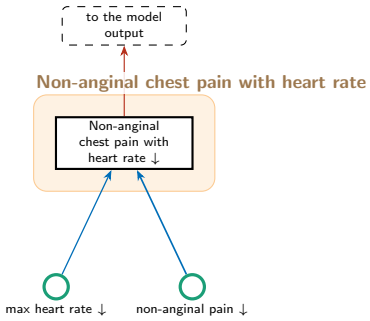


Figure 357: Non-anginal chest pain with heart rate: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part lowers the predicted risk for patients with non-anginal chest pain, and for them a higher maximum heart rate lowers it further, slowly at first, then faster. [C30, C32]

Evidence. It acts on 76 of 242 training patients, mostly patients with non-anginal chest pain, 15 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 5% without it and about 2% with it. [C34] Through it, non-anginal chest pain rather than any other chest pain type takes a typical patient's predicted risk from about 30% to about 17%. [C30] Through it, a higher maximum heart rate lowers the risk in an uneven way, for the typical patient it acts on from about 4% at 118 to about 2% at 175 beats per minute, flattening out near the bottom because the predicted risk cannot go much lower. [C31] Across the whole model, a higher maximum heart rate lowers the risk slowly at first, then faster, from about 7% to about 2% with non-anginal chest pain, against about 36% to about 27% for other chest pain types. [C32] Its formula reproduces the part exactly on all 242 training patients. [C35]
Evidence plots: Figure 358, Figure 359, Figure 360, Figure 361, Figure 362.
Figure 361 shows this part's own output, unit h1: it switches on at about 184.3 beats per minute of maximum heart rate, and is zero below.

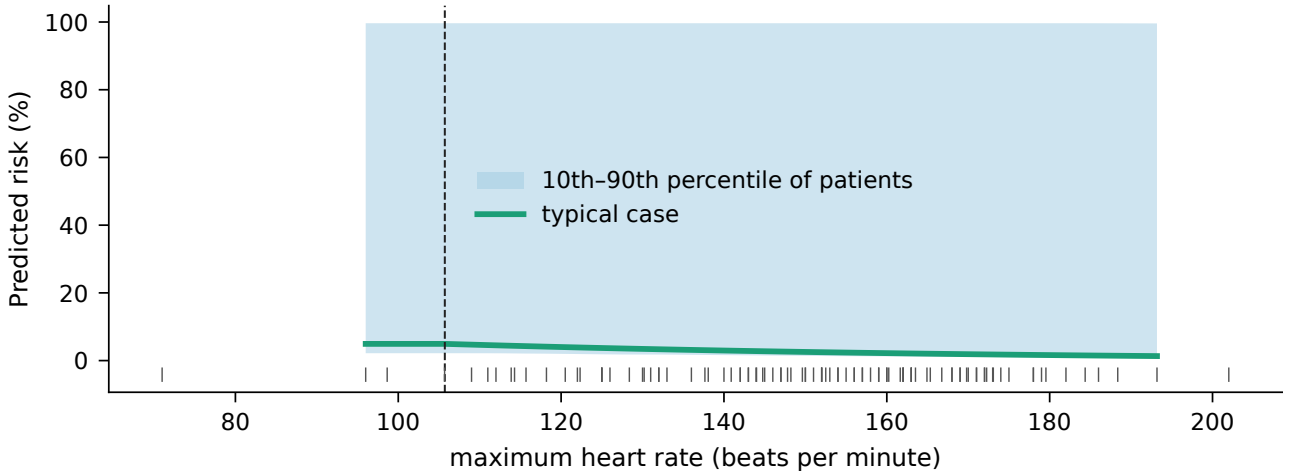


Figure 358: Non-anginal chest pain with heart rate: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual patients (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

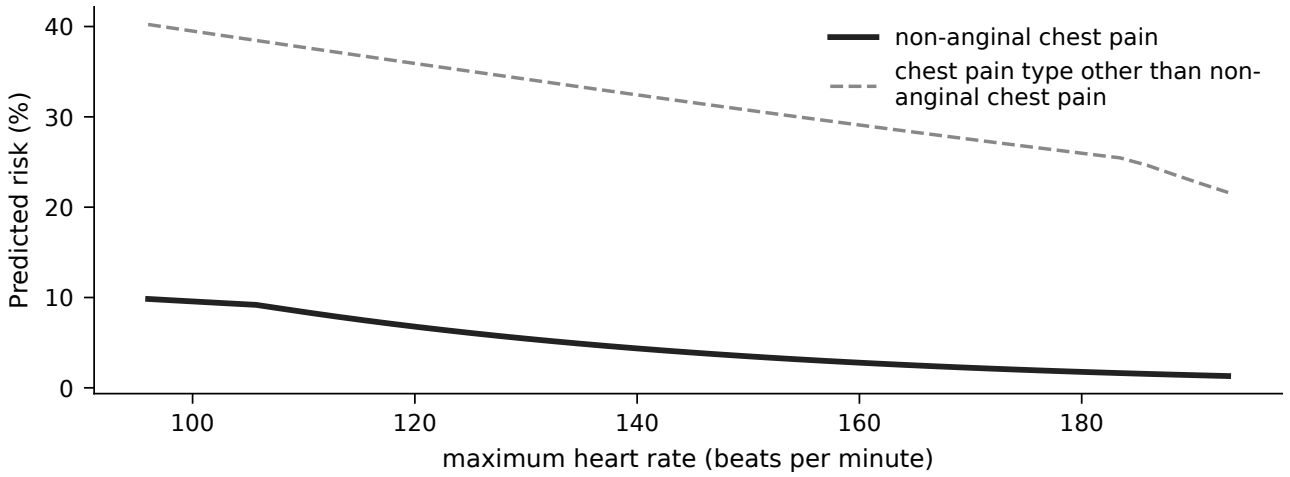


Figure 359: Non-anginal chest pain with heart rate: How the effect differs between groups: one curve per group, for the typical case of each group.

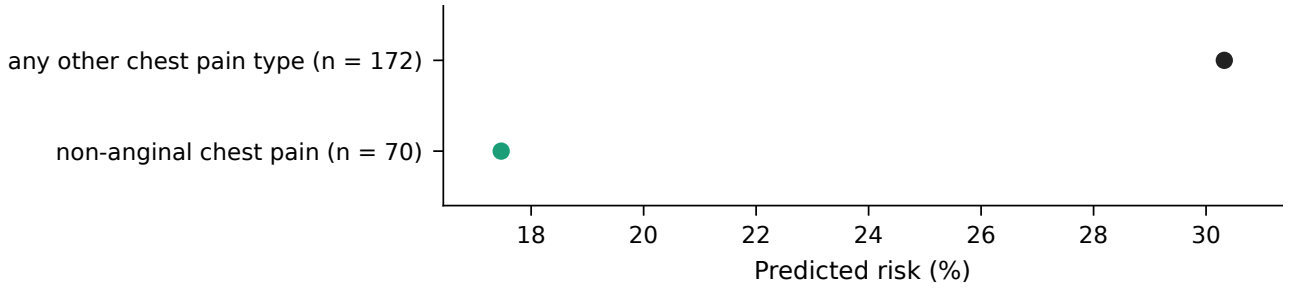


Figure 360: Non-anginal chest pain with heart rate: The predicted value for the typical case at each level, through this part (n = training patients; grey: fewer than 20).

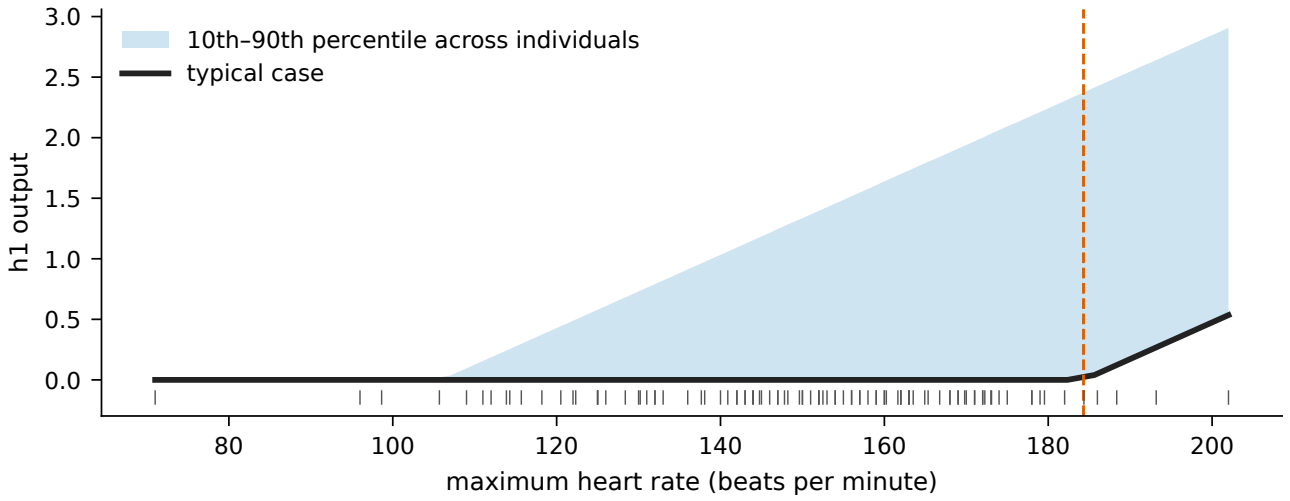


Figure 361: Non-anginal chest pain with heart rate: This part's own output, unit h1 (the output of 'Non-anginal chest pain with heart rate'), on its own scale (its activation), with every other input at the typical case; it is active for 76 of 242. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

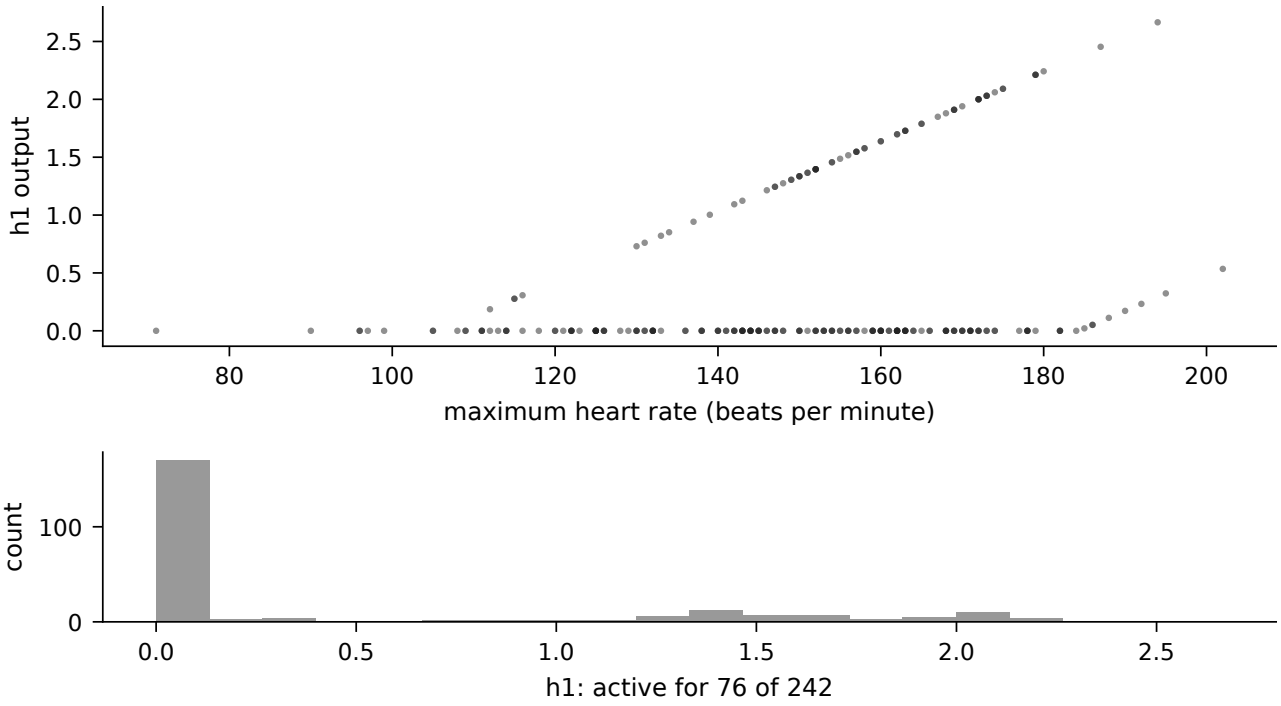


Figure 362: Non-anginal chest pain with heart rate: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.2.

9.2.3 Non-anginal chest pain with cholesterol

Part of: The whole model (12.2.2).
Figure 363 shows what feeds this part; Figure 364 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

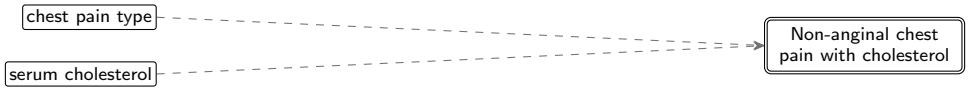


Figure 363: Non-anginal chest pain with cholesterol: what feeds it, one level down.

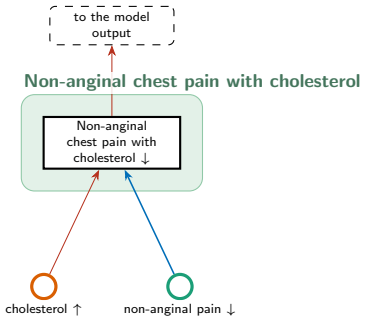


Figure 364: Non-anginal chest pain with cholesterol: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part lowers the predicted risk a little further for patients with non-anginal chest pain whose cholesterol stays below about 282 mg/dL. [C40, C39]

Evidence. It acts on 61 of 242 training patients, all of them with non-anginal chest pain, 13 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 3% without it and about 2% with it. [C40] Through it, higher cholesterol raises the predicted risk steadily up to about 282 mg/dL, where the part switches

off, then no further: for the typical patient it acts on, from about 2% at 193 mg/dL to about 3% at 308 mg/dL. [C37, C39] Through it, non-anginal chest pain rather than any other chest pain type takes a typical patient's predicted risk from about 30% to about 29%. [C36] Across the whole model, higher cholesterol raises the risk in an uneven way from about 2% to about 7% with non-anginal chest pain, against about 21% to about 48% for other chest pain types. [C38] Its formula reproduces the part exactly on all 242 training patients. [C41]
Evidence plots: Figure 365, Figure 366, Figure 367, Figure 368, Figure 369.
Figure 368 shows this part's own output, unit h3.

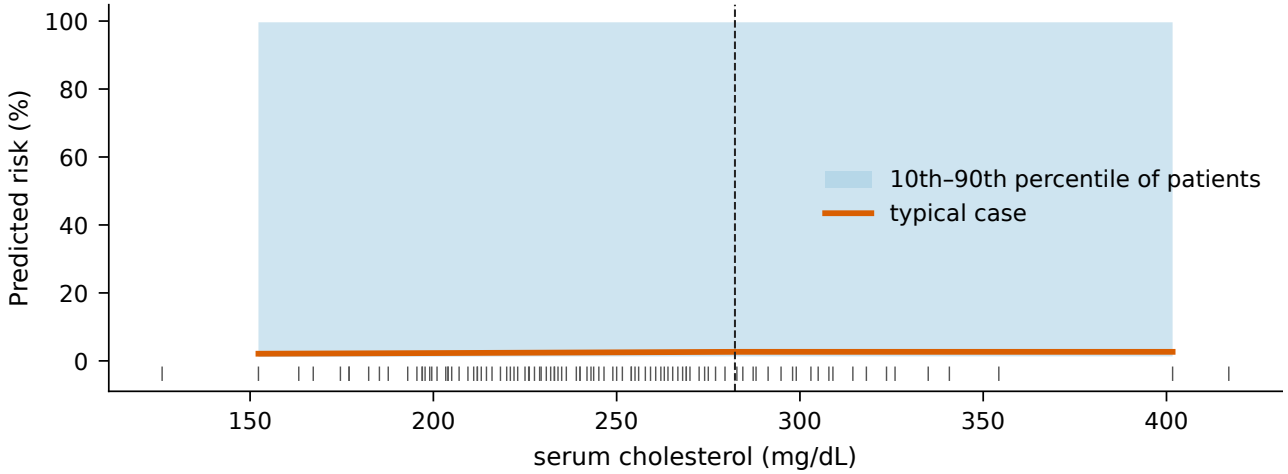


Figure 365: Non-anginal chest pain with cholesterol: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual patients (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

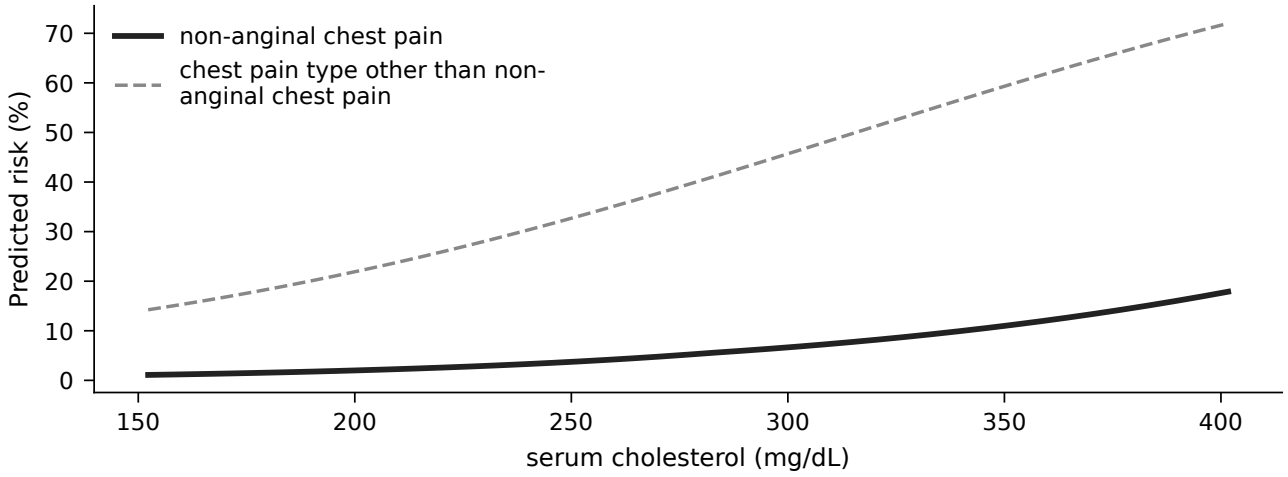


Figure 366: Non-anginal chest pain with cholesterol: How the effect differs between groups: one curve per group, for the typical case of each group.

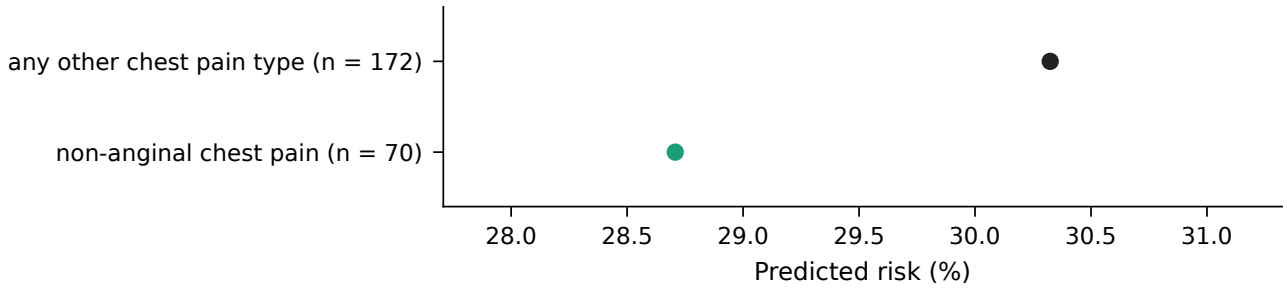


Figure 367: Non-anginal chest pain with cholesterol: The predicted value for the typical case at each level, through this part (n = training patients; grey: fewer than 20).

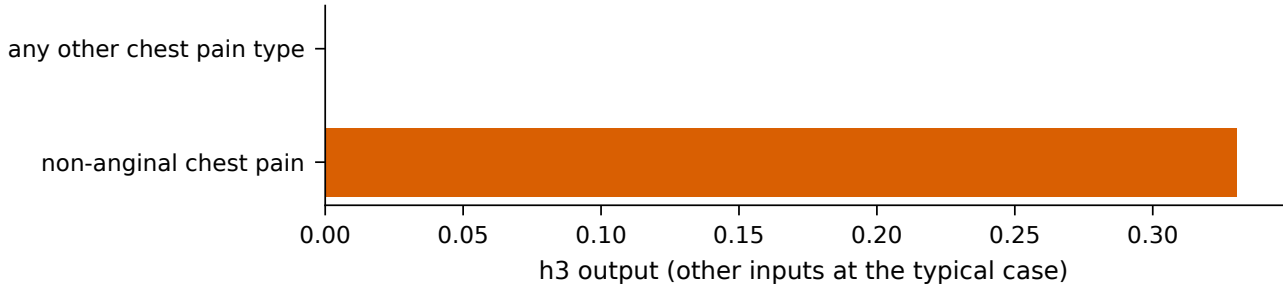


Figure 368: Non-anginal chest pain with cholesterol: This part's own output, unit h3 (the output of 'Non-anginal chest pain with cholesterol'), on its own scale (its activation), with every other input at the typical case; it is active for 61 of 242. Its output for each combination of the levels it reads.

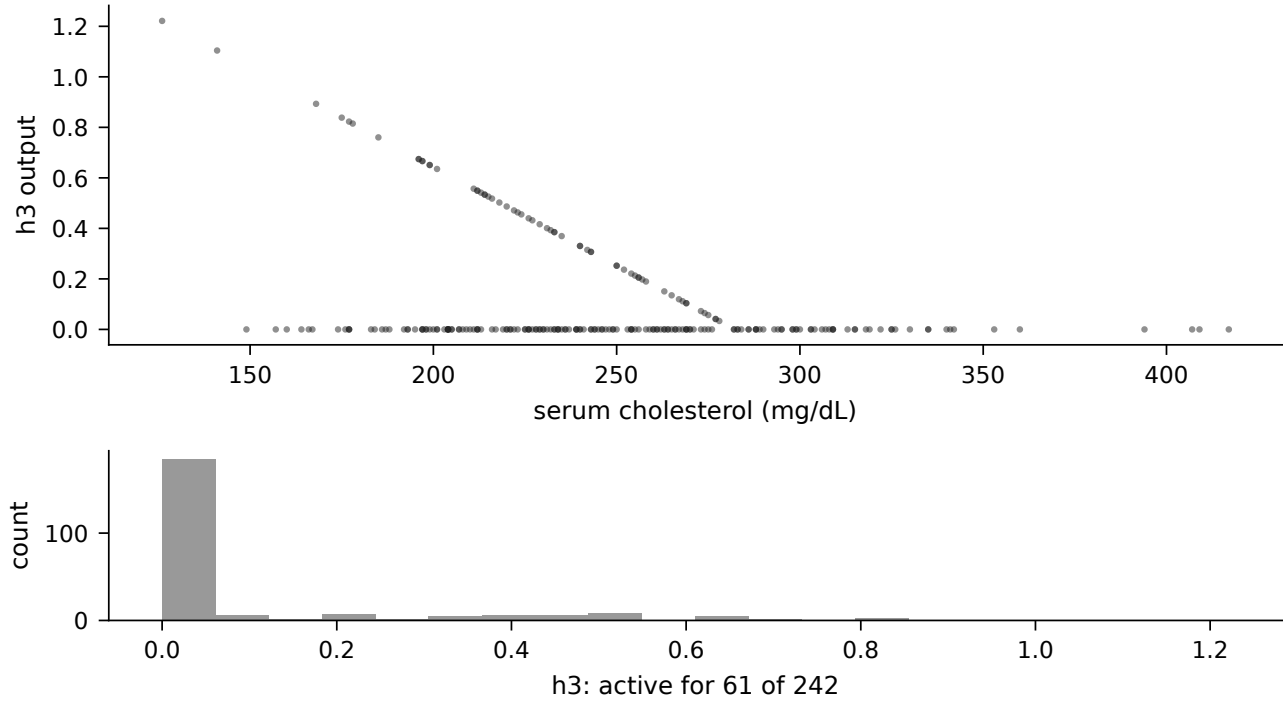


Figure 369: Non-anginal chest pain with cholesterol: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.2.

9.2.4 ST depression gate with little effect

Part of: The whole model (12.2.2).

Figure 370 shows what feeds this part; Figure 371 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

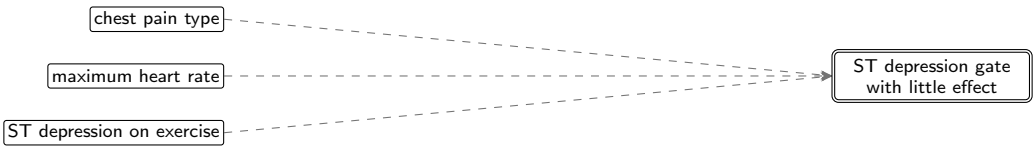


Figure 370: ST depression gate with little effect: what feeds it, one level down.

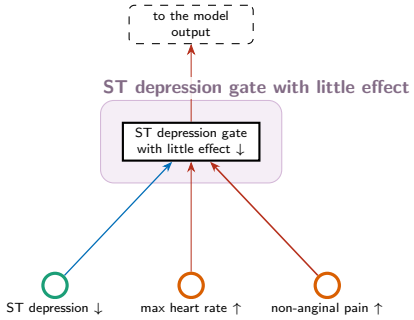


Figure 371: ST depression gate with little effect: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part barely changes the predictions. [C50]

Evidence. It switches on when ST depression on exercise minus about 0.019 × the maximum heart rate exceeds about -1.71, which never happens with non-anginal chest pain; 73 of 242 training patients are on, and 79% of them had heart disease, against 29% of the others. [C52] It changes the predicted risk by at most 0.5 percentage points: through it, ST depression on exercise and maximum heart rate barely move a typical patient's predicted risk, which stays at about 30%. [C50, C42, C43] Its formula reproduces the part exactly on all 242 training patients. [C51]
Evidence plots: Figure 372, Figure 373, Figure 374, Figure 375, Figure 376, Figure 377, Figure 378.

Worked example: reading this part. Figure 374 maps the part's own output over its main inputs. In the grey region it is switched off and adds nothing; past the black line it switches on, and the colour shows how much it adds to the score. The line is the rule in the claim (C52): it switches on when ST depression on exercise minus about 0.019 × maximum heart rate exceed about -1.71. 73 of 242 training patients lie on the 'on' side, and 79% of them had heart disease, against 29% of the others: the filled points crowd on the 'on' side. Figure 375 cuts the map into slices: the hinge moves as the other input changes, which is what a tilted line means. Figure 376 shows what this does to the final prediction for the typical case.
Figure 377 shows this part's own output, unit h2: over its two main inputs, with the line where it switches on.

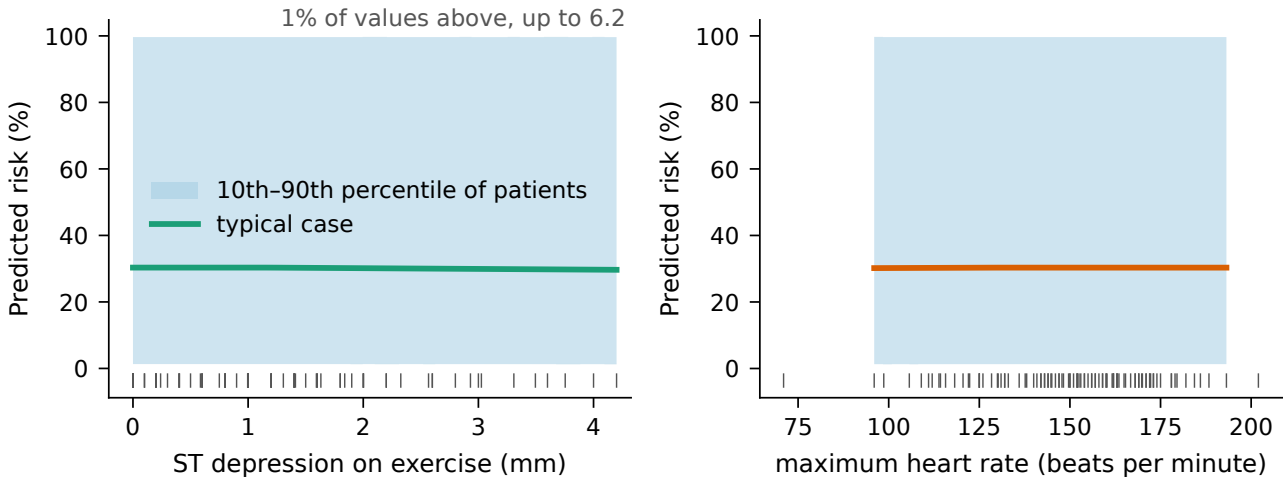


Figure 372: ST depression gate with little effect: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual patients (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

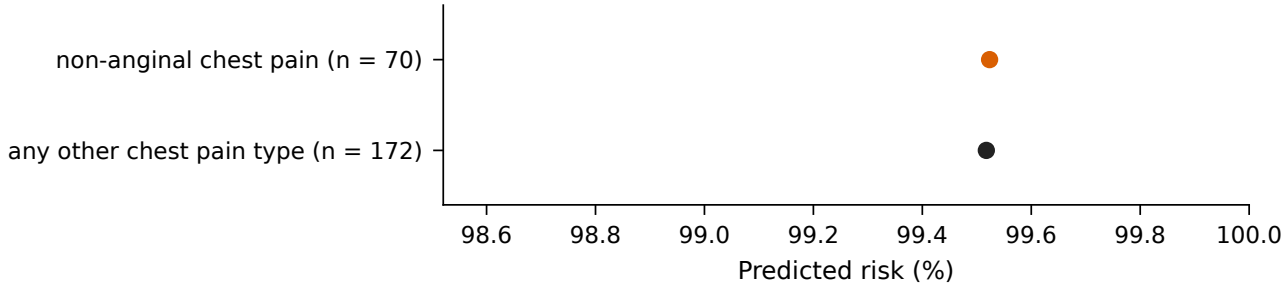


Figure 373: ST depression gate with little effect: The predicted value for the typical case at each level, through this part (n = training patients; grey: fewer than 20).

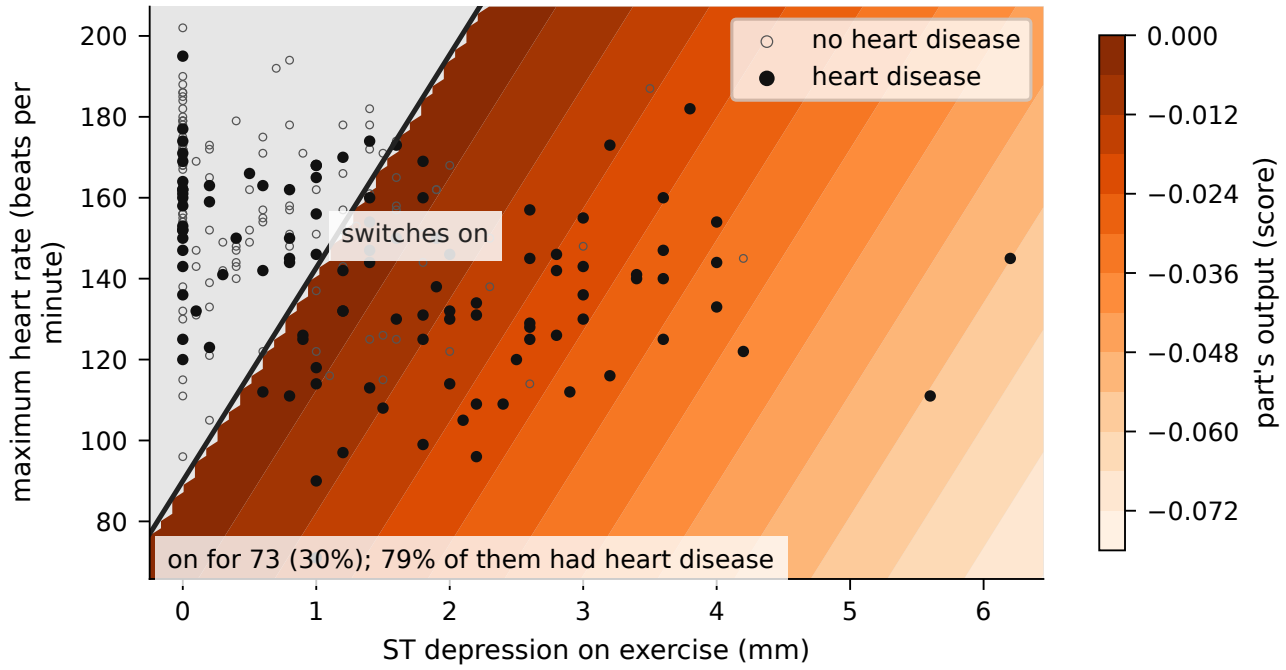


Figure 374: ST depression gate with little effect: The part's own output over its two numeric inputs, other inputs at the typical case: colour where it is on, flat grey where it is switched off, the black line where it switches on; training patients as hollow (no event) and filled (event) points, so the part can be seen catching the events (the predicted-chance overlay was harder to read).

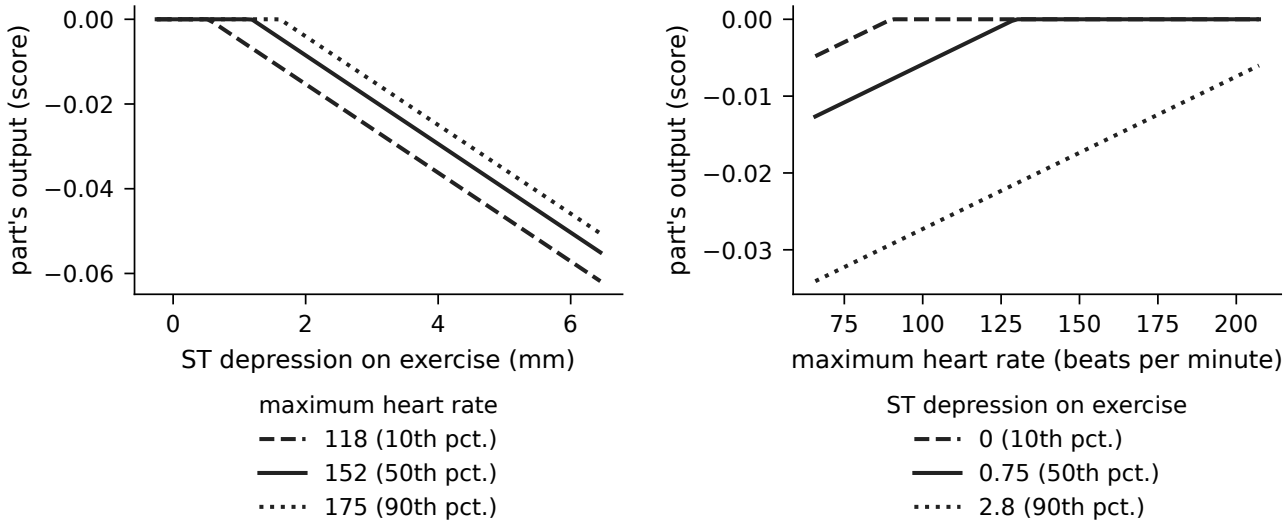


Figure 375: ST depression gate with little effect: Slices of the same map: the part's own output against each input at three values of the other, so the tilted boundary shows as a shifting hinge.

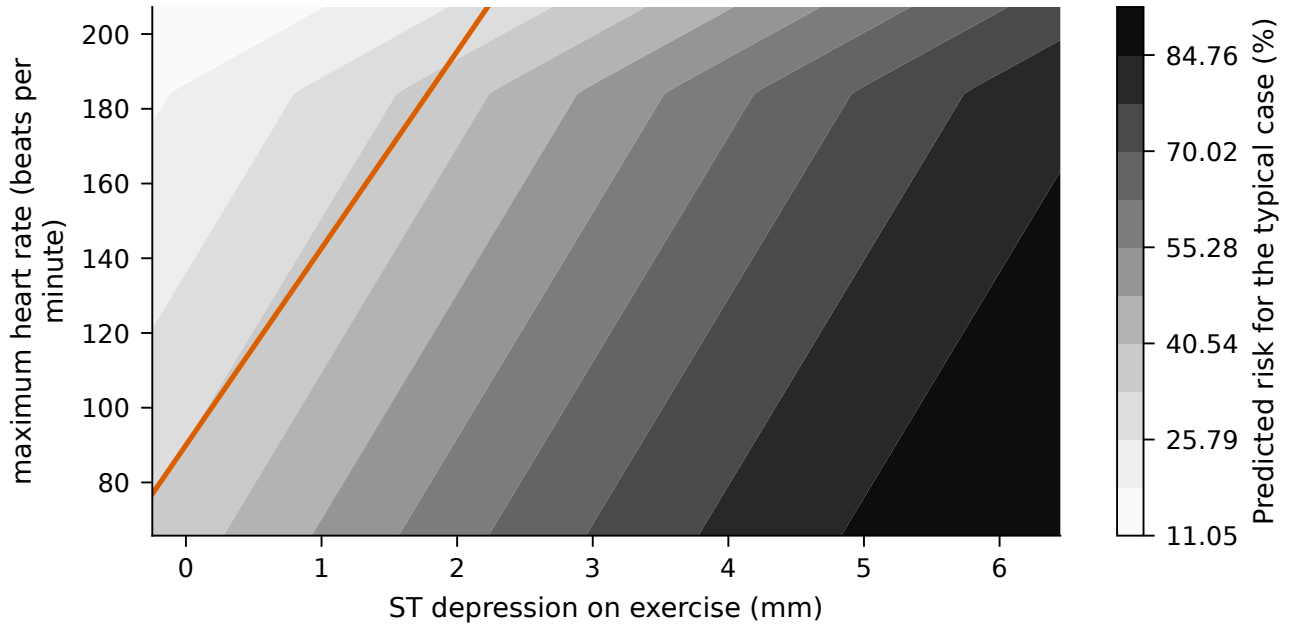


Figure 376: ST depression gate with little effect: The same map on the final scale: the predicted value for the typical case, with the switch-on line (orange).

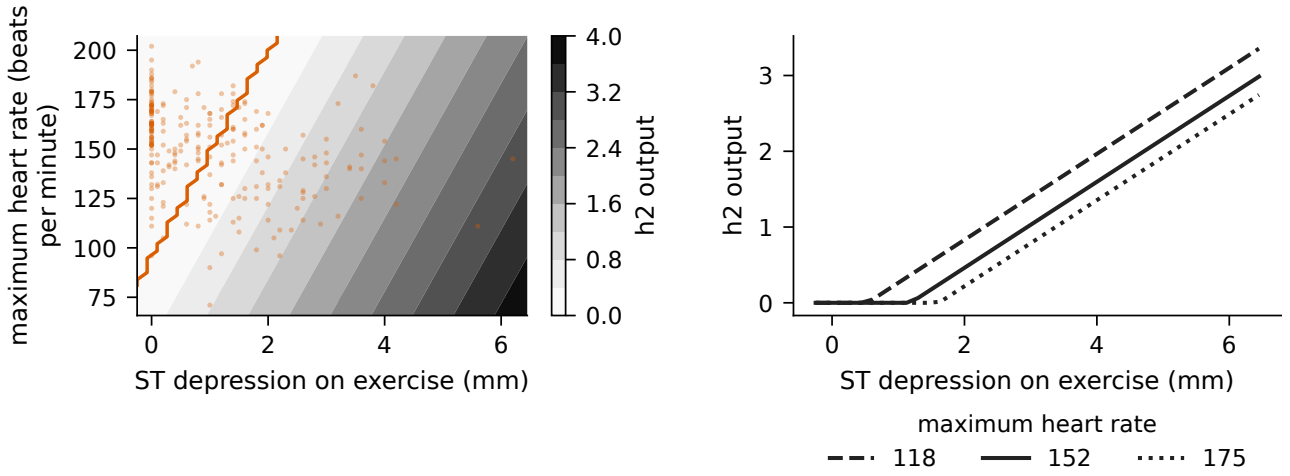


Figure 377: ST depression gate with little effect: This part's own output, unit h2 (the output of 'ST depression gate with little effect'), on its own scale (its activation), with every other input at the typical case; it is active for 73 of 242. Left: its output over its two main inputs (orange line: where it switches on; dots: the data); right: slices at three values of the second input.

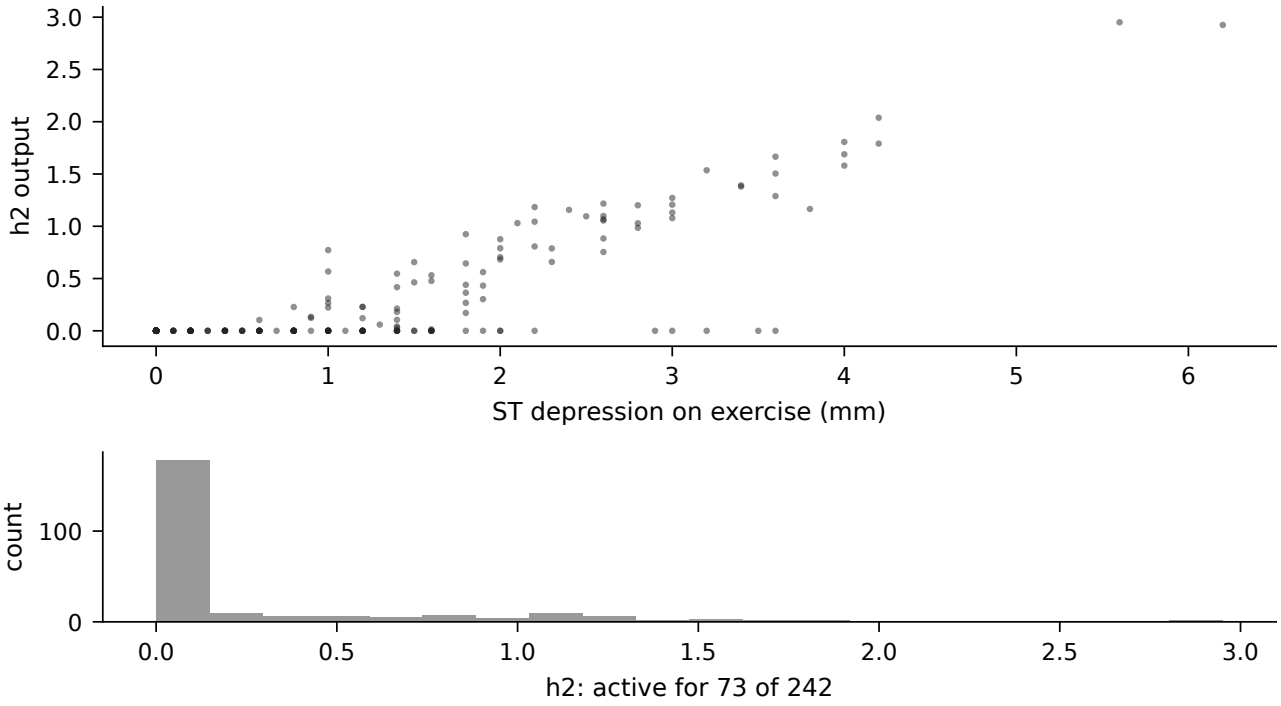


Figure 378: ST depression gate with little effect: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.2.

9.3 Not supported by the model

- C33: threshold claim about maximum heart rate reached during the exercise test in part ann_30: not supported by the model (the effect past the fold is too small or not consistent across patients).
- C45: interaction claim about ST depression during exercise (compared with rest) and chest pain type in part ann_31: not supported by the model (the group difference is below 0.1 log-odds).
- C46: interaction claim about ST depression during exercise (compared with rest) and maximum heart rate reached during the exercise test in part ann_31: not supported by the model (the group difference is below 0.1 log-odds).
- C47: threshold claim about ST depression during exercise (compared with rest) in part ann_31: not supported by the model (the fold changes the curve by only 0.011 log-odds compared with a straight line (needed 0.1); the effect past the fold is too small or not consistent across patients).
- C48: threshold claim about maximum heart rate reached during the exercise test in part ann_31: not supported by the model (the fold changes the curve by only 0.022 log-odds compared with a straight line (needed 0.1); the effect past the fold is too small or not consistent across patients).
- C49: subgroup claim about chest pain type and maximum heart rate reached during the exercise test in part ann_31: not supported by the model (its effect is negligible).

Technical detail: heart-s2-c3

The record behind each part: the original hypothesis, the closed form, and every claim with its estimate (log-odds and odds ratio), range across patients, n and test check.

T.1 The whole model

Annotation ann_33 (heart_disease_logit). Reads: age, sex, chest pain type, resting blood pressure, serum cholesterol, fasting blood sugar, resting ECG, maximum heart rate reached during the exercise test, exercise-induced angina, ST depression during exercise (compared with rest), slope of the ST segment at peak exercise, number of major vessels coloured by fluoroscopy, thallium stress test result.

Original hypothesis (unchanged). Root: p = sigmoid(z), z = 0.323 + LINEAR CORE - 0.511*nonanginal_thalach_gate - 0.018*oldpeak_gate_without_nonanginal - 0.235*nonanginal_low_chol_gate. LINEAR CORE (16 direct input->output edges, z-scored inputs): ca +1.779, cp:4 +0.869, sex:0 -0.760, slope:2 +0.610, oldpeak +0.559, chol +0.529, thal:7 +0.500, trestbps +0.493, slope:1 -0.488, cp:1 -0.393, sex:1 +0.347, thal:6 -0.282, cp:2 +0.273, fbs:1 -0.209, thalach -0.176, restecg:2 +0.053. Nine inputs are unused (age, fbs:0, restecg:0, restecg:1, exang:0, exang:1, slope:3, thal:3, thal:4; zero-effect entries of this root). Non-anginal pain (cp:3) has no direct edge; it acts only through the three children, all of which are switched by it. COMPOSITION: purely additive in the logit. The children share no hidden nodes and are coupled only through shared inputs (cp:3 feeds all three; thalach, oldpeak and chol also have direct edges). (1) nonanginal_thalach_gate: open for 97% of non-anginal-pain patients, lowers their risk, more with higher max heart rate; (2) nonanginal_low_chol_gate: open for 87% of non-anginal-pain patients (those with z-chol < 0.76), lowers their risk more the lower the cholesterol; (3) oldpeak_gate_without_nonanginal: a near-null oldpeak hinge for the other patients (readout -0.018). Together (1)+(2) give non-anginal pain an average logit shift of -0.85 and make thalach and chol slopes depend on chest-pain type. EVIDENCE (train, n=242): full model AUC 0.9381, acc 0.888 (compute_performance). Linear core alone AUC 0.934; logit SD 3.67 (core) vs 0.42 (all nonlinear readouts). Zero-ablating one child at a time: thalach gate 0.934, oldpeak gate 0.938, low-chol gate 0.938. Replacing all hidden readouts with one least-squares linear fit in their four inputs gives 0.938; an L2 logistic regression on all 26 inputs gives 0.936. Mean |SHAP| (train, all 242 rows): ca 0.161, cp:4 0.099, sex:0 0.073, slope:2 0.066, thal:7 0.054, slope:1 0.053, oldpeak 0.047, chol 0.041, trestbps 0.040, cp:3 (thalach gate copy) 0.029; unused inputs about 0.001 (estimator noise). Interpretation: logistic regression on 16 inputs plus a non-anginal-pain block whose nonlinearity is small (a linear refit loses nothing on train AUC). Counter-directional signs reported, not explained: cp:2 +0.27 and fbs:1 -0.21 in the core, the low-cholesterol protection for non-anginal pain. Compared with the 10-seed heart-s2 record (key 9083, gen 38, a different cluster of the same run): that genome has an age/heart-rate-reserve gate and oldpeak hinges and uses age; this one ignores age entirely and puts its only nonlinear effect on non-anginal chest pain.

Closed form. Read through its parts (their formulas are in their own sections).

Split inputs. To annotate the parts, the explanation split these inputs into copies (a refactoring step that does not change what the model computes); the diagrams draw each input once: cholesterol (2 copies); non-anginal pain (3 copies); ST depression (2 copies); max heart rate (3 copies).

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C01	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (whole model) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	30% to 96%	3.99	54.1	3.99 to 3.99	242	unclear; AUC gain 0.039 [-0.034, 0.120]	yes
C02	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	30% to 80%	2.20	9.04	2.20 to 2.20	115, 110	disagrees; AUC gain -0.005 [-0.050, 0.037]	yes
C03	direction: ST depression on exercise: 0.0 mm to 2.8 mm (direct contribution only, not through any part) *shape: steady; 100% of patients same; verified*	23% to 53%	1.32	3.74	1.32 to 1.32	242	unclear; AUC gain 0.011 [-0.01, 0.038]	yes
C04	direction: ST depression on exercise: 0.0 mm to 2.8 mm (whole model) *shape: steady; 100% of patients same; verified*	23% to 53%	1.31	3.69	1.30 to 1.32	242	unclear; AUC gain 0.011 [-0.01, 0.038]	yes
C05	direction: chest pain type: no chest pain (asymptomatic) to typical angina (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	30% to 2%	-3.11	0.04	-3.11 to -3.11	108, 22	agrees; AUC gain 0.038 [0.011, 0.072]	yes
C06	direction: chest pain type: no chest pain (asymptomatic) to typical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	30% to 2%	-3.11	0.04	-3.11 to -3.11	108, 22	agrees; AUC gain 0.055 [0.014, 0.101]	yes
C07	direction: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L) (direct contribution only, not through any part) *shape: steady; 100% of patients same; verified*	21% to 48%	1.26	3.54	1.26 to 1.26	242	disagrees; AUC gain -0.012 [-0.044, 0.013]	yes
C08	direction: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L) (whole model) *shape: steady; 100% of patients same; verified*	21% to 48%	1.31	3.71	1.26 to 1.43	242	disagrees; AUC gain -0.012 [-0.044, 0.013]	yes
C09	direction: chest pain type: no chest pain (asymptomatic) to non-anginal chest pain (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	30% to 3%	-2.51	0.08	-2.96 to -2.01	108, 70	agrees; AUC gain 0.055 [0.014, 0.101]	yes
C10	direction: sex: male to female (whole model) *shape: categorical; highest to lowest: male, female*	30% to 4%	-2.33	0.10	-2.33 to -2.33	159, 83	unclear; AUC gain 0.025 [-0.011, 0.071]	yes
C11	direction: resting blood pressure: 110 mm Hg to 155 mm Hg (whole model) *shape: steady; 100% of patients same; verified*	20% to 46%	1.22	3.39	1.22 to 1.22	242	unclear; AUC gain 0.004 [-0.019, 0.032]	yes
C12	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan to a reversible defect on the thallium scan (whole model) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan, a fixed defect on the thallium scan*	30% to 55%	1.03	2.81	1.03 to 1.03	137, 90	unclear; AUC gain 0.020 [-0.008, 0.054]	yes
C13	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a downsloping ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	30% to 54%	0.98	2.66	0.98 to 0.98	115, 17 (small n)	disagrees; AUC gain -0.005 [-0.050, 0.037]	yes
C14	direction: chest pain type: no chest pain (asymptomatic) to non-anginal chest pain (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	30% to 7%	-1.75	0.17	-1.75 to -1.75	108, 70	agrees; AUC gain 0.038 [0.011, 0.072]	yes
C15	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan to a fixed defect on the thallium scan (whole model) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan, a fixed defect on the thallium scan*	30% to 12%	-1.17	0.31	-1.17 to -1.17	137, 15 (small n)	unclear; AUC gain 0.020 [-0.008, 0.054]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C16	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	30% to 13%	-1.03	0.36	-1.03 to -1.03	108, 42	agrees; AUC gain 0.038 [0.011, 0.072]	yes
C17	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	30% to 13%	-1.03	0.36	-1.03 to -1.03	108, 42	agrees; AUC gain 0.055 [0.014, 0.101]	yes
C18	direction: fasting blood sugar: fasting blood sugar of 120 mg/dL or less to fasting blood sugar above 120 mg/dL (whole model) *shape: categorical; highest to lowest: fasting blood sugar of 120 mg/dL or less, fasting blood sugar above 120 mg/dL*	30% to 19%	-0.61	0.54	-0.61 to -0.61	209, 33	unclear; AUC gain 0.001 [-0.012, 0.016]	yes
C19	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (direct contribution only, not through any part) *shape: steady; 100% of patients same; verified*	36% to 27%	-0.45	0.64	-0.45 to -0.45	242	unclear; AUC gain 0.005 [-0.004, 0.020]	yes
C20	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (whole model) *shape: accelerating; 100% of patients same; verified*	36% to 27%	-0.70	0.50	-1.33 to -0.44	242	unclear; AUC gain 0.012 [-0.002, 0.034]	yes
C21	direction: resting ECG: resting ECG other than signs of left ventricular hypertrophy on the resting ECG to signs of left ventricular hypertrophy on the resting ECG (whole model) *shape: categorical; highest to lowest: signs of left ventricular hypertrophy on the resting ECG, resting ECG other than signs of left ventricular hypertrophy on the resting ECG*	30% to 33%	0.11	1.11	0.11 to 0.11	123, 119	disagrees; AUC gain -0.002 [-0.010, 0.000]	yes
C22	magnitude: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels, typical patient *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	30% to 96%	3.99	54.1		242	unclear; AUC gain 0.039 [-0.034, 0.120]	yes
C23	magnitude: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment, typical patient *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	30% to 80%	2.20	9.04		115, 110	disagrees; AUC gain -0.005 [-0.050, 0.037]	yes
C24	magnitude: ST depression on exercise: 0.0 mm to 2.8 mm, typical patient *shape: steady; 100% of patients same; verified*	23% to 53%	1.30	3.68		242	unclear; AUC gain 0.011 [-0.01, 0.038]	yes
C25	magnitude: chest pain type: no chest pain (asymptomatic) to typical angina, typical patient *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	30% to 2%	-3.11	0.04		108, 22	agrees; AUC gain 0.055 [0.014, 0.101]	yes
C26	magnitude: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L), typical patient *shape: steady; 100% of patients same; verified*	21% to 48%	1.26	3.54		242	disagrees; AUC gain -0.012 [-0.044, 0.013]	yes
C27	null: model ignores age (exact rule)		0.00			242	train only	yes
C28	null: model ignores exercise-induced angina (exact rule)		0.00			242	train only	yes
C29	structure: made of nonanginal_thalach_gate, oldpeak_gate_without_nonanginal, nonanginal_low_chol_gate and inputs sex, cp, trestbps, chol, fbs, restecg, thalach, oldpeak, slope, ca, thal; added					242	train only	yes

T.2 Non-anginal chest pain with heart rate

Annotation ann_30 (nonanginal_thalach_gate). Reads: chest pain type, maximum heart rate reached during the exercise test.

Original hypothesis (unchanged). Non-anginal-pain gate (protective). 2472 = ReLU(-0.348 + 1.076*cp:3 + 0.678*thalach), read out -0.511 into the logit. Hypothesis: this gate is how the model gives non-anginal chest pain (cp:3, which has no direct edge) a lower risk: it is open for nearly every non-anginal-pain patient and almost never otherwise, and once open a higher achieved max heart rate lowers the risk further (on top of thalach's small direct -0.176 edge). EVIDENCE (train, n=242): active for 97% of the 70 non-anginal-pain patients (closed only for z-thalach < -1.97) and 5% of the 172 others (only z-thalach > 1.53). Disease rate 20% when active vs 55% when inactive. For a non-anginal-pain patient it adds -0.684 - 0.346*z-thalach to the logit (mean -0.76). Mean |sensitivity| cp:3 0.34 > thalach 0.21. Zero-ablating its readout lowers train AUC 0.938 -> 0.934, the only nonlinear part that matters; a linear refit in (cp:3, thalach) recovers 0.938 (R2 0.87), so the gate acts almost as a cp:3 main effect plus a cp:3-specific thalach slope. Supported.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 4.44089e-16).

$$y_0 = \mathrm{relu}\big(-0.348 + 1.08 x_{cp:3_a} + 0.678 x_{thalach_b}\big)$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C30	direction: chest pain type: chest pain type other than non-anginal chest pain to non-anginal chest pain (through this part) *shape: categorical; highest to lowest: chest pain type other than non-anginal chest pain, non-anginal chest pain*	30% to 17%	-0.69	0.50	-1.07 to -0.19	172, 70	unclear; AUC gain 0.008 [-0.009, 0.028]	yes
C31	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (through this part) *shape: hinge at 106 beats per minute; 29% of patients same; fallback; risk saturates near 0%*	4% to 2% (typical patient this part acts on)	-0.25	0.78	-0.88 to 0.00	242	unclear; AUC gain 0.006 [0.000, 0.020]	yes
C32	interaction: effect of maximum heart rate (118 beats per minute to 175 beats per minute) by chest pain type: non-anginal chest pain minus chest pain type other than non-anginal chest pain; stronger in non-anginal chest pain *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	non-anginal chest pain: 7% to 2%; chest pain type other than non-anginal chest pain: 36% to 27%	-0.89	0.41		70, 172	agrees; AUC gain 0.008 [-0.009, 0.028]	yes
C33	threshold: maximum heart rate: fold at about 184 beats per minute (starts) for patients with no chest pain symptoms (asymptomatic) *shape: step at 190 beats per minute; 100% of patients same; verified; risk saturates near 100%*	99% to 99% (typical patient with a fold (patients with no chest pain symptoms (asymptomatic)))	0.00	1.00	0.00 to 0.00	108, 3 (small n)	unclear; AUC gain 0.006 [0.000, 0.020]	no
C34	subgroup: who this part acts on: 76 of 242 patients	5% to 2% (typical patient this part acts on (risk without the part, then with it))	-0.71	0.49	-1.06 to -0.14	76, 166	unclear; AUC gain 0.013 [-0.001, 0.038]	yes
C35	structure: its formula reproduces it on 242 training patients (largest difference 4.44089e-16)		4e-16			242	train only	yes

T.3 Non-anginal chest pain with cholesterol

Annotation ann_32 (nonanginal_low_chol_gate). Reads: chest pain type, serum cholesterol.

Original hypothesis (unchanged). Non-anginal-pain, low-cholesterol gate (protective, small). 2863 = ReLU(-1.255 + 0.983*cp:3 - 0.376*chol), read out -0.235. Hypothesis: open only for non-anginal-pain patients whose cholesterol is not high (z-chol below 0.76); it lowers their risk, by more the lower the cholesterol, so for these patients cholesterol's direct +0.529 edge is partly undone. EVIDENCE (train, n=242): active for 87% of the 70 non-anginal-pain patients and none of the others. Disease rate 21% when active vs 52% when inactive. Readout up to -0.29 logit units (SD 0.053). Mean |sensitivity| cp:3 0.25 > chol 0.09. Zero-ablating it leaves train AUC unchanged (0.9381 -> 0.9382): it shifts the level of the non-anginal-pain group rather than ranking patients. Supported but small.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 2.22045e-16).

$$y_0 = \mathrm{relu}\big(-1.25 + 0.983 x_{cp:3_c} - 0.376 x_{chol_b}\big)$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C36	direction: chest pain type: chest pain type other than non-anginal chest pain to non-anginal chest pain (through this part) *shape: categorical; highest to lowest: chest pain type other than non-anginal chest pain, non-anginal chest pain*	30% to 29%	-0.08	0.92	-0.16 to 0.00	172, 70	unclear; AUC gain 0.002 [0.000, 0.009]	yes
C37	direction: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L) (through this part) *shape: hinge at 282 mg/dL (7.3 mmol/L); 100% of patients same; verified; risk saturates near 0%; one point with C39 (median fold 282 mg/dL (7.3 mmol/L), typical patient 283 mg/dL (7.3 mmol/L), 10th-90th percentile 282 mg/dL (7.3 mmol/L) to 282 mg/dL (7.3 mmol/L))*	2% to 3% (typical patient this part acts on)	0.05	1.05	0.00 to 0.16	242	disagrees; AUC gain -0.001 [-0.006, 0.000]	yes
C38	interaction: effect of serum cholesterol (193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L)) by chest pain type: non-anginal chest pain minus chest pain type other than non-anginal chest pain; stronger in non-anginal chest pain *shape: steady; 100% of patients same; fallback; risk saturates near 0%*	non-anginal chest pain: 2% to 7%; chest pain type other than non-anginal chest pain: 21% to 48%	0.16	1.18		70, 172	agrees; AUC gain 0.002 [0.000, 0.009]	yes
C39	threshold: serum cholesterol: fold at about 282 mg/dL (7.3 mmol/L) (stops) for patients with non-anginal chest pain *shape: hinge at 282 mg/dL (7.3 mmol/L); 100% of patients same; verified; risk saturates near 0%; one point with C39 (median fold 282 mg/dL (7.3 mmol/L), typical patient 283 mg/dL (7.3 mmol/L), 10th-90th percentile 282 mg/dL (7.3 mmol/L) to 282 mg/dL (7.3 mmol/L))*	2% to 3% (typical patient with a fold)	0.19	1.21	0.19 to 0.19	70, 61	disagrees; AUC gain -0.001 [-0.006, 0.000]	yes
C40	subgroup: who this part acts on: 61 of 242 patients	3% to 2% (typical patient this part acts on (risk without the part, then with it))	-0.10	0.91	-0.18 to -0.02	61, 181	unclear; AUC gain 0.000 [0.000, 0.000]	yes
C41	structure: its formula reproduces it on 242 training patients (largest difference 2.22045e-16)		2e-16			242	train only	yes

T.4 ST depression gate with little effect

Annotation ann_31 (oldpeak_gate_without_nonanginal). Reads: chest pain type, maximum heart rate reached during the exercise test, ST depression during exercise (compared with rest).

Original hypothesis (unchanged). ST-depression gate for patients without non-anginal pain (negligible). 2596 = ReLU(-0.688 - 0.985*cp:3 - 0.241*thalach + 0.673*oldpeak), read out -0.018. Hypothesis: a hinge on ST depression (and lower max heart rate) for patients without non-anginal pain, whose readout is too small to change predictions. EVIDENCE (train, n=242): active for 42% of patients without non-anginal pain and never with it; disease rate 80% when active vs 29% when inactive, so it tracks risk, but its readout has SD 0.009 logit units against 3.67 for the linear core, and removing it changes train AUC by +0.0001. Its sign (slightly lowering the risk with more ST depression) runs against oldpeak's direct +0.559 edge, which dominates. Supported as a near-null part; a vestige rather than a mechanism.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 2.22045e-16).

$$y_0 = \mathrm{relu}\big(-0.688 + 0.673x_{oldpeak_b} - 0.241x_{thalach_c} - 0.985x_{cp:3_b}\big)$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C42	direction: ST depression on exercise: 0.0 mm to 2.8 mm (through this part) *shape: flat; 100% of patients same; verified*	30% to 30%	-0.01	0.99	-0.02 to 0.00	242	unclear; AUC gain 0.000 [0.000, 0.000]	yes
C43	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (through this part) *shape: flat; 100% of patients same; verified*	30% to 30%	0.004	1.00	0.00 to 0.01	242	unclear; AUC gain 0.000 [0.000, 0.000]	yes
C44	direction: chest pain type: chest pain type other than non-anginal chest pain to non-anginal chest pain (through this part) *shape: categorical; highest to lowest: non-anginal chest pain, chest pain type other than non-anginal chest pain*	>99% to >99% (typical patient this part acts on)	0.005	1.01	0.00 to 0.02	172, 70	unclear; AUC gain 0.000 [0.000, 0.000]	yes
C45	interaction: effect of ST depression on exercise (0.0 mm to 2.8 mm) by chest pain type: non-anginal chest pain minus chest pain type other than non-anginal chest pain; stronger in non-anginal chest pain *shape: steady; 100% of patients same; verified; risk saturates near 0%*	non-anginal chest pain: 2% to 8%; chest pain type other than non-anginal chest pain: 23% to 53%	0.02	1.02		70, 172	agrees; AUC gain 0.000 [0.000, 0.000]	no
C46	interaction: effect of ST depression on exercise (0.0 mm to 2.8 mm) by maximum heart rate: maximum heart rate above 152 beats per minute minus maximum heart rate at or below 152 beats per minute; stronger in maximum heart rate above 152 beats per minute *shape: steady; 100% of patients same; verified*	maximum heart rate above 152 beats per minute: 21% to 50%; maximum heart rate at or below 152 beats per minute: 26% to 56%	0.007	1.01		121, 121	agrees; AUC gain 0.000 [0.000, 0.000]	no
C47	threshold: ST depression on exercise: fold at about 1.0 mm (starts) for patients with no chest pain symptoms (asymptomatic) *shape: flat; 100% of patients same; verified; risk saturates near 100%*	99% to 99% (typical patient with a fold (patients with no chest pain symptoms (asymptomatic)))	-0.03	0.97	-0.03 to -0.02	106, 58	unclear; AUC gain 0.000 [0.000, 0.000]	no
C48	threshold: maximum heart rate: fold at about 90 beats per minute (stops) for patients with atypical angina *shape: flat; 100% of patients same; verified*	9% to 9% (typical patient with a fold (patients with atypical angina))	0.002	1.00	0.00 to 0.008	42, 3 (small n)	unclear; AUC gain 0.000 [0.000, 0.000]	no
C49	subgroup: who this part acts on: 73 of 242 patients	>99% to >99% (typical patient this part acts on (risk without the part, then with it))	-0.02	0.98	-0.03 to -0.002	73, 169	unclear; AUC gain 0.000 [0.000, 0.000]	no
C50	null: this part barely changes predictions (switched off)		0.005			242	train only	yes
C51	structure: its formula reproduces it on 242 training patients (largest difference 2.22045e-16)		2e-16			242	train only	yes
C52	boundary: boundary		1.00			242	train only	yes

The network before any explanation

Figure 379 (next page, landscape) is the trained network as it is, with no parts, no splits and no inserted units: inputs at the bottom, hidden units by depth, the output at the top, every connection drawn, every input column its own node (one-hot levels included).

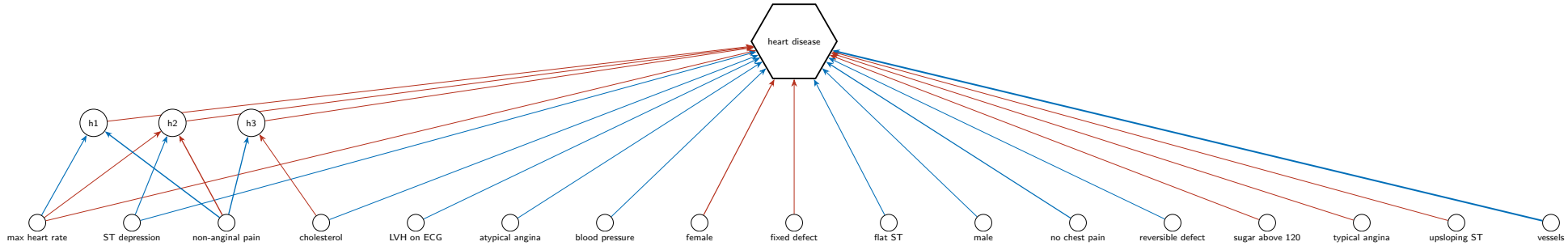


Figure 379: The network as trained, before any explanation, in full. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. Not used (no connection): age, sugar 120 or less, normal ECG, ST-T abnormal ECG, no exercise pain, exercise pain, downsloping ST, normal thallium, thallium code 4.

10 heart-s4-c1

Genome `c85804fa-08bd-4207-ac69-a31a8567473c`, explanation `10565a81-f55b-47f4-b51e-21641ec08556`. Split: train primary (n=242), test as a check (n=61). Claims: 47, verified 45, not supported 2.
AI-assisted draft for review (unverified)

Reference (typical) patient: the typical patient: the median training patient for every number and the most common training category for every category (not a real patient). age: 56 years; ca: 0 vessels; chol: 240 mg/dL (6.2 mmol/L); cp: no chest pain (asymptomatic); exang: no chest pain on exercise; fbs: fasting blood sugar of 120 mg/dL or less; oldpeak: 0.8 mm; restecg: a normal resting ECG; sex: male; slope: an upsloping ST segment; thal: a normal thallium scan; thalach: 152 beats per minute; trestbps: 130 mm Hg. Predicted risk of heart disease: 24%.

10.1 Summary

For a typical patient (a 56-year-old man with no chest pain symptoms and median test results) the model predicts a risk of heart disease of about 24%; it adds up most of its inputs independently, and four small parts add to that sum: one for a fixed defect on the thallium scan with normal fasting blood sugar, one for chest pain type, one for a downsloping ST segment and one for maximum heart rate. [C31] Figure 380 shows the model at this level: one box per paragraph below.

How to read the diagrams. Edge colour is the sign of that one connection: blue positive, red negative; thicker is a larger weight. The outline and arrow on an input (and on a part's output box) is its overall effect on the final prediction, read from the verified claims: orange and up arrow raises it, teal and down arrow lowers it, grey and double arrow depends (it goes both ways, or differs between levels). A negative connection into a part that itself raises the prediction means that input lowers the prediction; through a hinge the overall effect can only be read from the claims, not from the edges. Up arrow means towards heart disease; down arrow, away from it. A double outline marks an input (or a part's output) that acts at a fixed rate everywhere: it adds the same amount to the score per unit, so it is explained by its rate alone (see 'Linear factors'); a dashed double outline, at a fixed rate on the observed data. The normal outline is an input whose effect switches on and off somewhere. For example, no chest pain has a negative connection into 'Chest pain offset', a part that lowers the predicted risk of heart disease, so no chest pain raises it.

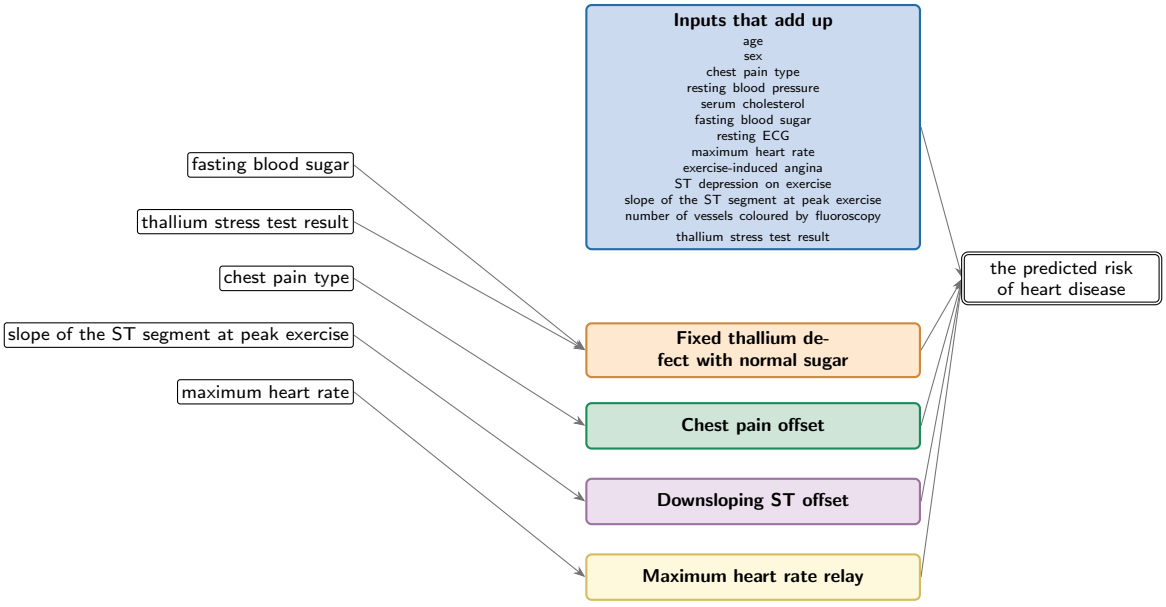


Figure 380: The model at the level of the summary: one box per part.

Inputs that add up. *Lowering the risk.* These add up independently, and these lower the risk: being a woman takes a typical patient's predicted risk from about 24% to about 1%; through this sum alone, compared with no chest pain symptoms, typical angina lowers it to about 2%, non-anginal chest pain to about 3% and atypical angina to about 14%, fasting blood sugar above 120 mg/dL lowers it to about 7%, and a higher maximum heart rate lowers it steadily, from about 27% at 118 to about 21% at 175 beats per minute; chest pain type, fasting blood sugar and maximum heart rate also act through the parts below. [C12, C13, C15, C22, C17, C24]
Raising the risk. Each extra vessel coloured by fluoroscopy adds about the same, from about 24% at 0 to about 98% at 2 vessels, flattening out near the top because the predicted risk cannot go much higher; through this sum

alone a flat rather than upsloping ST segment raises it to about 82% and any thallium result other than a normal scan to about 48%, and chest pain brought on by exercise raises it to about 36% (the ST segment and the thallium scan also act through the parts below). [C01, C02, C06, C21] Higher cholesterol (from about 14% at 193 mg/dL to about 44% at 308 mg/dL), more ST depression on exercise (from about 18% at 0 mm to about 42% at 2.8 mm), higher resting blood pressure (from about 14% at 110 mm Hg to about 41% at 155 mm Hg) and older age (from about 21% at 42 to about 25% at 66 years) raise it steadily. [C04, C08, C05, C25]
Rare results. Through this sum alone a downsloping ST segment raises it to about 39% (the model has little data on this, n = 17), and an ST-T wave abnormality on the resting ECG raises it to about 38% (the model has little data on this, n = 2). [C19, C20]

Fixed thallium defect with normal sugar. This part reads fasting blood sugar and the thallium scan and acts on 10 of 242 training patients, nearly all of them with a fixed defect on the thallium scan, 5 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 94% without it and about 24% with it (the model has little data on this, n = 10). [C35] Through it, a fixed defect on the thallium scan lowers a typical patient's predicted risk from about 24% to about 2% if fasting blood sugar is 120 mg/dL or less, but the other way if it is above 120 mg/dL, from about 7% to about 18% (the model has little data on this, n = 15). [C34] For the typical patient it acts on, fasting blood sugar above 120 mg/dL raises the risk through it from about 24% to about 94%. [C32]

Chest pain offset. This part reads only chest pain type and acts on the 134 of 242 training patients who have some chest pain symptom, 30 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 2% without it and about 1% with it. [C38] Through it, any chest pain type other than no chest pain symptoms lowers a typical patient's predicted risk from about 24% to about 12%. [C37]

Downsloping ST offset. This part reads only the slope of the ST segment and acts on the 17 of 242 training patients with a downsloping ST segment, 8 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 66% without it and about 73% with it (the model has little data on this, n = 17). [C41] Through it, a downsloping ST segment raises a typical patient's predicted risk from about 24% to about 29% (the model has little data on this, n = 17). [C40]

Maximum heart rate relay. This part reads only maximum heart rate and acts on 190 of 242 training patients, 71 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 21% without it and about 20% with it. [C45] Through it, a higher maximum heart rate changes nothing until about 131 beats per minute, then lowers the predicted risk steadily, for a typical patient from about 24% at 118 to about 23% at 175 beats per minute. [C43, C44]

The whole network. Figure 381 (next page, landscape) draws every unit and connection of the explained model, nested by part: shaded clusters are the parts of the model and darker boxes the parts inside them; circles are hidden units (h1, h2, ...); each cluster's rectangle is its output, and a dashed 'sum' stands for an addition the next unit does (drawn, not a unit of the model); the hexagon is the model's output. An input used by several parts is drawn under each of them (same label, dotted outline, the first copy marked with the number of copies): it is the same input, repeated for readability. Not used by the model (input columns with no connection): atypical angina, normal ECG, LVH on ECG, no exercise pain, thallium code 4, reversible defect.

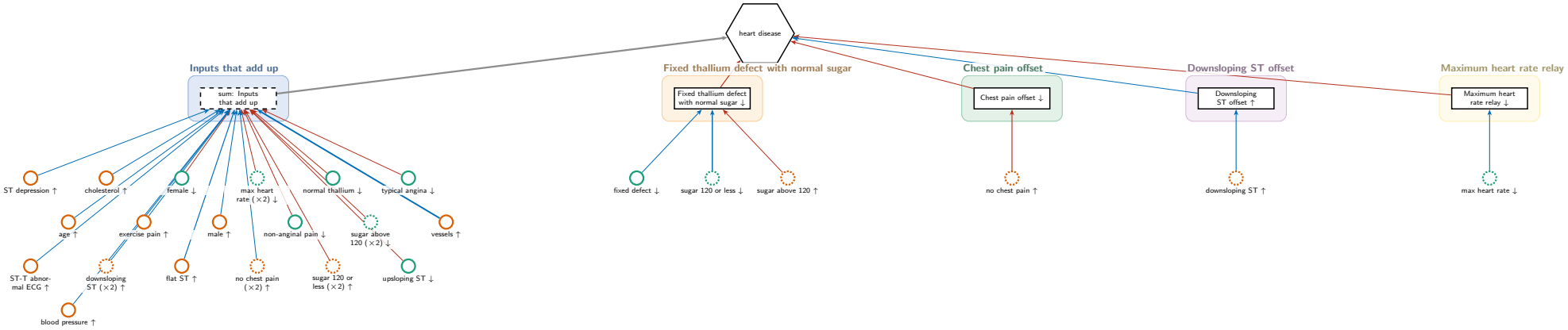


Figure 381: Every unit and connection of the explained model, nested by part. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Worth checking against what is known. Worth checking: going from 0 to 2 vessels coloured moves a typical patient's predicted risk from about 24% to about 98%. [C01] Worth checking: patients with typical angina get the lowest predicted risk of all chest pain types, below patients with no chest pain symptoms. [C10, C11, C16] Worth checking: the model treats fasting blood sugar above 120 mg/dL as lowering the risk of heart disease (for a typical patient from about 24% to about 7%). [C18] Worth checking: through one part, the model treats a fixed defect on the thallium scan as lowering the risk for patients with normal fasting blood sugar (for a typical patient from about 24% to about 2%), from only 15 training patients. [C34]

10.2 The model, read from the top down

Each part of the explanation in turn, from the whole model down to the smallest parts: what it does for the prediction (hypothesis), what was checked (evidence), a diagram one level down, and where to read on. The technical record of each part is in "Technical detail" at the end.

10.2.1 The whole model

Figure 382 shows what feeds this part; Figure 383 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

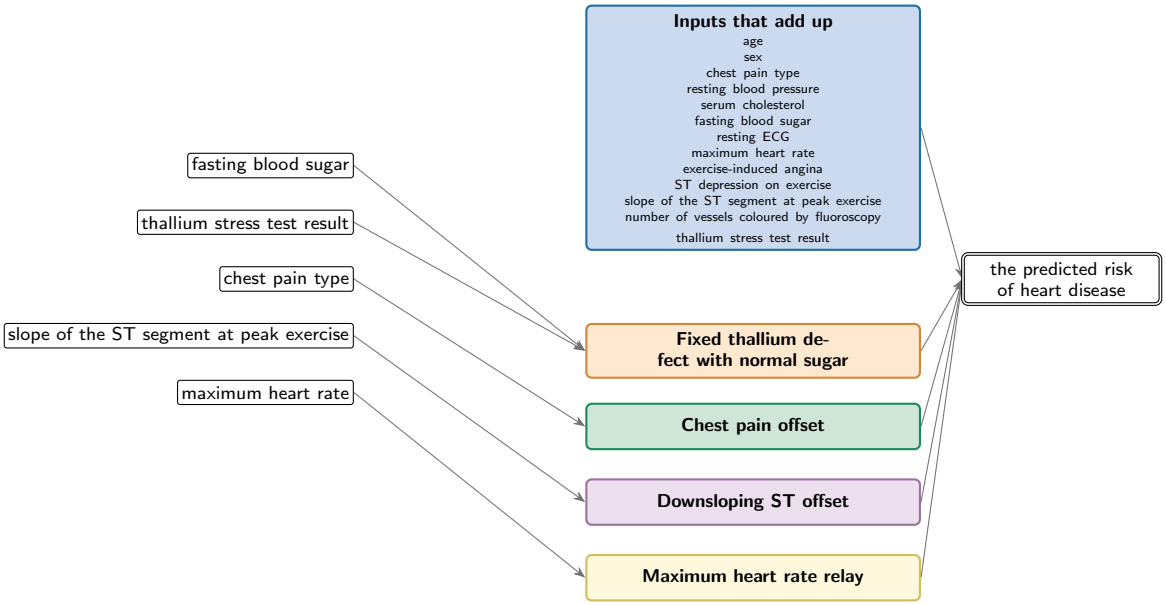


Figure 382: The whole model: what feeds it, one level down.

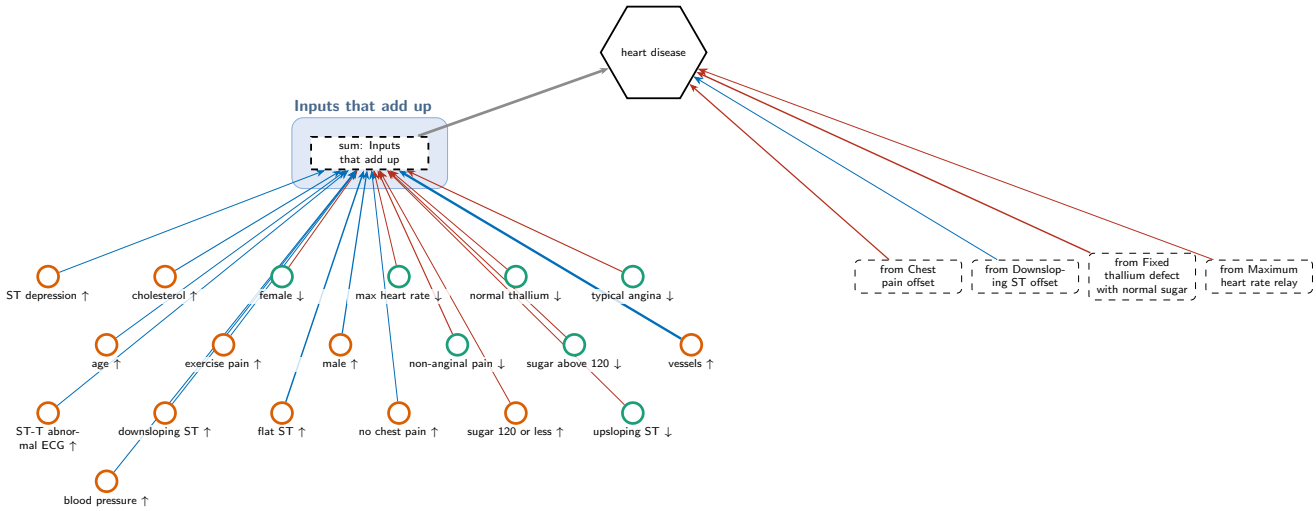


Figure 383: The whole model: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. The model adds up its inputs into one score: the vessels coloured (each extra vessel adding about the same), the ST segment, sex and chest pain type each move the predicted risk on their own, and four small parts add to that sum: one for a fixed thallium defect with normal fasting blood sugar, one for chest pain type, one for a downsloping ST segment and one for maximum heart rate. [C31, C01, C02, C12, C10]

Evidence. It is made up of the inputs that feed the score directly and four parts, the fixed thallium defect part, the chest pain offset, the downsloping ST offset and the maximum heart rate relay (each explained below); the score is their sum, which was checked on all 242 training patients. [C31] Changing one input at a time across the whole model, compared with no chest pain symptoms, typical angina lowers a typical patient's predicted risk from about 24% to about 1%, non-anginal chest pain to about 1% and atypical angina to about 7%, against about 2% for typical angina through the sum alone. [C10, C11, C16, C13] Across the whole model, fasting blood sugar above 120 mg/dL lowers it from about 24% to about 7%, the same as through the sum alone. [C18, C17] Across the whole model, a higher maximum heart rate lowers it slowly at first, then faster, from about 28% at 118 to about 21% at 175 beats per minute, against about 27% to about 21% through the sum alone, where it falls steadily. [C23, C24] Across the whole model, a downsloping rather than upsloping ST segment raises it to about 47% (the model has little data on this, n = 17), and a reversible defect on the thallium scan or a thallium result coded 4 raises it to about 48%. [C09, C07]
Evidence plots: Figure 384.

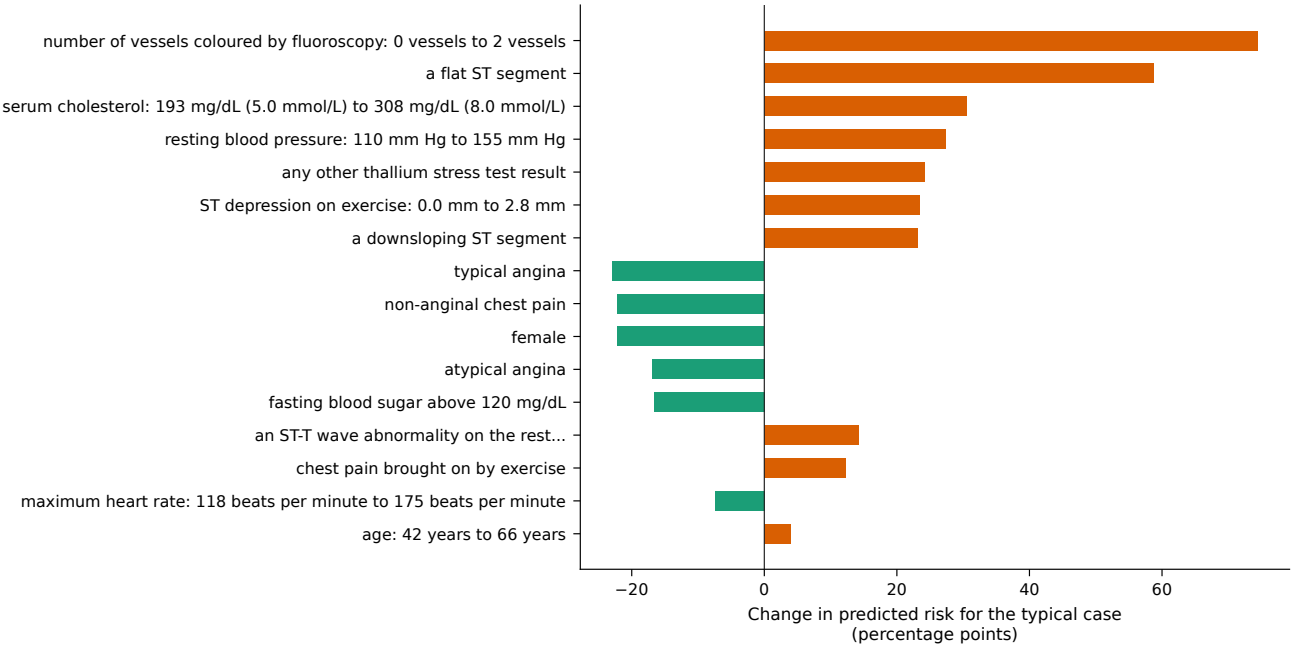


Figure 384: The whole model: Each input's change in the predicted value for the typical case across the whole model, largest first (orange: raises it, teal: lowers it).

Dive deeper: Fixed thallium defect with normal sugar (10.2.2); Chest pain offset (11.2.2); Downsloping ST offset (12.2.2); Maximum heart rate relay (12.2.3). Technical detail: T.4.

10.2.2 Fixed thallium defect with normal sugar

Part of: The whole model (12.2.4).
Figure 385 shows what feeds this part; Figure 386 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

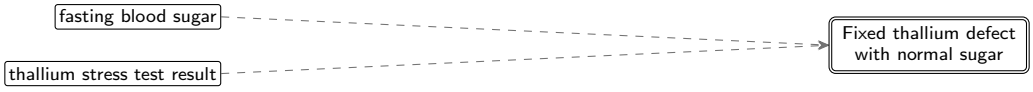


Figure 385: Fixed thallium defect with normal sugar: what feeds it, one level down.

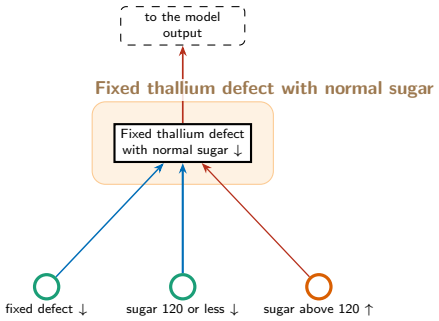


Figure 386: Fixed thallium defect with normal sugar: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part lowers the predicted risk for patients with a fixed defect on the thallium scan when their fasting blood sugar is 120 mg/dL or less (the model has little data on this, n = 15). [C34, C35]

Evidence. It acts on 10 of 242 training patients, nearly all of them with a fixed defect on the thallium scan, 5 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 94% without it and about 24% with it (the model has little data on this, n = 10). [C35] Through it, a fixed defect on the thallium scan lowers a typical patient's predicted risk from about 24% to below 1% (the model has little data on this, n = 15). [C33] Through it, a fixed defect lowers the risk from about 24% to about 2% if fasting blood sugar is 120 mg/dL or less, but raises it from about 7% to about 18% if it is above 120 mg/dL (the model has little data on this, n = 15). [C34] For the typical patient it acts on, fasting blood sugar above 120 mg/dL raises the risk through it from about 24% to about 94%. [C32] Its formula reproduces the part exactly on all 242 training patients. [C36]
Evidence plots: Figure 387, Figure 388, Figure 389.
Figure 388 shows this part's own output, unit h3.

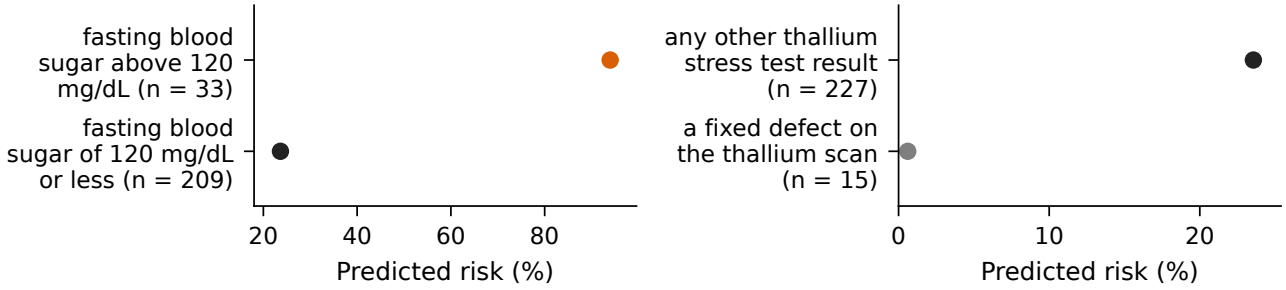


Figure 387: Fixed thallium defect with normal sugar: The predicted value for the typical case at each level, through this part (n = training patients; grey: fewer than 20).

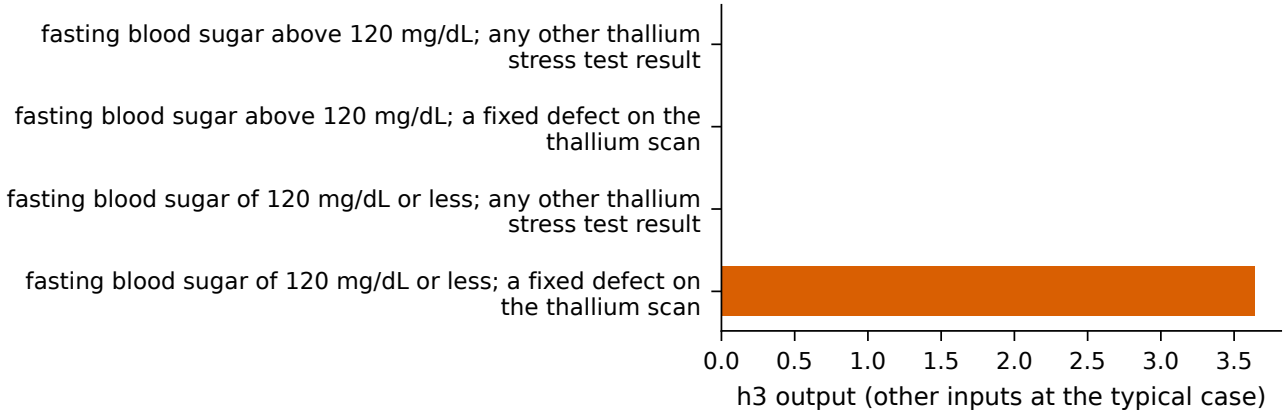


Figure 388: Fixed thallium defect with normal sugar: This part's own output, unit h3 (the output of 'Fixed thallium defect with normal sugar'), on its own scale (its activation), with every other input at the typical case; it is active for 10 of 242. Its output for each combination of the levels it reads.

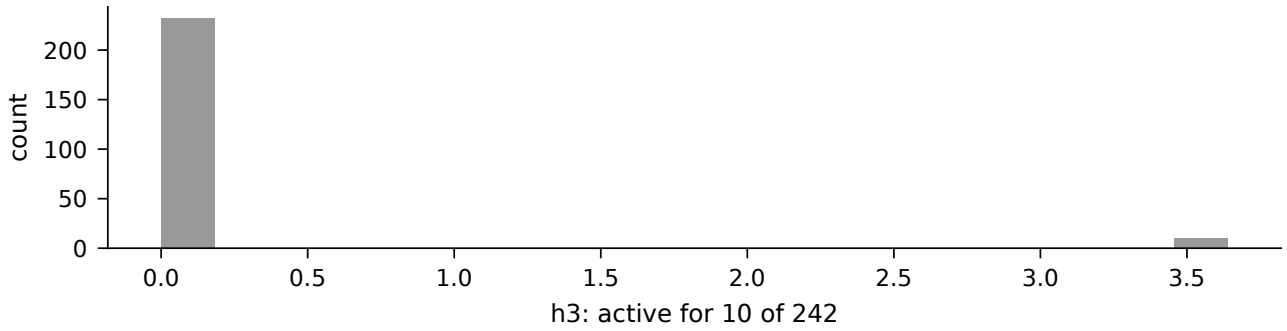


Figure 389: Fixed thallium defect with normal sugar: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.2.

10.2.3 Chest pain offset

Part of: The whole model (12.2.4).
Figure 390 shows what feeds this part; Figure 391 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).



Figure 390: Chest pain offset: what feeds it, one level down.

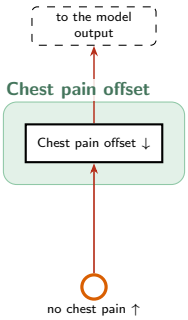


Figure 391: Chest pain offset: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part lowers the predicted risk for patients with any chest pain symptom, compared with patients who have none. [C37]

Evidence. It acts on the 134 of 242 training patients who have some chest pain symptom, 30 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 2% without it and about 1% with it. [C38] Through it, any chest pain type other than no chest pain symptoms takes a typical patient's predicted risk from about 24% to about 12%. [C37] Its formula reproduces the part exactly on all 242 training patients. [C39]
Evidence plots: Figure 392, Figure 393.

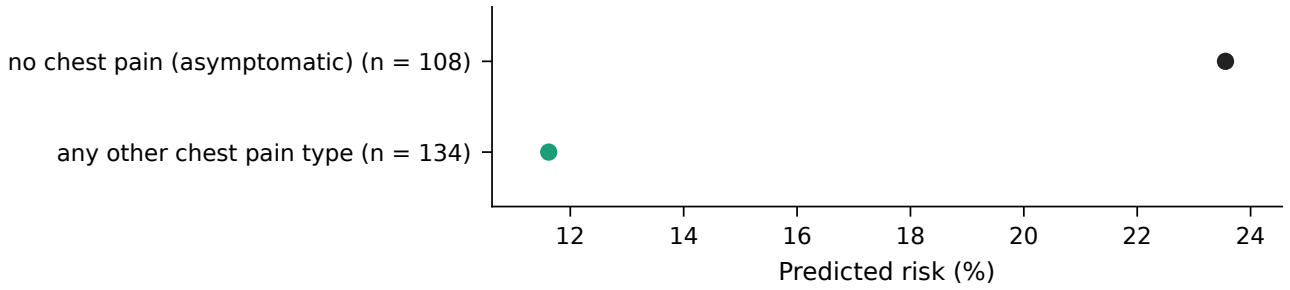


Figure 392: Chest pain offset: The predicted value for the typical case at each level, through this part (n = training patients; grey: fewer than 20).

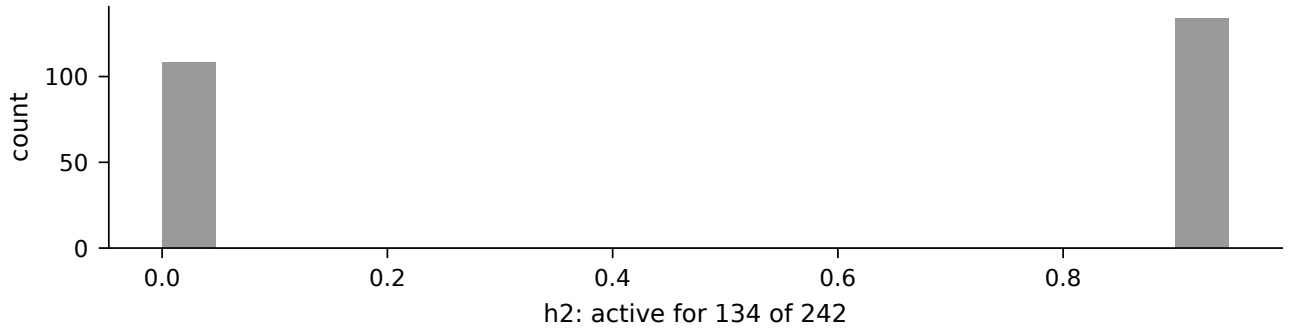


Figure 393: Chest pain offset: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.2.

10.2.4 Downsloping ST offset

Part of: The whole model (12.2.4).
Figure 394 shows what feeds this part; Figure 395 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).



Figure 394: Downsloping ST offset: what feeds it, one level down.

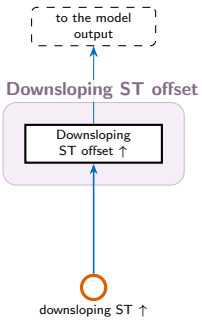


Figure 395: Downsloping ST offset: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part raises the predicted risk a little for patients with a downsloping ST segment (the model has little data on this, n = 17). [C40]

Evidence. It acts on the 17 of 242 training patients with a downsloping ST segment, 8 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 66% without it and about 73% with it (the model has little data on this, n = 17). [C41] Through it, a downsloping ST segment takes a typical patient's predicted risk from about 24% to about 29% (the model has little data on this, n = 17). [C40] Its formula reproduces the part exactly on all 242 training patients. [C42]
Evidence plots: Figure 396, Figure 397.

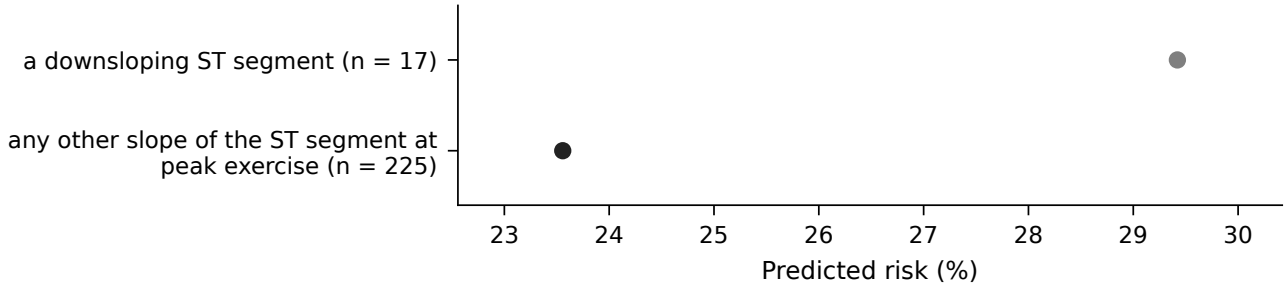


Figure 396: Downsloping ST offset: The predicted value for the typical case at each level, through this part (n = training patients; grey: fewer than 20).

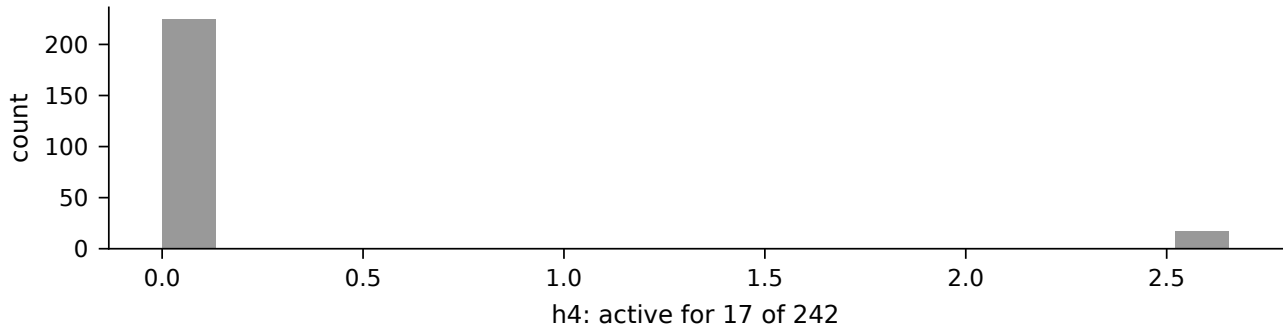


Figure 397: Downsloping ST offset: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.2.

10.2.5 Maximum heart rate relay

Part of: The whole model (12.2.4).
Figure 398 shows what feeds this part; Figure 399 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).



Figure 398: Maximum heart rate relay: what feeds it, one level down.

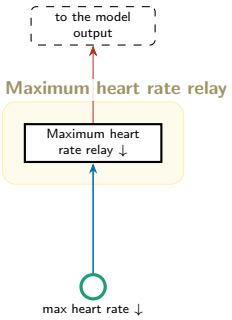


Figure 399: Maximum heart rate relay: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. Through this part, a higher maximum heart rate changes nothing until about 131 beats per minute, then lowers the predicted risk steadily. [C43, C44]

Evidence. It acts on 190 of 242 training patients, 71 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 21% without it and about 20% with it. [C45] Through it, a higher maximum heart rate changes nothing until about 131 beats per minute, then lowers the risk steadily, for a typical patient from about 24% at 118 to about 23% at 175 beats per minute. [C43, C44] Its formula reproduces the part exactly on all 242 training patients. [C47]
Evidence plots: Figure 400, Figure 401, Figure 402.
Figure 401 shows this part's own output, unit h1: it switches on at about 131.1 beats per minute of maximum heart rate, and is zero below.

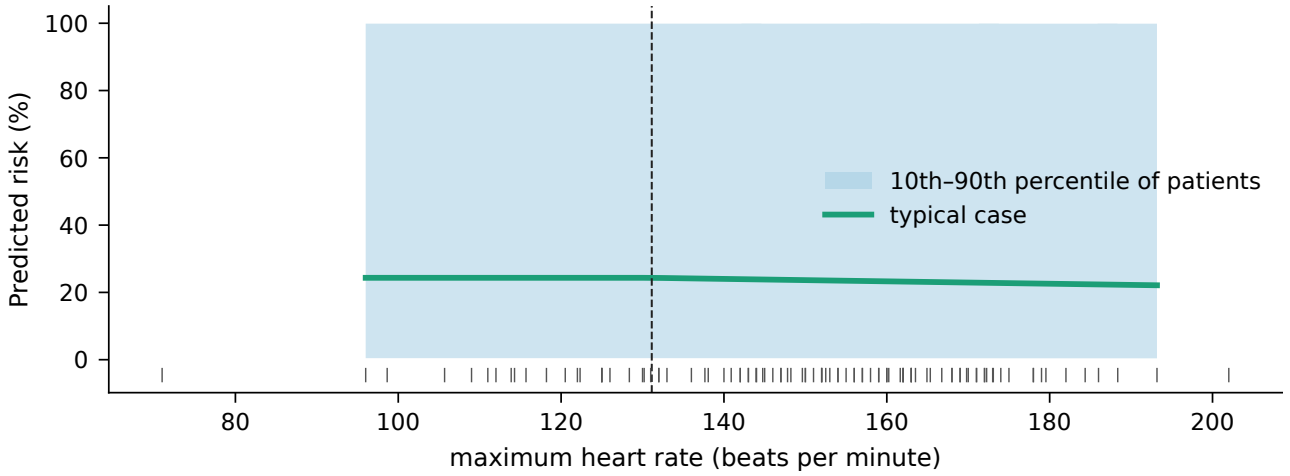


Figure 400: Maximum heart rate relay: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual patients (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

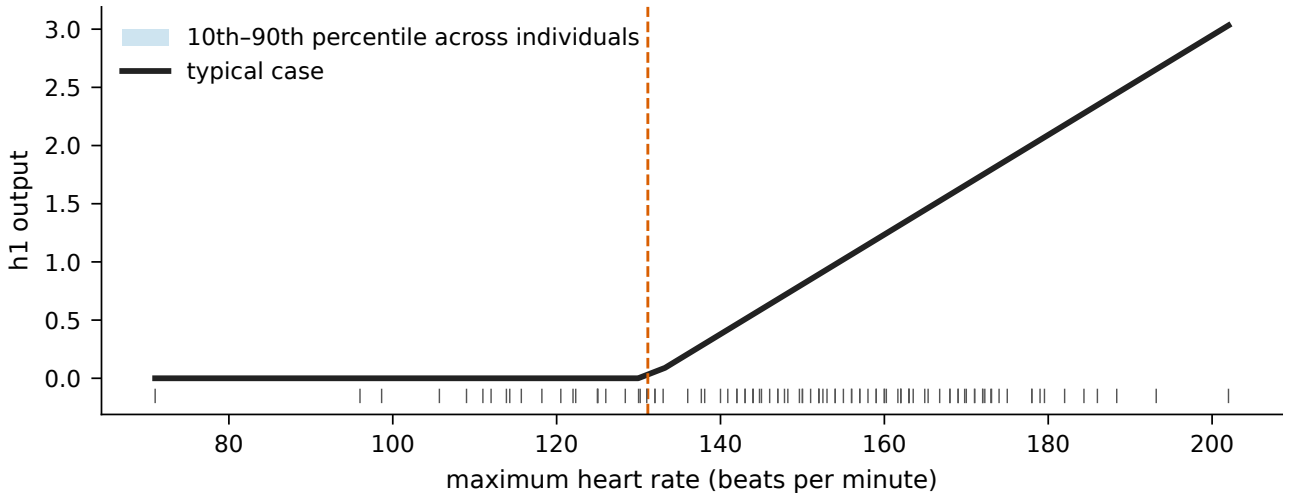


Figure 401: Maximum heart rate relay: This part's own output, unit h1 (the output of 'Maximum heart rate relay'), on its own scale (its activation), with every other input at the typical case; it is active for 190 of 242. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

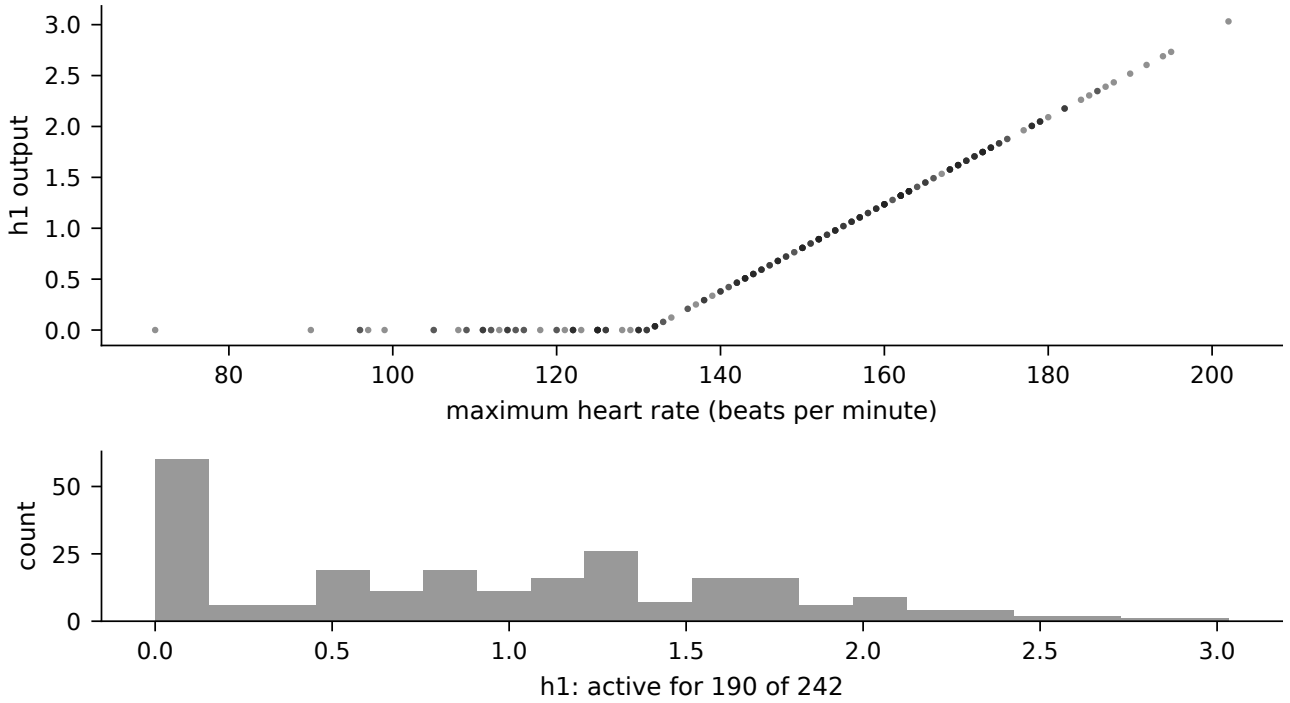


Figure 402: Maximum heart rate relay: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.3.

10.3 Not supported by the model

- C14: direction claim about thallium stress test result in the whole model: not supported by the model (14% of patients have the opposite sign (allowed: 10%)).
- C46: null claim about maximum heart rate reached during the exercise test in part ann_34: not supported by the model (predictions move by up to 0.026).

Technical detail: heart-s4-c1

The record behind each part: the original hypothesis, the closed form, and every claim with its estimate (log-odds and odds ratio), range across patients, n and test check.

T.1 The whole model

Annotation ann_35 (heart_disease_logit). Reads: age, sex, chest pain type, resting blood pressure, serum cholesterol, fasting blood sugar, resting ECG, maximum heart rate reached during the exercise test, exercise-induced angina, ST depression during exercise (compared with rest), slope of the ST segment at peak exercise, number of major vessels coloured by fluoroscopy, thallium stress test result.

Original hypothesis (unchanged). Root: p = sigmoid(z), z = 0.466 + LINEAR CORE - 1.077*fixed_defect_normal_sugar_gate - 0.899*no_asymptomatic_pain_indicator + 0.114*downsloping_st_indicator - 0.047*thalach_floor_relay. LINEAR CORE (19 direct input->output edges, z-scored inputs): ca +2.246, slope:2 +1.115, sex:1 +0.899, cp:3 -0.749, cp:1 -0.679, chol +0.671, trestbps +0.596, fbs:1 -0.583, sex:0 -0.559, thal:3 -0.541, oldpeak +0.491, cp:4 +0.308, exang:1 +0.282, slope:1 -0.238, thalach -0.125, fbs:0 -0.094, age +0.085, slope:3 +0.069, restecg:1 +0.062. Six inputs are unconnected (cp:2, restecg:0, restecg:2, exang:0, thal:4, thal:7; zero-effect entries of this root). thal:6 has no direct edge; it acts only through the fixed-defect gate. COMPOSITION: purely additive in the logit. Children: (1) fixed_defect_normal_sugar_gate, the one genuine interaction: an AND of fixed thallium defect and normal fasting sugar that lowers the logit by 3.92 for 10 patients; (2) no_asymptomatic_pain_indicator and (3) downsloping_st_indicator, binary ReLUs exactly affine on their one-hot inputs (re-parameterisations of the cp:4 and slope:3 linear terms; cp:4 total +1.47 logit, slope:3 total +0.57); (4) thalach_floor_relay, a near-inert hinge on max heart rate. The children share no hidden nodes; they are coupled only through inputs that also have direct edges (fbs:0, fbs:1, cp:4, slope:3, thalach). EVIDENCE (train, n=242): full model AUC 0.9423, acc 0.888 (compute_performance). Linear core alone (direct edges only) AUC 0.9328; logit SD 4.55 (core) vs 0.86 (all hidden readouts). Zero-ablating one child at a time: fixed-defect gate 0.9343, cp:4 indicator 0.9411, thalach relay 0.9416, slope:3 indicator 0.9421. Replacing all hidden readouts with one least-squares linear fit in their six inputs gives 0.9371; an L2 logistic regression on all 26 inputs gives 0.9355. Mean |SHAP| (train, all 242, probability scale): ca 0.172, slope:2 0.099, sex:1 0.069, cp:4 0.068, cp:3 0.056, thal:3 0.049, sex:0 0.042, chol 0.042. Interpretation: essentially logistic regression with one meaningful nonlinearity, the fixed-defect/normal-sugar gate, which cancels the core's over-prediction for that small group (10 patients). Counter-directional signs reported, not explained: fbs:1 -0.58 (high fasting sugar lowers the logit) and the gate itself making a fixed defect protective for normal-sugar patients. Same mechanism family as the 10-seed heart-s4 record (key 9650, gen 38): the same gate and linear core (ca +2.16 there), but here a cp:4 indicator and a slope:3 indicator replace that genome's cp:2 indicator, and the thalach relay feeds the output directly.

Closed form. Read through its parts (their formulas are in their own sections).

Split inputs. To annotate the parts, the explanation split these inputs into copies (a refactoring step that does not change what the model computes); the diagrams draw each input once: no chest pain (2 copies); sugar 120 or less (2 copies); sugar above 120 (2 copies); downsloping ST (2 copies); max heart rate (2 copies).

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C01	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (whole model) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	24% to 98%	5.04	154	5.04 to 5.04	242	unclear; AUC gain 0.044 [-0.038, 0.127]	yes
C02	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment (direct contribution only, not through any part) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	24% to 82%	2.72	15.1	2.72 to 2.72	115, 110	disagrees; AUC gain -0.014 [-0.061, 0.025]	yes
C03	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	24% to 82%	2.72	15.1	2.72 to 2.72	115, 110	disagrees; AUC gain -0.013 [-0.058, 0.026]	yes
C04	direction: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L) (whole model) *shape: steady; 100% of patients same; verified*	14% to 44%	1.60	4.97	1.60 to 1.60	242	disagrees; AUC gain -0.019 [-0.049, 0.002]	yes
C05	direction: resting blood pressure: 110 mm Hg to 155 mm Hg (whole model) *shape: steady; 100% of patients same; verified*	14% to 41%	1.48	4.37	1.48 to 1.48	242	unclear; AUC gain 0.002 [-0.018, 0.023]	yes
C06	direction: thallium stress test result: a normal thallium scan to thallium stress test result other than a normal thallium scan (direct contribution only, not through any part) *shape: categorical; highest to lowest: thallium stress test result other than a normal thallium scan, a normal thallium scan*	24% to 48%	1.09	2.98	1.09 to 1.09	135, 107	agrees; AUC gain 0.026 [0.005, 0.055]	yes
C07	direction: thallium stress test result: a normal thallium scan to thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan (whole model) *shape: categorical; highest to lowest: thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	24% to 48%	1.09	2.98	1.09 to 1.09	135, 92	unclear; AUC gain 0.005 [-0.028, 0.039]	yes
C08	direction: ST depression on exercise: 0.0 mm to 2.8 mm (whole model) *shape: steady; 100% of patients same; verified*	18% to 42%	1.16	3.19	1.16 to 1.16	242	unclear; AUC gain 0.005 [-0.01, 0.024]	yes
C09	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a downsloping ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	24% to 47%	1.05	2.85	1.05 to 1.05	115, 17 (small n)	disagrees; AUC gain -0.013 [-0.058, 0.026]	yes
C10	direction: chest pain type: no chest pain (asymptomatic) to typical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	24% to 1%	-3.84	0.02	-3.84 to -3.84	108, 22	agrees; AUC gain 0.057 [0.016, 0.105]	yes
C11	direction: chest pain type: no chest pain (asymptomatic) to non-anginal chest pain (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	24% to 1%	-3.12	0.04	-3.12 to -3.12	108, 70	agrees; AUC gain 0.057 [0.016, 0.105]	yes
C12	direction: sex: male to female (whole model) *shape: categorical; highest to lowest: male, female*	24% to 1%	-3.07	0.05	-3.07 to -3.07	159, 83	unclear; AUC gain 0.025 [-0.01, 0.069]	yes
C13	direction: chest pain type: no chest pain (asymptomatic) to typical angina (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	24% to 2%	-2.98	0.05	-2.98 to -2.98	108, 22	unclear; AUC gain 0.029 [0.000, 0.065]	yes
C14	direction: thallium stress test result: a normal thallium scan to a fixed defect on the thallium scan (whole model) *shape: categorical; highest to lowest: thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	24% to 2%	-2.29	0.10	-2.83 to 1.09	135, 15 (small n)	unclear; AUC gain 0.005 [-0.028, 0.039]	no

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C15	direction: chest pain type: no chest pain (asymptomatic) to non-anginal chest pain (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	24% to 3%	-2.27	0.10	-2.27 to -2.27	108, 70	unclear; AUC gain 0.029 [0.000, 0.065]	yes
C16	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	24% to 7%	-1.47	0.23	-1.47 to -1.47	108, 42	agrees; AUC gain 0.057 [0.016, 0.105]	yes
C17	direction: fasting blood sugar: fasting blood sugar of 120 mg/dL or less to fasting blood sugar above 120 mg/dL (direct contribution only, not through any part) *shape: categorical; highest to lowest: fasting blood sugar of 120 mg/dL or less, fasting blood sugar above 120 mg/dL*	24% to 7%	-1.43	0.24	-1.43 to -1.43	209, 33	disagrees; AUC gain -0.002 [-0.020, 0.015]	yes
C18	direction: fasting blood sugar: fasting blood sugar of 120 mg/dL or less to fasting blood sugar above 120 mg/dL (whole model) *shape: categorical; highest to lowest: fasting blood sugar of 120 mg/dL or less, fasting blood sugar above 120 mg/dL*	24% to 7%	-1.18	0.31	-1.43 to -1.43	209, 33	disagrees; AUC gain -0.005 [-0.025, 0.013]	yes
C19	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a downsloping ST segment (direct contribution only, not through any part) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	24% to 39%	0.75	2.11	0.75 to 0.75	115, 17 (small n)	disagrees; AUC gain -0.014 [-0.061, 0.025]	yes
C20	direction: resting ECG: resting ECG other than an ST-T wave abnormality on the resting ECG to an ST-T wave abnormality on the resting ECG (whole model) *shape: categorical; highest to lowest: an ST-T wave abnormality on the resting ECG, resting ECG other than an ST-T wave abnormality on the resting ECG*	24% to 38%	0.68	1.98	0.68 to 0.68	240, 2 (small n)	unclear; AUC gain 0.000 [0.000, 0.000]	yes
C21	direction: exercise-induced angina: no chest pain on exercise to chest pain brought on by exercise (whole model) *shape: categorical; highest to lowest: chest pain brought on by exercise, no chest pain on exercise*	24% to 36%	0.60	1.81	0.60 to 0.60	160, 82	unclear; AUC gain 0.009 [-0.002, 0.026]	yes
C22	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	24% to 14%	-0.62	0.54	-0.62 to -0.62	108, 42	unclear; AUC gain 0.029 [0.000, 0.065]	yes
C23	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (whole model) *shape: accelerating; 100% of patients same; verified*	28% to 21%	-0.41	0.67	-0.41 to -0.41	242	unclear; AUC gain 0.004 [-0.005, 0.016]	yes
C24	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (direct contribution only, not through any part) *shape: steady; 100% of patients same; verified*	27% to 21%	-0.32	0.73	-0.32 to -0.32	242	unclear; AUC gain 0.004 [0.000, 0.012]	yes
C25	direction: age: 42 years to 66 years (whole model) *shape: steady; 100% of patients same; verified*	21% to 25%	0.23	1.25	0.23 to 0.23	242	disagrees; AUC gain -0.002 [-0.009, 0.000]	yes
C26	magnitude: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels, typical patient *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	24% to 98%	5.04	154		242	unclear; AUC gain 0.044 [-0.038, 0.127]	yes
C27	magnitude: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment, typical patient *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	24% to 82%	2.72	15.1		115, 110	disagrees; AUC gain -0.013 [-0.058, 0.026]	yes
C28	magnitude: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L), typical patient *shape: steady; 100% of patients same; verified*	14% to 44%	1.60	4.97		242	disagrees; AUC gain -0.019 [-0.049, 0.002]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C29	magnitude: resting blood pressure: 110 mm Hg to 155 mm Hg, typical patient *shape: steady; 100% of patients same; verified*	14% to 41%	1.48	4.37		242	unclear; AUC gain 0.002 [-0.018, 0.023]	yes
C30	magnitude: thallium stress test result: a normal thallium scan to thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan, typical patient *shape: categorical; highest to lowest: thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	24% to 48%	1.09	2.98		135, 92	unclear; AUC gain 0.005 [-0.028, 0.039]	yes
C31	structure: made of fixed_defect_normal_sugar_gate, no_asymptomatic_pain_indicator, downsloping_st_indicator, thalach_floor_relay and inputs age, sex, cp, trestbps, chol, fbs, restecg, thalach, exang, oldpeak, slope, ca, thal; added					242	train only	yes

T.2 Fixed thallium defect with normal sugar

Annotation ann_31 (fixed_defect_normal_sugar_gate). Reads: fasting blood sugar, thallium stress test result.

Original hypothesis (unchanged). Fixed-defect AND normal-sugar gate (lowers risk). 2065 = ReLU(-0.703 + 1.576*fbs:0 - 0.683*fbs:1 + 0.885*thal:6), read out -1.077. Hypothesis: a discrete AND detector that is on only for patients with a fixed thallium defect (thal 6) and fasting blood sugar <= 120 mg/dL, and then lowers their logit by 3.92, cancelling the linear core's over-calling of this group (thal:6 has no direct edge; the core gives these patients a mean predicted probability of 0.83 while their disease rate is 50%). EVIDENCE (train, n=242): exact on-data reduction 2065 = 3.639*[thal:6=1 AND fbs=0] (fbs=1 drives the pre-activation to <= -6.39). Active for exactly the 10 patients in that group; disease rate 5/10. Zero-ablating it lowers train AUC 0.9423 -> 0.9343, the largest nonlinear effect; a linear refit in its three inputs recovers only to 0.937 (R2 0.69), because an AND cannot be written additively. Mean |sensitivity| fbs:0 0.065 > thal:6 0.037 > fbs:1 0.028 (small because the gate is off for 96%). Supported, but the correction rests on 10 training patients and may be partly noise-fitted. Same gate as the 10-seed heart-s4 record (gen 38), now also fed by fbs:0.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 4.44089e-16).

$$y_0 = \mathrm{relu}\big(-0.703 + 1.58 x_{fbs:0_b} + 0.885 x_{thal:6} - 0.683 x_{fbs:1_b}\big)$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C32	direction: fasting blood sugar: fasting blood sugar of 120 mg/dL or less to fasting blood sugar above 120 mg/dL (through this part) *shape: categorical; highest to lowest: fasting blood sugar above 120 mg/dL, fasting blood sugar of 120 mg/dL or less*	24% to 94% (typical patient this part acts on)	0.24	1.27	0.00 to 0.00	209, 33	disagrees; AUC gain -0.004 [-0.017, 0.000]	yes
C33	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan to a fixed defect on the thallium scan (through this part) *shape: categorical; highest to lowest: thallium stress test result other than a fixed defect on the thallium scan, a fixed defect on the thallium scan*	24% to 1%	-3.38	0.03	-3.92 to 0.00	227, 15 (small n)	disagrees; AUC gain -0.016 [-0.044, 0.001]	yes
C34	interaction: effect of thallium stress test result (thallium stress test result other than a fixed defect on the thallium scan to a fixed defect on the thallium scan) by fasting blood sugar: fasting blood sugar above 120 mg/dL minus fasting blood sugar of 120 mg/dL or less; stronger in fasting blood sugar of 120 mg/dL or less *shape: categorical; highest to lowest: thallium stress test result other than a fixed defect on the thallium scan, a fixed defect on the thallium scan*	fasting blood sugar above 120 mg/dL: 7% to 18%; fasting blood sugar of 120 mg/dL or less: 24% to 2%	3.92	50.3		33, 209, 15 (small n)	agrees; AUC gain -0.004 [-0.017, 0.000]	yes
C35	subgroup: who this part acts on: 10 of 242 patients	94% to 24% (typical patient this part acts on (risk without the part, then with it))	-3.92	0.02	-3.92 to -3.92	10, 232 (small n)	disagrees; AUC gain -0.016 [-0.044, 0.000]	yes
C36	structure: its formula reproduces it on 242 training patients (largest difference 4.44089e-16)		4e-16			242	train only	yes

T.3 Chest pain offset

Annotation ann_32 (no_asymptomatic_pain_indicator). Reads: chest pain type.

Original hypothesis (unchanged). Not-asymptomatic chest-pain indicator (lowers risk). 1754 = ReLU(-0.039 - 1.098*cp:4), read out -0.899. Hypothesis: a 0/1 indicator for chest pain other than asymptomatic (typical, atypical or non-anginal pain); because cp:4 is binary it is exactly affine on the data, so it only adds to the cp:4 direct edge (+0.308 per z-unit): together, asymptomatic chest pain raises the logit by 1.47 relative to the other pain types. EVIDENCE (train, n=242): on for exactly the 134 patients without asymptomatic pain (value 0.947, logit -0.852); disease rate 22% when on vs 71% when off. A linear replacement in cp:4 reproduces it exactly (R2 1.00, AUC unchanged 0.9423); zero-ablation (which also removes cp:4's main effect) gives 0.9411. Supported: a re-parameterisation of a linear cp:4 term, no interaction.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 0.0).

$$y_0 = \mathrm{relu}\big(-0.0387 - 1.1 x_{cp:4_b}\big)$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C37	direction: chest pain type: no chest pain (asymptomatic) to chest pain type other than no chest pain (asymptomatic) (through this part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), chest pain type other than no chest pain (asymptomatic)*	24% to 12%	-0.85	0.43	-0.85 to -0.85	108, 134	unclear; AUC gain 0.017 [-0.004, 0.043]	yes
C38	subgroup: who this part acts on: 134 of 242 patients	2% to 1% (typical patient this part acts on (risk without the part, then with it))	-0.85	0.43	-0.85 to -0.85	134, 108	unclear; AUC gain 0.017 [-0.004, 0.043]	yes
C39	structure: its formula reproduces it on 242 training patients (largest difference 0.0)		0.00			242	train only	yes

T.4 Downsloping ST offset

Annotation ann_33 (downsloping_st_indicator). Reads: slope of the ST segment at peak exercise.

Original hypothesis (unchanged). Downsloping-ST indicator (raises risk slightly). 3504 = ReLU(-1.020 + 1.010*slope:3), read out +0.114. Hypothesis: a 0/1 indicator for a downsloping peak-exercise ST segment; exactly affine on the binary input, adding +0.30 logit to slope:3's direct edge (total +0.57 logit for downsloping vs not). EVIDENCE (train, n=242): on for exactly the 17 downsloping patients; disease rate 47% vs 44%. Linear replacement exact (R2 1.00); zero-ablation 0.9423 -> 0.9421. Supported but nearly inert; rests on 17 patients.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 0.0).

$$y_0 = \mathrm{relu}\big(-1.02 + 1.01 x_{slope:3_b}\big)$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C40	direction: slope of the ST segment at peak exercise: slope of the ST segment at peak exercise other than a downsloping ST segment to a downsloping ST segment (through this part) *shape: categorical; highest to lowest: a downsloping ST segment, slope of the ST segment at peak exercise other than a downsloping ST segment*	24% to 29%	0.30	1.35	0.30 to 0.30	225, 17 (small n)	unclear; AUC gain 0.001 [0.000, 0.006]	yes
C41	subgroup: who this part acts on: 17 of 242 patients	66% to 73% (typical patient this part acts on (risk without the part, then with it))	0.30	1.35	0.30 to 0.30	17, 225 (small n)	unclear; AUC gain 0.001 [0.000, 0.006]	yes
C42	structure: its formula reproduces it on 242 training patients (largest difference 0.0)		0.00			242	train only	yes

T.5 Maximum heart rate relay

Annotation ann_34 (thalach_floor_relay). Reads: maximum heart rate reached during the exercise test.

Original hypothesis (unchanged). Max-heart-rate floor relay (lowers risk a little). 707 = ReLU(0.812 + 0.959*thalach), read out -0.047. Hypothesis: a hinge that adds a small extra protective slope in maximum heart rate achieved for all but the lowest ˜21% of thalach values (kink at z-thalach -0.85); combined with the direct edge (-0.125 per z) the thalach effect is -0.125 per z below the kink and -0.170 per z above it. EVIDENCE (train, n=242): active for 78.5%; disease rate 37% when active vs 69% when off; readout logit SD only 0.035. Zero-ablating it lowers train AUC 0.9423 -> 0.9416; a straight-line refit gives 0.9420 (R2 0.90). Supported: a near-linear, nearly inert extension of the thalach term.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 0.0).

$$y_0 = \mathrm{relu}\big(0.812 + 0.959 x_{thalach_b}\big)$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C43	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (through this part) *shape: hinge at 131 beats per minute; 100% of patients same; verified; one point with C44 (median fold 131 beats per minute, typical patient 130 beats per minute, 10th-90th percentile 131 beats per minute to 131 beats per minute)*	24% to 23%	-0.09	0.92	-0.09 to -0.09	242	unclear; AUC gain 0.000 [0.000, 0.000]	yes
C44	threshold: maximum heart rate: fold at about 131 beats per minute (starts) *shape: hinge at 131 beats per minute; 100% of patients same; verified; one point with C44 (median fold 131 beats per minute, typical patient 130 beats per minute, 10th-90th percentile 131 beats per minute to 131 beats per minute)*	24% to 23% (typical patient with a fold)	-0.10	0.90	-0.10 to -0.10	242, 190	unclear; AUC gain 0.000 [0.000, 0.000]	yes
C45	subgroup: who this part acts on: 190 of 242 patients	21% to 20% (typical patient this part acts on (risk without the part, then with it))	-0.06	0.95	-0.09 to -0.02	190, 52	unclear; AUC gain 0.000 [0.000, 0.000]	yes
C46	null: this part barely changes predictions (switched off)		0.03			242	train only	no
C47	structure: its formula reproduces it on 242 training patients (largest difference 0.0)		0.00			242	train only	yes

The network before any explanation

Figure 403 (next page, landscape) is the trained network as it is, with no parts, no splits and no inserted units: inputs at the bottom, hidden units by depth, the output at the top, every connection drawn, every input column its own node (one-hot levels included).

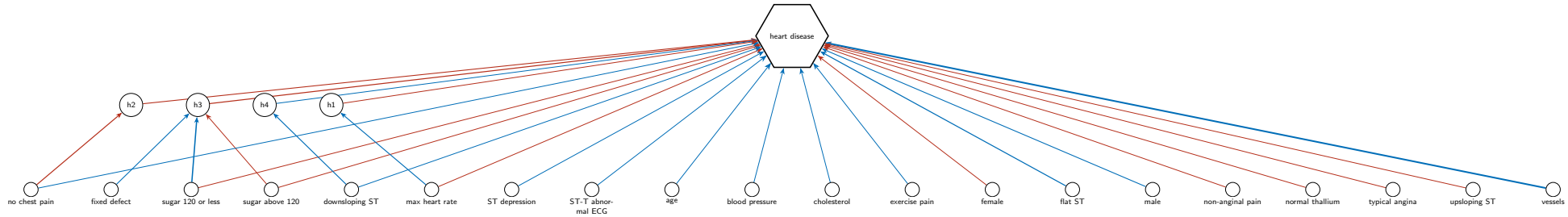


Figure 403: The network as trained, before any explanation, in full. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. Not used (no connection): atypical angina, normal ECG, LVH on ECG, no exercise pain, thallium code 4, reversible defect.

11 heart-s5-c1

Genome `4f9f15fb-446b-4560-9704-c7b1a83617b5`, explanation `39c7fe21-518b-4517-8765-7e6008dc7d98`. Split: train primary (n=242), test as a check (n=61). Claims: 52, verified 48, not supported 4.
AI-assisted draft for review (unverified)

Reference (typical) patient: the typical patient: the median training patient for every number and the most common training category for every category (not a real patient). age: 56 years; ca: 0 vessels; chol: 240 mg/dL (6.2 mmol/L); cp: no chest pain (asymptomatic); exang: no chest pain on exercise; fbs: fasting blood sugar of 120 mg/dL or less; oldpeak: 0.8 mm; restecg: a normal resting ECG; sex: male; slope: an upsloping ST segment; thal: a normal thallium scan; thalach: 152 beats per minute; trestbps: 130 mm Hg. Predicted risk of heart disease: 22%.

11.1 Summary

For a typical patient (a 56-year-old man with no chest pain symptoms and median test results) the model predicts a risk of heart disease of about 22%; it adds up most of its inputs independently, and a chain of three parts adds to that sum: a signal from ST depression on exercise, the thallium scan and the resting ECG that acts on 116 of 242 training patients, a blood pressure step that reads that signal, and an extra age term that acts on 85 patients. [C32, C40, C51] Figure 404 shows the model at this level: one box per paragraph below.

How to read the diagrams. Edge colour is the sign of that one connection: blue positive, red negative; thicker is a larger weight. The outline and arrow on an input (and on a part's output box) is its overall effect on the final prediction, read from the verified claims: orange and up arrow raises it, teal and down arrow lowers it, grey and double arrow depends (it goes both ways, or differs between levels). A negative connection into a part that itself raises the prediction means that input lowers the prediction; through a hinge the overall effect can only be read from the claims, not from the edges. Up arrow means towards heart disease; down arrow, away from it. A double outline marks an input (or a part's output) that acts at a fixed rate everywhere: it adds the same amount to the score per unit, so it is explained by its rate alone (see 'Linear factors'); a dashed double outline, at a fixed rate on the observed data. The normal outline is an input whose effect switches on and off somewhere. For example, age has a positive connection into 'Extra age term', a part that raises the predicted risk of heart disease, so age raises it.

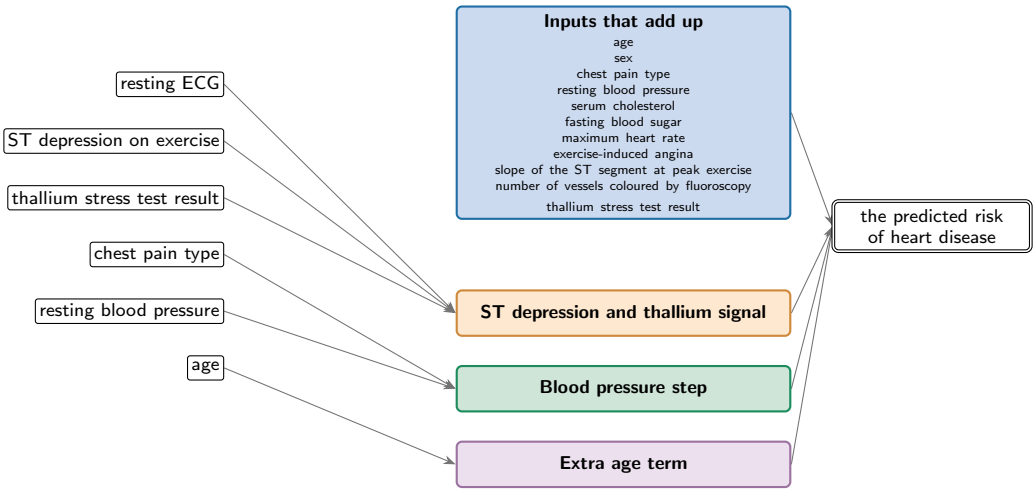


Figure 404: The model at the level of the summary: one box per part.

Inputs that add up. *Lowering the risk.* These add up independently, and these lower the risk: being a woman takes a typical patient's predicted risk from about 22% to about 3% and fasting blood sugar above 120 mg/dL to about 11%; a higher maximum heart rate lowers it steadily, from about 27% at 118 to about 18% at 175 beats per minute; and through this sum alone typical angina lowers it to about 2%, non-anginal chest pain to about 5% and atypical angina to about 11% (chest pain type also acts through the blood pressure step, below). [C12, C17, C22, C10, C13, C20]
Raising the risk. Each extra vessel coloured by fluoroscopy adds about the same, from about 22% at 0 to about 88% at 2 vessels, flattening out near the top because the predicted risk cannot go much higher; a flat rather than upsloping ST segment raises it to about 67%, higher cholesterol raises it steadily from about 14% at 193 mg/dL to about 37% at 308 mg/dL, and chest pain brought on by exercise raises it to about 33%. [C01, C04, C07, C16] Through this sum alone, higher resting blood pressure raises it steadily, from about 14% at 110 mm Hg to about 35% at

155 mm Hg, older age raises it steadily, from about 19% at 42 to about 24% at 66 years, and a reversible defect on the thallium scan raises it to about 35%; all three also act through the parts below. [C08, C24, C15]
Rare results. Some rarer results also count (the model has little data on this, n = 2 to 17): through this sum alone a thallium result coded 4 raises the risk to about 83% and a fixed defect on the thallium scan lowers it to about 11%, and a downsloping ST segment raises it to about 27%. [C02, C18, C25]

ST depression and thallium signal. This part reads ST depression on exercise, the thallium scan and the resting ECG, and acts on 116 of 242 training patients, 77 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 89% without it and about 96% with it. [C40] Through it, more ST depression adds to the predicted risk only once it rises above about 0.9 mm, rising in an uneven way: for a typical patient from about 22% at 0 mm to about 48% at 2.8 mm. [C33, C39] ST depression counts more with a reversible defect on the thallium scan (from about 36% to about 80%, against about 22% to about 48% otherwise) and with a normal resting ECG (from about 22% to about 48%, against about 22% to about 34% otherwise), in an uneven way. [C36, C37] A reversible defect on the thallium scan also raises the risk more with a normal resting ECG, from about 22% to about 47% against about 22% to about 36% otherwise. [C38]

Blood pressure step. This part reads that signal together with resting blood pressure and chest pain type, and barely changes the predictions: for the typical patient it acts on, atypical angina raises the predicted risk through it from about 69% to about 70%, and non-anginal chest pain leaves it at about 69%. [C42, C44]

Extra age term. This part acts on 85 of 242 training patients, 37 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 28% without it and about 30% with it. [C51] Through it, age adds nothing until about 53 years (between 53 and 66 years for different patients), then raises the risk steadily, for a typical patient from about 21% at 42 to about 24% at 66 years. [C49, C50]

The whole network. Figure 405 (next page, landscape) draws every unit and connection of the explained model, nested by part: shaded clusters are the parts of the model and darker boxes the parts inside them; circles are hidden units (h1, h2, ...); each cluster's rectangle is its output, and a dashed 'sum' stands for an addition the next unit does (drawn, not a unit of the model); the hexagon is the model's output. An input used by several parts is drawn under each of them (same label, dotted outline, the first copy marked with the number of copies): it is the same input, repeated for readability. Not used by the model (input columns with no connection): male, sugar above 120, ST-T abnormal ECG, LVH on ECG.

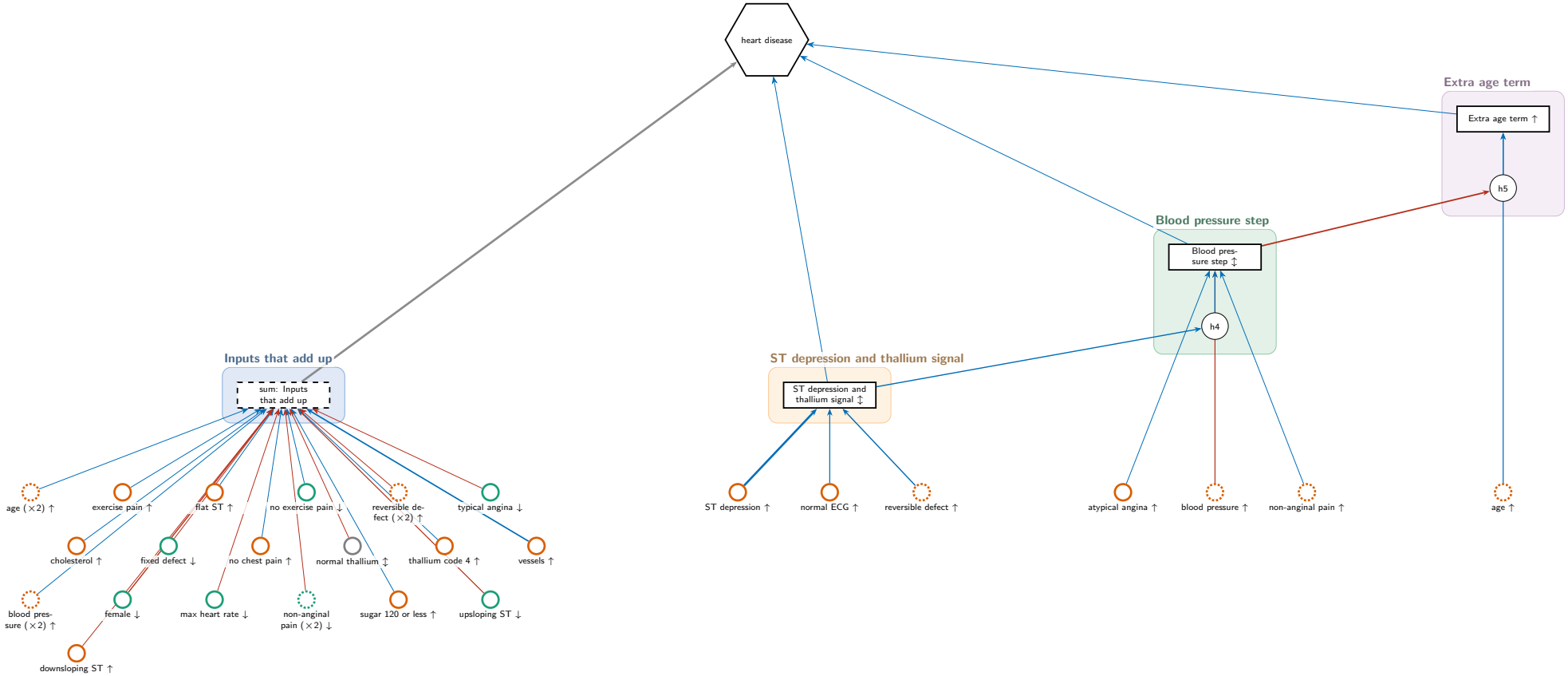


Figure 405: Every unit and connection of the explained model, nested by part. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Worth checking against what is known. Worth checking: the model treats fasting blood sugar above 120 mg/dL as lowering the risk of heart disease (for a typical patient from about 22% to about 11%). [C17] Worth checking: patients with typical angina get the lowest predicted risk of all chest pain types, below patients with no chest pain symptoms. [C11, C14, C21] Worth checking: the model treats a fixed defect on the thallium scan as lowering the risk compared with a normal scan, from only 15 training patients. [C19] Worth checking: the model gives a higher predicted risk to the 2 training patients whose thallium result is coded 4, a code the dataset documentation does not define. [C03] Worth checking: ST depression on exercise and a reversible thallium defect raise the predicted risk more when the resting ECG is normal than when it is not. [C37, C38]

11.2 The model, read from the top down

Each part of the explanation in turn, from the whole model down to the smallest parts: what it does for the prediction (hypothesis), what was checked (evidence), a diagram one level down, and where to read on. The technical record of each part is in "Technical detail" at the end.

11.2.1 The whole model

Figure 406 shows what feeds this part; Figure 407 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

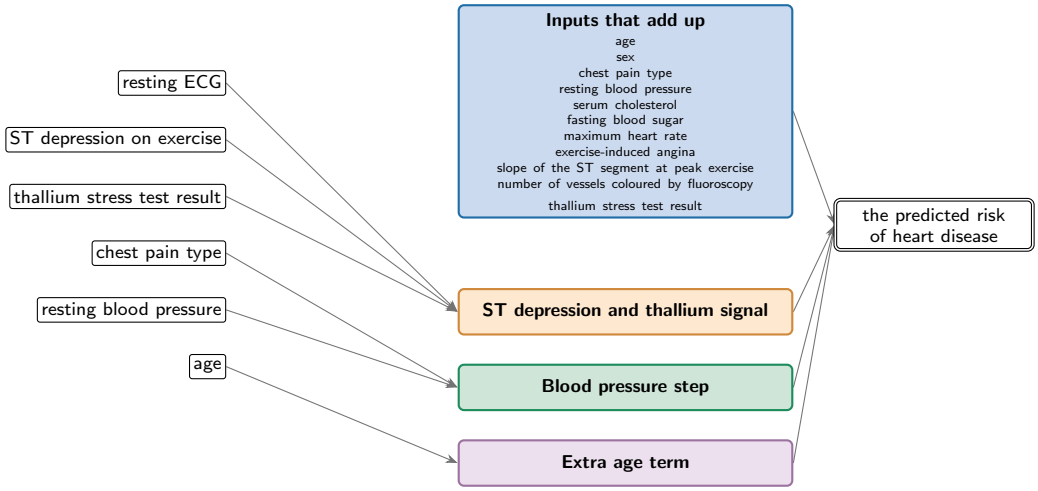


Figure 406: The whole model: what feeds it, one level down.

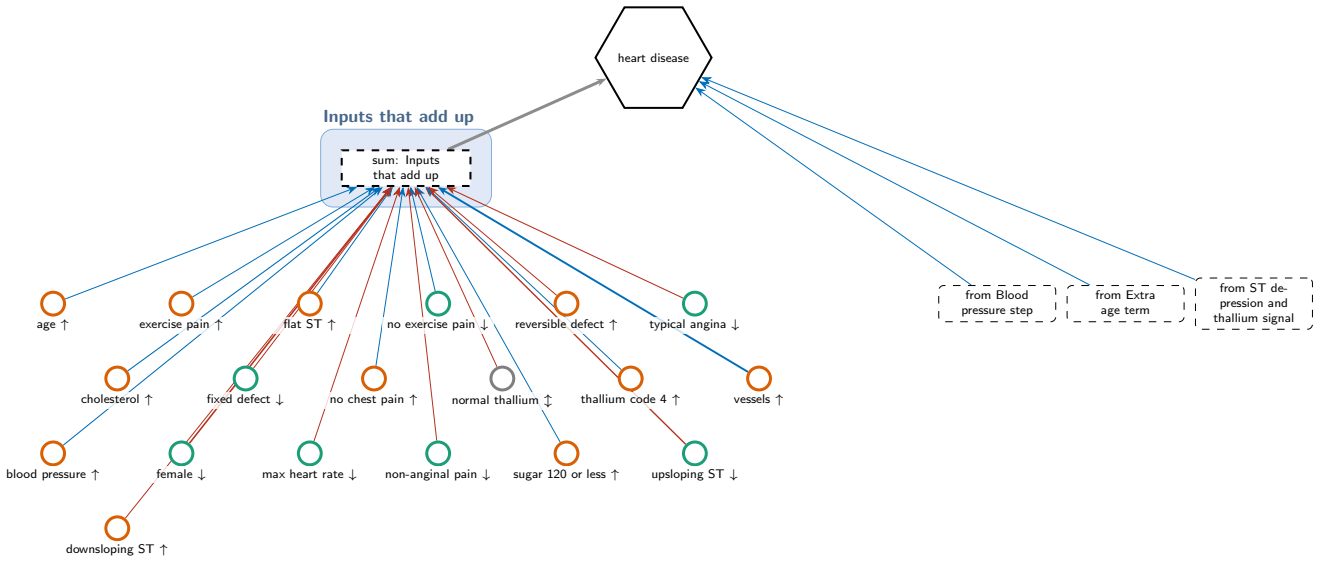


Figure 407: The whole model: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. The model adds up its inputs into one score: the vessels coloured (each extra vessel adding about the same), the ST segment, sex and chest pain type each move the predicted risk on their own, and a chain of

three parts adds to that sum: a signal from ST depression and the thallium scan, a blood pressure step that reads it, and an extra age term after that. [C32, C01, C04, C12, C10]

Evidence. It is made up of the inputs that feed the score directly and three parts, the ST depression and thallium signal, the blood pressure step and the extra age term (each explained below); the first part feeds the second and the second feeds the third, and the score is their sum, which was checked on all 242 training patients. [C32] ST depression on exercise and the resting ECG do not feed the score directly, so they act only through these parts. [C32, C33] Changing one input at a time across the whole model, more ST depression raises a typical patient's predicted risk in an uneven way, from about 22% at 0 mm to about 48% at 2.8 mm, and a reversible defect on the thallium scan takes it to about 47%, against about 35% through the sum alone. [C05, C06, C15] Across the whole model, higher resting blood pressure raises it in an uneven way, from about 14% at 110 mm Hg to about 35% at 155 mm Hg, and older age raises it in an uneven way, from about 18% at 42 to about 26% at 66 years. [C09, C23] Across the whole model, a fixed defect on the thallium scan lowers it to about 11% (the model has little data on this, n = 15), and a resting ECG other than normal barely changes it for a typical patient. [C19, C26]
Evidence plots: Figure 408.

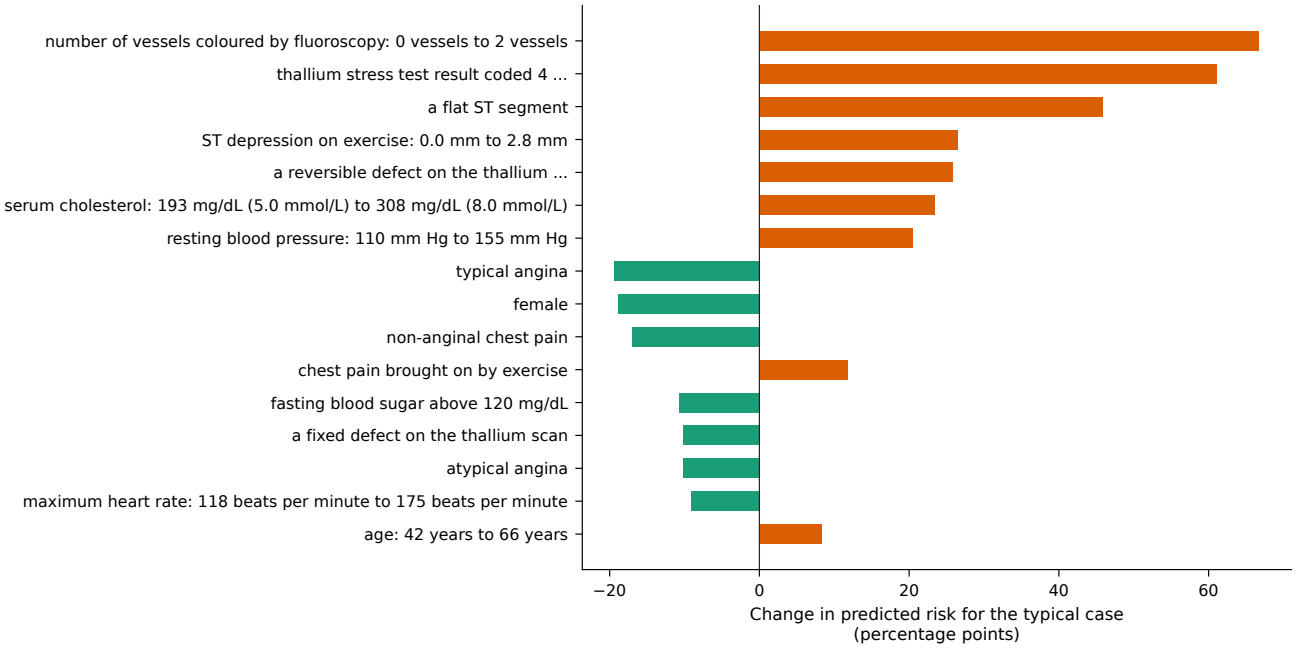


Figure 408: The whole model: Each input's change in the predicted value for the typical case across the whole model, largest first (orange: raises it, teal: lowers it).

Dive deeper: ST depression and thallium signal (11.2.2); Blood pressure step (12.2.2); Extra age term (12.2.3). Technical detail: T.4.

11.2.2 ST depression and thallium signal

Part of: The whole model (12.2.4).
Figure 409 shows what feeds this part; Figure 410 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

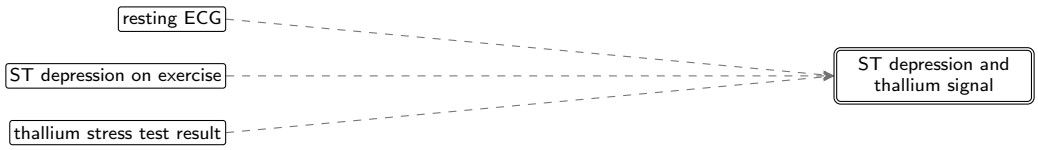


Figure 409: ST depression and thallium signal: what feeds it, one level down.

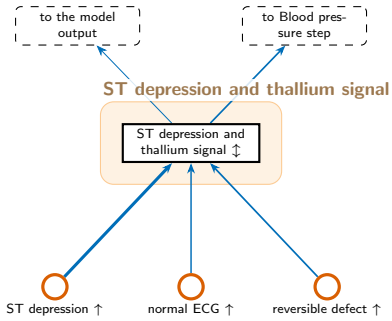


Figure 410: ST depression and thallium signal: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part raises the predicted risk with ST depression on exercise once it rises above about 0.9 mm, in an uneven way, and more so with a reversible defect on the thallium scan or a normal resting ECG. [C33, C39, C36, C37]

Evidence. It acts on 116 of 242 training patients, 77 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 89% without it and about 96% with it. [C40] Through it, more ST depression adds to the risk only once it rises above about 0.9 mm, rising in an uneven way: for a typical patient from about 22% at 0 mm to about 48% at 2.8 mm. [C33, C39] ST depression counts more with a reversible defect on the thallium scan (from about 36% to about 80%, against about 22% to about 48% otherwise) and with a normal resting ECG (from about 22% to about 48%, against about 22% to about 34% otherwise), in an uneven way. [C36, C37] Through it, a reversible defect on the thallium scan raises a typical patient's risk from about 22% to about 32%, and more with a normal resting ECG (to about 47%, against about 36% otherwise). [C34, C38] Its formula reproduces the part exactly on all 242 training patients. [C41]
Evidence plots: Figure 411, Figure 412, Figure 413, Figure 414, Figure 415.
Figure 414 shows this part's own output, unit h1: it switches on at about 0.9273 mm of ST depression on exercise, and is zero below.

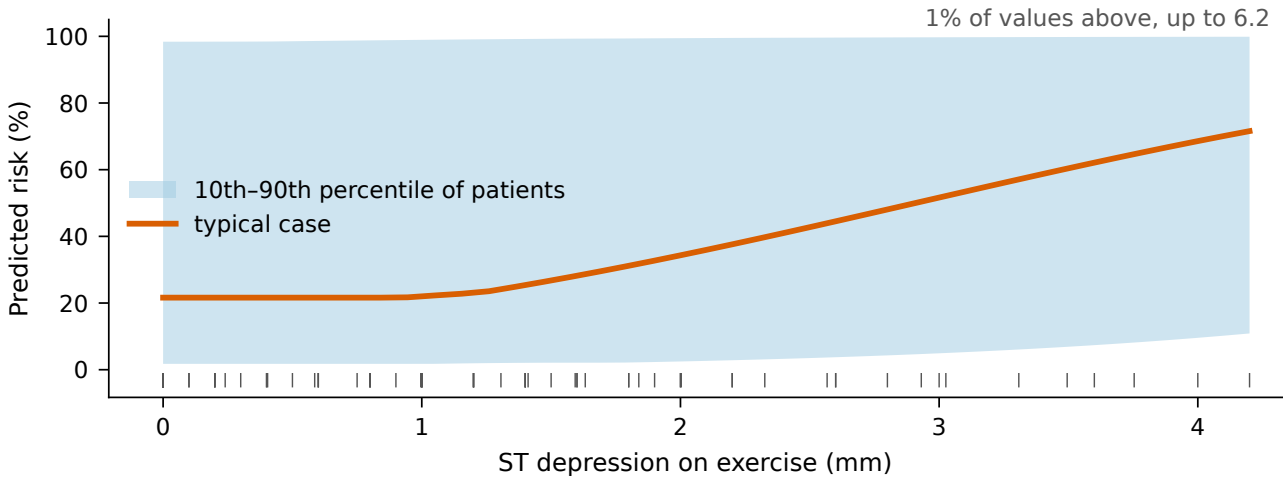


Figure 411: ST depression and thallium signal: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual patients (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

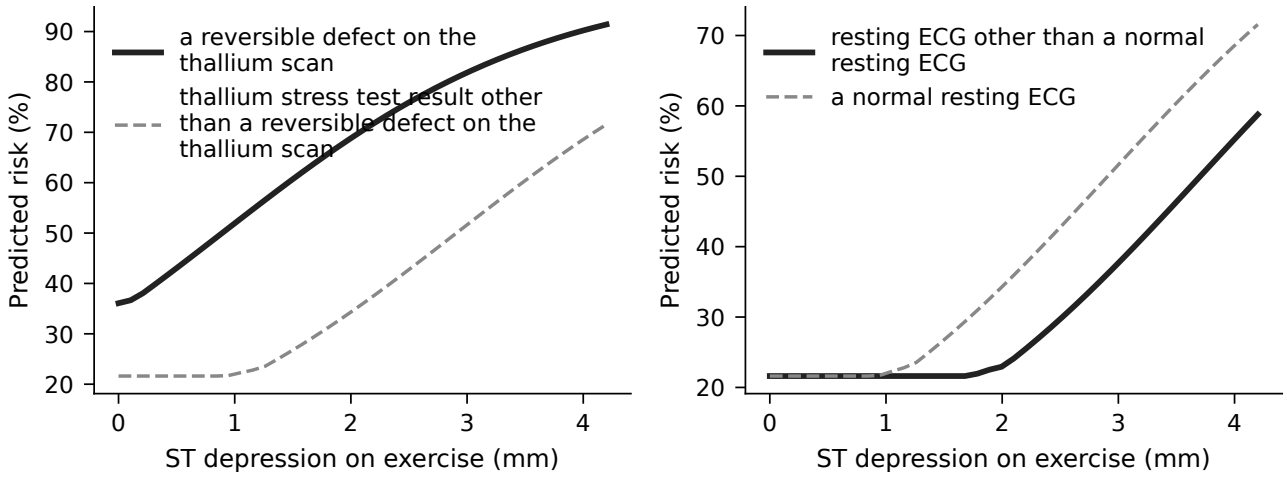


Figure 412: ST depression and thallium signal: How the effect differs between groups: one curve per group, for the typical case of each group.

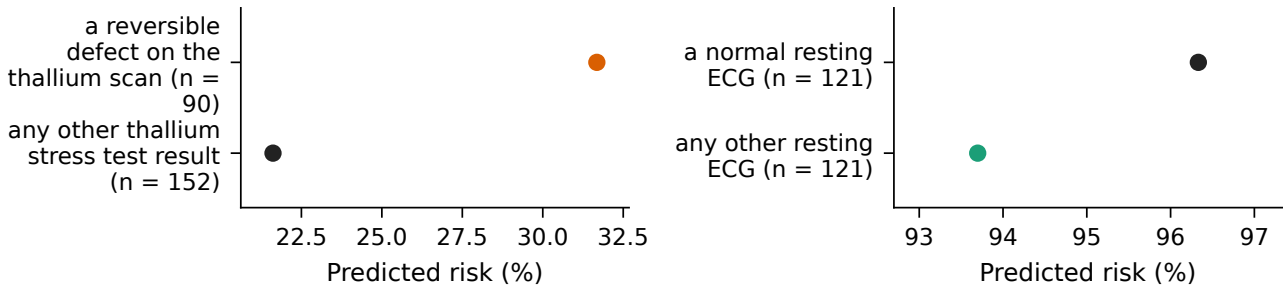


Figure 413: ST depression and thallium signal: The predicted value for the typical case at each level, through this part (n = training patients; grey: fewer than 20).

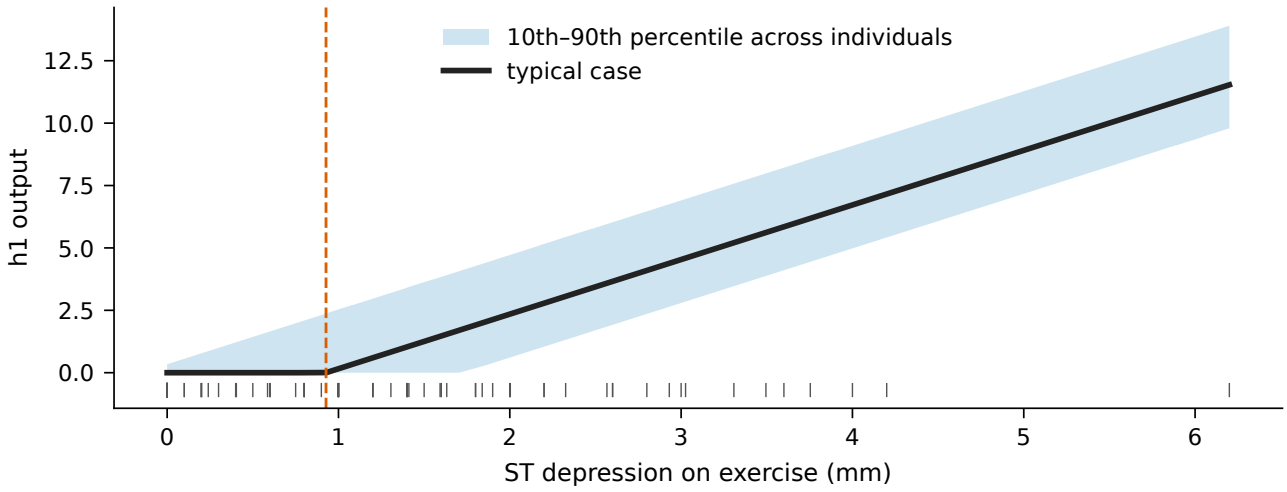


Figure 414: ST depression and thallium signal: This part's own output, unit h1 (the output of 'ST depression and thallium signal'), on its own scale (its activation), with every other input at the typical case; it is active for 116 of 242. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

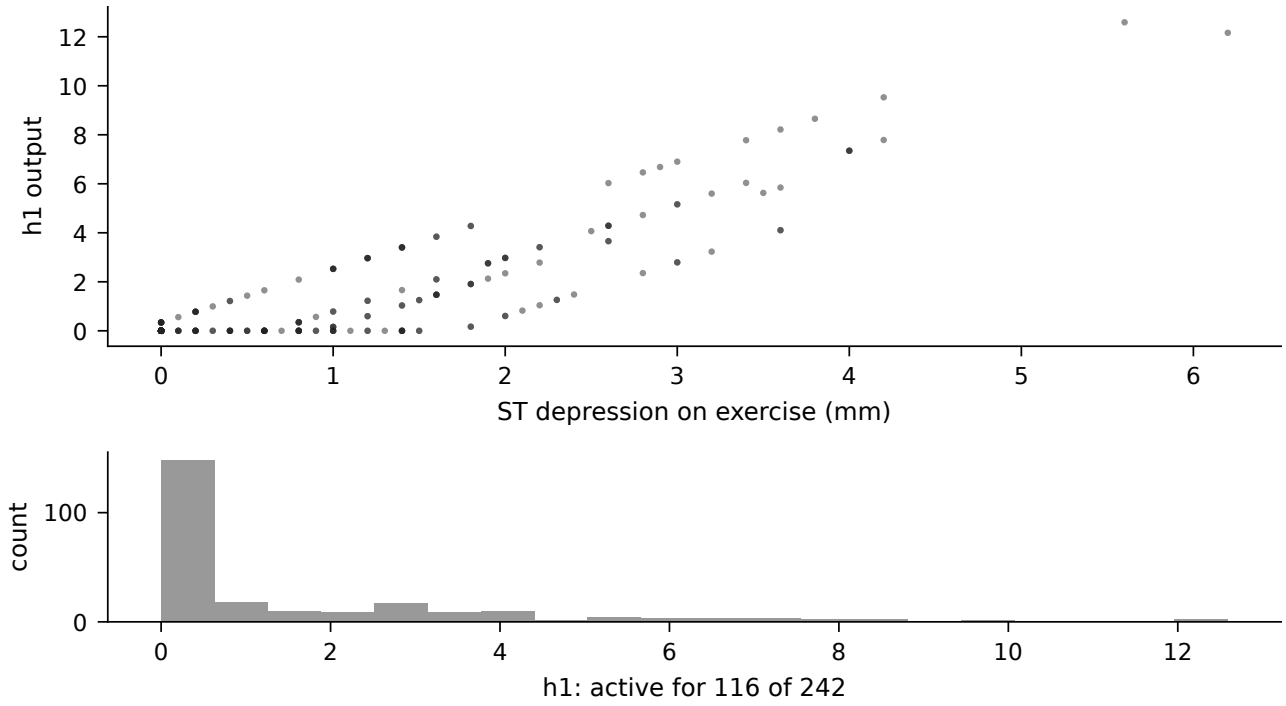


Figure 415: ST depression and thallium signal: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.2.

11.2.3 Blood pressure step

Part of: The whole model (12.2.4).
Figure 416 shows what feeds this part; Figure 417 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

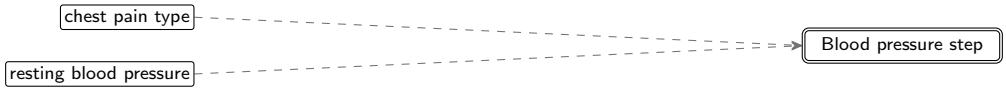


Figure 416: Blood pressure step: what feeds it, one level down.

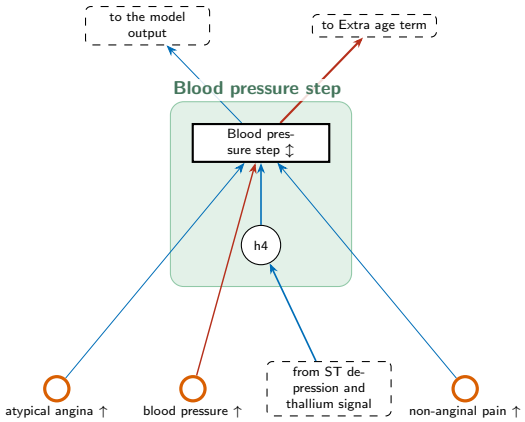


Figure 417: Blood pressure step: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part reads the ST depression and thallium signal together with resting blood pressure and chest pain type, and barely changes the predictions for the typical patient it acts on. [C42, C44]

Evidence. For the typical patient it acts on, atypical angina raises the predicted risk through it from about 69% to about 70%, and non-anginal chest pain leaves it at about 69%. [C42, C44] Its formula reproduces the part exactly on all 242 training patients. [C48]
Evidence plots: Figure 418, Figure 419, Figure 420.
Figure 419 shows this part's own output, unit h2: over its two main inputs, with the line where it switches on.

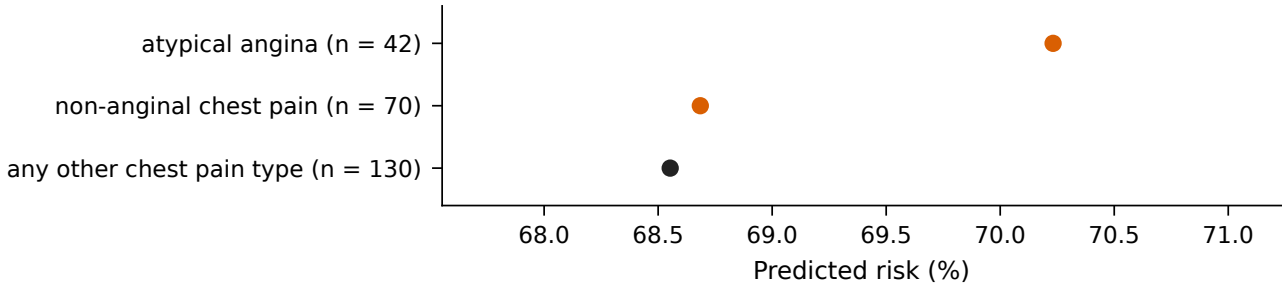


Figure 418: Blood pressure step: The predicted value for the typical case at each level, through this part (n = training patients; grey: fewer than 20).

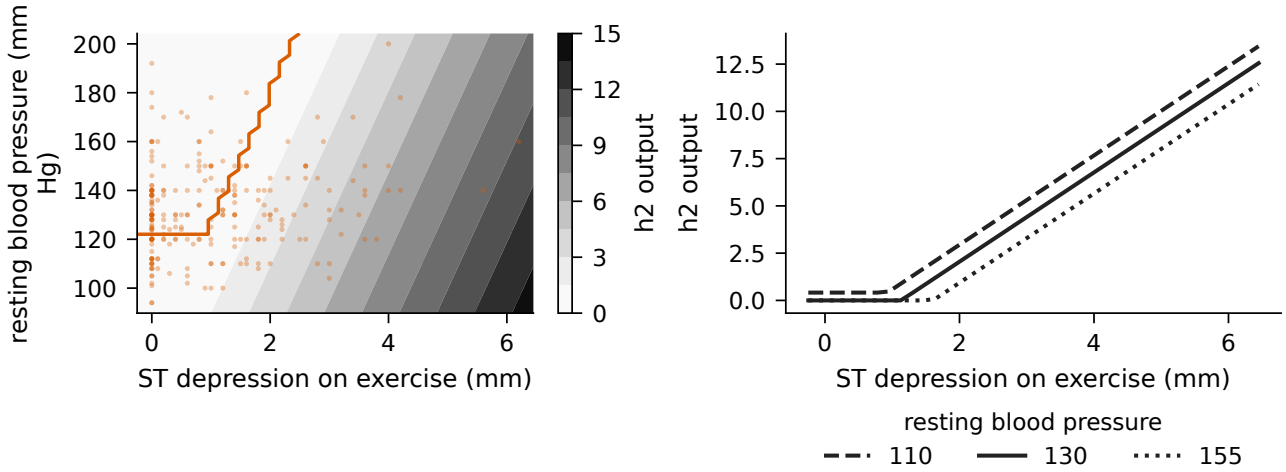


Figure 419: Blood pressure step: This part's own output, unit h2 (the output of 'Blood pressure step'), on its own scale (its activation), with every other input at the typical case; it is active for 140 of 242. Left: its output over its two main inputs (orange line: where it switches on; dots: the data); right: slices at three values of the second input.

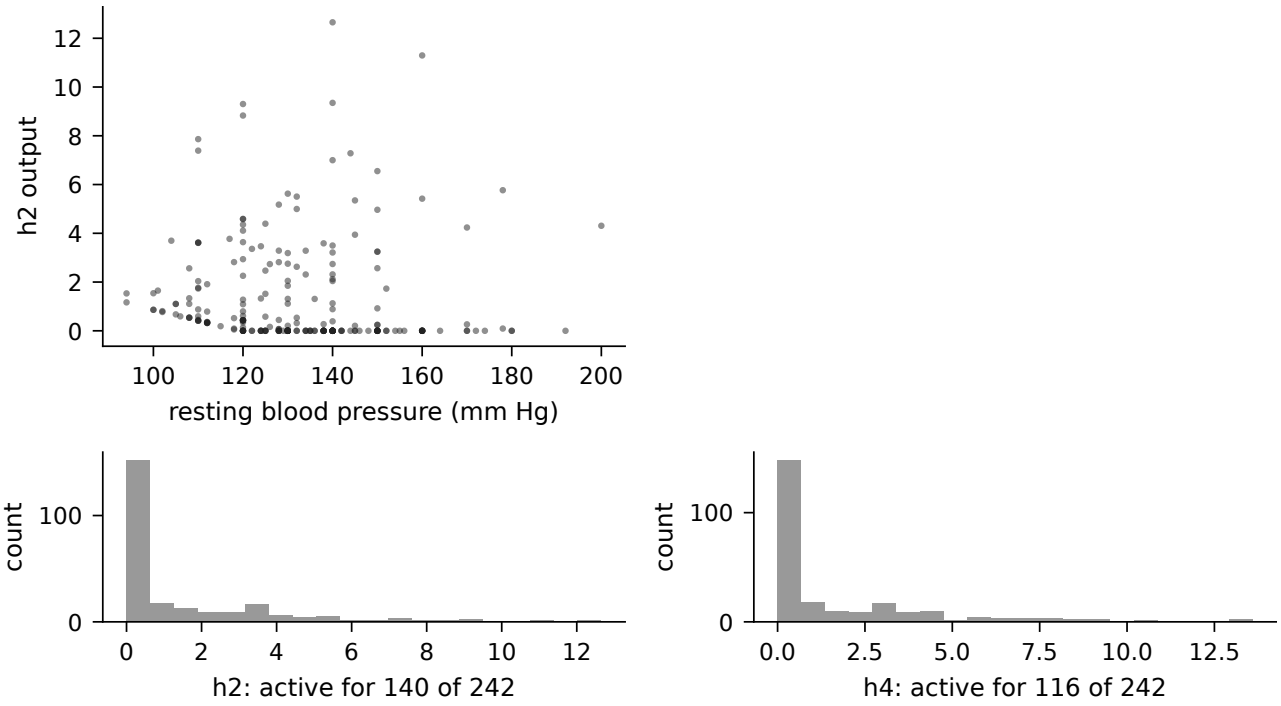


Figure 420: Blood pressure step: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.2.

11.2.4 Extra age term

Part of: The whole model (12.2.4).
Figure 421 shows what feeds this part; Figure 422 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).



Figure 421: Extra age term: what feeds it, one level down.

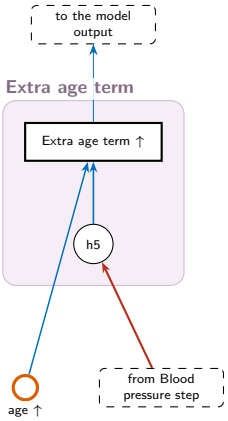


Figure 422: Extra age term: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part adds a little to the predicted risk for older patients: nothing until about 53 years, then steadily. [C49, C50]

Evidence. It acts on 85 of 242 training patients, 37 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 28% without it and about 30% with it. [C51] Through it, age adds nothing until about 53 years (between 53 and 66 years for different patients), then raises the risk steadily, for a typical patient from about 21% at 42 to about 24% at 66 years. [C49, C50] Its formula reproduces the part exactly on all 242 training patients. [C52]
Evidence plots: Figure 423, Figure 424, Figure 425.
Figure 424 shows this part's own output, unit h3: over its two main inputs, with the line where it switches on.

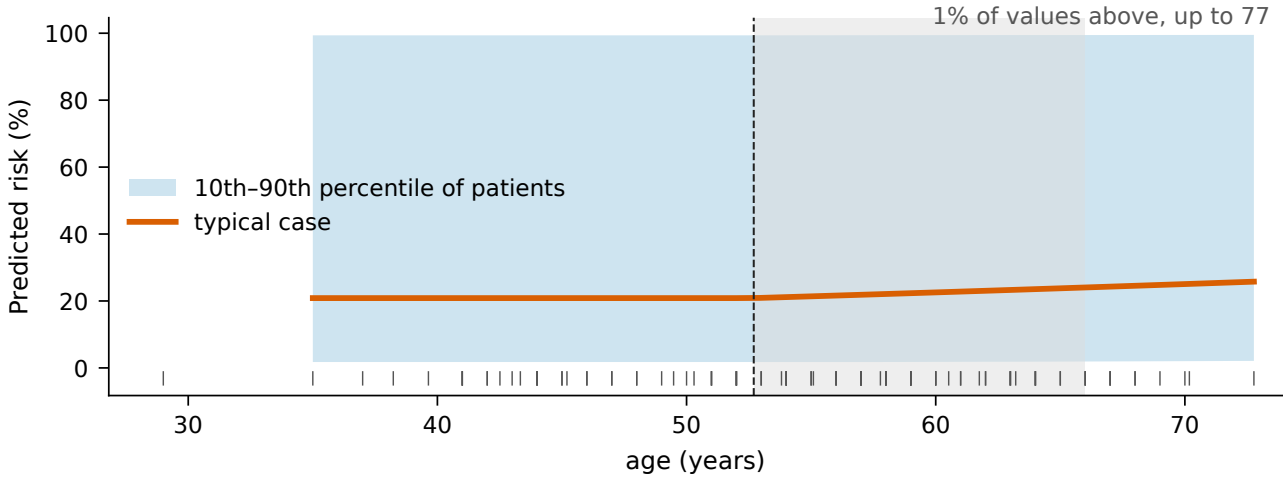


Figure 423: Extra age term: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual patients (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

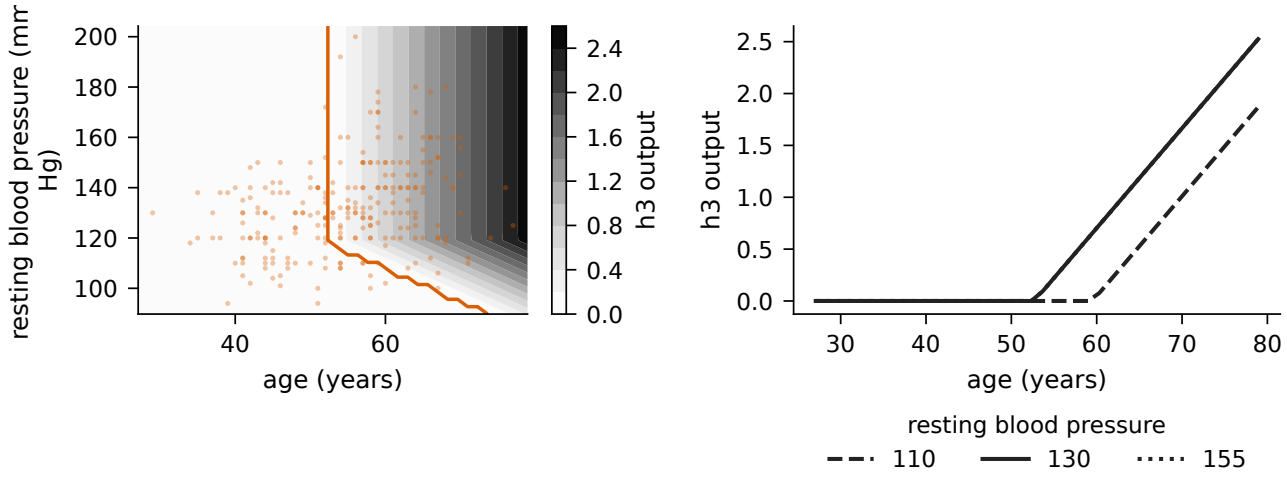


Figure 424: Extra age term: This part's own output, unit h3 (the output of 'Extra age term'), on its own scale (its activation), with every other input at the typical case; it is active for 85 of 242. Left: its output over its two main inputs (orange line: where it switches on; dots: the data); right: slices at three values of the second input.

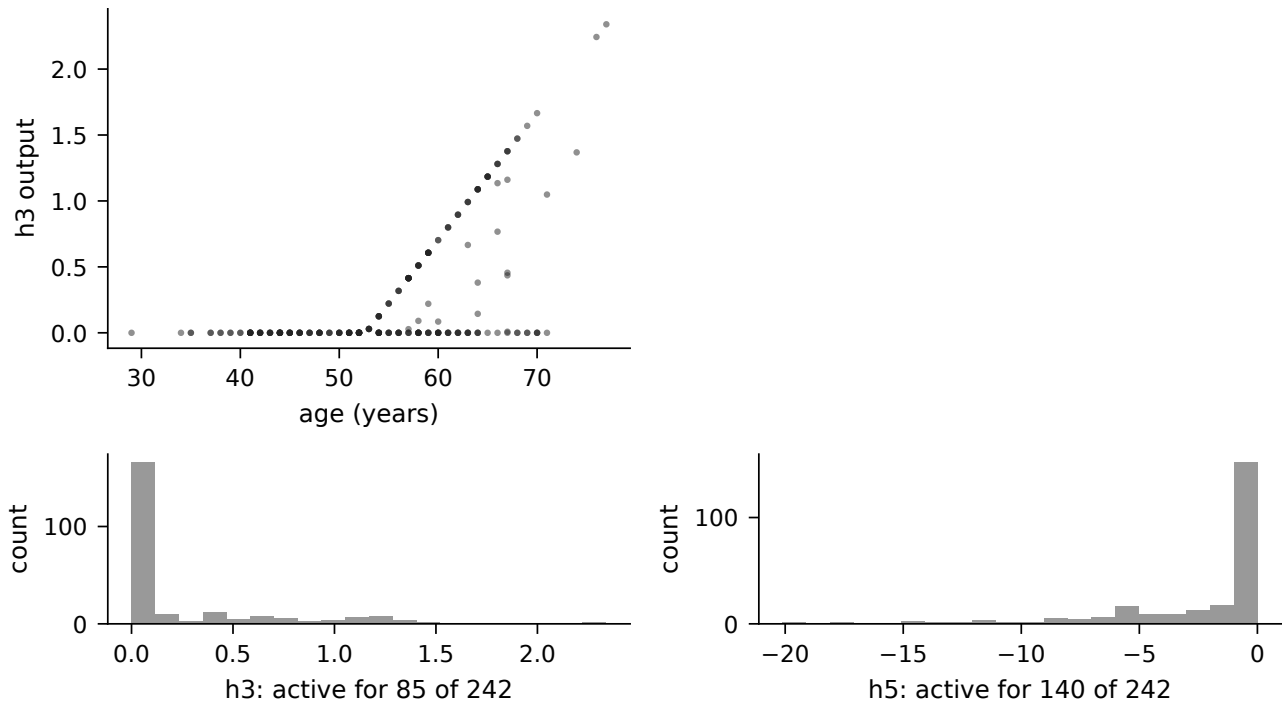


Figure 425: Extra age term: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.3.

11.3 Not supported by the model

- C43: direction claim about resting blood pressure in part ann_33: not supported by the model (17% of patients have the opposite sign (allowed: 10%)).
- C45: interaction claim about resting blood pressure and chest pain type in part ann_33: not supported by the model (the group difference is below 0.1 log-odds; the part's own counterfactual difference has the opposite sign).
- C46: threshold claim about resting blood pressure in part ann_33: not supported by the model (the effect past the fold is too small or not consistent across patients).
- C47: subgroup claim about chest pain type and resting blood pressure in part ann_33: not supported by the model (its effect goes both ways across the patients it acts on).

Technical detail: heart-s5-c1

The record behind each part: the original hypothesis, the closed form, and every claim with its estimate (log-odds and odds ratio), range across patients, n and test check.

T.1 The whole model

Annotation ann_35 (heart_disease_logit). Reads: age, sex, chest pain type, resting blood pressure, serum cholesterol, fasting blood sugar, resting ECG, maximum heart rate reached during the exercise test, exercise-induced angina, ST depression during exercise (compared with rest), slope of the ST segment at peak exercise, number of major vessels coloured by fluoroscopy, thallium stress test result.

Original hypothesis (unchanged). Root: p = sigmoid(z), z = -0.504 + LINEAR CORE + 0.143*ischaemia_gate + 0.170*bp_offset_gate + 0.143*age_gate_without_ischaemia. LINEAR CORE (19 direct input->output edges, z-scored inputs): ca +1.480, sex:0 -1.072, slope:1 -0.855, chol +0.546, cp:1 -0.493, trestbps +0.480, cp:3 -0.440, exang:1 +0.431, thal:3 -0.410, thal:6 -0.384, cp:4 +0.377, slope:3 -0.366, fbs:0 +0.278, thalach -0.207, thal:4 +0.183, slope:2 +0.153, exang:0 +0.149, age +0.117, thal:7 -0.077. Four inputs are unconnected (sex:1, fbs:1, restecg:1, restecg:2; zero-effect entries of this root). Oldpeak, restecg:0 and cp:2 have no direct edge and act only through the chain. COMPOSITION: serial and additive. The three children form a chain, each read out additively into z: ischaemia_gate (oldpeak, normal resting ECG, reversible defect) feeds bp_offset_gate (the ischaemia score minus a resting-BP offset), which in turn switches off age_gate_without_ischaemia (an extra age term for patients without ischaemia evidence). The chain therefore raises risk with ST depression, more steeply where resting BP is low, and adds a small age term where the chain is quiet. Identity nodes identity_290 and identity_1987 carry the chain links (204 -> 290, 290 -> 1987) so that each gate is its own leaf. EVIDENCE (train, n=242): full model AUC 0.9418, acc 0.872 (compute_performance). Linear core alone AUC 0.934; logit SD 3.14 (core) vs 0.67 (chain). Removing the whole chain (node 204 with downstream) gives 0.934; readout-only ablations: ischaemia 0.940, BP 0.939, age 0.941. Treating all three ReLUs as identities gives 0.936; one least-squares linear fit of the chain on its seven inputs gives 0.937 (R2 0.84); an L2 logistic regression on all 26 inputs gives 0.936. Mean |SHAP| (train, all 242): ca 0.144, sex:0 0.110, slope:1 0.100, thal:3 0.048, exang:1 0.047, chol 0.045, cp:4 0.044, cp:3 0.044. Interpretation: close to logistic regression plus one ischaemia correction: ST depression raises risk only above a threshold set by a normal resting ECG and a reversible defect, more so when resting BP is low; age gets a small extra weight when there is no such evidence. Counter-directional signs reported, not explained: restecg:0 positive in the ischaemia gate, trestbps negative in the BP gate, fbs:0 +0.28 directly. Same design as the 10-seed heart-s5 record (generation 37, key 9330: ischaemia gate -> resting-BP gate -> age term), here with three hidden nodes instead of six.

Closed form. Read through its parts (their formulas are in their own sections).

Split inputs. To annotate the parts, the explanation split these inputs into copies (a refactoring step that does not change what the model computes); the diagrams draw each input once: age (2 copies); non-anginal pain (2 copies); reversible defect (2 copies); blood pressure (2 copies).

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C01	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (whole model) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	22% to 88%	3.32	27.6	3.32 to 3.32	242	unclear; AUC gain 0.039 [-0.026, 0.111]	yes
C02	direction: thallium stress test result: a normal thallium scan to thallium stress test result coded 4 (meaning not recorded) (direct contribution only, not through any part) *shape: categorical; highest to lowest: thallium stress test result coded 4 (meaning not recorded), a reversible defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	22% to 83%	2.85	17.3	2.85 to 2.85	135, 2 (small n)	unclear; AUC gain 0.012 [-0.009, 0.035]	yes
C03	direction: thallium stress test result: a normal thallium scan to thallium stress test result coded 4 (meaning not recorded) (whole model) *shape: categorical; highest to lowest: thallium stress test result coded 4 (meaning not recorded), a reversible defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	22% to 83%	2.85	17.3	2.85 to 2.85	135, 2 (small n)	unclear; AUC gain 0.017 [-0.010, 0.052]	yes
C04	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	22% to 67%	2.02	7.53	2.02 to 2.02	115, 110	disagrees; AUC gain -0.019 [-0.070, 0.025]	yes
C05	direction: ST depression on exercise: 0.0 mm to 2.8 mm (whole model) *shape: accelerating; 48% of patients same; fallback*	22% to 48%	1.16	3.20	0.55 to 1.91	242	unclear; AUC gain 0.008 [-0.011, 0.031]	yes
C06	direction: thallium stress test result: a normal thallium scan to a reversible defect on the thallium scan (whole model) *shape: categorical; highest to lowest: thallium stress test result coded 4 (meaning not recorded), a reversible defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	22% to 47%	1.01	2.73	0.67 to 1.44	135, 90	unclear; AUC gain 0.017 [-0.010, 0.052]	yes
C07	direction: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L) (whole model) *shape: steady; 100% of patients same; verified*	14% to 37%	1.30	3.68	1.30 to 1.30	242	disagrees; AUC gain -0.022 [-0.063, 0.008]	yes
C08	direction: resting blood pressure: 110 mm Hg to 155 mm Hg (direct contribution only, not through any part) *shape: steady; 100% of patients same; verified*	14% to 35%	1.19	3.28	1.19 to 1.19	242	unclear; AUC gain 0.000 [-0.024, 0.022]	yes
C09	direction: resting blood pressure: 110 mm Hg to 155 mm Hg (whole model) *shape: steady; 43% of patients same; fallback*	14% to 35%	1.04	2.82	0.84 to 1.21	242	unclear; AUC gain 0.001 [-0.022, 0.023]	yes
C10	direction: chest pain type: no chest pain (asymptomatic) to typical angina (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	22% to 2%	-2.48	0.08	-2.48 to -2.48	108, 22	agrees; AUC gain 0.040 [0.008, 0.079]	yes
C11	direction: chest pain type: no chest pain (asymptomatic) to typical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	22% to 2%	-2.48	0.08	-2.48 to -2.48	108, 22	agrees; AUC gain 0.040 [0.008, 0.079]	yes
C12	direction: sex: male to female (whole model) *shape: categorical; highest to lowest: male, female*	22% to 3%	-2.26	0.10	-2.26 to -2.26	159, 83	unclear; AUC gain 0.024 [-0.008, 0.064]	yes
C13	direction: chest pain type: no chest pain (asymptomatic) to non-anginal chest pain (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	22% to 5%	-1.73	0.18	-1.73 to -1.73	108, 70	agrees; AUC gain 0.040 [0.008, 0.079]	yes
C14	direction: chest pain type: no chest pain (asymptomatic) to non-anginal chest pain (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	22% to 5%	-1.73	0.18	-1.73 to -1.72	108, 70	agrees; AUC gain 0.040 [0.008, 0.079]	yes
C15	direction: thallium stress test result: a normal thallium scan to a reversible defect on the thallium scan (direct contribution only, not through any part) *shape: categorical; highest to lowest: thallium stress test result coded 4 (meaning not recorded), a reversible defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	22% to 35%	0.67	1.95	0.67 to 0.67	135, 90	unclear; AUC gain 0.012 [-0.009, 0.035]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C16	direction: exercise-induced angina: no chest pain on exercise to chest pain brought on by exercise (whole model) *shape: categorical; highest to lowest: chest pain brought on by exercise, no chest pain on exercise*	22% to 33%	0.60	1.82	0.60 to 0.60	160, 82	unclear; AUC gain 0.012 [-0.003, 0.032]	yes
C17	direction: fasting blood sugar: fasting blood sugar of 120 mg/dL or less to fasting blood sugar above 120 mg/dL (whole model) *shape: categorical; highest to lowest: fasting blood sugar of 120 mg/dL or less, fasting blood sugar above 120 mg/dL*	22% to 11%	-0.81	0.44	-0.81 to -0.81	209, 33	disagrees; AUC gain -0.002 [-0.020, 0.015]	yes
C18	direction: thallium stress test result: a normal thallium scan to a fixed defect on the thallium scan (direct contribution only, not through any part) *shape: categorical; highest to lowest: thallium stress test result coded 4 (meaning not recorded), a reversible defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	22% to 11%	-0.77	0.46	-0.77 to -0.77	135, 15 (small n)	unclear; AUC gain 0.012 [-0.009, 0.035]	yes
C19	direction: thallium stress test result: a normal thallium scan to a fixed defect on the thallium scan (whole model) *shape: categorical; highest to lowest: thallium stress test result coded 4 (meaning not recorded), a reversible defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	22% to 11%	-0.77	0.46	-0.77 to -0.77	135, 15 (small n)	unclear; AUC gain 0.017 [-0.010, 0.052]	yes
C20	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	22% to 11%	-0.76	0.47	-0.76 to -0.76	108, 42	agrees; AUC gain 0.040 [0.008, 0.079]	yes
C21	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	22% to 11%	-0.72	0.49	-0.76 to -0.68	108, 42	agrees; AUC gain 0.040 [0.008, 0.079]	yes
C22	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (whole model) *shape: steady; 100% of patients same; verified*	27% to 18%	-0.53	0.59	-0.53 to -0.53	242	unclear; AUC gain 0.006 [0.000, 0.022]	yes
C23	direction: age: 42 years to 66 years (whole model) *shape: accelerating; 67% of patients same; fallback*	18% to 26%	0.41	1.50	0.31 to 0.49	242	disagrees; AUC gain -0.003 [-0.017, 0.013]	yes
C24	direction: age: 42 years to 66 years (direct contribution only, not through any part) *shape: steady; 100% of patients same; verified*	19% to 24%	0.31	1.36	0.31 to 0.31	242	disagrees; AUC gain -0.004 [-0.018, 0.012]	yes
C25	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a downsloping ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	22% to 27%	0.28	1.32	0.28 to 0.28	115, 17 (small n)	disagrees; AUC gain -0.019 [-0.070, 0.025]	yes
C26	direction: resting ECG: a normal resting ECG to resting ECG other than a normal resting ECG (whole model) *shape: categorical; highest to lowest: a normal resting ECG, resting ECG other than a normal resting ECG*	22% to 22%	-0.23	0.80	-0.57 to 0.00	121, 121	disagrees; AUC gain -0.002 [-0.015, 0.009]	yes
C27	magnitude: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels, typical patient *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	22% to 88%	3.32	27.6		242	unclear; AUC gain 0.039 [-0.026, 0.111]	yes
C28	magnitude: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment, typical patient *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	22% to 67%	2.02	7.53		115, 110	disagrees; AUC gain -0.019 [-0.070, 0.025]	yes
C29	magnitude: ST depression on exercise: 0.0 mm to 2.8 mm, typical patient *shape: accelerating; 48% of patients same; fallback*	22% to 48%	1.21	3.36		242	unclear; AUC gain 0.008 [-0.011, 0.031]	yes
C30	magnitude: thallium stress test result: a normal thallium scan to a reversible defect on the thallium scan, typical patient *shape: categorical; highest to lowest: thallium stress test result coded 4 (meaning not recorded), a reversible defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	22% to 47%	1.19	3.28		135, 90	unclear; AUC gain 0.017 [-0.010, 0.052]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C31	magnitude: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L), typical patient *shape: steady; 100% of patients same; verified*	14% to 37%	1.30	3.68		242	disagrees; AUC gain -0.022 [-0.063, 0.008]	yes
C32	structure: made of ischaemia_gate, bp_offset_gate, age_gate_without_ischaemia and inputs age, sex, cp, trestbps, chol, fbs, thalach, exang, slope, ca, thal; added; ann_32 feeds ann_33; ann_33 feeds ann_34					242	train only	yes

T.2 ST depression and thallium signal

Annotation ann_32 (ischaemia_gate). Reads: resting ECG, ST depression during exercise (compared with rest), thallium stress test result.

Original hypothesis (unchanged). Ischaemia-evidence gate (raises risk). 204 = ReLU(0.247 + 2.593*oldpeak + 0.871*restecg:0 + 1.145*thal:7). Hypothesis: a one-sided ST-depression score that opens above an oldpeak threshold set by two flags (always open with a normal resting ECG and a reversible thallium defect; above z-oldpeak -0.09 with a normal ECG only, -0.33 with a defect only, +0.58 with neither), and then raises the logit with oldpeak, directly (0.143 per unit) and through the blood-pressure gate (a further 0.183 per unit while that gate is open). EVIDENCE (train, n=242): active for 48% (100% of the 44 patients with both flags, 20% of the 75 with neither); corr with oldpeak 0.88; disease rate 66% when active vs 24% when inactive. Mean |sensitivity| oldpeak 1.24 > thal:7 0.55 > restecg:0 0.42. This is the main nonlinear effect: removing it with its downstream effects lowers train AUC 0.942 -> 0.934 (to the linear-core level); an affine replacement recovers 0.938. Supported. Oldpeak and restecg:0 have no direct edge; they act only here. The positive restecg:0 weight (normal ECG raises the score) is counter-intuitive and is reported, not explained.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 1.77636e-15).

$$y_0 = \mathrm{relu}\big(0.247 + 1.15x_{thal:7_b} + 2.59x_{oldpeak} + 0.87x_{restecg:0}\big)$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C33	direction: ST depression on exercise: 0.0 mm to 2.8 mm (through this part) *shape: accelerating; 48% of patients same; fallback*	22% to 48%	1.16	3.20	0.55 to 1.91	242	unclear; AUC gain 0.008 [-0.011, 0.031]	yes
C34	direction: thallium stress test result: thallium stress test result other than a reversible defect on the thallium scan to a reversible defect on the thallium scan (through this part) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result other than a reversible defect on the thallium scan*	22% to 32%	0.34	1.40	0.00 to 0.77	152, 90	unclear; AUC gain 0.005 [-0.009, 0.025]	yes
C35	direction: resting ECG: a normal resting ECG to resting ECG other than a normal resting ECG (through this part) *shape: categorical; highest to lowest: a normal resting ECG, resting ECG other than a normal resting ECG*	96% to 94% (typical patient this part acts on)	-0.23	0.80	-0.57 to 0.00	121, 121	disagrees; AUC gain -0.002 [-0.015, 0.009]	yes
C36	interaction: effect of ST depression on exercise (0.0 mm to 2.8 mm) by thallium stress test result: a reversible defect on the thallium scan minus thallium stress test result other than a reversible defect on the thallium scan; stronger in a reversible defect on the thallium scan *shape: steady; 37% of patients same; fallback*	a reversible defect on the thallium scan: 36% to 80%; thallium stress test result other than a reversible defect on the thallium scan: 22% to 48%	0.68	1.98		90, 152	agrees; AUC gain 0.005 [-0.009, 0.025]	yes
C37	interaction: effect of ST depression on exercise (0.0 mm to 2.8 mm) by resting ECG: resting ECG other than a normal resting ECG minus a normal resting ECG; stronger in a normal resting ECG *shape: accelerating; 33% of patients same; fallback*	resting ECG other than a normal resting ECG: 22% to 34%; a normal resting ECG: 22% to 48%	-0.59	0.55		121, 121	agrees; AUC gain -0.002 [-0.015, 0.009]	yes
C38	interaction: effect of thallium stress test result (thallium stress test result other than a reversible defect on the thallium scan to a reversible defect on the thallium scan) by resting ECG: resting ECG other than a normal resting ECG minus a normal resting ECG; stronger in a normal resting ECG *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result other than a reversible defect on the thallium scan*	resting ECG other than a normal resting ECG: 22% to 36%; a normal resting ECG: 22% to 47%	-0.11	0.90		121, 121	agrees; AUC gain -0.002 [-0.015, 0.009]	yes
C39	threshold: ST depression on exercise: fold at about 0.9 mm (starts) *shape: accelerating; 55% of patients same; fallback*	22% to 46% (typical patient with a fold)	1.50	4.47	1.00 to 1.86	198, 72	unclear; AUC gain 0.008 [-0.011, 0.031]	yes
C40	subgroup: who this part acts on: 116 of 242 patients	89% to 96% (typical patient this part acts on (risk without the part, then with it))	0.77	2.17	0.08 to 1.75	116, 126	unclear; AUC gain 0.016 [-0.005, 0.048]	yes
C41	structure: its formula reproduces it on 242 training patients (largest difference 1.77636e-15)		2e-15			242	train only	yes

T.3 Blood pressure step

Annotation ann_33 (bp_offset_gate). Reads: chest pain type, resting blood pressure.

Original hypothesis (unchanged). Blood-pressure offset gate on the ischaemia score (raises risk). 290 = ReLU(-0.495 + 1.079*ischaemia_gate - 0.815*trestbps + 0.176*cp:2 + 0.016*cp:3). Hypothesis: passes the ischaemia score on, minus an offset that grows with resting blood pressure; it adds 0.170 per unit to the logit and switches off the age gate. With the ischaemia gate closed it still opens for low resting blood pressure, so within this gate lower resting BP raises risk (0.170*0.815 = 0.139 per SD), partly offsetting trestbps's direct +0.48 edge. cp:2 (atypical angina) nudges it open; cp:3's weight 0.016 is negligible. EVIDENCE (train, n=242): active for 58% (83% with the ischaemia gate open, 35% without); with the ischaemia gate open, active for 100%, 87% and 59% of the low, middle and high resting-BP groups. Disease rate 51% when active vs 35% when inactive. Mean |sensitivity| ischaemia score 0.58 > trestbps 0.45 > cp:2 0.10 > cp:3 0.01. Readout ablation lowers train AUC 0.942 -> 0.939; an affine replacement fits it well (R2 0.97, AUC 0.941), so its nonlinearity matters little. Supported, small. The negative BP term inside a risk-raising gate is reported, not explained.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 8.88178e-16).

$$y_0 = \mathrm{relu}\big(-0.495 + 1.0x_{identity_290} + 0.0164x_{cp:3_b} + 0.176x_{cp:2} - 0.815x_{trestbps_b}\big)$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C42	direction: chest pain type: chest pain type other than atypical angina, non-anginal chest pain to atypical angina (through this part) *shape: categorical; highest to lowest: atypical angina, non-anginal chest pain, chest pain type other than atypical angina, non-anginal chest pain*	69% to 70% (typical patient this part acts on)	0.04	1.04	0.00 to 0.08	130, 42	unclear; AUC gain 0.000 [0.000, 0.000]	yes
C43	direction: resting blood pressure: 110 mm Hg to 155 mm Hg (through this part) *shape: hinge at 114 mm Hg; 45% of patients same; fallback*	22% to 22%	-0.15	0.86	-0.34 to 0.02	242	disagrees; AUC gain -0.003 [-0.015, 0.006]	no
C44	direction: chest pain type: chest pain type other than atypical angina, non-anginal chest pain to non-anginal chest pain (through this part) *shape: categorical; highest to lowest: atypical angina, non-anginal chest pain, chest pain type other than atypical angina, non-anginal chest pain*	69% to 69% (typical patient this part acts on)	0.003	1.00	0.00 to 0.006	130, 70	unclear; AUC gain 0.000 [0.000, 0.000]	yes
C45	interaction: effect of resting blood pressure (110 mm Hg to 155 mm Hg) by chest pain type: atypical angina minus chest pain type other than atypical angina, non-anginal chest pain; stronger in atypical angina *shape: accelerating; 90% of patients same; verified*	atypical angina: 8% to 20%; chest pain type other than atypical angina, non-anginal chest pain: 14% to 35%	0.04	1.04		42, 130	unclear; AUC gain 0.000 [0.000, 0.000]	no
C46	threshold: resting blood pressure: fold at about 128 mm Hg (stops) *shape: several pieces; 14% of patients same; fallback*	22% to 21% (typical patient with a fold)	-0.16	0.85	-0.42 to 0.03	203, 101	disagrees; AUC gain -0.003 [-0.015, 0.006]	no
C47	subgroup: who this part acts on: 140 of 242 patients	65% to 69% (typical patient this part acts on (risk without the part, then with it))	0.33	1.39	-0.01 to 0.84	140, 102	unclear; AUC gain 0.006 [-0.004, 0.022]	no
C48	structure: its formula reproduces it on 242 training patients (largest difference 8.88178e-16)		9e-16			242	train only	yes

T.4 Extra age term

Annotation ann_34 (age_gate_without_ischaemia). Reads: age.

Original hypothesis (unchanged). Age gate without ischaemia evidence (raises risk with age, small). 1987 = ReLU(0.203 + 0.875*age - 1.589*bp_offset_gate), read out +0.143. Hypothesis: an extra age term (0.125 per SD of age, on top of age's direct +0.117) that acts only while the BP gate is closed or nearly closed, i.e. for patients without much ischaemia evidence; any opening of the BP gate cancels it (each unit of the BP gate removes 0.227 from the logit through this node until it closes). EVIDENCE (train, n=242): active for 35% (65% when the BP gate is closed, rising with age: 0%, 80%, 100% of the youngest, middle and oldest groups; 14% when the BP gate is open). Disease rate barely differs (43% vs 45%). Mean |sensitivity| BP gate 0.35, age 0.31. Removing it lowers train AUC only 0.942 -> 0.941. Supported, near-negligible.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 1.11022e-16).

$$y_0 = \mathrm{relu}\big(0.203 + 1.0x_{identity_1987} + 0.875x_{age_b}\big)$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C49	direction: age: 42 years to 66 years (through this part) *shape: hinge at 53 years; 99% of patients same; verified; one point with C50 (median fold 53 years, typical patient 53 years, 10th-90th percentile 53 years to 66 years, wide)*	21% to 24%	0.10	1.11	0.00 to 0.18	242	unclear; AUC gain 0.000 [-0.01, 0.011]	yes
C50	threshold: age: fold at about 53 years (starts) *shape: hinge at 53 years; 99% of patients same; verified; one point with C50 (median fold 53 years, typical patient 53 years, 10th-90th percentile 53 years to 66 years, wide)*	20% to 24% (typical patient with a fold)	0.17	1.18	0.03 to 0.21	174, 85	unclear; AUC gain 0.000 [-0.01, 0.011]	yes
C51	subgroup: who this part acts on: 85 of 242 patients	28% to 30% (typical patient this part acts on (risk without the part, then with it))	0.10	1.11	0.01 to 0.20	85, 157	disagrees; AUC gain -0.002 [-0.01, 0.004]	yes
C52	structure: its formula reproduces it on 242 training patients (largest difference 1.11022e-16)		1e-16			242	train only	yes

The network before any explanation

Figure 426 (next page, landscape) is the trained network as it is, with no parts, no splits and no inserted units: inputs at the bottom, hidden units by depth, the output at the top, every connection drawn, every input column its own node (one-hot levels included).

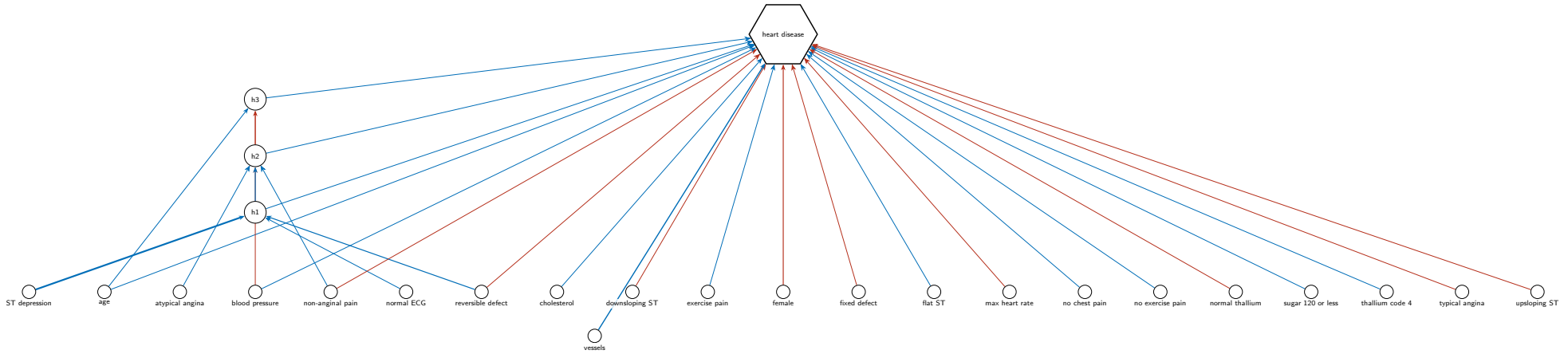


Figure 426: The network as trained, before any explanation, in full. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. Not used (no connection): male, sugar above 120, ST-T abnormal ECG, LVH on ECG.

12 heart-s5-c2

Genome `57c311c6-56f5-4c5d-a475-46433ef58bf2`, explanation `dabdb4bb-e227-46eb-a995-d4123373f234`. Split: train primary (n=242), test as a check (n=61). Claims: 67, verified 59, not supported 8.
AI-assisted draft for review (unverified)

Reference (typical) patient: the typical patient: the median training patient for every number and the most common training category for every category (not a real patient). age: 56 years; ca: 0 vessels; chol: 240 mg/dL (6.2 mmol/L); cp: no chest pain (asymptomatic); exang: no chest pain on exercise; fbs: fasting blood sugar of 120 mg/dL or less; oldpeak: 0.8 mm; restecg: a normal resting ECG; sex: male; slope: an upsloping ST segment; thal: a normal thallium scan; thalach: 152 beats per minute; trestbps: 130 mm Hg. Predicted risk of heart disease: 20%.

12.1 Summary

For a typical patient (a 56-year-old man with no chest pain symptoms and median test results) the model predicts a risk of heart disease of about 20%; it adds up most of its inputs independently, and five parts add to that sum: a signal from ST depression on exercise and the resting ECG that acts on 109 of 242 training patients, a blood pressure step and a damping term that both read that signal, an extra term for left ventricular hypertrophy without chest pain on exercise, and a very small term for high fasting blood sugar. [C34, C42, C53, C61] Figure 427 shows the model at this level: one box per paragraph below.

How to read the diagrams. Edge colour is the sign of that one connection: blue positive, red negative; thicker is a larger weight. The outline and arrow on an input (and on a part's output box) is its overall effect on the final prediction, read from the verified claims: orange and up arrow raises it, teal and down arrow lowers it, grey and double arrow depends (it goes both ways, or differs between levels). A negative connection into a part that itself raises the prediction means that input lowers the prediction; through a hinge the overall effect can only be read from the claims, not from the edges. Up arrow means towards heart disease; down arrow, away from it. A double outline marks an input (or a part's output) that acts at a fixed rate everywhere: it adds the same amount to the score per unit, so it is explained by its rate alone (see 'Linear factors'); a dashed double outline, at a fixed rate on the observed data. The normal outline is an input whose effect switches on and off somewhere. For example, exercise pain has a negative connection into 'Hypertrophy without exercise pain', a part that raises the predicted risk of heart disease, so exercise pain lowers it.

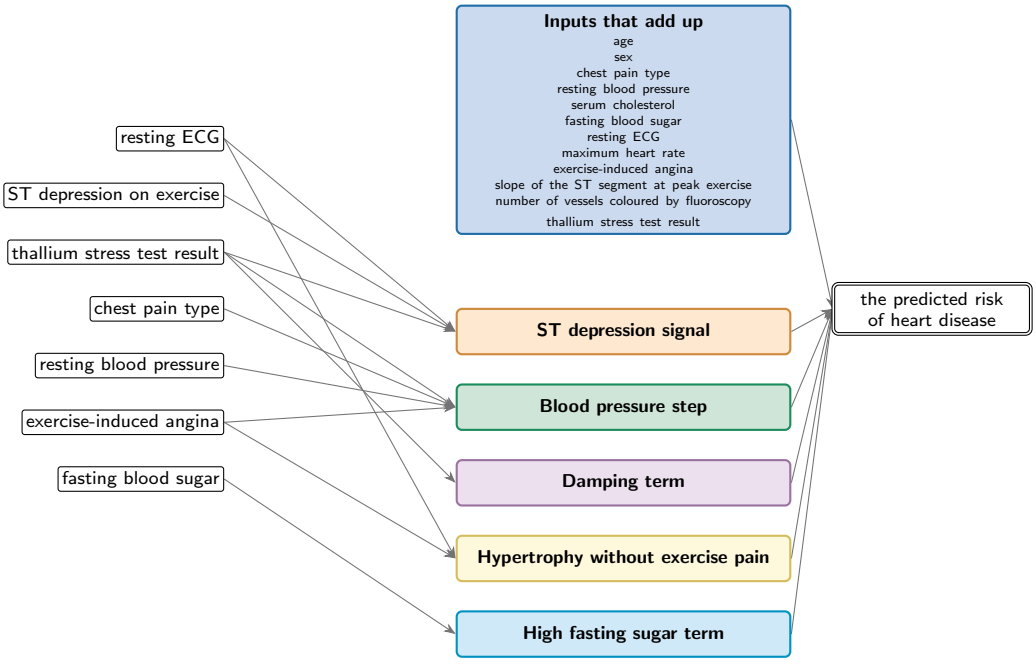


Figure 427: The model at the level of the summary: one box per part.

Inputs that add up. *Lowering the risk.* These add up independently, and these lower the risk: being a woman takes a typical patient's predicted risk from about 20% to about 3%; through this sum alone typical angina lowers it to about 2%, non-anginal chest pain to about 4%, fasting blood sugar above 120 mg/dL to about 10% and

atypical angina to about 11%; and a higher maximum heart rate lowers it steadily, from about 26% at 118 to about 16% at 175 beats per minute. [C08, C06, C12, C18, C20, C19]
Raising the risk. Each extra vessel coloured by fluoroscopy adds about the same, from about 20% at 0 to about 87% at 2 vessels, flattening out near the top because the predicted risk cannot go much higher; a flat rather than upsloping ST segment raises it to about 61%, higher cholesterol raises it steadily from about 13% at 193 mg/dL to about 34% at 308 mg/dL, and older age raises it steadily from about 17% at 42 to about 22% at 66 years. [C01, C02, C04, C27] Through this sum alone, higher resting blood pressure raises it steadily, from about 12% at 110 mm Hg to about 33% at 155 mm Hg, a thallium result other than a normal scan or a fixed defect raises it to about 36%, chest pain brought on by exercise raises it to about 33%, and a resting ECG other than normal raises it to about 25%; all four also act through the parts below. [C05, C09, C14, C26]
Rare results. Some rarer results also count (the model has little data on this, n = 15 to 17): through this sum alone a fixed defect on the thallium scan lowers the risk to about 9%, and a downsloping ST segment raises it to about 27%. [C17, C24]

ST depression signal. This part reads ST depression on exercise, the resting ECG and the thallium scan, and acts on 109 of 242 training patients, 65 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 76% without it and about 89% with it. [C42] Through it, more ST depression adds to the predicted risk only once it rises above about 1.3 mm (between 0.4 and 1.3 mm for different patients), rising in an uneven way: for a typical patient from about 16% at 0 mm to about 50% at 2.8 mm. [C35, C41] ST depression counts more with a normal resting ECG (from about 16% to about 50%) than with any other resting ECG (from about 25% to about 48%), in an uneven way. [C39] Through it, a resting ECG other than normal lowers a typical patient's risk from about 20% to about 16%, and a thallium result coded 4 shuts the signal (the model has little data on this, n = 2). [C36, C37]

Blood pressure step. This part reads that signal together with resting blood pressure, chest pain on exercise and chest pain type, and acts on 145 of 242 training patients, 73 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 51% without it and about 68% with it. [C53] Through it, higher resting blood pressure lowers the risk up to about 125 mm Hg (between 122 and 165 mm Hg for different patients), then no further, in an uneven way: for a typical patient from about 23% at 110 mm Hg to about 17% at 155 mm Hg. [C44, C52] Through it, chest pain on exercise lowers a typical patient's risk from about 20% to about 18%, and non-anginal chest pain to about 19%. [C45, C46]

Damping term. This part reads the same signal and the thallium scan, and acts on 234 of 242 training patients, 102 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 24% without it and about 20% with it. [C56] Through it, a fixed defect on the thallium scan raises a typical patient's risk from about 20% to about 23% (the model has little data on this, n = 15). [C55]

Hypertrophy without exercise pain. This part acts on 74 of 242 training patients, nearly all with left ventricular hypertrophy on the resting ECG and no chest pain on exercise, 30 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 25% without it and about 30% with it. [C61] Through it, left ventricular hypertrophy raises a typical patient's risk from about 20% to about 24%, and for the typical patient it acts on, chest pain on exercise lowers it from about 30% to about 25%. [C59, C58] Across the model, chest pain on exercise raises a typical patient's risk from about 20% to about 31% without left ventricular hypertrophy, and from about 25% to about 35% with it. [C60]

High fasting sugar term. This part acts on 33 of 242 training patients, nearly all with fasting blood sugar above 120 mg/dL, 15 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 24% without it and about 25% with it. [C65] It barely changes the predictions, by at most 2 percentage points. [C66]

The whole network. Figure 428 (next page, landscape) draws every unit and connection of the explained model, nested by part: shaded clusters are the parts of the model and darker boxes the parts inside them; circles are hidden units (h1, h2, ...); each cluster's rectangle is its output, and a dashed 'sum' stands for an addition the next unit does (drawn, not a unit of the model); the hexagon is the model's output. An input used by several parts is drawn under each of them (same label, dotted outline, the first copy marked with the number of copies): it is the same input, repeated for readability. Not used by the model (input columns with no connection): male, atypical angina, ST-T abnormal ECG, flat ST, reversible defect.

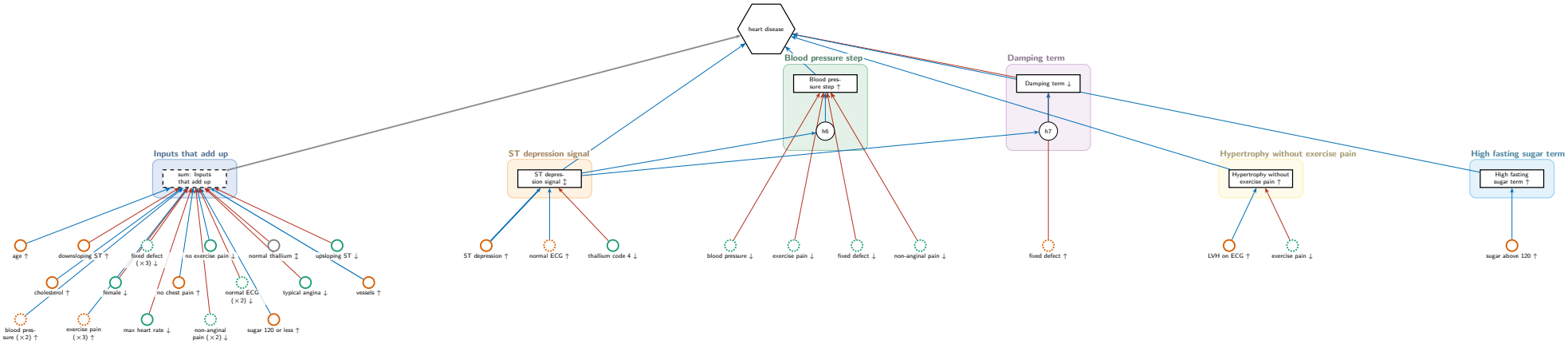


Figure 428: Every unit and connection of the explained model, nested by part. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Worth checking against what is known. Worth checking: the model treats fasting blood sugar above 120 mg/dL as lowering the risk of heart disease (for a typical patient from about 20% to about 11%). [C22] Worth checking: patients with typical angina get the lowest predicted risk of all chest pain types, below patients with no chest pain symptoms. [C07, C11, C21] Worth checking: the model treats a fixed defect on the thallium scan as lowering the risk compared with a normal scan, from only 15 training patients. [C23] Worth checking: ST depression on exercise raises the predicted risk more when the resting ECG is normal than when it is not. [C39] Worth checking: through the blood pressure step, higher resting blood pressure lowers the predicted risk, against its rise through the sum. [C44, C05] Worth checking: for the 2 training patients whose thallium result is coded 4, a code the dataset documentation does not define, the ST depression signal is shut. [C37]

12.2 The model, read from the top down

Each part of the explanation in turn, from the whole model down to the smallest parts: what it does for the prediction (hypothesis), what was checked (evidence), a diagram one level down, and where to read on. The technical record of each part is in "Technical detail" at the end.

12.2.1 The whole model

Figure 429 shows what feeds this part; Figure 430 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

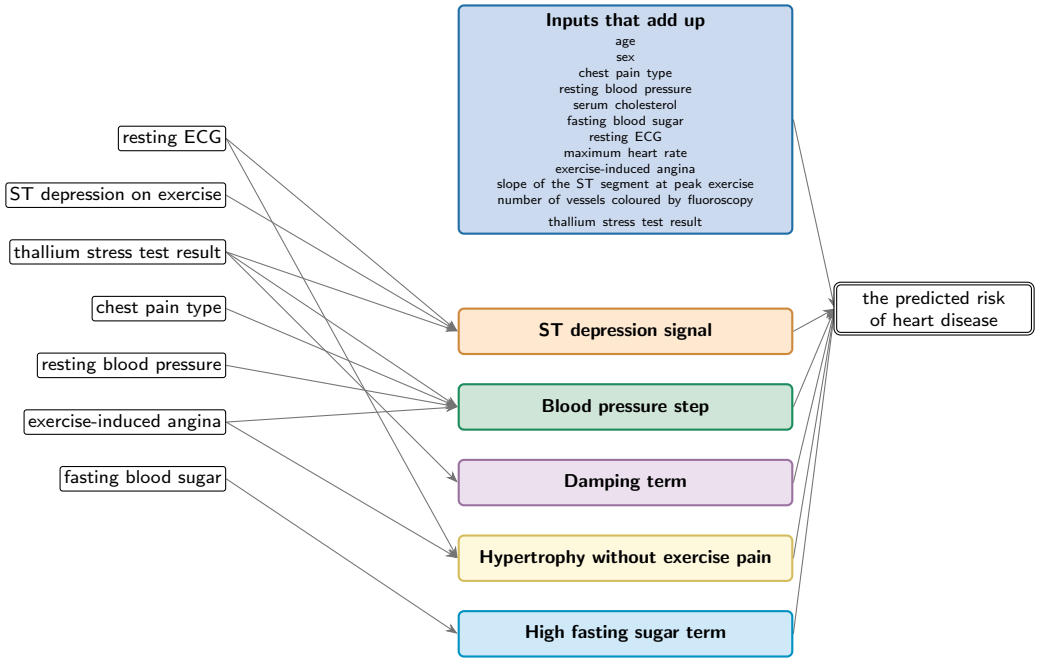


Figure 429: The whole model: what feeds it, one level down.

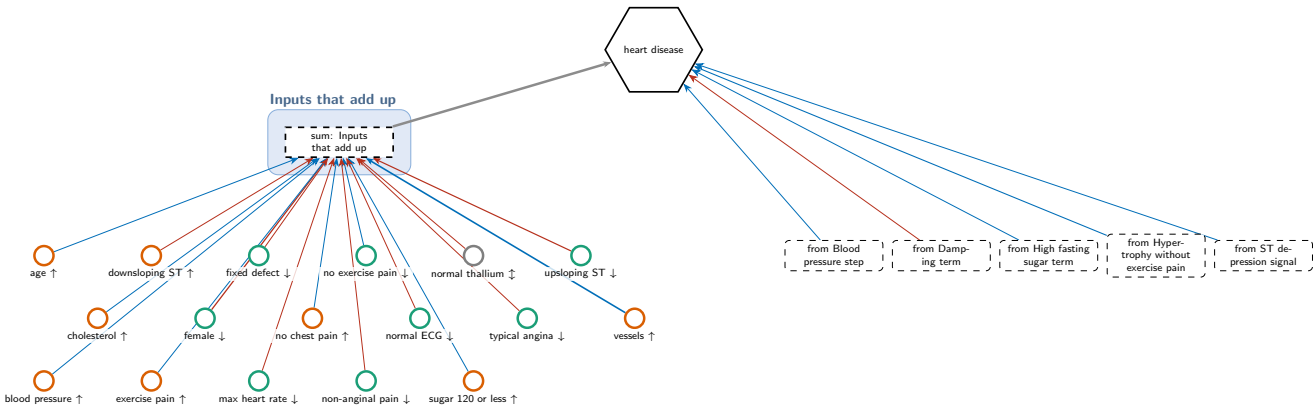


Figure 430: The whole model: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. The model adds up its inputs into one score: the vessels coloured (each extra vessel adding about the same), the ST segment, sex and chest pain type each move the predicted risk on their own, and five parts add to that sum: a signal from ST depression and the resting ECG, a blood pressure step and a damping term that read it, a hypertrophy term and a fasting sugar term. [C34, C01, C02, C08, C06]

Evidence. It is made up of the inputs that feed the score directly and five parts (each explained below); the first part feeds the second and the third, and the score is their sum, which was checked on all 242 training patients. [C34] ST depression on exercise does not feed the score directly, so it acts only through these parts. [C34, C35] Changing one input at a time across the whole model, more ST depression raises a typical patient's predicted risk in an uneven way, from about 16% at 0 mm to about 50% at 2.8 mm, and higher resting blood pressure raises it in an uneven way, from about 15% at 110 mm Hg to about 28% at 155 mm Hg; through the sum alone it rises steadily, from about 12% to about 33%. [C03, C13, C05] Across the whole model, chest pain on exercise raises it to about 31%, non-anginal chest pain lowers it to about 4% and fasting blood sugar above 120 mg/dL to about 11%. [C15, C11, C22] Across the whole model, a fixed defect on the thallium scan lowers it to about 11% (the model has little data on this, n = 15). [C23]
Evidence plots: Figure 431.

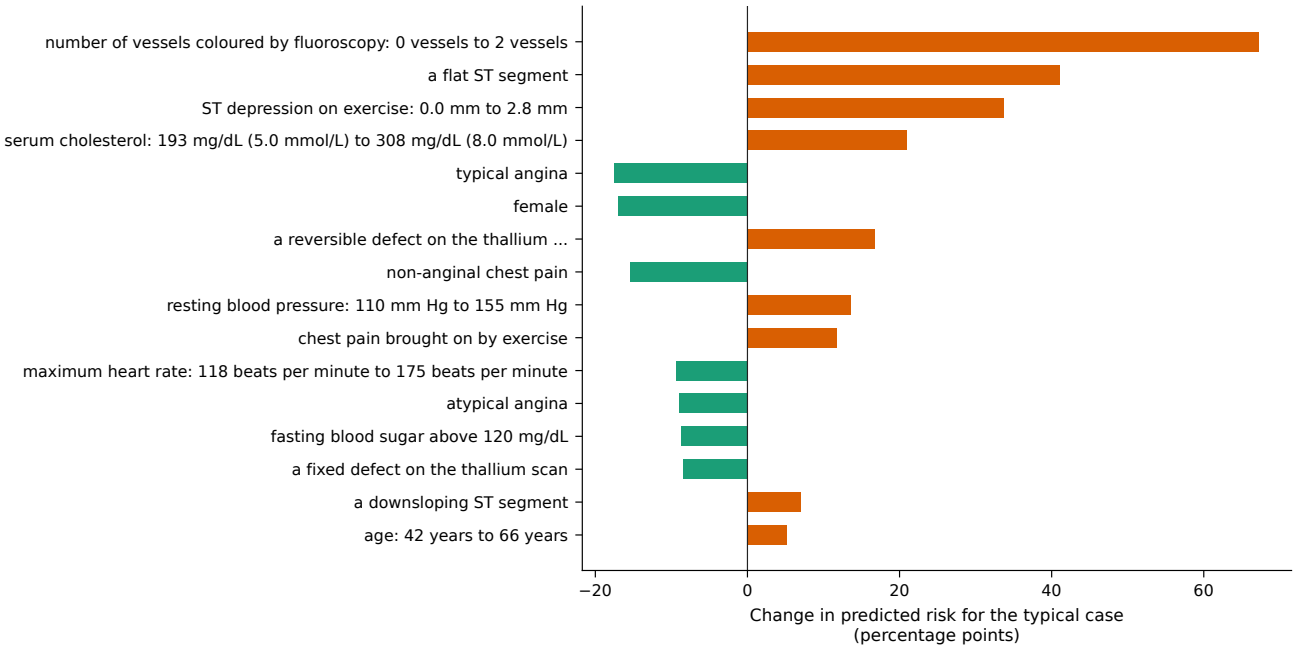


Figure 431: The whole model: Each input's change in the predicted value for the typical case across the whole model, largest first (orange: raises it, teal: lowers it).

Dive deeper: ST depression signal (12.2.2); Blood pressure step (12.2.3); Damping term (12.2.4); Hypertrophy without exercise pain (12.2.5); High fasting sugar term (12.2.6). Technical detail: T.1.

12.2.2 ST depression signal

Part of: The whole model (12.2.1).
Figure 432 shows what feeds this part; Figure 433 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

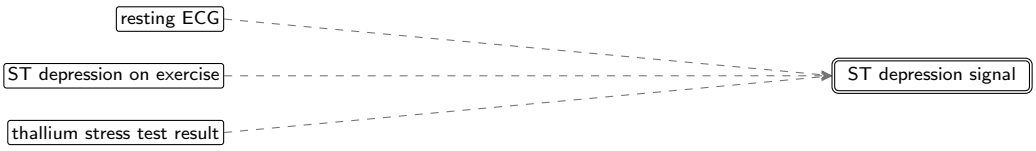


Figure 432: ST depression signal: what feeds it, one level down.

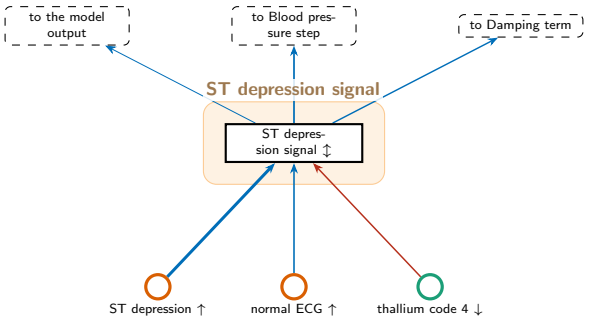


Figure 433: ST depression signal: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part raises the predicted risk with ST depression on exercise once it rises above about 1.3 mm, in an uneven way, and more so with a normal resting ECG. [C35, C41, C39]

Evidence. It acts on 109 of 242 training patients, 65 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 76% without it and about 89% with it. [C42] Through it, more ST depression adds to the risk only once it rises above about 1.3 mm (between 0.4 and 1.3 mm for different patients), rising in an uneven way: for a typical patient from about 16% at 0 mm to about 50% at 2.8 mm. [C35, C41] ST depression counts more with a normal resting ECG (from about 16% to about 50%) than with any other resting ECG (from about 25% to about 48%), in an uneven way. [C39] Through it, a resting ECG other than normal lowers a typical patient's risk from about 20% to about 16%, and a thallium result coded 4 shuts the signal (the model has little data on this, n = 2). [C36, C37] Its formula reproduces the part exactly on all 242 training patients. [C43]
Evidence plots: Figure 434, Figure 435, Figure 436, Figure 437, Figure 438.
Figure 437 shows this part's own output, unit h1: it switches on at about 0.3545 mm of ST depression on exercise, and is zero below.

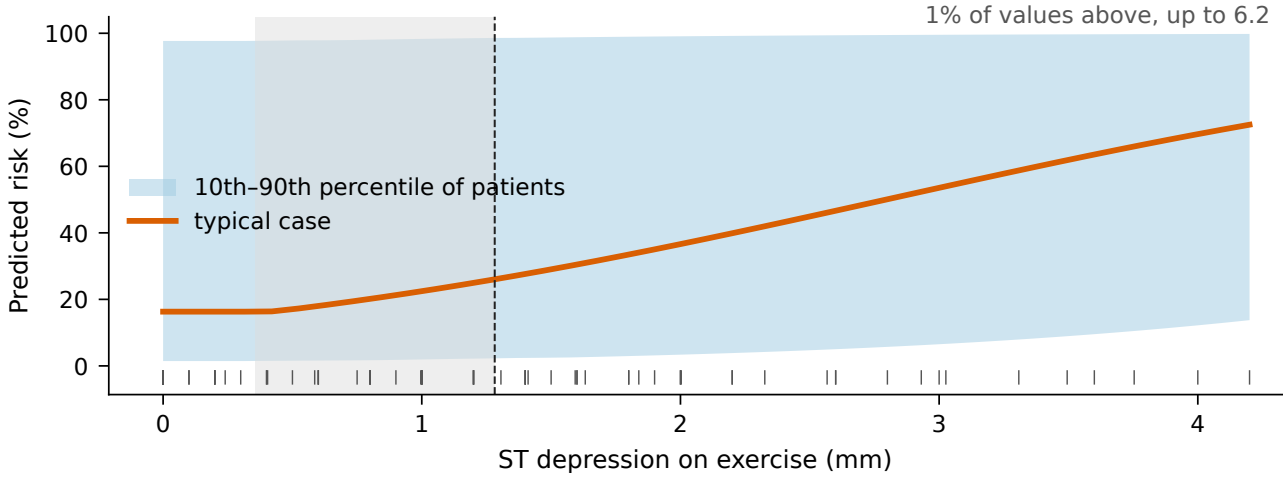


Figure 434: ST depression signal: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual patients (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

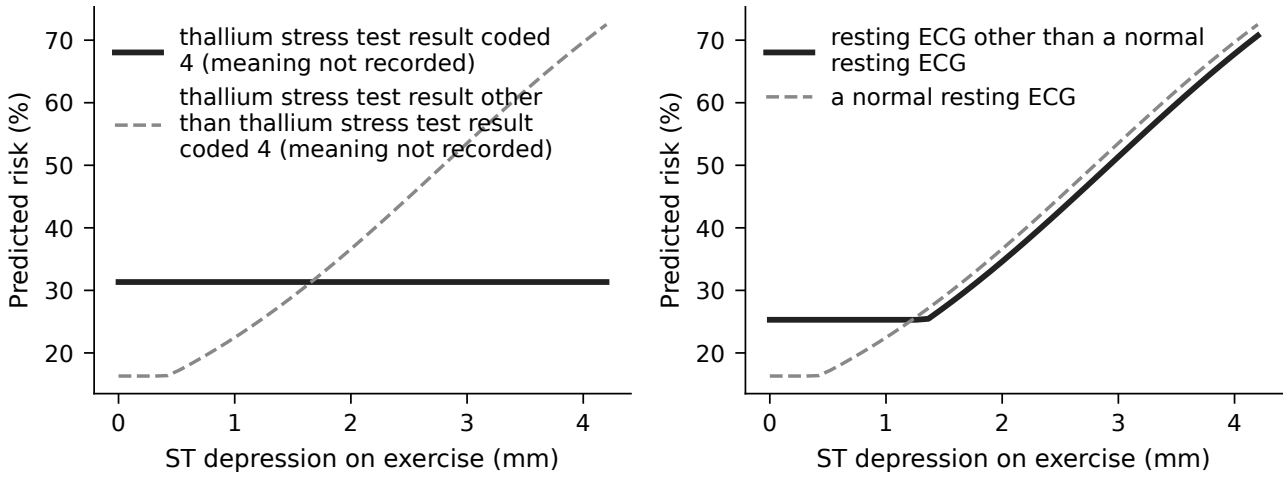


Figure 435: ST depression signal: How the effect differs between groups: one curve per group, for the typical case of each group.

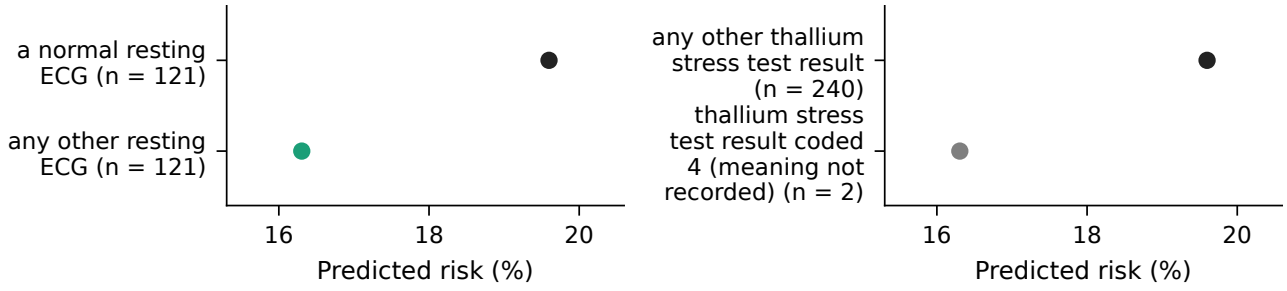


Figure 436: ST depression signal: The predicted value for the typical case at each level, through this part (n = training patients; grey: fewer than 20).

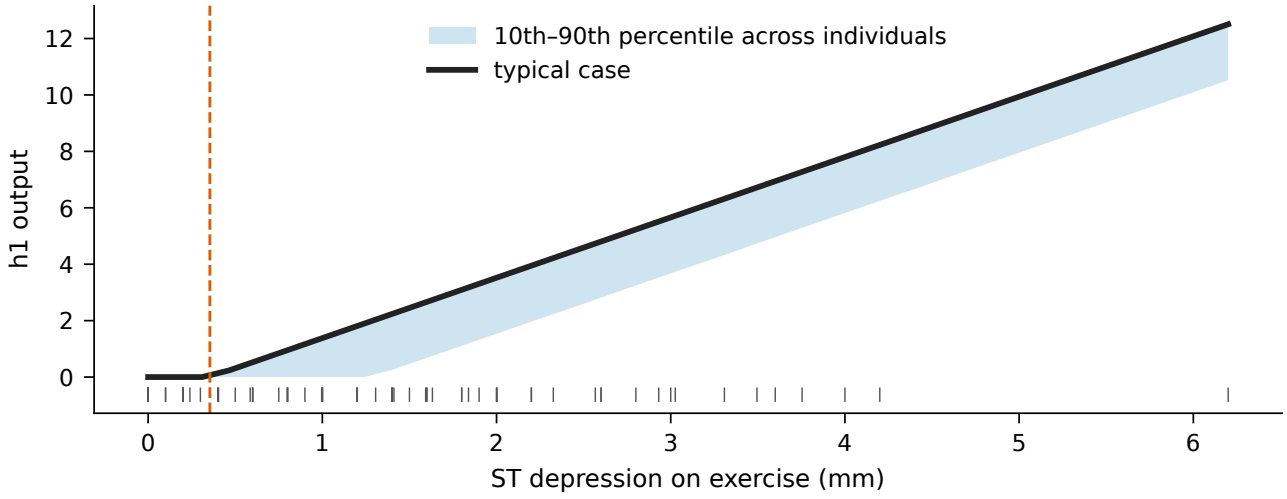


Figure 437: ST depression signal: This part's own output, unit h1 (the output of 'ST depression signal'), on its own scale (its activation), with every other input at the typical case; it is active for 109 of 242. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

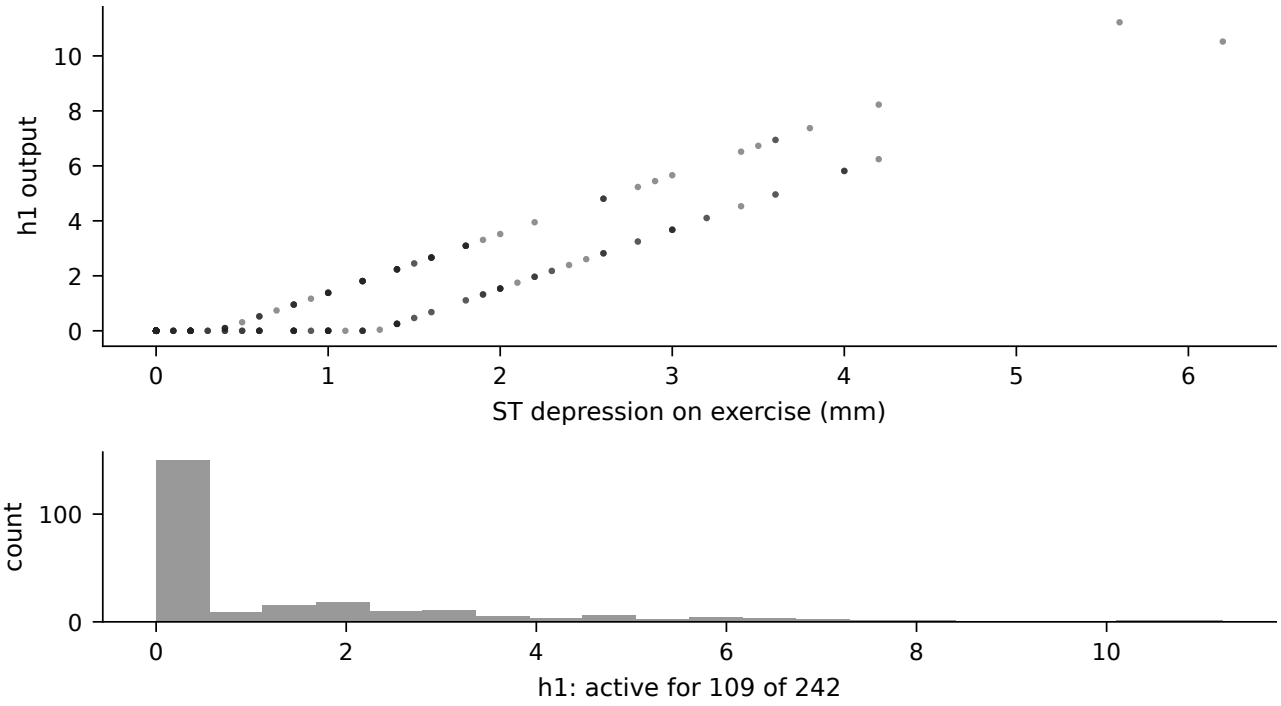


Figure 438: ST depression signal: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.2.

12.2.3 Blood pressure step

Part of: The whole model (12.2.1).
Figure 439 shows what feeds this part; Figure 440 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

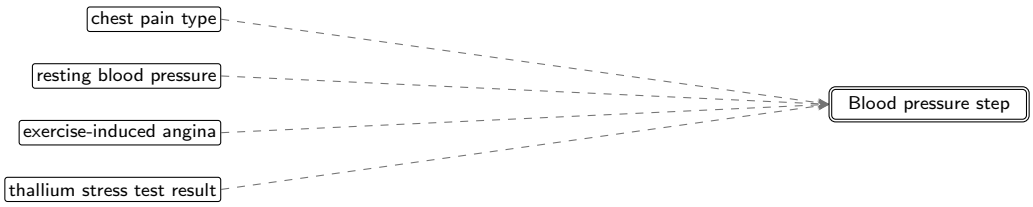


Figure 439: Blood pressure step: what feeds it, one level down.

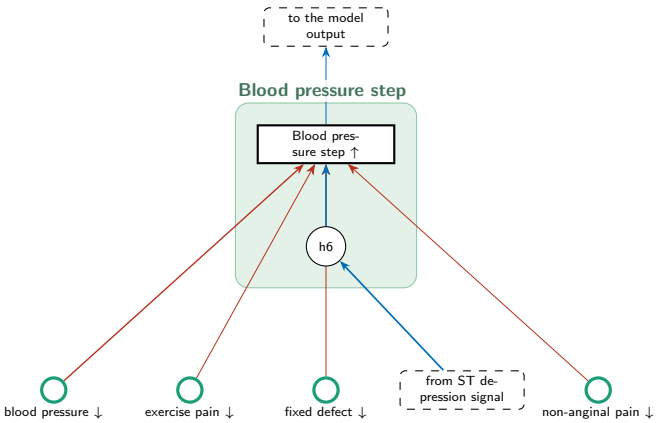


Figure 440: Blood pressure step: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part passes the ST depression signal on to the predicted risk, more for patients with lower resting blood pressure: through it, higher resting blood pressure lowers the risk in an uneven way up to about 125 mm Hg (between 122 mm Hg and 165 mm Hg for different patients), then no further. [C53, C44, C52]

Evidence. It acts on 145 of 242 training patients, 73 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 51% without it and about 68% with it. [C53] Through it, higher resting blood pressure lowers the risk up to about 125 mm Hg (between 122 and 165 mm Hg for different patients), then no further, in an uneven way: for a typical patient from about 23% at 110 mm Hg to about 17% at 155 mm Hg. [C44, C52] Through it, chest pain on exercise lowers a typical patient's risk from about 20% to about 18%, and non-anginal chest pain to about 19%. [C45, C46] Its formula reproduces the part exactly on all 242 training patients. [C54]
Evidence plots: Figure 441, Figure 442, Figure 443, Figure 444.
Figure 443 shows this part's own output, unit h2: over its two main inputs, with the line where it switches on.

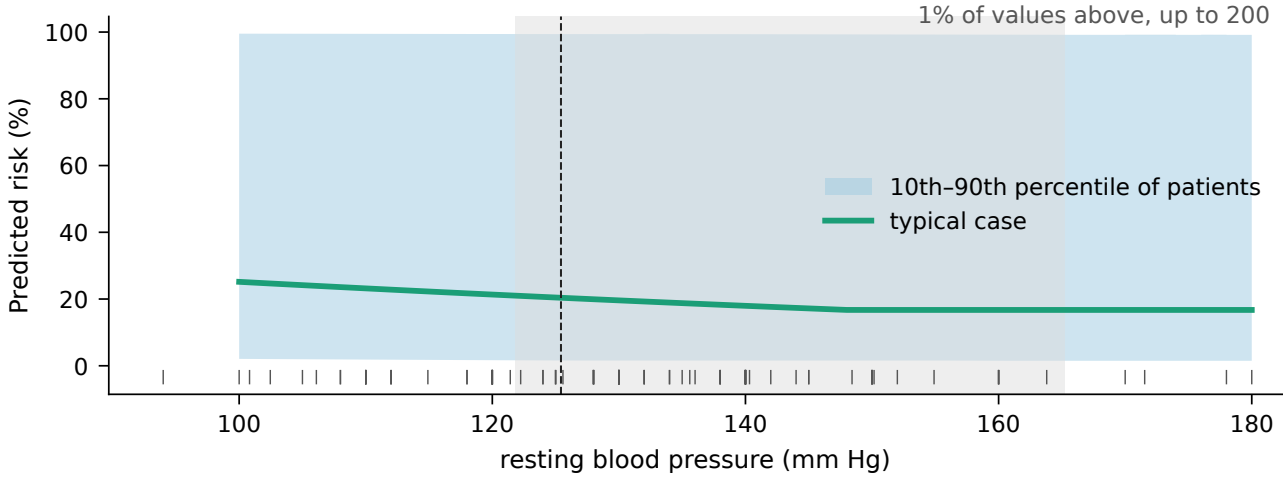


Figure 441: Blood pressure step: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual patients (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

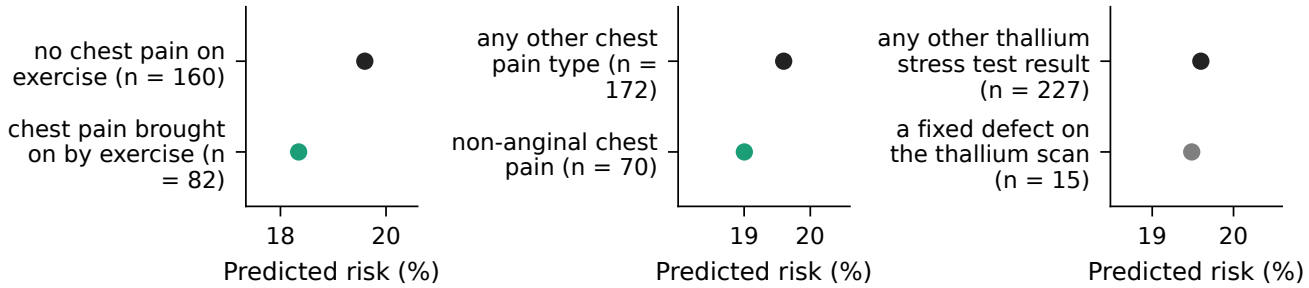


Figure 442: Blood pressure step: The predicted value for the typical case at each level, through this part (n = training patients; grey: fewer than 20).

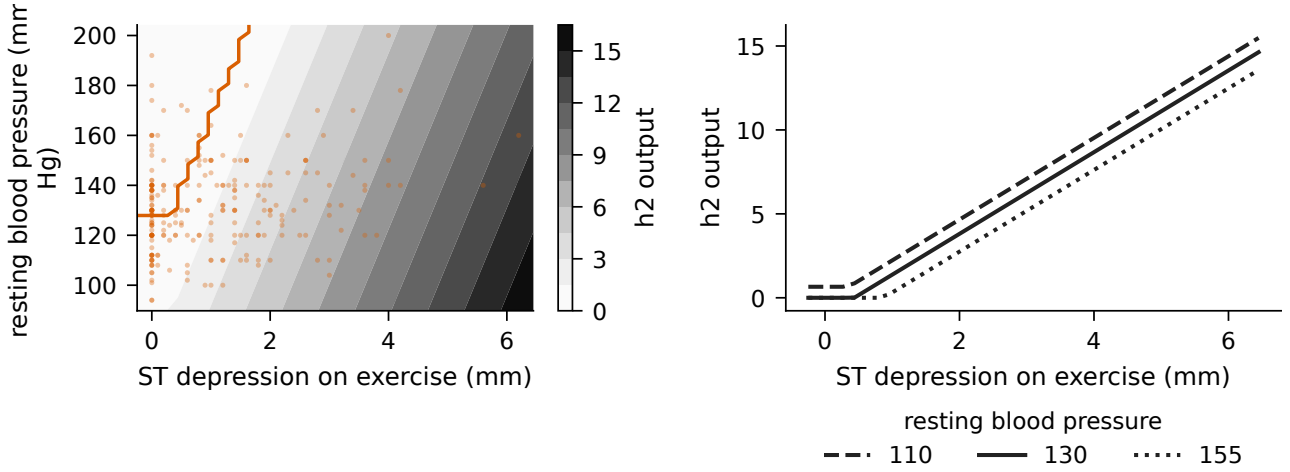


Figure 443: Blood pressure step: This part's own output, unit h2 (the output of 'Blood pressure step'), on its own scale (its activation), with every other input at the typical case; it is active for 145 of 242. Left: its output over its two main inputs (orange line: where it switches on; dots: the data); right: slices at three values of the second input.

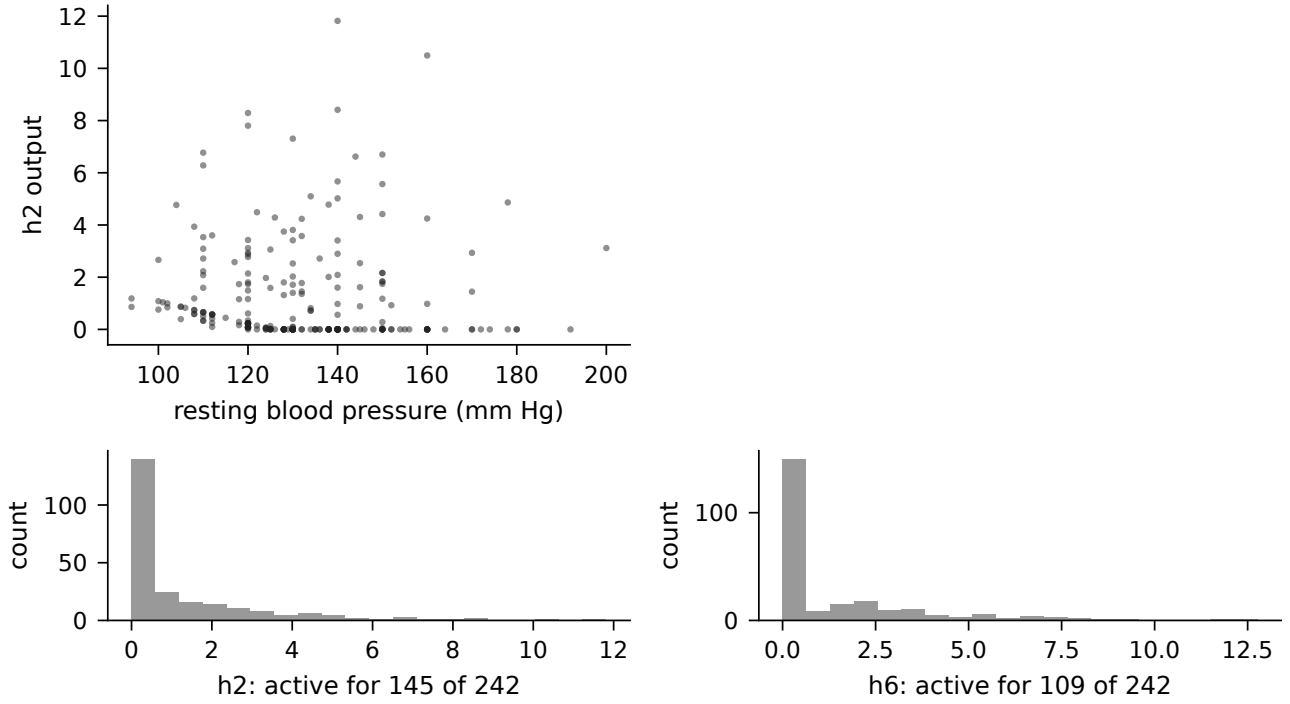


Figure 444: Blood pressure step: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.3.

12.2.4 Damping term

Part of: The whole model (12.2.1).
Figure 445 shows what feeds this part; Figure 446 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).



Figure 445: Damping term: what feeds it, one level down.

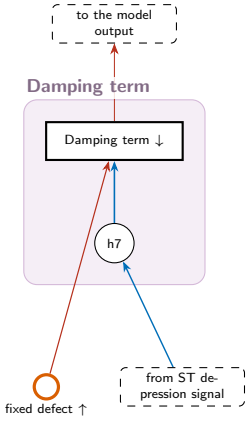


Figure 446: Damping term: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part takes back a little of the ST depression signal for nearly every patient and adds a little for a fixed defect on the thallium scan (the model has little data on this, n = 15). [C56, C55]

Evidence. It acts on 234 of 242 training patients, 102 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 24% without it and about 20% with it. [C56] Through it, a fixed defect on the thallium scan raises a typical patient's risk from about 20% to about 23% (the model has little data on this, n = 15). [C55] Its formula reproduces the part exactly on all 242 training patients. [C57]
Evidence plots: Figure 447, Figure 448, Figure 449.
Figure 448 shows this part's own output, unit h3.

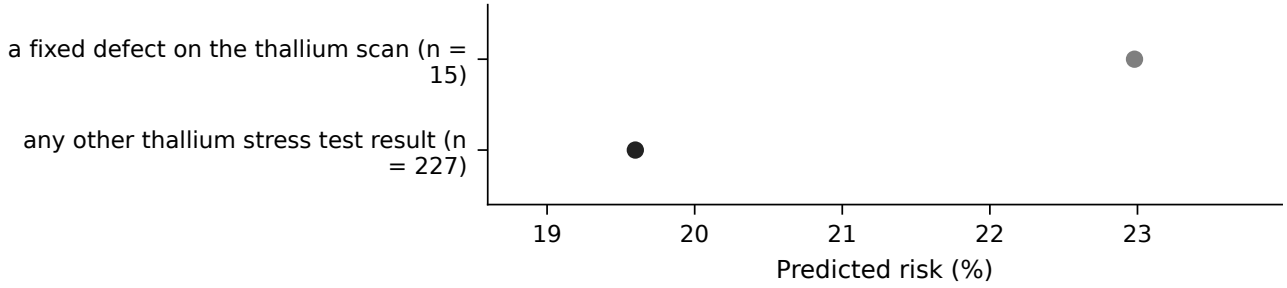


Figure 447: Damping term: The predicted value for the typical case at each level, through this part (n = training patients; grey: fewer than 20).

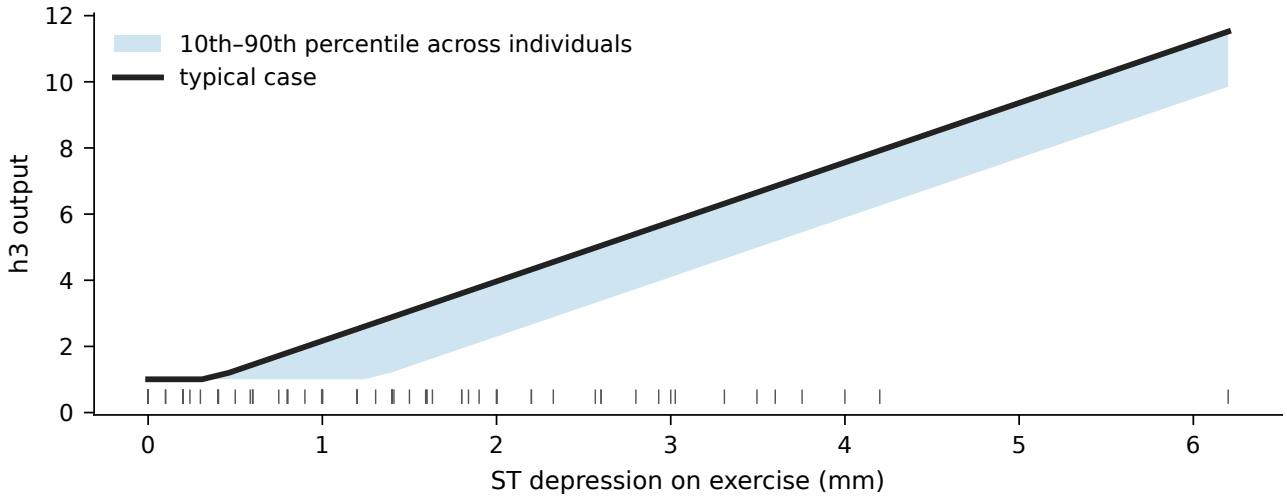


Figure 448: Damping term: This part's own output, unit h3 (the output of 'Damping term'), on its own scale (its activation), with every other input at the typical case; it is active for 234 of 242. Bold: the typical case; band: 10th to 90th percentile across individuals; ticks: the data.

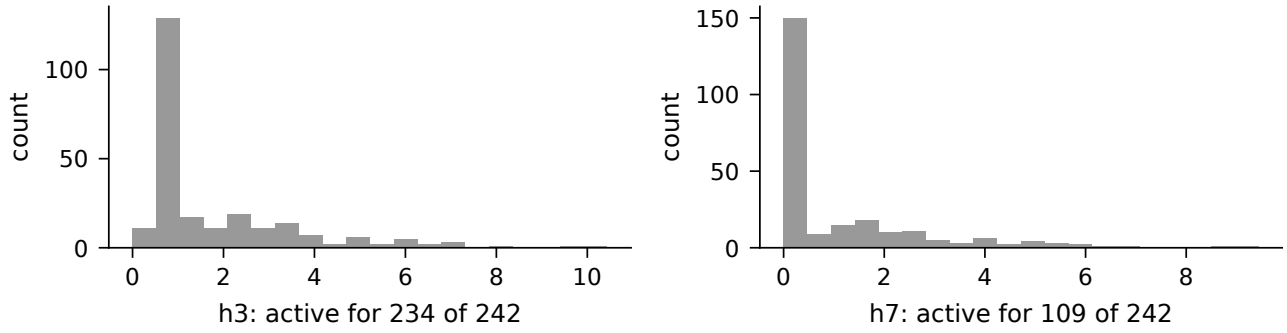


Figure 449: Damping term: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.4.

12.2.5 Hypertrophy without exercise pain

Part of: The whole model (12.2.1).
Figure 450 shows what feeds this part; Figure 451 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).



Figure 450: Hypertrophy without exercise pain: what feeds it, one level down.

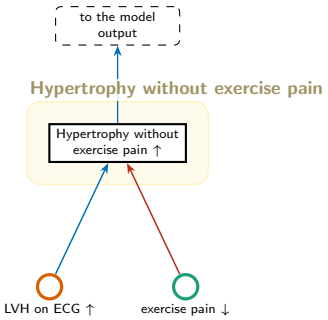


Figure 451: Hypertrophy without exercise pain: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part adds a little to the predicted risk for patients with left ventricular hypertrophy on the resting ECG and no chest pain on exercise. [C61, C59]

Evidence. It acts on 74 of 242 training patients, nearly all with left ventricular hypertrophy and no chest pain on exercise, 30 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 25% without it and about 30% with it. [C61] Through it, left ventricular hypertrophy raises a typical patient's risk from about 20% to about 24%, and for the typical patient it acts on, chest pain on exercise lowers it from about 30% to about 25%. [C59, C58] Across the model, chest pain on exercise raises a typical patient's risk from about 20% to about 31% without left ventricular hypertrophy, and from about 25% to about 35% with it. [C60] Its formula reproduces the part exactly on all 242 training patients. [C63]
Evidence plots: Figure 452, Figure 453, Figure 454.
Figure 453 shows this part's own output, unit h4.

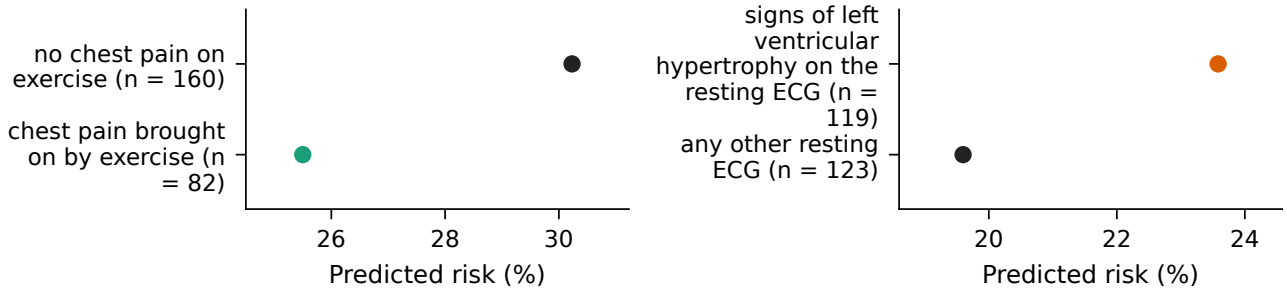


Figure 452: Hypertrophy without exercise pain: The predicted value for the typical case at each level, through this part (n = training patients; grey: fewer than 20).

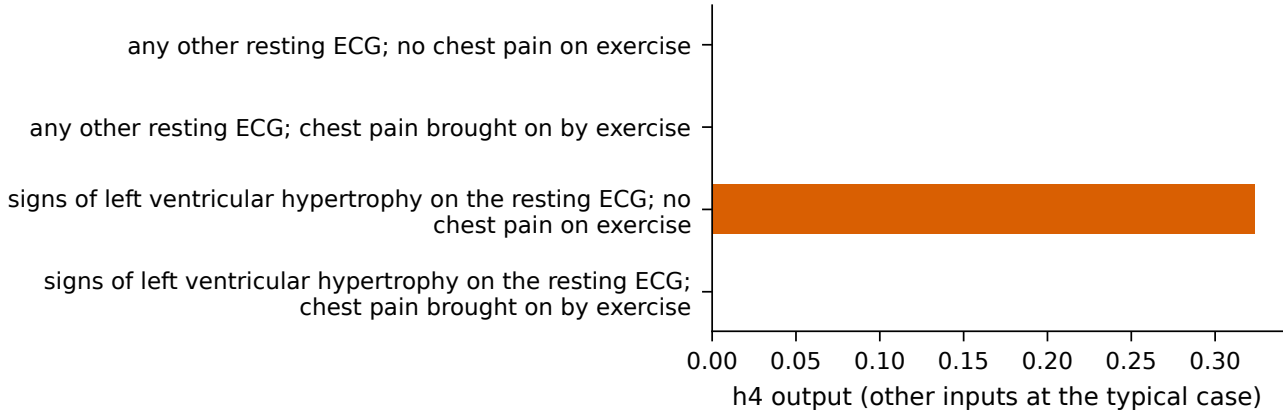


Figure 453: Hypertrophy without exercise pain: This part's own output, unit h4 (the output of 'Hypertrophy without exercise pain'), on its own scale (its activation), with every other input at the typical case; it is active for 74 of 242. Its output for each combination of the levels it reads.

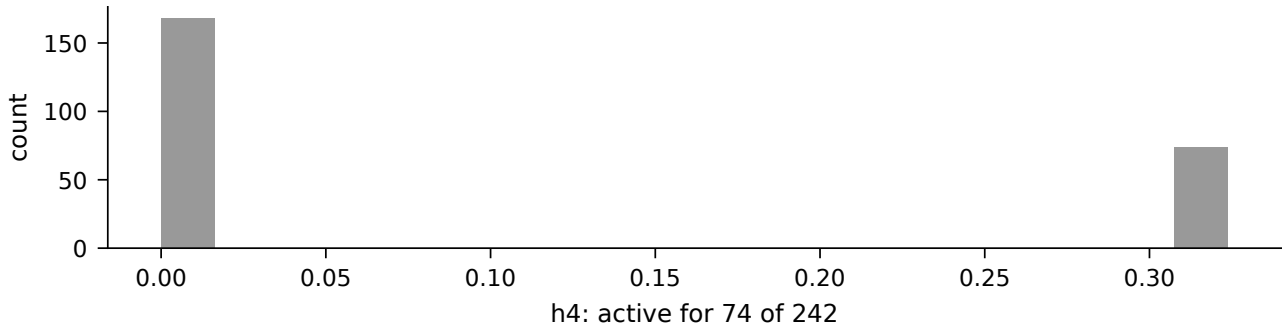


Figure 454: Hypertrophy without exercise pain: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.5.

12.2.6 High fasting sugar term

Part of: The whole model (12.2.1).
Figure 455 shows what feeds this part; Figure 456 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).



Figure 455: High fasting sugar term: what feeds it, one level down.

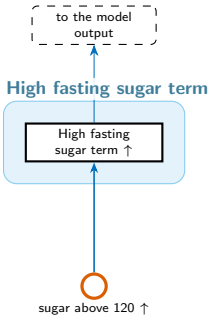


Figure 456: High fasting sugar term: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part adds a very small amount to the predicted risk for patients with fasting blood sugar above 120 mg/dL. [C65, C64]

Evidence. It acts on 33 of 242 training patients, nearly all with fasting blood sugar above 120 mg/dL, 15 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 24% without it and about 25% with it. [C65] Through it, fasting blood sugar above 120 mg/dL raises a typical patient's risk from about 20% to about 21%, and it changes the predictions by at most 2 percentage points. [C64, C66] Its formula reproduces the part exactly on all 242 training patients. [C67]
Evidence plots: Figure 457, Figure 458.

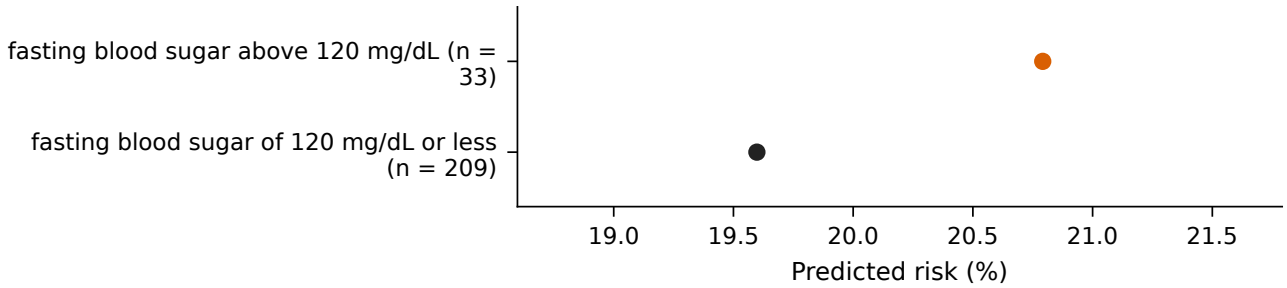


Figure 457: High fasting sugar term: The predicted value for the typical case at each level, through this part (n = training patients; grey: fewer than 20).

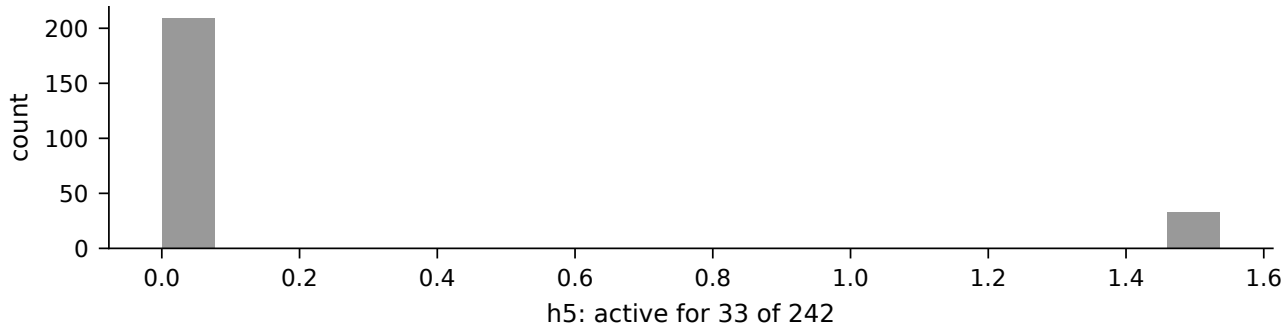


Figure 458: High fasting sugar term: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.6.

12.3 Not supported by the model

- C16: direction claim about thallium stress test result in the whole model: not supported by the model (15% of patients have the opposite sign (allowed: 10%)).
- C25: direction claim about resting ECG in the whole model: not supported by the model (34% of patients have the opposite sign (allowed: 10%)).
- C28: direction claim about resting ECG in the whole model: not supported by the model (41% of patients have the opposite sign (allowed: 10%)).
- C48: interaction claim about resting blood pressure and chest pain type in part ann_34: not supported by the model (the group difference is below 0.1 log-odds).
- C49: interaction claim about chest pain type and exercise-induced angina in part ann_34: not supported by the model (the group difference is below 0.1 log-odds; the part's own counterfactual difference has the opposite sign).
- C50: interaction claim about resting blood pressure and exercise-induced angina in part ann_34: not supported by the model (the group difference is below 0.1 log-odds; the part's own counterfactual difference has the opposite sign).
- C51: interaction claim about resting blood pressure and thallium stress test result in part ann_34: not supported by the model (the group difference is below 0.1 log-odds; the part's own counterfactual difference has the opposite sign).
- C62: null claim about resting ECG and exercise-induced angina in part ann_36: not supported by the model (predictions move by up to 0.059).

Technical detail: heart-s5-c2

The record behind each part: the original hypothesis, the closed form, and every claim with its estimate (log-odds and odds ratio), range across patients, n and test check.

T.1 The whole model

Annotation ann_38 (heart_disease_logit). Reads: age, sex, chest pain type, resting blood pressure, serum cholesterol, fasting blood sugar, resting ECG, maximum heart rate reached during the exercise test, exercise-induced angina, ST depression during exercise (compared with rest), slope of the ST segment at peak exercise, number of major vessels coloured by fluoroscopy, thallium stress test result.

Original hypothesis (unchanged). Root: p = sigmoid(z), z = -0.487 + LINEAR CORE + 0.136*ischaemia_gate + 0.252*bp_offset_gate - 0.118*ischaemia_damping_thal6 + 0.729*lvh_without_exercise_angina + 0.048*high_fasting_sugar. LINEAR CORE (17 direct input->output edges, z-scored inputs): ca +1.473, sex:0 -1.062, slope:1 -0.921, chol +0.522, trestbps +0.498, cp:1 -0.489, exang:1 +0.471, cp:3 -0.444, thal:3 -0.423, thal:6 -0.413, slope:3 -0.370, cp:4 +0.357, fbs:0 +0.264, thalach -0.226, restecg:0 -0.159, exang:0 +0.136, age +0.126. Five inputs are unconnected (sex:1, cp:2, restecg:1, slope:2, thal:7; zero-effect entries of this root). Oldpeak, thal:4, restecg:2 and fbs:1 have no direct edge and act only through the children. COMPOSITION: additive readouts, with a serial fan-out inside. ischaemia_gate (oldpeak above a resting-ECG-dependent threshold) is read out directly and also feeds bp_offset_gate (the ischaemia score minus a resting-BP offset, the largest nonlinear readout) and ischaemia_damping_thal6 (a near-linear term that takes back part of the ischaemia score and adds a fixed-defect term); identity nodes identity_290 and identity_1416 carry those two links so each gate is its own leaf. Two independent indicators add to z: lvh_without_exercise_angina (+0.236 for LVH on the resting ECG without exercise angina) and high_fasting_sugar (+0.074, negligible). EVIDENCE (train, n=242): full model AUC 0.9389, acc 0.880 (compute_performance). Linear core alone AUC 0.933; logit SD 3.19 (core) vs 0.57 (all hidden readouts). Removing the ischaemia gate with its downstream effects gives 0.932; readout-only ablations: ischaemia 0.938, BP 0.933, damping 0.938, LVH 0.939, fbs 0.939. Treating all five ReLUs as identities gives 0.917 (the gating matters); one least-squares linear fit of all hidden readouts on their inputs gives 0.938 (R2 0.84); an L2 logistic regression on all 26 inputs gives 0.936. Mean |SHAP| (train, all 242): ca 0.145, sex:0 0.109, slope:1 0.108, thal:3 0.050, oldpeak 0.047, cp:3 0.045, exang:1 0.044, chol 0.043. Interpretation: close to logistic regression plus one ischaemia correction: ST depression raises risk only above a threshold that is lower with a normal resting ECG, more steeply when resting BP is low; LVH without exercise angina adds a little. Counter-directional signs reported, not explained: restecg:0 positive in the ischaemia gate, trestbps negative in the BP gate, thal:6 (fixed defect) negative directly, fbs:0 +0.26 directly. Same core design as the 10-seed heart-s5 record (key 9330) and the sibling cluster lead heart-s5-c1 (ischaemia gate -> resting-BP gate), with a damping node, an LVH indicator and a fasting-sugar indicator instead of c1's age gate.

Closed form. Read through its parts (their formulas are in their own sections).

Split inputs. To annotate the parts, the explanation split these inputs into copies (a refactoring step that does not change what the model computes); the diagrams draw each input once: non-anginal pain (2 copies); exercise pain (3 copies); normal ECG (2 copies); fixed defect (3 copies); blood pressure (2 copies).

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C01	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (whole model) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	20% to 87%	3.30	27.2	3.30 to 3.30	242	unclear; AUC gain 0.025 [-0.049, 0.102]	yes
C02	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	20% to 61%	1.84	6.32	1.84 to 1.84	115, 110	disagrees; AUC gain -0.017 [-0.061, 0.016]	yes
C03	direction: ST depression on exercise: 0.0 mm to 2.8 mm (whole model) *shape: hinge at 0.4 mm; 70% of patients same; fallback*	16% to 50%	1.22	3.40	0.77 to 1.69	242	unclear; AUC gain 0.009 [-0.008, 0.033]	yes
C04	direction: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L) (whole model) *shape: steady; 100% of patients same; verified*	13% to 34%	1.25	3.48	1.25 to 1.25	242	disagrees; AUC gain -0.028 [-0.071, 0.002]	yes
C05	direction: resting blood pressure: 110 mm Hg to 155 mm Hg (direct contribution only, not through any part) *shape: steady; 100% of patients same; verified*	12% to 33%	1.23	3.43	1.23 to 1.23	242	unclear; AUC gain 0.006 [-0.024, 0.037]	yes
C06	direction: chest pain type: no chest pain (asymptomatic) to typical angina (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	20% to 2%	-2.42	0.09	-2.42 to -2.42	108, 22	agrees; AUC gain 0.032 [0.008, 0.062]	yes
C07	direction: chest pain type: no chest pain (asymptomatic) to typical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	20% to 2%	-2.42	0.09	-2.42 to -2.42	108, 22	agrees; AUC gain 0.032 [0.008, 0.062]	yes
C08	direction: sex: male to female (whole model) *shape: categorical; highest to lowest: male, female*	20% to 3%	-2.24	0.11	-2.24 to -2.24	159, 83	unclear; AUC gain 0.022 [-0.011, 0.065]	yes
C09	direction: thallium stress test result: a normal thallium scan to thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan (direct contribution only, not through any part) *shape: categorical; highest to lowest: thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	20% to 36%	0.85	2.34	0.85 to 0.85	135, 92	unclear; AUC gain 0.013 [-0.011, 0.041]	yes
C10	direction: thallium stress test result: a normal thallium scan to a reversible defect on the thallium scan (whole model) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result coded 4 (meaning not recorded), a normal thallium scan, a fixed defect on the thallium scan*	20% to 36%	0.85	2.34	0.85 to 0.85	135, 90	unclear; AUC gain 0.014 [-0.009, 0.042]	yes
C11	direction: chest pain type: no chest pain (asymptomatic) to non-anginal chest pain (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	20% to 4%	-1.72	0.18	-1.74 to -1.70	108, 70	agrees; AUC gain 0.032 [0.008, 0.062]	yes
C12	direction: chest pain type: no chest pain (asymptomatic) to non-anginal chest pain (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	20% to 4%	-1.70	0.18	-1.70 to -1.70	108, 70	agrees; AUC gain 0.032 [0.008, 0.062]	yes
C13	direction: resting blood pressure: 110 mm Hg to 155 mm Hg (whole model) *shape: accelerating; 75% of patients same; fallback*	15% to 28%	0.97	2.63	0.75 to 1.15	242	unclear; AUC gain 0.005 [-0.017, 0.031]	yes
C14	direction: exercise-induced angina: no chest pain on exercise to chest pain brought on by exercise (direct contribution only, not through any part) *shape: categorical; highest to lowest: chest pain brought on by exercise, no chest pain on exercise*	20% to 33%	0.71	2.03	0.71 to 0.71	160, 82	agrees; AUC gain 0.012 [0.001, 0.030]	yes
C15	direction: exercise-induced angina: no chest pain on exercise to chest pain brought on by exercise (whole model) *shape: categorical; highest to lowest: chest pain brought on by exercise, no chest pain on exercise*	20% to 31%	0.55	1.73	0.39 to 0.71	160, 82	unclear; AUC gain 0.008 [0.000, 0.022]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C16	direction: thallium stress test result: a normal thallium scan to thallium stress test result coded 4 (meaning not recorded) (whole model) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result coded 4 (meaning not recorded), a normal thallium scan, a fixed defect on the thallium scan*	20% to 31%	0.52	1.69	-0.22 to 0.85	135, 2 (small n)	unclear; AUC gain 0.014 [-0.009, 0.042]	no
C17	direction: thallium stress test result: a normal thallium scan to a fixed defect on the thallium scan (direct contribution only, not through any part) *shape: categorical; highest to lowest: thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	20% to 9%	-0.86	0.42	-0.86 to -0.86	135, 15 (small n)	unclear; AUC gain 0.013 [-0.011, 0.041]	yes
C18	direction: fasting blood sugar: fasting blood sugar of 120 mg/dL or less to fasting blood sugar above 120 mg/dL (direct contribution only, not through any part) *shape: categorical; highest to lowest: fasting blood sugar of 120 mg/dL or less, fasting blood sugar above 120 mg/dL*	20% to 10%	-0.77	0.46	-0.77 to -0.77	209, 33	disagrees; AUC gain -0.004 [-0.020, 0.009]	yes
C19	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (whole model) *shape: steady; 100% of patients same; verified*	26% to 16%	-0.57	0.56	-0.57 to -0.57	242	unclear; AUC gain 0.006 [0.000, 0.022]	yes
C20	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	20% to 11%	-0.72	0.49	-0.72 to -0.72	108, 42	agrees; AUC gain 0.032 [0.008, 0.062]	yes
C21	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	20% to 11%	-0.72	0.49	-0.72 to -0.72	108, 42	agrees; AUC gain 0.032 [0.008, 0.062]	yes
C22	direction: fasting blood sugar: fasting blood sugar of 120 mg/dL or less to fasting blood sugar above 120 mg/dL (whole model) *shape: categorical; highest to lowest: fasting blood sugar of 120 mg/dL or less, fasting blood sugar above 120 mg/dL*	20% to 11%	-0.69	0.50	-0.69 to -0.69	209, 33	disagrees; AUC gain -0.003 [-0.019, 0.009]	yes
C23	direction: thallium stress test result: a normal thallium scan to a fixed defect on the thallium scan (whole model) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result coded 4 (meaning not recorded), a normal thallium scan, a fixed defect on the thallium scan*	20% to 11%	-0.69	0.50	-0.75 to -0.58	135, 15 (small n)	unclear; AUC gain 0.014 [-0.009, 0.042]	yes
C24	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a downsloping ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	20% to 27%	0.40	1.48	0.40 to 0.40	115, 17 (small n)	disagrees; AUC gain -0.017 [-0.061, 0.016]	yes
C25	direction: resting ECG: a normal resting ECG to signs of left ventricular hypertrophy on the resting ECG (whole model) *shape: categorical; highest to lowest: signs of left ventricular hypertrophy on the resting ECG, an ST-T wave abnormality on the resting ECG, a normal resting ECG*	20% to 25%	0.21	1.24	-0.32 to 0.55	121, 119	disagrees; AUC gain -0.003 [-0.023, 0.013]	no
C26	direction: resting ECG: a normal resting ECG to resting ECG other than a normal resting ECG (direct contribution only, not through any part) *shape: categorical; highest to lowest: resting ECG other than a normal resting ECG, a normal resting ECG*	20% to 25%	0.32	1.37	0.32 to 0.32	121, 121	unclear; AUC gain 0.000 [-0.016, 0.015]	yes
C27	direction: age: 42 years to 66 years (whole model) *shape: steady; 100% of patients same; verified*	17% to 22%	0.33	1.39	0.33 to 0.33	242	disagrees; AUC gain -0.005 [-0.023, 0.01]	yes
C28	direction: resting ECG: a normal resting ECG to an ST-T wave abnormality on the resting ECG (whole model) *shape: categorical; highest to lowest: signs of left ventricular hypertrophy on the resting ECG, an ST-T wave abnormality on the resting ECG, a normal resting ECG*	20% to 21%	0.06	1.06	-0.32 to 0.32	121, 2 (small n)	disagrees; AUC gain -0.003 [-0.023, 0.013]	no

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C29	magnitude: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels, typical patient *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	20% to 87%	3.30	27.2		242	unclear; AUC gain 0.025 [-0.049, 0.102]	yes
C30	magnitude: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment, typical patient *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	20% to 61%	1.84	6.32		115, 110	disagrees; AUC gain -0.017 [-0.061, 0.016]	yes
C31	magnitude: ST depression on exercise: 0.0 mm to 2.8 mm, typical patient *shape: hinge at 0.4 mm; 70% of patients same; fallback*	16% to 50%	1.64	5.15		242	unclear; AUC gain 0.009 [-0.008, 0.033]	yes
C32	magnitude: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L), typical patient *shape: steady; 100% of patients same; verified*	13% to 34%	1.25	3.48		242	disagrees; AUC gain -0.028 [-0.071, 0.002]	yes
C33	magnitude: chest pain type: no chest pain (asymptomatic) to typical angina, typical patient *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	20% to 2%	-2.42	0.09		108, 22	agrees; AUC gain 0.032 [0.008, 0.062]	yes
C34	structure: made of ischaemia_gate, bp_offset_gate, ischaemia_damping_thal6, lvh_without_exercise_angina, high_fasting_sugar and inputs age, sex, cp, trestbps, chol, fbs, restecg, thalach, exang, slope, ca, thal; added; ann_33 feeds ann_34; ann_33 feeds ann_35					242	train only	yes

T.2 ST depression signal

Annotation ann_33 (ischaemia_gate). Reads: resting ECG, ST depression during exercise (compared with rest), thallium stress test result.

Original hypothesis (unchanged). Ischaemia-evidence gate (raises risk). 204 = ReLU(0.385 + 2.536*oldpeak + 0.992*restecg:0 - 0.876*thal:4). Hypothesis: a one-sided ST-depression score that opens above an oldpeak threshold set by the resting ECG (z-oldpeak -0.57 with a normal ECG, +0.21 without) and then rises with oldpeak; it adds 0.136 per unit to the logit directly, a further 0.286 per unit through the BP gate while that gate is open, and -0.099 per unit through the damping node, so about 0.32 logit per unit in all. EVIDENCE (train, n=242): active for 45% (51% with a normal ECG, 40% without); corr with oldpeak 0.92; disease rate 60% when active vs 32% when inactive. Mean |sensitivity| oldpeak 1.14 > restecg:0 0.45 > thal:4 0.39 (thal:4 has 2 patients). This is the main nonlinear effect: removing it with its downstream effects lowers train AUC 0.939 -> 0.932 (below the linear-core level 0.933); an affine replacement recovers 0.938. Supported. Oldpeak and thal:4 have no direct edge and act only here. The positive restecg:0 weight (a normal ECG raises the score; restecg:0's direct edge is -0.16) is counter-intuitive and is reported, not explained.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 1.77636e-15).

$$y_0 = \mathrm{relu}\big(0.385 + 0.992x_{restecg:0_b} + 2.54x_{oldpeak} - 0.876x_{thal:4}\big)$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C35	direction: ST depression on exercise: 0.0 mm to 2.8 mm (through this part) *shape: hinge at 1.3 mm; 70% of patients same; fallback; one point with C41 (median fold 1.3 mm, typical patient 0.4 mm, 10th-90th percentile 0.4 mm to 1.3 mm, wide)*	16% to 50%	1.22	3.40	0.77 to 1.69	242	unclear; AUC gain 0.009 [-0.008, 0.033]	yes
C36	direction: resting ECG: a normal resting ECG to resting ECG other than a normal resting ECG (through this part) *shape: categorical; highest to lowest: a normal resting ECG, resting ECG other than a normal resting ECG*	20% to 16%	-0.26	0.77	-0.64 to 0.00	121, 121	disagrees; AUC gain -0.002 [-0.016, 0.012]	yes
C37	direction: thallium stress test result: thallium stress test result other than thallium stress test result coded 4 (meaning not recorded) to thallium stress test result coded 4 (meaning not recorded) (through this part) *shape: categorical; highest to lowest: thallium stress test result other than thallium stress test result coded 4 (meaning not recorded), thallium stress test result coded 4 (meaning not recorded)*	20% to 16%	-0.33	0.72	-1.07 to 0.00	240, 2 (small n)	disagrees; AUC gain -0.001 [-0.006, 0.000]	yes
C38	interaction: effect of ST depression on exercise (0.0 mm to 2.8 mm) by thallium stress test result: thallium stress test result coded 4 (meaning not recorded) minus thallium stress test result other than thallium stress test result coded 4 (meaning not recorded); stronger in thallium stress test result other than thallium stress test result coded 4 (meaning not recorded) *shape: hinge at 0.4 mm; 74% of patients same; fallback*	thallium stress test result coded 4 (meaning not recorded): 31% to 31%; thallium stress test result other than thallium stress test result coded 4 (meaning not recorded): 16% to 50%	-1.23	0.29		2, 240 (small n)	unclear; AUC gain -0.001 [-0.006, 0.000]	yes
C39	interaction: effect of ST depression on exercise (0.0 mm to 2.8 mm) by resting ECG: resting ECG other than a normal resting ECG minus a normal resting ECG; stronger in a normal resting ECG *shape: hinge at 0.4 mm; 57% of patients same; fallback*	resting ECG other than a normal resting ECG: 25% to 48%; a normal resting ECG: 16% to 50%	-0.68	0.50		121, 121	agrees; AUC gain -0.002 [-0.016, 0.012]	yes
C40	interaction: effect of thallium stress test result (thallium stress test result other than thallium stress test result coded 4 (meaning not recorded) to thallium stress test result coded 4 (meaning not recorded)) by resting ECG: resting ECG other than a normal resting ECG minus a normal resting ECG; stronger in resting ECG other than a normal resting ECG *shape: categorical; highest to lowest: thallium stress test result other than thallium stress test result coded 4 (meaning not recorded), thallium stress test result coded 4 (meaning not recorded)*	resting ECG other than a normal resting ECG: 25% to 44%; a normal resting ECG: 20% to 31%	0.14	1.16		121, 121, 2 (small n)	unclear; AUC gain -0.002 [-0.016, 0.012]	yes
C41	threshold: ST depression on exercise: fold at about 1.3 mm (starts) *shape: hinge at 1.3 mm; 70% of patients same; fallback; one point with C41 (median fold 1.3 mm, typical patient 0.4 mm, 10th-90th percentile 0.4 mm to 1.3 mm, wide)*	16% to 62% (typical patient with a fold)	1.70	5.47	1.25 to 2.17	242, 109	unclear; AUC gain 0.009 [-0.008, 0.033]	yes
C42	subgroup: who this part acts on: 109 of 242 patients	76% to 89% (typical patient this part acts on (risk without the part, then with it))	0.74	2.09	0.02 to 1.62	109, 133	unclear; AUC gain 0.006 [-0.014, 0.032]	yes
C43	structure: its formula reproduces it on 242 training patients (largest difference 1.77636e-15)		2e-15			242	train only	yes

T.3 Blood pressure step

Annotation ann_34 (bp_offset_gate). Reads: chest pain type, resting blood pressure, exercise-induced angina, thallium stress test result.

Original hypothesis (unchanged). Blood-pressure offset gate on the ischaemia score (raises risk). 290 = ReLU(-0.447 + 1.137*ischaemia_gate - 0.773*trestbps - 0.153*exang:1 - 0.069*cp:3 - 0.007*thal:6), read out +0.252. Hypothesis: passes the ischaemia score on, minus an offset that grows with resting blood pressure (and slightly with exercise angina and non-anginal pain). With the ischaemia gate closed it still opens for low resting blood pressure, so within this gate lower resting BP raises risk (0.252*0.773 = 0.195 per SD while open), partly offsetting trestbps's direct +0.50 edge. EVIDENCE (train, n=242): active for 60% (87% with the ischaemia gate open, 38% without); with the ischaemia gate open, active for 100%, 85% and 77% of the low, middle and high resting-BP groups. Disease rate 50% when active vs 35% when inactive. Mean |sensitivity| ischaemia score 0.60 > trestbps 0.46 > exang:1 0.09 > cp:3 0.04 > thal:6 0.004. It is the largest nonlinear readout (logit SD 0.50): readout ablation lowers train AUC 0.939 -> 0.933; an affine replacement fits it well (R2 0.97, AUC 0.939), so its nonlinearity matters little. Supported. The negative BP term inside a risk-raising gate is reported, not explained.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 1.77636e-15).

$$y_0 = \mathrm{relu}\big(-0.447 + 1.0x_{identity_290} - 0.0688x_{cp:3_b} \\ - 0.153x_{exang:1_b} - 0.00683x_{thal:6_b} - 0.773x_{trestbps_b}\big)$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C44	direction: resting blood pressure: 110 mm Hg to 155 mm Hg (through this part) *shape: hinge at 125 mm Hg; 75% of patients same; fallback; one point with C52 (median fold 125 mm Hg, typical patient 148 mm Hg, 10th-90th percentile 122 mm Hg to 165 mm Hg, wide)*	23% to 17%	-0.27	0.77	-0.48 to -0.08	242	disagrees; AUC gain -0.003 [-0.014, 0.004]	yes
C45	direction: exercise-induced angina: no chest pain on exercise to chest pain brought on by exercise (through this part) *shape: categorical; highest to lowest: no chest pain on exercise, chest pain brought on by exercise*	20% to 18%	-0.04	0.96	-0.08 to 0.00	160, 82	disagrees; AUC gain -0.001 [-0.006, 0.000]	yes
C46	direction: chest pain type: chest pain type other than non-anginal chest pain to non-anginal chest pain (through this part) *shape: categorical; highest to lowest: chest pain type other than non-anginal chest pain, non-anginal chest pain*	20% to 19%	-0.02	0.98	-0.04 to 0.00	172, 70	unclear; AUC gain 0.000 [0.000, 0.000]	yes
C47	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan to a fixed defect on the thallium scan (through this part) *shape: categorical; highest to lowest: thallium stress test result other than a fixed defect on the thallium scan, a fixed defect on the thallium scan*	20% to 19%	-0.004	1.00	-0.007 to 0.00	227, 15 (small n)	unclear; AUC gain 0.000 [0.000, 0.000]	yes
C48	interaction: effect of resting blood pressure (110 mm Hg to 155 mm Hg) by chest pain type: non-anginal chest pain minus chest pain type other than non-anginal chest pain; stronger in non-anginal chest pain *shape: accelerating; 84% of patients same; fallback; risk saturates near 0%*	non-anginal chest pain: 3% to 7%; chest pain type other than non-anginal chest pain: 15% to 28%	0.06	1.06		70, 172	unclear; AUC gain 0.000 [0.000, 0.000]	no
C49	interaction: effect of chest pain type (chest pain type other than non-anginal chest pain to non-anginal chest pain) by exercise-induced angina: chest pain brought on by exercise minus no chest pain on exercise; stronger in chest pain brought on by exercise *shape: categorical; highest to lowest: chest pain type other than non-anginal chest pain, non-anginal chest pain*	chest pain brought on by exercise: 31% to 7%; no chest pain on exercise: 20% to 4%	-0.001	1.00		82, 160	unclear; AUC gain -0.001 [-0.006, 0.000]	no
C50	interaction: effect of resting blood pressure (110 mm Hg to 155 mm Hg) by exercise-induced angina: chest pain brought on by exercise minus no chest pain on exercise; stronger in no chest pain on exercise *shape: accelerating; 81% of patients same; fallback*	chest pain brought on by exercise: 25% to 45%; no chest pain on exercise: 15% to 28%	-0.05	0.95		82, 160	unclear; AUC gain -0.001 [-0.006, 0.000]	no
C51	interaction: effect of resting blood pressure (110 mm Hg to 155 mm Hg) by thallium stress test result: a fixed defect on the thallium scan minus thallium stress test result other than a fixed defect on the thallium scan; stronger in thallium stress test result other than a fixed defect on the thallium scan *shape: accelerating; 76% of patients same; fallback*	a fixed defect on the thallium scan: 8% to 17%; thallium stress test result other than a fixed defect on the thallium scan: 15% to 28%	-0.05	0.95		15, 227 (small n)	unclear; AUC gain 0.000 [0.000, 0.000]	no
C52	threshold: resting blood pressure: fold at about 125 mm Hg (stops) for patients without chest pain on exercise *shape: hinge at 125 mm Hg; 92% of patients same; verified; risk saturates near 0%; one point with C52 (median fold 125 mm Hg, typical patient 122 mm Hg, 10th-90th percentile 122 mm Hg to 165 mm Hg, wide)*	4% to 3% (typical patient with a fold (patients without chest pain on exercise))	-0.30	0.74	-0.63 to -0.17	143, 78	disagrees; AUC gain -0.003 [-0.014, 0.004]	yes
C53	subgroup: who this part acts on: 145 of 242 patients	51% to 68% (typical patient this part acts on (risk without the part, then with it))	0.52	1.69	0.03 to 1.22	145, 97	unclear; AUC gain 0.004 [-0.013, 0.024]	yes
C54	structure: its formula reproduces it on 242 training patients (largest difference 1.77636e-15)		2e-15			242	train only	yes

T.4 Damping term

Annotation ann_35 (ischaemia_damping_thal6). Reads: thallium stress test result.

Original hypothesis (unchanged). Ischaemia damping with a fixed-defect term (near-linear, small). 1416 = ReLU(0.853 + 0.841*ischaemia_gate - 0.594*thal:6), read out -0.118. Hypothesis: almost always open, so it acts as a linear correction: it subtracts 0.099 per unit of the ischaemia score from the logit (damping the ischaemia gate's direct +0.136) and adds 0.070 per SD of thal:6 (about +0.29 logit for a fixed thallium defect, against thal:6's direct -0.41 edge). It closes only for fixed-defect patients whose ischaemia score is below 1.73. EVIDENCE (train, n=242): active for 97% (closed for 8 of the 15 fixed-defect patients). Mean |sensitivity| ischaemia score 0.97, thal:6 0.57. Readout ablation: train AUC 0.939 -> 0.938; affine replacement R2 0.99, AUC 0.939. Supported; effectively linear.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 2.22045e-16).

$$y_0 = \mathrm{relu}\big(0.853 + 1.0x_{identity_1416} - 0.594x_{thal:6_c}\big)$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C55	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan to a fixed defect on the thallium scan (through this part) *shape: categorical; highest to lowest: a fixed defect on the thallium scan, thallium stress test result other than a fixed defect on the thallium scan*	20% to 23%	0.18	1.20	0.12 to 0.29	227, 15 (small n)	unclear; AUC gain 0.002 [0.000, 0.01]	yes
C56	subgroup: who this part acts on: 234 of 242 patients	24% to 20% (typical patient this part acts on (risk without the part, then with it))	-0.23	0.79	-0.48 to -0.12	234, 8	unclear; AUC gain 0.000 [-0.011, 0.01]	yes
C57	structure: its formula reproduces it on 242 training patients (largest difference 2.22045e-16)		2e-16			242	train only	yes

T.5 Hypertrophy without exercise pain

Annotation ann_36 (lvh_without_exercise_angina). Reads: resting ECG, exercise-induced angina.

Original hypothesis (unchanged). LVH without exercise angina (AND indicator, raises risk, small). 1611 = ReLU(-1.261 + 0.887*restecg:2 - 0.953*exang:1), read out +0.729. Hypothesis: a binary AND: open (constant 0.324, +0.236 logit) only for patients with left ventricular hypertrophy on the resting ECG and no exercise-induced angina, zero otherwise. EVIDENCE (train, n=242): active for exactly the 74 patients with LVH and no exercise angina; their disease rate is 41%, against 20% for patients with neither. Mean |sensitivity| exang:1 0.29, restecg:2 0.27. Readout ablation: train AUC 0.939 -> 0.939 (-0.0003). restecg:2 acts only here. Supported; near-negligible for ranking.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 0.0).

$$y_0 = \mathrm{relu}\big(-1.26 + 0.887 x_{restecg:2} - 0.953 x_{exang:1_c}\big)$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C58	direction: exercise-induced angina: no chest pain on exercise to chest pain brought on by exercise (through this part) *shape: categorical; highest to lowest: no chest pain on exercise, chest pain brought on by exercise*	30% to 25% (typical patient this part acts on)	-0.12	0.89	-0.24 to 0.00	160, 82	disagrees; AUC gain -0.003 [-0.013, 0.000]	yes
C59	direction: resting ECG: resting ECG other than signs of left ventricular hypertrophy on the resting ECG to signs of left ventricular hypertrophy on the resting ECG (through this part) *shape: categorical; highest to lowest: signs of left ventricular hypertrophy on the resting ECG, resting ECG other than signs of left ventricular hypertrophy on the resting ECG*	20% to 24%	0.16	1.17	0.00 to 0.24	123, 119	unclear; AUC gain 0.001 [-0.01, 0.012]	yes
C60	interaction: effect of exercise-induced angina (no chest pain on exercise to chest pain brought on by exercise) by resting ECG: signs of left ventricular hypertrophy on the resting ECG minus resting ECG other than signs of left ventricular hypertrophy on the resting ECG; stronger in resting ECG other than signs of left ventricular hypertrophy on the resting ECG *shape: categorical; highest to lowest: chest pain brought on by exercise, no chest pain on exercise*	signs of left ventricular hypertrophy on the resting ECG: 25% to 35%; resting ECG other than signs of left ventricular hypertrophy on the resting ECG: 20% to 31%	-0.22	0.80		119, 123	agrees; AUC gain 0.001 [-0.01, 0.012]	yes
C61	subgroup: who this part acts on: 74 of 242 patients	25% to 30% (typical patient this part acts on (risk without the part, then with it))	0.24	1.27	0.24 to 0.24	74, 168	disagrees; AUC gain -0.002 [-0.016, 0.006]	yes
C62	null: this part barely changes predictions (switched off)		0.06			242	train only	no
C63	structure: its formula reproduces it on 242 training patients (largest difference 0.0)		0.00			242	train only	yes

T.6 High fasting sugar term

Annotation ann_37 (high_fasting_sugar). Reads: fasting blood sugar.

Original hypothesis (unchanged). High fasting blood sugar indicator (negligible). 3884 = ReLU(-0.980 + fbs:1), read out +0.048: +0.074 logit for the 33 patients with fasting blood sugar > 120 mg/dl. Hypothesis: a small additive term for high fasting sugar (exactly linear on the two values of fbs:1); together with fbs:0's direct +0.26 edge, high fasting sugar overall lowers the logit (fbs:0 is the normal-sugar flag). EVIDENCE (train, n=242): active for exactly the 33 high-sugar patients; disease rate 45% vs 44%. Readout ablation changes train AUC by less than 0.0001; affine replacement R2 1.0. Supported; negligible.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 0.0).

$$y_0 = \mathrm{relu}\big(-0.98 + 1.0 x_{fbs:1}\big)$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C64	direction: fasting blood sugar: fasting blood sugar of 120 mg/dL or less to fasting blood sugar above 120 mg/dL (through this part) *shape: categorical; highest to lowest: fasting blood sugar above 120 mg/dL, fasting blood sugar of 120 mg/dL or less*	20% to 21%	0.07	1.08	0.07 to 0.07	209, 33	unclear; AUC gain 0.000 [-0.005, 0.004]	yes
C65	subgroup: who this part acts on: 33 of 242 patients	24% to 25% (typical patient this part acts on (risk without the part, then with it))	0.07	1.08	0.07 to 0.07	33, 209	unclear; AUC gain 0.000 [-0.005, 0.004]	yes
C66	null: this part barely changes predictions (switched off)		0.02			242	train only	yes
C67	structure: its formula reproduces it on 242 training patients (largest difference 0.0)		0.00			242	train only	yes

The network before any explanation

Figure 459 (next page, landscape) is the trained network as it is, with no parts, no splits and no inserted units: inputs at the bottom, hidden units by depth, the output at the top, every connection drawn, every input column its own node (one-hot levels included).

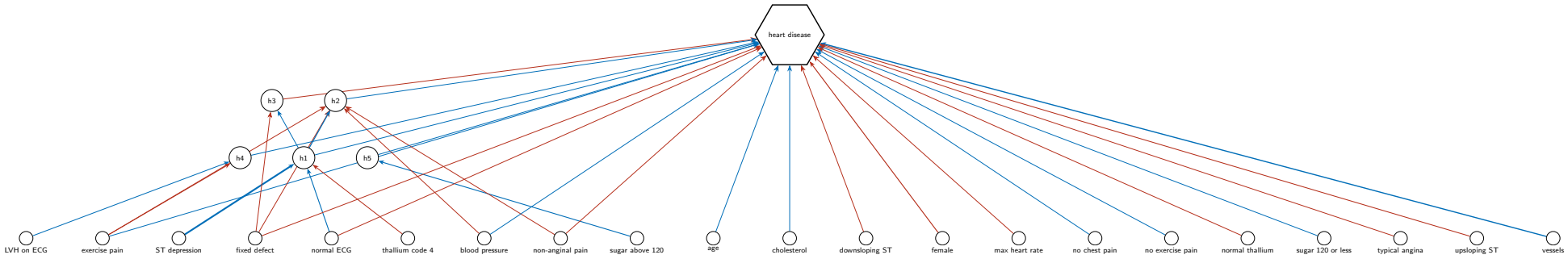


Figure 459: The network as trained, before any explanation, in full. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. Not used (no connection): male, atypical angina, ST-T abnormal ECG, flat ST, reversible defect.

13 heart-s5-c3

Genome `96fb45fb-972d-4aaa-b6d5-01fc63fbc4e6`, explanation `1b3f669d-9fb5-4126-b6e4-35535db7b3b1`. Split: train primary (n=242), test as a check (n=61). Claims: 36, verified 33, not supported 3.
AI-assisted draft for review (unverified)

Reference (typical) patient: the typical patient: the median training patient for every number and the most common training category for every category (not a real patient). age: 56 years; ca: 0 vessels; chol: 240 mg/dL (6.2 mmol/L); cp: no chest pain (asymptomatic); exang: no chest pain on exercise; fbs: fasting blood sugar of 120 mg/dL or less; oldpeak: 0.8 mm; restecg: a normal resting ECG; sex: male; slope: an upsloping ST segment; thal: a normal thallium scan; thalach: 152 beats per minute; trestbps: 130 mm Hg. Predicted risk of heart disease: 18%.

13.1 Summary

For a typical patient (a 56-year-old man with no chest pain symptoms and median test results) the model predicts a risk of heart disease of about 18%; it adds up most of its inputs independently, and two parts add to that sum: a signal from ST depression on exercise and the resting ECG that acts on 105 of 242 training patients, and an age term that acts on 148. [C25, C30, C34] Figure 460 shows the model at this level: one box per paragraph below.

How to read the diagrams. Edge colour is the sign of that one connection: blue positive, red negative; thicker is a larger weight. The outline and arrow on an input (and on a part's output box) is its overall effect on the final prediction, read from the verified claims: orange and up arrow raises it, teal and down arrow lowers it, grey and double arrow depends (it goes both ways, or differs between levels). A negative connection into a part that itself raises the prediction means that input lowers the prediction; through a hinge the overall effect can only be read from the claims, not from the edges. Up arrow means towards heart disease; down arrow, away from it. A double outline marks an input (or a part's output) that acts at a fixed rate everywhere: it adds the same amount to the score per unit, so it is explained by its rate alone (see 'Linear factors'); a dashed double outline, at a fixed rate on the observed data. The normal outline is an input whose effect switches on and off somewhere. For example, ST depression has a positive connection into 'ST depression signal', a part that raises the predicted risk of heart disease, so ST depression raises it.

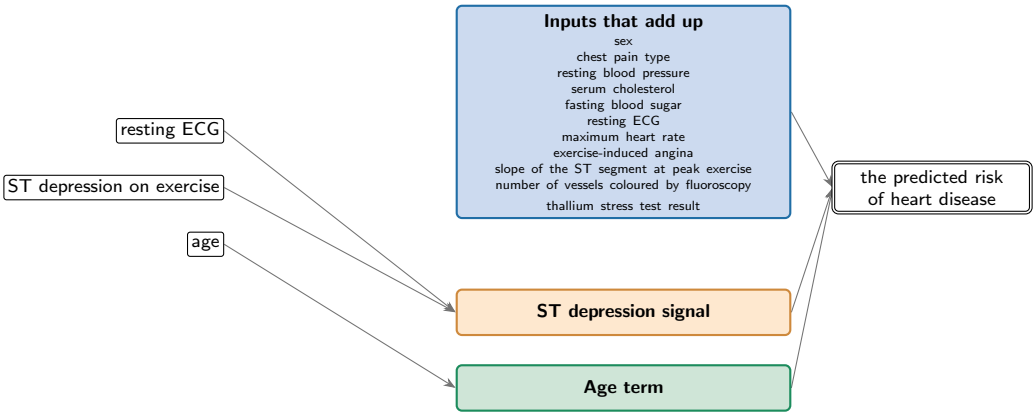


Figure 460: The model at the level of the summary: one box per part.

Inputs that add up. *Lowering the risk.* These add up independently, and these lower the risk: being a woman takes a typical patient's predicted risk from about 18% to about 2%, typical angina to about 2%, non-anginal chest pain to about 4% and atypical angina to about 10%; fasting blood sugar above 120 mg/dL lowers it to about 9%; and a higher maximum heart rate lowers it steadily, from about 23% at 118 to about 15% at 175 beats per minute. [C10, C09, C11, C15, C12, C14]
Raising the risk. Each extra vessel coloured by fluoroscopy adds about the same, from about 18% at 0 to about 86% at 2 vessels, flattening out near the top because the predicted risk cannot go much higher; a flat rather than upsloping ST segment raises it to about 56%, a reversible defect on the thallium scan to about 38% and chest pain brought on by exercise to about 35%; higher cholesterol raises it steadily, from about 11% at 193 mg/dL to about 33% at 308 mg/dL, and higher resting blood pressure steadily, from about 12% at 110 mm Hg to about 30% at 155 mm Hg. [C01, C04, C06, C08, C05, C07] Through this sum alone, a resting ECG other than normal raises it from about 18% to about 24%; the resting ECG also acts through the ST depression signal (below). [C16]
Rare results. Some rarer results also count (the model has little data on this, n = 2 to 17): a thallium result coded 4 raises the risk to about 79%, a fixed defect on the thallium scan lowers it to about 9%, and a downsloping ST segment leaves it at about 19%. [C02, C13, C19]

ST depression signal. This part reads ST depression on exercise and the resting ECG, and acts on 105 of 242 training patients, 66 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 76% without it and about 91% with it. [C30] Through it, more ST depression adds nothing until about 0.5 mm, then raises the predicted risk steadily: for a typical patient from about 16% at 0 mm to about 60% at 2.8 mm. [C26] ST depression counts more with a normal resting ECG, nothing until about 0.5 mm and then steadily, from about 16% to about 60%, against about 21% to about 48% with any other resting ECG. [C28] Through it, a resting ECG other than normal lowers a typical patient's risk from about 18% to about 16%. [C27]

Age term. This part acts on 148 of 242 training patients, 81 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 91% without it and about 92% with it. [C34] Through it, age adds nothing until about 53 years, then raises the risk steadily, for a typical patient from about 18% at 42 to about 19% at 66 years. [C32, C33]

The whole network. Figure 461 (next page, landscape) draws every unit and connection of the explained model, nested by part: shaded clusters are the parts of the model and darker boxes the parts inside them; circles are hidden units (h1, h2, ...); each cluster's rectangle is its output, and a dashed 'sum' stands for an addition the next unit does (drawn, not a unit of the model); the hexagon is the model's output. An input used by several parts is drawn under each of them (same label, dotted outline, the first copy marked with the number of copies): it is the same input, repeated for readability. Not used by the model (input columns with no connection): male, atypical angina, sugar above 120, ST-T abnormal ECG, LVH on ECG, no exercise pain, flat ST.

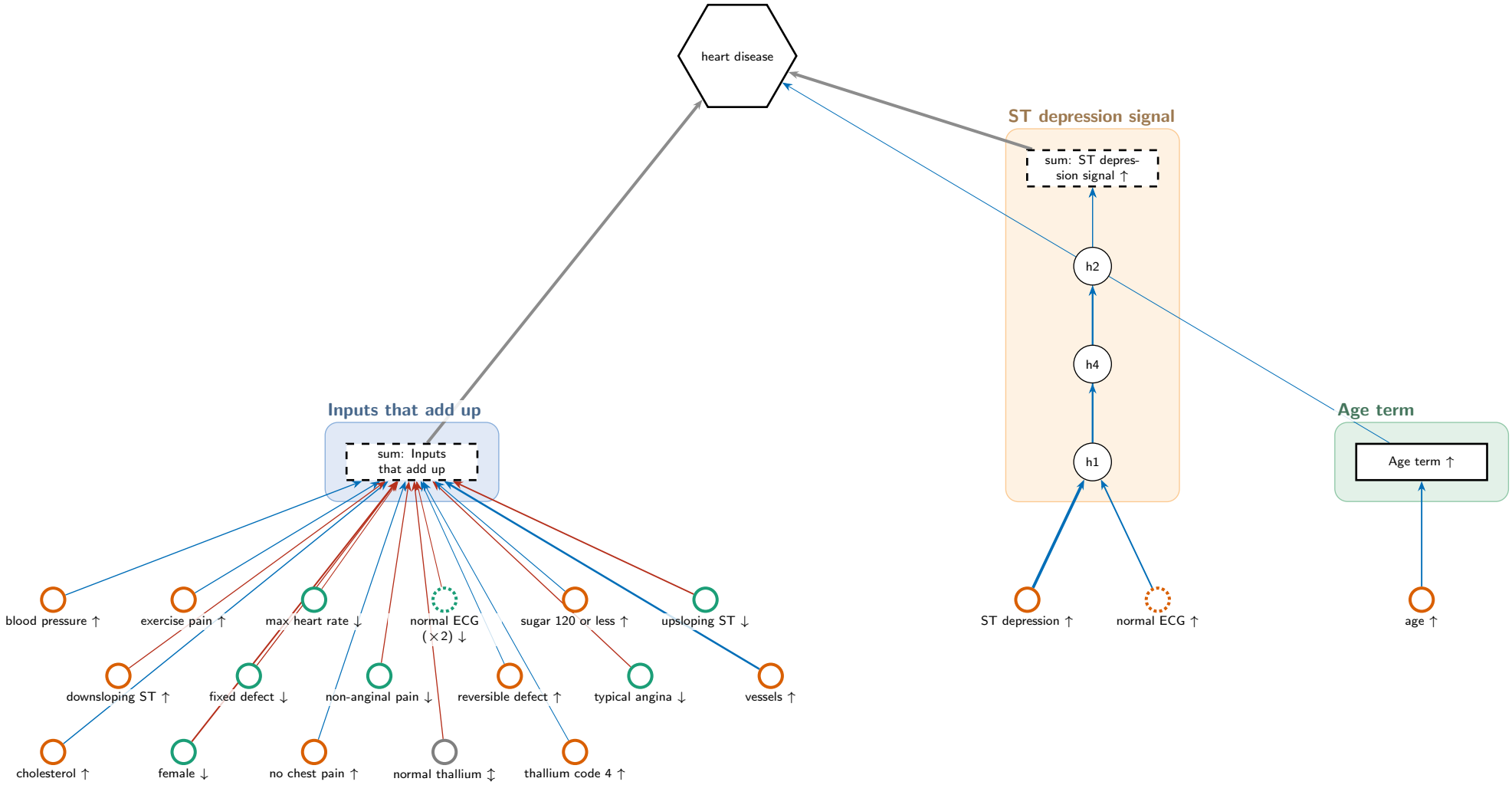


Figure 461: Every unit and connection of the explained model, nested by part. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Worth checking against what is known. Worth checking: the model treats fasting blood sugar above 120 mg/dL as lowering the risk of heart disease (for a typical patient from about 18% to about 9%). [C12] Worth checking: patients with typical angina get the lowest predicted risk of all chest pain types, below patients with no chest pain symptoms. [C09, C11, C15] Worth checking: the model treats a fixed defect on the thallium scan as lowering the risk compared with a normal scan, from only 15 training patients. [C13] Worth checking: the model gives a higher predicted risk to the 2 training patients whose thallium result is coded 4, a code the dataset documentation does not define. [C02] Worth checking: ST depression on exercise raises the predicted risk more when the resting ECG is normal than when it is not. [C28]

13.2 The model, read from the top down

Each part of the explanation in turn, from the whole model down to the smallest parts: what it does for the prediction (hypothesis), what was checked (evidence), a diagram one level down, and where to read on. The technical record of each part is in "Technical detail" at the end.

13.2.1 The whole model

Figure 462 shows what feeds this part; Figure 463 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

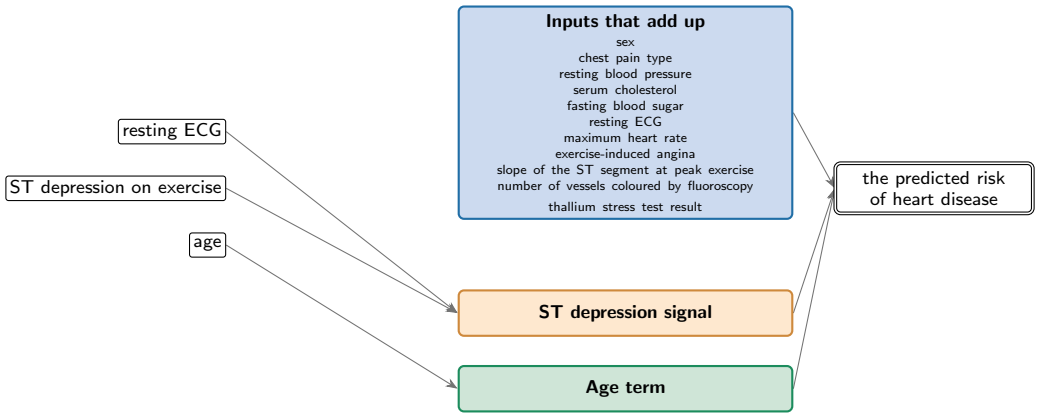


Figure 462: The whole model: what feeds it, one level down.

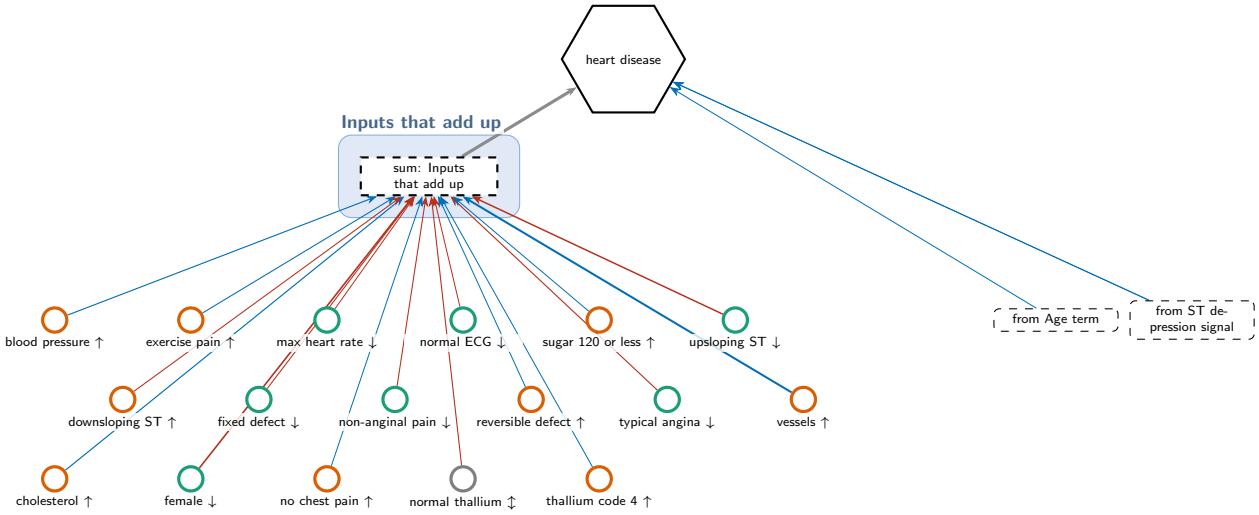


Figure 463: The whole model: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. The model adds up its inputs into one score: the vessels coloured (each extra vessel adding about the same), the ST segment, sex and chest pain type each move the predicted risk on their own, and two parts add to that sum: a signal from ST depression shaped by the resting ECG, and an age term. [C25, C01, C04, C10, C09]

Evidence. It is made up of the inputs that feed the score directly and two parts, the ST depression signal and the age term (each explained below), and the score is their sum, which was checked on all 242 training patients. [C25] ST depression on exercise and age do not feed the score directly, so they act only through these parts. [C25] Changing one input at a time across the whole model, more ST depression adds nothing until about 0.5 mm, then raises a typical patient's predicted risk steadily, from about 16% at 0 mm to about 60% at 2.8 mm. [C03] Across the whole model, older age adds nothing until about 53 years, then raises it steadily, from about 18% at 42 to about 19% at 66 years. [C18]
Evidence plots: Figure 464.

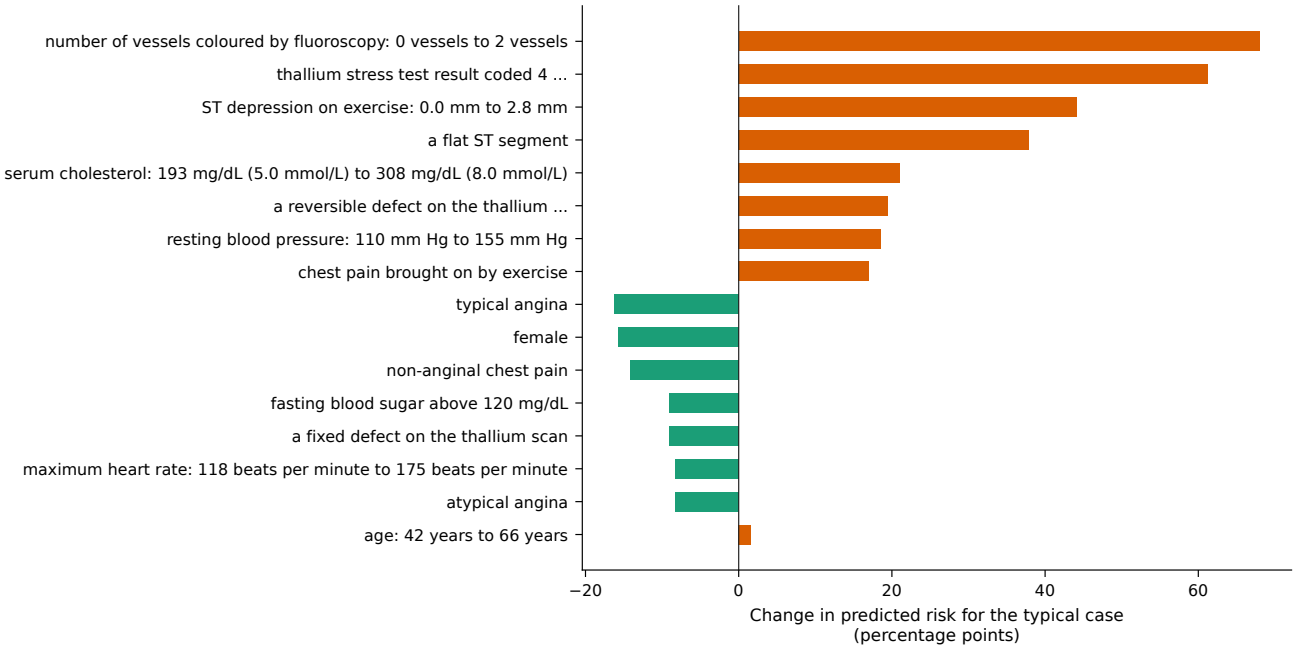


Figure 464: The whole model: Each input's change in the predicted value for the typical case across the whole model, largest first (orange: raises it, teal: lowers it).

Dive deeper: ST depression signal (13.2.2); Age term (13.2.3). Technical detail: T.1.

13.2.2 ST depression signal

Part of: The whole model (13.2.1).
Figure 465 shows what feeds this part; Figure 466 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).



Figure 465: ST depression signal: what feeds it, one level down.

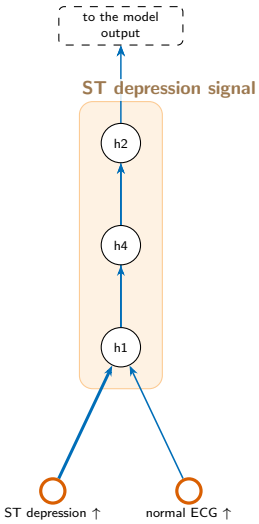


Figure 466: ST depression signal: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part raises the predicted risk with ST depression on exercise, adding nothing until about 0.5 mm and then steadily, and more so with a normal resting ECG. [C26, C28]

Evidence. It acts on 105 of 242 training patients, 66 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 76% without it and about 91% with it. [C30] Through it, more ST depression adds nothing until about 0.5 mm, then raises the predicted risk steadily: for a typical patient from about 16% at 0 mm to about 60% at 2.8 mm. [C26] ST depression counts more with a normal resting ECG, nothing until about 0.5 mm and then steadily, from about 16% to about 60%, against about 21% to about 48% with any other resting ECG. [C28] Through it, a resting ECG other than normal lowers a typical patient's risk from about 18% to about 16%. [C27] Its formula reproduces the part exactly on all 242 training patients. [C31]
Evidence plots: Figure 467, Figure 468, Figure 469, Figure 470, Figure 471, Figure 472.
Figure 470 shows this part's own output, unit h1: it switches on at about 0.4548 mm of ST depression on exercise, and is zero below.
Figure 471 shows this part's own output, unit h2: it switches on at about 0.5774 mm of ST depression on exercise, and is zero below.

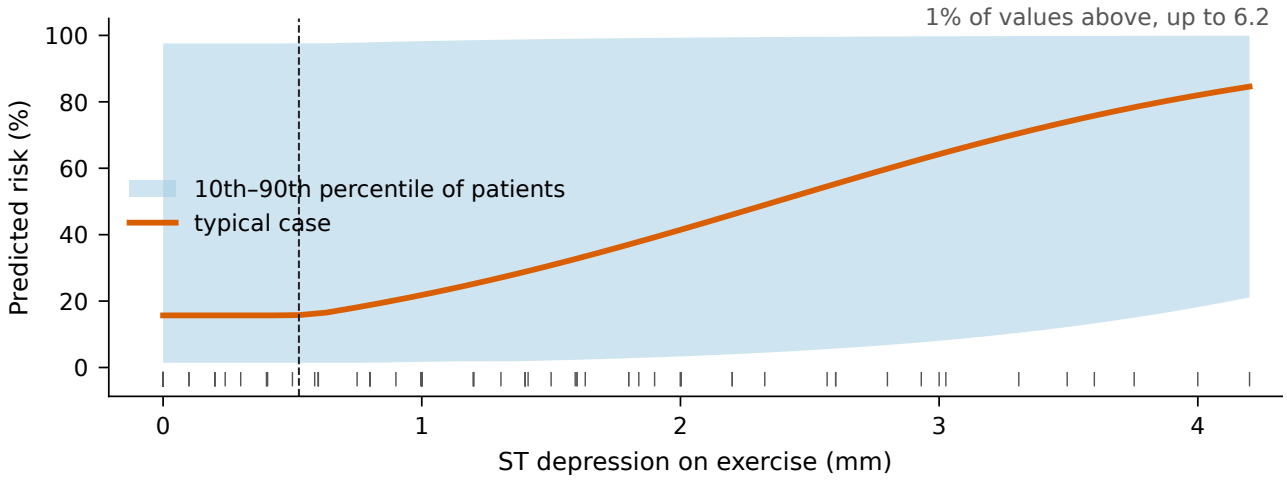


Figure 467: ST depression signal: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual patients (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

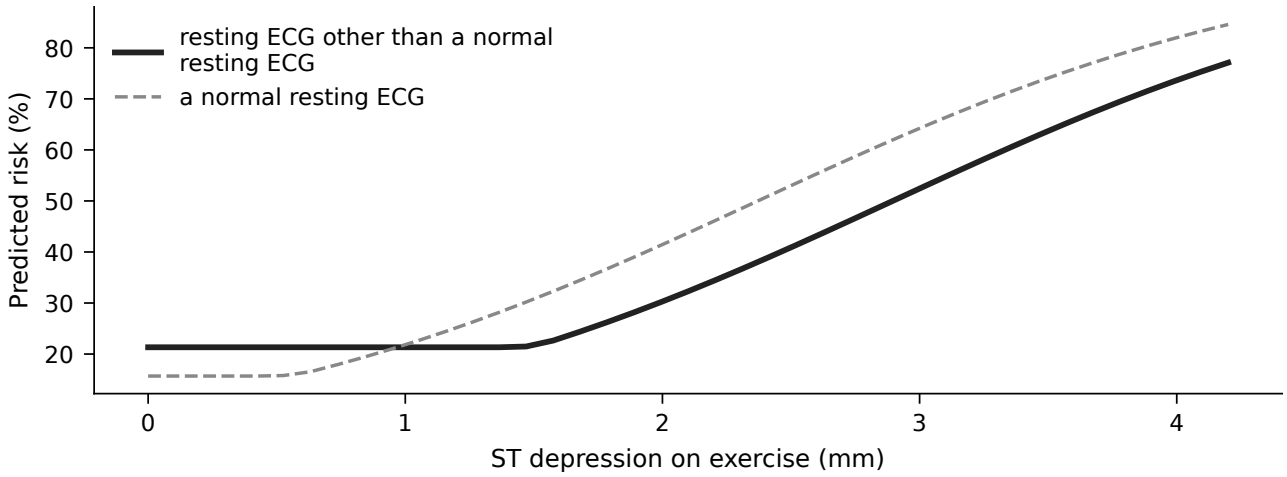


Figure 468: ST depression signal: How the effect differs between groups: one curve per group, for the typical case of each group.

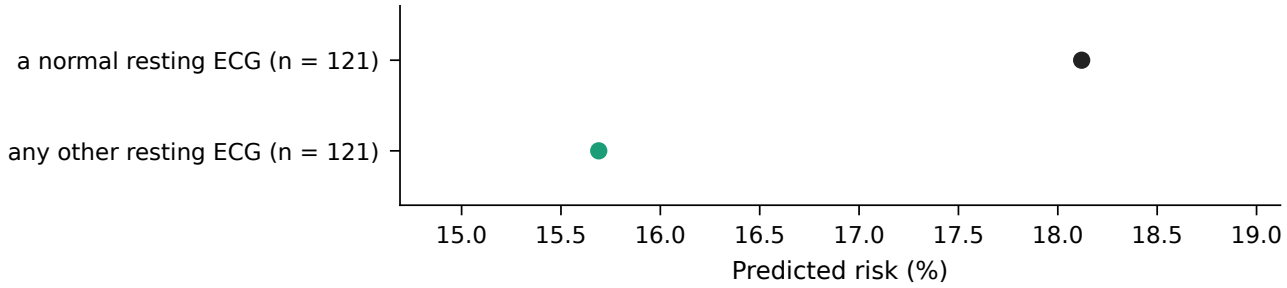


Figure 469: ST depression signal: The predicted value for the typical case at each level, through this part (n = training patients; grey: fewer than 20).

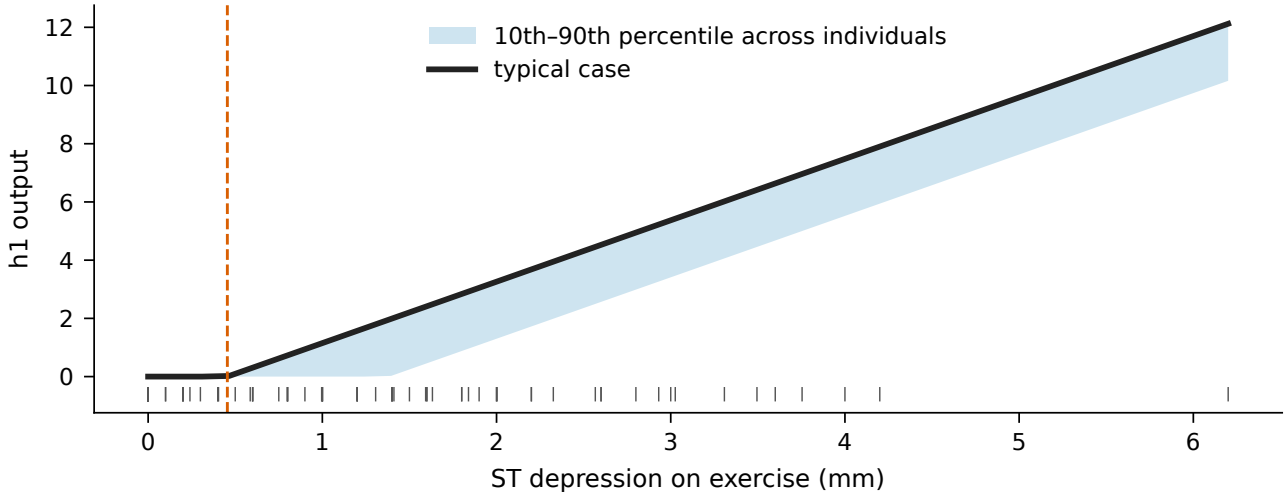


Figure 470: ST depression signal: This part's own output, unit h1 (the output of 'ST depression signal'), on its own scale (its activation), with every other input at the typical case; it is active for 105 of 242. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

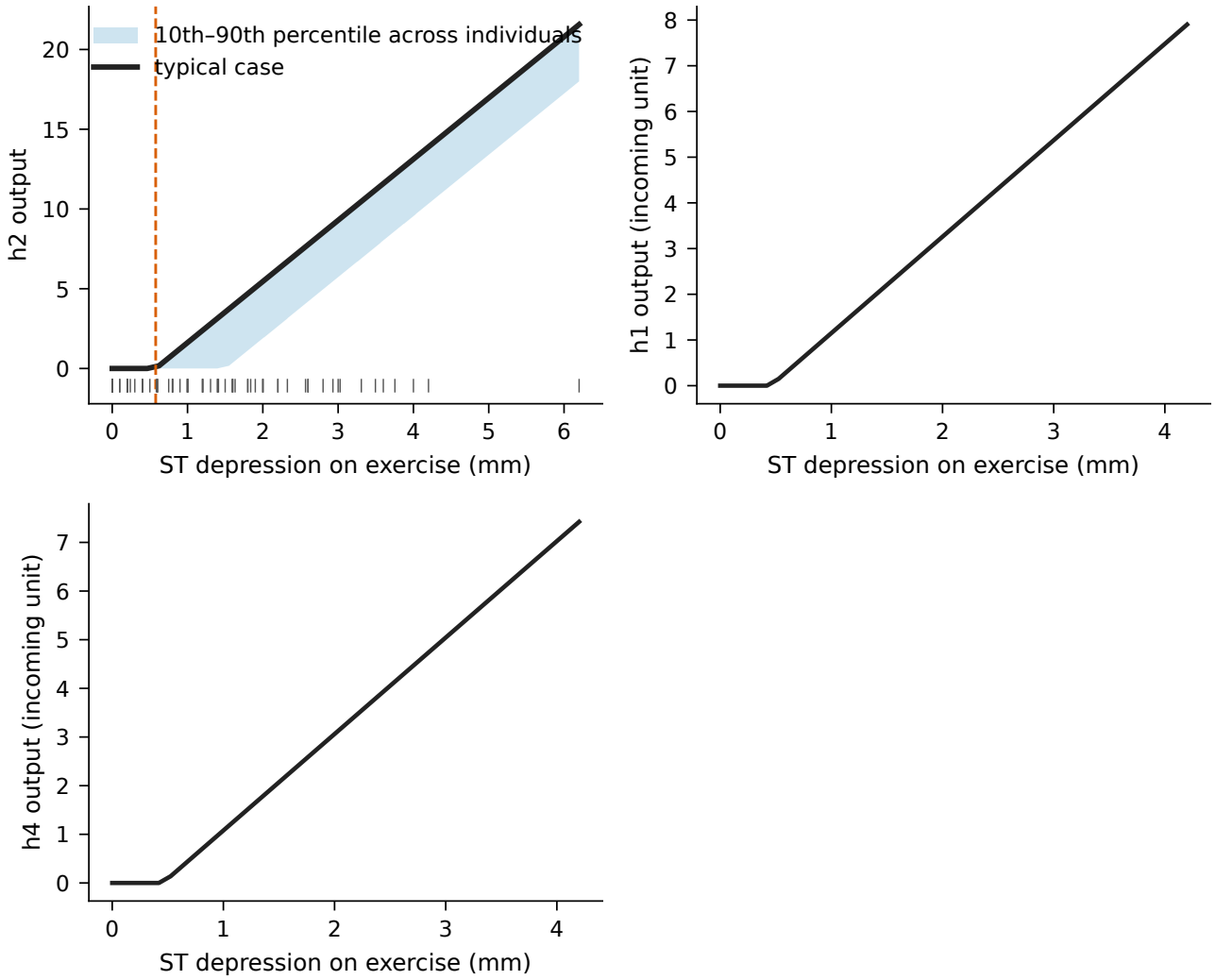


Figure 471: ST depression signal: This part's own output, unit h2 (combines 2 non-linear units), on its own scale (its activation), with every other input at the typical case; it is active for 96 of 242. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

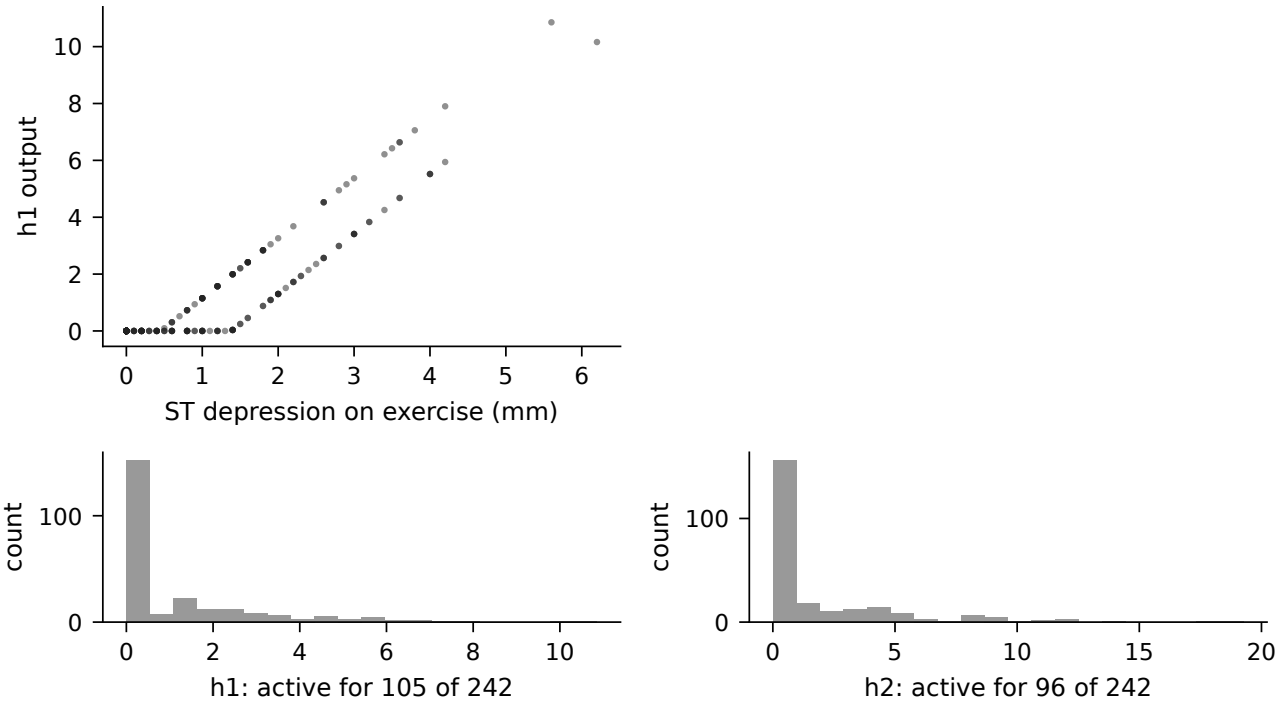


Figure 472: ST depression signal: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.2.

13.2.3 Age term

Part of: The whole model (13.2.1).
Figure 473 shows what feeds this part; Figure 474 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

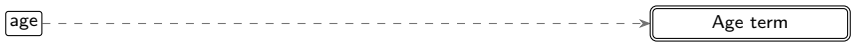


Figure 473: Age term: what feeds it, one level down.

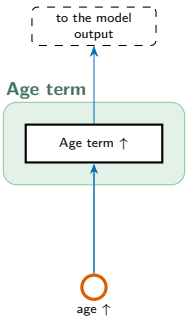


Figure 474: Age term: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part adds a little to the predicted risk for older patients: nothing until about 53 years, then steadily. [C32, C33]

Evidence. It acts on 148 of 242 training patients, 81 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 91% without it and about 92% with it. [C34] Through it, age adds nothing until about 53 years, then raises the risk steadily, for a typical patient from about 18% at 42 to about 19% at 66 years. [C32, C33] Its formula reproduces the part exactly on all 242 training patients. [C36]
Evidence plots: Figure 475, Figure 476, Figure 477.

Figure 476 shows this part's own output, unit h3: it switches on at about 52.81 years of age, and is zero below.

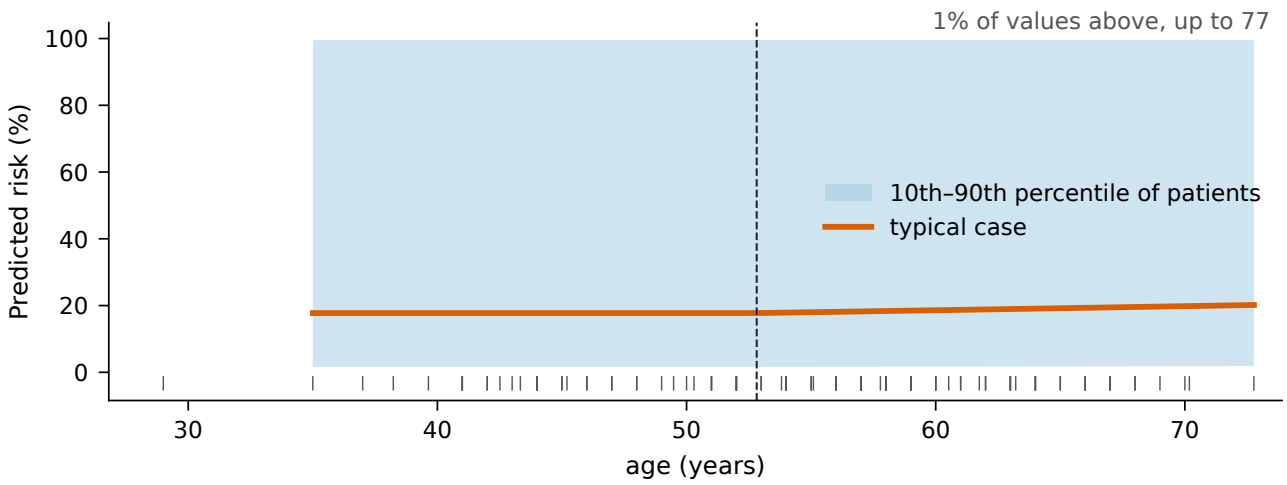


Figure 475: Age term: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual patients (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

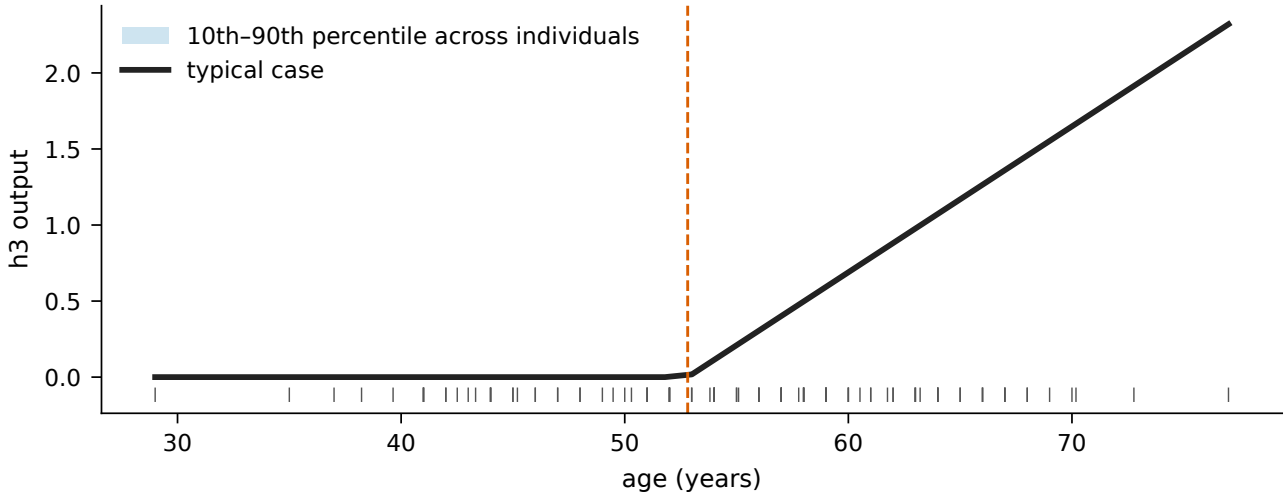


Figure 476: Age term: This part's own output, unit h3 (the output of 'Age term'), on its own scale (its activation), with every other input at the typical case; it is active for 148 of 242. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

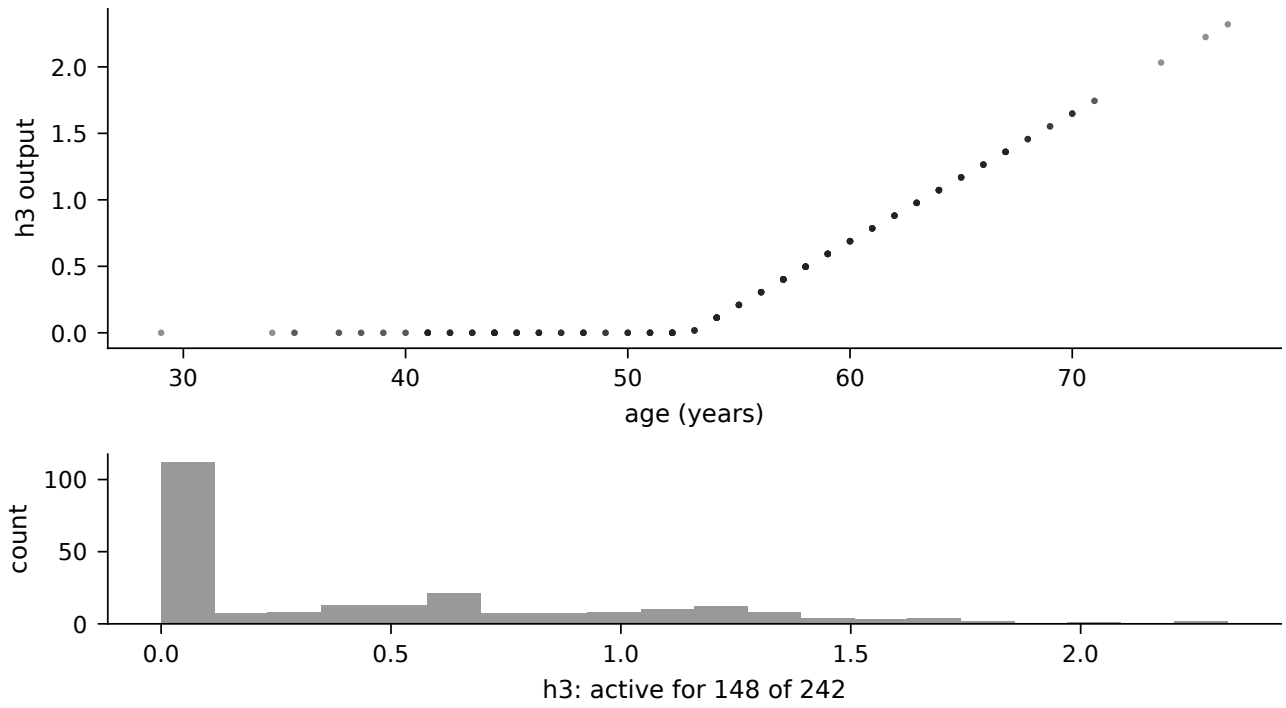


Figure 477: Age term: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.3.

13.3 Not supported by the model

- C17: direction claim about resting ECG in the whole model: not supported by the model (44% of patients have the opposite sign (allowed: 10%)).
- C29: threshold claim about ST depression during exercise (compared with rest) in part ann_27: not supported by the model (the fold changes the curve by only 0.095 log-odds compared with a straight line (needed 0.1)).
- C35: null claim about age in part ann_28: not supported by the model (predictions move by up to 0.032).

Technical detail: heart-s5-c3

The record behind each part: the original hypothesis, the closed form, and every claim with its estimate (log-odds and odds ratio), range across patients, n and test check.

T.1 The whole model

Annotation ann_29 (heart_disease_logit). Reads: age, sex, chest pain type, resting blood pressure, serum cholesterol, fasting blood sugar, resting ECG, maximum heart rate reached during the exercise test, exercise-induced angina, ST depression during exercise (compared with rest), slope of the ST segment at peak exercise, number of major vessels coloured by fluoroscopy, thallium stress test result.

Original hypothesis (unchanged). Root: p = sigmoid(z), z = -0.536 + LINEAR CORE + 0.049*ischaemia_chain[204] + 0.216*ischaemia_chain[290] + 0.082*age_hinge. LINEAR CORE (17 direct input->output edges, z-scored inputs): ca +1.485, sex:0 -1.042, slope:1 -0.873, chol +0.551, cp:1 -0.494, trestbps +0.483, thal:3 -0.471, cp:3 -0.442, slope:3 -0.434, exang:1 +0.423, thal:6 -0.421, cp:4 +0.354, fbs:0 +0.274, thalach -0.213, restecg:0 -0.188, thal:4 +0.172, thal:7 +0.025. Seven inputs are unconnected (sex:1, cp:2, fbs:1, restecg:1, restecg:2, exang:0, slope:2; zero-effect entries of this root). Oldpeak and age have no direct edge and act only through the children; restecg:0 acts both directly and through the ischaemia chain (split into restecg:0_a direct and restecg:0_b). COMPOSITION: additive. The two children are read out additively into z: ischaemia_chain (two exits: a one-sided ST-depression score opened earlier by a normal resting ECG, and a steeper second stage of the same score) and age_hinge (an age term that starts just below the median age). Together the chain gives oldpeak a hinge-shaped effect: about 0.12 logit per SD between its two thresholds and about 1.10 per SD above the second (z-oldpeak -0.39 with a normal ECG, +0.40 otherwise). EVIDENCE (train, n=242): full model AUC 0.9376, acc 0.888 (compute_performance). Linear core alone AUC 0.9317; logit SD 3.22 (core) vs 0.80 (nonlinear readouts). Removing the whole chain (204 with downstream) gives 0.9317; readout-only ablations: 204 0.9380, 290 0.9341, age 0.9371. All ReLUs as identities 0.9304; one least-squares linear fit of all nonlinear readouts on restecg:0, oldpeak and age gives 0.9355 (R2 0.88); an L2 logistic regression on all 26 inputs gives 0.9355. Mean |SHAP| (kernel, all 242 train rows): ca 0.141, sex:0 0.104, slope:1 0.099, oldpeak 0.060, thal:3 0.054, exang:1 0.045, chol 0.044, cp:3 0.043, cp:4 0.040, trestbps 0.040. Interpretation: close to logistic regression plus one ST-depression correction that switches on above a threshold (lower for patients with a normal resting ECG) and a small age term above median age. Counter-directional signs reported, not explained: a normal resting ECG lowers the oldpeak threshold inside the chain (raising risk), fbs:0 (normal fasting sugar) +0.27 directly, thal:7 (reversible defect) only +0.025. Compared with the 10-seed heart-s5 record (generation 37, key 9330: ischaemia gate -> resting-BP gate -> age term) and its cluster-1 sibling: the same ischaemia gate on oldpeak and restecg:0, but here the second stage is a steeper copy of the score (no resting-BP offset) and the age term is a free-standing hinge (not switched off by the chain).

Closed form. Read through its parts (their formulas are in their own sections).

Split inputs. To annotate the parts, the explanation split these inputs into copies (a refactoring step that does not change what the model computes); the diagrams draw each input once: normal ECG (2 copies).

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C01	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (whole model) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	18% to 86%	3.33	28.0	3.33 to 3.33	242	unclear; AUC gain 0.029 [-0.039, 0.100]	yes
C02	direction: thallium stress test result: a normal thallium scan to thallium stress test result coded 4 (meaning not recorded) (whole model) *shape: categorical; highest to lowest: thallium stress test result coded 4 (meaning not recorded), a reversible defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	18% to 79%	2.85	17.4	2.85 to 2.85	135, 2 (small n)	unclear; AUC gain 0.020 [-0.012, 0.058]	yes
C03	direction: ST depression on exercise: 0.0 mm to 2.8 mm (whole model) *shape: hinge at 0.5 mm; 100% of patients same; verified*	16% to 60%	1.65	5.20	1.22 to 2.08	242	unclear; AUC gain 0.023 [-0.001, 0.055]	yes
C04	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	18% to 56%	1.75	5.75	1.75 to 1.75	115, 110	disagrees; AUC gain -0.024 [-0.071, 0.009]	yes
C05	direction: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L) (whole model) *shape: steady; 100% of patients same; verified*	11% to 33%	1.32	3.73	1.32 to 1.32	242	disagrees; AUC gain -0.019 [-0.058, 0.008]	yes
C06	direction: thallium stress test result: a normal thallium scan to a reversible defect on the thallium scan (whole model) *shape: categorical; highest to lowest: thallium stress test result coded 4 (meaning not recorded), a reversible defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	18% to 38%	1.00	2.72	1.00 to 1.00	135, 90	unclear; AUC gain 0.020 [-0.012, 0.058]	yes
C07	direction: resting blood pressure: 110 mm Hg to 155 mm Hg (whole model) *shape: steady; 100% of patients same; verified*	12% to 30%	1.19	3.30	1.19 to 1.19	242	unclear; AUC gain 0.008 [-0.018, 0.037]	yes
C08	direction: exercise-induced angina: no chest pain on exercise to chest pain brought on by exercise (whole model) *shape: categorical; highest to lowest: chest pain brought on by exercise, no chest pain on exercise*	18% to 35%	0.89	2.44	0.89 to 0.89	160, 82	agrees; AUC gain 0.022 [0.003, 0.049]	yes
C09	direction: chest pain type: no chest pain (asymptomatic) to typical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	18% to 2%	-2.43	0.09	-2.43 to -2.43	108, 22	agrees; AUC gain 0.033 [0.008, 0.066]	yes
C10	direction: sex: male to female (whole model) *shape: categorical; highest to lowest: male, female*	18% to 2%	-2.19	0.11	-2.19 to -2.19	159, 83	unclear; AUC gain 0.025 [-0.007, 0.066]	yes
C11	direction: chest pain type: no chest pain (asymptomatic) to non-anginal chest pain (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	18% to 4%	-1.69	0.18	-1.69 to -1.69	108, 70	agrees; AUC gain 0.033 [0.008, 0.066]	yes
C12	direction: fasting blood sugar: fasting blood sugar of 120 mg/dL or less to fasting blood sugar above 120 mg/dL (whole model) *shape: categorical; highest to lowest: fasting blood sugar of 120 mg/dL or less, fasting blood sugar above 120 mg/dL*	18% to 9%	-0.80	0.45	-0.80 to -0.80	209, 33	unclear; AUC gain 0.000 [-0.013, 0.013]	yes
C13	direction: thallium stress test result: a normal thallium scan to a fixed defect on the thallium scan (whole model) *shape: categorical; highest to lowest: thallium stress test result coded 4 (meaning not recorded), a reversible defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	18% to 9%	-0.80	0.45	-0.80 to -0.80	135, 15 (small n)	unclear; AUC gain 0.020 [-0.012, 0.058]	yes
C14	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (whole model) *shape: steady; 100% of patients same; verified*	23% to 15%	-0.54	0.58	-0.54 to -0.54	242	agrees; AUC gain 0.014 [0.002, 0.035]	yes
C15	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	18% to 10%	-0.71	0.49	-0.71 to -0.71	108, 42	agrees; AUC gain 0.033 [0.008, 0.066]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C16	direction: resting ECG: a normal resting ECG to resting ECG other than a normal resting ECG (direct contribution only, not through any part) *shape: categorical; highest to lowest: resting ECG other than a normal resting ECG, a normal resting ECG*	18% to 24%	0.38	1.46	0.38 to 0.38	121, 121	unclear; AUC gain 0.005 [-0.005, 0.024]	yes
C17	direction: resting ECG: a normal resting ECG to resting ECG other than a normal resting ECG (whole model) *shape: categorical; highest to lowest: resting ECG other than a normal resting ECG, a normal resting ECG*	18% to 21%	0.02	1.02	-0.49 to 0.38	121, 121	disagrees; AUC gain 0.003 [-0.013, 0.021]	no
C18	direction: age: 42 years to 66 years (whole model) *shape: hinge at 53 years; 100% of patients same; verified; matches the fold at 53 years*	18% to 19%	0.10	1.11	0.10 to 0.10	242	disagrees; AUC gain -0.003 [-0.017, 0.004]	yes
C19	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a downsloping ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	18% to 19%	0.05	1.05	0.05 to 0.05	115, 17 (small n)	disagrees; AUC gain -0.024 [-0.071, 0.009]	yes
C20	magnitude: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels, typical patient *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	18% to 86%	3.33	28.0		242	unclear; AUC gain 0.029 [-0.039, 0.100]	yes
C21	magnitude: ST depression on exercise: 0.0 mm to 2.8 mm, typical patient *shape: hinge at 0.5 mm; 100% of patients same; verified*	16% to 60%	2.08	8.01		242	unclear; AUC gain 0.023 [-0.001, 0.055]	yes
C22	magnitude: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment, typical patient *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	18% to 56%	1.75	5.75		115, 110	disagrees; AUC gain -0.024 [-0.071, 0.009]	yes
C23	magnitude: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L), typical patient *shape: steady; 100% of patients same; verified*	11% to 33%	1.32	3.73		242	disagrees; AUC gain -0.019 [-0.058, 0.008]	yes
C24	magnitude: thallium stress test result: a normal thallium scan to a reversible defect on the thallium scan, typical patient *shape: categorical; highest to lowest: thallium stress test result coded 4 (meaning not recorded), a reversible defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	18% to 38%	1.00	2.72		135, 90	unclear; AUC gain 0.020 [-0.012, 0.058]	yes
C25	structure: made of ischaemia_chain, age_hinge and inputs sex, cp, trestbps, chol, fbs, restecg, thalach, exang, slope, ca, thal; added					242	train only	yes

T.2 ST depression signal

Annotation ann_27 (ischaemia_chain). Reads: resting ECG, ST depression during exercise (compared with rest).

Original hypothesis (unchanged). Ischaemia chain (raises risk with ST depression, in two stages). 204 = ReLU(0.245 + 2.501*oldpeak + 0.980*restecg:0); 3487 = ReLU(0.940*204 - 0.002) (a near-copy of 204); 290 = ReLU(-0.468 + 1.008*204 + 0.862*3487) = ReLU(1.818*204 - 0.470) whenever 204 > 0.0023. Readouts: logit += 0.049*204 + 0.216*290. Hypothesis: an ST-depression score that opens above an oldpeak threshold set by the resting ECG (z-oldpeak -0.49 with a normal ECG, +0.29 otherwise) and then a second, much steeper stage once the score passes 0.259 (z-oldpeak -0.39 with a normal ECG, +0.40 otherwise). Net effect on the logit: about 0.12 per SD of oldpeak between the two thresholds and about 1.10 per SD above the second, i.e. a hinge in oldpeak whose position moves about 0.78 SD lower with a normal resting ECG. EVIDENCE (train, n=242): 204 active for 105/242 (58 of 121 with a normal ECG, 47 of 121 without); 290 active for 96/242; disease rate 66% when 290 is active vs 30% when not; corr(204, oldpeak) 0.91. Sensitivities (mean |d exit/d entry|): 290: oldpeak 1.82, restecg:0 0.71; 204: oldpeak 1.09, restecg:0 0.43. Ablation: removing 204 with its downstream (the whole chain) lowers train AUC 0.9376 -> 0.9317, the linear-core level; removing the steepener readout (290) alone 0.9341; removing 204's own readout 0.9380 (no loss). An affine replacement of 290 on 204 fits almost perfectly (R2 0.998, AUC 0.9376): the steepening is a hinge that the data barely probe on its flat side. Supported. Oldpeak has no direct edge and acts only here. Counter-intuitive sign reported, not explained: a normal resting ECG (restecg:0) lowers the oldpeak threshold (raising risk) inside the chain, partly offset by its direct -0.188 edge.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 1.77636e-15).

$$
\begin{aligned}
y_0 &= \operatorname{relu}\big(0.245 + 2.5x_{oldpeak} + 0.98x_{restecg:0_b}\big)\\
y_1 &= \operatorname{relu}\big(-0.468 + 0.862\operatorname{relu}\big(-0.00219\\
&\qquad + 0.94\operatorname{relu}\big(0.245 + 2.5x_{oldpeak} + 0.98x_{restecg:0_b}\big)\big)\\
&\qquad + 1.01\operatorname{relu}\big(0.245 + 2.5x_{oldpeak} + 0.98x_{restecg:0_b}\big)\big)
\end{aligned}
$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C26	direction: ST depression on exercise: 0.0 mm to 2.8 mm (through this part) *shape: hinge at 0.5 mm; 100% of patients same; verified*	16% to 60%	1.65	5.20	1.22 to 2.08	242	unclear; AUC gain 0.023 [-0.001, 0.055]	yes
C27	direction: resting ECG: a normal resting ECG to resting ECG other than a normal resting ECG (through this part) *shape: categorical; highest to lowest: a normal resting ECG, resting ECG other than a normal resting ECG*	18% to 16%	-0.36	0.70	-0.86 to 0.00	121, 121	unclear; AUC gain 0.006 [-0.009, 0.028]	yes
C28	interaction: effect of ST depression on exercise (0.0 mm to 2.8 mm) by resting ECG: resting ECG other than a normal resting ECG minus a normal resting ECG; stronger in a normal resting ECG *shape: hinge at 0.5 mm; 100% of patients same; verified*	resting ECG other than a normal resting ECG: 21% to 48%; a normal resting ECG: 16% to 60%	-0.86	0.42		121, 121	agrees; AUC gain 0.006 [-0.009, 0.028]	yes
C29	threshold: ST depression on exercise: fold at about 0.9 mm (starts) *shape: hinge at 0.5 mm; 100% of patients same; verified*	16% to 74% (typical patient with a fold)	2.30	9.93	1.86 to 2.73	242, 105	unclear; AUC gain 0.023 [-0.001, 0.055]	no
C30	subgroup: who this part acts on: 105 of 242 patients	76% to 91% (typical patient this part acts on (risk without the part, then with it))	1.03	2.80	0.03 to 2.33	105, 137	unclear; AUC gain 0.026 [-0.001, 0.062]	yes
C31	structure: its formula reproduces it on 242 training patients (largest difference 1.77636e-15)		2e-15			242	train only	yes

T.3 Age term

Annotation ann_28 (age_hinge). Reads: age.

Original hypothesis (unchanged). Age hinge (raises risk with age above a threshold, small). 1987 = ReLU(0.191 + 0.872*age), read out +0.082. Hypothesis: age's only route to the output; nothing below z-age -0.22 (just under the median age), then +0.072 logit per SD of age. EVIDENCE (train, n=242): active for 148/242; disease rate 55% when active vs 28% when not; line plot correlation 0.90; mean |sensitivity| 0.53. Removing it lowers train AUC 0.9376 -> 0.9371; an affine replacement (R2 0.80) gives 0.9376. Supported, near-negligible.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 0.0).

$$y_0 = \mathrm{relu}\big(0.191 + 0.872 x_{age}\big)$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C32	direction: age: 42 years to 66 years (through this part) *shape: hinge at 53 years; 100% of patients same; verified; one point with C33 (median fold 53 years, typical patient 53 years, 10th-90th percentile 53 years to 53 years)*	18% to 19%	0.10	1.11	0.10 to 0.10	242	disagrees; AUC gain -0.003 [-0.017, 0.004]	yes
C33	threshold: age: fold at about 53 years (starts) *shape: hinge at 53 years; 100% of patients same; verified; one point with C33 (median fold 53 years, typical patient 53 years, 10th-90th percentile 53 years to 53 years)*	18% to 20% (typical patient with a fold)	0.12	1.13	0.12 to 0.12	242, 148	disagrees; AUC gain -0.003 [-0.017, 0.004]	yes
C34	subgroup: who this part acts on: 148 of 242 patients	91% to 92% (typical patient this part acts on (risk without the part, then with it))	0.06	1.07	0.009 to 0.12	148, 94	disagrees; AUC gain -0.003 [-0.017, 0.004]	yes
C35	null: this part barely changes predictions (switched off)		0.03			242	train only	no
C36	structure: its formula reproduces it on 242 training patients (largest difference 0.0)		0.00			242	train only	yes

The network before any explanation

Figure 478 (next page, landscape) is the trained network as it is, with no parts, no splits and no inserted units: inputs at the bottom, hidden units by depth, the output at the top, every connection drawn, every input column its own node (one-hot levels included).

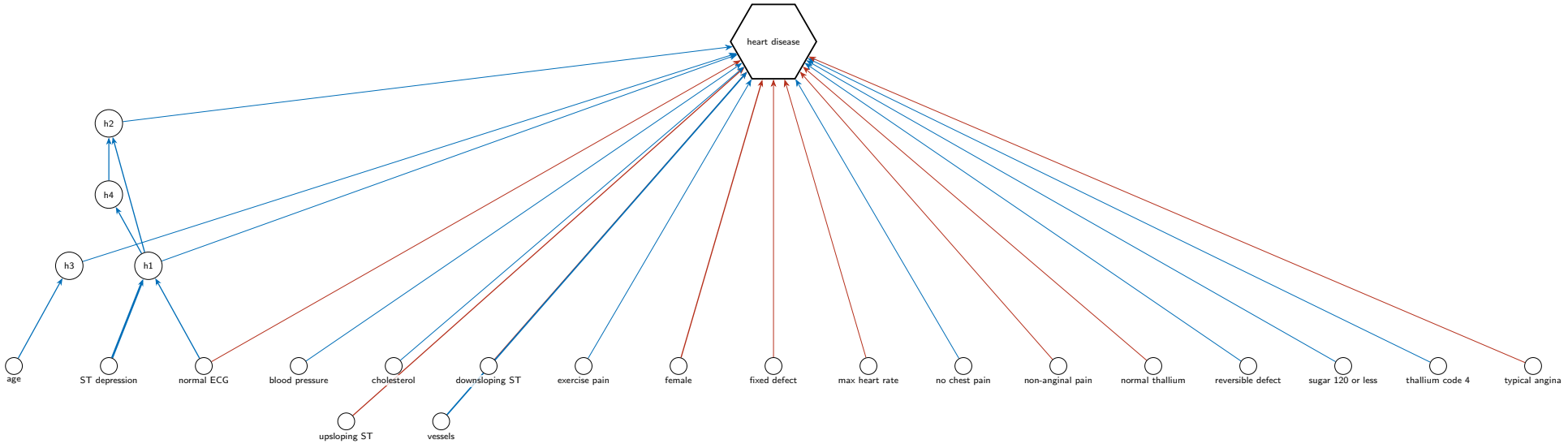


Figure 478: The network as trained, before any explanation, in full. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. Not used (no connection): male, atypical angina, sugar above 120, ST-T abnormal ECG, LVH on ECG, no exercise pain, flat ST.